

Multivariable Calculus

A Computational and Visual Introduction

Complete Book: Chapters 1–27

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June 15, 2026

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Chapter 1

From Single-Variable Calculus to Multivariable Thinking

Chapter overview

Single-variable calculus studies functions such as $f(x)$, where one input number produces one output number. Multivariable calculus studies functions whose inputs may be points, vectors, measurements, parameters, or data records:

$$f : \mathbb{R}^n \longrightarrow \mathbb{R}, \quad x = (x_1, x_2, \dots, x_n) \longmapsto f(x).$$

This change looks small, but it changes the whole subject. In one-variable calculus, we move left or right along a line. In multivariable calculus, we can move in infinitely many directions. In one-variable calculus, the graph of $y = f(x)$ is a curve. In two-variable calculus, the graph of $z = f(x, y)$ is a surface. In higher dimensions, the graph may be impossible to draw directly, but the same ideas remain: local approximation, rate of change, accumulation, optimization, and geometric structure.

This chapter is a bridge. It reviews core ideas of single-variable calculus and explains how they become the language of multivariable calculus, applied mathematics, statistics, data science, and machine learning.

Main idea

Multivariable calculus is the calculus of **many inputs**. The output may represent temperature, height, cost, probability density, potential energy, risk, error, or loss. The input may represent spatial coordinates, model parameters, data features, or the state of a system.

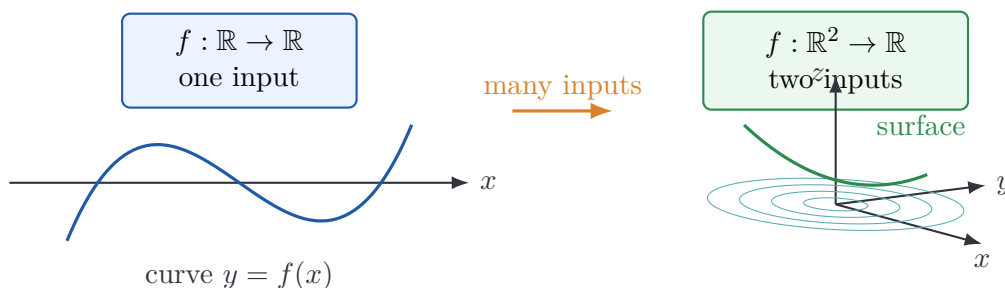


Figure 1.1: The central shift: from a one-dimensional input line to a multidimensional input space.

Learning goals

After studying this chapter, you should be able to:

1. Explain the difference between $f : \mathbb{R} \rightarrow \mathbb{R}$, $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, and $f : \mathbb{R}^n \rightarrow \mathbb{R}$.
2. Interpret a multivariable input as a point, vector, data record, parameter vector, or state.
3. Describe a function using formulas, tables, graphs, level sets, and computation.
4. Explain why derivatives become directional and why gradients are central.
5. Explain why integrals become accumulation over regions, volumes, and probability spaces.
6. Recognize the role of multivariable calculus in optimization, statistics, machine learning, and applied mathematics.

1.1 Why multivariable calculus?

In single-variable calculus, a typical function has the form

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad x \mapsto f(x).$$

For example, $f(x) = x^2$ takes one number x and returns one number x^2 . This is already powerful. We can ask for slopes, tangent lines, areas under curves, maximum values, and differential equations.

But modern quantitative problems rarely depend on only one variable. A temperature model may depend on longitude, latitude, altitude, and time:

$$T = T(x, y, z, t).$$

A statistical loss function may depend on many model parameters:

$$L = L(\beta_0, \beta_1, \dots, \beta_p).$$

A neural network may have millions or billions of parameters, and training means minimizing a function of all those parameters:

$$\text{loss} = L(w_1, w_2, \dots, w_N).$$

A probability density for two continuous random variables is a function

$$p : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad (x, y) \mapsto p(x, y),$$

and a velocity field assigns a vector to each point in space:

$$F : \mathbb{R}^3 \rightarrow \mathbb{R}^3.$$

So the core move is

$$\text{one input} \longrightarrow \text{many inputs.}$$

Bridge to MS preparation

For applied mathematics, statistics, data science, AI, and machine learning, the most important mental shift is to see formulas as functions on spaces. A loss function is a surface or landscape. A density is a surface or high-dimensional object whose total mass is one. A dynamical system is motion through a state space. A machine learning model is a map from features to predictions.

1.2 Inputs, outputs, and spaces

A function is a rule that assigns to each input exactly one output. The input set is the **domain**. The output set is the **codomain**. In calculus, the most common spaces are Euclidean spaces:

$$\mathbb{R}, \quad \mathbb{R}^2, \quad \mathbb{R}^3, \quad \mathbb{R}^n.$$

The space $\mathbb{R}^n$ consists of ordered n -tuples of real numbers:

$$\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{R}\}.$$

The same n -tuple may be interpreted in different ways depending on the model.

Mathematical object	Possible interpretation
(x, y)	A point in the plane, longitude and latitude, two features of an observation
(x, y, z)	A point in space, three physical coordinates, RGB color values
$(x_1, \dots, x_n)$	A data vector, a parameter vector, a state vector, a list of measurements
$f(x, y)$	Height, temperature, cost, density, loss, utility
$F(x, y, z)$	Vector field, force field, velocity field, gradient flow

Scalar-valued and vector-valued functions

A **scalar-valued function** has real-number output:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}.$$

Examples include

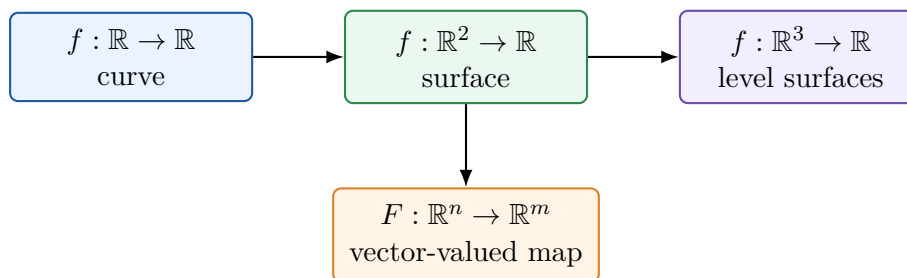
$$f(x, y) = x^2 + y^2, \quad T(x, y, z) = x^2 e^y + \ln z, \quad L(w, b) = \frac{1}{m} \sum_{i=1}^m (wx_i + b - y_i)^2.$$

A **vector-valued function** has vector output:

$$F : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

Examples include

$$r(t) = \langle \cos t, \sin t, t \rangle, \quad F(x, y) = \langle -y, x \rangle, \quad G(x, y, z) = \langle yz, xz, xy \rangle.$$



The output type determines the geometry: curves, surfaces, level sets, vector fields, transformations, and systems.

Figure 1.2: A small taxonomy of functions that appear in multivariable calculus.

1.3 A comparison with single-variable calculus

The table below gives a roadmap for the course.

Single-variable idea	Multivariable version	Modern interpretation
Number x	Vector $x = (x_1, \dots, x_n)$	Feature vector or parameter vector
Function $f(x)$	Function $f(x_1, \dots, x_n)$	Loss, density, energy, response surface
Graph $y = f(x)$	Surface $z = f(x, y)$ or high-dimensional graph	Landscape
Slope $f'(a)$	Partial derivatives and gradient $\nabla f(a)$	Sensitivity and learning signal
Tangent line	Tangent plane or linear approximation	Local linear model
Critical point $f'(a) = 0$	Critical point $\nabla f(a) = 0$	Candidate optimum or saddle point
Integral $\int_a^b f(x) dx$	Double or triple integral	Accumulation, mass, probability
Substitution	Change of variables and Jacobian	Transformation of densities and coordinates
Fundamental theorem of calculus	Green, Stokes, divergence theorem	Conservation and flow laws

The unifying theme is that calculus studies **change** and **accumulation**. Multivariable calculus studies change and accumulation when inputs live in spaces with dimension larger than one.

1.4 Four ways to represent a function

A modern calculus course should move comfortably among several representations:

1. **Verbal:** a description in words.
2. **Numerical:** a table, dataset, or sampled values.
3. **Visual:** a graph, contour plot, heatmap, vector field, or surface.
4. **Algebraic:** a formula.
5. **Computational:** a function implemented in code.

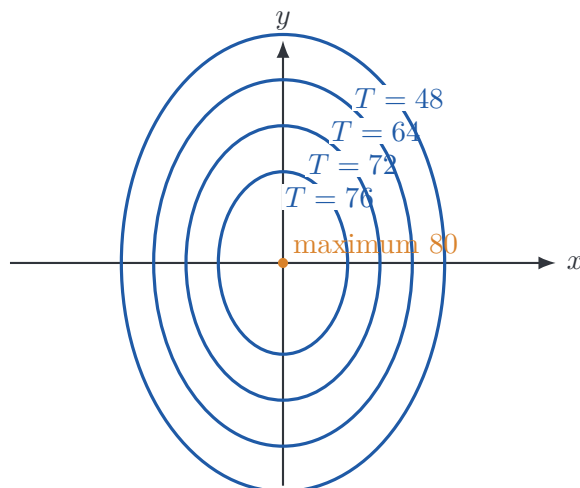
Example: temperature on a plate

Suppose the temperature on a rectangular metal plate is modeled by

$$T(x, y) = 80 - 2x^2 - y^2.$$

The same object can be represented in several ways.

Representation	Description
Verbal	Temperature is highest near the origin and decreases quadratically away from it.
Algebraic	$T(x, y) = 80 - 2x^2 - y^2$
Visual	Surface plot or contour plot
Numerical	Table of values on a grid of (x, y) points
Computational	A Python function <code>T(x, y)</code>



Level curves of $T(x, y) = 80 - 2x^2 - y^2$. Each ellipse contains points with equal temperature.

Figure 1.3: Contour plot for the temperature model. The ellipses get larger as the temperature decreases.

The contour plot does not draw the full surface in three dimensions. Instead, it draws curves where the function has constant value. These curves are called **level curves**. They are among the most important visualization tools in multivariable calculus.

1.5 Graphs, level curves, and level sets

For a function of one variable $f : \mathbb{R} \rightarrow \mathbb{R}$, the graph is a subset of $\mathbb{R}^2$:

$$\{(x, y) \in \mathbb{R}^2 : y = f(x)\}.$$

For a function of two variables $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, the graph is a subset of $\mathbb{R}^3$:

$$\{(x, y, z) \in \mathbb{R}^3 : z = f(x, y)\}.$$

For example,

$$f(x, y) = x^2 + y^2$$

has graph

$$z = x^2 + y^2,$$

an elliptic paraboloid.

A **level curve** is a curve where the output is constant:

$$f(x, y) = c.$$

For $f(x, y) = x^2 + y^2$, the level curves are

$$x^2 + y^2 = c,$$

which are circles centered at the origin when $c > 0$.

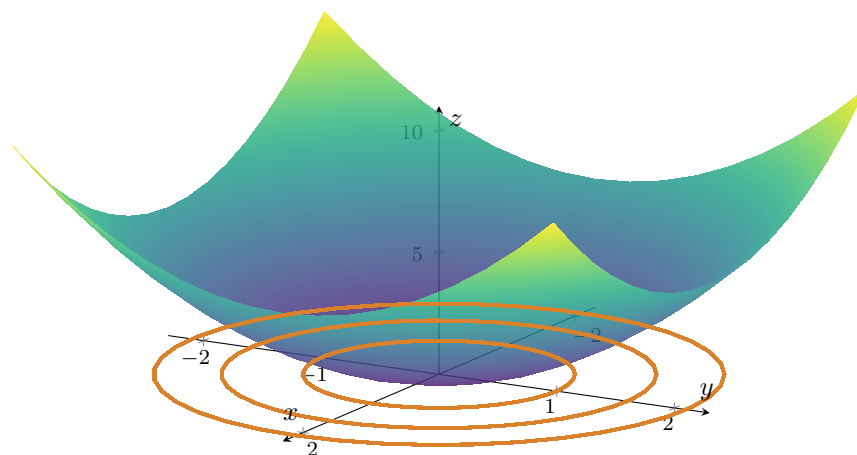


Figure 1.4: The surface $z = x^2 + y^2$ together with circular level curves projected onto the xy -plane.

Why level sets matter

In higher dimensions, full graphs are usually impossible to draw. Level sets remain useful. For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, a level set has the form

$$\{x \in \mathbb{R}^n : f(x) = c\}.$$

In statistics, level sets of a likelihood or log-likelihood show parameter values with equal fit. In machine learning, level sets of a loss function show parameter values with equal training error.

1.6 Derivatives become directional

In one-variable calculus, the derivative is the limit

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

This measures the instantaneous rate of change as x moves along the real line.

For a function $f(x, y)$, the input point (x, y) can move in many directions. From (a, b) , it can move in the x direction, the y direction, diagonally, along a curve, or along any unit vector u .

This leads to several related ideas:

- **Partial derivatives:** rates of change along coordinate directions.
- **Directional derivatives:** rates of change along arbitrary directions.
- **Gradient:** the vector that organizes all directional derivative information.
- **Jacobian:** the matrix that represents the derivative of a vector-valued function.

For a scalar-valued function $f(x, y)$, the gradient is

$$\nabla f(x, y) = \left\langle \frac{\partial f}{\partial x}(x, y), \frac{\partial f}{\partial y}(x, y) \right\rangle.$$

For $f(x, y, z)$,

$$\nabla f(x, y, z) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle.$$

The gradient points in the direction of fastest increase. This is why gradients are central in optimization and machine learning.

A first gradient computation

Let

$$f(x, y) = x^2 + 2y^2.$$

Then

$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 4y,$$

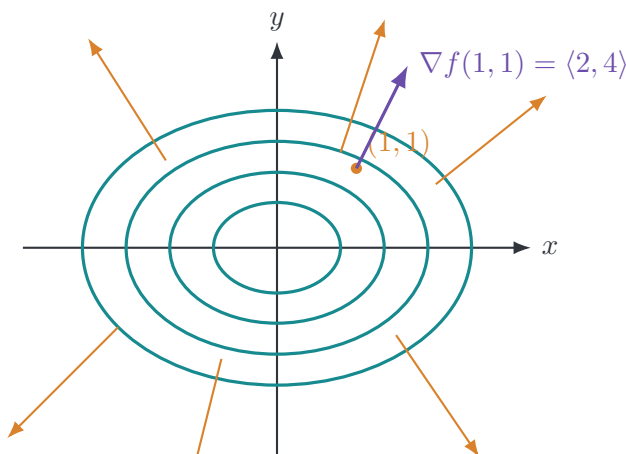
so

$$\nabla f(x, y) = \langle 2x, 4y \rangle.$$

At the point $(1, 1)$,

$$\nabla f(1, 1) = \langle 2, 4 \rangle.$$

Thus near $(1, 1)$, the function increases fastest in the direction of $\langle 2, 4 \rangle$.



For $f(x, y) = x^2 + 2y^2$, gradients are perpendicular to the elliptical level curves and point toward larger function values.

Figure 1.5: Gradient vectors organize directional change.

1.7 Integrals become accumulation over regions

In one-variable calculus, the definite integral

$$\int_a^b f(x) dx$$

accumulates values of f over an interval. In multivariable calculus, we accumulate over regions and solids:

$$\iint_R f(x, y) dA, \quad \iiint_E f(x, y, z) dV.$$

If $f(x, y) \geq 0$, then

$$\iint_R f(x, y) dA$$

can be interpreted as the volume under the surface $z = f(x, y)$ above the region R .

If $\rho(x, y, z)$ is a density, then

$$\iiint_E \rho(x, y, z) dV$$

is the mass of the object occupying the solid region E .

If $p(x, y)$ is a joint probability density, then

$$\iint_R p(x, y) dA$$

is the probability that (X, Y) lies in the region R .

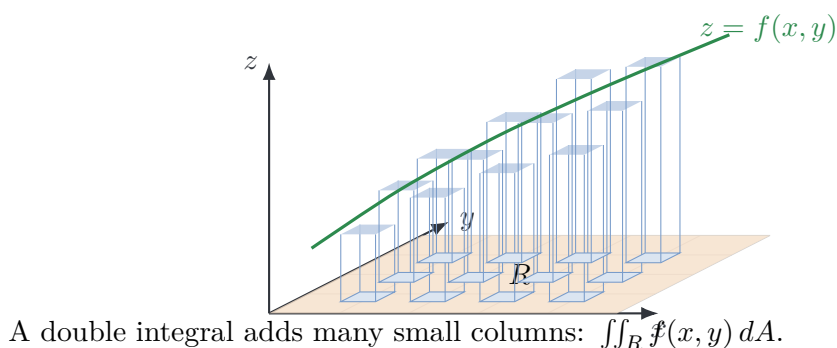


Figure 1.6: A double integral as accumulated volume above a region.

Statistics preview

For a joint density $p(x, y)$, total probability means

$$\iint_{\mathbb{R}^2} p(x, y) dA = 1.$$

Expected values, variances, covariances, and conditional distributions are all built from multivariable integrals.

1.8 The geometry of optimization

Optimization is one of the main bridges from calculus to statistics and machine learning. In one variable, we minimize or maximize a curve $f(x)$. In several variables, we minimize or maximize a surface or high-dimensional landscape $f(x_1, \dots, x_n)$.

A basic unconstrained optimization problem is

$$\min_{x \in \mathbb{R}^n} f(x).$$

For a differentiable function, candidate optima often satisfy

$$\nabla f(x) = 0.$$

In machine learning, the function f is often a **loss function**. Training a model means finding parameters that make the loss small.

Example: least-squares loss for a line

Suppose we have data points (x_i, y_i) and want to fit a line

$$\hat{y} = wx + b.$$

The mean squared error is

$$L(w, b) = \frac{1}{m} \sum_{i=1}^m (wx_i + b - y_i)^2.$$

This is a function of two variables, w and b . The input point (w, b) is not a physical point in space; it is a point in **parameter space**.

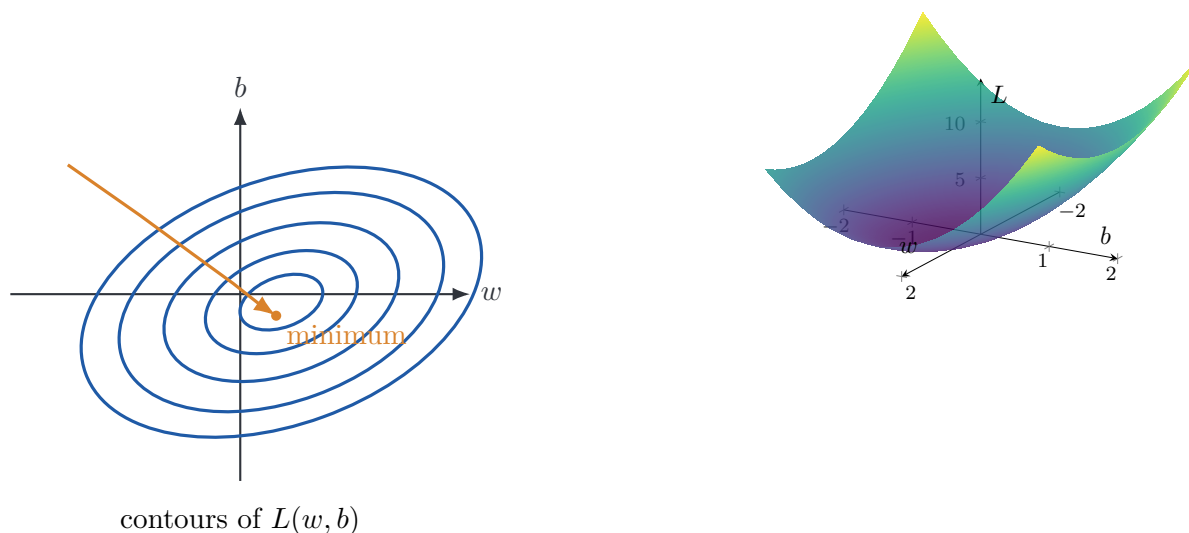


Figure 1.7: A least-squares loss surface can be viewed as a bowl-shaped landscape in parameter space.

The goal is to move downhill on this surface. In later chapters, gradient descent will update parameters by

$$(w, b) \leftarrow (w, b) - \eta \nabla L(w, b),$$

where $\eta > 0$ is a step size called the learning rate.

AI connection

Training a neural network is conceptually similar to minimizing $L(w, b)$, but the parameter vector may have millions of components. The derivative is not just a slope. It is a high-dimensional gradient, and the chain rule becomes backpropagation.

1.9 Local and global viewpoints

Calculus has both local and global ideas.

A **local** idea describes what happens near one point. Examples include tangent lines, tangent planes, derivatives, gradients, linear approximation, Hessian matrices, local maxima, local minima, and saddle points.

A **global** idea describes an entire interval, region, surface, or space. Examples include total area, total volume, total mass, total probability, line integrals around closed curves, flux through a surface, and global maximum and minimum values.

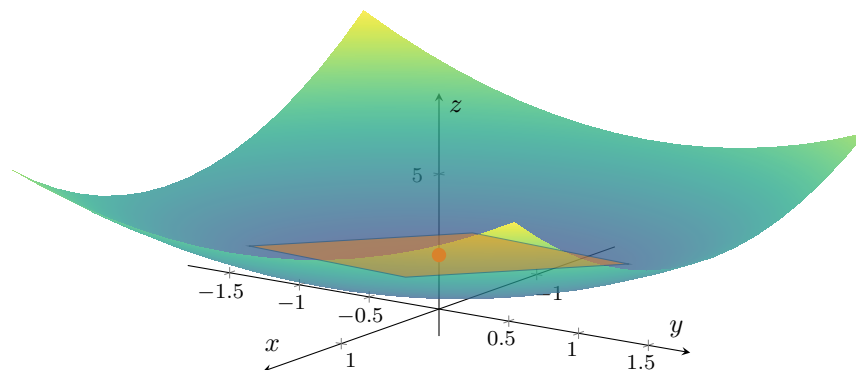


Figure 1.8: A local tangent plane approximates a surface near one point, while global questions concern the whole surface or region.

Both viewpoints are essential. A gradient is local, but gradient descent uses many local steps to solve a global problem. A density may be defined pointwise, but probabilities require integrating the density over a region. A vector field may assign a vector at each point, but flux and circulation describe global behavior.

1.10 High-dimensional thinking

Most of this book builds geometric intuition in $\mathbb{R}^2$ and $\mathbb{R}^3$. But the real goal is to understand ideas that work in $\mathbb{R}^n$.

A data point with n features can be written as

$$x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n.$$

A linear prediction model may have the form

$$\hat{y} = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n.$$

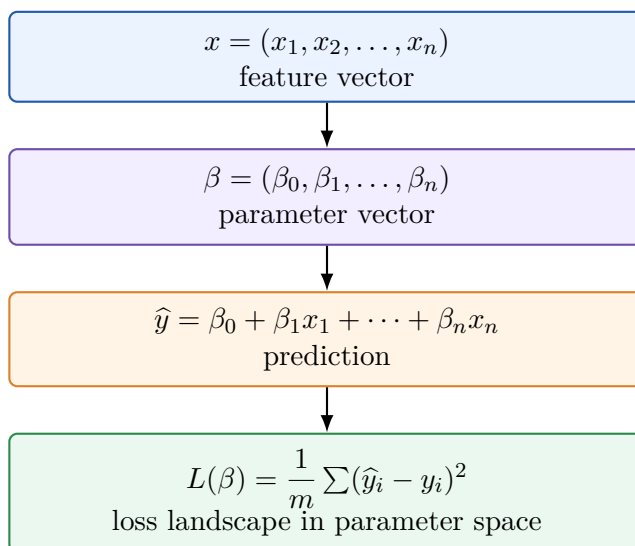
A loss function for the parameters $\beta = (\beta_0, \beta_1, \dots, \beta_n)$ may be written as

$$L(\beta) = \frac{1}{m} \sum_{i=1}^m (\beta_0 + \beta_1 x_{i1} + \dots + \beta_n x_{in} - y_i)^2.$$

This is a scalar-valued function on a high-dimensional parameter space. Even when we cannot draw the graph, we can still compute

$$\nabla L(\beta), \quad \nabla^2 L(\beta), \quad \min_{\beta} L(\beta).$$

These are multivariable calculus objects.



High-dimensional calculus uses the same objects: vectors, functions, gradients, Hessians, integrals, and approximations.

Figure 1.9: A high-dimensional model pipeline: features, parameters, predictions, and loss.

Common misconception

High-dimensional calculus is not a different subject. It is the same subject with fewer pictures and more structure. We use lower-dimensional pictures to build intuition, then rely on vectors, matrices, gradients, Jacobians, Hessians, and integrals to work in general dimensions.

1.11 Worked examples

1. Classifying functions by domain and output

Classify each function by its domain and output type.

1. $f(x) = x^2 + 1$.
2. $g(x, y) = \sqrt{x^2 + y^2}$.
3. $T(x, y, z) = x^2 e^y + \ln z$.
4. $r(t) = \langle \cos t, \sin t, t \rangle$.
5. $F(x, y) = \langle -y, x \rangle$.

Solution.

1. $f : \mathbb{R} \rightarrow \mathbb{R}$ is a scalar-valued function of one variable.
2. $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a scalar-valued function of two variables.
3. T is a scalar-valued function of three variables. Its natural domain is $z > 0$ because $\ln z$ is defined only for positive z .
4. $r : \mathbb{R} \rightarrow \mathbb{R}^3$ is a vector-valued function of one variable. It parameterizes a curve in space.
5. $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a vector field on the plane.

2. Reading a level curve

Let

$$f(x, y) = x^2 + 2y^2.$$

Describe the level curve $f(x, y) = 8$.

Solution. The level curve is

$$x^2 + 2y^2 = 8.$$

Dividing by 8 gives

$$\frac{x^2}{8} + \frac{y^2}{4} = 1.$$

This is an ellipse centered at the origin. The x -intercepts are $\pm\sqrt{8} = \pm 2\sqrt{2}$, and the y -intercepts are ± 2 .

Interpretation. On this ellipse, every point has the same function value 8. If f is a loss function, all parameter pairs on this ellipse have equal loss. If f is height, all points on this ellipse have equal elevation.

3. From a table to partial rates of change

Suppose $T(x, y)$ is temperature on a plate. The table below gives approximate values.

$T(x, y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	80	78	72
$x = 1$	76	74	68
$x = 2$	64	62	56

Estimate the rate of change in the x direction near $(1, 1)$ using centered differences.

Solution. Holding $y = 1$ fixed, use the values at $x = 0$ and $x = 2$:

$$\frac{T(2, 1) - T(0, 1)}{2 - 0} = \frac{62 - 78}{2} = -8.$$

So near $(1, 1)$, the temperature decreases by about 8 units per unit increase in x , with y held fixed.

Interpretation. This is the numerical idea behind a partial derivative. Instead of needing a formula, we estimate local change from nearby data.

4. A loss function as a multivariable function

Let one data point be $(x, y) = (2, 5)$, and consider the prediction model

$$\hat{y} = wx + b.$$

The squared error for this one point is

$$L(w, b) = (2w + b - 5)^2.$$

Find all parameter pairs (w, b) with zero loss.

Solution. Zero loss means

$$(2w + b - 5)^2 = 0.$$

Thus

$$2w + b - 5 = 0,$$

or

$$b = 5 - 2w.$$

So all points (w, b) on the line $b = 5 - 2w$ give perfect prediction for this single data point.

Interpretation. A single data point does not determine a unique line. With more data points, the loss surface changes, and the best parameters usually become more constrained.

1.12 Python mini-lab: from curves to surfaces to loss landscapes

This mini-lab previews the computational style of the book. Each full chapter can also have a separate notebook in the `labs/` folder.

Task A. A one-variable curve

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 x = np.linspace(-3, 3, 400)
5 y = x**3 - 3*x
6
7 plt.figure()
8 plt.plot(x, y)
9 plt.axhline(0, linewidth=0.8)
10 plt.axvline(0, linewidth=0.8)
11 plt.xlabel("x")
12 plt.ylabel("f(x)")
13 plt.title("One-variable function: f(x)=x^3-3x")
14 plt.show()
    
```

Task B. A two-variable contour plot

```

1 x = np.linspace(-3, 3, 250)
    
```

```

2 y = np.linspace(-3, 3, 250)
3 X, Y = np.meshgrid(x, y)
4 Z = X**2 - Y**2
5
6 plt.figure()
7 cs = plt.contour(X, Y, Z, levels=21)
8 plt.clabel(cs, inline=True, fontsize=8)
9 plt.xlabel("x")
10 plt.ylabel("y")
11 plt.gca().set_aspect("equal")
12 plt.title("Level curves of f(x,y)=x^2-y^2")
13 plt.show()

```

Task C. A first numerical gradient

For

$$f(x, y) = x^2 + 2y^2,$$

we know exactly that

$$\nabla f(x, y) = \langle 2x, 4y \rangle.$$

The following code approximates the gradient at a point.

```

1 def f_xy(x, y):
2     return x**2 + 2*y**2
3
4 def numerical_gradient(f, x0, y0, h=1e-5):
5     df_dx = (f(x0 + h, y0) - f(x0 - h, y0)) / (2*h)
6     df_dy = (f(x0, y0 + h) - f(x0, y0 - h)) / (2*h)
7     return np.array([df_dx, df_dy])
8
9 point = (1.0, 1.0)
10 print("Numerical gradient:", numerical_gradient(f_xy, *point))
11 print("Exact gradient:      ", np.array([2*point[0], 4*point[1]]))

```

Task D. A first gradient descent experiment

```

1 def f_val(x, y):
2     return x**2 + 2*y**2
3
4 def grad_f(x, y):
5     return np.array([2*x, 4*y])
6
7 eta = 0.12
8 point = np.array([2.5, 2.0])
9 path = [point.copy()]
10
11 for _ in range(25):
12     point = point - eta * grad_f(point[0], point[1])
13     path.append(point.copy())
14

```

```

15 path = np.array(path)
16
17 x = np.linspace(-3, 3, 250)
18 y = np.linspace(-3, 3, 250)
19 X, Y = np.meshgrid(x, y)
20 Z = f_val(X, Y)
21
22 plt.figure()
23 plt.contour(X, Y, Z, levels=20)
24 plt.plot(path[:, 0], path[:, 1], marker="o")
25 plt.xlabel("x")
26 plt.ylabel("y")
27 plt.gca().set_aspect("equal")
28 plt.title("Gradient descent for f(x,y)=x^2+2y^2")
29 plt.show()

```

1.13 Skills checklist

By the end of this chapter, you should be able to do the following.

Skill	Can you do it?
Identify the domain and codomain of a function	☐
Distinguish scalar-valued and vector-valued functions	☐
Explain why $f(x, y)$ has a surface graph	☐
Interpret a level curve $f(x, y) = c$	☐
Explain why multivariable derivatives depend on direction	☐
Connect gradients to steepest increase and optimization	☐
Connect double integrals to volume, mass, and probability	☐
Interpret a loss function as a function on parameter space	☐
Use Python to plot curves, contours, surfaces, and simple gradients	☐

1.14 Practice problems

Conceptual problems

1. Give three examples of functions of one variable and three examples of functions of several variables. For each example, identify the input and output.
2. Explain the difference between a point in physical space and a point in parameter space.
3. Why can $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ not be graphed in ordinary three-dimensional space in the same way as $f : \mathbb{R}^2 \rightarrow \mathbb{R}$?
4. Explain why a contour map is useful even when a surface plot is available.
5. In your own words, explain why derivatives become more complicated when the input has more than one dimension.

Computation problems

6. Classify each function by domain and codomain:
 - (a) $f(x) = e^{-x^2}$
 - (b) $g(x, y) = x^2 - y^2$
 - (c) $h(x, y, z) = \sqrt{x^2 + y^2 + z^2}$
 - (d) $r(t) = \langle t, t^2, t^3 \rangle$
 - (e) $F(x, y) = \langle x + y, x - y \rangle$
7. For $f(x, y) = x^2 + y^2$, describe the level curves $f(x, y) = 1$, $f(x, y) = 4$, and $f(x, y) = 9$.
8. For $f(x, y) = x^2 + 4y^2$, describe the level curve $f(x, y) = 16$.
9. Compute $\nabla f(x, y)$ for $f(x, y) = 3x^2 + xy + y^2$.
10. Compute $\nabla T(x, y, z)$ for $T(x, y, z) = x^2 e^y + \ln z$.
11. Let $L(w, b) = (2w + b - 5)^2 + (w + b - 3)^2$. Compute $L(1, 1)$ and $L(2, 1)$.
12. Let $p(x, y) = C e^{-(x^2 + y^2)}$. Explain what condition C must satisfy if p is to be a probability density on $\mathbb{R}^2$.

Python problems

13. Modify the contour plot for $f(x, y) = x^2 + y^2$ to plot $f(x, y) = x^2 + 4y^2$. Describe how the level curves change.
14. Write a Python function for $f(x, y) = \sin(xy)$. Plot its contour map on $[-3, 3] \times [-3, 3]$.
15. Numerically approximate the gradient of $f(x, y) = \sin x + \cos y$ at $(0, 0)$ using finite differences.
16. Create a least-squares loss surface for data generated from $y = 2x + 1$ with small random noise.
17. Run gradient descent on $f(x, y) = x^2 + y^2$ from three different starting points. Plot all three paths.

Discussion and writing problems

18. Write a short paragraph explaining the phrase: “A loss function is a landscape.”
19. Write a short paragraph explaining why multiple integrals are important for probability and statistics.
20. Choose one application area: physics, economics, biology, statistics, AI, or machine learning. Describe one natural function of several variables in that area.

1.15 Chapter summary

Single-variable calculus studies functions with one input. Multivariable calculus studies functions whose inputs live in spaces such as $\mathbb{R}^2$, $\mathbb{R}^3$, and $\mathbb{R}^n$. This makes the geometry richer and the applications broader.

The main ideas of the book are already visible:

- A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ can represent a surface, density, loss, risk, energy, or potential.
- When there are many input directions, derivatives become partial derivatives, directional derivatives, gradients, Jacobians, and Hessians.
- Integrals become accumulation over regions, solids, surfaces, and probability spaces.
- Optimization becomes the study of landscapes in parameter space.
- Python helps us visualize, approximate, and experiment with objects that are difficult or impossible to draw by hand.

The next chapters build the geometric foundation: Euclidean space, vectors, dot products, projections, lines, planes, cross products, and curves. These objects are the language needed for functions, derivatives, integrals, and vector calculus.

Chapter 2

Euclidean Space $\mathbb{R}^n$

Chapter overview

Multivariable calculus begins by changing what an input is. In single-variable calculus, an input is one number x . In multivariable calculus, an input may be a point (x, y) in the plane, a point (x, y, z) in space, or a long vector

$$(x_1, x_2, \dots, x_n)$$

whose components store measurements, features, parameters, or decisions. The common home for all of these objects is **Euclidean space**.

This chapter builds the geometric language used throughout the book. We will study points, coordinates, distance, projections, spheres, balls, and equations that describe regions in two, three, and higher dimensions. The same ideas will later support tangent planes, gradients, multiple integrals, optimization, probability densities, and loss functions.

Main idea

Euclidean space $\mathbb{R}^n$ is the setting in which numbers become geometry. A point can be a location, a vector can be a displacement, a distance can measure similarity, and an equation or inequality can describe a geometric object or a data region.

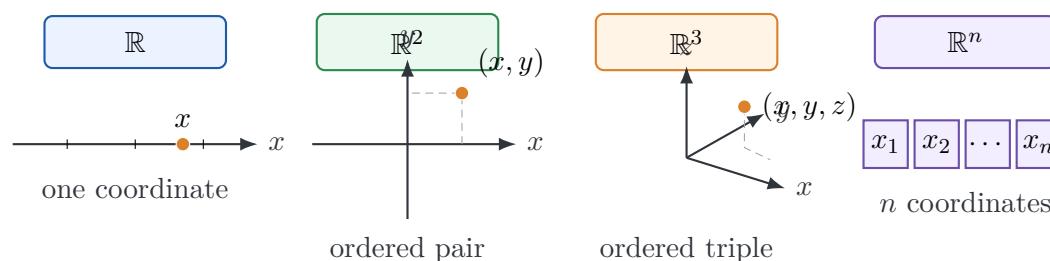


Figure 2.1: The same coordinate idea extends from the real line to the plane, space, and high-dimensional feature vectors.

Learning goals

After studying this chapter, you should be able to:

1. describe $\mathbb{R}^2$, $\mathbb{R}^3$, and $\mathbb{R}^n$ as spaces of ordered tuples;
2. distinguish between coordinates, points, vectors, and displacement;
3. compute distances in $\mathbb{R}^n$;
4. project points in $\mathbb{R}^3$ onto coordinate planes;
5. recognize how the same equation may describe different objects in different ambient spaces;
6. write equations of spheres, balls, and common regions;
7. explain why high-dimensional Euclidean geometry is essential for statistics, data science, artificial intelligence, and applied mathematics.

2.1 From the real line to $\mathbb{R}^n$

The real line $\mathbb{R}$ consists of real numbers. A point on the line is described by one coordinate:

$$x \in \mathbb{R}.$$

The plane $\mathbb{R}^2$ consists of ordered pairs:

$$(x, y) \in \mathbb{R}^2.$$

Three-dimensional space $\mathbb{R}^3$ consists of ordered triples:

$$(x, y, z) \in \mathbb{R}^3.$$

More generally, n -dimensional Euclidean space is the set

$$\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{R} \text{ for } i = 1, \dots, n\}.$$

Each number x_i is called a **coordinate** or **component**.

Euclidean space

For a positive integer n , Euclidean n -space is the set of ordered n -tuples of real numbers:

$$\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{R}\}.$$

The origin is

$$0 = (0, 0, \dots, 0).$$

A useful mental model

Think of $\mathbb{R}^n$ as a universal coordinate system. In geometry, a point in $\mathbb{R}^3$ may be a location in space. In statistics, a point in $\mathbb{R}^n$ may be one observation with n measured features. In machine learning, a point in $\mathbb{R}^n$ may be a vector of model weights.

Examples

- $(3, -2)$ is a point in $\mathbb{R}^2$.
- $(1, 0, -4)$ is a point in $\mathbb{R}^3$.
- $(2, -1, 0, 5)$ is a point in $\mathbb{R}^4$.
- $(x_1, x_2, \dots, x_{100})$ can represent a data record with 100 numerical features.
- $(w_0, w_1, \dots, w_p)$ can represent the parameters of a linear regression model.

We can draw $\mathbb{R}^2$ and $\mathbb{R}^3$, but we usually cannot directly draw $\mathbb{R}^n$ for large n . The power of Euclidean geometry is that the formulas still work.

2.2 Points, vectors, and displacement

A point records position. A vector records displacement, direction, or magnitude. The same ordered tuple may be used for either purpose, depending on context.

If

$$A = (a_1, a_2, \dots, a_n), \quad B = (b_1, b_2, \dots, b_n),$$

then the displacement from A to B is

$$\overrightarrow{AB} = B - A = (b_1 - a_1, b_2 - a_2, \dots, b_n - a_n).$$

Displacement vector

The displacement vector from A to B is the vector that moves the initial point A to the terminal point B :

$$\overrightarrow{AB} = B - A.$$

For example, if $A = (1, -2, 3)$ and $B = (2, 1, 5)$, then

$$\overrightarrow{AB} = (2 - 1, 1 - (-2), 5 - 3) = (1, 3, 2).$$

Point versus vector

The notation $(1, 3, 2)$ can mean either a point or a vector. As a point, it is a location. As a vector, it is a displacement. The algebra is the same, but the interpretation is different.

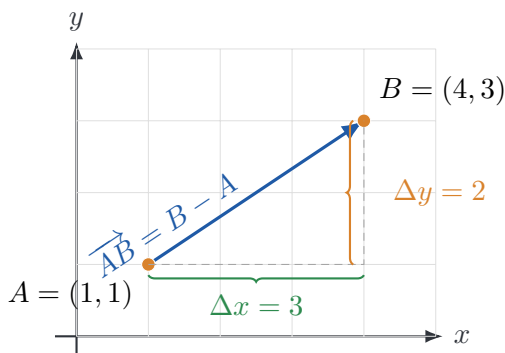


Figure 2.2: A displacement vector from A to B in the plane.

2.3 Coordinates, axes, planes, and octants

In $\mathbb{R}^2$, the coordinate axes are the x -axis and the y -axis. The two axes divide the plane into four quadrants.

In $\mathbb{R}^3$, the coordinate axes are the x -, y -, and z -axes. We use a right-handed coordinate system. The three coordinate planes are

$$xy\text{-plane} : z = 0,$$

$$yz\text{-plane} : x = 0,$$

$$xz\text{-plane} : y = 0.$$

The coordinate planes divide $\mathbb{R}^3$ into eight octants. The first octant is

$$x \geq 0, \quad y \geq 0, \quad z \geq 0.$$

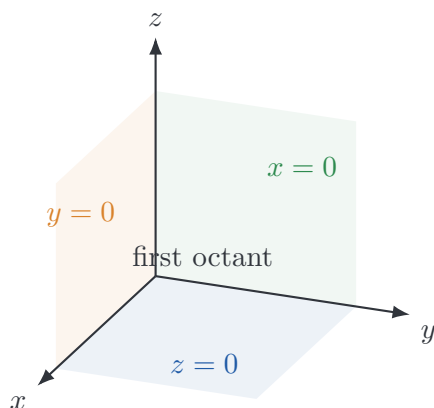


Figure 2.3: The coordinate planes in $\mathbb{R}^3$ and the first octant.

Projection onto coordinate planes

For a point $P = (a, b, c)$ in $\mathbb{R}^3$:

$$\text{projection onto the } xy\text{-plane} = (a, b, 0),$$

$$\text{projection onto the } yz\text{-plane} = (0, b, c),$$

$$\text{projection onto the } xz\text{-plane} = (a, 0, c).$$

The idea is simple: to project onto a coordinate plane, set the missing coordinate equal to zero.

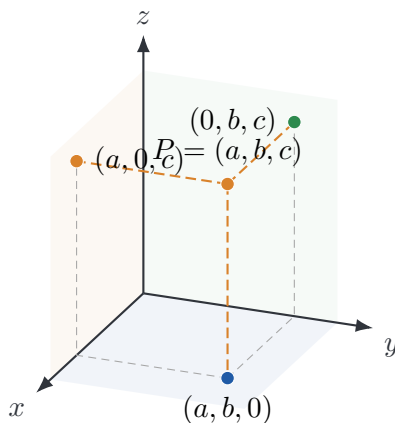


Figure 2.4: A point in $\mathbb{R}^3$ and its projections onto the coordinate planes.

Worked example: projections

Let $P = (5, -2, 7)$. Then

$$P_{xy} = (5, -2, 0),$$

$$P_{yz} = (0, -2, 7),$$

$$P_{xz} = (5, 0, 7).$$

The xy -projection removes height. The yz -projection removes the x -coordinate. The xz -projection removes the y -coordinate.

2.4 Distance in Euclidean space

Distance is the most important measurement in Euclidean geometry. It is also fundamental in statistics and machine learning, where distance measures similarity, error, residual size, and model change.

For points

$$A = (a_1, a_2, \dots, a_n), \quad B = (b_1, b_2, \dots, b_n)$$

in $\mathbb{R}^n$, the Euclidean distance between A and B is

$$d(A, B) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \dots + (a_n - b_n)^2}.$$

Equivalently,

$$d(A, B) = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}.$$

In $\mathbb{R}^3$, this becomes

$$d(A, B) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}.$$

Euclidean distance

For $A = (a_1, \dots, a_n)$ and $B = (b_1, \dots, b_n)$ in $\mathbb{R}^n$,

$$d(A, B) = \|A - B\| = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}.$$

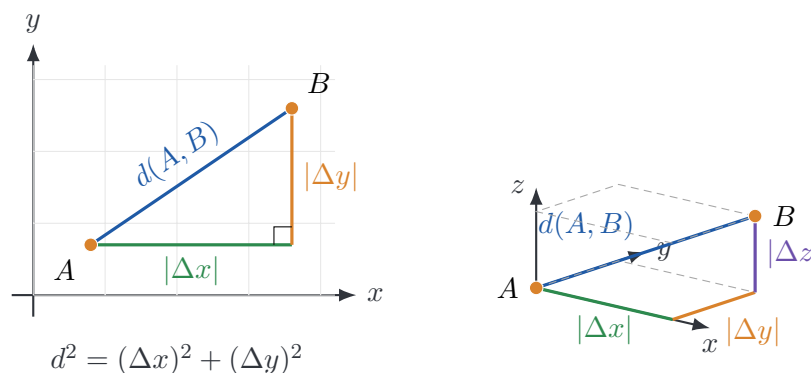


Figure 2.5: Distance in $\mathbb{R}^2$ and $\mathbb{R}^3$ comes from the Pythagorean theorem.

Worked example: distance in $\mathbb{R}^3$

Find the distance from $P = (1, 2, 3)$ to $Q = (3, 4, 2)$.

$$\begin{aligned} d(P, Q) &= \sqrt{(1-3)^2 + (2-4)^2 + (3-2)^2} \\ &= \sqrt{4 + 4 + 1} \\ &= 3. \end{aligned}$$

Worked example: distance in $\mathbb{R}^4$

Find the distance between

$$a = (2, -1, 0, 4), \quad b = (-1, 3, 5, 2).$$

Then

$$\begin{aligned} d(a, b) &= \sqrt{(2-(-1))^2 + (-1-3)^2 + (0-5)^2 + (4-2)^2} \\ &= \sqrt{3^2 + (-4)^2 + (-5)^2 + 2^2} \\ &= \sqrt{54} = 3\sqrt{6}. \end{aligned}$$

2.5 Spheres and balls

A **sphere** is the set of points at a fixed distance from a center. A **ball** is the set of points within a fixed distance from a center.

In $\mathbb{R}^3$, the sphere with center $C = (h, k, \ell)$ and radius r has equation

$$(x - h)^2 + (y - k)^2 + (z - \ell)^2 = r^2.$$

If the center is the origin, this becomes

$$x^2 + y^2 + z^2 = r^2.$$

In $\mathbb{R}^n$, the sphere with center $c = (c_1, \dots, c_n)$ and radius r is

$$\sum_{i=1}^n (x_i - c_i)^2 = r^2.$$

The corresponding closed ball is

$$\sum_{i=1}^n (x_i - c_i)^2 \leq r^2,$$

and the corresponding open ball is

$$\sum_{i=1}^n (x_i - c_i)^2 < r^2.$$

Sphere and ball

Let $c \in \mathbb{R}^n$ and $r > 0$.

$$S_r(c) = \{x \in \mathbb{R}^n : d(x, c) = r\},$$

$$B_r(c) = \{x \in \mathbb{R}^n : d(x, c) < r\},$$

$$\overline{B}_r(c) = \{x \in \mathbb{R}^n : d(x, c) \leq r\}.$$

The set $S_r(c)$ is the boundary sphere, $B_r(c)$ is the open ball, and $\overline{B}_r(c)$ is the closed ball.

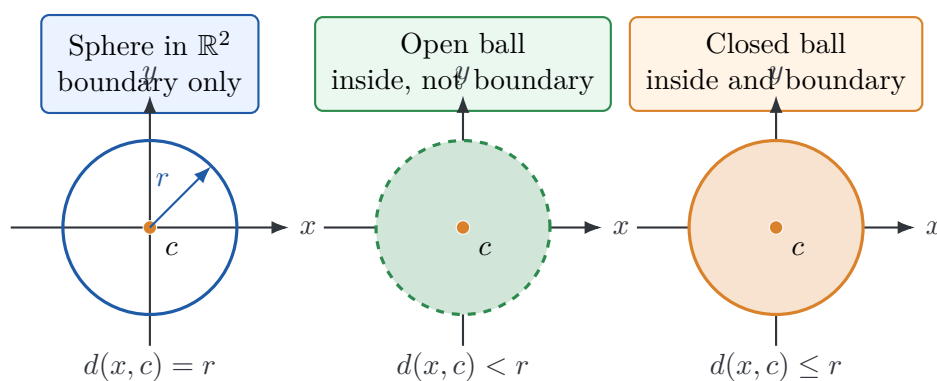


Figure 2.6: The difference between a sphere, an open ball, and a closed ball. In $\mathbb{R}^2$, the sphere is a circle.

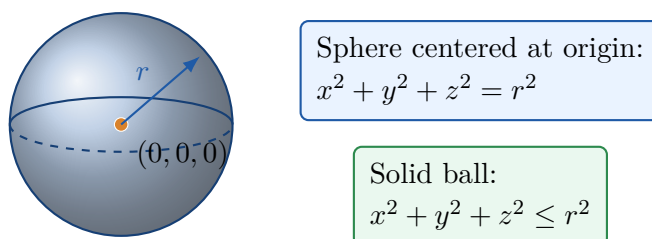


Figure 2.7: A sphere centered at the origin in $\mathbb{R}^3$ and the corresponding solid ball.

Boundary versus region

The equation $x^2 + y^2 + z^2 = 9$ describes only the boundary sphere of radius 3. The inequality $x^2 + y^2 + z^2 \leq 9$ describes the solid ball inside and including the boundary.

Worked example: equation of a sphere

Find the equation of the sphere with center $(2, -5, 1)$ and radius 2.

Using $(x - h)^2 + (y - k)^2 + (z - \ell)^2 = r^2$, we get

$$(x - 2)^2 + (y + 5)^2 + (z - 1)^2 = 4.$$

Worked example: diameter endpoints

Find the equation of the sphere for which one diameter has endpoints

$$A = (2, -1, 1), \quad B = (-2, -3, 2).$$

The center is the midpoint of the diameter:

$$C = \frac{A + B}{2} = \left(0, -2, \frac{3}{2}\right).$$

The diameter length is

$$|AB| = \sqrt{(2 - (-2))^2 + (-1 - (-3))^2 + (1 - 2)^2} = \sqrt{21}.$$

Therefore $r = \sqrt{21}/2$, and the equation is

$$x^2 + (y + 2)^2 + \left(z - \frac{3}{2}\right)^2 = \frac{21}{4}.$$

2.6 Equations depend on the ambient space

An equation does not describe an object until we know the space in which it lives. For example:

- In $\mathbb{R}^2$, the equation $y = 2$ is a line.
- In $\mathbb{R}^3$, the equation $y = 2$ is a plane parallel to the xz -plane.
- In $\mathbb{R}^3$, the equation $z = 2$ is a plane parallel to the xy -plane.

Similarly, the equation

$$x^2 + y^2 = 4$$

has different meanings in different spaces. In $\mathbb{R}^2$, it is a circle of radius 2 centered at the origin. In $\mathbb{R}^3$, it is a cylinder of radius 2 around the z -axis, because z is free. If we add $z = 3$, then

$$x^2 + y^2 = 4, \quad z = 3$$

describes a circle of radius 2 lying in the horizontal plane $z = 3$.

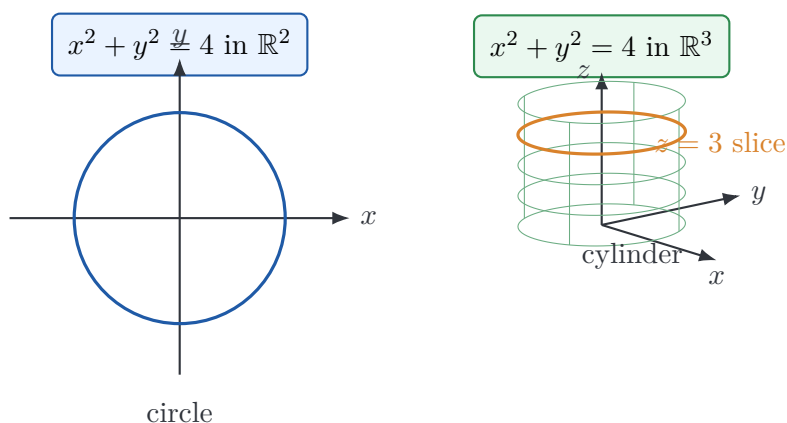


Figure 2.8: The same equation can describe different objects in different ambient spaces.

Worked example: the surface $x = y$

Describe the surface $x = y$ in $\mathbb{R}^3$.

The equation $x = y$ places one restriction on three variables. The variable z is free. Therefore the surface is a plane through the z -axis. Points on the plane have the form

$$(x, y, z) = (t, t, s),$$

where s and t are real numbers.

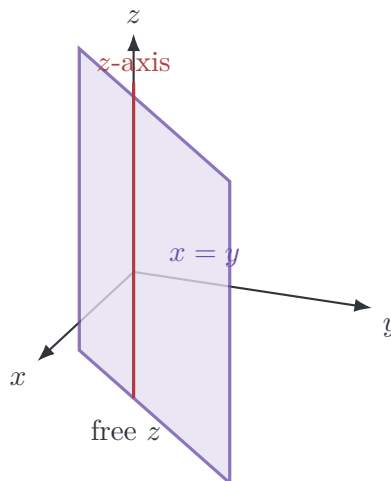


Figure 2.9: The equation $x = y$ describes a plane in $\mathbb{R}^3$.

2.7 Regions described by inequalities

Equations describe boundaries. Inequalities describe regions.

The inequality

$$x^2 + y^2 + z^2 \leq 9$$

describes the solid ball of radius 3 centered at the origin. The inequality

$$x^2 + y^2 + z^2 > 9$$

describes all points outside that ball.

Worked example: completing the square

Describe the region

$$x^2 + y^2 + z^2 > 2z.$$

Move all terms to one side and complete the square in z :

$$\begin{aligned} x^2 + y^2 + z^2 &> 2z, \\ x^2 + y^2 + z^2 - 2z &> 0, \\ x^2 + y^2 + (z - 1)^2 - 1 &> 0, \\ x^2 + y^2 + (z - 1)^2 &> 1. \end{aligned}$$

This is the set of all points whose distance from $(0, 0, 1)$ is greater than 1. Thus the region is the outside of the sphere of radius 1 centered at $(0, 0, 1)$. The boundary sphere itself is not included because the inequality is strict.

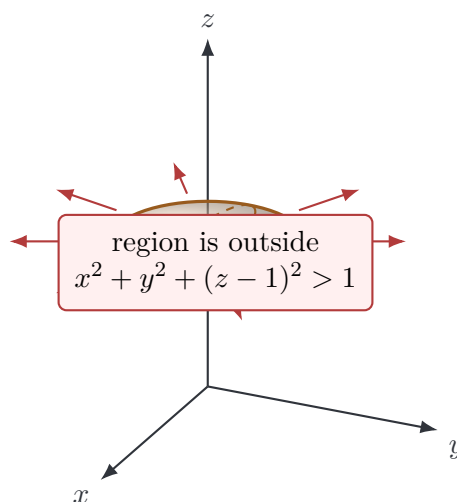


Figure 2.10: Boundary sphere for the region $x^2 + y^2 + z^2 > 2z$. The arrows indicate the outside region.

Common mistake

The inequality $x^2 + y^2 + (z - 1)^2 > 1$ is not the sphere. It is the outside region. The sphere is the boundary where equality holds.

2.8 Open balls, closed balls, and neighborhoods

In one-variable calculus, limits are described using intervals around a point. In multivariable calculus, the natural replacement is a ball.

For $a \in \mathbb{R}^n$ and $r > 0$, the **open ball** centered at a with radius r is

$$B_r(a) = \{x \in \mathbb{R}^n : d(x, a) < r\}.$$

The **closed ball** centered at a with radius r is

$$\overline{B}_r(a) = \{x \in \mathbb{R}^n : d(x, a) \leq r\}.$$

The boundary is the sphere

$$S_r(a) = \{x \in \mathbb{R}^n : d(x, a) = r\}.$$

These objects are essential for defining limits, continuity, open sets, and local behavior.

Preview of limits

When we say (x, y) approaches (a, b) , we mean the distance $\sqrt{(x - a)^2 + (y - b)^2}$ approaches 0. This is why Euclidean distance appears immediately in the definition of a multivariable limit.

2.9 High-dimensional Euclidean space and data geometry

A data set with m observations and n features can be represented as a matrix

$$X = \begin{bmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_m^T & - \end{bmatrix},$$

where each row $x_i \in \mathbb{R}^n$ is a data vector.

Euclidean distance is often used to measure similarity:

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^n (x_{ik} - x_{jk})^2}.$$

This idea appears in nearest-neighbor classification, clustering, anomaly detection, least-squares regression, gradient-based optimization, and measuring error between predicted and observed vectors.

However, high-dimensional geometry can behave differently from low-dimensional geometry. Random points in $[0, 1]^n$ tend to become farther apart on average as n increases, and distances can also concentrate.

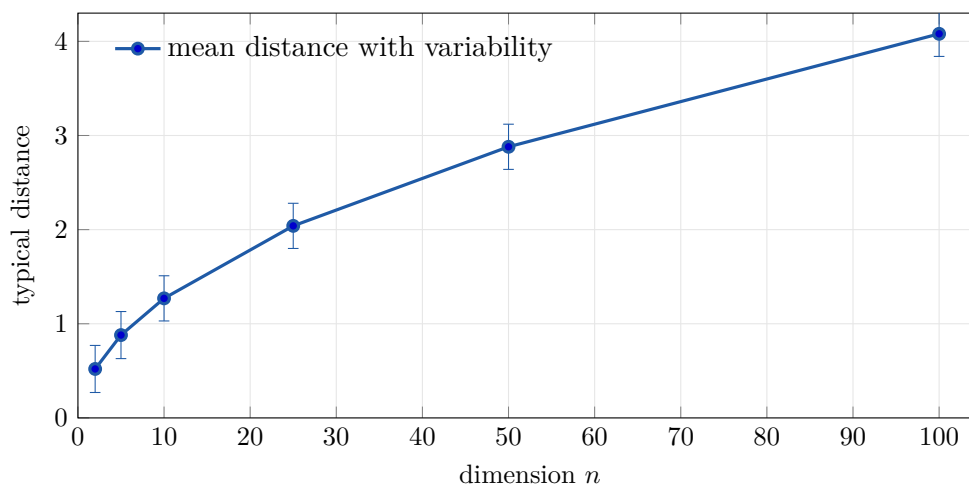


Figure 2.11: Typical distances between random points increase and become more concentrated as dimension grows.

Connection to artificial intelligence

Modern systems represent images, text, users, and documents as high-dimensional vectors. Euclidean geometry gives the first language for comparing such objects, but high-dimensional geometry also warns us that intuition from the plane may be misleading.

2.10 Standard geometric objects in $\mathbb{R}^n$

The following objects appear repeatedly in multivariable calculus and applied mathematics.

Object	Description	Typical equation or inequality
Line in $\mathbb{R}^2$	one-dimensional straight set	$ax + by = c$
Plane in $\mathbb{R}^3$	two-dimensional flat surface	$ax + by + cz = d$
Hyperplane in $\mathbb{R}^n$	codimension-one flat set	$a_1x_1 + \cdots + a_nx_n = b$
Sphere	fixed distance from a center	$\sum_i (x_i - c_i)^2 = r^2$
Ball	distance at most r from a center	$\sum_i (x_i - c_i)^2 \leq r^2$
Cylinder in $\mathbb{R}^3$	one variable free	$x^2 + y^2 = r^2$
First octant	nonnegative coordinates in $\mathbb{R}^3$	$x \geq 0, y \geq 0, z \geq 0$

Table 2.1: Common geometric objects and their algebraic descriptions.

Hyperplanes and machine learning

A hyperplane in $\mathbb{R}^n$ has equation

$$w_1x_1 + w_2x_2 + \cdots + w_nx_n = b.$$

In machine learning, this is the basic form of a linear decision boundary. For example, a classifier may assign one label when

$$w \cdot x - b > 0$$

and another label when

$$w \cdot x - b < 0.$$

This is one reason multivariable calculus and linear algebra are both central to data science.

2.11 Worked examples

Example: detecting collinearity using distances

Determine whether the points

$$A = (2, 4, 2), \quad B = (3, 7, -2), \quad C = (1, 3, 3)$$

lie on a straight line.

Compute the three distances:

$$\begin{aligned} |AB| &= \sqrt{(3-2)^2 + (7-4)^2 + (-2-2)^2} = \sqrt{26}, \\ |AC| &= \sqrt{(1-2)^2 + (3-4)^2 + (3-2)^2} = \sqrt{3}, \\ |BC| &= \sqrt{(1-3)^2 + (3-7)^2 + (3-(-2))^2} = \sqrt{45}. \end{aligned}$$

For three points to lie on a line, one of the three distances must equal the sum of the other two. Here this does not happen. Therefore the points do not lie on a straight line.

A more algebraic method is to compare displacement vectors:

$$\overrightarrow{AB} = (1, 3, -4), \quad \overrightarrow{AC} = (-1, -1, 1).$$

These are not scalar multiples of each other, so the three points are not collinear.

Example: intersections with coordinate planes

Consider the sphere

$$(x-2)^2 + (y+5)^2 + (z-1)^2 = 4.$$

Describe its intersection with the coordinate planes.

For the xy -plane, set $z = 0$:

$$(x-2)^2 + (y+5)^2 + 1 = 4.$$

Thus

$$(x-2)^2 + (y+5)^2 = 3,$$

which is a circle in the plane $z = 0$.

For the xz -plane, set $y = 0$:

$$(x-2)^2 + 25 + (z-1)^2 = 4.$$

This has no solution, so the sphere does not intersect the xz -plane.

For the yz -plane, set $x = 0$:

$$4 + (y + 5)^2 + (z - 1)^2 = 4.$$

Thus

$$(y + 5)^2 + (z - 1)^2 = 0,$$

so the intersection is the single point $(0, -5, 1)$.

Check the signs

The center of $(x - 2)^2 + (y + 5)^2 + (z - 1)^2 = 4$ is $(2, -5, 1)$, not $(2, 5, 1)$. The sign inside the square is opposite the coordinate of the center.

2.12 Computational practice: nearest points

Distance is the basic ingredient in nearest-neighbor search. Given a query point and several data points, the nearest neighbor is the point with smallest Euclidean distance.

Listing 2.1: Nearest-neighbor search using Euclidean distance.

```

1 import numpy as np
2
3 rng = np.random.default_rng(7)
4 X = rng.normal(size=(20, 2))
5 query = np.array([0.75, 0.25])
6
7 distances = np.linalg.norm(X - query, axis=1)
8 nearest_index = np.argmin(distances)
9 nearest_point = X[nearest_index]
10
11 nearest_index, nearest_point, distances[nearest_index]
```

The same idea extends to high-dimensional vectors. For example, two images may be compared by first converting them to vectors and then measuring the distance between those vectors.

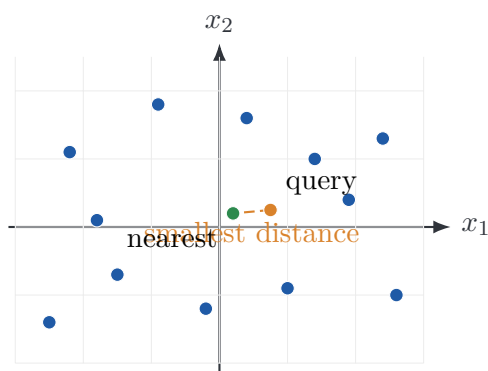


Figure 2.12: Nearest-neighbor search compares a query point to data points using Euclidean distance.

2.13 Common mistakes and how to avoid them

Mistake 1: confusing a circle and a sphere

The equation

$$x^2 + y^2 = 4$$

is a circle in $\mathbb{R}^2$ but a cylinder in $\mathbb{R}^3$. Always identify the ambient space.

Mistake 2: forgetting free variables

The equation $z = 2$ in $\mathbb{R}^3$ does not describe one point. The variables x and y are free, so it describes a plane.

Mistake 3: using equality when the problem asks for a region

The equation

$$x^2 + y^2 + z^2 = 1$$

describes a sphere. The inequality

$$x^2 + y^2 + z^2 \leq 1$$

describes a solid ball.

Mistake 4: sign errors in sphere centers

The sphere

$$(x - h)^2 + (y - k)^2 + (z - \ell)^2 = r^2$$

has center (h, k, ℓ) . Thus $(y + 5)^2$ corresponds to $k = -5$.

2.14 Summary

Concept	Formula or meaning
Euclidean space	$\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{R}\}$
Displacement	$\overrightarrow{AB} = B - A$
Distance	$d(A, B) = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}$
Coordinate projections	$P_{xy} = (a, b, 0)$, $P_{yz} = (0, b, c)$, $P_{xz} = (a, 0, c)$ for $P = (a, b, c)$
Sphere in $\mathbb{R}^3$	$(x - h)^2 + (y - k)^2 + (z - \ell)^2 = r^2$
Closed ball in $\mathbb{R}^n$	$\sum_{i=1}^n (x_i - c_i)^2 \leq r^2$
Open ball in $\mathbb{R}^n$	$\sum_{i=1}^n (x_i - c_i)^2 < r^2$
Hyperplane in $\mathbb{R}^n$	$a_1 x_1 + \dots + a_n x_n = b$

Table 2.2: Core formulas from the chapter.

Practice problems

1. Find the displacement vector from $A = (-1, 4, 2)$ to $B = (3, -2, 5)$.
2. Find the projections of $P = (4, -3, 6)$ onto the xy -, yz -, and xz -planes.
3. Compute the distance between $A = (2, 0, -1)$ and $B = (-1, 4, 3)$.
4. Find the equation of the sphere with center $(-2, 1, 4)$ and radius 5.
5. Find the center and radius of the sphere $(x + 3)^2 + (y - 2)^2 + (z + 1)^2 = 16$.
6. Describe the object $x^2 + y^2 = 9$ in $\mathbb{R}^2$ and in $\mathbb{R}^3$.
7. Describe the region $x^2 + y^2 + z^2 \leq 4z$.
8. Decide whether $A = (1, 2, 3)$, $B = (3, 6, 9)$, and $C = (-1, -2, -3)$ are collinear.
9. Write a sentence explaining why an equation with one constraint in $\mathbb{R}^3$ often describes a surface rather than a curve.
10. Give a data-science example where a point in $\mathbb{R}^n$ is not a physical location.

Chapter 3

Vectors, Norms, Dot Products, and Projections

Chapter overview

Chapter 2 introduced Euclidean space as the natural home for points such as (x, y) , (x, y, z) , and $(x_1, \dots, x_n)$. This chapter develops the algebra and geometry of **vectors** in that space.

A vector can represent a displacement, a velocity, a force, a row of data, a direction of change, a model parameter, or a feature embedding. In multivariable calculus, vectors are the language used to describe motion, tangent lines, tangent planes, gradients, flux, optimization, and linear approximation. In statistics and machine learning, vectors are also used to store observations, coefficients, residuals, gradients, and learned representations.

Main idea

A vector has both magnitude and direction. The norm measures magnitude, the dot product measures alignment, and projection separates a vector into a part along a chosen direction and a part perpendicular to that direction.

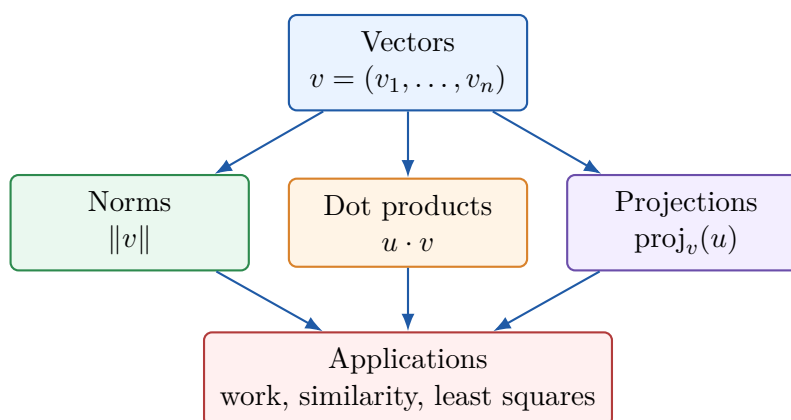


Figure 3.1: The main objects of the chapter and how they connect.

Learning goals

After studying this chapter, you should be able to:

1. perform vector addition, subtraction, and scalar multiplication in $\mathbb{R}^n$;
2. compute the Euclidean norm of a vector and construct unit vectors;
3. interpret distance as the norm of a displacement vector;
4. compute dot products and use them to find angles;
5. recognize orthogonal vectors and understand why orthogonality means zero dot product;
6. project one vector onto another and decompose a vector into parallel and perpendicular parts;
7. use the dot product to compute work and cosine similarity;
8. connect vector geometry to data science, statistics, optimization, and AI.

3.1 Vectors as displacement, direction, and data

Definition

A vector in $\mathbb{R}^n$ is an ordered list

$$v = (v_1, v_2, \dots, v_n).$$

Geometrically, we often draw a vector as an arrow. Algebraically, we compute with its components.

The same component list can represent different kinds of objects.

Context	Vector meaning
Geometry	displacement, direction, normal vector
Physics	force, velocity, acceleration
Statistics	data point, residual vector, coefficient vector
Machine learning	feature vector, weight vector, gradient vector
Optimization	search direction, update direction

If $A = (a_1, \dots, a_n)$ and $B = (b_1, \dots, b_n)$ are points, the displacement from A to B is

$$\overrightarrow{AB} = B - A = (b_1 - a_1, \dots, b_n - a_n).$$

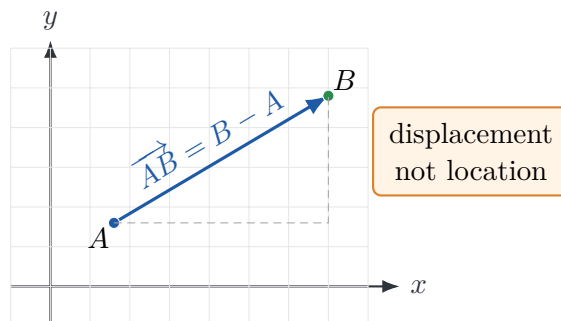


Figure 3.2: A displacement vector records how to move from one point to another.

Example 3.1: A displacement vector in space

Let

$$A = (1, -2, 3), \quad B = (2, 1, 5).$$

Then

$$\vec{AB} = (2 - 1, 1 - (-2), 5 - 3) = (1, 3, 2).$$

This vector tells us how to move from A to B : increase x by 1, increase y by 3, and increase z by 2.

Point versus vector

The ordered triple $(1, 3, 2)$ can be interpreted as a point or as a vector. As a point, it is a location. As a vector, it is a displacement. In calculus and applied mathematics, the interpretation matters as much as the arithmetic.

3.2 Vector addition and scalar multiplication

If

$$u = (u_1, \dots, u_n), \quad v = (v_1, \dots, v_n),$$

then vector addition is defined componentwise:

$$u + v = (u_1 + v_1, \dots, u_n + v_n).$$

If c is a real number, scalar multiplication is

$$cv = (cv_1, \dots, cv_n).$$

Geometrically:

- $u + v$ means follow displacement u and then displacement v ;
- $v - u$ means the displacement from the endpoint of u to the endpoint of v ;
- cv stretches v by the factor $|c|$;
- if $c < 0$, then cv also reverses direction.

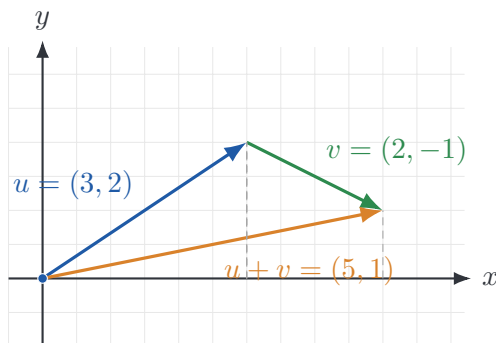


Figure 3.3: Vector addition by following one displacement after another.

Example 3.2: Net displacement

A particle is displaced first by $(3, 2)$ and then by $(2, -1)$. Its net displacement is

$$(3, 2) + (2, -1) = (5, 1).$$

The final location depends on the initial location, but the net displacement does not.

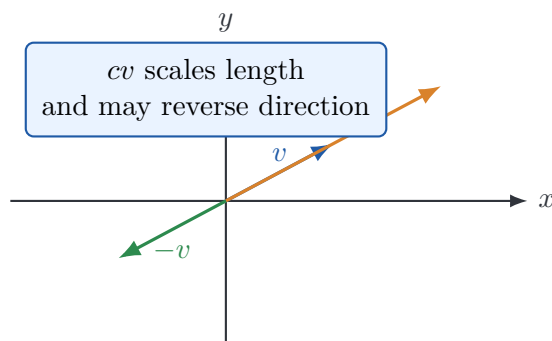


Figure 3.4: Scalar multiplication changes the magnitude of a vector, and negative scalars reverse its direction.

3.3 Norm, length, and unit vectors

Definition

The **Euclidean norm** of a vector $v = (v_1, \dots, v_n)$ is

$$\|v\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}.$$

A vector u is a **unit vector** if $\|u\| = 1$.

This generalizes the Pythagorean theorem. In $\mathbb{R}^2$, it gives the length of an arrow in the plane. In $\mathbb{R}^3$, it gives the length of an arrow in space. In $\mathbb{R}^n$, it gives the standard size of a data vector or parameter vector.

If $v \neq 0$, then the unit vector in the direction of v is

$$\hat{v} = \frac{v}{\|v\|}.$$

The vector v can be decomposed as

$$v = \|v\| \hat{v}.$$

This formula says that a nonzero vector is determined by two pieces of information: its magnitude and its direction.

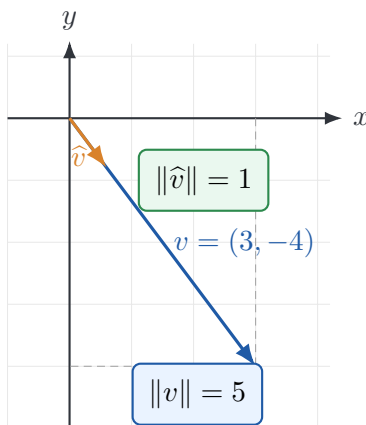


Figure 3.5: A vector and its unit direction have the same direction but different lengths.

Example 3.3: Unit vector parallel to $(3, -4)$

Let $v = (3, -4)$. Then

$$\|v\| = \sqrt{3^2 + (-4)^2} = 5.$$

Therefore a unit vector parallel to v and pointing in the same direction is

$$\hat{v} = \frac{1}{5}(3, -4) = \left(\frac{3}{5}, -\frac{4}{5}\right).$$

A unit vector parallel to v but pointing in the opposite direction is

$$-\hat{v} = \left(-\frac{3}{5}, \frac{4}{5}\right).$$

Example 3.4: Unit vectors perpendicular to $(3, -4)$

In the plane, if $v = (a, b)$, then $(-b, a)$ and $(b, -a)$ are perpendicular to v . For $v = (3, -4)$, two perpendicular vectors are

$$(4, 3), \quad (-4, -3).$$

Both have length 5, so the corresponding unit vectors are

$$\left(\frac{4}{5}, \frac{3}{5}\right), \quad \left(-\frac{4}{5}, -\frac{3}{5}\right).$$

Common mistake

Do not confuse a vector with its norm. The vector $(3, -4)$ has direction and magnitude. The norm $\|(3, -4)\| = 5$ is only a number.

3.4 Distance as norm of displacement

The distance between two points is the length of the displacement vector connecting them. If

$$A = (a_1, \dots, a_n), \quad B = (b_1, \dots, b_n),$$

then

$$\text{dist}(A, B) = \|B - A\| = \sqrt{(b_1 - a_1)^2 + \dots + (b_n - a_n)^2}.$$

This is one of the most important bridges between geometry and computation. In data science, distance is used in nearest-neighbor methods, clustering, anomaly detection, and similarity search.

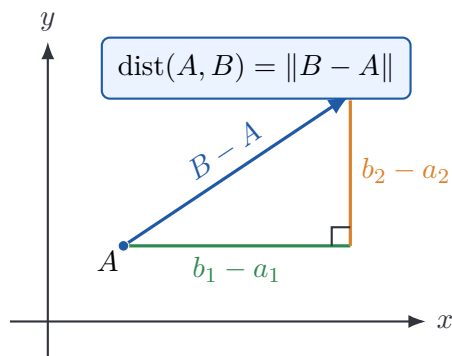


Figure 3.6: Distance is the norm of displacement. In two dimensions this is the Pythagorean theorem.

3.5 Dot product

Definition

The **dot product** of two vectors

$$u = (u_1, \dots, u_n), \quad v = (v_1, \dots, v_n)$$

is

$$u \cdot v = u_1 v_1 + u_2 v_2 + \dots + u_n v_n.$$

The dot product takes two vectors and returns one number. It measures alignment.

The geometric identity is

$$u \cdot v = \|u\| \|v\| \cos \theta,$$

where θ is the angle between u and v . Therefore, for nonzero vectors,

$$\cos \theta = \frac{u \cdot v}{\|u\| \|v\|}.$$

This formula is central in geometry, statistics, and machine learning.

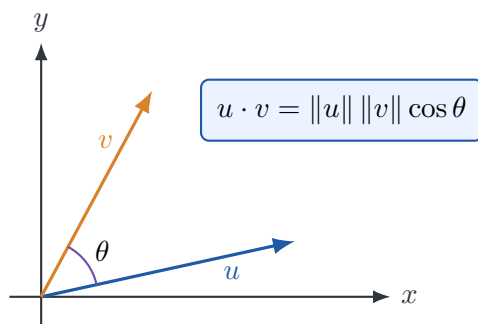


Figure 3.7: The dot product connects algebraic components with the angle between vectors.

Suppose u and v are nonzero.

- If $u \cdot v > 0$, the angle is acute and the vectors point generally in the same direction.
- If $u \cdot v = 0$, the angle is 90° and the vectors are orthogonal.
- If $u \cdot v < 0$, the angle is obtuse and the vectors point generally in opposite directions.

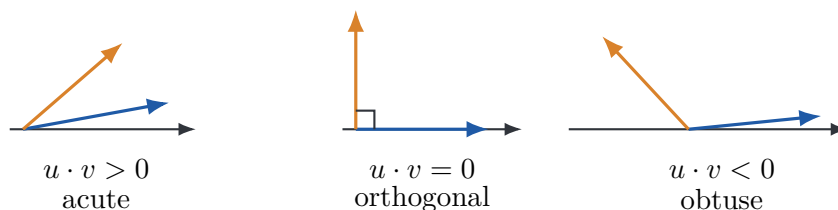


Figure 3.8: The sign of the dot product gives qualitative information about the angle.

Example 3.5: Angle between two vectors

Let

$$F = (1, 2), \quad v = (-1, 3).$$

Then

$$F \cdot v = (1)(-1) + (2)(3) = 5,$$

and

$$\|F\| = \sqrt{1^2 + 2^2} = \sqrt{5}, \quad \|v\| = \sqrt{(-1)^2 + 3^2} = \sqrt{10}.$$

Thus

$$\cos \theta = \frac{F \cdot v}{\|F\| \|v\|} = \frac{5}{\sqrt{5}\sqrt{10}} = \frac{1}{\sqrt{2}}.$$

So

$$\theta = \cos^{-1} \left(\frac{1}{\sqrt{2}} \right) = \frac{\pi}{4}.$$

The angle is 45° , so the two vectors are strongly aligned.

3.6 Orthogonality

Definition

Two vectors u and v are **orthogonal** if

$$u \cdot v = 0.$$

Orthogonality generalizes perpendicularity. In $\mathbb{R}^2$ and $\mathbb{R}^3$, it means the vectors meet at a right angle. In $\mathbb{R}^n$, where we may not be able to draw the vectors, the equation $u \cdot v = 0$ is still meaningful.

Example 3.6: Checking orthogonality

Let

$$u = (2, -1, 3), \quad v = (1, 5, 1).$$

Then

$$u \cdot v = (2)(1) + (-1)(5) + (3)(1) = 2 - 5 + 3 = 0.$$

Therefore u and v are orthogonal.

Bridge to applications

Orthogonality is one of the main organizing ideas of applied mathematics. It appears in least squares, projections, Fourier series, PCA, QR factorization, residual analysis, and the geometry of independent directions in data.

3.7 Projection onto a direction

Projection answers a basic question:

How much of one vector points in the direction of another vector?

Let $v \neq 0$. The vector projection of u onto v is

$$\text{proj}_v(u) = \frac{u \cdot v}{v \cdot v} v.$$

The scalar component of u in the direction of v is

$$\text{comp}_v(u) = \frac{u \cdot v}{\|v\|}.$$

If $\hat{v} = v / \|v\|$ is the unit vector in the direction of v , then

$$\text{proj}_v(u) = (u \cdot \hat{v}) \hat{v}.$$

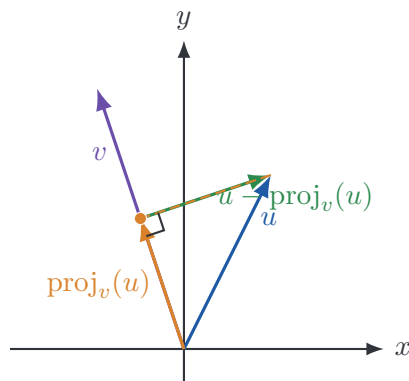


Figure 3.9: Projection decomposes a vector into a part parallel to v and a residual perpendicular to v .

Example 3.7: Projecting $(1, 2)$ onto $(-1, 3)$

Let

$$F = (1, 2), \quad v = (-1, 3).$$

Then

$$F \cdot v = 5, \quad v \cdot v = (-1)^2 + 3^2 = 10.$$

Therefore

$$\text{proj}_v(F) = \frac{F \cdot v}{v \cdot v} v = \frac{5}{10}(-1, 3) = \left(-\frac{1}{2}, \frac{3}{2}\right).$$

This is the part of F that lies in the direction of v .

Projection creates an orthogonal decomposition

Let

$$p = \text{proj}_v(u), \quad r = u - p.$$

Then

$$u = p + r,$$

where p is parallel to v and r is orthogonal to v .

To check this, compute

$$\begin{aligned} r \cdot v &= (u - p) \cdot v \\ &= u \cdot v - \left(\frac{u \cdot v}{v \cdot v} v \right) \cdot v \\ &= u \cdot v - \frac{u \cdot v}{v \cdot v} (v \cdot v) = 0. \end{aligned}$$

Bridge to least squares

In linear regression, the fitted values are projections of the response vector onto a column space. The residual vector is orthogonal to every column of the design matrix. This is why dot products and projections are essential for statistics.

3.8 Work as a dot product

If a constant force F moves an object through a displacement d , then the work done by the force is

$$W = F \cdot d.$$

Only the component of the force in the direction of motion contributes to work.

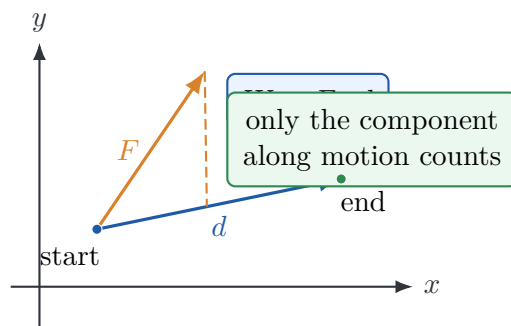


Figure 3.10: Work is the dot product of force and displacement.

Example 3.8: Work by a constant force

A constant force

$$F = (-1, -3, 4)$$

is applied while an object moves from

$$A = (1, 2, 0)$$

to

$$B = (2, 0, 1).$$

The displacement is

$$d = B - A = (1, -2, 1).$$

The work is

$$W = F \cdot d = (-1)(1) + (-3)(-2) + (4)(1) = -1 + 6 + 4 = 9.$$

The positive result means that the force has a net component in the direction of motion.

3.9 Cosine similarity and data vectors

For nonzero vectors u and v , the quantity

$$\frac{u \cdot v}{\|u\| \|v\|}$$

is the cosine of the angle between them. In data science and AI, this value is often called **cosine similarity**.

Cosine similarity focuses on direction rather than magnitude:

- similar direction: value near 1;
- orthogonal or unrelated direction: value near 0;
- opposite direction: value near -1 .

This is useful when the magnitude of a vector is less important than its pattern. For example, in text analysis, two documents may have different lengths but similar word-frequency directions.

Example 3.9: Cosine similarity of data vectors

Suppose two observations are

$$x = (2, 1, 0, 3), \quad y = (4, 2, 0, 6).$$

Then $y = 2x$, so the two observations point in exactly the same direction. Their cosine similarity is 1.

Now compare

$$x = (2, 1, 0, 3), \quad z = (0, 3, 4, 0).$$

The dot product is

$$x \cdot z = (2)(0) + (1)(3) + (0)(4) + (3)(0) = 3,$$

so the cosine similarity is positive but not close to 1.

	x	y	z	w
x	1.00	1.00	0.12	0.71
y	1.00	1.00	0.12	0.71
z	0.12	0.12	1.00	0.36
w	0.71	0.71	0.36	1.00

cosine similarity matrix

Figure 3.11: Cosine similarity measures direction rather than length. A value near 1 means similar direction.

3.10 High-dimensional geometry: most random directions are nearly orthogonal

High-dimensional geometry can be surprising. In low dimensions, random vectors often have visibly meaningful angles. In high dimensions, randomly chosen directions are usually close to orthogonal. This fact helps explain why high-dimensional data can be difficult to visualize and why algorithms must be designed carefully.

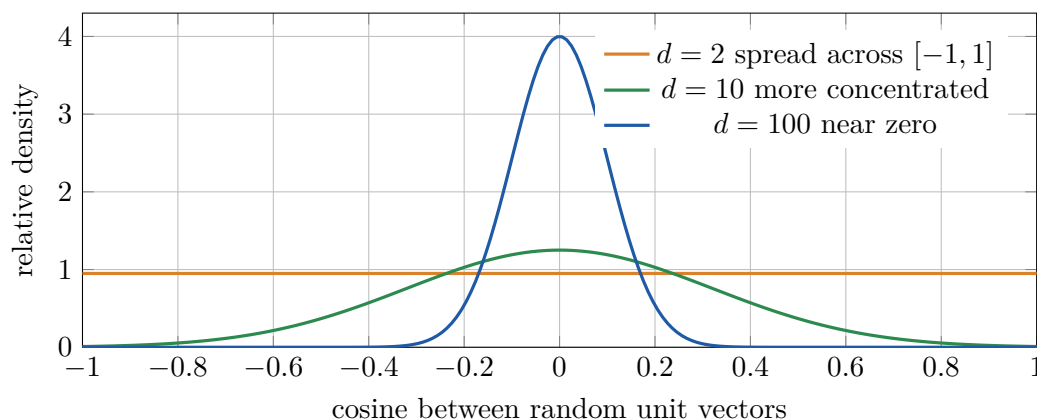


Figure 3.12: Cosines between random unit vectors concentrate near zero in high dimensions. The plotted curves are schematic density shapes emphasizing the concentration effect.

Modern interpretation

In high-dimensional data spaces, distance and angle can behave differently from our three-dimensional intuition. This is one reason why normalization, projection, regularization, and dimensionality reduction are central in statistics and machine learning.

3.11 Norms beyond the Euclidean norm

The Euclidean norm is the main norm used in this chapter, but other norms are important in applied mathematics and data science.

For $v = (v_1, \dots, v_n)$:

$$\|v\|_1 = |v_1| + \dots + |v_n|$$

is the ℓ^1 norm, and

$$\|v\|_\infty = \max\{|v_1|, \dots, |v_n|\}$$

is the ℓ^∞ norm. The Euclidean norm is

$$\|v\|_2 = \sqrt{v_1^2 + \dots + v_n^2}.$$

Different norms encode different ideas of size.

Norm	Formula	Common interpretation
$\ v\ _1$	sum of absolute values	sparsity, total variation, Manhattan distance
$\ v\ _2$	square root of sum of squares	Euclidean length, energy, least squares
$\ v\ _\infty$	maximum absolute component	worst-case component size

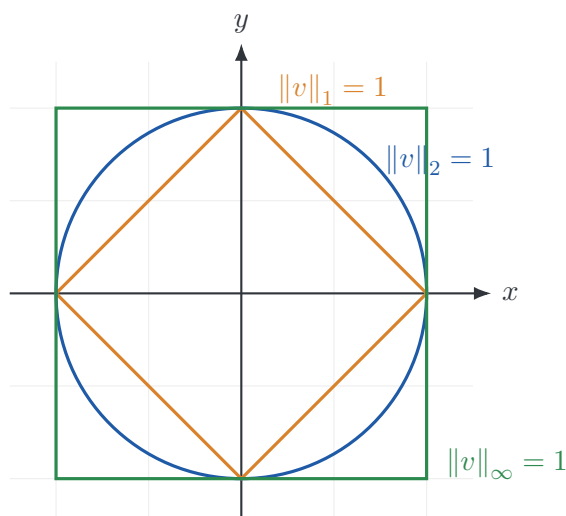


Figure 3.13: Different norms produce different unit balls in the plane.

Why this matters later

Least-squares regression is built from the ℓ^2 norm. Lasso regression is built from the ℓ^1 norm. Robust and worst-case methods often use ℓ^∞ -type constraints. Norms are not just geometric objects; they shape models.

3.12 Computational implementation of core vector operations

The formulas in this chapter are simple enough to compute by hand, but they also form the basic operations used by numerical linear algebra and machine learning libraries.

```

1 import numpy as np
2
3 u = np.array([1.0, 2.0])
4 v = np.array([-1.0, 3.0])
5
6 norm_u = np.linalg.norm(u)
7 norm_v = np.linalg.norm(v)
8 dot_uv = u @ v
9 cos_theta = dot_uv / (norm_u * norm_v)
10 projection = (dot_uv / (v @ v)) * v
11 residual = u - projection
12
13 print("||u|| =", norm_u)
14 print("||v|| =", norm_v)
15 print("u dot v =", dot_uv)
16 print("cos(theta) =", cos_theta)
17 print("projection of u onto v =", projection)
18 print("residual =", residual)
19 print("residual dot v =", residual @ v)
    
```

The last line should be approximately zero. This confirms that the residual $u - \text{proj}_v(u)$ is orthogonal to v .

3.13 Worked example summary

Example 3.10: Full vector analysis

Let

$$u = (2, -1, 2), \quad v = (1, 2, 2).$$

Step 1: Compute the norms.

$$\|u\| = \sqrt{2^2 + (-1)^2 + 2^2} = 3,$$

and

$$\|v\| = \sqrt{1^2 + 2^2 + 2^2} = 3.$$

Step 2: Compute the dot product.

$$u \cdot v = (2)(1) + (-1)(2) + (2)(2) = 2 - 2 + 4 = 4.$$

Step 3: Find the angle.

$$\cos \theta = \frac{u \cdot v}{\|u\| \|v\|} = \frac{4}{9}, \quad \theta = \cos^{-1} \left(\frac{4}{9} \right).$$

Step 4: Project u onto v .

$$\text{proj}_v(u) = \frac{u \cdot v}{v \cdot v} v = \frac{4}{9} (1, 2, 2) = \left(\frac{4}{9}, \frac{8}{9}, \frac{8}{9} \right).$$

Step 5: Interpret. The vector u has a component in the direction of v and a residual component perpendicular to v . This is the same geometric idea that later appears in least-squares approximation.

3.14 Mini-project: cosine similarity for short documents

A simple way to represent a document is by a vector of word counts. Suppose our vocabulary is

(calculus, data, matrix, gradient).

Three short documents might have count vectors

$$d_1 = (3, 1, 0, 2), \quad d_2 = (2, 1, 0, 3), \quad d_3 = (0, 3, 4, 0).$$

Cosine similarity compares their directions.

```
1 import numpy as np
2
3 docs = np.array([
```

```

4     [3, 1, 0, 2],
5     [2, 1, 0, 3],
6     [0, 3, 4, 0]
7 ], dtype=float)
8
9 names = ["doc1", "doc2", "doc3"]
10 unit_docs = docs / np.linalg.norm(docs, axis=1, keepdims=True)
11 sim = unit_docs @ unit_docs.T
12
13 for i, name_i in enumerate(names):
14     for j, name_j in enumerate(names):
15         print(f"{name_i} vs {name_j}: {sim[i, j]:.3f}")

```

Documents d_1 and d_2 are more similar to each other than either is to d_3 . This is a small example of the geometry behind search, clustering, and embedding models.

3.15 Practice problems

Conceptual questions

1. Explain the difference between a point and a vector using an example in $\mathbb{R}^2$.
2. Why does multiplying a vector by a negative scalar reverse its direction?
3. What does it mean for two vectors in $\mathbb{R}^n$ to be orthogonal?
4. Why is the dot product a measure of alignment?
5. Explain why cosine similarity ignores magnitude.
6. Why is projection important for least-squares approximation?

Computational practice

1. Let $v = (6, 8)$. Find $\|v\|$ and a unit vector in the direction of v .
2. Let $u = (1, 2, 3)$ and $v = (4, 0, -1)$. Compute $u + v$, $u - v$, $3u$, and $-2v$.
3. Find the displacement vector from $A = (2, -1, 4)$ to $B = (5, 3, -2)$.
4. Compute the distance between $A = (2, -1, 4)$ and $B = (5, 3, -2)$.
5. Compute the dot product of $u = (2, -1, 3)$ and $v = (1, 5, 1)$. Are they orthogonal?
6. Find the angle between $u = (1, 1)$ and $v = (1, -1)$.
7. Find the projection of $u = (3, 4)$ onto $v = (1, 0)$.
8. Find the projection of $u = (1, 2)$ onto $v = (-1, 3)$.
9. A force $F = (2, -1, 4)$ moves an object from $A = (0, 1, 2)$ to $B = (3, 0, 5)$. Compute the work done.
10. Let $x = (1, 0, 2, 2)$ and $y = (2, 0, 4, 4)$. Compute their cosine similarity.

Applied and data-oriented problems

1. Suppose two students have feature vectors $x = (80, 90, 70)$ and $y = (40, 45, 35)$ representing scores on three assessments. Compute their cosine similarity. Interpret why this similarity may be high even though the magnitudes differ.
2. Suppose $r = y - \hat{y}$ is a residual vector in a regression problem. What does it mean geometrically if r is orthogonal to the fitted-value vector $\hat{y}$?
3. In a nearest-neighbor method, why might scaling the features change the computed distances?
4. Find a vector perpendicular to $(5, -2)$. Then normalize it.
5. Let $w = (w_1, w_2, \dots, w_n)$ be a model parameter vector. Explain why $\|w\|_2$ can be interpreted as a measure of model size.

Challenge problems

1. Prove that if $v \neq 0$, then $u - \text{proj}_v(u)$ is orthogonal to v .
2. Prove the Cauchy-Schwarz inequality in the special case of $\mathbb{R}^2$:

$$|u \cdot v| \leq \|u\| \|v\|.$$

3. Show that the projection formula minimizes the distance from u to the line spanned by v :

$$\min_{c \in \mathbb{R}} \|u - cv\|.$$

4. Let u and v be nonzero vectors. Show that their cosine similarity is unchanged if u and v are multiplied by positive constants.
5. In $\mathbb{R}^n$, show that if u is orthogonal to $v_1, \dots, v_k$, then u is orthogonal to every linear combination $c_1 v_1 + \dots + c_k v_k$.

Chapter 4

Lines, Planes, Hyperplanes, and Parametric Geometry

Chapter overview

In single-variable calculus, the most important local geometric object is the **tangent line**. In multivariable calculus, the corresponding objects are **lines**, **planes**, and more generally **hyperplanes**. These are not only geometric objects in $\mathbb{R}^2$ and $\mathbb{R}^3$; they are also the basic language of linear models, constraints, optimization, regression, classification, tangent spaces, and numerical algorithms.

A line can describe a path of motion. A plane can describe a constraint surface or a local approximation. A hyperplane can separate two classes of data. A line segment can describe interpolation between data points. A parametric surface can describe a plane patch, a sphere, a cylinder, or the local tangent plane to a curved surface.

Main idea

A geometric object is usually easier to understand when we describe it using points and vectors. A point tells us **where** the object is. A vector tells us **how to move** along it or **which direction** is **perpendicular** to it.

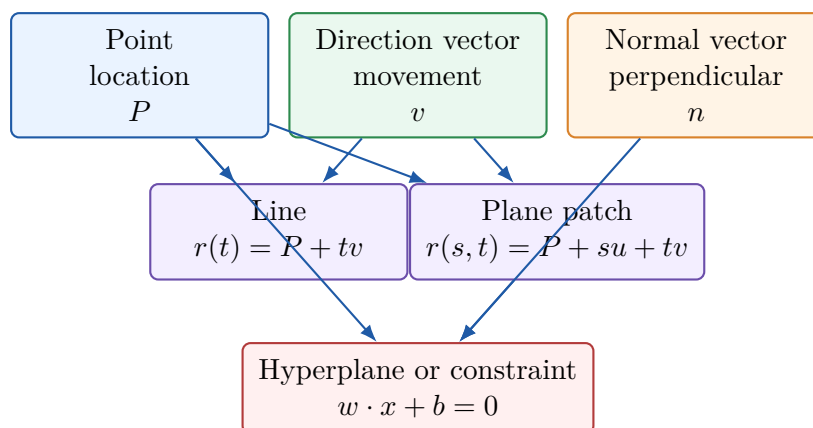


Figure 4.1: Parametric geometry is built from points, direction vectors, and normal vectors.

We will move between three complementary descriptions:

1. **Parametric form**, useful for tracing an object.
2. **Normal form**, useful for testing whether a point lies on an object.
3. **Standard scalar form**, useful for algebra and computation.

Learning goals

After studying this chapter, you should be able to:

1. write vector, parametric, and symmetric equations for lines;
2. parameterize line segments between two points;
3. write point-normal and scalar equations of planes in $\mathbb{R}^3$;
4. parameterize planes using two direction vectors;
5. find intersections among lines and planes;
6. compute angles and distances involving lines, planes, and hyperplanes;
7. interpret hyperplanes as linear constraints and decision boundaries;
8. use computation to visualize lines, planes, hyperplanes, and simple geometric algorithms.

4.1 Why parametric geometry matters

A line in the plane is often introduced as

$$y = mx + b.$$

This is useful, but it has two limitations. First, it does not describe vertical lines. Second, it does not generalize naturally to curves, lines in $\mathbb{R}^3$, planes in $\mathbb{R}^3$, or hyperplanes in $\mathbb{R}^n$.

The vector point of view is more general:

- a line is determined by one point and one nonzero direction vector;
- a plane is determined by either one point and one normal vector, or one point and two nonparallel direction vectors;
- a hyperplane in $\mathbb{R}^n$ is determined by one point and one nonzero normal vector.

This is the same geometry that appears in calculus when we construct tangent lines, tangent planes, and local linear approximations.

4.2 Lines in the plane

4.2.1 Slope form

A nonvertical line in $\mathbb{R}^2$ through (x_0, y_0) with slope m has equation

$$y - y_0 = m(x - x_0).$$

This form is familiar from algebra and single-variable calculus. But it hides the vector structure of the line. The slope m means that a direction vector for the line is

$$v = \langle 1, m \rangle.$$

Thus the same line can be written as

$$\langle x, y \rangle = \langle x_0, y_0 \rangle + t \langle 1, m \rangle.$$

Here t is a parameter. As t changes, the point moves along the line.

4.2.2 Vector form

More generally, if $P = (x_0, y_0)$ is a point and $v = \langle a, b \rangle$ is a nonzero direction vector, then the line through P parallel to v is

$$r(t) = r_0 + tv,$$

or

$$\langle x, y \rangle = \langle x_0, y_0 \rangle + t \langle a, b \rangle.$$

In coordinates,

$$x = x_0 + at, \quad y = y_0 + bt.$$

The same idea works in $\mathbb{R}^3$ and $\mathbb{R}^n$.

4.2.3 Point-normal form in the plane

A line in $\mathbb{R}^2$ can also be described by a normal vector. If $n = \langle A, B \rangle$ is perpendicular to a line and $P_0 = (x_0, y_0)$ is on the line, then any point $P = (x, y)$ on the line satisfies

$$n \cdot (P - P_0) = 0.$$

Therefore

$$A(x - x_0) + B(y - y_0) = 0.$$

Expanding gives the standard form

$$Ax + By + C = 0.$$

Note

For a line in $\mathbb{R}^2$, a **direction vector** points along the line, while a **normal vector** points perpendicular to the line. If $v = \langle a, b \rangle$ is a direction vector, then $n = \langle -b, a \rangle$ and $n = \langle b, -a \rangle$ are normal vectors.

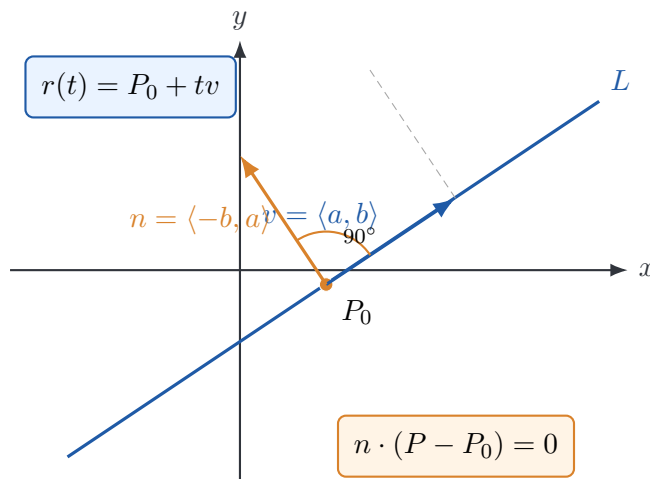


Figure 4.2: The same line can be described by a point and direction vector or by a point and normal vector.

Example 4.1: Same line, three descriptions

Find vector, parametric, and standard equations for the line through $P = (2, -1)$ with direction vector $v = \langle 3, 2 \rangle$.

The vector equation is

$$r(t) = \langle 2, -1 \rangle + t \langle 3, 2 \rangle.$$

The parametric equations are

$$x = 2 + 3t, \quad y = -1 + 2t.$$

A normal vector is $n = \langle -2, 3 \rangle$. Therefore

$$-2(x - 2) + 3(y + 1) = 0.$$

Expanding gives

$$-2x + 3y + 7 = 0.$$

4.3 Lines in $\mathbb{R}^n$

The vector equation of a line is dimension independent. In $\mathbb{R}^n$, a line through a point

$$p = (p_1, p_2, \dots, p_n)$$

with nonzero direction vector

$$v = (v_1, v_2, \dots, v_n)$$

is

$$r(t) = p + tv, \quad -\infty < t < \infty.$$

In coordinates,

$$x_1 = p_1 + tv_1, \quad x_2 = p_2 + tv_2, \quad \dots, \quad x_n = p_n + tv_n.$$

The parameter t is not always time, but it often behaves like time: as t increases, the point moves along the line at constant velocity v .

4.3.1 Lines in $\mathbb{R}^3$

If $P = (x_0, y_0, z_0)$ and $v = \langle a, b, c \rangle$, then

$$\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t \langle a, b, c \rangle.$$

Equivalently,

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct.$$

If a, b, c are nonzero, the symmetric form is

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

If one component of the direction vector is zero, the corresponding coordinate is constant and should not be placed in the denominator.

Example 4.2: A line through two points

Find a vector equation and symmetric equation for the line through

$$P = (-5, 5, 4), \quad Q = (7, -1, 2).$$

A direction vector is

$$v = Q - P = \langle 12, -6, -2 \rangle.$$

Thus

$$r(t) = \langle -5, 5, 4 \rangle + t \langle 12, -6, -2 \rangle.$$

The parametric equations are

$$x = -5 + 12t, \quad y = 5 - 6t, \quad z = 4 - 2t.$$

The symmetric form is

$$\frac{x + 5}{12} = \frac{y - 5}{-6} = \frac{z - 4}{-2}.$$

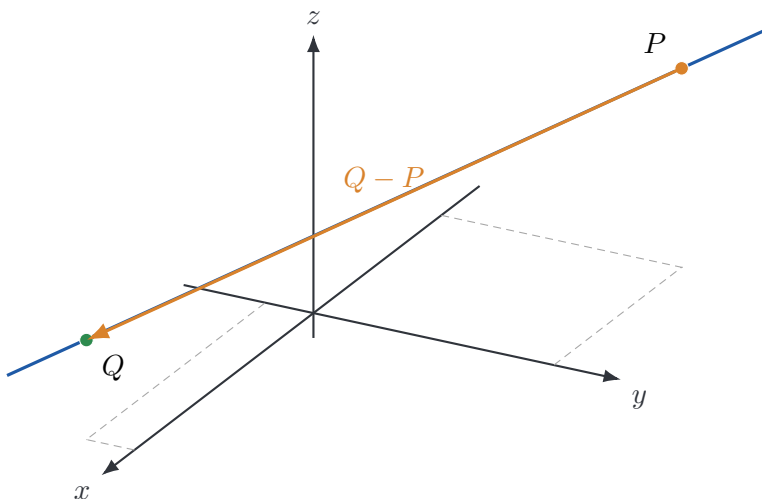


Figure 4.3: A line in space through two points. The displacement $Q - P$ gives a direction vector.

4.4 Line segments and interpolation

The full line through P and Q extends infinitely in both directions. The **line segment** from P to Q is obtained by restricting the parameter:

$$r(t) = P + t(Q - P), \quad 0 \leq t \leq 1.$$

At $t = 0$, we get P . At $t = 1$, we get Q . At $t = 1/2$, we get the midpoint.

This formula is one of the most useful formulas in applied mathematics. It describes linear interpolation between two states, two observations, two images, two parameter vectors, or two model weights.

Main idea

The line segment from P to Q is

$$r(t) = (1 - t)P + tQ, \quad 0 \leq t \leq 1.$$

This is the same as $P + t(Q - P)$, but the form $(1 - t)P + tQ$ makes the interpolation interpretation clear.

Example 4.3: Interpolating between two model parameter vectors

Suppose a simple model has parameter vector

$$w_0 = (1, -2, 0.5)$$

before training and

$$w_1 = (3, 1, -1)$$

after training. The path

$$w(t) = (1 - t)w_0 + tw_1, \quad 0 \leq t \leq 1,$$

linearly interpolates between the two models. This does not mean gradient descent moves along this exact path. It only gives a useful way to compare intermediate parameter vectors.

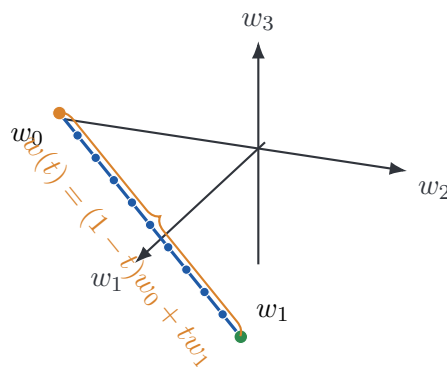


Figure 4.4: Linear interpolation traces the straight segment between two parameter vectors.

4.5 Planes in $\mathbb{R}^3$: point-normal form

A plane in $\mathbb{R}^3$ is determined by a point $P_0 = (x_0, y_0, z_0)$ and a nonzero normal vector

$$n = \langle a, b, c \rangle.$$

A point $P = (x, y, z)$ lies on the plane exactly when $P - P_0$ is perpendicular to n . Therefore

$$n \cdot (P - P_0) = 0.$$

This gives

$$\langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0.$$

Equivalently,

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

Expanding gives standard form

$$ax + by + cz + d = 0.$$

Note

In the equation $ax + by + cz + d = 0$, the vector $n = \langle a, b, c \rangle$ is perpendicular to the plane. This is why the coefficients of x , y , and z are geometrically meaningful.

Example 4.4: Equation of a plane from point and normal vector

Find the equation of the plane through $P_0 = (1, 2, 3)$ with normal vector $n = \langle 4, 5, 6 \rangle$.

Use the point-normal equation:

$$4(x - 1) + 5(y - 2) + 6(z - 3) = 0.$$

Expanding,

$$4x + 5y + 6z - 32 = 0.$$

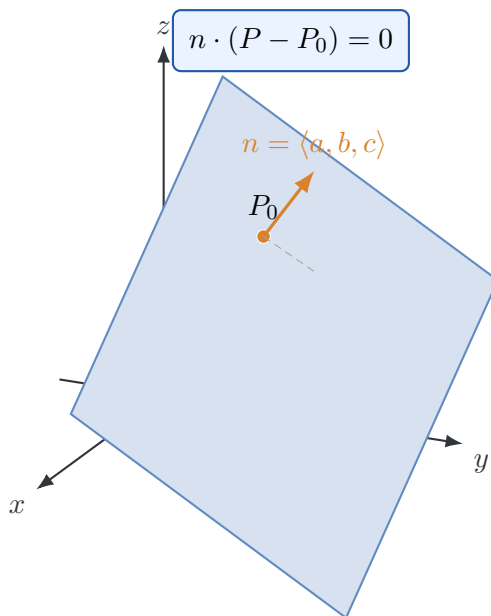


Figure 4.5: A plane is the set of points whose displacement from P_0 is perpendicular to the normal vector n .

4.6 Planes from three points

Three noncollinear points determine a plane. Suppose P, Q, R are three points in $\mathbb{R}^3$. Two direction vectors in the plane are

$$Q - P, \quad R - P.$$

If the points are not collinear, then these vectors are not parallel. A normal vector is

$$n = (Q - P) \times (R - P).$$

Then the plane is

$$n \cdot (\langle x, y, z \rangle - P) = 0.$$

Example 4.5: Plane through three points

Find the plane through

$$P = (1, 2, 3), \quad Q = (2, 2, 5), \quad R = (3, 4, 1).$$

Compute

$$Q - P = \langle 1, 0, 2 \rangle, \quad R - P = \langle 2, 2, -2 \rangle.$$

A normal vector is

$$(Q - P) \times (R - P) = \langle -4, 6, 2 \rangle.$$

We may divide by 2 and use $n = \langle -2, 3, 1 \rangle$. Therefore the plane is

$$-2(x - 1) + 3(y - 2) + (z - 3) = 0.$$

Expanding,

$$-2x + 3y + z - 7 = 0.$$

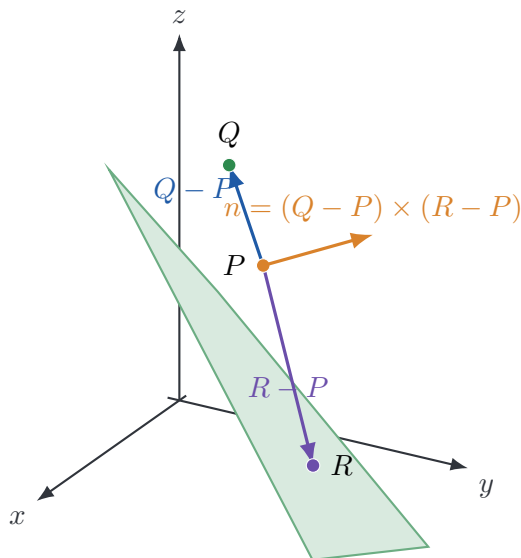


Figure 4.6: Three noncollinear points determine two independent directions and hence a normal vector by the cross product.

4.7 Parametric form of a plane

A plane may also be described by one point and two nonparallel direction vectors. If P_0 is a point and u, v are nonparallel vectors, then

$$r(s, t) = P_0 + su + tv$$

is the plane through P_0 spanned by u and v .

This is the two-parameter analogue of a line:

- a line has one parameter: $r(t) = P_0 + tv$;
- a plane has two parameters: $r(s, t) = P_0 + su + tv$.

The normal vector to the plane is

$$n = u \times v.$$

If $u \times v = 0$, then u and v are parallel, and the formula does not describe a genuine plane.

Example 4.6: Parametric and scalar plane equations

Let

$$P_0 = (1, 0, 2), \quad u = \langle 1, 2, 0 \rangle, \quad v = \langle 0, 1, 3 \rangle.$$

A parametric equation of the plane is

$$r(s, t) = \langle 1, 0, 2 \rangle + s \langle 1, 2, 0 \rangle + t \langle 0, 1, 3 \rangle.$$

In coordinates,

$$x = 1 + s, \quad y = 2s + t, \quad z = 2 + 3t.$$

A normal vector is

$$u \times v = \langle 6, -3, 1 \rangle.$$

Therefore a scalar equation is

$$6(x - 1) - 3(y - 0) + (z - 2) = 0,$$

or

$$6x - 3y + z - 8 = 0.$$

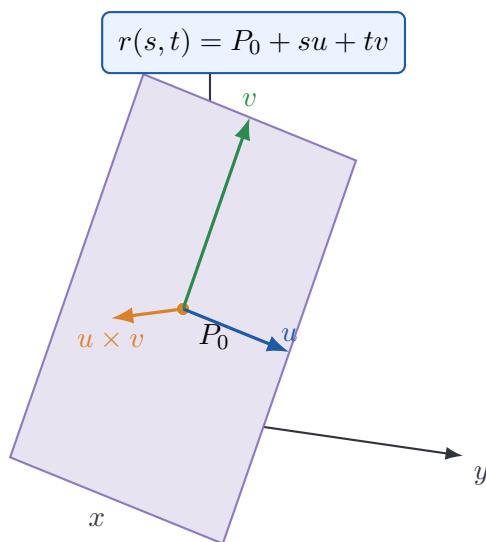


Figure 4.7: A parametric plane is swept out by moving from P_0 in two independent directions.

4.8 Intersections

Intersections are where algebra meets geometry. In calculus and applied mathematics, intersection problems appear when we solve constraints, find feasible sets, or describe where surfaces meet.

4.8.1 Line-plane intersection

Suppose a line is given by

$$r(t) = P + tv$$

and a plane is given by

$$n \cdot (x - P_0) = 0.$$

Substitute $x = r(t)$ into the plane equation:

$$n \cdot (P + tv - P_0) = 0.$$

This is a linear equation in t :

$$n \cdot (P - P_0) + t(n \cdot v) = 0.$$

If $n \cdot v \neq 0$, there is one intersection point:

$$t = -\frac{n \cdot (P - P_0)}{n \cdot v}.$$

If $n \cdot v = 0$, then the line is parallel to the plane. It either never intersects the plane or lies entirely in the plane.

Example 4.7: A line intersects a plane

Find the intersection point of

$$x = 1 - 2t, \quad y = 3 + t, \quad z = 2 + 4t$$

with the plane

$$2x - y + z = 0.$$

Substitute the line into the plane:

$$2(1 - 2t) - (3 + t) + (2 + 4t) = 0.$$

Simplifying gives

$$1 - t = 0,$$

so $t = 1$. Therefore

$$(x, y, z) = (-1, 4, 6).$$

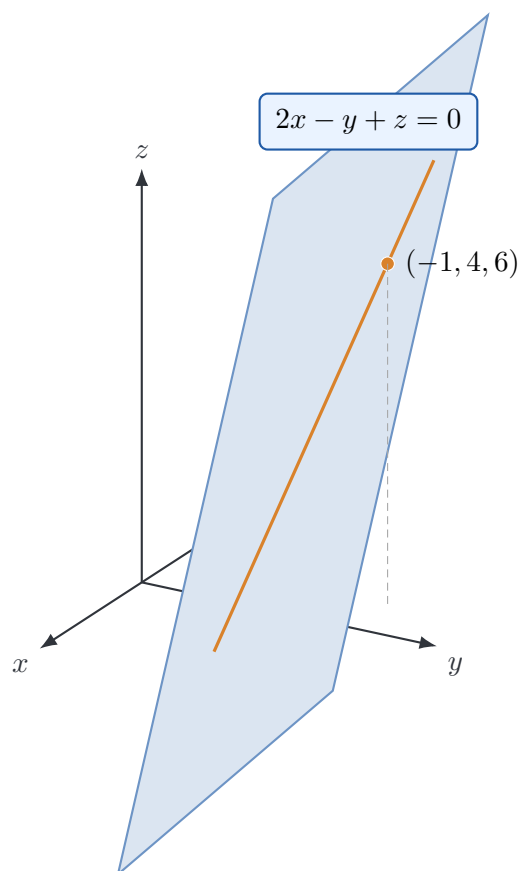


Figure 4.8: A line meeting a plane at exactly one point. Algebraically, substitution produces the parameter value.

4.8.2 Intersection of two planes

Two nonparallel planes in $\mathbb{R}^3$ intersect in a line. Suppose

$$n_1 \cdot x = d_1, \quad n_2 \cdot x = d_2.$$

If n_1 and n_2 are not parallel, then the direction vector of the intersection line is

$$v = n_1 \times n_2.$$

To find a point on the line, solve the two plane equations simultaneously. Since there are two equations in three unknowns, one variable can usually be chosen freely.

Example 4.8: Intersection of two planes

Find a parametric equation for the intersection of

$$x + y + z = 3$$

and

$$2x + 3y + z = 8.$$

Subtract the first equation from the second:

$$x + 2y = 5.$$

Let $z = t$. Then from the first equation,

$$x + y = 3 - t.$$

Solving

$$x + 2y = 5, \quad x + y = 3 - t$$

gives

$$y = 2 + t, \quad x = 1 - 2t.$$

So the intersection line is

$$r(t) = \langle 1, 2, 0 \rangle + t \langle -2, 1, 1 \rangle.$$

The direction vector can also be found from the cross product of normal vectors:

$$\langle 1, 1, 1 \rangle \times \langle 2, 3, 1 \rangle = \langle -2, 1, 1 \rangle.$$

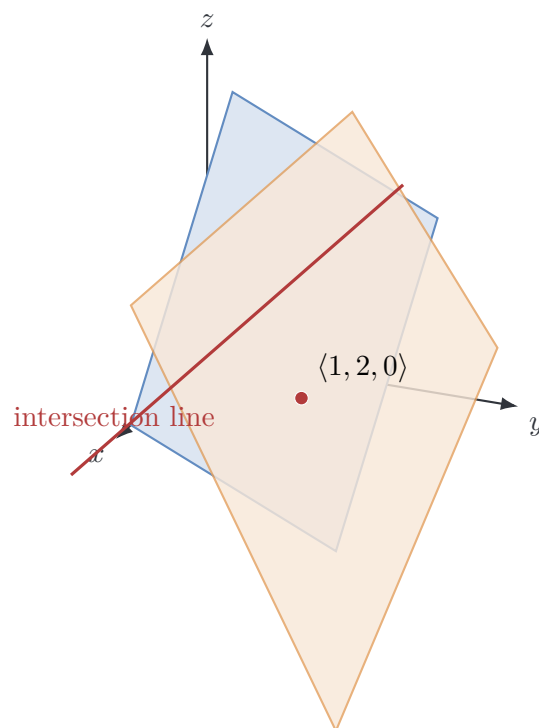


Figure 4.9: Two nonparallel planes intersect in a line whose direction is $n_1 \times n_2$.

4.9 Angles involving lines and planes

4.9.1 Angle between two lines

If two lines have direction vectors u and v , the angle θ between the lines satisfies

$$\cos \theta = \frac{|u \cdot v|}{\|u\| \|v\|}.$$

The absolute value gives the acute angle between the lines.

4.9.2 Angle between two planes

If two planes have normal vectors n_1 and n_2 , then the angle between the planes is the acute angle between their normal vectors:

$$\cos \theta = \frac{|n_1 \cdot n_2|}{\|n_1\| \|n_2\|}.$$

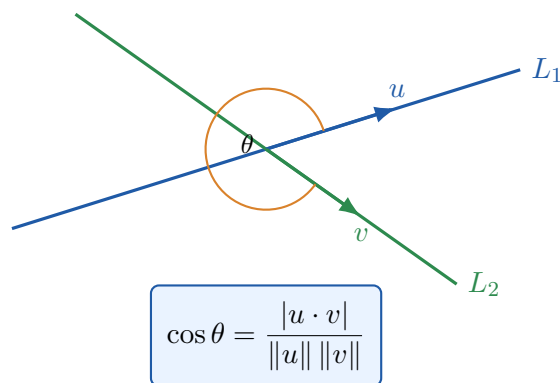


Figure 4.10: Angles between lines are computed from their direction vectors. Angles between planes are computed from their normal vectors.

Example 4.9: Angle between two planes

Find the angle between

$$x + 2y + z = 5$$

and

$$x + y = 8.$$

The normal vectors are

$$n_1 = \langle 1, 2, 1 \rangle, \quad n_2 = \langle 1, 1, 0 \rangle.$$

Then

$$\cos \theta = \frac{|n_1 \cdot n_2|}{\|n_1\| \|n_2\|} = \frac{3}{\sqrt{6}\sqrt{2}} = \frac{\sqrt{3}}{2}.$$

Therefore

$$\theta = \frac{\pi}{6}.$$

4.10 Distances to lines, planes, and hyperplanes

Distance formulas are important because they turn geometry into quantitative measurement. In data science, the distance from a data point to a hyperplane is a signed confidence score for a linear classifier. In optimization, it measures violation of a linear constraint.

4.10.1 Distance from a point to a plane

Suppose the plane is

$$ax + by + cz + d = 0.$$

The distance from $P = (x_0, y_0, z_0)$ to the plane is

$$\text{dist}(P, \Pi) = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

The signed distance is

$$\frac{ax_0 + by_0 + cz_0 + d}{\sqrt{a^2 + b^2 + c^2}}.$$

4.10.2 Distance from a point to a line in $\mathbb{R}^3$

If a line is $r(t) = P_0 + tv$ and P is a point, then

$$\text{dist}(P, L) = \frac{\|(P - P_0) \times v\|}{\|v\|}.$$

This formula comes from the area of a parallelogram. The numerator is the area of the parallelogram with sides $P - P_0$ and v , and the base length is $\|v\|$.

Example 4.10: Distance from a point to a plane

Find the distance from $P = (1, 2, 3)$ to

$$2x - y + 2z - 5 = 0.$$

Use the formula:

$$\text{dist}(P, \Pi) = \frac{|2(1) - 2 + 2(3) - 5|}{\sqrt{2^2 + (-1)^2 + 2^2}} = \frac{1}{3}.$$

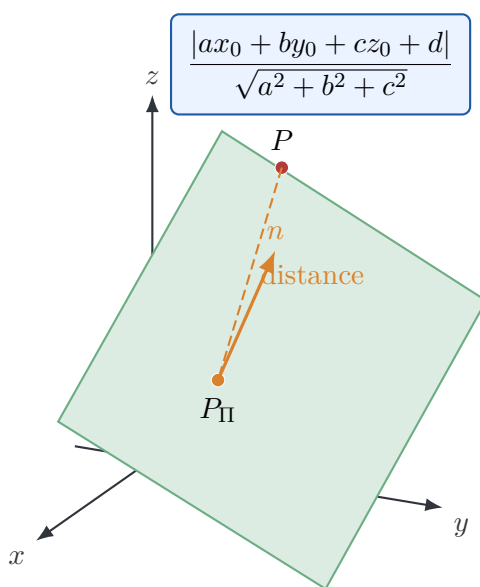


Figure 4.11: The point-to-plane distance is the length of the perpendicular segment.

4.11 Hyperplanes in $\mathbb{R}^n$

A **hyperplane** in $\mathbb{R}^n$ is the set of points satisfying one linear equation:

$$w \cdot x + b = 0,$$

where $w \neq 0$ is a vector in $\mathbb{R}^n$ and b is a scalar.

The vector w is normal to the hyperplane. The hyperplane has dimension $n - 1$. For example:

- in $\mathbb{R}^2$, a hyperplane is a line;

- in $\mathbb{R}^3$, a hyperplane is a plane;
- in $\mathbb{R}^{100}$, a hyperplane is a 99-dimensional flat object.

We usually cannot draw high-dimensional hyperplanes, but the algebra remains the same.

The two sides of the hyperplane are half-spaces:

$$w \cdot x + b > 0, \quad w \cdot x + b < 0.$$

The signed distance from x to the hyperplane is

$$\frac{w \cdot x + b}{\|w\|}.$$

4.11.1 Hyperplanes and linear classifiers

A linear classifier assigns a label according to the sign of

$$w \cdot x + b.$$

This is the geometric core of logistic regression, support vector machines, perceptrons, and the final linear layer of many neural networks.

Hyperplane as a decision boundary

A classifier of the form

$$\hat{y} = \text{sign}(w \cdot x + b)$$

separates feature space into two half-spaces. The boundary between the two predicted classes is the hyperplane $w \cdot x + b = 0$.

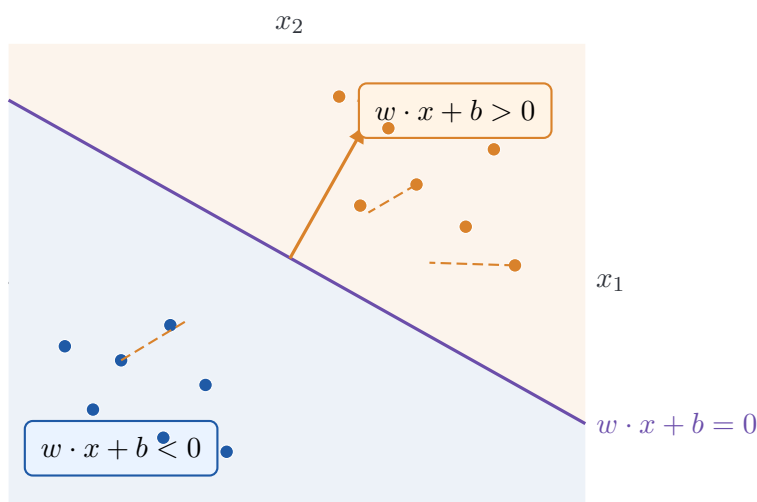


Figure 4.12: Signed distances to a line used as a linear decision boundary in two features.

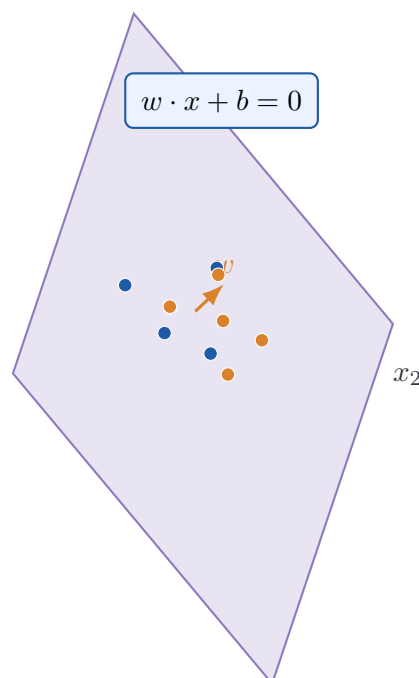


Figure 4.13: A plane can serve as a linear decision boundary in three-dimensional feature space.

4.12 Affine subspaces and constraints

A line, plane, or hyperplane is an example of an **affine subspace**. The word affine means that the object may not pass through the origin.

A linear subspace has the form

$$\text{span}\{v_1, \dots, v_k\}.$$

An affine subspace has the form

$$p + \text{span}\{v_1, \dots, v_k\}.$$

Examples:

- A line through the origin is a one-dimensional linear subspace.
- A line not through the origin is a one-dimensional affine subspace.
- A plane through the origin is a two-dimensional linear subspace.
- A plane not through the origin is a two-dimensional affine subspace.

In optimization, linear equality constraints form affine subspaces. For example,

$$Ax = b$$

describes an affine set when it is consistent. The geometry of constraints is the foundation for Lagrange multipliers, constrained least squares, and convex optimization.

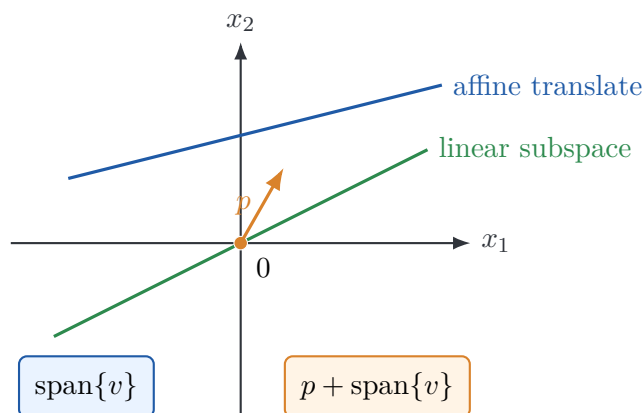


Figure 4.14: An affine subspace is a translate of a linear subspace.

4.13 Computational templates

The following functions are useful for computational geometry.

```

1 import numpy as np
2
3 def line_point(P, v, t):
4     """Return P + t v."""
5     return np.asarray(P, dtype=float) + t * np.asarray(v, dtype=float)
6
7
8 def plane_signed_distance(P, normal, d):
9     """Signed distance from P to normal dot x + d = 0."""
10    P = np.asarray(P, dtype=float)
11    normal = np.asarray(normal, dtype=float)
12    return (normal @ P + d) / np.linalg.norm(normal)
13
14
15 def project_point_to_plane(P, normal, d):
16     """Orthogonal projection of P onto normal dot x + d = 0."""
17    P = np.asarray(P, dtype=float)
18    normal = np.asarray(normal, dtype=float)
19    return P - ((normal @ P + d) / (normal @ normal)) * normal
20
21
22 def line_plane_intersection(P, v, normal, d):
23     """Intersection of P + t v and normal dot x + d = 0, if unique."""
24    P = np.asarray(P, dtype=float)
25    v = np.asarray(v, dtype=float)
26    normal = np.asarray(normal, dtype=float)
27    denom = normal @ v
28    if abs(denom) < 1e-12:
29        return None
30    t = -(normal @ P + d) / denom
31    return P + t * v
    
```

The same templates appear in many numerical algorithms. Projection onto a hyperplane is used

in constrained optimization and feasibility methods. Signed distance to a hyperplane is used in classification. Line-plane intersection is used in graphics, computational geometry, and ray tracing.

4.14 Problem-solving strategies

Strategy 1: Identify the data type

Ask: are you given points, direction vectors, normal vectors, or equations?

- Two points usually give a direction vector by subtraction.
- A plane equation gives a normal vector from its coefficients.
- A parametric line gives a direction vector from the coefficients of t .
- A parametric plane gives two direction vectors from the coefficients of the two parameters.

Strategy 2: Choose the form that matches the task

- To **trace** a line or plane, use parametric form.
- To **test membership**, use scalar or normal form.
- To **find intersections**, substitute parametric equations into scalar equations.
- To **compute angles**, use dot products between direction vectors or normal vectors.
- To **compute distances**, use projections and normal vectors.

Strategy 3: Check dimensions

A common source of mistakes is using a formula in the wrong dimension.

- A line in $\mathbb{R}^2$ can be represented by one scalar equation.
- A plane in $\mathbb{R}^3$ can be represented by one scalar equation.
- A line in $\mathbb{R}^3$ usually needs either a parametric equation or two scalar equations.
- A hyperplane in $\mathbb{R}^n$ is represented by one scalar equation.

Strategy 4: Check degeneracy

Some formulas require nonzero or nonparallel vectors.

- A direction vector for a line cannot be zero.
- Two direction vectors for a plane cannot be parallel.
- Three points determine a plane only if they are not collinear.
- A symmetric line equation cannot divide by a zero component.

4.15 Worked review examples

Example 4.11: Line perpendicular to a plane

Find the line through $(1, 2, 3)$ perpendicular to

$$5x + y - 2z = 4.$$

A normal vector to the plane is

$$n = \langle 5, 1, -2 \rangle.$$

A line perpendicular to the plane has this normal vector as a direction vector. Therefore

$$r(t) = \langle 1, 2, 3 \rangle + t \langle 5, 1, -2 \rangle.$$

In coordinates,

$$x = 1 + 5t, \quad y = 2 + t, \quad z = 3 - 2t.$$

Example 4.12: Is a line parallel to a plane?

Show that the line

$$r(t) = \langle 1, 4, 5 \rangle + t \langle 1, 1, 1 \rangle$$

is parallel to the plane

$$4x - 9y + 5z - 8 = 0.$$

The direction vector of the line is

$$v = \langle 1, 1, 1 \rangle.$$

The normal vector of the plane is

$$n = \langle 4, -9, 5 \rangle.$$

Compute

$$v \cdot n = 4 - 9 + 5 = 0.$$

Thus the line direction is perpendicular to the plane normal, so the line is parallel to the plane. To decide whether the line lies in the plane or is disjoint from it, plug in a point on the line. At $(1, 4, 5)$,

$$4(1) - 9(4) + 5(5) - 8 = -15 \neq 0.$$

So the line is parallel to the plane but does not lie in the plane.

Example 4.13: A hyperplane in four dimensions

Let

$$w = \langle 2, -1, 3, 4 \rangle, \quad b = -5.$$

The hyperplane in $\mathbb{R}^4$ is

$$2x_1 - x_2 + 3x_3 + 4x_4 - 5 = 0.$$

For the point

$$x = (1, 0, 2, -1),$$

we get

$$w \cdot x + b = 2(1) - 0 + 3(2) + 4(-1) - 5 = -1.$$

So the point lies on the negative side of the hyperplane. Its signed distance is

$$\frac{-1}{\sqrt{2^2 + (-1)^2 + 3^2 + 4^2}} = -\frac{1}{\sqrt{30}}.$$

4.16 Python lab preview

The full lab for this chapter should ask students to:

1. write functions for lines, planes, and hyperplanes;
2. plot lines and planes in $\mathbb{R}^2$ and $\mathbb{R}^3$;
3. compute line-plane intersections;
4. compute distances to hyperplanes;
5. build a simple linear classifier from a given hyperplane;
6. visualize how changing w and b changes the decision boundary.

The goal is not just to draw pictures. The goal is to make the formulas computationally useful.

4.17 Exercises

Conceptual questions

1. Explain the difference between a direction vector and a normal vector.
2. Why does $r(t) = P + tv$ describe a line when $v \neq 0$?
3. Why does $r(s, t) = P + su + tv$ describe a plane only when u and v are not parallel?
4. What is the geometric meaning of the coefficients a, b, c in $ax + by + cz + d = 0$?
5. What does the sign of $w \cdot x + b$ tell us about a point relative to a hyperplane?

Skills practice

6. Find vector and parametric equations for the line through $(2, -1, 3)$ with direction vector $\langle 4, 0, -2 \rangle$.
7. Find the line segment from $P = (1, 2, 3)$ to $Q = (4, 5, 7)$.
8. Find the plane through $(1, 2, 3)$ with normal vector $\langle -1, 4, 2 \rangle$.
9. Find the plane through $(0, 1, 2)$, $(1, 0, 3)$, and $(2, 1, 5)$.
10. Find the angle between the planes $x - y + z = 2$ and $2x + y - z = 5$.
11. Find the intersection of the line $x = 1 + t$, $y = 2 - t$, $z = 3 + 2t$ with the plane $x + y + z = 10$.
12. Determine whether the line $r(t) = \langle 1, 0, 2 \rangle + t \langle 2, -1, 3 \rangle$ is parallel to the plane $2x + y - z = 4$.

Applications and computation

13. Write a Python function that returns the signed distance from a point in $\mathbb{R}^n$ to the hyperplane $w \cdot x + b = 0$.
14. Generate random points in $\mathbb{R}^2$ and classify them by the sign of $2x_1 - x_2 + 1$.
15. For the hyperplane $x_1 + x_2 + x_3 = 1$, compute the signed distances of the points $(1, 0, 0)$, $(1, 1, 1)$, and $(0, 0, 0)$.
16. A linear regression model with two features has prediction $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$. Explain why the graph of $\hat{y}$ as a function of (x_1, x_2) is a plane.
17. In logistic regression, the decision boundary is $w \cdot x + b = 0$. Explain why this boundary is a line in two features and a plane in three features.

Challenge problems

18. Derive the point-to-plane distance formula using projection onto the normal vector.
19. Let P, Q, R be three points in $\mathbb{R}^3$. Prove that they are collinear if and only if $(Q - P) \times (R - P) = 0$.
20. Let $H = \{x \in \mathbb{R}^n : w \cdot x + b = 0\}$. Show that H is a translate of the subspace $\{x \in \mathbb{R}^n : w \cdot x = 0\}$ whenever H is nonempty.
21. Suppose A is a $k \times n$ matrix and $b \in \mathbb{R}^k$. Explain why the solution set of $Ax = b$ is an affine subspace when it is nonempty.
22. Let H be a hyperplane $w \cdot x + b = 0$. Derive the formula for the orthogonal projection of a point p onto H .

4.18 Chapter summary

The central objects of this chapter are the flat geometric objects of multivariable calculus.

- A line is described by $r(t) = P + tv$.
- A line segment is described by $r(t) = (1 - t)P + tQ$ for $0 \leq t \leq 1$.
- A plane in $\mathbb{R}^3$ can be written as $n \cdot (x - P_0) = 0$.
- A plane can also be parameterized as $r(s, t) = P_0 + su + tv$.
- A hyperplane in $\mathbb{R}^n$ is described by $w \cdot x + b = 0$.
- Normal vectors are used for scalar equations, distances, and decision boundaries.
- Direction vectors are used for parametric equations and tracing objects.
- Intersections are usually found by substitution or by solving linear systems.

These ideas prepare us for cross products, curves, tangent planes, gradients, optimization, constrained problems, and linear models in statistics and machine learning.

Chapter 5

Cross Products, Curves, and Motion in Space

Chapter overview

In the previous chapters we learned how to describe points, vectors, lines, planes, and hyperplanes. This chapter adds two geometric tools that are essential for the rest of multivariable calculus.

The first tool is the **cross product**. Unlike the dot product, which returns a scalar, the cross product returns a vector. It encodes orientation, normal directions, area, and volume. It is the natural algebraic language for finding normal vectors to planes, surface orientations, torque, curl, and flux.

The second tool is the calculus of **vector-valued functions**. A curve in the plane or in space can be described by a position vector

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle.$$

Differentiating this vector gives velocity. Differentiating again gives acceleration. Integrating speed gives arc length. Differentiating the unit tangent vector gives curvature. These ideas connect geometry, physics, optimization paths, gradient descent trajectories, and data streams evolving through time.

Main idea

This chapter is a bridge from vector geometry to vector calculus. Cross products give normal directions and oriented area; vector-valued functions describe paths and motion. Later, line integrals, vector fields, flux, curl, Stokes' theorem, and the divergence theorem will all use these ideas.

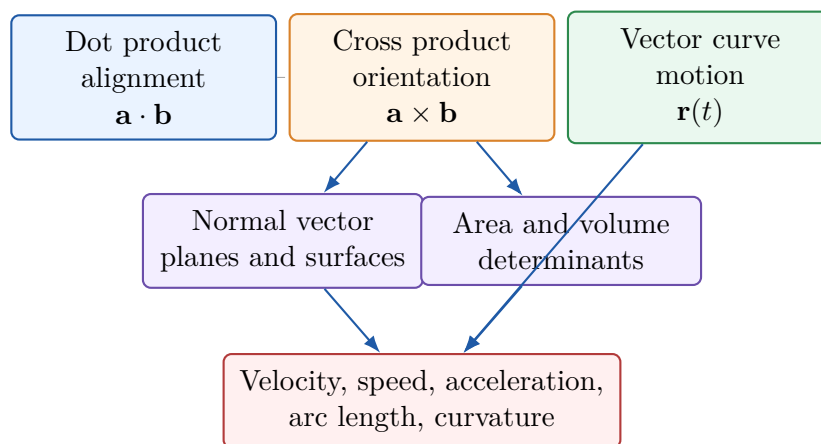


Figure 5.1: The chapter moves from algebraic orientation to geometric motion.

Learning goals

After studying this chapter, you should be able to:

1. compute cross products using determinants and component formulas;
2. interpret the direction of $\mathbf{a} \times \mathbf{b}$ using the right-hand rule;
3. compute areas using $\|\mathbf{a} \times \mathbf{b}\|$;
4. compute volumes using the scalar triple product;
5. describe curves using vector-valued functions;
6. compute velocity, speed, and acceleration;
7. compute arc length and reparameterize simple curves by arc length;
8. compute unit tangent vectors and curvature;
9. connect curves and motion to modern data and optimization examples.

5.1 Why the cross product is needed

The dot product answers a question about **alignment**:

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta.$$

It tells us whether two vectors point in similar, opposite, or perpendicular directions. But in three dimensions we often need a vector that is **perpendicular to two given vectors**. For example:

- a plane is often described by a normal vector;
- the orientation of a surface is described by a normal vector;
- the area of a parallelogram spanned by two vectors is measured by a perpendicular vector;
- the torque from a force is perpendicular to both the radius vector and the force vector;

- the curl of a vector field is written using a cross product with the differential operator ∇ .

The cross product is designed for exactly these tasks.

5.2 Definition and computation of the cross product

Cross product in $\mathbb{R}^3$

Let

$$\mathbf{a} = \langle a_1, a_2, a_3 \rangle, \quad \mathbf{b} = \langle b_1, b_2, b_3 \rangle.$$

The **cross product** $\mathbf{a} \times \mathbf{b}$ is the vector

$$\mathbf{a} \times \mathbf{b} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle.$$

It is often remembered using determinant notation:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

Expanding along the first row gives

$$\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}.$$

Only in three dimensions for this course

The cross product in this familiar vector form is a special operation for three-dimensional vectors. In higher-dimensional linear algebra, related ideas are expressed using determinants, exterior products, orientations, and differential forms.

Example 5.1: Compute a cross product

Let

$$\mathbf{a} = \langle 1, 2, 3 \rangle, \quad \mathbf{b} = \langle 3, 2, 1 \rangle.$$

Then

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \langle 2 \cdot 1 - 3 \cdot 2, 3 \cdot 3 - 1 \cdot 1, 1 \cdot 2 - 2 \cdot 3 \rangle \\ &= \langle -4, 8, -4 \rangle. \end{aligned}$$

This vector is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$:

$$\langle -4, 8, -4 \rangle \cdot \langle 1, 2, 3 \rangle = -4 + 16 - 12 = 0,$$

$$\langle -4, 8, -4 \rangle \cdot \langle 3, 2, 1 \rangle = -12 + 16 - 4 = 0.$$

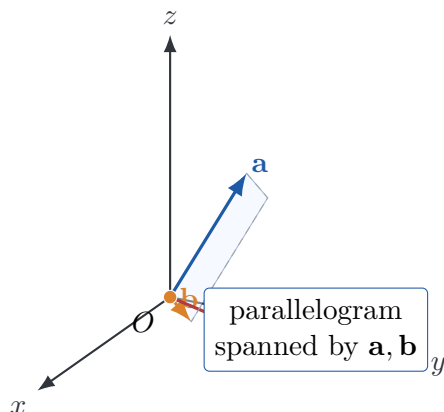


Figure 5.2: The cross product $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. The vector is shown scaled for readability.

5.3 Basic properties of the cross product

For vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ in $\mathbb{R}^3$ and scalars λ , the cross product satisfies

$$\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a}),$$

$$\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c},$$

$$(\lambda \mathbf{a}) \times \mathbf{b} = \lambda(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \times (\lambda \mathbf{b}),$$

$$\mathbf{a} \times \mathbf{a} = \mathbf{0}.$$

The first identity is especially important: **order matters**. Reversing the order reverses the direction.

5.3.1 Standard basis orientation

The standard basis vectors satisfy

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}, \quad \mathbf{j} \times \mathbf{k} = \mathbf{i}, \quad \mathbf{k} \times \mathbf{i} = \mathbf{j}.$$

Reversing the order changes the sign:

$$\mathbf{j} \times \mathbf{i} = -\mathbf{k}, \quad \mathbf{k} \times \mathbf{j} = -\mathbf{i}, \quad \mathbf{i} \times \mathbf{k} = -\mathbf{j}.$$

This orientation is called the **right-hand rule**. If the fingers of your right hand rotate from $\mathbf{a}$ toward $\mathbf{b}$, then your thumb points in the direction of $\mathbf{a} \times \mathbf{b}$.

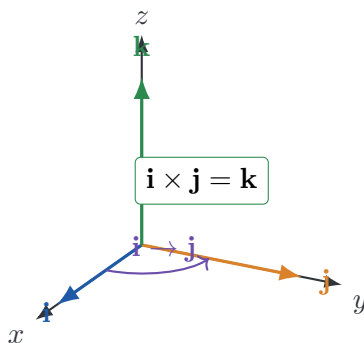


Figure 5.3: The standard right-handed orientation: rotating from $\mathbf{i}$ to $\mathbf{j}$ sends the thumb toward $\mathbf{k}$.

5.4 Area from the cross product

The magnitude of the cross product has a geometric meaning:

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta,$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$. This is the area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$.

Therefore

$$\text{Area of parallelogram} = \|\mathbf{a} \times \mathbf{b}\|, \quad \text{Area of triangle} = \frac{1}{2} \|\mathbf{a} \times \mathbf{b}\|.$$

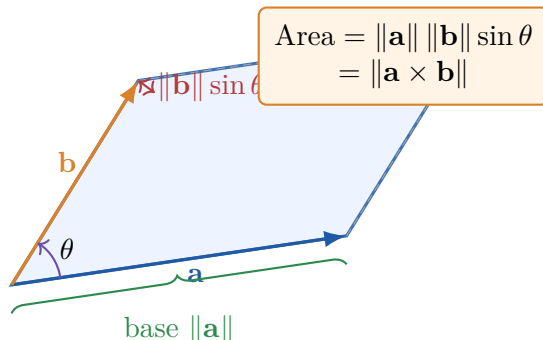


Figure 5.4: The cross product magnitude is base times height, so it gives the parallelogram area.

Example 5.2: Area of a triangle

Find the area of the triangle with vertices

$$O = (0, 0, 0), \quad A = (1, 2, 3), \quad B = (3, 2, 1).$$

The two side vectors from O are

$$\overrightarrow{OA} = \langle 1, 2, 3 \rangle, \quad \overrightarrow{OB} = \langle 3, 2, 1 \rangle.$$

From Example 5.1,

$$\overrightarrow{OA} \times \overrightarrow{OB} = \langle -4, 8, -4 \rangle.$$

Thus

$$\|\vec{OA} \times \vec{OB}\| = \sqrt{(-4)^2 + 8^2 + (-4)^2} = \sqrt{96} = 4\sqrt{6}.$$

So the area of the triangle is

$$\frac{1}{2}(4\sqrt{6}) = 2\sqrt{6}.$$

5.5 Normal vectors and planes

A plane in $\mathbb{R}^3$ can be described using a point $P_0 = (x_0, y_0, z_0)$ and a normal vector

$$\mathbf{n} = \langle A, B, C \rangle.$$

The equation is

$$\mathbf{n} \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0,$$

or in standard form,

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0.$$

If a plane is spanned by two nonparallel vectors $\mathbf{u}$ and $\mathbf{v}$, then a normal vector is

$$\mathbf{n} = \mathbf{u} \times \mathbf{v}.$$

Example 5.3: Plane through three points

Find an equation of the plane through

$$P = (1, 2, 3), \quad Q = (2, 2, 5), \quad R = (3, 4, 1).$$

First compute two direction vectors in the plane:

$$\vec{PQ} = Q - P = \langle 1, 0, 2 \rangle, \quad \vec{PR} = R - P = \langle 2, 2, -2 \rangle.$$

A normal vector is

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 2 \\ 2 & 2 & -2 \end{vmatrix} = \langle -4, 6, 2 \rangle.$$

We may divide by 2 and use $\mathbf{n} = \langle -2, 3, 1 \rangle$. Using $P = (1, 2, 3)$, the plane is

$$-2(x - 1) + 3(y - 2) + (z - 3) = 0,$$

so

$$-2x + 3y + z - 7 = 0.$$

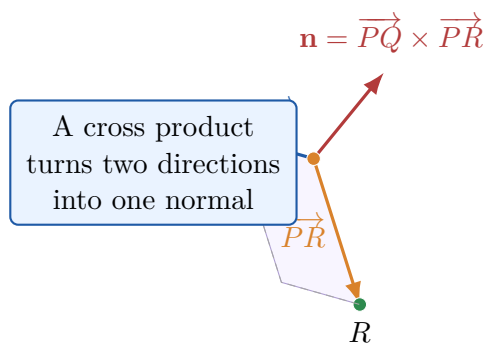


Figure 5.5: A cross product gives a normal vector to the plane through three non-collinear points.

5.6 Scalar triple product and volume

Scalar triple product

The **scalar triple product** of $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ is

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}).$$

Geometrically,

$$|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$$

is the volume of the parallelepiped determined by $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.

Using a determinant,

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

If the scalar triple product is zero, then the three vectors are coplanar.

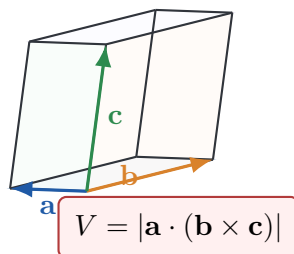


Figure 5.6: The scalar triple product measures signed volume. Its absolute value is the volume of the parallelepiped.

Example 5.4: Test coplanarity

Let

$$\mathbf{a} = \langle -3, 0, 6 \rangle, \quad \mathbf{b} = \langle 4, 1, -7 \rangle, \quad \mathbf{c} = \langle 1, -2, -4 \rangle.$$

Compute

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} -3 & 0 & 6 \\ 4 & 1 & -7 \\ 1 & -2 & -4 \end{vmatrix}.$$

First,

$$\begin{aligned} \mathbf{b} \times \mathbf{c} &= \langle 1(-4) - (-7)(-2), (-7)(1) - 4(-4), 4(-2) - 1(1) \rangle \\ &= \langle -18, 9, -9 \rangle. \end{aligned}$$

Then

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \langle -3, 0, 6 \rangle \cdot \langle -18, 9, -9 \rangle = 54 + 0 - 54 = 0.$$

Therefore the vectors are coplanar.

5.7 Vector-valued functions and space curves

Vector-valued function

A **vector-valued function** takes a scalar input and returns a vector:

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle.$$

The image of $\mathbf{r}(t)$ is a curve in space. The parameter t may represent time, but it can also simply be a variable that traces the curve.

Example 5.5: A helix

The curve

$$\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$$

is a helix. Its projection onto the xy -plane is the unit circle, while the z -coordinate increases linearly.

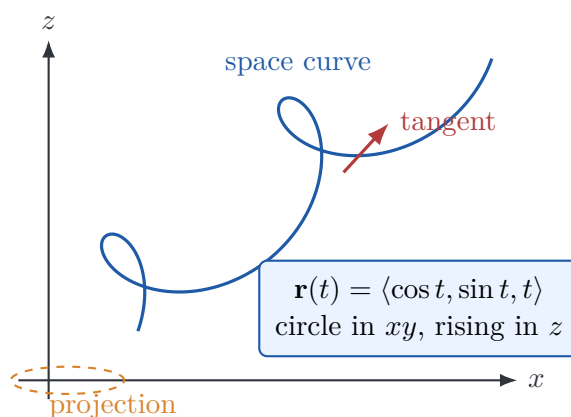


Figure 5.7: The helix winds around the z -axis while rising at a constant rate. The picture uses a projected view.

5.8 Derivatives and integrals of vector-valued functions

The derivative of

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$$

is computed component by component:

$$\mathbf{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle.$$

Similarly,

$$\int_a^b \mathbf{r}(t) dt = \left\langle \int_a^b x(t) dt, \int_a^b y(t) dt, \int_a^b z(t) dt \right\rangle.$$

The product rules are analogous to ordinary calculus:

$$\frac{d}{dt} (\mathbf{a}(t) \cdot \mathbf{b}(t)) = \mathbf{a}'(t) \cdot \mathbf{b}(t) + \mathbf{a}(t) \cdot \mathbf{b}'(t),$$

$$\frac{d}{dt} (\mathbf{a}(t) \times \mathbf{b}(t)) = \mathbf{a}'(t) \times \mathbf{b}(t) + \mathbf{a}(t) \times \mathbf{b}'(t).$$

Example 5.6: Differentiate a vector function

Let

$$\mathbf{r}(t) = \left\langle t^3, \frac{1}{2}t, t^2 - 4t \right\rangle.$$

Then

$$\mathbf{r}'(t) = \left\langle 3t^2, \frac{1}{2}, 2t - 4 \right\rangle, \quad \mathbf{r}''(t) = \langle 6t, 0, 2 \rangle.$$

At $t = 1$,

$$\mathbf{r}'(1) = \left\langle 3, \frac{1}{2}, -2 \right\rangle, \quad \mathbf{r}''(1) = \langle 6, 0, 2 \rangle.$$

5.9 Motion in space: velocity, speed, and acceleration

Suppose a particle moves through space with position vector

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle.$$

The **velocity vector** is

$$\mathbf{v}(t) = \mathbf{r}'(t).$$

The **speed** is the magnitude of the velocity vector:

$$\text{speed} = \|\mathbf{v}(t)\| = \|\mathbf{r}'(t)\|.$$

The **acceleration vector** is

$$\mathbf{a}(t) = \mathbf{v}'(t) = \mathbf{r}''(t).$$

Vector versus scalar quantities

Velocity is a vector because it has magnitude and direction. Speed is a scalar because it is only the magnitude of velocity.

Example 5.7: Motion on a helix

Let

$$\mathbf{r}(t) = \langle \sin t, \cos t, t \rangle.$$

Then

$$\mathbf{v}(t) = \mathbf{r}'(t) = \langle \cos t, -\sin t, 1 \rangle,$$

and

$$\mathbf{a}(t) = \mathbf{r}''(t) = \langle -\sin t, -\cos t, 0 \rangle.$$

The speed is

$$\|\mathbf{v}(t)\| = \sqrt{\cos^2 t + \sin^2 t + 1} = \sqrt{2}.$$

So the particle moves with constant speed even though its direction is changing.

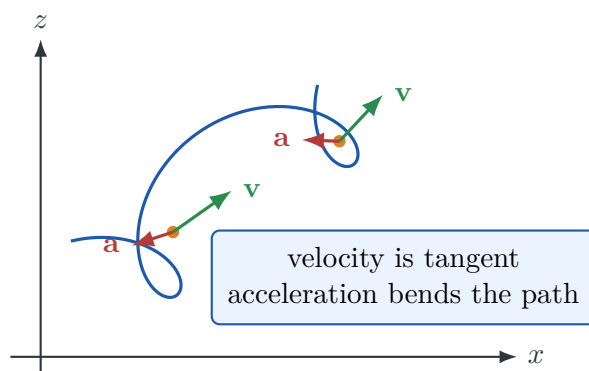


Figure 5.8: On a helix, velocity is tangent to the curve while acceleration points inward toward the axis.

5.10 Recovering velocity and position from acceleration

Because

$$\mathbf{a}(t) = \mathbf{v}'(t),$$

we can recover velocity by integration:

$$\mathbf{v}(t) = \mathbf{v}(0) + \int_0^t \mathbf{a}(u) du.$$

Because

$$\mathbf{v}(t) = \mathbf{r}'(t),$$

we can recover position by another integration:

$$\mathbf{r}(t) = \mathbf{r}(0) + \int_0^t \mathbf{v}(u) du.$$

Example 5.8: From acceleration to position

A particle starts at

$$\mathbf{r}(0) = \langle 0, 0, 1 \rangle$$

with initial velocity

$$\mathbf{v}(0) = \mathbf{i} + \mathbf{k} = \langle 1, 0, 1 \rangle.$$

Its acceleration is

$$\mathbf{a}(t) = \langle 2t, 1, 6t \rangle.$$

Then

$$\mathbf{v}(t) = \langle 1, 0, 1 \rangle + \int_0^t \langle 2u, 1, 6u \rangle du = \langle 1 + t^2, t, 1 + 3t^2 \rangle.$$

Now integrate again:

$$\begin{aligned} \mathbf{r}(t) &= \langle 0, 0, 1 \rangle + \int_0^t \langle 1 + u^2, u, 1 + 3u^2 \rangle du \\ &= \left\langle t + \frac{t^3}{3}, \frac{t^2}{2}, 1 + t + t^3 \right\rangle. \end{aligned}$$

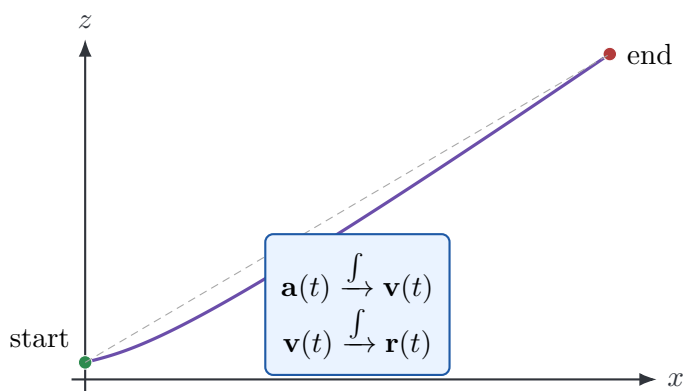


Figure 5.9: Integrating acceleration gives velocity; integrating velocity gives position. The plotted curve shows the x - z projection of the trajectory.

5.11 Arc length

Suppose a curve is given by

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle, \quad a \leq t \leq b,$$

and assume the curve is traversed exactly once. The length of the curve is

$$L = \int_a^b \|\mathbf{r}'(t)\| dt.$$

Equivalently,

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt.$$

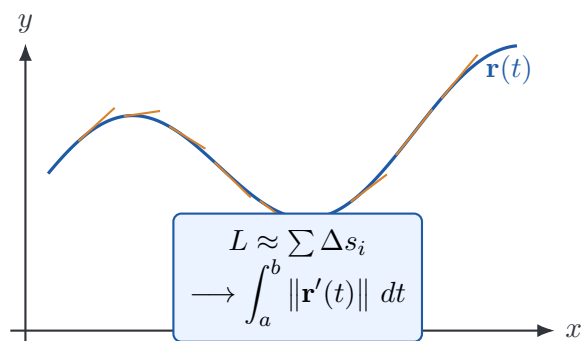


Figure 5.10: Arc length is obtained as the limit of many small straight-line approximations.

Example 5.9: Length of a circular helix

Let

$$\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle, \quad 0 \leq t \leq 2\pi.$$

Then

$$\mathbf{r}'(t) = \langle -\sin t, \cos t, 1 \rangle,$$

so

$$\|\mathbf{r}'(t)\| = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}.$$

Therefore

$$L = \int_0^{2\pi} \sqrt{2} \, dt = 2\sqrt{2}\pi.$$

5.12 Arc length parameterization

The arc length function from $t = a$ to t is

$$s(t) = \int_a^t \|\mathbf{r}'(u)\| \, du.$$

When $s(t)$ can be solved for t as a function of s , we can reparameterize the curve by arc length. This is useful because the new parameter directly measures distance traveled along the curve.

Example 5.10: Reparameterize a helix by arc length

For

$$\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle, \quad t \geq 0,$$

we have $\|\mathbf{r}'(t)\| = \sqrt{2}$. Thus

$$s(t) = \int_0^t \sqrt{2} \, du = \sqrt{2}t.$$

Solving for t gives

$$t = \frac{s}{\sqrt{2}}.$$

Therefore the arc length parameterization is

$$\mathbf{r}(s) = \left\langle \cos \left(\frac{s}{\sqrt{2}} \right), \sin \left(\frac{s}{\sqrt{2}} \right), \frac{s}{\sqrt{2}} \right\rangle.$$

With this parameterization,

$$\left\| \frac{d\mathbf{r}}{ds} \right\| = 1.$$

That is why an arc length parameterization is also called a **unit-speed parameterization**.

5.13 Unit tangent vector

For a smooth curve with $\mathbf{r}'(t) \neq \mathbf{0}$, the **unit tangent vector** is

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}.$$

This vector points in the direction of motion, but has length 1.

For the helix

$$\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle,$$

we have

$$\mathbf{T}(t) = \frac{1}{\sqrt{2}} \langle -\sin t, \cos t, 1 \rangle.$$

5.14 Curvature

Curvature measures how quickly a curve changes direction. A straight line has curvature 0. A small circle has larger curvature than a large circle.

The conceptual definition is

$$\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\|,$$

where s is arc length. A practical formula is

$$\kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}.$$

For many computations in $\mathbb{R}^3$, the most convenient formula is

$$\kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}.$$

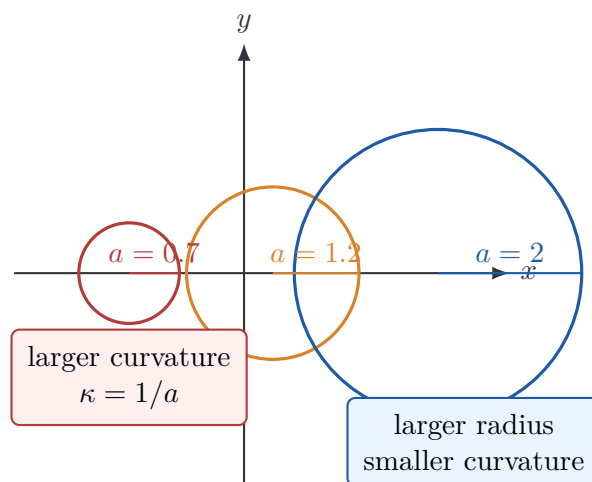


Figure 5.11: For a circle of radius a , the curvature is $\kappa = 1/a$.

Example 5.11: Curvature of a circle

Let a circle of radius a be parameterized by

$$\mathbf{r}(t) = \langle a \cos t, a \sin t, 0 \rangle.$$

Then

$$\mathbf{r}'(t) = \langle -a \sin t, a \cos t, 0 \rangle, \quad \mathbf{r}''(t) = \langle -a \cos t, -a \sin t, 0 \rangle.$$

We have $\|\mathbf{r}'(t)\| = a$ and

$$\|\mathbf{r}'(t) \times \mathbf{r}''(t)\| = a^2.$$

Therefore

$$\kappa(t) = \frac{a^2}{a^3} = \frac{1}{a}.$$

Thus a circle with larger radius has smaller curvature.

Example 5.12: Curvature of the circular helix

Let

$$\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle.$$

Then

$$\mathbf{r}'(t) = \langle -\sin t, \cos t, 1 \rangle, \quad \mathbf{r}''(t) = \langle -\cos t, -\sin t, 0 \rangle.$$

Compute the cross product:

$$\mathbf{r}'(t) \times \mathbf{r}''(t) = \langle \sin t, -\cos t, 1 \rangle.$$

Thus

$$\|\mathbf{r}'(t) \times \mathbf{r}''(t)\| = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}, \quad \|\mathbf{r}'(t)\| = \sqrt{2}.$$

Therefore

$$\kappa(t) = \frac{\sqrt{2}}{(\sqrt{2})^3} = \frac{1}{2}.$$

The helix has constant curvature.

5.15 Finding when speed is minimized

In applications, a trajectory may have variable speed. Since speed is nonnegative, minimizing speed is equivalent to minimizing the square of speed.

Example 5.13: Minimum speed

Let

$$\mathbf{r}(t) = \langle t^2 + 1, 3t, t^2 - 4t \rangle.$$

Then

$$\mathbf{v}(t) = \mathbf{r}'(t) = \langle 2t, 3, 2t - 4 \rangle.$$

The speed is

$$\|\mathbf{v}(t)\| = \sqrt{(2t)^2 + 3^2 + (2t - 4)^2}.$$

Instead of minimizing this square root, minimize

$$\|\mathbf{v}(t)\|^2 = 4t^2 + 9 + (2t - 4)^2.$$

Simplify:

$$\|\mathbf{v}(t)\|^2 = 4t^2 + 9 + 4t^2 - 16t + 16 = 8t^2 - 16t + 25.$$

This quadratic is minimized when

$$t = \frac{-(-16)}{2 \cdot 8} = 1.$$

Therefore the speed is minimized at $t = 1$.

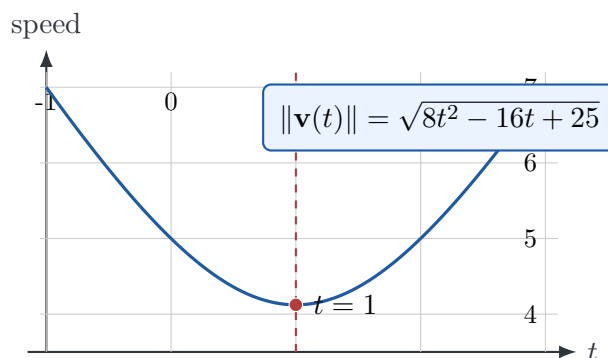


Figure 5.12: Minimizing speed is often easier by minimizing speed squared.

5.16 Modern viewpoint: trajectories in data and optimization

A vector-valued function does not need to describe a physical particle. It can describe any time-indexed or parameter-indexed state.

Gradient descent as a curve

In machine learning, an optimization algorithm generates a sequence of model parameters

$$\theta_0, \theta_1, \theta_2, \dots,$$

where each θ_k may be a vector in a high-dimensional space. If we interpolate these points, we can view training as a curve in parameter space. The velocity of this curve describes how fast the parameters are changing. The curvature describes how sharply the path bends. A highly curved path may indicate unstable optimization, poor scaling, or a loss landscape with narrow valleys.

Data streams as vector-valued functions

A time series with several measurements at each time can be represented as

$$\mathbf{x}(t) = \langle x_1(t), x_2(t), \dots, x_n(t) \rangle.$$

Then $\mathbf{x}'(t)$ measures the instantaneous change of all features. This viewpoint appears in dynamical systems, finance, biology, robotics, and sensor data.

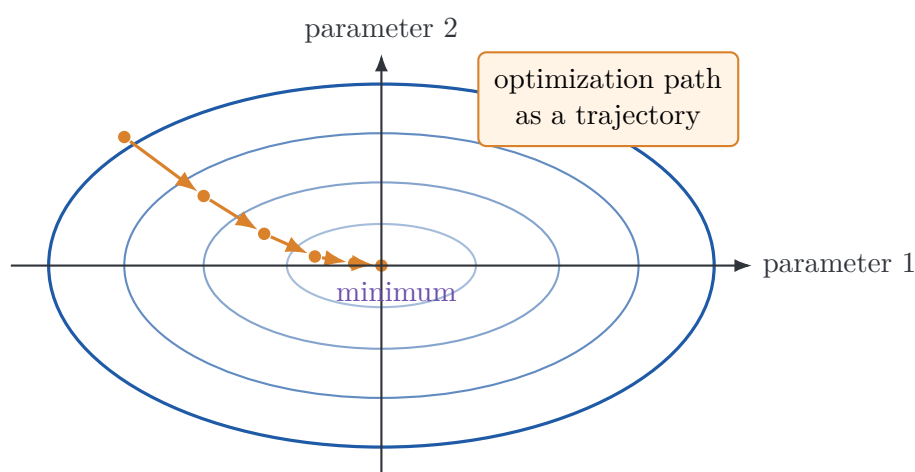


Figure 5.13: An optimization path can be viewed as a vector-valued trajectory in parameter space.

5.17 Summary of core formulas

Concept	Formula
Cross product	$\mathbf{a} \times \mathbf{b} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle$
Orthogonality	$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = 0, (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{b} = 0$
Parallelogram area	$\ \mathbf{a} \times \mathbf{b}\ $
Triangle area	$\frac{1}{2} \ \mathbf{a} \times \mathbf{b}\ $
Scalar triple product	$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$
Parallelepiped volume	$ \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) $
Curve	$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$
Velocity	$\mathbf{v}(t) = \mathbf{r}'(t)$
Speed	$\ \mathbf{v}(t)\ = \ \mathbf{r}'(t)\ $
Acceleration	$\mathbf{a}(t) = \mathbf{r}''(t)$
Arc length	$L = \int_a^b \ \mathbf{r}'(t)\ dt$
Arc length function	$s(t) = \int_a^t \ \mathbf{r}'(u)\ du$
Unit tangent	$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\ \mathbf{r}'(t)\ }$
Curvature	$\kappa = \left\ \frac{d\mathbf{T}}{ds} \right\ $
Curvature formula	$\kappa(t) = \frac{\ \mathbf{r}'(t) \times \mathbf{r}''(t)\ }{\ \mathbf{r}'(t)\ ^3}$

Table 5.1: Core formulas for cross products, curves, and motion.

5.18 Practice problems

Conceptual questions

1. Explain why $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.
2. Why does reversing the order of a cross product reverse its direction?
3. Why does $\|\mathbf{a} \times \mathbf{b}\|$ represent area?
4. What does it mean if $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$?
5. Explain the difference between velocity and speed.
6. Why is arc length parameterization called unit-speed parameterization?
7. Why is curvature defined using $d\mathbf{T}/ds$ rather than simply $d\mathbf{T}/dt$?

Computational exercises

1. Compute $\langle 2, -1, 4 \rangle \times \langle 0, 3, 1 \rangle$.

2. Find the area of the parallelogram spanned by $\langle 1, 2, 0 \rangle$ and $\langle 3, -1, 4 \rangle$.
3. Find the area of the triangle with vertices $P = (1, 0, 2)$, $Q = (3, 1, 1)$, and $R = (0, 2, 4)$.
4. Find a normal vector to the plane through $P = (1, 1, 0)$, $Q = (2, 0, 1)$, and $R = (0, 3, 2)$.
5. Find an equation of the plane through the points in Problem 4.
6. Compute the volume of the parallelepiped determined by $\mathbf{a} = \langle 1, 0, 2 \rangle$, $\mathbf{b} = \langle 2, 1, 1 \rangle$, and $\mathbf{c} = \langle 0, 3, 1 \rangle$.
7. Determine whether $\langle 1, 2, 3 \rangle$, $\langle 2, 4, 6 \rangle$, and $\langle 0, 1, 1 \rangle$ are coplanar.
8. Let $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$. Find $\mathbf{v}(t)$, $\mathbf{a}(t)$, and speed.
9. Let $\mathbf{r}(t) = \langle \cos(2t), \sin(2t), 3t \rangle$. Find speed.
10. Find the arc length of $\mathbf{r}(t) = \langle 3t, 4t, 12t \rangle$ for $0 \leq t \leq 2$.
11. Find the arc length of $\mathbf{r}(t) = \langle \cos t, \sin t, 2t \rangle$ for $0 \leq t \leq 2\pi$.
12. Compute the curvature of $\mathbf{r}(t) = \langle \cos t, \sin t, 0 \rangle$.
13. Compute the curvature of $\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t, 0 \rangle$.
14. Find when the speed of $\mathbf{r}(t) = \langle t^2, t^2 - 2t, 3t \rangle$ is minimized.

Computational exploration

1. Write a function that computes the cross product from the component formula.
2. Verify numerically that $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ for several random examples.
3. Plot the curve $\mathbf{r}(t) = \langle \cos t, \sin t, 0.2t \rangle$ for $0 \leq t \leq 8\pi$.
4. Plot velocity vectors at five points on the curve in the previous problem.
5. Numerically approximate arc length for $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ on $0 \leq t \leq 1$.
6. Use finite differences to approximate velocity and acceleration from sampled position data.
7. Simulate a gradient descent path for a quadratic function and plot the trajectory on contour lines.

5.19 Chapter project: tracking motion from data

Suppose a sensor records the position of an object at equally spaced times. The data are noisy, so you do not know $\mathbf{r}(t)$ exactly. Your goal is to estimate velocity, speed, acceleration, and curvature from the data.

A possible workflow:

1. Load or simulate position data $\mathbf{r}(t_k)$.
2. Estimate velocity using finite differences.
3. Estimate speed as the norm of velocity.

4. Estimate acceleration using finite differences again.
5. Estimate curvature using

$$\kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}.$$

6. Plot the trajectory and color it by speed or curvature.

This project connects calculus to robotics, sports tracking, physics simulations, autonomous systems, and time-dependent data analysis.

5.20 What comes next

The cross product and space curves are not isolated topics. They prepare us for the main constructions of vector calculus. In later chapters:

- line integrals use $ds = \|\mathbf{r}'(t)\| dt$;
- work integrals use $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)$;
- surface normals often come from cross products;
- curl is remembered as $\nabla \times \mathbf{F}$;
- Stokes' theorem relates circulation around a boundary curve to curl through a surface.

The ideas of this chapter therefore form the geometric foundation for the second half of the book.

Chapter 6. Functions of Several Variables

Scalar fields, domains, surfaces, level sets, and data landscapes

Single-variable calculus studies functions of the form $y = f(x)$, where one real input produces one real output. Multivariable calculus studies functions such as

$$z = f(x, y), \quad w = f(x, y, z), \quad F = f(x_1, x_2, \dots, x_n).$$

The input is no longer a point on a line. It may be a point in the plane, a point in space, or a vector in a high-dimensional parameter space. The output is still a scalar: a height, temperature, density, probability, loss, likelihood, energy, cost, or risk.

This chapter develops the geometric language for these objects. We learn how to describe domains, draw surfaces, read contour maps, study level curves and level surfaces, and use lower-dimensional views to understand high-dimensional functions.

Role in the book

The next several chapters study limits, derivatives, gradients, optimization, and multiple integrals. All of those topics depend on the viewpoint developed here: a multivariable function is a scalar field over a region of space.

Learning goals

After studying this chapter, you should be able to:

1. define functions of two, three, and n variables;
2. distinguish domain, codomain, and range;
3. find domains defined by square roots, logarithms, denominators, and geometric restrictions;
4. interpret the graph of $z = f(x, y)$ as a surface above a domain in the xy -plane;
5. recognize planes, hemispheres, paraboloids, and saddle surfaces;
6. compute and interpret level curves $f(x, y) = c$;
7. interpret level surfaces $f(x, y, z) = c$;
8. use slices, contour plots, heatmaps, and surface plots to visualize functions;
9. connect multivariable functions to probability, statistics, optimization, data science, and machine learning.

1 Functions of several variables

A *function of two variables* is a rule that assigns to each input pair (x, y) in a domain $D \subseteq \mathbb{R}^2$ exactly one real number $f(x, y)$:

$$f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}.$$

The set D is the *domain*. The set of all output values is the *range*:

$$\text{Range}(f) = \{f(x, y) : (x, y) \in D\}.$$

A *function of three variables* is a rule

$$f : E \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}, \quad (x, y, z) \mapsto f(x, y, z).$$

More generally, a *function of n variables* is a rule

$$f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}, \quad x = (x_1, x_2, \dots, x_n) \mapsto f(x_1, x_2, \dots, x_n).$$

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is often called a *scalar field* because it assigns a scalar value to each point in space.

Examples from science, statistics, and AI

Function	Input	Output	Interpretation
$T(x, y)$	longitude and latitude	temperature	weather map
$p(x, y, z)$	point in space	pressure	fluid or atmosphere
$\rho(x, y, z)$	point in space	density	mass distribution
$L(\beta_0, \beta_1)$	model parameters	squared error loss	linear regression
$\ell(\theta)$	statistical parameter	log-likelihood	inference
$J(w_1, \dots, w_n)$	model weights	training objective	machine learning

Modern viewpoint

In data science and AI, the most important functions are often high-dimensional. A neural-network loss function may depend on millions or billions of parameters. We cannot draw its graph, but the ideas of domains, level sets, slices, gradients, and local approximations remain central.

2 Domains as regions

Finding the domain is often the first step in studying a multivariable function. The algebraic restrictions are familiar from single-variable calculus, but now they describe regions in the plane or space.

Expression	Restriction
$\sqrt{g(x, y)}$	$g(x, y) \geq 0$
$\ln(g(x, y))$	$g(x, y) > 0$
$\frac{1}{g(x, y)}$	$g(x, y) \neq 0$
$\arcsin(g(x, y))$	$-1 \leq g(x, y) \leq 1$

Worked example 2.1: Square root and denominator

Find the domain of

$$f(x, y) = \frac{\sqrt{y - x}}{y}.$$

The square root requires $y - x \geq 0$, so $y \geq x$. The denominator requires $y \neq 0$. Therefore

$$D = \{(x, y) \in \mathbb{R}^2 : y \geq x, y \neq 0\}.$$

This is the half-plane above the line $y = x$, with the portion of the line $y = 0$ removed.

Worked example 2.2: Parabolic domain with excluded lines

Find the domain of

$$g(x, y) = \frac{\sqrt{x^2 - y}}{x^2 - 4}.$$

The square root requires $x^2 - y \geq 0$, so $y \leq x^2$. The denominator requires $x^2 - 4 \neq 0$, so $x \neq 2$ and $x \neq -2$. Thus

$$D = \{(x, y) \in \mathbb{R}^2 : y \leq x^2, x \neq \pm 2\}.$$

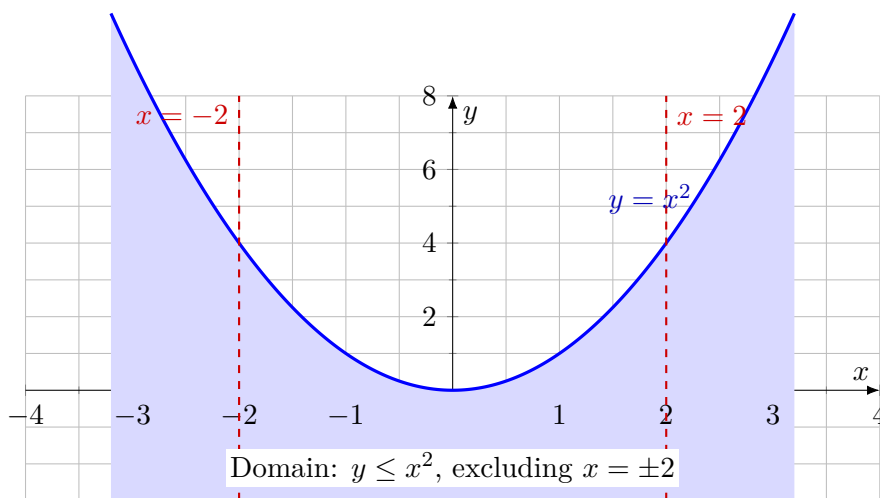


Figure 1: The domain of $\sqrt{x^2 - y}/(x^2 - 4)$ is the region below the parabola $y = x^2$, with the vertical lines $x = \pm 2$ removed.

Worked example 2.3: Logarithmic disk

Find the domain of

$$h(x, y) = \ln(9 - x^2 - y^2).$$

The logarithm requires $9 - x^2 - y^2 > 0$, or equivalently,

$$x^2 + y^2 < 9.$$

Thus the domain is the open disk of radius 3 centered at the origin.

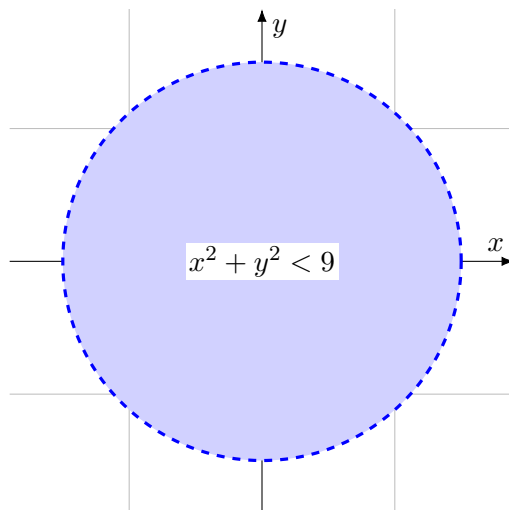


Figure 2: The domain of $\ln(9 - x^2 - y^2)$ is the open disk $x^2 + y^2 < 9$. The boundary circle is not included.

Common mistake

For $\ln(9 - x^2 - y^2)$ the boundary circle $x^2 + y^2 = 9$ is *not* included, because $\ln(0)$ is undefined.
For $\sqrt{9 - x^2 - y^2}$ the boundary is included, because $\sqrt{0} = 0$.

3 Graphs of functions of two variables

For a function $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$, the *graph* is the set of points in $\mathbb{R}^3$ given by

$$\{(x, y, z) : (x, y) \in D, z = f(x, y)\}.$$

We usually write the graph as $z = f(x, y)$. The graph is a surface above or below the domain in the xy -plane.

Worked example 3.1: A plane as a graph

Let

$$f(x, y) = 6 - 2x - 3y.$$

Then

$$z = 6 - 2x - 3y,$$

or equivalently,

$$2x + 3y + z = 6.$$

This is a plane. It has x -intercept 3, y -intercept 2, and z -intercept 6.

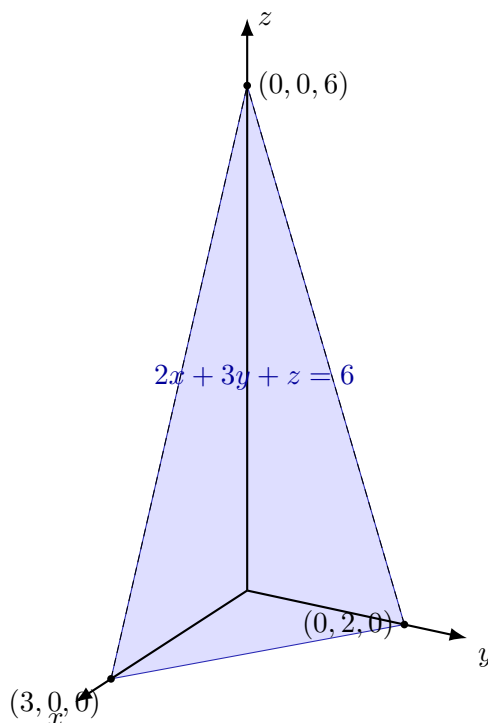


Figure 3: The graph of $f(x, y) = 6 - 2x - 3y$ is a plane. Its intercepts make the geometry transparent.

Worked example 3.2: Upper hemisphere

Let

$$f(x, y) = \sqrt{4 - x^2 - y^2}.$$

The domain is determined by $4 - x^2 - y^2 \geq 0$, so $x^2 + y^2 \leq 4$. The graph satisfies

$$z = \sqrt{4 - x^2 - y^2}.$$

Squaring both sides gives

$$x^2 + y^2 + z^2 = 4.$$

Because $z \geq 0$, the graph is the upper hemisphere of radius 2 centered at the origin. The range is $0 \leq z \leq 2$.

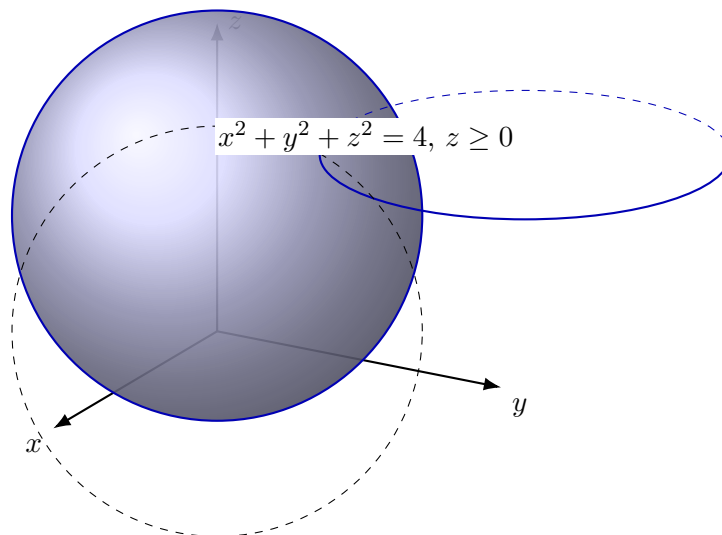


Figure 4: The graph of $f(x, y) = \sqrt{4 - x^2 - y^2}$ is the upper hemisphere of radius 2.

Worked example 3.3: Elliptic paraboloid

Let

$$f(x, y) = x^2 + 9y^2.$$

The domain is all of $\mathbb{R}^2$. Since squares are nonnegative, the range is $z \geq 0$. The graph

$$z = x^2 + 9y^2$$

is an *elliptic paraboloid*. It opens upward, and it rises faster in the y -direction than in the x -direction because the coefficient of y^2 is larger.

Worked example 3.4: A saddle surface

Let

$$f(x, y) = x^2 - y^2.$$

The graph

$$z = x^2 - y^2$$

is a saddle surface. Along the x -axis, where $y = 0$, it becomes $z = x^2$, which opens upward. Along the y -axis, where $x = 0$, it becomes $z = -y^2$, which opens downward.

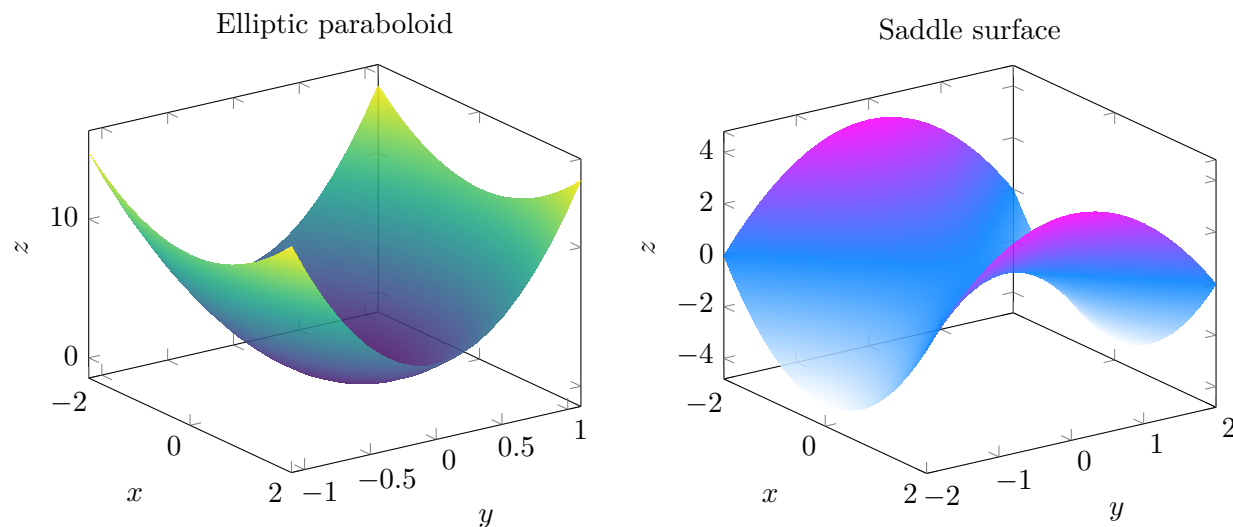


Figure 5: Two basic surfaces: the elliptic paraboloid $z = x^2 + 9y^2$ and the saddle surface $z = x^2 - y^2$.

4 Slices and traces

A surface can be understood by taking lower-dimensional slices. For $z = f(x, y)$:

- fixing $y = y_0$ gives the curve $z = f(x, y_0)$ in a vertical plane parallel to the xz -plane;
- fixing $x = x_0$ gives the curve $z = f(x_0, y)$ in a vertical plane parallel to the yz -plane;
- fixing $z = c$ gives the level curve $f(x, y) = c$ in the xy -plane.

Slices connect single-variable calculus to multivariable calculus. Partial derivatives in later chapters measure slopes of coordinate slices.

Worked example 4.1: Slices of an elliptic paraboloid

For

$$z = x^2 + 9y^2,$$

fixing $y = 0$ gives $z = x^2$, while fixing $x = 0$ gives $z = 9y^2$. The second slice is steeper.

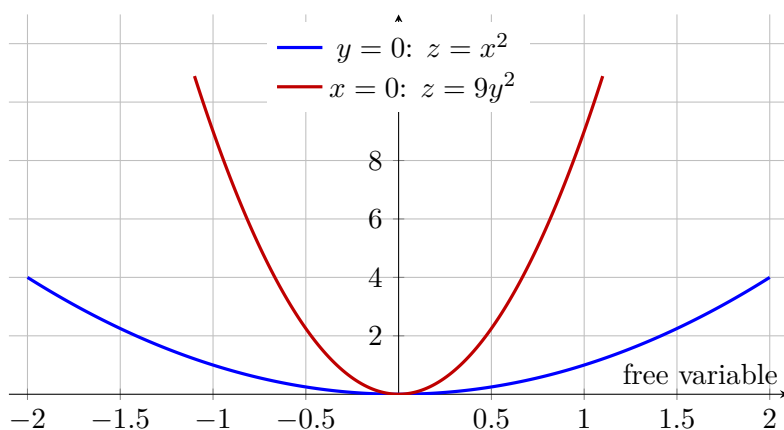


Figure 6: Coordinate slices of $z = x^2 + 9y^2$ show different rates of change in the x and y directions.

5 Level curves and contour maps

A *level curve* of a function $f(x, y)$ is a curve of the form

$$f(x, y) = c,$$

where c is a constant in the range of f .

Geometrically, the level curve $f(x, y) = c$ is obtained by intersecting the surface $z = f(x, y)$ with the horizontal plane $z = c$ and projecting the intersection down to the xy -plane. Contour maps are collections of level curves.

How to read contours

Where level curves are close together, the function changes rapidly. Where level curves are far apart, the function changes slowly.

Worked example 5.1: Circular level curves

Let

$$f(x, y) = x^2 + y^2.$$

The level curve $f(x, y) = c$ is

$$x^2 + y^2 = c.$$

For $c > 0$, this is a circle centered at the origin with radius $\sqrt{c}$.

Worked example 5.2: Elliptic level curves

Let

$$f(x, y) = x^2 + 9y^2.$$

The level curve $f(x, y) = c$ is

$$x^2 + 9y^2 = c.$$

For $c > 0$, these are ellipses.

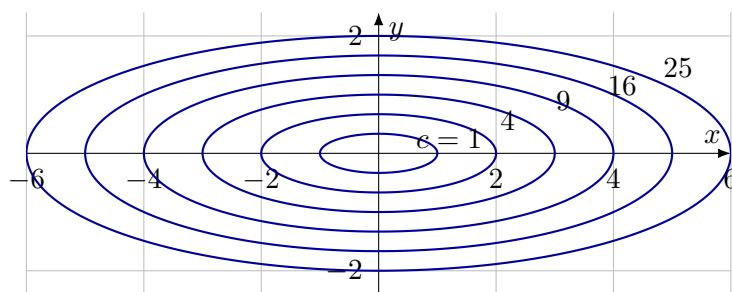


Figure 7: Level curves of $f(x, y) = x^2 + 9y^2$ are ellipses. Larger level values produce larger ellipses.

Worked example 5.3: Exponential level curves

Let

$$f(x, y) = xe^{4y}.$$

A level curve $f(x, y) = c$ satisfies

$$xe^{4y} = c.$$

If $x \neq 0$ and $c/x > 0$, then

$$y = \frac{1}{4} \ln\left(\frac{c}{x}\right).$$

This shows that level curves need not be lines, circles, parabolas, or ellipses.

6 Heatmaps and data surfaces

A multivariable function may be known through data rather than a formula. For example, a temperature field may be measured on a grid of locations, or a loss function may be evaluated at many parameter values.

A *heatmap* colors the plane according to function values. A *contour plot* draws level curves. Both are useful when a full surface plot is difficult to read.

7 Functions of three variables and level surfaces

A function of three variables has the form $f(x, y, z)$. Its graph would be $w = f(x, y, z)$, which lives in four-dimensional space. Since we cannot directly draw a four-dimensional graph, we usually study *level surfaces*:

$$f(x, y, z) = c.$$

Worked example 7.1: Spherical level surfaces

Let

$$f(x, y, z) = x^2 + y^2 + z^2.$$

The level surface $f(x, y, z) = c$ is

$$x^2 + y^2 + z^2 = c.$$

For $c > 0$, this is a sphere of radius $\sqrt{c}$ centered at the origin.

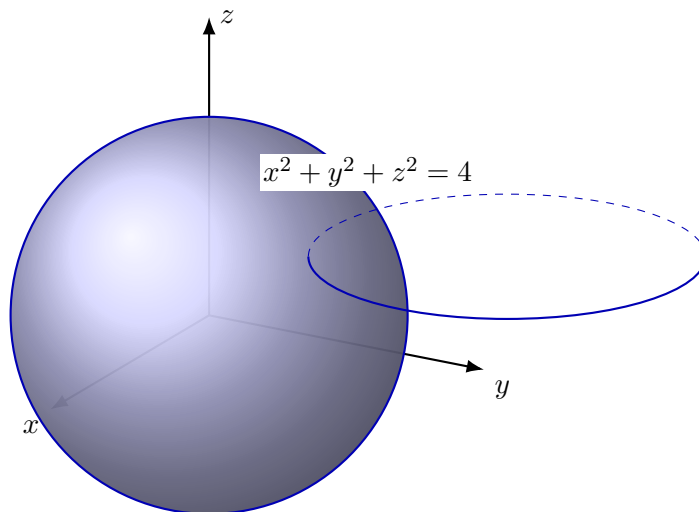


Figure 8: A level surface of $f(x, y, z) = x^2 + y^2 + z^2$ is a sphere.

Worked example 7.2: Plane level surfaces

Let

$$f(x, y, z) = 2x - y + 3z.$$

The level surface $f(x, y, z) = c$ is

$$2x - y + 3z = c.$$

For each value of c , this is a plane. Different values of c give parallel planes.

Worked example 7.3: Ellipsoid level surfaces

Let

$$f(x, y, z) = \frac{x^2}{4} + \frac{y^2}{9} + z^2.$$

The level surface $f(x, y, z) = 1$ is

$$\frac{x^2}{4} + \frac{y^2}{9} + z^2 = 1,$$

an ellipsoid with semi-axis lengths 2, 3, and 1.

8 Functions on high-dimensional spaces

In statistics and machine learning, a function often has many variables. Suppose a regression model predicts

$$\hat{y}_i = \beta_0 + \beta_1 x_i.$$

The least-squares loss is

$$L(\beta_0, \beta_1) = \sum_{i=1}^m (y_i - (\beta_0 + \beta_1 x_i))^2.$$

This is a function $L : \mathbb{R}^2 \rightarrow \mathbb{R}$. For a model with p predictors, the parameter vector is

$$\beta = (\beta_0, \beta_1, \dots, \beta_p),$$

and the loss becomes a function $L : \mathbb{R}^{p+1} \rightarrow \mathbb{R}$.

We cannot draw this graph when p is large, but we can still study slices, contours, gradients, and local quadratic approximations.

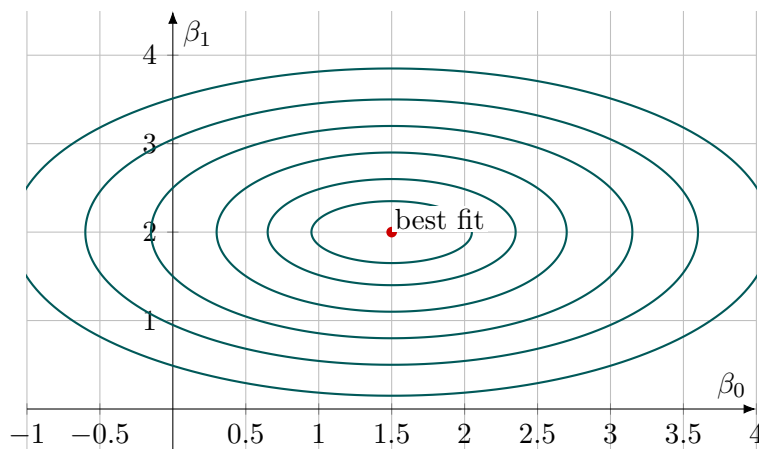


Figure 9: A two-parameter least-squares loss landscape. Smaller contours indicate lower loss.

Likelihood functions

In statistics, a likelihood function assigns to each parameter value a number measuring how compatible that parameter is with the observed data. For example, for a normal model with known variance, the log-likelihood has the form

$$\ell(\mu) = -\frac{1}{2\sigma^2} \sum_{i=1}^m (x_i - \mu)^2 + \text{constant}.$$

For a multivariate model, θ may have many components, and the log-likelihood becomes

$$\ell(\theta) : \mathbb{R}^n \rightarrow \mathbb{R}.$$

Thus optimization in statistics is also multivariable calculus.

9 Representations of a multivariable function

Representation	Meaning	Example
Verbal	description in words	temperature depends on longitude and latitude
Algebraic	formula	$f(x, y) = x^2 + 9y^2$
Numerical	table or data grid	measured temperatures on a map
Visual	surface, contour, heatmap	graph of $z = f(x, y)$
Algorithmic	code that computes values	simulation or machine learning model

Modern applied mathematics often moves among all five. Data are numerical, models are algebraic or algorithmic, and interpretation requires visual and verbal reasoning.

10 Common surfaces and level sets

Equation	Shape	Key feature
$z = ax + by + c$	plane	linear surface
$z = x^2 + y^2$	circular paraboloid	opens upward
$z = x^2 + 9y^2$	elliptic paraboloid	different curvature in different directions
$z = x^2 - y^2$	saddle	rises in one direction, falls in another
$x^2 + y^2 + z^2 = r^2$	sphere	level surface of distance squared
$x^2 + y^2 = r^2$	cylinder	no restriction on z
$z = \sqrt{r^2 - x^2 - y^2}$	upper hemisphere	graph over a disk
$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$	ellipsoid	stretched sphere

Recognizing these shapes will help with optimization, multiple integration, and vector calculus.

11 Worked synthesis: one function, many views

Consider

$$f(x, y) = e^{-x^2 - y^2}.$$

Domain. The exponential function is defined for every real number, so the domain is all of $\mathbb{R}^2$.

Range. Since $x^2 + y^2 \geq 0$, we have $-x^2 - y^2 \leq 0$, so

$$0 < f(x, y) \leq 1.$$

The maximum value is 1 at $(0, 0)$.

Graph. The graph is a bell-shaped surface centered at the origin.

Level curves. A level curve satisfies

$$e^{-x^2 - y^2} = c.$$

For $0 < c \leq 1$, taking natural logarithms gives

$$x^2 + y^2 = -\ln c.$$

Thus the level curves are circles centered at the origin.

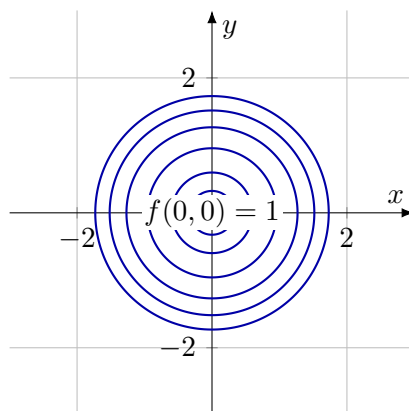


Figure 10: The function $e^{-x^2-y^2}$ has circular level curves, reflecting its radial symmetry.

This example is a first glimpse of the geometry behind multivariate normal densities, radial basis functions, kernel methods, and smooth energy landscapes.

12 Common mistakes

Mistake 1: Confusing graph and domain

The domain of $f(x, y)$ is a region in the xy -plane. The graph of f is a surface in xyz -space. For example, the domain of $\sqrt{4 - x^2 - y^2}$ is a disk, while the graph is a hemisphere.

Mistake 2: Forgetting strict inequalities

For logarithms, the input must be positive, not merely nonnegative. Thus $\ln(4 - x^2 - y^2)$ has domain $x^2 + y^2 < 4$, not $x^2 + y^2 \leq 4$.

Mistake 3: Thinking all level curves are circles

Level curves may be lines, ellipses, hyperbolas, logarithmic curves, or more complicated curves. Their shape depends on the function.

13 Practice problems

1. Find and sketch the domain of $f(x, y) = \sqrt{x + y}$.
2. Find and sketch the domain of $f(x, y) = \sqrt{x + y}/(x - y)$.
3. Find and sketch the domain of $f(x, y) = \ln(4 - x^2 - y^2)$.
4. Find the domain and range of $f(x, y) = \sqrt{9 - x^2 - y^2}$.
5. Describe the graph of $z = 3x - 2y + 5$.
6. Describe the graph of $z = x^2 + 4y^2$.

7. Describe the graph of $z = x^2 - y^2$.
8. Find the level curves of $f(x, y) = x^2 + y^2$.
9. Find the level curves of $f(x, y) = x^2 + 4y^2$.
10. Find the level curves of $f(x, y) = xy$.
11. Find the level curve of $f(x, y) = e^{-x^2-y^2}$ corresponding to $c = 1/2$.
12. Find the level surface of $f(x, y, z) = x^2 + y^2 + z^2$ through the point $(1, 2, 2)$.
13. Describe the level surfaces of $f(x, y, z) = x + y + z$.
14. Describe the level surface $x^2/4 + y^2/9 + z^2 = 1$.
15. Give an example of a function $L : \mathbb{R}^n \rightarrow \mathbb{R}$ that appears in statistics, machine learning, economics, physics, or engineering.

14 Applied and computational exercises

1. Plot the surface $z = \sin x \cos y$ on $[-2\pi, 2\pi] \times [-2\pi, 2\pi]$.
2. Draw contour plots for $f(x, y) = x^2 - y^2$.
3. Generate a grid of values for $f(x, y) = e^{-x^2-y^2}$ and display it as a heatmap.
4. For $f(x, y) = x^2 + 4y^2$, compare the contours for levels 1, 4, 9, and 16.
5. Simulate noisy data from a line $y = \beta_0 + \beta_1 x + \varepsilon$ and plot the loss contours in the (β_0, β_1) -plane.
6. Choose a function of three variables and describe at least three of its level surfaces.

15 Chapter summary

A function of several variables assigns one scalar output to each input point in a multidimensional domain:

$$f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}.$$

The main ideas of this chapter are:

- domains of multivariable functions are regions in $\mathbb{R}^n$;
- the graph of $z = f(x, y)$ is a surface in $\mathbb{R}^3$;
- level curves $f(x, y) = c$ show where a two-variable function is constant;
- level surfaces $f(x, y, z) = c$ play the same role for functions of three variables;
- slices and traces reduce multivariable functions to single-variable views;
- high-dimensional functions in statistics and AI are studied through contours, slices, heatmaps, gradients, and optimization paths.

In the next chapter, we study limits and continuity. The key new issue is that points can approach a target point from infinitely many directions.

Chapter 7. Limits and Continuity in Several Variables

Path dependence, continuity, and local stability

In single-variable calculus, the limit

$$\lim_{x \rightarrow a} f(x)$$

asks what happens as x approaches a from the left and from the right. In multivariable calculus, the limit

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y)$$

asks what happens as the point (x,y) approaches (a,b) from every possible direction and along every possible path. This is the first major conceptual difference between single-variable and multivariable calculus.

This chapter is about learning how to decide when a multivariable limit exists, when it fails to exist, and why continuity is the local stability condition that makes differentiation and integration behave well.

Connection

This chapter develops limits and continuity of functions of several variables, including the distance-based definition, path tests, examples such as $\frac{x^2-y^2}{x^2+y^2}$ and $\frac{xy^2}{x^2+y^4}$, and continuous examples such as $e^{-xy} \sin(x+y)$ and $\frac{\sin(x^2+y^2)}{x^2+y^2}$.

Learning goals

After studying this chapter, you should be able to:

1. explain the meaning of $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$;
2. use direct substitution when continuity applies;
3. show that a limit does not exist by finding two paths with different limiting values;
4. understand why checking only lines is sometimes not enough;
5. prove simple limits using estimates, polar coordinates, and the squeeze theorem;
6. define continuity for functions of several variables;
7. identify common sources of discontinuity;
8. use numerical and graphical evidence responsibly;
9. connect continuity to stability in statistics, data science, and machine learning.

1 Why multivariable limits are different

A single-variable limit has only two basic directions of approach:

$$x \rightarrow a^- \quad \text{and} \quad x \rightarrow a^+.$$

A two-variable limit has infinitely many possible approaches:

$$(x, y) \rightarrow (a, b).$$

The point (x, y) can approach (a, b) along a horizontal line, a vertical line, a slanted line, a parabola, a spiral, or a more complicated curve. For the limit to exist, *all paths must lead to the same value*.

Core idea

To prove a multivariable limit exists, we must control all possible approaches. To prove a multivariable limit does not exist, it is enough to find two approaches that give different answers.

A useful analogy is hiking toward the summit of a mountain. If all trails leading to the summit approach the same height, the height has a well-defined limit. If different trails approach different heights near the same map location, then the surface does not have a well-defined limiting height there.

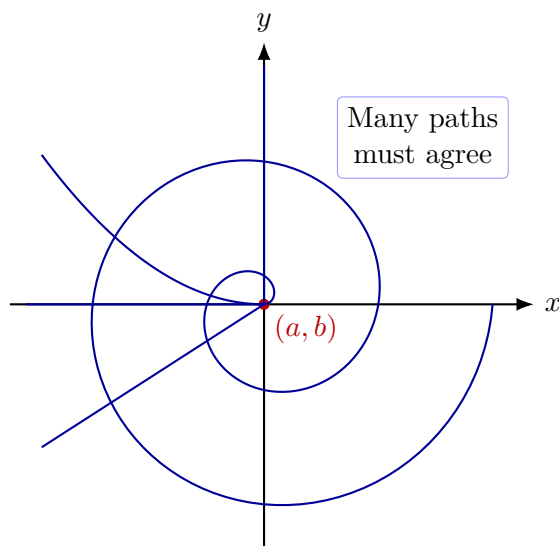


Figure 1: In two variables, a point can approach the target along infinitely many paths.

2 The formal definition

Let f be a function of two variables whose domain contains points arbitrarily close to (a, b) . We write

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$$

if the values $f(x, y)$ become arbitrarily close to L whenever (x, y) is sufficiently close to (a, b) , but not equal to (a, b) .

Using distance, this means:

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \implies |f(x, y) - L| < \varepsilon.$$

Here ε measures closeness in output, and δ measures closeness in input.

Practical interpretation

The formal definition says that the output can be forced close to L by taking the input point close enough to (a, b) , regardless of the direction of approach.

The same definition works in $\mathbb{R}^n$. If $x = (x_1, \dots, x_n)$ and $a = (a_1, \dots, a_n)$, then

$$\|x - a\| = \sqrt{(x_1 - a_1)^2 + \dots + (x_n - a_n)^2}.$$

Then

$$\lim_{x \rightarrow a} f(x) = L$$

means that $f(x)$ is close to L whenever $\|x - a\|$ is small and nonzero.

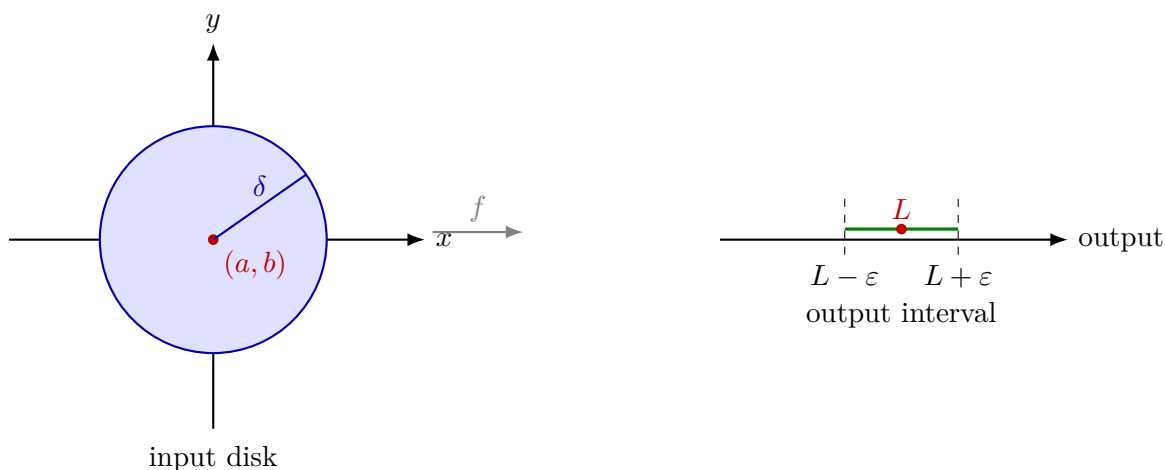


Figure 2: The ε - δ definition controls all points in a small punctured disk around the input point.

3 Easy limits by continuity

Many limits are easy because the function is continuous. Polynomials, exponential functions, trigonometric functions, and compositions of continuous functions are continuous wherever they are defined.

Example 3.1: Direct substitution

Evaluate

$$\lim_{(x,y) \rightarrow (0,0)} e^{-xy} \sin(x+y).$$

Both e^{-xy} and $\sin(x+y)$ are continuous, so their product is continuous. Therefore the limit is obtained by substitution:

$$\lim_{(x,y) \rightarrow (0,0)} e^{-xy} \sin(x+y) = e^0 \sin(0) = 0.$$

Example 3.2: A rational function at a regular point

Evaluate

$$\lim_{(x,y) \rightarrow (1,2)} \frac{x + y^2}{x^2 - y^2}.$$

The denominator at $(1, 2)$ is

$$1^2 - 2^2 = -3 \neq 0.$$

So the function is continuous at $(1, 2)$, and

$$\lim_{(x,y) \rightarrow (1,2)} \frac{x + y^2}{x^2 - y^2} = \frac{1 + 2^2}{1^2 - 2^2} = -\frac{5}{3}.$$

When substitution works

If a formula is built from continuous operations and is defined at the target point, substitution is usually the first method to try.

4 Path tests for nonexistence

To show a limit does not exist, find two paths approaching the same point that give different limiting values.

Example 4.1: A classic path-dependent limit

Consider

$$f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}.$$

Along the path $y = 0$,

$$f(x, 0) = \frac{x^2}{x^2} = 1,$$

so $\lim_{x \rightarrow 0} f(x, 0) = 1$. Along the path $x = 0$,

$$f(0, y) = \frac{-y^2}{y^2} = -1,$$

so $\lim_{y \rightarrow 0} f(0, y) = -1$. The two paths give different values. Therefore

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$$

does not exist.

Along a general line $y = mx$ with $x \neq 0$,

$$f(x, mx) = \frac{x^2 - m^2 x^2}{x^2 + m^2 x^2} = \frac{1 - m^2}{1 + m^2}.$$

The result depends on m , the slope of the path. That is a strong sign that the full two-variable limit does not exist.

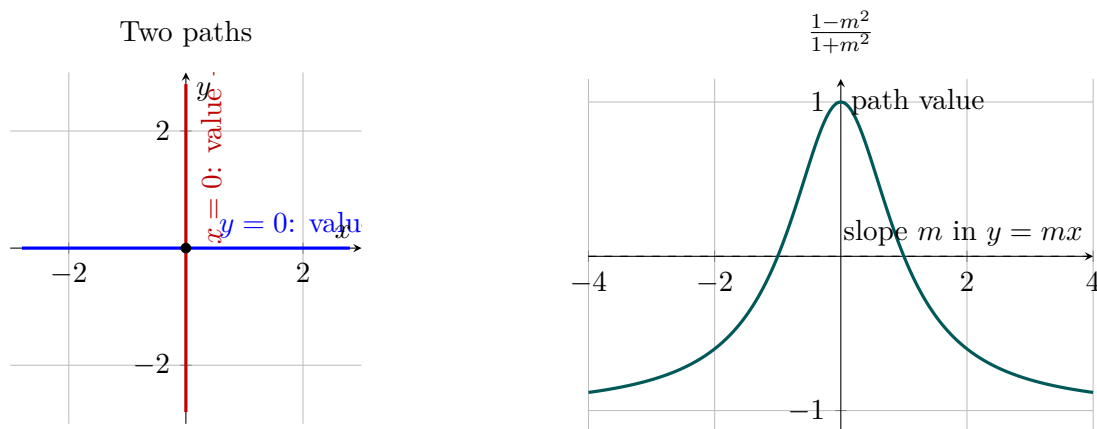


Figure 3: Different straight-line paths give different limiting values for $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$.

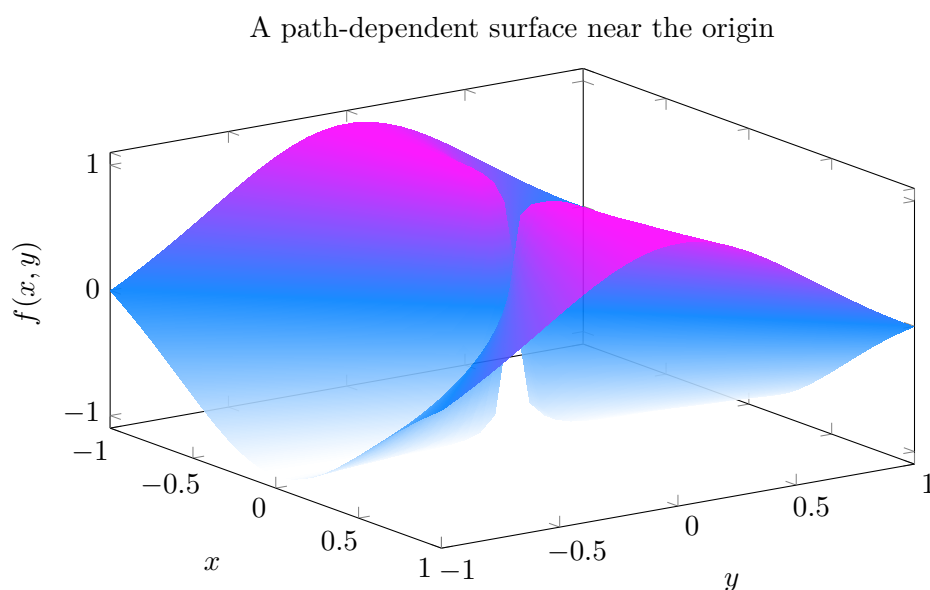


Figure 4: Near the origin, the surface approaches different heights from different directions.

5 Checking only lines is not enough

Sometimes every straight-line path gives the same value, but the limit still does not exist. This is one of the most important warnings in multivariable calculus.

Example 5.1: Lines are misleading

Consider

$$g(x, y) = \frac{xy^2}{x^2 + y^4}.$$

Along a straight line $y = mx$,

$$g(x, mx) = \frac{x(m^2x^2)}{x^2 + m^4x^4} = \frac{m^2x}{1 + m^4x^2} \rightarrow 0.$$

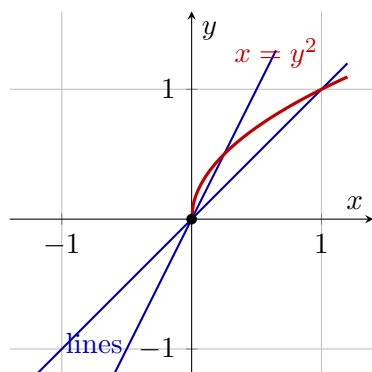
So every line $y = mx$ gives limiting value 0.

But along the curved path $x = y^2$,

$$g(y^2, y) = \frac{y^2y^2}{(y^2)^2 + y^4} = \frac{y^4}{2y^4} = \frac{1}{2}.$$

Therefore the full limit does not exist.

Straight and curved paths



Path values

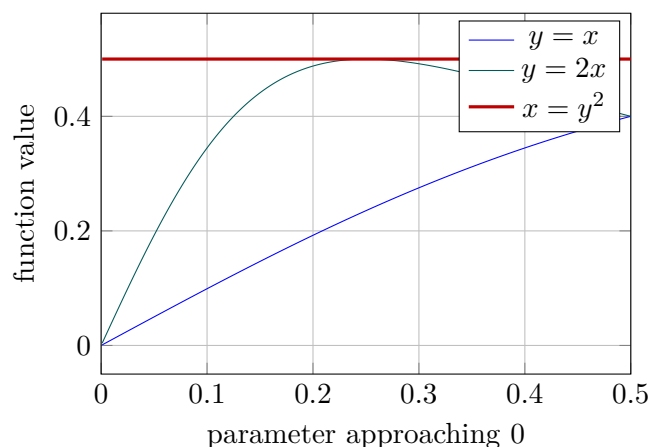


Figure 5: Line paths can all approach zero while a curved path approaches a different value.

Common mistake

Showing that several paths give the same value does not prove a limit exists. It only provides evidence. To prove existence, we need an argument that controls all paths.

6 Proving limits with estimates

To prove that a limit exists, we often use inequalities. The idea is to bound the complicated expression by something simpler that goes to zero.

A common strategy near $(0,0)$ is to use

$$r = \sqrt{x^2 + y^2}.$$

Then

$$|x| \leq r, \quad |y| \leq r.$$

If we can show

$$|f(x, y) - L| \leq Cr^p$$

for constants $C > 0$ and $p > 0$, then $f(x, y) \rightarrow L$ as $(x, y) \rightarrow (0, 0)$.

Example 6.1: A squeeze theorem proof

Show that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0.$$

For $(x, y) \neq (0, 0)$,

$$\left| \frac{x^2 y}{x^2 + y^2} \right| = \frac{|x|^2 |y|}{x^2 + y^2}.$$

Since $x^2 \leq x^2 + y^2$, we have

$$\frac{|x|^2}{x^2 + y^2} \leq 1.$$

Therefore

$$\left| \frac{x^2 y}{x^2 + y^2} \right| \leq |y|.$$

As $(x, y) \rightarrow (0, 0)$, we have $|y| \rightarrow 0$. By the squeeze theorem,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0.$$

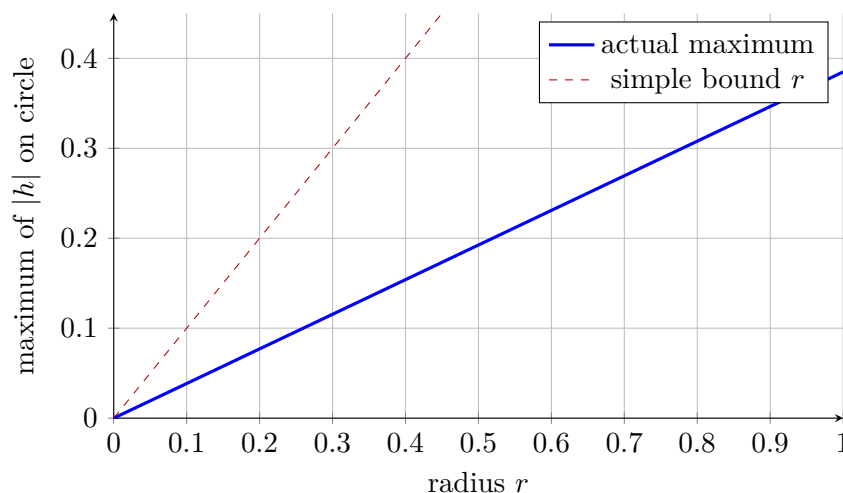


Figure 6: For $h(x, y) = \frac{x^2 y}{x^2 + y^2}$, the maximum absolute value on circles of radius r decreases linearly toward zero.

7 Polar coordinates as a limit tool

For limits at the origin in the plane, polar coordinates are often useful:

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Then $(x, y) \rightarrow (0, 0)$ means $r \rightarrow 0$. The angle θ may change in any way. If the expression becomes something that goes to L independently of θ , then the limit is L .

Example 7.1: A polar-coordinate proof

Show that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}} = 0.$$

Using $x = r \cos \theta$ and $y = r \sin \theta$,

$$\frac{xy}{\sqrt{x^2 + y^2}} = \frac{r^2 \cos \theta \sin \theta}{r} = r \cos \theta \sin \theta.$$

Since $|\cos \theta \sin \theta| \leq 1$,

$$|r \cos \theta \sin \theta| \leq r.$$

As $r \rightarrow 0$, this tends to 0. Therefore

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}} = 0.$$

A limit controlled by distance to the origin

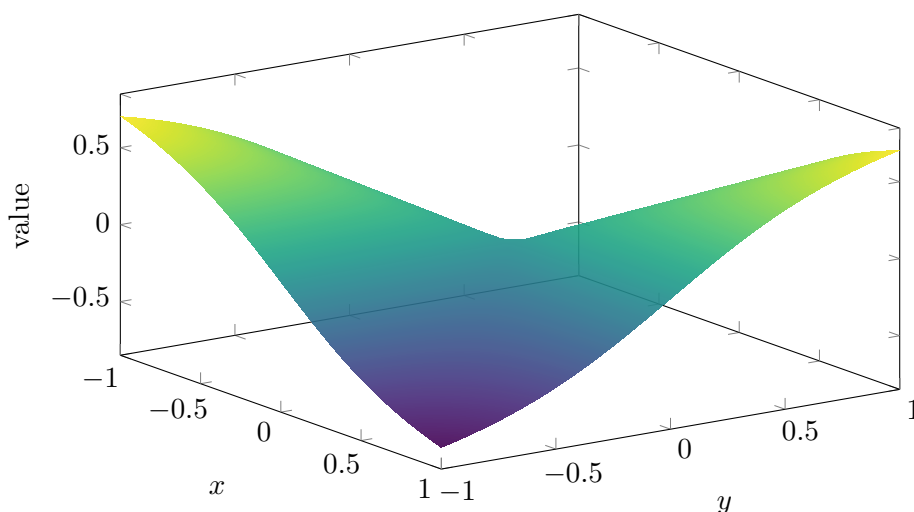


Figure 7: The polar form $r \cos \theta \sin \theta$ is controlled by the radius r .

8 A removable discontinuity

Consider

$$q(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}.$$

The expression is undefined at $(0, 0)$, but the limit exists. Let

$$u = x^2 + y^2.$$

As $(x, y) \rightarrow (0, 0)$, we have $u \rightarrow 0^+$. Therefore

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2} = \lim_{u \rightarrow 0^+} \frac{\sin u}{u} = 1.$$

The function is not continuous at the origin because it is not defined there. But we can make it continuous by defining

$$q(0, 0) = 1.$$

This is called a *removable discontinuity*.

A removable discontinuity filled in continuously

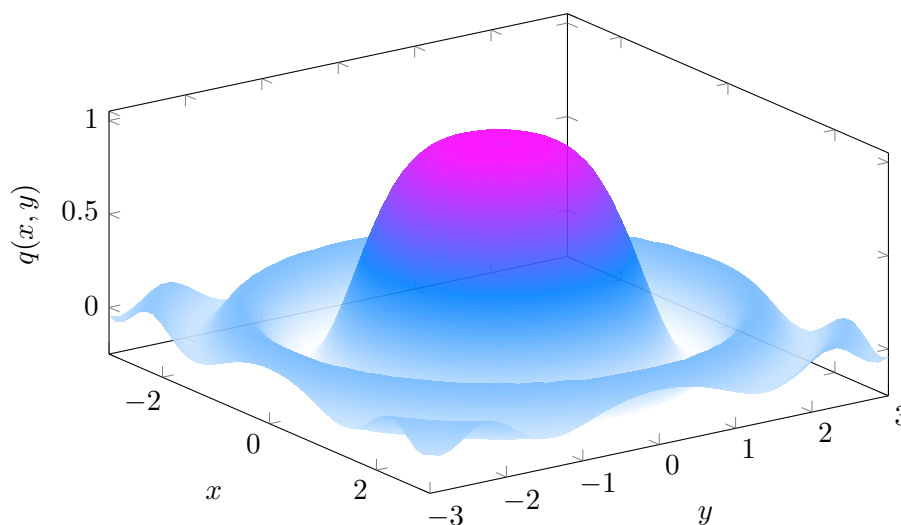


Figure 8: The function $\frac{\sin(x^2+y^2)}{x^2+y^2}$ has limiting value 1 at the origin.

9 Continuity in several variables

A function $f(x, y)$ is *continuous at* (a, b) if

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b).$$

This requires three things:

1. $f(a, b)$ is defined;
2. $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ exists;
3. the limit equals the function value.

A function is continuous on a region D if it is continuous at every point of D .

Common sources of discontinuity

Source	Example	What can go wrong?
Denominator zero	$\frac{1}{x^2 + y^2}$	infinite blow-up near the origin
Path dependence	$\frac{x^2 - y^2}{x^2 + y^2}$	different paths give different values
Removable hole	$\frac{\sin(x^2 + y^2)}{x^2 + y^2}$	formula undefined at one point, but limit exists
Domain boundary	$\sqrt{1 - x^2 - y^2}$	function exists only on a disk
Piecewise rule	threshold or classifier	jump discontinuity across a boundary

Continuity and modeling

A continuous model changes outputs gradually when inputs are changed slightly. This is essential for numerical stability, optimization, interpolation, and sensitivity analysis.

10 Continuity on domains and boundaries

Some functions are continuous on their natural domains even though the domain has a boundary. For example,

$$f(x, y) = \sqrt{1 - x^2 - y^2}$$

is defined on the disk

$$D = \{(x, y) : x^2 + y^2 \leq 1\}.$$

It is continuous at every point of this disk relative to its domain. Near boundary points, we only approach through points in the domain.

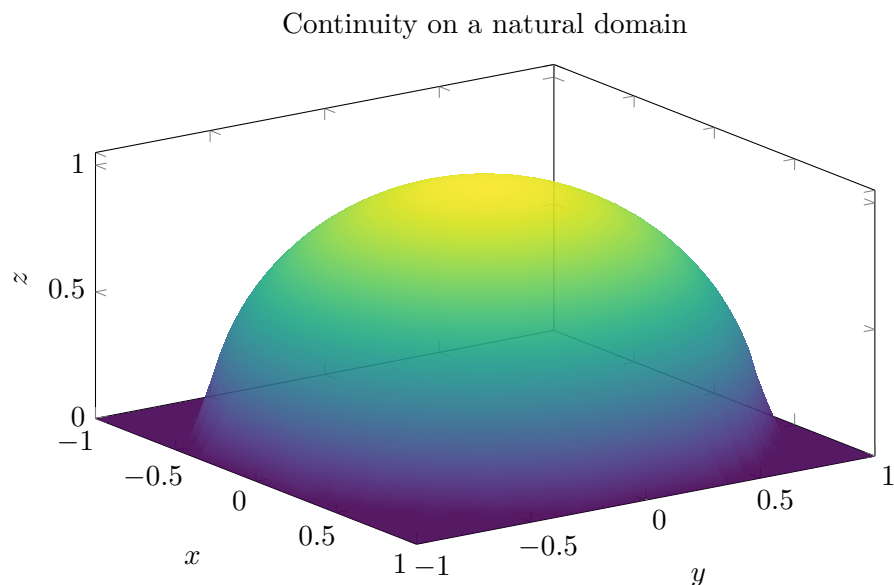


Figure 9: The upper hemisphere is continuous on the closed disk $x^2 + y^2 \leq 1$.

This distinction matters in applications. A probability density may be defined only on a simplex, a loss function may be meaningful only on feasible parameters, and a physical model may be valid only inside a spatial region.

11 Numerical experiments with limits

Numerical exploration can suggest whether a limit exists, but it cannot by itself prove a limit. A computer can sample many paths, but it cannot sample all paths.

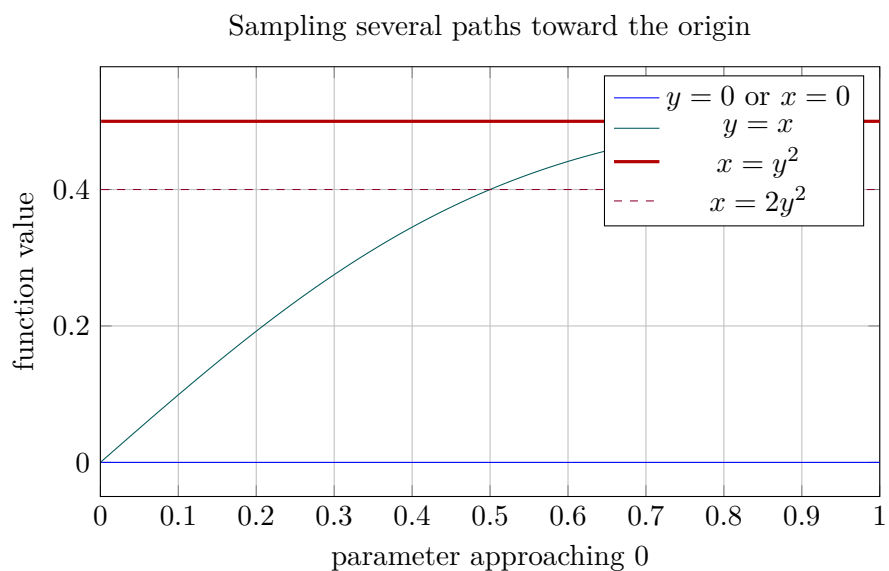


Figure 10: Numerical path sampling is useful evidence but not a proof.

Computational caution

A graph can miss a narrow ridge, a sharp spike, or a curved path where behavior changes. Use computation to develop intuition, then use algebra to justify the conclusion.

12 Higher-dimensional continuity

For a function

$$f : \mathbb{R}^n \rightarrow \mathbb{R},$$

continuity at $a \in \mathbb{R}^n$ means

$$\lim_{x \rightarrow a} f(x) = f(a).$$

The same distance idea applies:

$$\|x - a\| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

For example, the squared norm function

$$f(x) = \|x\|^2 = x_1^2 + \cdots + x_n^2$$

is continuous on $\mathbb{R}^n$. The logistic regression loss, least-squares loss, and many smooth neural-network losses are also continuous functions of their parameters.

Example 12.1: Least-squares loss is continuous

Let A be a fixed data matrix and b a fixed response vector. Define

$$L(w) = \|Aw - b\|^2.$$

This is a polynomial function of the components of w , so it is continuous. Small changes in w cause small changes in $L(w)$. This is one reason gradient-based methods can be used to minimize least-squares loss.

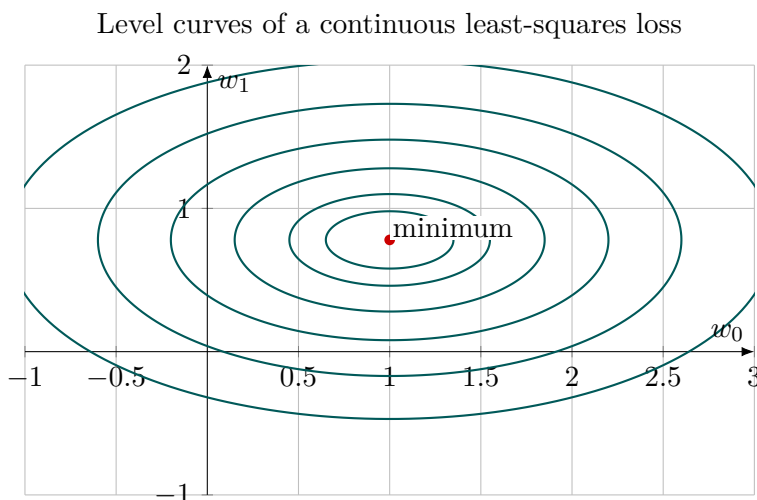


Figure 11: A least-squares loss surface is continuous and has smooth level curves.

13 Modern applications

Statistics

In statistics, likelihood and loss functions are often functions of parameters. Continuity allows us to use local approximations, optimization methods, and asymptotic arguments. For example, if a likelihood changes continuously with parameters, then nearby parameter values produce nearby likelihood values.

Data science

A data record may be a vector

$$x = (x_1, \dots, x_n).$$

A prediction model may be a function

$$f(x_1, \dots, x_n).$$

Continuity means that small changes in the input features produce small changes in the prediction. This is desirable for many regression models.

Machine learning and AI

Training a model usually means minimizing a loss function

$$L(\theta).$$

Continuity of L is the first requirement for meaningful optimization. Differentiability, which comes next, is stronger. A function can be continuous without being differentiable, but it cannot be differentiable without being continuous.

Bridge to gradients

The next chapters build on continuity. Partial derivatives, tangent planes, gradients, and optimization all require careful local behavior near a point.

14 Worked examples summary

Example type	Limit	Reason
Exists by continuity	$\lim_{(x,y) \rightarrow (1,2)} \frac{x+y^2}{x^2-y^2} = -\frac{5}{3}$	denominator is nonzero at $(1, 2)$
Does not exist by two paths	$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2-y^2}{x^2+y^2}$	$y = 0$ gives 1, while $x = 0$ gives -1
Lines are not enough	$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2+y^4}$	every line gives 0, but $x = y^2$ gives $1/2$
Exists by squeezing	$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^2+y^2} = 0$	absolute value is bounded above by $ y $

15 Practice problems

Concept checks

1. Explain why multivariable limits are harder than single-variable limits.
2. What does it mean for a function $f(x, y)$ to be continuous at (a, b) ?
3. Why can two different path limits prove that a limit does not exist?
4. Why does checking only horizontal and vertical paths usually not prove that a limit exists?
5. Explain the difference between a removable discontinuity and a path-dependent discontinuity.

Computational skill problems

6. Evaluate by substitution, if possible:

$$\lim_{(x,y) \rightarrow (2,-1)} (x^2y + e^{xy}).$$

7. Evaluate by substitution, if possible:

$$\lim_{(x,y) \rightarrow (1,1)} \frac{x^2 + y^2}{x + y}.$$

8. Determine whether the limit exists:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2}{x^2 + y^2}.$$

9. Determine whether the limit exists:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}.$$

10. Determine whether the limit exists:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 + y^2}.$$

11. Use polar coordinates to evaluate:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 1} + 1}.$$

12. Show that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 + y^3}{x^2 + y^2} = 0.$$

13. Show that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{\sqrt{x^2 + y^2}} = 0.$$

14. Determine where $f(x, y) = \frac{x + y}{x^2 + y^2 - 1}$ is continuous.

15. Determine where $g(x, y) = \sqrt{4 - x^2 - y^2}$ is continuous.

Modeling and interpretation problems

16. A temperature function $T(x, y)$ is continuous on a city map. Explain what continuity means physically.
17. A classifier is defined by $C(x, y) = 1$ if $x + y \geq 0$ and $C(x, y) = 0$ if $x + y < 0$. Where is this function discontinuous?
18. Let $L(w) = \|Aw - b\|^2$. Explain why L is continuous as a function of the parameter vector w .
19. Give an example of a model where discontinuity is expected or useful.
20. Give an example of a model where continuity is important for stability.

Computational practice

21. Test the paths $y = 2x$, $y = -2x$, and $y = 3x$ for $\frac{x^2 - y^2}{x^2 + y^2}$.
22. Test several paths for a new function of your choice.
23. Visualize the function $f(x, y) = \frac{xy}{x^2 + y^2}$ near the origin.
24. Numerically estimate the maximum of $\left| \frac{x^2 y}{x^2 + y^2} \right|$ on circles of radius r for several values of r .
25. Create a contour plot of a continuous loss function and explain why the level curves are useful.

16 Lab preview

The lab for this chapter focuses on three skills:

1. visualizing surfaces near singular points;
2. sampling limits along many paths;
3. distinguishing numerical evidence from mathematical proof.

A useful workflow is:

1. plot the surface or contour map;
2. test simple paths such as $y = 0$, $x = 0$, and $y = mx$;
3. test curved paths if needed;
4. use algebra to prove the final result.

Chapter summary

A multivariable limit requires agreement along all possible paths. Direct substitution works when a function is continuous at the point. To show that a limit does not exist, it is enough to find two paths with different limiting values. To show that a limit exists, we need a path-independent argument, often using estimates, polar coordinates, or the squeeze theorem.

Continuity means that the value of a function agrees with its local limiting behavior. This idea is central not only to calculus, but also to numerical modeling, optimization, statistics, data science, and machine learning.

Chapter 8. Partial Derivatives and the Gradient

Coordinate rates of change, sensitivity, and local information

A function of one variable has one basic input direction: move left or move right on the number line. A function of several variables has many input directions. The first step is to measure change in coordinate directions.

For a function $f(x, y)$, the partial derivative f_x measures the instantaneous rate of change when x changes and y is held fixed. The partial derivative f_y measures the instantaneous rate of change when y changes and x is held fixed. Together they form the *gradient vector*

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle.$$

The gradient is the local direction-information vector for a scalar field. It tells us how the output changes to first order near a point, how level curves are oriented, and how a loss function changes when model parameters change.

Teaching-material connection

This chapter develops the material on partial derivatives, geometric traces, numerical derivative estimates, second partial derivatives, Clairaut's theorem, and the gradient vector. The main idea is that partial derivatives reduce multivariable differentiation to single-variable differentiation by holding all other variables fixed.

Learning goals

After studying this chapter, you should be able to:

1. define $f_x(a, b)$ and $f_y(a, b)$ using limits;
2. interpret partial derivatives as slopes of coordinate traces of a surface;
3. compute partial derivatives by treating other variables as constants;
4. estimate partial derivatives from data or finite differences;
5. compute second partial derivatives and mixed partial derivatives;
6. use Clairaut's theorem when mixed partial derivatives are continuous;
7. compute the gradient of functions on $\mathbb{R}^2$, $\mathbb{R}^3$, and $\mathbb{R}^n$;
8. explain why the gradient is perpendicular to level curves and level surfaces;
9. connect gradients with sensitivity analysis, image processing, statistics, and machine learning.

1 From one derivative to many partial derivatives

For a single-variable function $y = f(x)$, the derivative at $x = a$ is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

Geometrically, this is the slope of the tangent line at $x = a$. For a two-variable function $z = f(x, y)$, the graph is a surface. At a point $(a, b, f(a, b))$, there are many possible curves on the surface passing through that point. The easiest ones are the two coordinate traces:

- the trace obtained by fixing $y = b$ and allowing x to vary;
- the trace obtained by fixing $x = a$ and allowing y to vary.

The slopes of these traces are the partial derivatives.

Core idea

A partial derivative is an ordinary derivative applied to a slice of a multivariable function.

2 Definitions of f_x and f_y

Let f be a function of two variables. The partial derivative of f with respect to x at (a, b) is

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h},$$

provided the limit exists. The partial derivative of f with respect to y at (a, b) is

$$f_y(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h},$$

provided the limit exists.

Equivalently, if $g(x) = f(x, b)$, then $f_x(a, b) = g'(a)$. If $k(y) = f(a, y)$, then $f_y(a, b) = k'(b)$.

The notation is flexible:

$$f_x = \frac{\partial f}{\partial x} = \frac{\partial z}{\partial x} = D_x f, \quad f_y = \frac{\partial f}{\partial y} = \frac{\partial z}{\partial y} = D_y f.$$

How to compute partial derivatives

To compute f_x , treat y as a constant and differentiate with respect to x . To compute f_y , treat x as a constant and differentiate with respect to y .

3 Geometric interpretation: slopes of traces

Suppose the surface is $z = f(x, y)$ and we focus on the point $(a, b, f(a, b))$. The trace in the plane $y = b$ is the single-variable curve $z = f(x, b)$. Its tangent slope at $x = a$ is $f_x(a, b)$. The trace in the plane $x = a$ is $z = f(a, y)$, and its tangent slope at $y = b$ is $f_y(a, b)$.

Example 3.1: A surface and its coordinate traces

Let

$$f(x, y) = 4 - \frac{x^2}{4} - y^2.$$

This is a downward-opening elliptic paraboloid. At the point $(1, 1)$,

$$f_x(x, y) = -\frac{x}{2}, \quad f_y(x, y) = -2y.$$

Thus

$$f_x(1, 1) = -\frac{1}{2}, \quad f_y(1, 1) = -2.$$

Interpretation:

- if $y = 1$ is held fixed and x increases near $x = 1$, the height decreases at rate $1/2$;
- if $x = 1$ is held fixed and y increases near $y = 1$, the height decreases at rate 2 .

Surface with coordinate traces

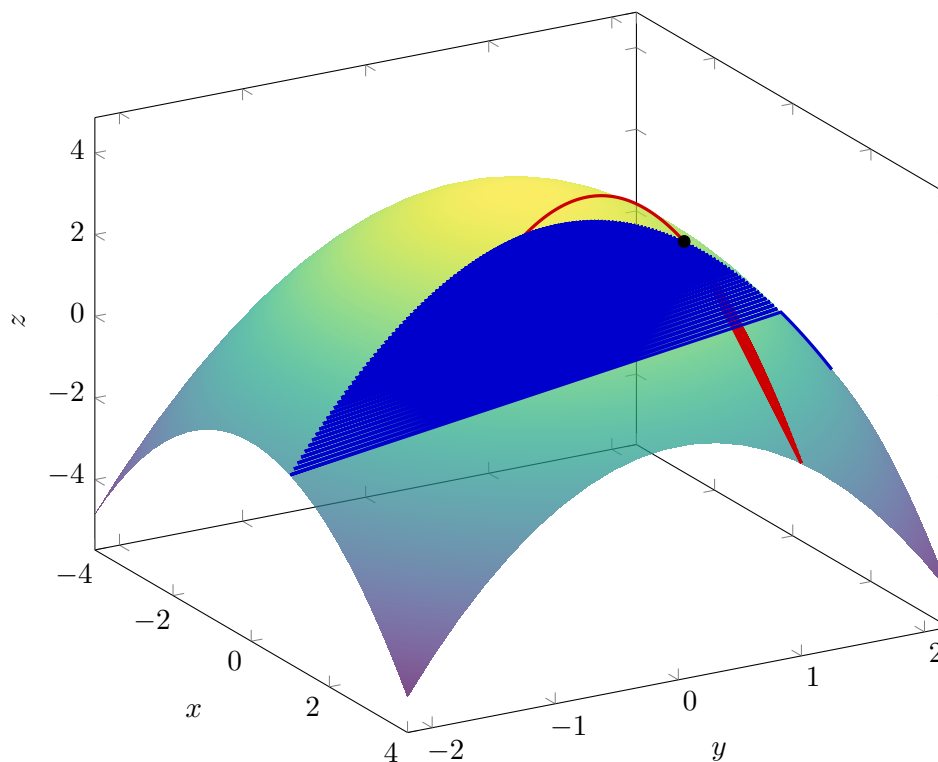


Figure 1: Coordinate traces of $f(x, y) = 4 - x^2/4 - y^2$ through the point $(1, 1, 2.75)$. The partial derivatives are the slopes of these traces.

4 Computing partial derivatives

Example 4.1: Polynomial, trigonometric, and exponential terms

Let

$$f(x, y) = x^3 + x^2 \sin y - xe^y.$$

To compute f_x , hold y constant:

$$f_x(x, y) = 3x^2 + 2x \sin y - e^y.$$

To compute f_y , hold x constant:

$$f_y(x, y) = x^2 \cos y - xe^y.$$

At $(1, 2)$,

$$f_x(1, 2) = 3 + 2 \sin 2 - e^2, \quad f_y(1, 2) = \cos 2 - e^2.$$

Example 4.2: A radial function

Let

$$f(x, y) = \sqrt{x^2 + y^2}.$$

For $(x, y) \neq (0, 0)$,

$$f_x(x, y) = \frac{x}{\sqrt{x^2 + y^2}}, \quad f_y(x, y) = \frac{y}{\sqrt{x^2 + y^2}}.$$

At $(1, 2)$,

$$f_x(1, 2) = \frac{1}{\sqrt{5}}, \quad f_y(1, 2) = \frac{2}{\sqrt{5}}.$$

At the origin, these formulas do not apply because the denominator vanishes. In fact, $f_x(0, 0)$ and $f_y(0, 0)$ exist, but the function is not differentiable there. Partial derivatives give coordinate-direction information, but not always the whole story.

5 Partial derivatives from data

In applications, we often do not know a symbolic formula for f . We may only have sampled values. Then partial derivatives are estimated by finite differences. For example,

$$f_x(1, 2) \approx \frac{f(1.1, 2) - f(1, 2)}{0.1}, \quad f_y(1, 2) \approx \frac{f(1, 2.15) - f(1, 2)}{0.15}.$$

This idea is fundamental in numerical differentiation, image processing, computer vision, and machine learning.

Example 5.1: Finite-difference estimates

For $f(x, y) = x^3 + x^2 \sin y - xe^y$, forward differences provide simple estimates for $f_x(1, 2)$ and $f_y(1, 2)$. A more accurate approximation often uses centered differences:

$$f_x(a, b) \approx \frac{f(a + h, b) - f(a - h, b)}{2h}.$$

The same idea works for f_y .

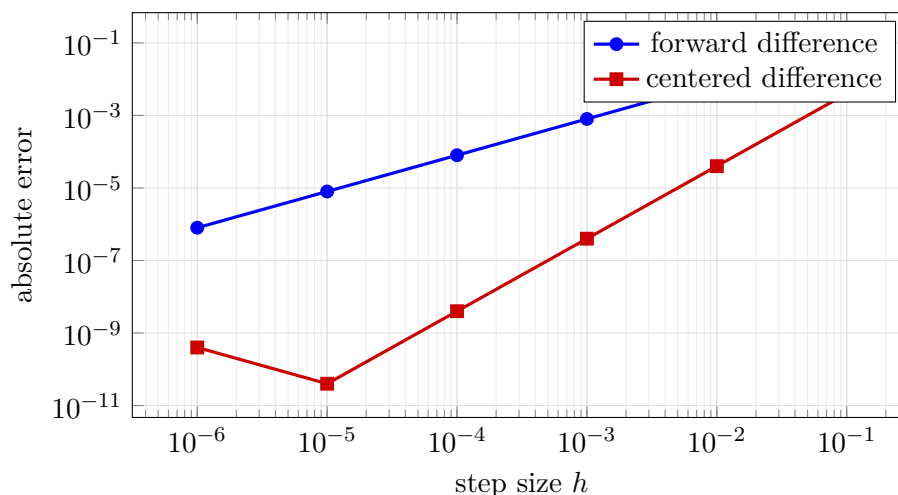


Figure 2: Centered differences usually converge faster than forward differences for smooth functions. The red curve drops more rapidly as h decreases.

6 Image processing interpretation

An image can be viewed as a function

$$I(x, y) = \text{pixel intensity at location } (x, y).$$

Then I_x measures horizontal change and I_y measures vertical change. Large values of $|I_x|$ or $|I_y|$ often indicate edges. This is the same calculus idea, but applied to a matrix of data instead of a symbolic formula.

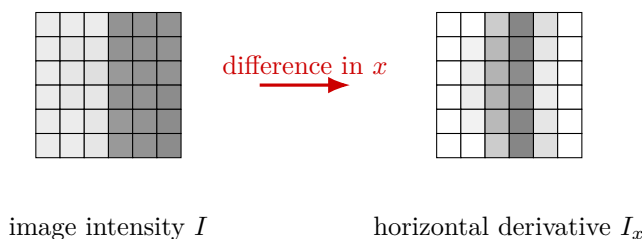


Figure 3: Finite differences applied to image-like data. Horizontal derivatives reveal sharp transitions, which are often edges.

Modern interpretation

In symbolic calculus, f_x and f_y are formulas. In data analysis, they are often approximations computed from samples, grids, or pixels. The concept is the same: measure local change in coordinate directions.

7 Second partial derivatives

If $z = f(x, y)$, then the four second partial derivatives are

$$\begin{aligned} f_{xx} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right), & f_{xy} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right), \\ f_{yx} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right), & f_{yy} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right). \end{aligned}$$

The derivatives f_{xy} and f_{yx} are the *mixed partial derivatives*.

Example 7.1: Second partials and mixed partials

Let

$$f(x, y) = x^2 + xy^3 + \sin(xy).$$

First derivatives:

$$f_x = 2x + y^3 + y \cos(xy), \quad f_y = 3xy^2 + x \cos(xy).$$

Second derivatives:

$$\begin{aligned} f_{xx} &= 2 - y^2 \sin(xy), \\ f_{xy} &= 3y^2 + \cos(xy) - xy \sin(xy), \\ f_{yx} &= 3y^2 + \cos(xy) - xy \sin(xy), \\ f_{yy} &= 6xy - x^2 \sin(xy). \end{aligned}$$

Here the mixed partial derivatives are equal.

8 Clairaut's theorem

Clairaut's theorem

Suppose f is defined on a disk containing (a, b) . If f_{xy} and f_{yx} are both continuous on that disk, then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

In most elementary examples involving polynomials, exponentials, logarithms, and trigonometric functions on regions where the formulas are defined, the continuity hypothesis is satisfied. Then the order of mixed partial differentiation can be switched.

9 The gradient vector

The first partial derivatives form a vector. For $f(x, y)$,

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle.$$

For $f(x, y, z)$,

$$\nabla f(x, y, z) = \langle f_x(x, y, z), f_y(x, y, z), f_z(x, y, z) \rangle.$$

More generally, for $f(x_1, \dots, x_n)$,

$$\nabla f(x_1, \dots, x_n) = \left\langle \frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right\rangle.$$

The symbol ∇ is pronounced *del* or *nabla*.

Example 9.1: Gradient of a two-variable function

Let

$$f(x, y) = \sin(xy) + e^y.$$

Then

$$f_x = y \cos(xy), \quad f_y = x \cos(xy) + e^y.$$

Therefore

$$\nabla f(x, y) = \langle y \cos(xy), x \cos(xy) + e^y \rangle.$$

At $(0, 1)$,

$$\nabla f(0, 1) = \langle 1, e \rangle.$$

Example 9.2: Gradient of a three-variable function

Let

$$f(x, y, z) = xyz.$$

Then

$$f_x = yz, \quad f_y = xz, \quad f_z = xy.$$

Thus

$$\nabla f(x, y, z) = \langle yz, xz, xy \rangle.$$

At $(1, 2, 3)$,

$$\nabla f(1, 2, 3) = \langle 6, 3, 2 \rangle.$$

10 The gradient and level curves

Level curves have equations $f(x, y) = c$. The gradient ∇f is perpendicular to the level curve at points where $\nabla f \neq 0$.

Suppose $r(t) = \langle x(t), y(t) \rangle$ is a curve that stays on the level curve $f(x, y) = c$. Then $f(x(t), y(t)) = c$. Differentiating gives

$$\nabla f(x(t), y(t)) \cdot r'(t) = 0.$$

The vector $r'(t)$ is tangent to the level curve. Therefore the gradient is perpendicular to the tangent direction.

Geometric meaning of the gradient

The gradient points normal to level curves and level surfaces. It tells us which direction cuts most directly across the contours.

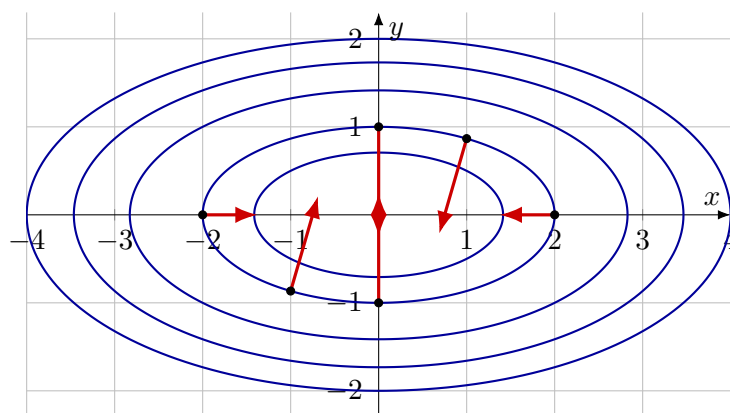


Figure 4: Gradient vectors are perpendicular to level curves of $H(x, y) = 4 - x^2/4 - y^2$.

11 Gradient fields

For each point (x, y) , the gradient produces a vector $\nabla f(x, y)$. The collection of these vectors is called a *gradient field*. Gradient fields show local directions of increase and form an important special class of vector fields.

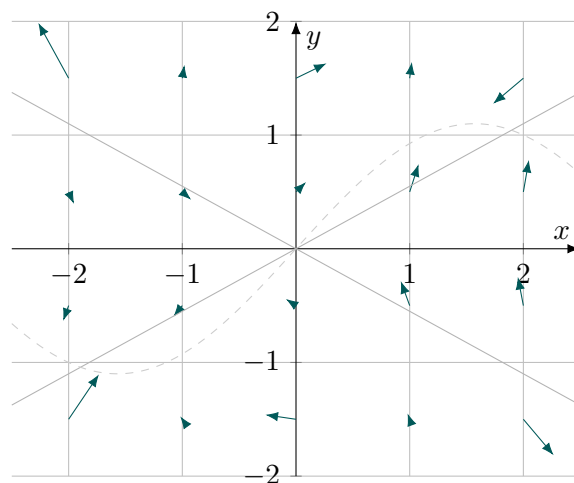


Figure 5: A gradient field for a smooth scalar function. The arrows encode local partial derivative information.

12 Directional derivatives as a preview

The next major idea is that we can measure rate of change in any unit direction u . If

$$u = \langle a, b \rangle$$

is a unit vector, the directional derivative is

$$D_u f(x, y) = \nabla f(x, y) \cdot u.$$

This formula says that the gradient contains all first-order directional information. The coordinate partial derivatives are special cases:

$$D_{\mathbf{i}}f = f_x, \quad D_{\mathbf{j}}f = f_y.$$

Thus the gradient is the vector that converts directions into rates of change.

Example 12.1: Directional derivative from a gradient

For

$$f(x, y) = \sin(xy) + e^y,$$

we found

$$\nabla f(0, 1) = \langle 1, e \rangle.$$

Let $v = \langle 2, 3 \rangle$. The corresponding unit vector is

$$u = \frac{v}{\|v\|} = \left\langle \frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}} \right\rangle.$$

Therefore

$$D_u f(0, 1) = \langle 1, e \rangle \cdot \left\langle \frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}} \right\rangle = \frac{2 + 3e}{\sqrt{13}}.$$

13 Gradients in statistics and machine learning

In statistics and machine learning, a model often has parameters

$$\theta = (\theta_1, \dots, \theta_n),$$

and we minimize a loss function $L(\theta_1, \dots, \theta_n)$. The partial derivative $\partial L / \partial \theta_k$ measures how sensitive the loss is to parameter θ_k while the other parameters are held fixed. The gradient $\nabla L(\theta)$ collects all parameter sensitivities into one vector.

For least-squares linear regression with data (x_i, y_i) and model $\hat{y}_i = \beta_0 + \beta_1 x_i$, the loss is

$$L(\beta_0, \beta_1) = \sum_{i=1}^m (y_i - (\beta_0 + \beta_1 x_i))^2.$$

The partial derivatives are

$$\frac{\partial L}{\partial \beta_0} = -2 \sum_{i=1}^m (y_i - (\beta_0 + \beta_1 x_i)),$$

and

$$\frac{\partial L}{\partial \beta_1} = -2 \sum_{i=1}^m x_i (y_i - (\beta_0 + \beta_1 x_i)).$$

These two numbers form the gradient in the parameter plane.

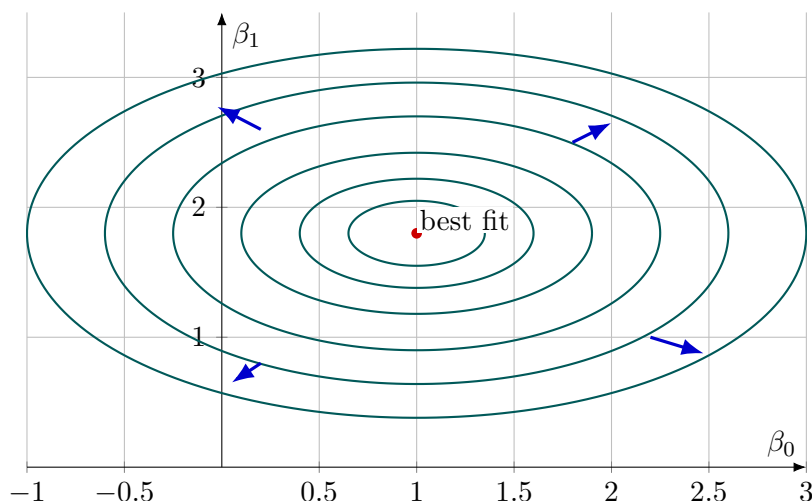


Figure 6: The gradient of a least-squares loss points toward increasing loss; the negative gradient points downhill.

14 Common mistakes

Mistake 1: Differentiating every variable at once

When computing f_x , only x varies. All other variables are constants. For example, if $f(x, y) = x^2y$, then $f_x = 2xy$, not $2xy + x^2$.

Mistake 2: Confusing partial derivatives with total change

The partial derivative $f_x(a, b)$ measures change only in the x direction. It does not describe motion along an arbitrary path unless the path moves only in the x direction.

Mistake 3: Forgetting to normalize a direction vector

In the directional derivative formula $D_u f = \nabla f \cdot u$, the vector u must be a unit vector. If a problem gives a nonunit direction vector, divide by its length first.

Mistake 4: Assuming mixed partials are always equal

Mixed partials are equal under hypotheses such as continuity near the point. For standard smooth examples this is usually true, but the theorem has a condition.

15 Chapter exercises

Conceptual questions

1. Explain in words what $f_x(a, b)$ means.

2. Explain why $f_x(a, b)$ is the slope of a trace of the surface $z = f(x, y)$.
3. Explain the difference between $f_x(a, b)$ and $f_y(a, b)$.
4. Why is a partial derivative not the same thing as a directional derivative in every direction?
5. Explain why ∇f is perpendicular to a level curve.
6. In an image, what do large values of $|I_x|$ and $|I_y|$ usually indicate?
7. In a loss function $L(\theta_1, \theta_2)$, what does $\partial L / \partial \theta_1$ mean?
8. Explain why the gradient is useful even when the function has many variables.

Skill problems

9. Find f_x and f_y for $f(x, y) = x^2y + 3xy^2 - y$.
10. Find f_x and f_y for $f(x, y) = e^{xy} + \ln(x + y)$ on its domain.
11. Find f_x and f_y for $f(x, y) = \cos(x^2 + y^2)$.
12. Find $\nabla f(1, 2)$ for $f(x, y) = x^3y + y^2$.
13. Find $\nabla f(0, 1)$ for $f(x, y) = \sin(xy) + e^y$.
14. Find $\nabla f(1, 2, 3)$ for $f(x, y, z) = x^2y + yz^3$.
15. Find all four second partial derivatives of $f(x, y) = x^2 + xy^3 + \sin(xy)$.
16. Verify Clairaut's theorem for $f(x, y) = e^{xy} + x^2y^3$.
17. Estimate $f_x(2, 1)$ using $f(2, 1) = 5.0$ and $f(2.1, 1) = 5.32$.
18. Estimate $f_y(2, 1)$ using $f(2, 1) = 5.0$ and $f(2, 1.2) = 4.7$.

Applied and computational problems

19. Let $H(x, y) = 1100 - 2x^2 - y^2$ model the height of a hill in meters. Compute H_x and H_y , and interpret them.
20. For the same hill, compute $\nabla H(10, 20)$ and explain what the signs of its components mean.
21. Let $T(x, y, z) = xe^y + 0.5z^2$. Compute $\nabla T(1, 0, 3)$.
22. Let I be an image intensity function. Explain how finite differences approximate I_x and I_y .
23. For least-squares regression with two parameters, derive the gradient of

$$L(\beta_0, \beta_1) = \sum_{i=1}^m (y_i - (\beta_0 + \beta_1 x_i))^2.$$

24. Write code to approximate $\nabla f(1, 2)$ for $f(x, y) = x^3 + x^2 \sin y - xe^y$ using centered differences.
25. Generate a contour plot of $f(x, y) = x^2 + 4y^2$ and draw gradient vectors at several points.
26. Compute numerical gradients on a grid for $f(x, y) = \sin x \cos y$.

Challenge problems

27. Let $f(x, y) = \sqrt{x^2 + y^2}$. Show that $f_x(0, 0)$ and $f_y(0, 0)$ exist, but explain why this does not mean the surface has a tangent plane at the origin.
28. Find a function for which f_x and f_y exist at the origin but the function is not continuous at the origin.
29. Suppose $f(x, y) = g(x^2 + y^2)$ for a differentiable single-variable function g . Find ∇f .
30. Suppose $f(x, y) = h(ax + by)$ for a differentiable single-variable function h . Find ∇f and describe the direction of the gradient.
31. Let $f(x, y, z) = F(x^2 + y^2 + z^2)$. Find ∇f and describe its direction geometrically.
32. In a high-dimensional model, the gradient has one component per parameter. Explain why this makes gradient methods scalable compared with searching every direction separately.

16 Python lab and interactive page

The Python lab for this chapter focuses on numerical and visual differentiation:

- estimating partial derivatives from function values;
- comparing forward, backward, and centered differences;
- plotting coordinate traces of surfaces;
- visualizing gradients on contour plots;
- computing image gradients;
- computing gradients of least-squares losses.

Resources:

- Chapter 8 Python lab: `../labs/Chapter08_Partial_Derivatives_and_the_Gradient.ipynb`
- Chapter 8 interactive HTML page: `../htmls/chapter-08.html`

17 Chapter summary

Partial derivatives are the coordinate-direction derivatives of a multivariable function. They are ordinary derivatives applied to slices of the function. For $f(x, y)$, the two first partial derivatives are f_x and f_y . For a function of n variables, there are n first partial derivatives.

The gradient packages these partial derivatives into a vector:

$$\nabla f = \left\langle \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right\rangle.$$

This vector is perpendicular to level sets and contains the first-order sensitivity information of the function. In modern applications, the gradient is the basic object behind numerical differentiation, image processing, optimization, maximum likelihood, backpropagation, and gradient-based learning.

Chapter 9. Linear Approximation, Tangent Planes, and Differentials

The local linear model behind tangent planes, sensitivity, and uncertainty

The central message of this chapter is simple and powerful:

a differentiable multivariable function is approximately linear near a point.

In single-variable calculus, the tangent line is the best first-order approximation to a curve near a point. In multivariable calculus, the same idea becomes the *tangent plane* for a graph $z = f(x, y)$ and the *tangent hyperplane* for a function of many variables.

Algebraically, the approximation is built from partial derivatives. Geometrically, it is the flat object that best matches the surface near the point. Computationally, it is the first local model used in numerical methods, optimization, uncertainty propagation, and statistical approximation.

Core principle

A differentiable multivariable function is approximately linear near a point.

Teaching-material connection

This chapter develops tangent planes, linearization, differential approximation, differentiability, total differentials, and tangent planes to level surfaces. The examples $z = 3x^2 + y^2$, gradient-based approximation, and the level-surface problem $F(x, y, z) = x^3 + y^2e^z + xyz$ are used as core models.

Learning goals

After studying this chapter, you should be able to:

1. explain why tangent planes are multivariable tangent lines;
2. find the tangent plane to a surface $z = f(x, y)$ at a point;
3. compute the linearization $L(x, y)$ of a function $f(x, y)$ at (a, b) ;
4. use linearization to estimate nearby function values;
5. write linear approximation in gradient-dot-product form;
6. explain the difference between partial derivatives existing and differentiability;
7. compute total differentials and use them for error propagation;
8. find tangent planes to level surfaces $F(x, y, z) = c$;
9. interpret local linear models in statistics, data science, and machine learning;
10. use computation to visualize tangent planes and approximation errors.

1 Review: tangent lines and local linearity

For a single-variable function $y = f(x)$, the tangent line at $x = a$ is

$$L(x) = f(a) + f'(a)(x - a).$$

The tangent line is useful because when x is close to a ,

$$f(x) \approx L(x).$$

If $\Delta x = x - a$, then

$$\Delta y = f(x) - f(a) \approx f'(a)\Delta x.$$

The multivariable version is

$$\Delta z \approx f_x(a, b)\Delta x + f_y(a, b)\Delta y.$$

The right side is linear in the input changes Δx and Δy .

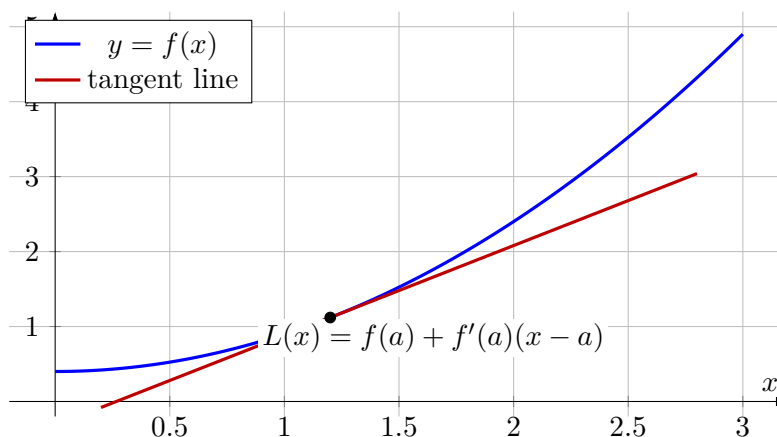


Figure 1: In one variable, a tangent line is the local linear model. In two variables, the tangent line becomes a tangent plane.

Local does not mean global

A tangent line or tangent plane usually gives a good approximation only near the point of tangency. Far away from the point, nonlinear curvature becomes important.

2 Tangent planes to graphs $z = f(x, y)$

Let $z = f(x, y)$ be differentiable near (a, b) , and let

$$P = (a, b, f(a, b))$$

be the point on the graph. The tangent plane at P is

$$z - f(a, b) = f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Equivalently,

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

The function on the right is the *linearization* of f at (a, b) :

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

The approximation

$$f(x, y) \approx L(x, y)$$

is called the *linear approximation* or *tangent plane approximation*.

At the point (a, b) , the tangent plane matches three pieces of first-order information: the height $f(a, b)$, the x -direction slope $f_x(a, b)$, and the y -direction slope $f_y(a, b)$. A useful normal vector to the tangent plane is

$$\langle f_x(a, b), f_y(a, b), -1 \rangle.$$

3 Worked example: tangent plane to an elliptic paraboloid

Example 3.1: Tangent plane to $z = 3x^2 + y^2$

Find the tangent plane to

$$z = f(x, y) = 3x^2 + y^2$$

at the point $(1, 2, 7)$.

First check the height:

$$f(1, 2) = 3(1)^2 + (2)^2 = 7.$$

The partial derivatives are

$$f_x(x, y) = 6x, \quad f_y(x, y) = 2y.$$

At $(1, 2)$,

$$f_x(1, 2) = 6, \quad f_y(1, 2) = 4.$$

Therefore the tangent plane is

$$z - 7 = 6(x - 1) + 4(y - 2),$$

or

$$z = 6x + 4y - 7.$$

Thus the linearization is

$$L(x, y) = 6x + 4y - 7.$$

To estimate $f(1.1, 1.9)$,

$$L(1.1, 1.9) = 6(1.1) + 4(1.9) - 7 = 7.2.$$

The exact value is

$$f(1.1, 1.9) = 3(1.1)^2 + (1.9)^2 = 7.24.$$

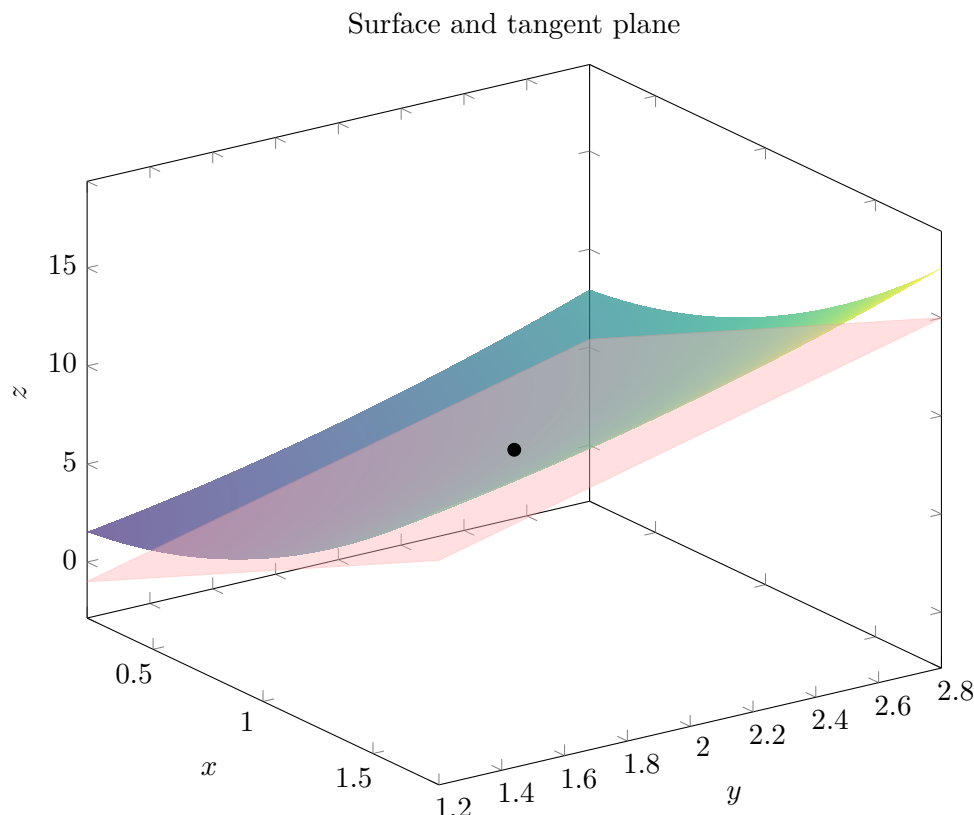


Figure 2: The graph of $z = 3x^2 + y^2$ and its tangent plane $z = 6x + 4y - 7$ at $(1, 2, 7)$.

4 Linear approximation in gradient form

Let

$$\Delta x = x - a, \quad \Delta y = y - b, \quad \Delta \mathbf{x} = \langle \Delta x, \Delta y \rangle.$$

Then

$$f(x, y) - f(a, b) \approx \nabla f(a, b) \cdot \Delta \mathbf{x}.$$

Equivalently,

$$\Delta z \approx \nabla f(a, b) \cdot \Delta \mathbf{x}.$$

For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the same formula is

$$f(\mathbf{x}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}).$$

Example 4.1: Gradient-dot-displacement estimate

Suppose

$$f(2, 1) = 13, \quad \nabla f(2, 1) = \langle 12, 2 \rangle.$$

Estimate $f(2.1, 0.9)$.

The input displacement is

$$\Delta \mathbf{x} = \langle 2.1 - 2, 0.9 - 1 \rangle = \langle 0.1, -0.1 \rangle.$$

Therefore

$$\Delta z \approx \langle 12, 2 \rangle \cdot \langle 0.1, -0.1 \rangle = 12(0.1) + 2(-0.1) = 1.$$

So

$$f(2.1, 0.9) \approx 13 + 1 = 14.$$

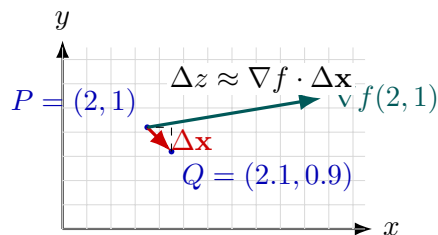


Figure 3: Linear approximation uses the dot product of the gradient and the displacement vector.

5 Approximation error and curvature

The linearization keeps only first-order information. The error comes from second-order and higher-order behavior. For

$$f(x, y) = 3x^2 + y^2,$$

the tangent plane at $(1, 2)$ is

$$L(x, y) = 6x + 4y - 7.$$

The error is

$$E(x, y) = f(x, y) - L(x, y) = 3(x - 1)^2 + (y - 2)^2.$$

This is always nonnegative because the surface is convex and lies above its tangent plane. The error is small near $(1, 2)$ and grows quadratically as we move away.

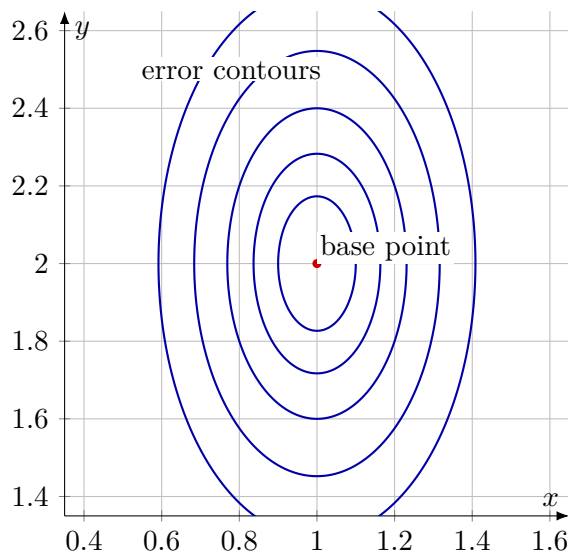


Figure 4: The linearization error is small near the base point and grows quadratically as we move away.

Connection to later approximation

Taylor approximation makes this precise. The second-order term involves the Hessian matrix. Linear approximation is first-order Taylor approximation; quadratic approximation is second-order Taylor approximation.

6 Differentiability

A function $f(x, y)$ is *differentiable* at (a, b) if its change can be written as a linear part plus an error that becomes negligible compared with the size of the input change. One way to state this is

$$\Delta z = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y,$$

where

$$\varepsilon_1 \rightarrow 0, \quad \varepsilon_2 \rightarrow 0$$

as $(\Delta x, \Delta y) \rightarrow (0, 0)$. Equivalently,

$$f(a + \Delta x, b + \Delta y) = f(a, b) + f_x(a, b)\Delta x + f_y(a, b)\Delta y + \text{small error}.$$

The important idea is not just that the partial derivatives exist. The important idea is that together they produce a good linear model near the point.

Partial derivatives alone are not enough

A function may have partial derivatives at a point but fail to be differentiable there. Differentiability is stronger than the existence of coordinate-direction slopes.

Sufficient condition for differentiability

If f_x and f_y exist near (a, b) and are continuous at (a, b) , then f is differentiable at (a, b) .

7 A cautionary example

Consider

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2 + y^2}}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

At the origin, the coordinate partial derivatives exist:

$$f_x(0, 0) = 0, \quad f_y(0, 0) = 0.$$

So the candidate linearization is $L(x, y) = 0$. However, along the line $y = x$,

$$f(x, x) = \frac{x^2}{\sqrt{2x^2}} = \frac{|x|}{\sqrt{2}}.$$

This is not negligible compared with the distance $\sqrt{x^2 + x^2} = \sqrt{2}|x|$. Therefore the function is not differentiable at the origin.

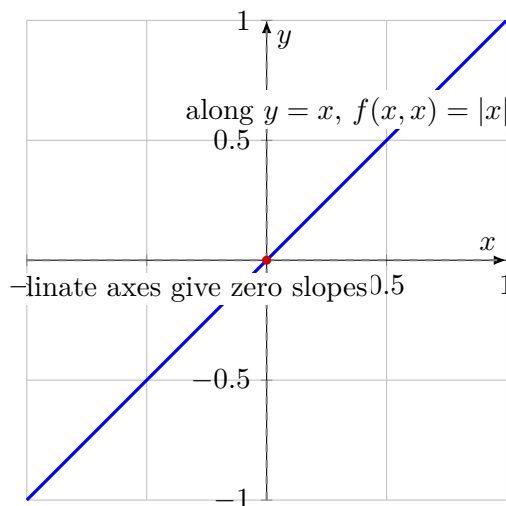


Figure 5: A function can have coordinate partial derivatives at a point but still fail to be differentiable there.

8 Tangent planes to level surfaces

Many important surfaces are not given as graphs $z = f(x, y)$. Instead, they are given implicitly as level surfaces

$$F(x, y, z) = c.$$

Suppose $P = (a, b, c_0)$ is a point on the level surface. If

$$\nabla F(P) \neq \mathbf{0},$$

then the gradient vector is normal to the level surface at P . Therefore the tangent plane is

$$\nabla F(P) \cdot \langle x - a, y - b, z - c_0 \rangle = 0.$$

In coordinates, if

$$\nabla F(P) = \langle F_x(P), F_y(P), F_z(P) \rangle,$$

then the tangent plane is

$$F_x(P)(x - a) + F_y(P)(y - b) + F_z(P)(z - c_0) = 0.$$

If a curve $\mathbf{r}(t)$ lies on the level surface, then $F(\mathbf{r}(t)) = c$. Differentiating with respect to t gives

$$\nabla F(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 0.$$

Thus the gradient is perpendicular to every tangent direction on the surface.

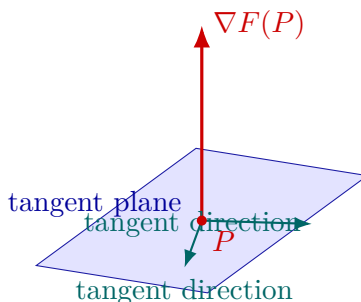


Figure 6: For a level surface, the gradient $\nabla F(P)$ is normal to the tangent plane.

9 Worked example: linearization and tangent plane to a level surface

Example 9.1: Linearization and tangent plane for a level surface

Let

$$F(x, y, z) = x^3 + y^2 e^z + xyz$$

and let $P = (1, 2, 0)$.

First evaluate the function:

$$F(1, 2, 0) = 1^3 + 2^2 e^0 + 1 \cdot 2 \cdot 0 = 5.$$

The partial derivatives are

$$F_x = 3x^2 + yz,$$

$$F_y = 2ye^z + xz,$$

$$F_z = y^2 e^z + xy.$$

At $P = (1, 2, 0)$,

$$F_x(P) = 3, \quad F_y(P) = 4, \quad F_z(P) = 6.$$

So the linearization is

$$L(x, y, z) = 5 + 3(x - 1) + 4(y - 2) + 6z.$$

Equivalently,

$$L(x, y, z) = 3x + 4y + 6z - 6.$$

Now estimate:

$$F(1.1, 1.9, 0) \approx L(1.1, 1.9, 0) = 3(1.1) + 4(1.9) - 6 = 4.9.$$

The level surface through P is $F(x, y, z) = 5$. Its tangent plane is

$$3(x - 1) + 4(y - 2) + 6z = 0,$$

or

$$3x + 4y + 6z - 11 = 0.$$

10 Total differentials

For a differentiable function $z = f(x, y)$, define dx and dy as independent small changes in the input variables. The *total differential* is

$$dz = f_x(x, y) dx + f_y(x, y) dy.$$

At a specific point (a, b) , this becomes

$$dz = f_x(a, b) dx + f_y(a, b) dy.$$

The differential is the linear approximation to the actual change:

$$\Delta z \approx dz.$$

For a function of three variables $w = f(x, y, z)$,

$$dw = f_x dx + f_y dy + f_z dz.$$

More generally, for $f : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$df = \nabla f(\mathbf{x}) \cdot d\mathbf{x}.$$

Why differentials matter

Deltas such as Δx and Δy represent actual changes. Differentials such as dx and dy are independent input changes used in the linear model. The differential dz is the predicted output change from the tangent plane.

11 Error propagation and sensitivity analysis

In applied problems, inputs are often measured with error. Linear approximation estimates how input error affects output error. Suppose

$$z = f(x, y)$$

and the possible measurement errors are

$$|dx| \leq \epsilon_x, \quad |dy| \leq \epsilon_y.$$

Then a first-order estimate for the output error is

$$|dz| \leq |f_x(x, y)|\epsilon_x + |f_y(x, y)|\epsilon_y.$$

If dx and dy are random errors with covariance matrix Σ , then the approximate variance of the output is

$$\text{Var}(df) \approx \nabla f(\mathbf{x})^T \Sigma \nabla f(\mathbf{x}).$$

This is the core idea behind the delta method in statistics.

Example 11.1: Volume of a cylinder

The volume of a cylinder is

$$V(r, h) = \pi r^2 h.$$

The differential is

$$dV = 2\pi r h dr + \pi r^2 dh.$$

At $r = 3$ and $h = 10$,

$$dV = 60\pi dr + 9\pi dh.$$

If $dr = 0.02$ and $dh = -0.05$, then

$$dV \approx 60\pi(0.02) + 9\pi(-0.05) = 0.75\pi \approx 2.36.$$

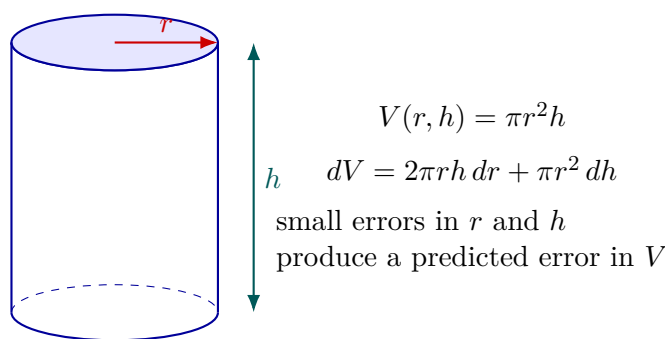


Figure 7: Linear error propagation predicts how measurement errors in radius and height affect cylinder volume.

12 Linearization as a local model in statistics and machine learning

Many statistical and machine-learning models depend on parameters. A simple example is a two-parameter linear prediction model

$$\hat{y} = \beta_0 + \beta_1 x.$$

A more complicated model may have many parameters:

$$\hat{y} = f(x; \theta_1, \theta_2, \dots, \theta_n).$$

Near a parameter vector θ_0 , linear approximation gives

$$f(x; \theta) \approx f(x; \theta_0) + \nabla_{\theta} f(x; \theta_0) \cdot (\theta - \theta_0).$$

This is the local linear model used in sensitivity analysis, uncertainty propagation, first-order optimization, local approximation of nonlinear regression models, and gradient-based learning.

Local linearization of a sigmoid

The logistic sigmoid is

$$\sigma(t) = \frac{1}{1 + e^{-t}}.$$

If $t = \beta_0 + \beta_1 x$, then

$$p(x; \beta_0, \beta_1) = \sigma(\beta_0 + \beta_1 x).$$

For fixed x , this is a function of the parameters β_0 and β_1 . Its gradient tells how the predicted probability changes when the parameters change.

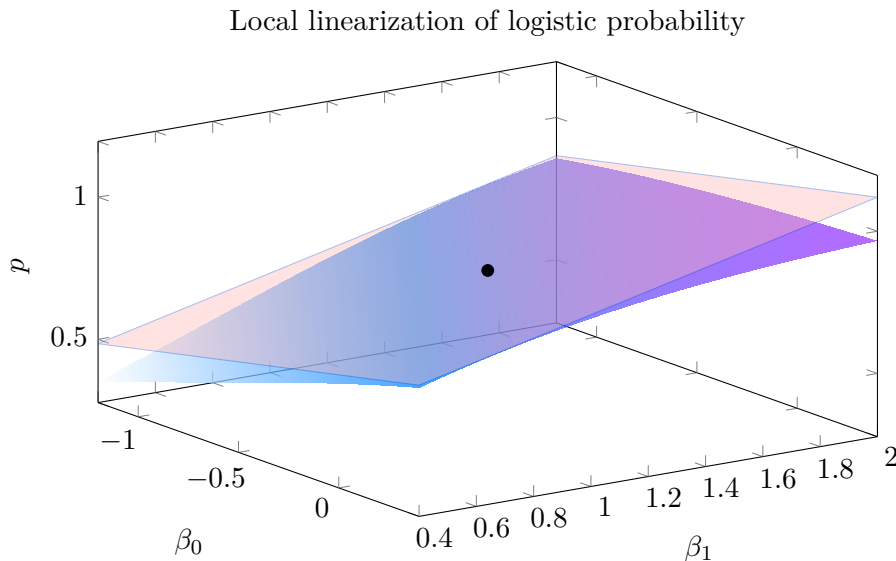


Figure 8: A nonlinear probability model is locally approximated by its tangent plane in parameter space.

13 Numerical linearization from finite differences

In real applications, we may not have a symbolic formula for the gradient. We may only have a black-box function that can be evaluated. Then we can approximate the gradient numerically. For a function $f(x, y)$, central finite differences give

$$f_x(a, b) \approx \frac{f(a + h, b) - f(a - h, b)}{2h},$$

$$f_y(a, b) \approx \frac{f(a, b + h) - f(a, b - h)}{2h}.$$

Then the numerical linearization is

$$L(x, y) = f(a, b) + \hat{f}_x(a, b)(x - a) + \hat{f}_y(a, b)(y - b).$$

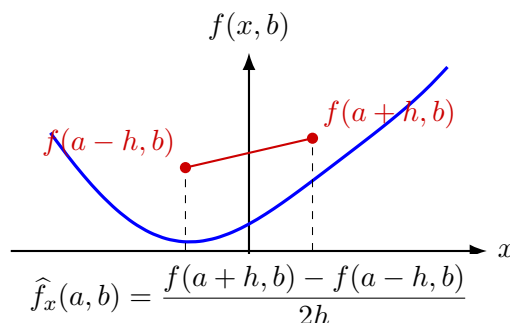


Figure 9: Finite differences approximate the gradient used in a local linear model.

14 Vector-valued and matrix viewpoints

The scalar-valued formula

$$f(\mathbf{x}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a})$$

has a natural extension to vector-valued functions. If

$$\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m,$$

then the best first-order approximation is

$$\mathbf{F}(\mathbf{x}) \approx \mathbf{F}(\mathbf{a}) + J_{\mathbf{F}}(\mathbf{a})(\mathbf{x} - \mathbf{a}),$$

where $J_{\mathbf{F}}(\mathbf{a})$ is the *Jacobian matrix*. The key idea is that derivatives in many variables are linear maps. The gradient-dot-product formula is the scalar-output version of this principle.

15 Common mistakes

Mistake 1: using the wrong base point

The tangent plane formula must use the point of tangency (a, b) and the height $f(a, b)$. Do not plug in the nearby point where you want the estimate.

Mistake 2: mixing up graph surfaces and level surfaces

For a graph $z = f(x, y)$, use

$$z - f(a, b) = f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

For a level surface $F(x, y, z) = c$, use

$$\nabla F(P) \cdot \langle x - a, y - b, z - c_0 \rangle = 0.$$

Mistake 3: thinking partial derivatives guarantee differentiability

Partial derivatives measure coordinate-direction slopes. Differentiability means there is a good linear approximation in all directions at once.

16 Summary of key formulas

For $z = f(x, y)$:

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Tangent plane to $z = f(x, y)$:

$$z - f(a, b) = f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Gradient form:

$$f(x, y) - f(a, b) \approx \nabla f(a, b) \cdot \langle x - a, y - b \rangle.$$

Total differential:

$$dz = f_x dx + f_y dy.$$

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$:

$$f(\mathbf{x}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}).$$

Tangent plane to $F(x, y, z) = c$ at $P = (a, b, c_0)$:

$$F_x(P)(x - a) + F_y(P)(y - b) + F_z(P)(z - c_0) = 0.$$

17 Practice problems

Conceptual problems

1. In your own words, explain why a tangent plane is a local linear model.
2. What information does the tangent plane to $z = f(x, y)$ match at the base point?
3. Explain why the gradient of F is normal to the level surface $F(x, y, z) = c$.
4. Explain the difference between Δz and dz .
5. Why can a function have partial derivatives but fail to be differentiable?

Skill problems

6. Find the tangent plane to $z = x^2 + 3y^2$ at $(1, -1, 4)$.
7. Find the linearization of $f(x, y) = e^{xy} + x^2y$ at $(1, 0)$.
8. Use a linear approximation to estimate $\sqrt{(4.1)^2 + (2.9)^2}$.
9. Suppose $f(3, 2) = 10$ and $\nabla f(3, 2) = \langle -2, 5 \rangle$. Estimate $f(2.9, 2.2)$.
10. Find dz for $z = x^2 \sin y + e^{xy}$.

Level-surface problems

11. Find the tangent plane to $x^2 + y^2 + z^2 = 14$ at $(1, 2, 3)$.
12. Find the tangent plane to $x^3 + y^2e^z + xyz = 5$ at $(1, 2, 0)$.
13. Find the tangent plane to $\ln(x + y + z) = 0$ at $(1, 0, 0)$.
14. Find a normal vector to the level surface $F(x, y, z) = x^2y + yz^2$ at $(1, 2, 1)$.
15. Find the linearization of $F(x, y, z) = x^2 + y^2 + z^2$ at $(1, 2, 2)$.

Applied and computational problems

16. The volume of a rectangular box is $V = xyz$. Use differentials to estimate the change in volume when $(x, y, z) = (10, 5, 2)$ changes by $(dx, dy, dz) = (0.1, -0.05, 0.02)$.
17. The BMI formula is $B(W, H) = 703W/H^2$, where W is weight in pounds and H is height in inches. Use differentials to estimate how small errors in W and H affect BMI.
18. For $f(x, y) = \sin(xy) + x^2$, use finite differences to estimate the tangent plane at $(1, 1)$.
19. For a prediction model $p(\beta_0, \beta_1) = 1/(1 + e^{-(\beta_0 + 2\beta_1)})$, find the linearization at $(0, 1)$.
20. Choose a nonlinear function of two variables, plot the surface and tangent plane, and describe where the approximation is accurate.

18 Python lab and interactive page

The Python lab for this chapter should ask students to build local linear models from symbolic formulas, numerical finite differences, and data-based approximations. Recommended lab activities:

1. visualize the surface $z = 3x^2 + y^2$ and its tangent plane at $(1, 2)$;
2. compare exact values with linearized values on a grid;
3. compute and visualize approximation error;
4. estimate gradients numerically using finite differences;
5. find tangent planes to level surfaces using gradients;
6. simulate error propagation for measurement uncertainty;
7. explore local linearization of a logistic probability model.

Resources:

- Notebook lab: Chapter 9 linear approximation lab.
- Interactive HTML page: `chapter-09.html`.

19 Chapter summary

Linear approximation is the first major synthesis point of multivariable calculus. Partial derivatives become a gradient. The gradient becomes a local linear model. The local linear model becomes a tangent plane, a differential, a sensitivity estimate, and a computational approximation.

For graphs $z = f(x, y)$, the tangent plane is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

For level surfaces $F(x, y, z) = c$, the tangent plane is determined by the normal vector $\nabla F(P)$:

$$\nabla F(P) \cdot (\mathbf{x} - P) = 0.$$

In applied mathematics and statistics, this chapter is the foundation for error propagation, the delta method, local optimization, and first-order approximation of nonlinear models. In machine learning and AI, it is the idea behind using gradients to approximate and improve complicated models one small step at a time.

Chapter 10. Chain Rule, Jacobians, and Automatic Differentiation

Compositions, sensitivity, computational graphs, and backpropagation

The chain rule is the rule for differentiating a function built from other functions. In single-variable calculus, if $y = f(u)$ and $u = g(t)$, then

$$\frac{dy}{dt} = \frac{dy}{du} \frac{du}{dt}.$$

In multivariable calculus, the same idea becomes more powerful because each intermediate quantity may depend on several variables, and each output may have several components. The derivative is no longer just one number. It may be a gradient vector, a Jacobian matrix, or a linear transformation.

This chapter is one of the main bridges from classical multivariable calculus to modern computation. The chain rule explains how input changes propagate through a model, why Jacobian matrices describe local sensitivity, how coordinate transformations are differentiated, and why backpropagation in neural networks is repeated use of the chain rule.

Core principle

The multivariable chain rule says that derivatives propagate through compositions by multiplying the correct local linear maps.

Teaching-material connection

This chapter develops partial derivatives, gradients, tangent-plane approximation, directional derivatives, and implicit differentiation into a systematic treatment of chain rules and Jacobians. The same notation later supports change of variables, optimization, statistics, machine learning, and vector calculus.

Learning goals

After studying this chapter, you should be able to:

1. recognize compositions of one-variable, two-variable, and vector-valued functions;
2. apply the chain rule when $z = f(x, y)$ and $x = x(t)$, $y = y(t)$;
3. apply the chain rule when $z = f(x, y)$ and $x = x(s, t)$, $y = y(s, t)$;
4. compute Jacobian matrices for maps $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$;
5. write the multivariable chain rule in matrix form;
6. interpret the Jacobian as the local linear map of a nonlinear function;
7. compute and verify Jacobians numerically using finite differences;
8. understand computational graphs as organized compositions;
9. explain reverse-mode automatic differentiation and backpropagation at a conceptual

level.

1 Why composition matters

A function of several variables often appears indirectly. For example, suppose

$$z = f(x, y) = x^2y + \sin y,$$

but x and y themselves depend on time:

$$x = t^2, \quad y = e^t.$$

Then z is ultimately a function of t :

$$z(t) = f(x(t), y(t)) = t^4 e^t + \sin(e^t).$$

We could substitute first and then differentiate. But in applications, substitution may be inconvenient or impossible. The chain rule computes the derivative by following the dependencies.

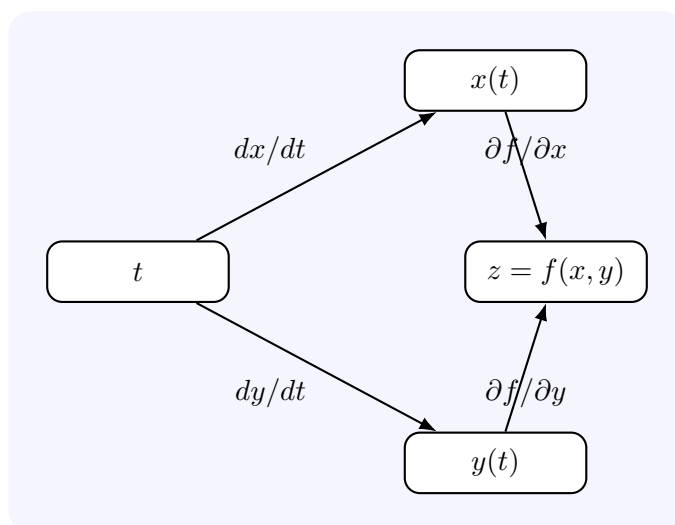


Figure 1: A composed function passes information from input variables through intermediate variables to the final output.

The input t affects x and y . Then x and y affect z . Therefore a small change in t changes z through two routes:

$$t \rightarrow x \rightarrow z, \quad t \rightarrow y \rightarrow z.$$

The chain rule adds the contributions from all routes.

2 Chain rule along a curve

Suppose $z = f(x, y)$ is differentiable, and $x = x(t)$ and $y = y(t)$ are differentiable functions of t . Then

$$z(t) = f(x(t), y(t))$$

has derivative

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

All partial derivatives are evaluated at $(x(t), y(t))$.

Equivalently, using the gradient,

$$\frac{dz}{dt} = \nabla f(x(t), y(t)) \cdot \frac{d}{dt} \langle x(t), y(t) \rangle.$$

Thus

Chain rule along a curve

$$\frac{d}{dt} f(r(t)) = \nabla f(r(t)) \cdot r'(t).$$

The rate of change of a scalar field along a curve is the gradient dotted with the velocity vector.

Worked example 10.0: Chain rule along a curve

Let

$$f(x, y) = x^2 y + \sin y, \quad x(t) = t^2, \quad y(t) = e^t.$$

Find dz/dt at $t = 0$.

The partial derivatives are

$$f_x(x, y) = 2xy, \quad f_y(x, y) = x^2 + \cos y.$$

At $t = 0$,

$$x(0) = 0, \quad y(0) = 1,$$

and

$$x'(0) = 0, \quad y'(0) = 1.$$

Therefore

$$\left. \frac{dz}{dt} \right|_{t=0} = f_x(0, 1)x'(0) + f_y(0, 1)y'(0) = 0 \cdot 0 + \cos(1) \cdot 1 = \cos(1).$$

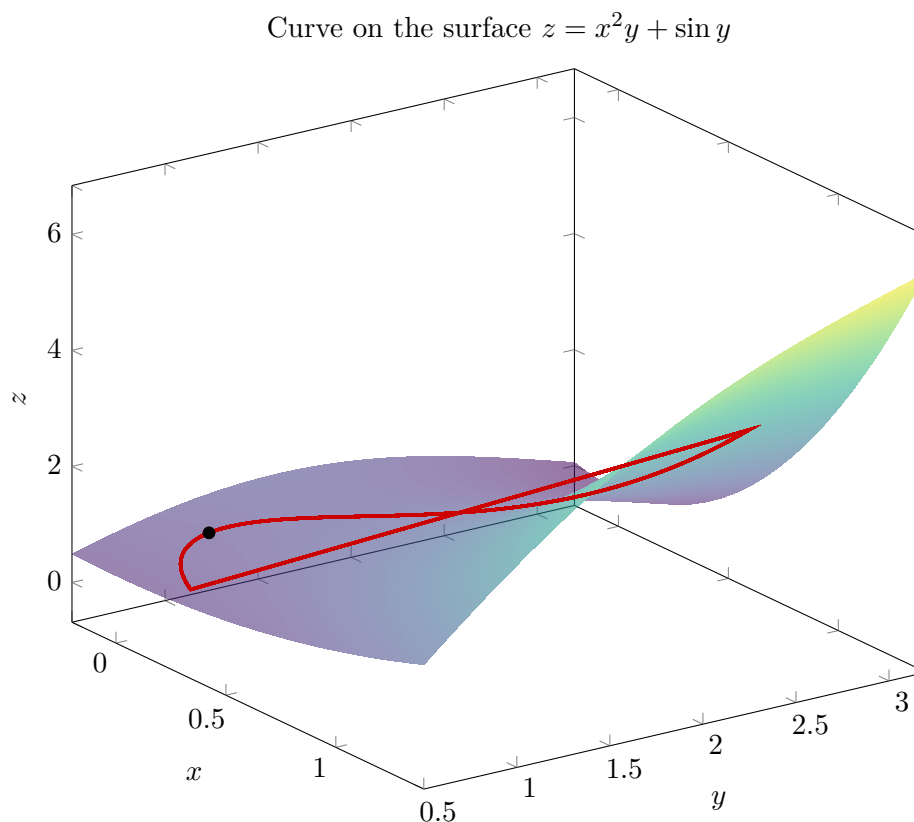


Figure 2: The chain rule measures the rate at which a surface changes along a parameterized curve in the domain.

3 The tree form of the chain rule

Now suppose

$$z = f(x, y), \quad x = x(s, t), \quad y = y(s, t).$$

Then z is a function of two variables s and t :

$$z(s, t) = f(x(s, t), y(s, t)).$$

The partial derivatives of z are

$$\frac{\partial z}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s},$$

and

$$\frac{\partial z}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t}.$$

A useful memory device is

$$(s, t) \longrightarrow (x, y) \longrightarrow z.$$

To differentiate z with respect to one input variable, follow every path from that variable to z , multiply derivatives along each path, and add the path contributions.

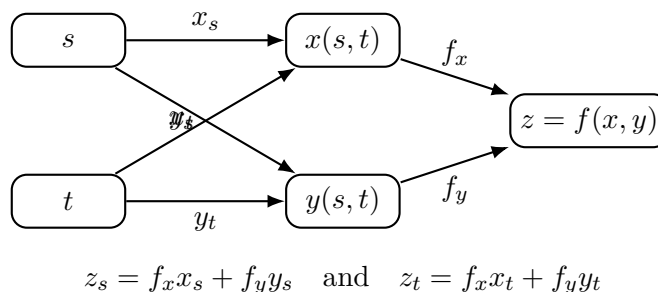


Figure 3: The tree form of the chain rule adds the derivative contribution from every dependency path.

Worked example 10.1: Two-parameter chain rule

Let

$$f(x, y) = x^2 + y^3, \quad x = s + t, \quad y = st.$$

Then $f_x = 2x$ and $f_y = 3y^2$. Also

$$x_s = 1, \quad x_t = 1, \quad y_s = t, \quad y_t = s.$$

Therefore

$$z_s = 2x \cdot 1 + 3y^2 \cdot t = 2x + 3ty^2,$$

and

$$z_t = 2x \cdot 1 + 3y^2 \cdot s = 2x + 3sy^2.$$

Substituting $x = s + t$ and $y = st$ gives

$$z_s = 2(s + t) + 3t(s^2 t^2), \quad z_t = 2(s + t) + 3s(s^2 t^2).$$

4 Jacobian matrices

Let

$$F : \mathbb{R}^n \rightarrow \mathbb{R}^m,$$

where

$$F(x_1, \dots, x_n) = \begin{bmatrix} F_1(x_1, \dots, x_n) \\ \vdots \\ F_m(x_1, \dots, x_n) \end{bmatrix}.$$

The *Jacobian matrix* of F is

$$J_F(x) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} & \dots & \frac{\partial F_1}{\partial x_n} \\ \frac{\partial F_2}{\partial x_1} & \frac{\partial F_2}{\partial x_2} & \dots & \frac{\partial F_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \frac{\partial F_m}{\partial x_2} & \dots & \frac{\partial F_m}{\partial x_n} \end{bmatrix}.$$

The entry in row i , column j tells how the i th output changes when the j th input changes.

Dimensions matter

If $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$, then J_F is an $m \times n$ matrix. It maps small input displacements $\Delta x \in \mathbb{R}^n$ to small output displacements $\Delta F \in \mathbb{R}^m$:

$$\Delta F \approx J_F(x) \Delta x.$$

Worked example 10.2: A Jacobian matrix

Let

$$F(x, y) = \begin{bmatrix} x^2 y \\ e^{x+y} \\ \sin(xy) \end{bmatrix}.$$

Then $F : \mathbb{R}^2 \rightarrow \mathbb{R}^3$. Its Jacobian is a 3×2 matrix:

$$J_F(x, y) = \begin{bmatrix} 2xy & x^2 \\ e^{x+y} & e^{x+y} \\ y \cos(xy) & x \cos(xy) \end{bmatrix}.$$

At $(1, 0)$,

$$J_F(1, 0) = \begin{bmatrix} 0 & 1 \\ e & e \\ 0 & 1 \end{bmatrix}.$$

Thus if (x, y) moves by a small vector $\Delta x = \langle h, k \rangle$, then

$$\Delta F \approx \begin{bmatrix} 0 & 1 \\ e & e \\ 0 & 1 \end{bmatrix} \begin{bmatrix} h \\ k \end{bmatrix} = \begin{bmatrix} k \\ e(h+k) \\ k \end{bmatrix}.$$

5 The Jacobian as a local linear map

For a scalar function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the derivative at a point is encoded by the gradient. For a vector-valued function $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the derivative is encoded by the Jacobian. Near a point a ,

$$F(a + h) \approx F(a) + J_F(a)h.$$

This means that the nonlinear function behaves like an affine linear map near a .

Worked example 10.3: Local effect of a nonlinear map

Let

$$F(x, y) = \begin{bmatrix} x + y \\ x^2 - y \end{bmatrix}.$$

Then

$$J_F(x, y) = \begin{bmatrix} 1 & 1 \\ 2x & -1 \end{bmatrix}.$$

At $a = (1, 1)$,

$$J_F(1, 1) = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}.$$

Therefore a small square around $(1, 1)$ in the input plane is approximately mapped to a parallelogram in the output plane.

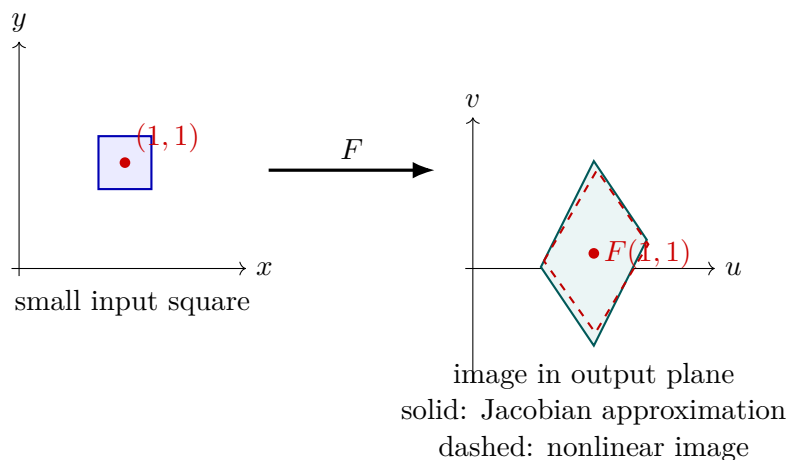


Figure 4: Near a point, a nonlinear map is approximated by its Jacobian linear map.

6 Matrix form of the chain rule

Suppose

$$F : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad G : \mathbb{R}^m \rightarrow \mathbb{R}^p.$$

Then the composition $G \circ F : \mathbb{R}^n \rightarrow \mathbb{R}^p$ has Jacobian

Matrix chain rule

$$J_{G \circ F}(x) = J_G(F(x))J_F(x).$$

The order is important: first apply F , then apply G , but the Jacobian matrices multiply as outer derivative times inner derivative.

This is the multivariable version of $(g \circ f)'(x) = g'(f(x))f'(x)$.

Worked example 10.4: Matrix chain rule

Let

$$F(s, t) = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} s + t \\ st \end{bmatrix}, \quad G(x, y) = x^2 + y^3.$$

The Jacobian of F is

$$J_F(s, t) = \begin{bmatrix} 1 & 1 \\ t & s \end{bmatrix}.$$

The Jacobian of G is the row vector

$$J_G(x, y) = [2x \quad 3y^2].$$

Therefore

$$J_{G \circ F}(s, t) = [2x \quad 3y^2] \begin{bmatrix} 1 & 1 \\ t & s \end{bmatrix} = [2x + 3ty^2 \quad 2x + 3sy^2].$$

This matches the tree form of the chain rule.

7 Gradients through linear maps

A very common situation in data science and machine learning is

$$L(w) = \phi(Aw + b),$$

where $w \in \mathbb{R}^n$ is a parameter vector, A is a matrix, b is a vector, and ϕ is a scalar loss function. Let

$$u = Aw + b.$$

Then the chain rule gives

Gradient through a linear map

$$\nabla_w L(w) = A^T \nabla_u \phi(u), \quad u = Aw + b.$$

The value moves forward through $u = Aw + b$. The gradient moves backward through A^T .

Worked example 10.5: Least-squares gradient

Let

$$L(w) = \frac{1}{2} \|Xw - y\|^2,$$

where X is a data matrix, w is a parameter vector, and y is an observed response vector. Define $r = Xw - y$. Then

$$L = \frac{1}{2} r^T r.$$

The gradient with respect to r is $\nabla_r L = r$. Since $r = Xw - y$, the chain rule gives

$$\nabla_w L = X^T (Xw - y).$$

This is one of the most important gradients in applied mathematics, statistics, and machine learning.

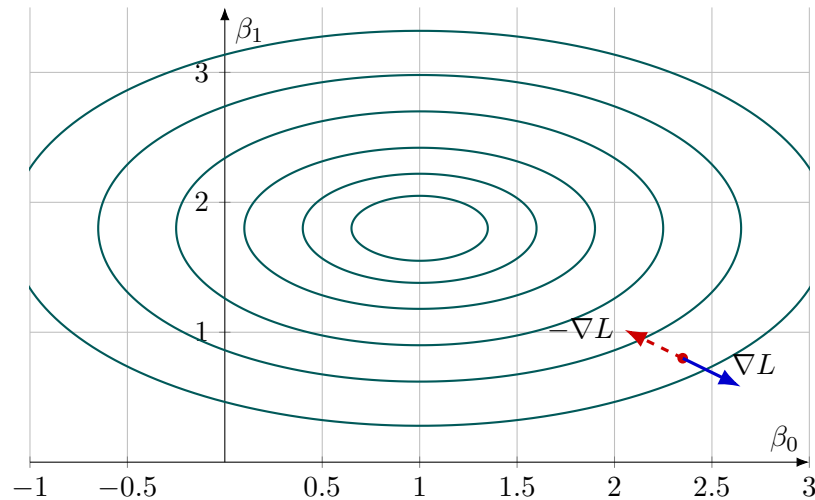


Figure 5: The least-squares gradient points toward increasing loss, while gradient descent moves in the opposite direction.

8 Numerical Jacobians by finite differences

In applications, we may have a function implemented as code but not have a closed formula. We can approximate the Jacobian using finite differences. For $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the j th column of the Jacobian can be approximated by

$$\frac{F(x + he_j) - F(x - he_j)}{2h},$$

where e_j is the j th standard basis vector.

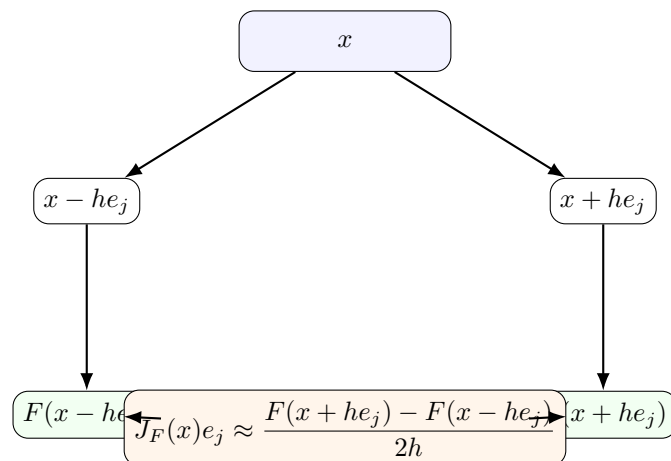


Figure 6: A centered finite difference estimates one column of the Jacobian at a time.

Finite differences are useful, but they can be slow or unstable for very large models. Automatic differentiation gives a more efficient and accurate way to compute derivatives of functions defined by code.

9 Computational graphs

A *computational graph* breaks a complicated function into elementary operations. Each node stores a value. Each edge represents dependence.

Consider

$$f(x, y) = (xy + \sin x)^2.$$

We can compute it step by step:

$$a = xy, \quad b = \sin x, \quad c = a + b, \quad f = c^2.$$

The chain rule can be applied locally at every step.

Worked example 10.6: Forward sensitivity

For $f(x, y) = (xy + \sin x)^2$, set

$$a = xy, \quad b = \sin x, \quad c = a + b, \quad f = c^2.$$

Then

$$\frac{\partial f}{\partial c} = 2c, \quad \frac{\partial c}{\partial a} = 1, \quad \frac{\partial c}{\partial b} = 1,$$

and

$$\frac{\partial a}{\partial x} = y, \quad \frac{\partial a}{\partial y} = x, \quad \frac{\partial b}{\partial x} = \cos x.$$

Therefore

$$\frac{\partial f}{\partial x} = 2c(y + \cos x), \quad \frac{\partial f}{\partial y} = 2cx.$$

Since $c = xy + \sin x$,

$$\nabla f(x, y) = \langle 2(xy + \sin x)(y + \cos x), 2x(xy + \sin x) \rangle.$$

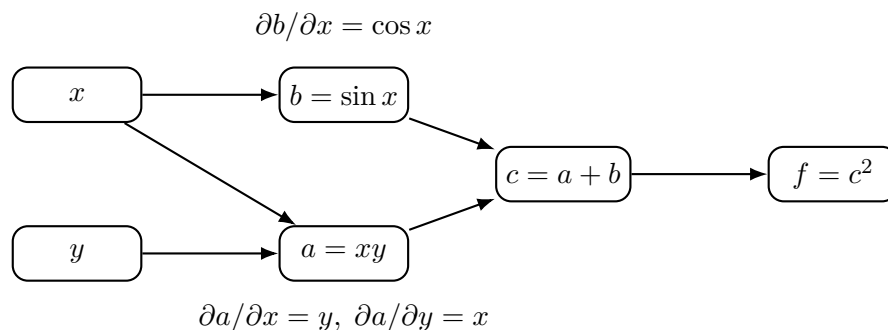


Figure 7: A computational graph decomposes a function into elementary operations where local derivatives are easy to compute.

10 Reverse-mode automatic differentiation

Automatic differentiation is not symbolic differentiation and not finite differences. It computes exact derivatives of the operations performed by a program, up to floating-point rounding.

There are two main modes:

- *Forward mode* propagates derivatives from inputs to outputs.
- *Reverse mode* propagates derivatives from output back to inputs.

Reverse mode is especially efficient when there is one scalar output and many inputs, which is exactly the situation for many loss functions:

$$L(\theta_1, \dots, \theta_n) : \mathbb{R}^n \rightarrow \mathbb{R}.$$

Backpropagation is reverse-mode automatic differentiation applied to neural networks.

Worked example 10.7: Reverse accumulation

Again let

$$f = (xy + \sin x)^2.$$

At each node, reverse mode stores an *adjoint*, meaning the derivative of the final output with respect to that node. Start with

$$\bar{f} = \frac{\partial f}{\partial f} = 1.$$

Since $f = c^2$,

$$\bar{c} = \bar{f} \frac{\partial f}{\partial c} = 2c.$$

Since $c = a + b$,

$$\bar{a} = \bar{c}, \quad \bar{b} = \bar{c}.$$

Since $a = xy$ and $b = \sin x$,

$$\bar{x} = \bar{a}y + \bar{b} \cos x, \quad \bar{y} = \bar{a}x.$$

This gives the same gradient as before, but the organization is different: compute values forward, then gradients backward.

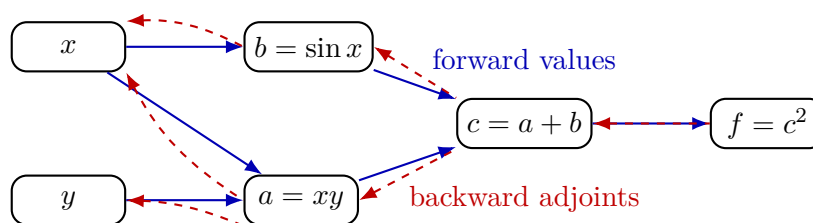


Figure 8: Reverse mode first evaluates the graph forward, then propagates adjoints backward from the scalar output.

11 A tiny neural-network example

Consider a very small model with one input x , one weight w , one bias b , and sigmoid activation

$$\hat{y} = \sigma(wx + b), \quad \sigma(u) = \frac{1}{1 + e^{-u}}.$$

For a target value y , use the squared loss

$$L(w, b) = \frac{1}{2}(\hat{y} - y)^2.$$

The computation is

$$u = wx + b, \quad \hat{y} = \sigma(u), \quad L = \frac{1}{2}(\hat{y} - y)^2.$$

The chain rule gives

$$\frac{\partial L}{\partial w} = (\hat{y} - y)\sigma'(u)x, \quad \frac{\partial L}{\partial b} = (\hat{y} - y)\sigma'(u),$$

where

$$\sigma'(u) = \sigma(u)(1 - \sigma(u)).$$

This is backpropagation in its smallest possible form.

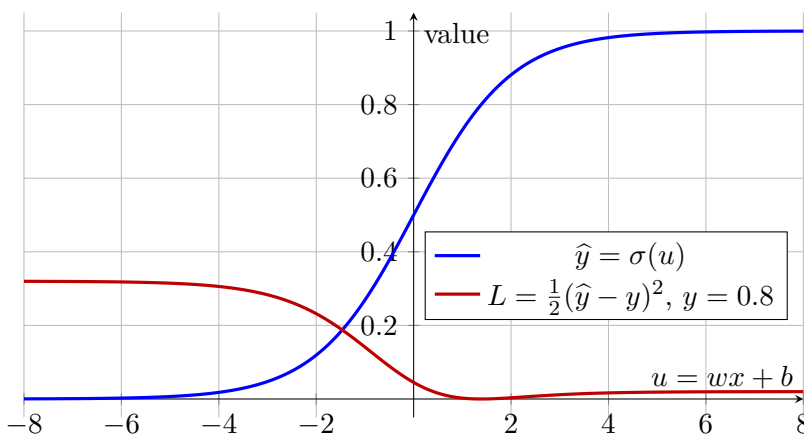


Figure 9: A one-neuron model is a composition of an affine map, a nonlinear activation, and a loss function.

12 Chain rule and coordinate transformations

Coordinate transformations are also compositions. For polar coordinates,

$$x = r \cos \theta, \quad y = r \sin \theta.$$

If $z = f(x, y)$, then

$$z(r, \theta) = f(r \cos \theta, r \sin \theta).$$

The chain rule gives

$$\frac{\partial z}{\partial r} = f_x \cos \theta + f_y \sin \theta,$$

and

$$\frac{\partial z}{\partial \theta} = f_x(-r \sin \theta) + f_y(r \cos \theta).$$

The Jacobian matrix of the polar coordinate map

$$\Phi(r, \theta) = \begin{bmatrix} r \cos \theta \\ r \sin \theta \end{bmatrix}$$

is

$$J_{\Phi}(r, \theta) = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}.$$

Its determinant is

$$\det J_{\Phi}(r, \theta) = r.$$

This determinant explains why the area element becomes

$$dA = r \, dr \, d\theta.$$

Thus the chain rule for derivatives and the change-of-variables rule for integrals are closely connected.

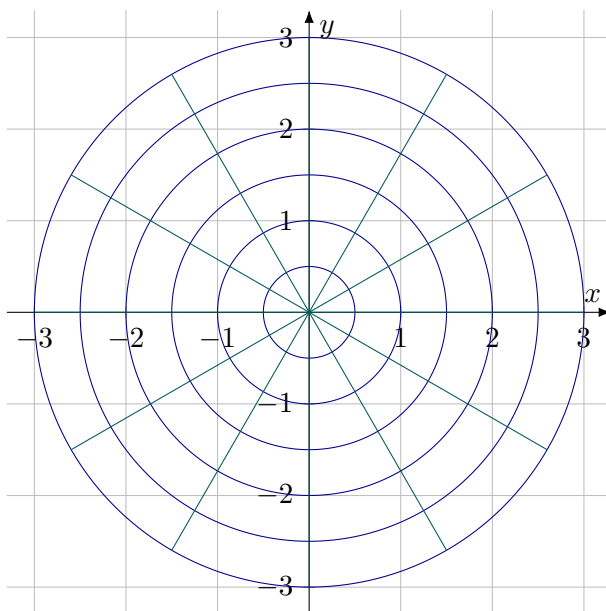


Figure 10: The polar coordinate map bends a rectangular grid in parameter space into radial and circular coordinate curves.

13 Implicit differentiation as a chain rule

Suppose $z = z(x, y)$ is defined implicitly by

$$F(x, y, z) = 0.$$

Then $F(x, y, z(x, y))$ is identically zero. Differentiating with respect to x gives

$$F_x + F_z \frac{\partial z}{\partial x} = 0.$$

Therefore, if $F_z \neq 0$,

$$\boxed{\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}}.$$

Similarly,

$$\boxed{\frac{\partial z}{\partial y} = -\frac{F_y}{F_z}}.$$

This is the same chain rule idea, but now one variable is hidden inside an equation.

Worked example 10.8: Implicit partial derivatives

Suppose $z = f(x, y)$ is defined by

$$x^3 + y^3 + z^3 + 2xyz = 3.$$

Let

$$F(x, y, z) = x^3 + y^3 + z^3 + 2xyz - 3.$$

Then

$$F_x = 3x^2 + 2yz, \quad F_y = 3y^2 + 2xz, \quad F_z = 3z^2 + 2xy.$$

Thus

$$\frac{\partial z}{\partial x} = -\frac{3x^2 + 2yz}{3z^2 + 2xy}, \quad \frac{\partial z}{\partial y} = -\frac{3y^2 + 2xz}{3z^2 + 2xy}.$$

14 Common mistakes

Mistake 1: Forgetting all paths

If $z = f(x, y)$ and both x and y depend on t , then dz/dt has two contributions:

$$\frac{dz}{dt} = f_x x'(t) + f_y y'(t).$$

Do not keep only one path.

Mistake 2: Multiplying Jacobians in the wrong order

For $G \circ F$, the derivative is

$$J_{G \circ F}(x) = J_G(F(x))J_F(x),$$

not $J_F(x)J_G(F(x))$. Check matrix dimensions.

Mistake 3: Confusing gradient and Jacobian

For a scalar-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the Jacobian is a $1 \times n$ row vector, while the gradient is usually written as an $n \times 1$ column vector or as a vector in $\mathbb{R}^n$. The information is the same, but the orientation matters in matrix formulas.

Mistake 4: Treating automatic differentiation as numerical differentiation

Finite differences estimate derivatives by evaluating nearby points. Automatic differentiation applies the chain rule exactly to the operations in a program. These are different methods.

15 Summary of core formulas

Chain rule along a curve

If $z = f(x(t), y(t))$, then

$$\frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt}, \quad \frac{d}{dt} f(r(t)) = \nabla f(r(t)) \cdot r'(t).$$

Chain rule with two parameters

If $z = f(x(s, t), y(s, t))$, then

$$z_s = f_x x_s + f_y y_s, \quad z_t = f_x x_t + f_y y_t.$$

Jacobian matrix and linear approximation

For $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$,

$$J_F(x) = \begin{bmatrix} \frac{\partial F_i}{\partial x_j} \end{bmatrix}, \quad F(a + h) \approx F(a) + J_F(a)h.$$

Matrix chain rule

$$J_{G \circ F}(x) = J_G(F(x)) J_F(x).$$

Gradient through a linear map

If $L(w) = \phi(Aw + b)$, then

$$\nabla_w L = A^T \nabla_u \phi(u), \quad u = Aw + b.$$

Implicit differentiation

If $F(x, y, z(x, y)) = 0$, then

$$z_x = -\frac{F_x}{F_z}, \quad z_y = -\frac{F_y}{F_z}.$$

16 Practice problems

Conceptual problems

1. Explain why the chain rule is a rule about composition.
2. Explain why the chain rule adds contributions from different paths in a dependency graph.
3. What is the dimension of the Jacobian matrix for a map $F : \mathbb{R}^5 \rightarrow \mathbb{R}^3$?

4. Explain the difference between a gradient and a Jacobian.
5. Why does reverse-mode automatic differentiation work well for scalar loss functions with many parameters?
6. Explain why A^T appears when backpropagating a gradient through $u = Aw + b$.

Computational and algebraic problems

7. Let $z = x^2 + xy + y^2$, where $x = t^2$ and $y = \sin t$. Find dz/dt .
8. Let $z = e^{xy}$, where $x = s + t$ and $y = s - t$. Find z_s and z_t .
9. Let $F(x, y) = \langle x^2 + y, xy, \sin(x + y) \rangle$. Compute $J_F(x, y)$.
10. Let $F(x, y) = \langle x + y, x - y \rangle$ and $G(u, v) = u^2 + v^2$. Compute $J_{G \circ F}$ using matrix multiplication.
11. Let $F(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$ define $z = z(x, y)$ near a point with $z \neq 0$. Find z_x and z_y .
12. Let $L(w) = \frac{1}{2} \|Aw - y\|^2$. Derive $\nabla_w L = A^T(Aw - y)$.
13. For $f(x, y) = (x^2 + y)^3$, draw a computational graph and compute ∇f by the chain rule.
14. For $\hat{y} = \sigma(w_1 x_1 + w_2 x_2 + b)$ and $L = \frac{1}{2} (\hat{y} - y)^2$, compute $\partial L / \partial w_1$, $\partial L / \partial w_2$, and $\partial L / \partial b$.
15. Use finite differences to approximate the Jacobian of $F(x, y) = \langle e^{xy}, x^2 - y^2 \rangle$ at $(1, 2)$.

Challenge problems

16. Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be $F(r, \theta) = \langle r \cos \theta, r \sin \theta \rangle$. Compute J_F and explain geometrically why $|\det J_F| = r$.
17. Let $F(x) = Ax$ and $G(u) = \frac{1}{2} \|u - y\|^2$. Use the matrix chain rule to derive the least-squares gradient.
18. Suppose $H = G \circ F$ where $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $G : \mathbb{R}^m \rightarrow \mathbb{R}$. Show that $\nabla H(x) = J_F(x)^T \nabla G(F(x))$ when gradients are written as column vectors.
19. Write a short paragraph explaining how backpropagation is related to the chain rule.
20. Consider $f(x, y, z) = \sin(xy) + z^2$ and a curve $r(t) = \langle t, e^t, \cos t \rangle$. Compute $d(f \circ r)/dt$ at $t = 0$.

17 Python lab and interactive page

- Notebook lab: `Chapter10_Chain_Rule_Jacobians_and_Automatic_Differentiation.ipynb`
- Interactive HTML page: `../htmls/chapter-10.html`

18 Chapter summary

The chain rule is the calculus of composition. In its simplest form, it differentiates $f(g(t))$. In multivariable calculus, it differentiates functions passing through curves, surfaces, coordinate transformations, and computational graphs. The Jacobian matrix organizes these derivative calculations as linear maps. The matrix chain rule

$$J_{G \circ F}(x) = J_G(F(x))J_F(x)$$

is the central formula of the chapter.

For modern applied mathematics, statistics, and machine learning, the most important lesson is that derivatives can be propagated through models. Least-squares gradients, logistic-regression gradients, neural-network backpropagation, coordinate transformations, and sensitivity analysis are all expressions of the same idea: compute local derivatives at each step and combine them by the chain rule.

Chapter 11. Directional Derivatives, Hessians, and Optimization

From local rates of change to modern loss landscapes

Chapter goal. In this chapter we turn derivatives into decisions. A gradient tells us how a function changes when we move in a chosen direction. A Hessian tells us the local curvature of a surface or a loss function. Together, they give a practical language for maxima, minima, saddle points, and numerical optimization.

A function of one variable has only two local directions: left and right. A function of two or more variables has infinitely many directions. At a point on a surface $z = f(x, y)$, we can move east, west, north, south, northeast, or along any unit direction u . The *directional derivative* measures the rate of change in that direction.

This idea is central in applied mathematics, statistics, data science, and machine learning. A statistical loss function or negative log-likelihood is usually a function of many parameters. To minimize it, we must know which direction decreases the loss most rapidly. The answer is built from the gradient and refined by the Hessian.

Learning objectives

After studying this chapter, you should be able to:

1. compute directional derivatives using the formula $D_u f = \nabla f \cdot u$;
2. explain why the gradient points in the direction of steepest increase;
3. find and classify critical points of functions of two variables;
4. use the Hessian matrix and the second derivative test;
5. distinguish local extrema, absolute extrema, and saddle points;
6. solve absolute optimization problems on closed bounded regions;
7. implement gradient descent and Newton's method for two-variable functions;
8. interpret optimization as the geometric core of modern model fitting.

1 Directional derivatives

Let $f(x, y)$ be a differentiable function and let $u = \langle a, b \rangle$ be a unit vector. The *directional derivative* of f at (x_0, y_0) in the direction u is

$$D_u f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}.$$

This is the rate of change of f as we move from (x_0, y_0) in the direction u .

The important computational formula is

$$D_u f(x, y) = \nabla f(x, y) \cdot u.$$

For a function of three variables, if $u = \langle a, b, c \rangle$ is a unit vector, then

$$D_u f(x, y, z) = \nabla f(x, y, z) \cdot u.$$

Always normalize the direction vector

The direction vector in a directional derivative must be a *unit vector*. If the problem gives a nonzero vector v , first compute

$$u = \frac{v}{\|v\|}.$$

Then use $D_u f = \nabla f \cdot u$.

Worked example 1.1: A directional derivative from a non-unit vector

Let

$$f(x, y) = x^2 + 3xy - 2y^4.$$

Find $D_u f(0, 1)$ in the direction of $v = \langle 1, 2 \rangle$.

First compute the gradient:

$$\nabla f(x, y) = \langle 2x + 3y, 3x - 8y^3 \rangle.$$

At $(0, 1)$,

$$\nabla f(0, 1) = \langle 3, -8 \rangle.$$

The vector $v = \langle 1, 2 \rangle$ has length $\sqrt{5}$, so

$$u = \frac{1}{\sqrt{5}} \langle 1, 2 \rangle.$$

Therefore

$$D_u f(0, 1) = \langle 3, -8 \rangle \cdot \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle = \frac{3 - 16}{\sqrt{5}} = -\frac{13}{\sqrt{5}}.$$

The negative sign means that f is decreasing in the direction $\langle 1, 2 \rangle$ at $(0, 1)$.

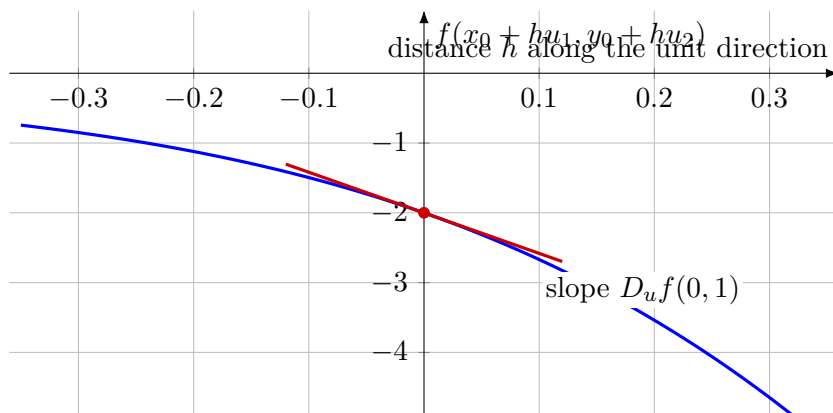


Figure 1: A directional derivative is the slope of a one-variable slice through a surface.

2 The gradient as the direction of steepest increase

The formula $D_u f = \nabla f \cdot u$ has a powerful geometric interpretation. If u is a unit vector, then

$$D_u f = \nabla f \cdot u = \|\nabla f\| \|u\| \cos \theta = \|\nabla f\| \cos \theta,$$

where θ is the angle between ∇f and u . Since $-1 \leq \cos \theta \leq 1$,

$$-\|\nabla f\| \leq D_u f \leq \|\nabla f\|.$$

Thus the maximum directional derivative is $\|\nabla f\|$, attained in the direction of ∇f , and the minimum directional derivative is $-\|\nabla f\|$, attained in the opposite direction.

Steepest directions

At a point where $\nabla f \neq 0$,

$$\text{steepest ascent direction} = \frac{\nabla f}{\|\nabla f\|}, \quad \text{steepest descent direction} = -\frac{\nabla f}{\|\nabla f\|}.$$

This is the geometric reason gradient descent works: to decrease a differentiable function rapidly, move approximately in the direction $-\nabla f$.

Worked example 2.1: Fastest increase of temperature

Suppose the temperature at a point (x, y, z) is

$$T(x, y, z) = xe^y + \frac{1}{2}z^2.$$

The gradient is

$$\nabla T(x, y, z) = \langle e^y, xe^y, z \rangle.$$

At $(1, 0, 3)$,

$$\nabla T(1, 0, 3) = \langle 1, 1, 3 \rangle.$$

Therefore the direction of fastest increase is

$$\frac{1}{\sqrt{11}} \langle 1, 1, 3 \rangle,$$

and the maximum rate of increase is $\|\nabla T(1, 0, 3)\| = \sqrt{11}$.

Worked example 2.2: Climbing a hill

Suppose a hill is modeled by

$$z = f(x, y) = 1100 - 2x^2 - y^2,$$

where x and y are measured in meters. At $(10, 20)$,

$$\nabla f(x, y) = \langle -4x, -2y \rangle, \quad \nabla f(10, 20) = \langle -40, -40 \rangle.$$

The hill rises fastest in the southwest direction

$$\frac{1}{\sqrt{2}} \langle -1, -1 \rangle,$$

and the maximum rate of increase is $40\sqrt{2}$.

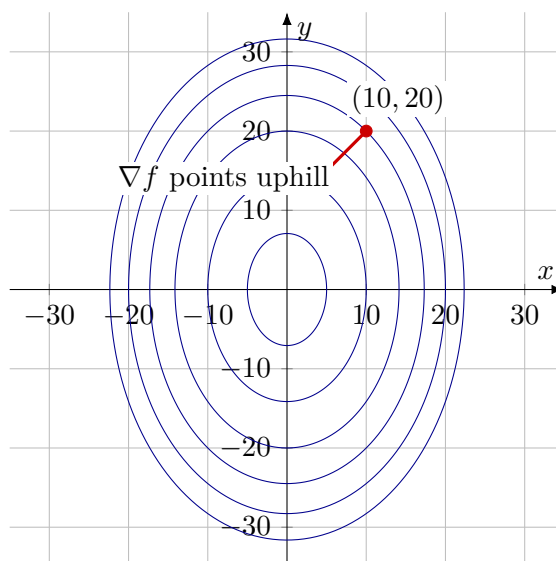


Figure 2: The gradient on a contour map points in the direction of fastest increase.

3 Gradients and level curves

A level curve of f is a curve $f(x, y) = c$. If a differentiable curve $r(t) = \langle x(t), y(t) \rangle$ stays on the level curve, then $f(x(t), y(t)) = c$. Differentiating gives

$$\nabla f(x(t), y(t)) \cdot r'(t) = 0.$$

So the gradient is perpendicular to the tangent vector of the level curve.

Gradient and level curves

At a regular point of a level curve, ∇f is normal to the level curve. Closely spaced level curves usually indicate a large gradient magnitude.

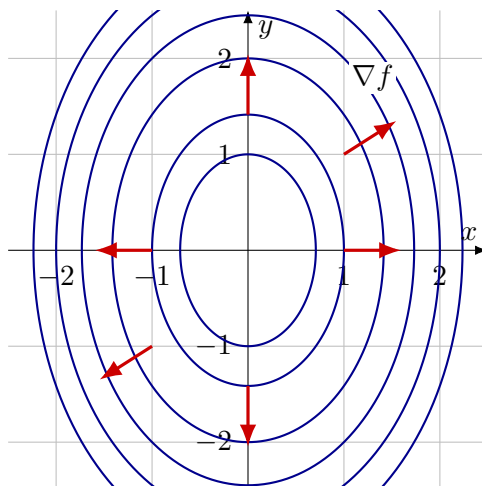


Figure 3: Gradient vectors are perpendicular to level curves and point toward increasing values.

4 Critical points and local extrema

A function $f(x, y)$ has a *local maximum* at (a, b) if $f(x, y) \leq f(a, b)$ for all points sufficiently close to (a, b) . It has a *local minimum* at (a, b) if $f(x, y) \geq f(a, b)$ for all points sufficiently close to (a, b) .

A point (a, b) is called a *critical point* if

$$\nabla f(a, b) = 0,$$

or if one of the first partial derivatives fails to exist.

First derivative test for possible extrema

If f has a local maximum or local minimum at (a, b) and the first partial derivatives exist there, then

$$f_x(a, b) = 0, \quad f_y(a, b) = 0.$$

So local extrema can occur only at critical points, boundary points, or points where differentiability fails.

Worked example 4.1: A quadratic local minimum

Let

$$f(x, y) = x^2 + y^2 - 4x - 2y + 18.$$

The gradient is

$$\nabla f(x, y) = \langle 2x - 4, 2y - 2 \rangle.$$

Setting $\nabla f = 0$ gives $x = 2$ and $y = 1$. Completing the square,

$$f(x, y) = (x - 2)^2 + (y - 1)^2 + 13.$$

So f has a local and absolute minimum at $(2, 1)$, with minimum value 13.

Worked example 4.2: A saddle point

Let $f(x, y) = y^2 - x^2$. Then

$$\nabla f(x, y) = \langle -2x, 2y \rangle,$$

so the only critical point is $(0, 0)$. Along the y -axis, $f(0, y) = y^2 \geq 0$, while along the x -axis, $f(x, 0) = -x^2 \leq 0$. Thus $(0, 0)$ is neither a local maximum nor a local minimum. It is a *saddle point*.

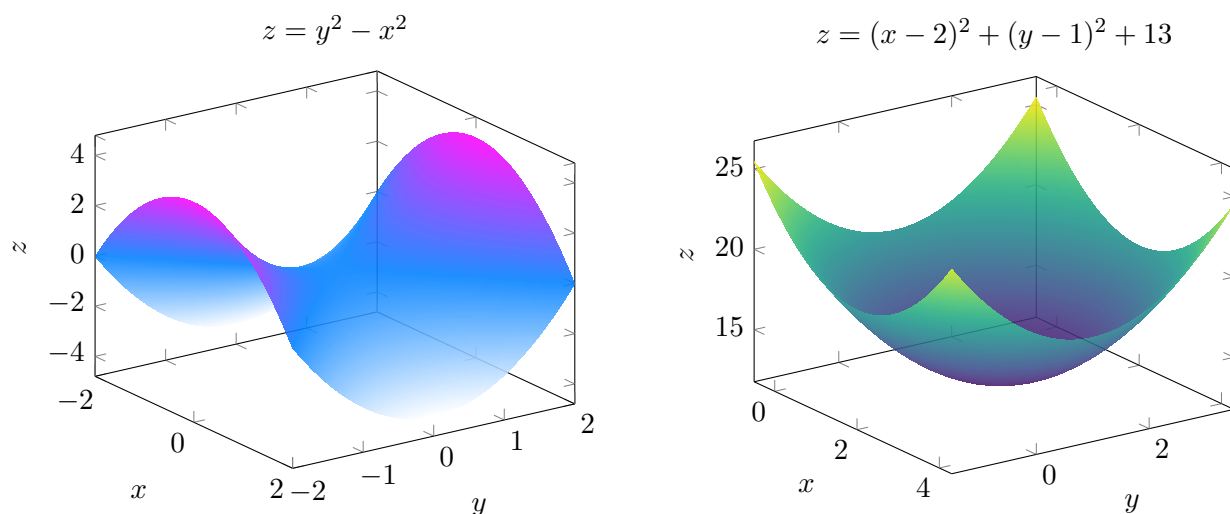


Figure 4: A saddle point has mixed curvature, while a local minimum curves upward in all directions.

5 The Hessian matrix and second derivative test

The second derivative information of a function $f(x, y)$ is organized into the *Hessian matrix*

$$H_f(x, y) = \begin{bmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{yx}(x, y) & f_{yy}(x, y) \end{bmatrix}.$$

When $f_{xy} = f_{yx}$, this matrix is symmetric. Near a critical point (a, b) , the quadratic part of the Taylor approximation is

$$\frac{1}{2} \begin{bmatrix} x - a & y - b \end{bmatrix} H_f(a, b) \begin{bmatrix} x - a \\ y - b \end{bmatrix}.$$

So the Hessian describes local curvature.

For two-variable functions, define

$$D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2.$$

This is the determinant of the Hessian matrix.

Second derivative test

Suppose $f_x(a, b) = 0$, $f_y(a, b) = 0$, and the second partial derivatives are continuous near (a, b) .

- If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then f has a local minimum at (a, b) .
- If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$, then f has a local maximum at (a, b) .
- If $D(a, b) < 0$, then f has a saddle point at (a, b) .
- If $D(a, b) = 0$, the test is inconclusive.

The determinant works because the Hessian represents the local quadratic shape. Positive curvature in all directions gives a local minimum, negative curvature in all directions gives a local maximum, and mixed signs give a saddle. This is the two-dimensional version of the eigenvalue test for quadratic forms.

Worked example 5.1: Classifying critical points

Let

$$f(x, y) = x^3 + y^3 - 3xy + 4.$$

The first derivatives are

$$f_x = 3x^2 - 3y, \quad f_y = 3y^2 - 3x.$$

Setting them equal to zero gives $y = x^2$ and $x = y^2$. Hence $x = x^4$, so $x = 0$ or $x = 1$. The critical points are $(0, 0)$ and $(1, 1)$.

The second derivatives are

$$f_{xx} = 6x, \quad f_{yy} = 6y, \quad f_{xy} = -3.$$

Thus $D(x, y) = 36xy - 9$. At $(0, 0)$, $D = -9 < 0$, so $(0, 0)$ is a saddle point. At $(1, 1)$, $D = 27 > 0$ and $f_{xx} = 6 > 0$, so $(1, 1)$ is a local minimum. The minimum value is $f(1, 1) = 3$.

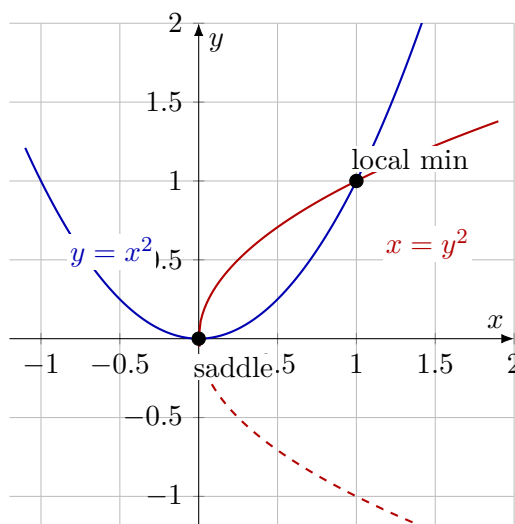


Figure 5: Critical points of $x^3 + y^3 - 3xy + 4$ occur where $y = x^2$ and $x = y^2$ intersect.

Worked example 5.2: Four critical points

Let

$$f(x, y) = x^3 + y^3 + 3x^2 - 3y^2 - 7.$$

The first derivatives are $f_x = 3x(x + 2)$ and $f_y = 3y(y - 2)$. Therefore

$$x = 0 \text{ or } x = -2, \quad y = 0 \text{ or } y = 2.$$

The critical points are $(0, 0)$, $(0, 2)$, $(-2, 0)$, and $(-2, 2)$. Since $f_{xx} = 6x + 6$, $f_{yy} = 6y - 6$, and $f_{xy} = 0$, the determinant is $D = f_{xx}f_{yy}$.

- $(0, 0)$: $D = 6(-6) < 0$, saddle point.
- $(0, 2)$: $D = 6(6) > 0$ and $f_{xx} > 0$, local minimum.
- $(-2, 0)$: $D = (-6)(-6) > 0$ and $f_{xx} < 0$, local maximum.
- $(-2, 2)$: $D = (-6)(6) < 0$, saddle point.

6 Absolute extrema on closed bounded regions

Local extrema describe what happens near a point. Absolute extrema describe the largest and smallest values on an entire region.

Extreme value principle

If f is continuous on a closed and bounded region $D \subset \mathbb{R}^2$, then f attains both an absolute maximum and an absolute minimum on D .

To find absolute extrema on a closed bounded region:

1. find critical points inside D ;
2. find extrema on the boundary of D ;
3. compare all candidate values.

Worked example 6.1: Extrema on a rectangle

Find the absolute maximum and minimum of

$$f(x, y) = x^2 - 2xy + 2y$$

on the rectangle $0 \leq x \leq 3$, $0 \leq y \leq 2$.

Interior critical points satisfy

$$f_x = 2x - 2y = 0, \quad f_y = -2x + 2 = 0,$$

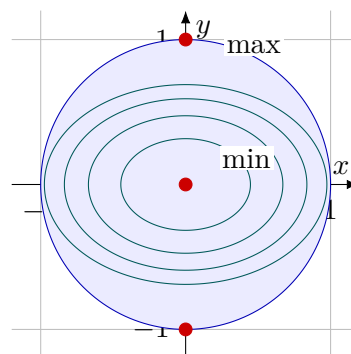
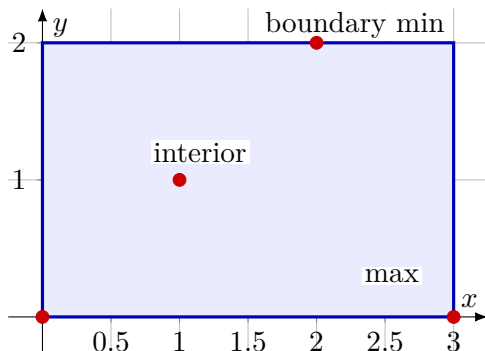
so $x = 1$ and $y = 1$. The interior value is $f(1, 1) = 1$.

On the four boundary sides:

$$x = 0 : f = 2y, \quad x = 3 : f = 9 - 4y,$$

$$y = 0 : f = x^2, \quad y = 2 : f = (x - 2)^2.$$

Comparing all candidates, the absolute maximum value is 9, attained at $(3, 0)$, and the absolute minimum value is 0, attained at $(0, 0)$ and $(2, 2)$.



Rectangle: compare interior and boundary candidates.

Disk: check the center and the boundary circle.

Figure 6: Absolute extrema on closed bounded regions require checking both interior critical points and boundary candidates.

Worked example 6.2: Extrema on a disk

Find the extreme values of

$$f(x, y) = x^2 + 2y^2 + 2$$

on the disk $x^2 + y^2 \leq 1$. The gradient is $\nabla f = \langle 2x, 4y \rangle$, so the only interior critical point is $(0, 0)$, where $f(0, 0) = 2$.

On the boundary $x^2 + y^2 = 1$,

$$f(x, y) = x^2 + 2y^2 + 2 = (1 - y^2) + 2y^2 + 2 = 3 + y^2.$$

Since $0 \leq y^2 \leq 1$, the boundary values satisfy $3 \leq f \leq 4$. Therefore the absolute minimum is 2 at $(0, 0)$, and the absolute maximum is 4 at $(0, 1)$ and $(0, -1)$.

7 Gradient descent

Many optimization problems in statistics and machine learning cannot be solved exactly by hand. Instead, we use iterative algorithms. The most basic method is *gradient descent*. To minimize a differentiable function $L(\theta)$, start with an initial guess θ_0 and update by

$$\theta_{k+1} = \theta_k - \eta \nabla L(\theta_k),$$

where $\eta > 0$ is the learning rate.

The idea is simple: ∇L points in the direction of steepest increase, so $-\nabla L$ points in the direction of steepest decrease. The learning rate controls the step size.

Worked example 7.1: Gradient descent on a quadratic loss

Consider

$$L(x, y) = 3(x - 1)^2 + (y + 2)^2 + xy.$$

The gradient is

$$\nabla L(x, y) = \langle 6(x - 1) + y, 2(y + 2) + x \rangle.$$

The critical point solves

$$6(x - 1) + y = 0, \quad 2(y + 2) + x = 0.$$

Solving gives

$$(x, y) = \left(\frac{16}{11}, -\frac{30}{11} \right).$$

Gradient descent approaches this point when the learning rate is chosen appropriately.

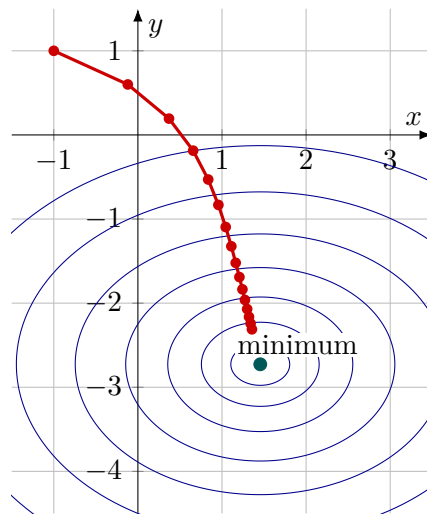


Figure 7: Gradient descent follows the negative gradient direction toward a local minimum.

8 Least squares as an optimization problem

Suppose we have data points (x_i, y_i) and model them with a line

$$\hat{y}_i = \beta_0 + \beta_1 x_i.$$

The least-squares loss is

$$L(\beta_0, \beta_1) = \sum_{i=1}^n [y_i - (\beta_0 + \beta_1 x_i)]^2.$$

This is a function of two variables: the intercept β_0 and the slope β_1 . The best-fit line is the point in parameter space where this loss is minimized.

The gradient is

$$\frac{\partial L}{\partial \beta_0} = -2 \sum_{i=1}^n [y_i - (\beta_0 + \beta_1 x_i)],$$

and

$$\frac{\partial L}{\partial \beta_1} = -2 \sum_{i=1}^n x_i [y_i - (\beta_0 + \beta_1 x_i)].$$

Gradient descent updates the parameters by moving opposite this gradient.

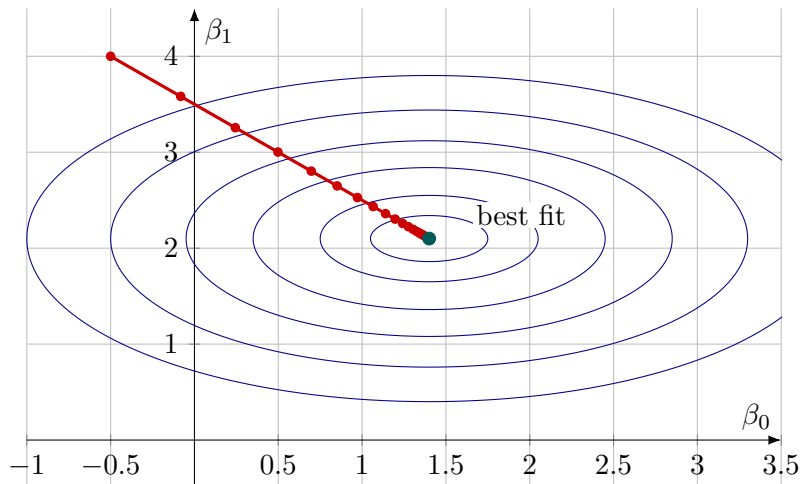


Figure 8: Least-squares fitting minimizes a loss function over parameter space.

9 Newton's method in several variables

Gradient descent uses first derivative information. Newton's method also uses second derivative information. For a function $L(\theta)$ with gradient $\nabla L(\theta)$ and Hessian $H_L(\theta)$, Newton's update is

$$\theta_{k+1} = \theta_k - H_L(\theta_k)^{-1} \nabla L(\theta_k).$$

Equivalently, solve the linear system

$$H_L(\theta_k) s_k = -\nabla L(\theta_k)$$

and set $\theta_{k+1} = \theta_k + s_k$.

Newton's method can be very fast near a local minimum when the Hessian is positive definite. However, it can behave poorly near saddle points or when the Hessian is singular or indefinite. In large-scale machine learning, full Newton steps may be too expensive, so many algorithms use approximations to Hessian information.

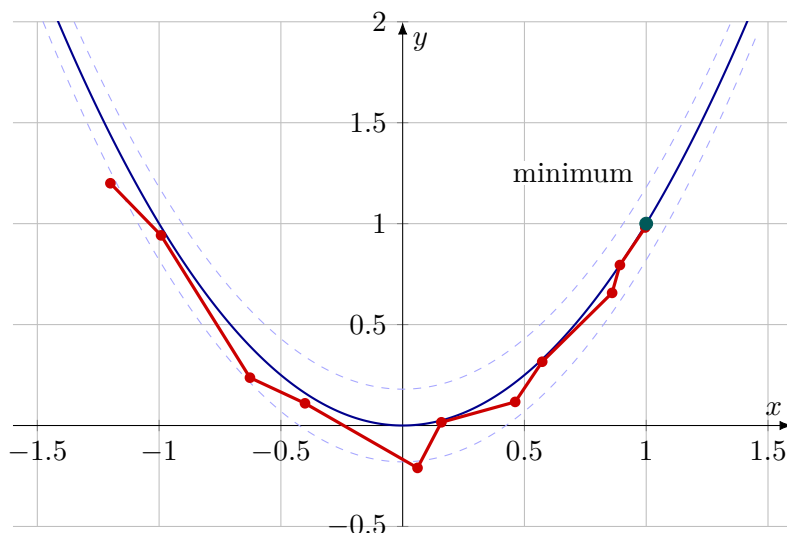


Figure 9: Newton-type steps use curvature information and can move differently from gradient descent.

10 Practical optimization workflow

For a smooth function $L(\theta)$, a standard optimization workflow is:

1. **Model the objective.** Decide what function should be minimized or maximized.
2. **Compute the gradient.** Use calculus, automatic differentiation, or finite differences.
3. **Find critical points.** Solve $\nabla L = 0$ when possible.
4. **Use the Hessian.** Classify local behavior when the Hessian is available.
5. **Check boundaries or constraints.** Local critical points are not the whole story.
6. **Run numerical optimization.** Use gradient descent, Newton's method, or another iterative method.
7. **Interpret the answer.** In applications, the meaning of the optimizer matters as much as the numerical value.

When an explicit gradient is unavailable, we can approximate it numerically:

$$\frac{\partial f}{\partial x_j}(x) \approx \frac{f(x + he_j) - f(x - he_j)}{2h}.$$

This is useful for checking analytic gradients and for debugging optimization code.

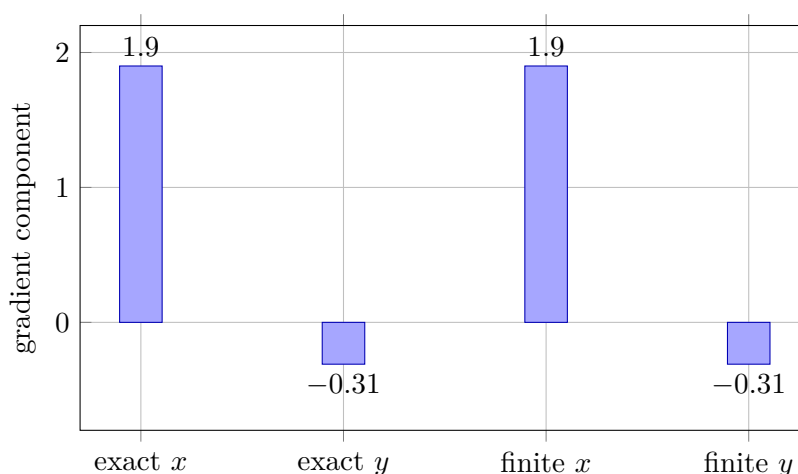


Figure 10: Finite differences approximate the gradient when symbolic derivatives are unavailable.

11 Common mistakes

Frequent errors

1. Forgetting to normalize the direction vector. Directional derivative formulas require a unit vector.
2. Assuming every critical point is an extremum. Saddle points are common in several variables.
3. Using the second derivative test when $D = 0$. The test gives no conclusion in this case.
4. Ignoring the boundary. Absolute extrema on closed regions may occur on the boundary.
5. Choosing a learning rate blindly. In gradient descent, too large a learning rate may cause divergence or oscillation.
6. Confusing local and global behavior. A local minimum may not be the absolute minimum.

12 Chapter summary

The directional derivative measures the rate of change of a function in a chosen direction:

$$D_u f = \nabla f \cdot u.$$

The gradient is the direction of steepest increase, and the negative gradient is the direction of steepest decrease. Critical points occur where the gradient vanishes or where derivatives fail to exist. The Hessian matrix provides second derivative information, and the determinant

$$D = f_{xx}f_{yy} - f_{xy}^2$$

classifies nondegenerate critical points in two variables.

For absolute extrema on closed bounded regions, critical points are not enough: the boundary must also be checked. In computational settings, gradient descent and Newton's method turn these ideas into algorithms for fitting models and minimizing loss functions.

13 Exercises

Conceptual questions

1. Explain in words what $D_u f(a, b)$ measures.
2. Why must u be a unit vector in the definition of directional derivative?
3. Explain why the gradient points perpendicular to level curves.
4. What is the difference between a critical point and a local minimum?
5. Why does $D < 0$ in the second derivative test indicate a saddle point?
6. Why must boundary points be checked when finding absolute extrema?
7. Explain the relationship between gradient descent and steepest descent.
8. Why can Newton's method fail near a saddle point?

Computational practice

1. Let $f(x, y) = x^2 + xy + y^2$. Compute $D_u f(1, 2)$ in the direction of $v = \langle 3, 4 \rangle$.
2. Let $f(x, y) = e^{xy} + y^3$. Find the direction of fastest increase at $(1, 0)$ and the maximum rate of increase.
3. Let $f(x, y, z) = x^2y + yz^2$. Compute the directional derivative at $(1, 2, -1)$ in the direction of $v = \langle 1, 0, 2 \rangle$.
4. Find and classify the critical points of $f(x, y) = x^2 + y^2 - 6x + 4y + 20$.
5. Find and classify the critical points of $f(x, y) = x^2 - y^2 + 4x + 2y$.
6. Find and classify the critical points of $f(x, y) = x^3 + y^3 - 6xy$.
7. Find and classify the critical points of $f(x, y) = x^4 + y^4 - 4xy$.
8. Find the absolute maximum and minimum of $f(x, y) = x + y$ on the square $0 \leq x \leq 1$, $0 \leq y \leq 2$.
9. Find the absolute maximum and minimum of $f(x, y) = x^2 + y^2 - 2x$ on the disk $x^2 + y^2 \leq 4$.
10. Run gradient descent on $L(x, y) = (x - 3)^2 + 4(y + 1)^2$ from $(0, 0)$ with several learning rates. Describe what changes.

Applied and data-oriented problems

1. A model has loss $L(w_1, w_2) = 4(w_1 - 2)^2 + (w_2 + 1)^2 + w_1w_2$. Find the critical point and classify it.

2. For least squares with data (x_i, y_i) , derive the gradient of $L(\beta_0, \beta_1) = \sum_i [y_i - (\beta_0 + \beta_1 x_i)]^2$.
3. Suppose a log-likelihood $\ell(\theta_1, \theta_2)$ has Hessian negative definite at a critical point. What does that tell you about the likelihood locally?
4. A neural-network training loss has a point where $\nabla L = 0$ but the Hessian has both positive and negative eigenvalues. What kind of point is this?
5. Write a function that approximates the gradient of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ using central differences.

14 Python lab

The Chapter 11 Python lab asks you to:

1. compute directional derivatives numerically and analytically;
2. visualize gradient vectors on contour maps;
3. classify critical points using the Hessian determinant;
4. implement gradient descent on quadratic and least-squares losses;
5. compare gradient descent with a damped Newton method.

15 Looking ahead

This chapter focused on unconstrained optimization and boundary checks. The next chapter develops constrained optimization systematically. The key geometric idea will be that, at a constrained optimum, the gradient of the objective is parallel to the gradient of the constraint.

Chapter 12. Constrained Optimization and Lagrange Multipliers

Optimizing under equations, budgets, geometry, and model constraints

Many important optimization problems are not free optimization problems. A model may need to satisfy a budget, a probability vector must have total mass 1, a point may be required to lie on a curve or surface, and a physical system may preserve mass or energy. This chapter develops the method of Lagrange multipliers, a geometric and computational tool for optimizing a function subject to one or more constraints.

In unconstrained optimization, an interior local maximum or minimum of a differentiable function usually satisfies $\nabla f = 0$. Constrained optimization is different. If the point must stay on a curve such as $x^2 + y^2 = 10$, the variables cannot move independently. At an optimal constrained point, a level curve of f is tangent to the constraint curve. Equivalently, the normal vectors are parallel.

Learning objectives

After studying this chapter, you should be able to:

1. distinguish unconstrained and constrained optimization problems;
2. explain the geometric meaning of the condition $\nabla f = \lambda \nabla g$;
3. solve constrained optimization problems with one equality constraint;
4. combine interior critical points with boundary candidates for absolute extrema;
5. solve distance, area, volume, and budget problems using constraints;
6. use Lagrange multipliers with two equality constraints;
7. recognize when the method needs regularity assumptions such as $\nabla g \neq 0$;
8. interpret constrained optimization in statistics, machine learning, and applied mathematics.

1 What makes an optimization problem constrained?

A typical unconstrained optimization problem asks for the maximum or minimum of $f(x, y)$ over a region where (x, y) can move freely. A constrained optimization problem asks for the maximum or minimum of $f(x, y)$ subject to an equation such as

$$g(x, y) = k.$$

The function f is the *objective function*. The equation $g(x, y) = k$ is the *constraint*. The set of all points satisfying the constraint is the *constraint set*.

For example,

$$f(x, y) = 2x + 6y, \quad x^2 + y^2 = 10$$

asks us to optimize a linear function along a circle. We are not allowed to move anywhere in the plane; we may only move along the circle.

The optimization question has two parts

Every constrained optimization problem should be read as:

1. **Objective:** What quantity are we trying to maximize or minimize?
2. **Constraint:** What equations or restrictions must the variables satisfy?

Most mistakes come from confusing the objective with the constraint.

2 The geometry of tangency

Suppose $f(x, y)$ is to be optimized subject to $g(x, y) = k$. The constraint is a level curve of g . The objective also has level curves $f(x, y) = c$.

At a constrained maximum or minimum, the level curve of f usually just touches the constraint curve. If the two curves crossed transversely, then moving along the constraint would move f both upward and downward, so the point would not be an extremum.

Gradients are normal to level curves. Therefore, at a regular constrained extremum,

$$\nabla f \text{ is parallel to } \nabla g.$$

This means that there is a scalar λ such that

$$\nabla f = \lambda \nabla g.$$

The number λ is called a *Lagrange multiplier*.

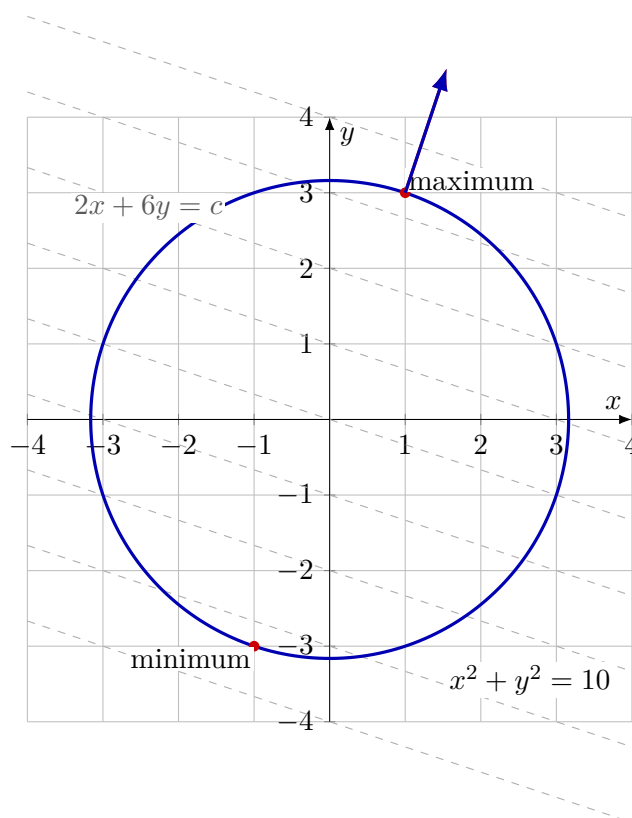


Figure 1: At a constrained extremum, a level curve of the objective is tangent to the constraint curve. The normal directions are parallel.

3 The method of Lagrange multipliers: one constraint

Let f and g be differentiable functions of x and y . To find extrema of $f(x, y)$ subject to $g(x, y) = k$, solve the system

$$\nabla f(x, y) = \lambda \nabla g(x, y), \quad g(x, y) = k.$$

Equivalently,

$$\begin{cases} f_x = \lambda g_x, \\ f_y = \lambda g_y, \\ g(x, y) = k. \end{cases}$$

For a function of three variables subject to one constraint $g(x, y, z) = k$, solve

$$\begin{cases} f_x = \lambda g_x, \\ f_y = \lambda g_y, \\ f_z = \lambda g_z, \\ g(x, y, z) = k. \end{cases}$$

Regularity assumption

The standard Lagrange multiplier condition assumes that the constraint is regular at the point, meaning $\nabla g \neq 0$. If $\nabla g = 0$ somewhere on the constraint set, the method may miss an extremum or produce incomplete information. Such points must be checked separately.

Algorithm: equality-constrained extrema

To optimize f subject to $g = k$:

1. Compute ∇f and ∇g .
2. Set up $\nabla f = \lambda \nabla g$ together with $g = k$.
3. Solve for the variables and λ .
4. Evaluate f at all candidate points.
5. If the constraint set is closed and bounded and f is continuous, the largest value is the absolute maximum and the smallest value is the absolute minimum.

4 A first worked example: a linear objective on a circle

Worked example 4.1: A linear objective on a circle

Find the maximum and minimum values of

$$f(x, y) = 2x + 6y$$

subject to $x^2 + y^2 = 10$.

Let $g(x, y) = x^2 + y^2$. Then

$$\nabla f = \langle 2, 6 \rangle, \quad \nabla g = \langle 2x, 2y \rangle.$$

The Lagrange equations are

$$\langle 2, 6 \rangle = \lambda \langle 2x, 2y \rangle, \quad x^2 + y^2 = 10.$$

Thus

$$2 = 2\lambda x, \quad 6 = 2\lambda y.$$

Assuming $\lambda \neq 0$,

$$x = \frac{1}{\lambda}, \quad y = \frac{3}{\lambda}.$$

Substitution into the constraint gives

$$\frac{1}{\lambda^2} + \frac{9}{\lambda^2} = 10,$$

so $\lambda^2 = 1$. Therefore $\lambda = 1$ gives $(x, y) = (1, 3)$, while $\lambda = -1$ gives $(x, y) = (-1, -3)$. Since

$$f(1, 3) = 20, \quad f(-1, -3) = -20,$$

the maximum value is 20 and the minimum value is -20 .

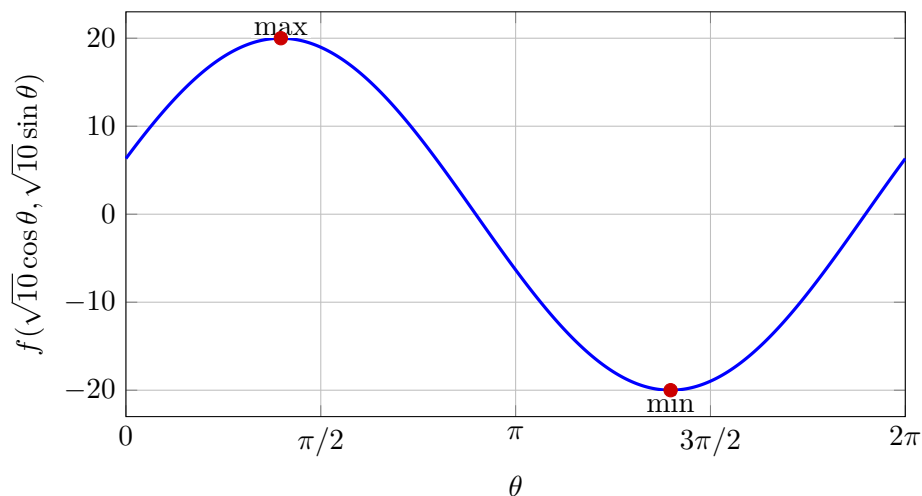


Figure 2: The objective restricted to the circle becomes a one-variable function of the angle θ .

5 Why $\nabla f = \lambda \nabla g$ means no feasible first-order change

Suppose $r(t)$ is a path that stays on the constraint set, so $g(r(t)) = k$. Differentiating gives

$$\nabla g(r(t)) \cdot r'(t) = 0.$$

Thus the velocity vector $r'(t)$ is tangent to the constraint set. At a constrained extremum, $f(r(t))$ should have derivative zero for every feasible tangent direction:

$$\frac{d}{dt}f(r(t)) = \nabla f(r(t)) \cdot r'(t) = 0.$$

So ∇f must be perpendicular to every tangent direction. But ∇g is also perpendicular to every tangent direction. Therefore, in the one-constraint case, ∇f and ∇g must be parallel.

6 Distance problems

Distance problems are among the cleanest applications of Lagrange multipliers.

Worked example 6.1: Closest point on a parabola

Find the point on $y = x^2$ closest to $(0, 1)$.

Minimize the squared distance

$$f(x, y) = x^2 + (y - 1)^2$$

subject to

$$g(x, y) = y - x^2 = 0.$$

The gradients are

$$\nabla f = \langle 2x, 2(y - 1) \rangle, \quad \nabla g = \langle -2x, 1 \rangle.$$

The Lagrange equations are

$$\begin{cases} 2x = -2\lambda x, \\ 2(y - 1) = \lambda, \\ y = x^2. \end{cases}$$

The first equation gives $x(1 + \lambda) = 0$. If $x = 0$, then $y = 0$. If $\lambda = -1$, then $2(y - 1) = -1$, so $y = 1/2$ and hence $x = \pm 1/\sqrt{2}$.

The candidate points are

$$(0, 0), \quad \left(\frac{1}{\sqrt{2}}, \frac{1}{2}\right), \quad \left(-\frac{1}{\sqrt{2}}, \frac{1}{2}\right).$$

Since

$$f(0, 0) = 1, \quad f\left(\pm \frac{1}{\sqrt{2}}, \frac{1}{2}\right) = \frac{3}{4},$$

the closest points are

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{2}\right) \quad \text{and} \quad \left(-\frac{1}{\sqrt{2}}, \frac{1}{2}\right).$$

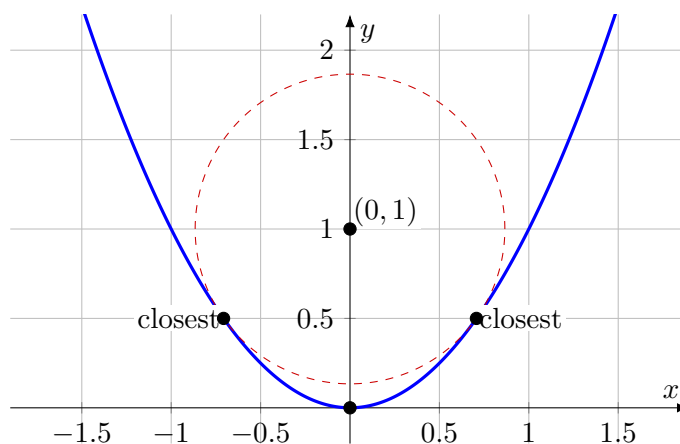


Figure 3: The closest points occur where a distance circle centered at $(0, 1)$ is tangent to the parabola.

7 Absolute extrema on closed bounded regions

For a closed bounded region D , absolute extrema of a continuous function f can occur:

1. at interior critical points, where $\nabla f = 0$;
2. on smooth boundary pieces, where Lagrange multipliers apply;
3. at corners or nonsmooth boundary points.

Interior-boundary-corner checklist

To find absolute maxima and minima on a closed bounded region:

1. Find all interior critical points.
2. Analyze each boundary piece.
3. Use Lagrange multipliers on curved or implicit boundary pieces.
4. Check corners and endpoints.
5. Evaluate f at every candidate.

Worked example 7.1: A quadratic function on a disk

Find the maximum and minimum of

$$f(x, y) = x^2 + 2y^2 + 2$$

on the disk $D = \{(x, y) : x^2 + y^2 \leq 1\}$.

Interior critical points satisfy

$$\nabla f = \langle 2x, 4y \rangle = 0,$$

so the only interior critical point is $(0, 0)$, where $f(0, 0) = 2$.

On the boundary $x^2 + y^2 = 1$, let $g(x, y) = x^2 + y^2$. The Lagrange equations are

$$\langle 2x, 4y \rangle = \lambda \langle 2x, 2y \rangle, \quad x^2 + y^2 = 1.$$

The boundary candidates are $(\pm 1, 0)$ and $(0, \pm 1)$. Evaluating,

$$f(\pm 1, 0) = 3, \quad f(0, \pm 1) = 4, \quad f(0, 0) = 2.$$

Therefore, the absolute minimum is 2 at $(0, 0)$, and the absolute maximum is 4 at $(0, \pm 1)$.

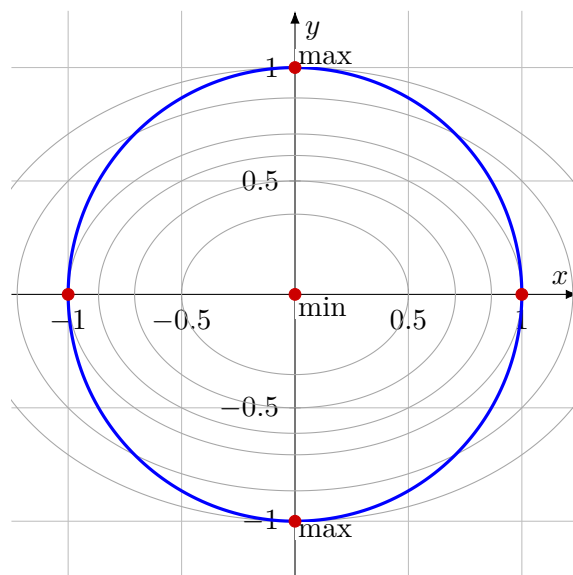


Figure 4: Absolute extrema on a disk require checking both the interior and the boundary.

8 Area and volume under constraints

Many modeling problems involve maximizing or minimizing a geometric quantity while preserving a resource.

Worked example 8.1: Maximum area with fixed perimeter

A rectangle has perimeter 12. Find the dimensions that maximize its area.

Let x and y be side lengths. The area is $A(x, y) = xy$. The perimeter constraint is $2x + 2y = 12$. Compute

$$\nabla A = \langle y, x \rangle, \quad \nabla g = \langle 2, 2 \rangle.$$

The Lagrange equations give

$$\langle y, x \rangle = \lambda \langle 2, 2 \rangle, \quad 2x + 2y = 12.$$

Thus $y = 2\lambda$ and $x = 2\lambda$, so $x = y$. From $2x + 2y = 12$, we get $x = y = 3$. The maximum-area rectangle is a square of side length 3, and the maximum area is 9.

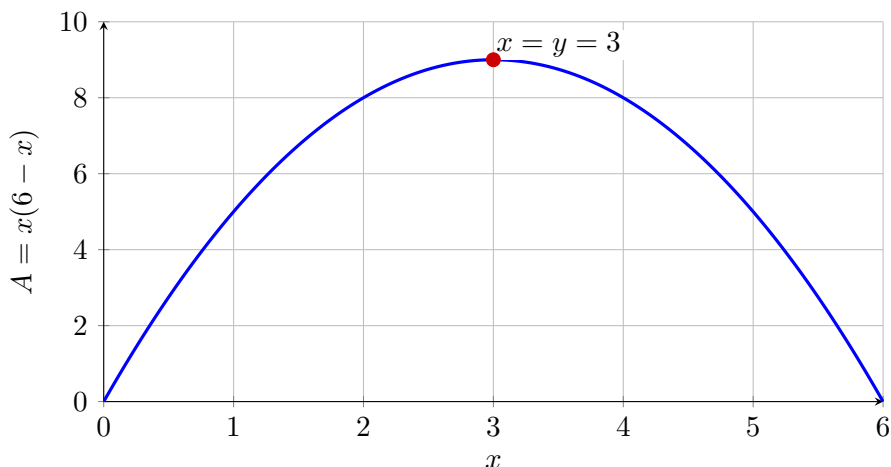


Figure 5: With fixed perimeter 12, the rectangle area xy is maximized when $x = y = 3$.

Worked example 8.2: Minimizing material for a box with fixed volume

A box with no top has volume 4. Find the dimensions that minimize the cardboard area. Let x , y , and z be the length, width, and height. The volume constraint is $xyz = 4$, and the cardboard area is

$$A(x, y, z) = xy + 2xz + 2yz.$$

Solving the constraint for z gives $z = 4/(xy)$. Hence

$$A(x, y) = xy + \frac{8}{y} + \frac{8}{x}.$$

Set partial derivatives equal to zero:

$$A_x = y - \frac{8}{x^2} = 0, \quad A_y = x - \frac{8}{y^2} = 0.$$

By symmetry, $x = y$. Then $x = 8/x^2$, so $x^3 = 8$ and $x = 2$. Thus $y = 2$, and the volume constraint gives $z = 1$. The minimal cardboard dimensions are $2 \times 2 \times 1$.

9 Lagrange multipliers in three variables

For one constraint in three variables, the method is exactly the same:

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z), \quad g(x, y, z) = k.$$

Worked example 9.1: Closest point on a plane

Find the point on the plane $2x + y - z = 1$ closest to $(0, 1, 3)$.

Minimize squared distance

$$f(x, y, z) = x^2 + (y - 1)^2 + (z - 3)^2$$

subject to $g(x, y, z) = 2x + y - z = 1$. Then

$$\nabla f = \langle 2x, 2(y-1), 2(z-3) \rangle, \quad \nabla g = \langle 2, 1, -1 \rangle.$$

The Lagrange equations give

$$2x = 2\lambda, \quad 2(y-1) = \lambda, \quad 2(z-3) = -\lambda.$$

Thus

$$x = \lambda, \quad y = 1 + \frac{\lambda}{2}, \quad z = 3 - \frac{\lambda}{2}.$$

Substitution into $2x + y - z = 1$ gives $3\lambda - 2 = 1$, so $\lambda = 1$. Therefore,

$$(x, y, z) = \left(1, \frac{3}{2}, \frac{5}{2}\right).$$

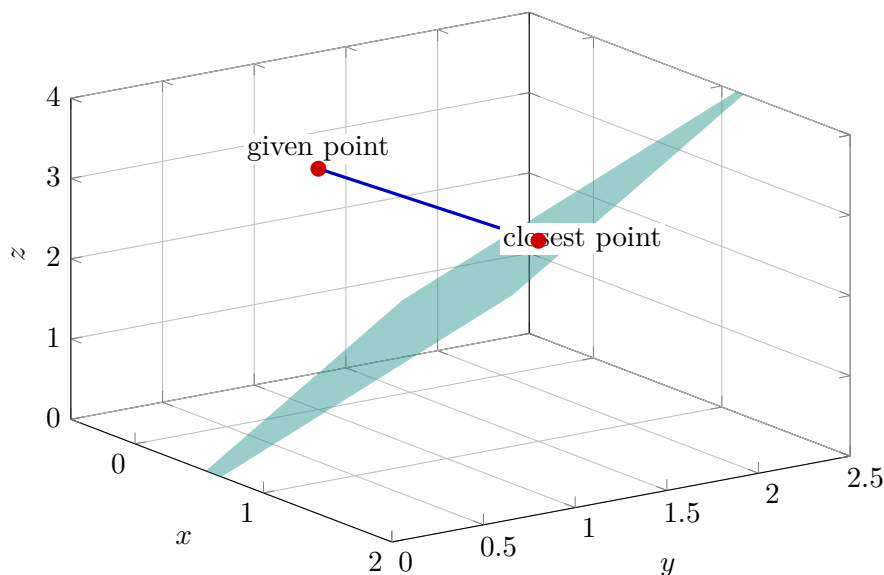


Figure 6: The closest point on a plane is obtained by moving from the given point in the normal direction.

10 Hot and cold spots in a solid ball

Constrained optimization is often combined with interior optimization. Suppose the temperature in a solid ball of radius 2 is

$$T(x, y, z) = x^2 + y^2 + 2z^2 - 2x + 2y.$$

The solid ball is $x^2 + y^2 + z^2 \leq 4$. To find the hottest and coldest spots, check:

1. interior critical points, where $\nabla T = 0$;
2. boundary points on the sphere $x^2 + y^2 + z^2 = 4$.

Interior critical points satisfy

$$\nabla T = \langle 2x - 2, 2y + 2, 4z \rangle = 0,$$

so the interior critical point is $(1, -1, 0)$, and it lies inside the ball because $1^2 + (-1)^2 + 0^2 = 2 < 4$.

On the boundary, solve

$$\begin{cases} 2x - 2 = 2\lambda x, \\ 2y + 2 = 2\lambda y, \\ 4z = 2\lambda z, \\ x^2 + y^2 + z^2 = 4. \end{cases}$$

This illustrates a common feature of three-variable Lagrange multiplier problems: algebra may split into cases. The method gives candidate points; then we evaluate the temperature at all candidates and compare.

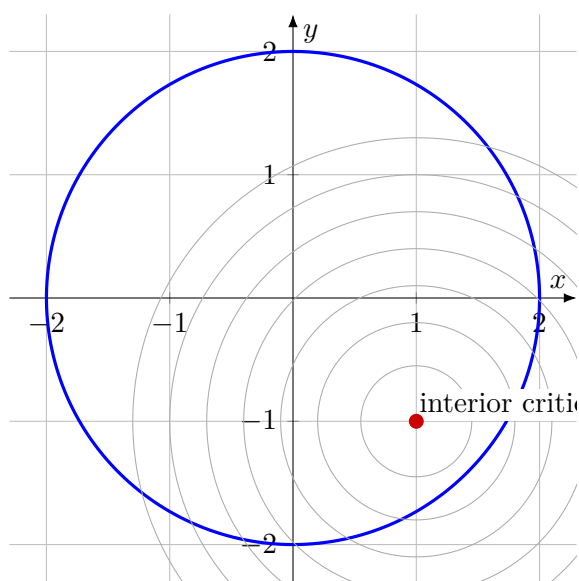


Figure 7: A slice $z = 0$ of the temperature field. The disk boundary and interior critical point must both be checked.

11 Multiple constraints

Suppose $f(x, y, z)$ is to be optimized subject to two constraints:

$$g(x, y, z) = k, \quad h(x, y, z) = c.$$

The constraint set is often a curve in space: the intersection of two surfaces. At a regular constrained extremum, ∇f is perpendicular to the tangent direction of the constraint curve. The gradients ∇g and ∇h are also perpendicular to that tangent direction. Therefore,

$$\nabla f = \lambda \nabla g + \mu \nabla h.$$

The system is

$$\begin{cases} f_x = \lambda g_x + \mu h_x, \\ f_y = \lambda g_y + \mu h_y, \\ f_z = \lambda g_z + \mu h_z, \\ g(x, y, z) = k, \\ h(x, y, z) = c. \end{cases}$$

Geometric meaning of two constraints

With one constraint in $\mathbb{R}^3$, the feasible set is usually a surface. With two constraints in $\mathbb{R}^3$, the feasible set is usually a curve. At an extremum on that curve, ∇f must be orthogonal to the curve's tangent direction, so it must be a linear combination of the two constraint normals.

Worked example 11.1: Two constraints

Find the extreme values of

$$f(x, y, z) = xy + yz$$

subject to

$$xy = 1, \quad y^2 + z^2 = 9.$$

Let $g(x, y, z) = xy$ and $h(x, y, z) = y^2 + z^2$. Then

$$\nabla f = \langle y, x + z, y \rangle, \quad \nabla g = \langle y, x, 0 \rangle, \quad \nabla h = \langle 0, 2y, 2z \rangle.$$

The constraints give $x = 1/y$ and $z^2 = 9 - y^2$, so on the constraint set,

$$f = xy + yz = 1 + yz.$$

Since $y^2 + z^2 = 9$, the product yz lies between $-9/2$ and $9/2$. Therefore

$$f_{\max} = 1 + \frac{9}{2} = \frac{11}{2}, \quad f_{\min} = 1 - \frac{9}{2} = -\frac{7}{2}.$$

The corresponding points are obtained from $x = 1/y$ and the sign choices for y and z .

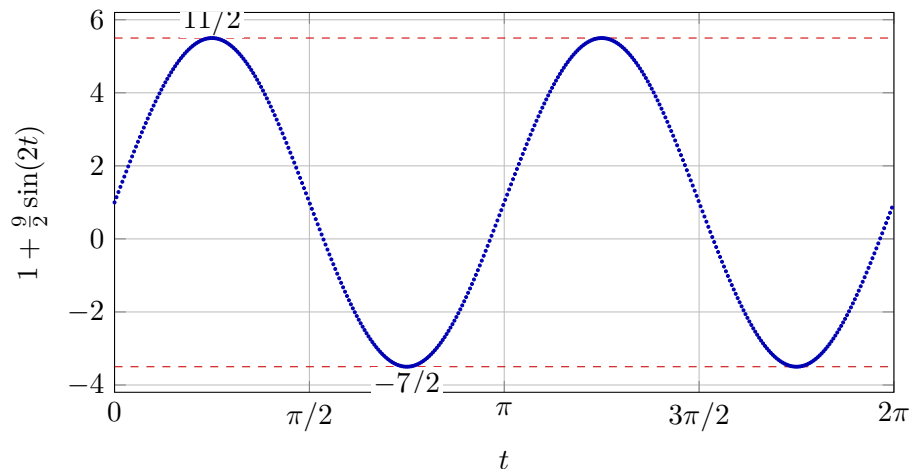


Figure 8: With two constraints, the feasible set can be parameterized; the objective varies along the feasible curve.

12 Lagrange multipliers and machine learning constraints

Many machine-learning and statistics problems are constrained optimization problems.

Probability simplex

A probability vector $p = (p_1, \dots, p_n)$ must satisfy

$$p_i \geq 0, \quad \sum_{i=1}^n p_i = 1.$$

The equality constraint is $g(p) = \sum_i p_i = 1$. When maximizing entropy

$$H(p) = - \sum_{i=1}^n p_i \log p_i$$

subject to $\sum_i p_i = 1$, the Lagrange multiplier equation leads to the uniform distribution when there are no additional constraints.

Normalized parameters

Some models impose a normalization constraint such as $\|w\|^2 = 1$. Optimizing a quadratic form

$$f(w) = w^T A w$$

subject to $w^T w = 1$ leads to

$$A w = \lambda w,$$

which is an eigenvalue problem. This is one reason eigenvectors appear naturally in principal component analysis, spectral methods, and constrained variance maximization.

Regularization versus hard constraints

A constrained problem might look like

$$\min_w L(w) \quad \text{subject to} \quad \|w\|^2 \leq c.$$

A related unconstrained problem is

$$\min_w L(w) + \alpha \|w\|^2.$$

The constrained version uses a hard budget on model complexity. The regularized version uses a penalty. Lagrange multipliers explain why these two viewpoints are closely related.

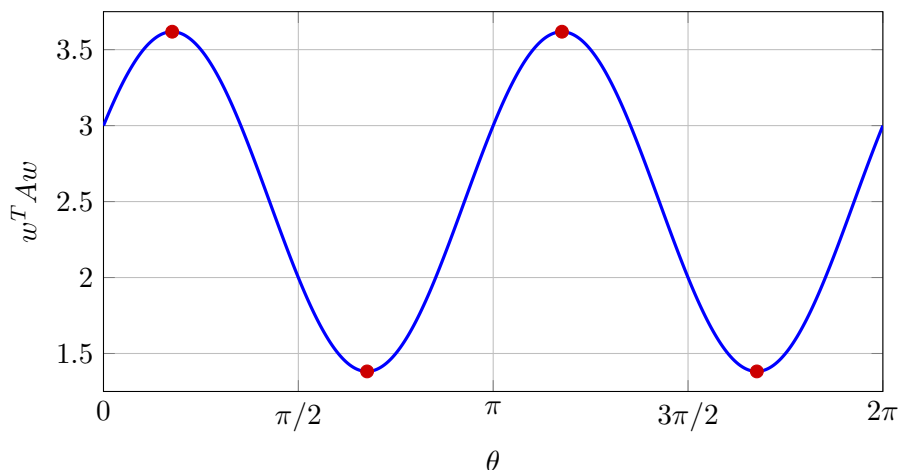


Figure 9: Optimizing a quadratic form on the unit circle leads to eigenvector directions.

13 Numerical search on a constraint

Even when exact algebra is difficult, the geometry remains useful. A curve constraint can often be parameterized, sampled, or solved numerically.

For example, consider

$$f(x, y) = x^2 - xy + 2y^2$$

on the ellipse

$$\frac{x^2}{4} + y^2 = 1.$$

A parameterization is

$$x = 2 \cos t, \quad y = \sin t.$$

Then the constrained problem becomes the one-variable problem

$$F(t) = f(2 \cos t, \sin t).$$

This parameterization method does not replace Lagrange multipliers, but it is an excellent way to check answers and build intuition.

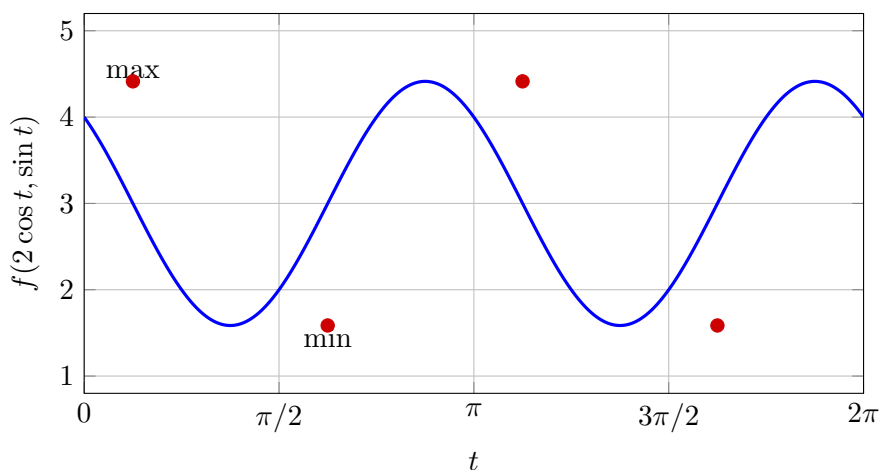


Figure 10: A constrained problem can be checked numerically by parameterizing the constraint.

14 When Lagrange multipliers may fail or need care

The equation $\nabla f = \lambda \nabla g$ is powerful, but it is not magic. It requires a smooth constraint and a nonzero constraint gradient.

Be careful in the following situations:

1. **The constraint gradient vanishes.** If $\nabla g = 0$ at a feasible point, check that point separately.
2. **The constraint has corners or cusps.** A nonsmooth boundary may not have a well-defined normal vector.
3. **The feasible set is not closed or not bounded.** Absolute extrema may fail to exist.
4. **The constraint is an inequality.** Lagrange multipliers handle equality constraints directly. Inequality constraints require checking interiors, active boundaries, and sometimes more advanced conditions.
5. **Solving the equations gives candidates, not conclusions.** Always evaluate the objective and compare values when looking for absolute extrema.

Candidate points are not automatically extrema

Solving the Lagrange equations gives possible constrained extrema. It does not automatically classify them. For absolute extrema, evaluate f at all candidates and compare values. For local classification, additional second-order constrained tests or local analysis may be needed.

15 Summary

The central idea of constrained optimization is that variables are not free to move in all directions. At an optimal feasible point, all allowable first-order changes fail to increase or decrease the objective in the relevant direction.

For one equality constraint $g(x) = k$, the condition is

$$\nabla f = \lambda \nabla g.$$

For two equality constraints in three variables,

$$g(x, y, z) = k, \quad h(x, y, z) = c,$$

the condition is

$$\nabla f = \lambda \nabla g + \mu \nabla h.$$

The method is used across mathematics, statistics, economics, physics, engineering, machine learning, and artificial intelligence. It explains why optimum solutions often occur where level sets are tangent, why normalization constraints lead to eigenvalue problems, and why penalties and constraints are two sides of the same modeling idea.

16 Exercises

Conceptual exercises

1. Explain why $\nabla f = 0$ is not usually true at a constrained maximum or minimum.
2. Draw a picture showing why the level curve of f should be tangent to the constraint curve $g = k$ at a constrained extremum.
3. What does the condition $\nabla g \neq 0$ mean geometrically?
4. In a closed disk, why must one check both interior critical points and the boundary?
5. Explain the difference between a hard constraint and a penalty term in an optimization problem.

Computational exercises

6. Use Lagrange multipliers to find the maximum and minimum of $f(x, y) = x + 2y$ subject to $x^2 + y^2 = 5$.
7. Use Lagrange multipliers to find the maximum and minimum of $f(x, y) = xy$ subject to $x^2 + y^2 = 8$.
8. Find the point on the line $3x + 4y = 12$ closest to the origin.
9. Find the point on the plane $x + y + z = 1$ closest to $(2, 0, 0)$.
10. Find the maximum area of a rectangle with perimeter 20.
11. Find the dimensions of a closed rectangular box of volume 64 with minimum surface area.
12. Find the maximum and minimum of $f(x, y) = x^2 + 4y^2$ on the circle $x^2 + y^2 = 1$.
13. Find the maximum and minimum of $f(x, y) = x^2 + y^2$ subject to $x + 2y = 4$.
14. Find the constrained critical points of $f(x, y) = x^3 + y^2$ subject to $3x + 2y = 0$.
15. Find the closest point on the parabola $y = x^2$ to the point $(0, 2)$.

Applied and computational exercises

16. A company has a budget constraint $4L + 9K = 360$, where L is labor and K is capital. Suppose production is $P(L, K) = L^{1/2}K^{1/2}$. Use Lagrange multipliers to maximize production.
17. A probability vector (p_1, p_2, p_3) satisfies $p_1 + p_2 + p_3 = 1$ and $p_i > 0$. Use Lagrange multipliers to maximize $H(p) = -\sum p_i \log p_i$.
18. Maximize $w^T A w$ subject to $w^T w = 1$ for $A = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$. Interpret the answer geometrically.
19. A cylindrical can has fixed volume V . Use constrained optimization to find the radius and height that minimize surface area.
20. A point must lie on the sphere $x^2 + y^2 + z^2 = 9$. Find the maximum and minimum of $f(x, y, z) = x + 2y + 2z$.
21. Numerically parameterize the circle $x^2 + y^2 = 10$ and confirm the extrema of $f(x, y) = 2x + 6y$.
22. Parameterize the ellipse $x^2/9 + y^2/4 = 1$ and numerically optimize $f(x, y) = xy$.
23. Use a grid search on the unit disk to approximate the maximum and minimum of $f(x, y) = x^2 + 2y^2 + 2$. Compare with the exact solution.

Chapter 13. Double Integrals and Riemann Sums

Area accumulation, volume, numerical approximation, and iterated integration

Chapter goal. A single integral adds small contributions along a line. A double integral adds small contributions over a two-dimensional region. This chapter develops double integrals from Riemann sums, explains their geometric meaning, and shows how Fubini's theorem turns a two-dimensional accumulation problem into two ordinary one-variable integrations.

The ideas in this chapter are important bridges from calculus to applied mathematics, statistics, data science, and machine learning. A double integral may represent volume, mass, charge, expected value, average temperature, total risk, probability over a region, or total loss accumulated over a two-parameter domain.

$$\text{total amount} \approx \sum \text{value at a sample point} \times \text{small area}.$$

When the rectangles become very small, the approximation becomes an integral.

Learning objectives

After studying this chapter, you should be able to:

1. explain a double integral as the limit of double Riemann sums;
2. compute area elements $dA = \Delta x \Delta y$ on rectangular partitions;
3. interpret $\iint_R f(x, y) dA$ as volume when $f \geq 0$;
4. interpret double integrals as signed accumulation when f changes sign;
5. approximate double integrals using sample points and the midpoint rule;
6. evaluate double integrals over rectangles using Fubini's theorem;
7. compute average values over rectangles;
8. connect double integrals to density, mass, numerical integration, and probability.

1 From one-dimensional area to two-dimensional accumulation

Recall the definite integral of a continuous function $f(x)$ on an interval $[a, b]$:

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_i f(x_i^*) \Delta x.$$

Geometrically, if $f(x) \geq 0$, this is the area under the curve $y = f(x)$ above the interval $[a, b]$. Computationally, it means:

1. divide the interval into small subintervals;
2. choose a sample point in each subinterval;
3. multiply height by width;

4. add the pieces;
5. pass to the limit.

A double integral follows the same pattern, but the base region is two-dimensional. Instead of small intervals, we use small rectangles. Instead of length Δx , each small piece has area

$$\Delta A = \Delta x \Delta y.$$

If the height above (x, y) is $f(x, y)$, then a small rectangular column has approximate volume

$$f(x_{ij}^*, y_{ij}^*) \Delta A.$$

Adding all such columns gives a double Riemann sum.

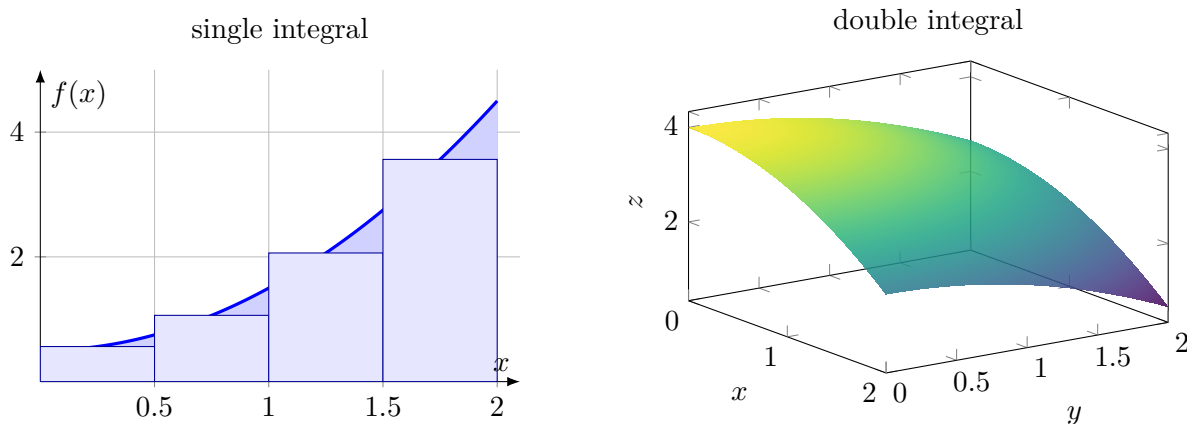


Figure 1: A single integral accumulates over an interval; a double integral accumulates over a region.

The main shift

A single integral adds over a line-like set. A double integral adds over an area-like set. The algebra looks similar, but the geometry changes from intervals to regions and from rectangles to columns.

2 Rectangular regions and partitions

For the first part of the theory, we focus on a rectangular region

$$R = [a, b] \times [c, d] = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}.$$

Divide $[a, b]$ into m equal subintervals and $[c, d]$ into n equal subintervals. Then

$$\Delta x = \frac{b-a}{m}, \quad \Delta y = \frac{d-c}{n}, \quad \Delta A = \Delta x \Delta y.$$

This divides R into mn small rectangles R_{ij} . In each R_{ij} , choose a sample point (x_{ij}^*, y_{ij}^*) . The associated double Riemann sum is

$$\sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A.$$

If $f(x, y) \geq 0$, this sum estimates the volume under $z = f(x, y)$ and above R .

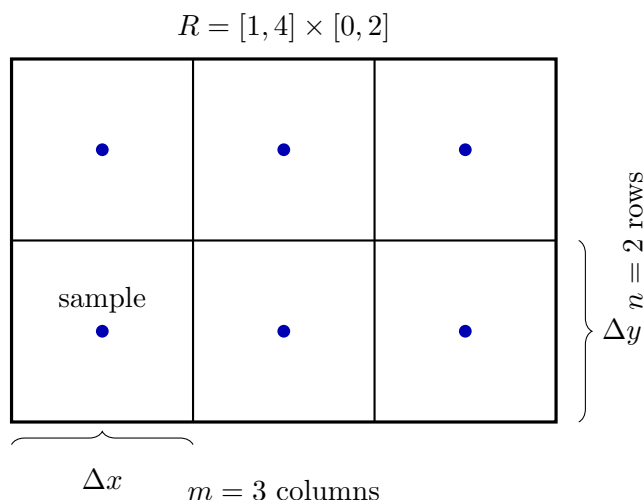


Figure 2: A rectangle divided into subrectangles. Each subrectangle contributes one term to a double Riemann sum.

3 The double integral on a rectangle

Let f be a function defined on a rectangle $R = [a, b] \times [c, d]$. If the double Riemann sums approach a single limiting value as the partition becomes finer, independent of the sample points, then f is integrable on R . The limit is called the *double integral* of f over R .

Definition: double integral over a rectangle

For a sufficiently nice function f on R ,

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A.$$

Here dA represents a small piece of area. In rectangular coordinates, it is approximated by $\Delta A = \Delta x \Delta y$.

The notation dA reminds us that the integral is adding over area. Later, when we change coordinates, the expression for dA will change. In rectangular coordinates,

$$dA = dx dy \quad \text{or} \quad dA = dy dx.$$

Volume interpretation

If $f(x, y) \geq 0$ on R , then

$$\iint_R f(x, y) dA$$

is the volume of the solid below the surface $z = f(x, y)$ and above the base region R .

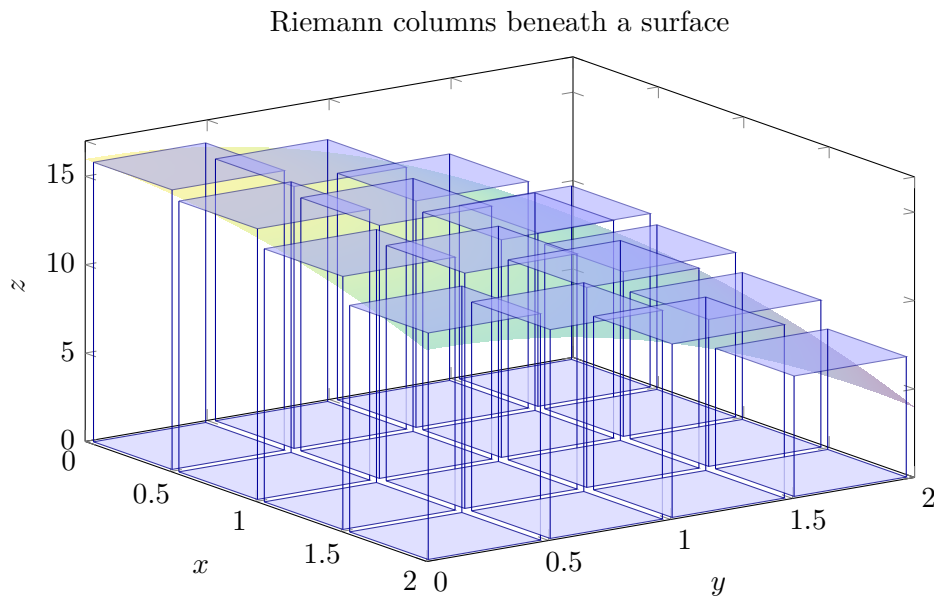


Figure 3: A double Riemann sum approximates volume by rectangular columns beneath $z = 16 - x^2 - 2y^2$.

4 Worked example: upper-right Riemann sum

Example 4.1: Upper-right sum for $f(x, y) = x^2y$

Consider

$$f(x, y) = x^2y, \quad R = [1, 4] \times [0, 2].$$

Use $m = 3$ subdivisions in the x -direction and $n = 2$ subdivisions in the y -direction. Then

$$\Delta x = \frac{4-1}{3} = 1, \quad \Delta y = \frac{2-0}{2} = 1, \quad \Delta A = 1.$$

Using upper-right corners as sample points gives

$$x = 2, 3, 4, \quad y = 1, 2.$$

Therefore

$$\begin{aligned} \sum_{i=1}^3 \sum_{j=1}^2 f(x_i^*, y_j^*) \Delta A &= f(2, 1) + f(3, 1) + f(4, 1) + f(2, 2) + f(3, 2) + f(4, 2) \\ &= 4 + 9 + 16 + 8 + 18 + 32 \\ &= 87. \end{aligned}$$

Since x^2y increases as x and y increase on this rectangle, upper-right corners overestimate the integral.

x	y	$f(x, y) = x^2y$	ΔA	contribution
2	1	4	1	4
3	1	9	1	9
4	1	16	1	16
2	2	8	1	8
3	2	18	1	18
4	2	32	1	32
Total				87

5 The midpoint rule for double integrals

The *midpoint rule* uses the center of each subrectangle as the sample point. For a rectangle divided into m by n equal subrectangles,

$$\iint_R f(x, y) dA \approx \sum_{i=1}^m \sum_{j=1}^n f(\bar{x}_i, \bar{y}_j) \Delta A,$$

where $\bar{x}_i$ and $\bar{y}_j$ are midpoint coordinates.

Example 5.1: Midpoint rule for x^2y

Again let

$$f(x, y) = x^2y, \quad R = [1, 4] \times [0, 2], \quad m = 3, \quad n = 2.$$

The midpoint sample values are

$$x = 1.5, 2.5, 3.5, \quad y = 0.5, 1.5.$$

Thus

$$\iint_R x^2y dA \approx \sum_{x \in \{1.5, 2.5, 3.5\}} \sum_{y \in \{0.5, 1.5\}} x^2y = 44.5.$$

The exact value, found later in this chapter, is 42, so the midpoint estimate is much closer than the upper-right estimate 87.

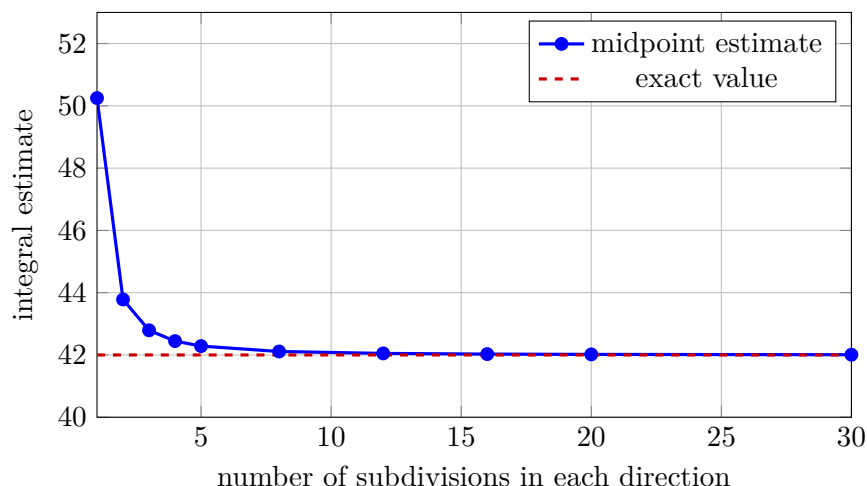


Figure 4: Midpoint Riemann sums converge to the exact double integral as the grid becomes finer.

Why midpoint sampling is often good

The midpoint rule samples each rectangle at its center. For smooth functions, this often balances overestimates and underestimates better than using only corners. This is one reason midpoint-type ideas appear in numerical integration, finite-volume methods, and simulation.

6 Fubini's theorem on rectangles

Riemann sums define the double integral, but limits of double sums are rarely the easiest way to compute exact answers. The key computational result is Fubini's theorem.

Fubini's theorem for rectangles

If f is continuous on $R = [a, b] \times [c, d]$, then

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy.$$

The double integral can be computed by integrating one variable at a time.

The inner integral treats the other variable as constant. For example,

$$\int_1^4 x^2 y dx = y \int_1^4 x^2 dx = y \left[\frac{x^3}{3} \right]_1^4 = 21y.$$

Then

$$\int_0^2 21y dy = \left[\frac{21y^2}{2} \right]_0^2 = 42.$$

Therefore

$$\iint_{[1,4] \times [0,2]} x^2 y dA = 42.$$

The same answer is obtained in the other order:

$$\int_1^4 \int_0^2 x^2 y \, dy \, dx = \int_1^4 2x^2 \, dx = 42.$$

The order changes the intermediate expressions, but not the final value.

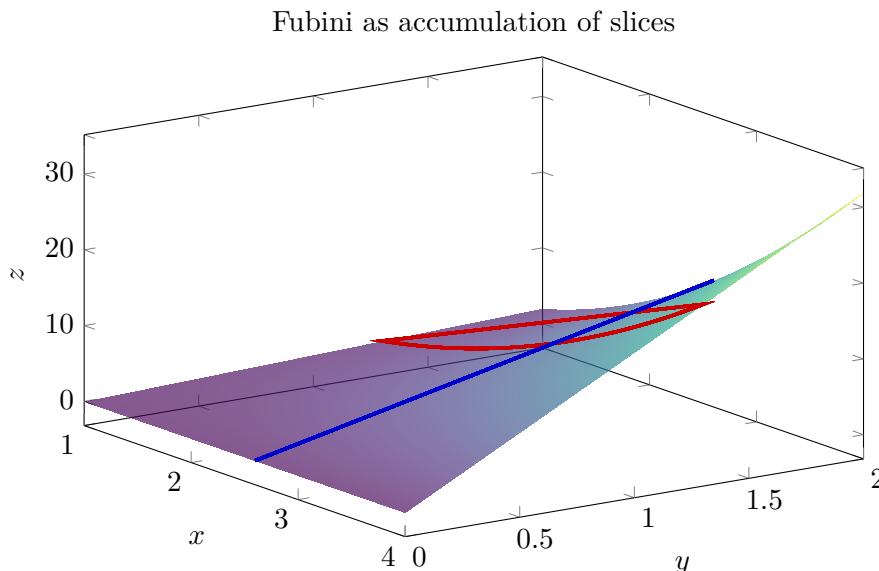


Figure 5: Fubini's theorem computes a double integral by accumulating one-dimensional slices of the surface.

7 Separability and product structure

Sometimes a function separates into a product of a function of x and a function of y :

$$f(x, y) = g(x)h(y).$$

On a rectangle, Fubini's theorem gives

$$\iint_{[a,b] \times [c,d]} g(x)h(y) \, dA = \left(\int_a^b g(x) \, dx \right) \left(\int_c^d h(y) \, dy \right).$$

For example,

$$\iint_{[0,\pi] \times [0,1]} y \sin x \, dA = \left(\int_0^\pi \sin x \, dx \right) \left(\int_0^1 y \, dy \right) = 2 \cdot \frac{1}{2} = 1.$$

This product structure appears frequently in probability. If two random variables are independent, their joint density often factors as $p(x, y) = p_X(x)p_Y(y)$.

8 Signed accumulation

If $f(x, y)$ takes both positive and negative values, then

$$\iint_R f(x, y) \, dA$$

is no longer ordinary volume. It is *signed volume*: positive parts above the xy -plane count positively, and negative parts below the xy -plane count negatively. For example, on a symmetric rectangle,

$$\iint_{[-1,1] \times [0,2]} xy \, dA = 0,$$

because the positive and negative contributions cancel.

signed accumulation for $z = xy$

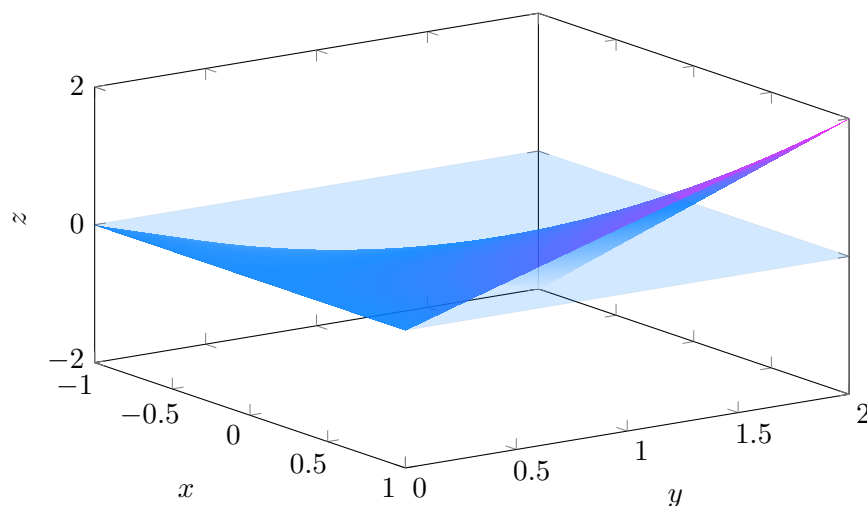


Figure 6: A double integral can measure signed accumulation when the surface has positive and negative parts.

Volume versus signed volume

If $f \geq 0$, the double integral is a volume. If f changes sign, the double integral is signed volume. To compute total physical volume between the surface and the plane, one would integrate $|f(x, y)|$ instead.

9 Average value over a rectangle

The average value of a one-variable function over $[a, b]$ is

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) \, dx.$$

The average value of a two-variable function over a region R is

$$f_{\text{avg}} = \frac{1}{\text{Area}(R)} \iint_R f(x, y) \, dA.$$

For a rectangle $R = [a, b] \times [c, d]$,

$$\text{Area}(R) = (b-a)(d-c).$$

Example 9.1: Average height

Let

$$f(x, y) = 16 - x^2 - 2y^2, \quad R = [0, 2] \times [0, 2].$$

The double integral is

$$\iint_R (16 - x^2 - 2y^2) dA = \int_0^2 \int_0^2 (16 - x^2 - 2y^2) dy dx = \frac{136}{3}.$$

Since $\text{Area}(R) = 4$, the average value is

$$f_{\text{avg}} = \frac{1}{4} \cdot \frac{136}{3} = \frac{34}{3}.$$

In a physical interpretation, this is the average height of the surface over the rectangular base.

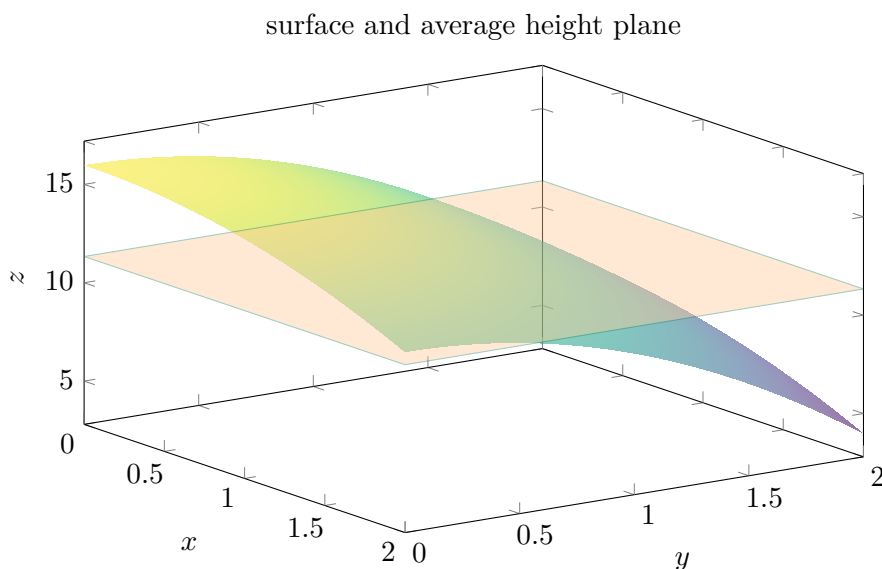


Figure 7: The average value is the constant height that gives the same total volume over the base region.

10 Mass and density on a rectangular plate

Suppose a thin rectangular plate occupies a region R in the xy -plane. If its density at (x, y) is $\rho(x, y)$, then the mass of a small rectangle is approximately

$$\rho(x_{ij}^*, y_{ij}^*) \Delta A.$$

The total mass is therefore

$$M = \iint_R \rho(x, y) dA.$$

Example 10.1: Mass of a rectangular plate

Let

$$\rho(x, y) = 2 + x + y, \quad R = [0, 3] \times [0, 2].$$

Then

$$\begin{aligned} M &= \int_0^3 \int_0^2 (2 + x + y) \, dy \, dx \\ &= \int_0^3 \left[2y + xy + \frac{y^2}{2} \right]_0^2 \, dx \\ &= \int_0^3 (6 + 2x) \, dx = 27. \end{aligned}$$

The double integral is not just a geometric volume. It is a general accumulation tool: density times area gives mass, and the integral adds these contributions.

density $\rho(x, y) = 2 + x + y$ on a plate

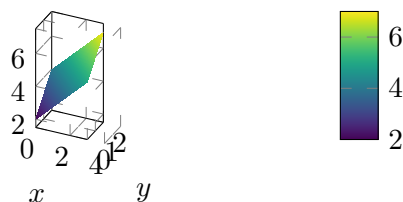


Figure 8: A density field over a rectangular plate. Integrating density over area gives mass.

11 Numerical double integration

In applications, exact symbolic integration may not be available. A function may be known only through data, simulation, or a black-box model. Then Riemann sums become algorithms.

A reusable midpoint rule for rectangles has the form

$$\text{estimate} = \sum_{i=1}^m \sum_{j=1}^n f(\bar{x}_i, \bar{y}_j) \Delta x \Delta y.$$

For high-dimensional applied mathematics and machine learning, this point of view matters. Integrals are often approximated numerically, and sums over grids, samples, or data points are discrete versions of continuous accumulation.

12 Monte Carlo viewpoint

A double integral over a rectangle can also be estimated by random sampling. If R has area A and (X, Y) is sampled uniformly from R , then

$$\iint_R f(x, y) \, dA = A \mathbb{E}[f(X, Y)].$$

This gives the Monte Carlo estimator

$$\iint_R f(x, y) dA \approx A \cdot \frac{1}{N} \sum_{k=1}^N f(X_k, Y_k).$$

Monte Carlo integration is usually less accurate than grid-based integration in low dimensions, but it becomes central in statistics, Bayesian computation, uncertainty quantification, and high-dimensional integration.

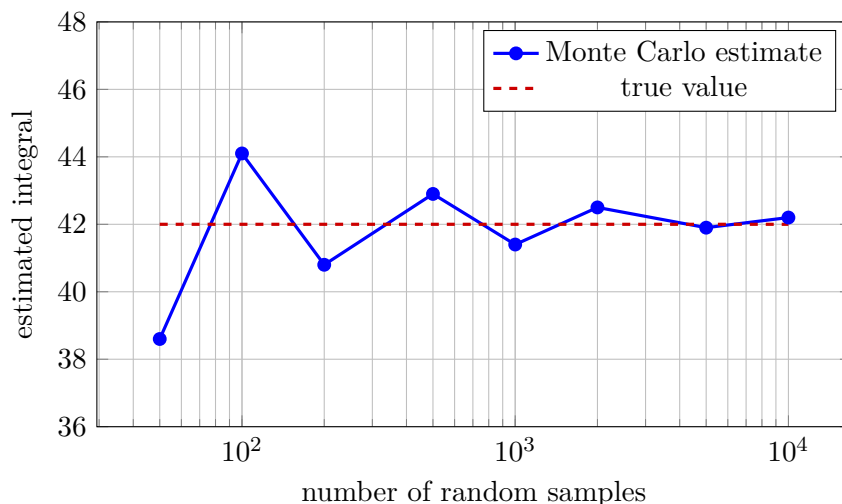


Figure 9: Monte Carlo estimates fluctuate randomly but approach the true integral as the number of samples increases.

13 Double integrals and probability on rectangles

If $p(x, y)$ is a joint probability density on a region R , then probabilities are double integrals:

$$P((X, Y) \in A) = \iint_A p(x, y) dA.$$

For a rectangular event $A = [a, b] \times [c, d]$, we compute

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d p(x, y) dy dx.$$

This is one reason double integrals are essential preparation for statistics. Later chapters will develop non-rectangular regions, change of variables, and multivariate normal densities in more detail.

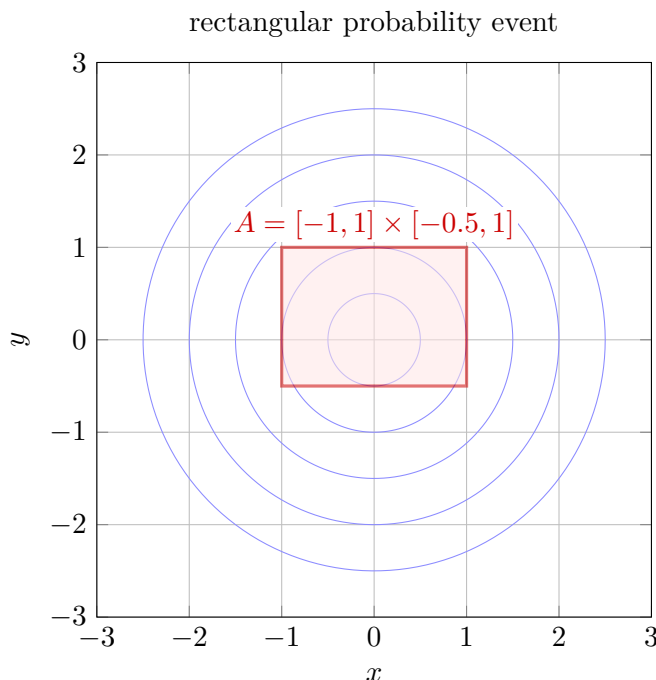


Figure 10: Probability over a rectangle is the double integral of a density over that rectangle.

14 Common mistakes

Mistake 1: forgetting ΔA

A Riemann sum must have the form

$$\sum f(x_{ij}^*, y_{ij}^*) \Delta A.$$

The values of f alone do not approximate an integral. They must be multiplied by area.

Mistake 2: confusing the order of integration

The expression

$$\int_a^b \int_c^d f(x, y) dy dx$$

means: integrate with respect to y first, then integrate the resulting function of x with respect to x .

Mistake 3: assuming every double integral is volume

A double integral is ordinary volume only when $f \geq 0$. Otherwise it is signed accumulation.

Mistake 4: using Fubini without checking the region

In this chapter, the main region is a rectangle. In the next chapter, we will study general regions where bounds may depend on variables.

15 Chapter exercises**Conceptual problems**

1. Explain in words why the term $f(x_{ij}^*, y_{ij}^*)\Delta A$ represents a small volume when $f \geq 0$.
2. What is the difference between a double Riemann sum and a double integral?
3. Why does dA represent area rather than length?
4. Explain why $\iint_R f(x, y) dA$ may be negative if f changes sign.
5. Describe the difference between a midpoint estimate and an upper-right corner estimate.

Computational problems

6. Let $f(x, y) = 3x + 2y$ on $R = [0, 2] \times [1, 3]$. Use $m = 2$, $n = 2$, and upper-right sample points to approximate $\iint_R f dA$.
7. Repeat Problem 6 using midpoint sample points.
8. Compute the exact value of $\iint_R (3x + 2y) dA$.
9. Evaluate $\iint_{[0,1] \times [0,2]} xy^2 dA$.
10. Evaluate $\iint_{[1,3] \times [0,\pi]} x \cos y dA$.
11. Evaluate $\iint_{[-1,1] \times [-1,1]} x^3 y^2 dA$ without doing a long calculation. Explain the cancellation.
12. Find the average value of $f(x, y) = x + y$ on $[0, 4] \times [0, 2]$.
13. Find the mass of a rectangular plate $[0, 2] \times [0, 3]$ with density $\rho(x, y) = 1 + x + 2y$.
14. Find the volume below $z = 9 - x^2 - y^2$ and above $[0, 1] \times [0, 2]$.
15. Compute $\iint_{[0,1] \times [0,1]} e^{x+y} dA$.

Modeling and interpretation problems

16. A temperature field on a rectangular plate is $T(x, y) = 20 + 3x - y$. Write a double integral for the average temperature over $[0, 4] \times [0, 2]$ and evaluate it.
17. A risk surface over two model parameters is $L(\theta_1, \theta_2) = \theta_1^2 + 4\theta_2^2$. Compute the average risk over $[-1, 1] \times [-1, 1]$.
18. Suppose $p(x, y) = cxy$ on $[0, 1] \times [0, 1]$. Find c so that p is a probability density.
19. For the density in Problem 18, compute $P(0 \leq X \leq 1/2, 0 \leq Y \leq 1)$.

20. Explain why numerical integration is important when $f(x, y)$ is only available through simulation or data.

Computational lab problems

21. Write a midpoint-rule function to approximate $\iint_{[0,2] \times [0,2]} (16 - x^2 - 2y^2) dA$.
22. Compare midpoint estimates for several subdivisions in each direction.
23. Use random sampling to estimate $\iint_{[1,4] \times [0,2]} x^2 y dA$.
24. Plot the error of the midpoint rule as the number of subdivisions increases.
25. Plot the surface $z = x^2 y$ above $[1, 4] \times [0, 2]$ and add a coarse grid of columns.

16 Python lab and interactive page

- Notebook lab: `../labs/Chapter13_Double_Integrals_and_Riemann_Sums.ipynb`
- Interactive HTML page: `../htmls/chapter-13.html`

17 Chapter summary

A double integral is the limit of double Riemann sums. On a rectangle, the approximation

$$\sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

becomes the exact quantity

$$\iint_R f(x, y) dA.$$

When $f \geq 0$, the integral is volume. When f changes sign, it is signed accumulation. When f is a density, it computes mass or probability. Fubini's theorem makes double integrals computable by reducing them to iterated one-variable integrals:

$$\iint_{[a,b] \times [c,d]} f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy.$$

This chapter prepares the next step: double integrals over non-rectangular regions, where the bounds themselves encode geometry.

Chapter 14. Double Integrals over General Regions

Type I and Type II regions, reversing order, area, mass, and probability

Rectangles are the training ground for double integrals, but most meaningful regions are not rectangles. This chapter develops double integrals over curved and irregular plane regions. The main skill is translating a geometric region into correct limits of integration, choosing an efficient order, and interpreting the result as area, volume, mass, average value, or probability. A double integral over a general region has the same conceptual meaning as before:

$$\iint_D f(x, y) dA.$$

It adds the values of f over a two-dimensional region D . The new difficulty is that D may be bounded by curves, so the inner limits often depend on the outer variable.

Learning objectives

After studying this chapter, you should be able to:

1. describe Type I regions using inequalities $a \leq x \leq b$, $p(x) \leq y \leq q(x)$;
2. describe Type II regions using inequalities $c \leq y \leq d$, $g(y) \leq x \leq h(y)$;
3. set up and evaluate double integrals over nonrectangular regions;
4. sketch a region from integral limits and write integral limits from a sketch;
5. reverse the order of integration when it simplifies the computation;
6. compute area using $\iint_D 1 dA$;
7. split a complicated region into simpler pieces;
8. interpret double integrals as volume, mass, average value, and probability.

1 Why rectangles are not enough

In the previous chapter, the base region was usually a rectangle $R = [a, b] \times [c, d]$. For such a region, Fubini's theorem gives

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy.$$

But many natural regions are bounded by curves: disks, triangular feasible sets, regions under density curves, parameter constraints, geographic regions, and decision regions in classification.

For example, the region below the parabola $y = 4 - x^2$ and above the line $y = 3x$ is not a rectangle. Its vertical height changes as x changes.

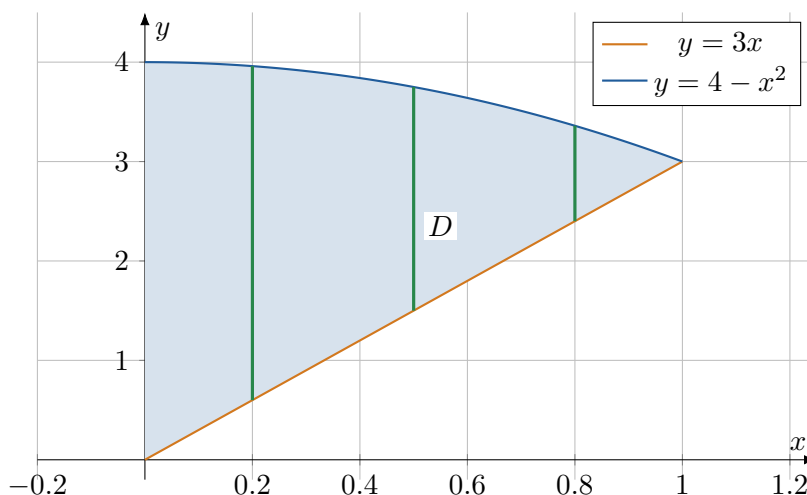


Figure 1: A nonrectangular region bounded by a parabola and a line. The vertical height of the region changes with x .

The key skill is to decide how to slice the region. A *vertical slice* leads to limits $p(x) \leq y \leq q(x)$. A *horizontal slice* leads to limits $g(y) \leq x \leq h(y)$. The same region may be easy in one direction and difficult in the other.

2 Type I regions: vertical slicing

A plane region D is called a *Type I region* if it can be described as

$$D = \{(x, y) : a \leq x \leq b, p(x) \leq y \leq q(x)\},$$

where p and q are continuous functions and $p(x) \leq q(x)$ for $a \leq x \leq b$.

Geometrically, we move from left to right. For each fixed x , the variable y begins at the lower curve $y = p(x)$ and ends at the upper curve $y = q(x)$.

Double integral over a Type I region

If $D = \{(x, y) : a \leq x \leq b, p(x) \leq y \leq q(x)\}$ and f is continuous on D , then

$$\iint_D f(x, y) dA = \int_a^b \int_{p(x)}^{q(x)} f(x, y) dy dx.$$

The inner integral adds along a vertical slice. The outer integral adds all slices from $x = a$ to $x = b$.

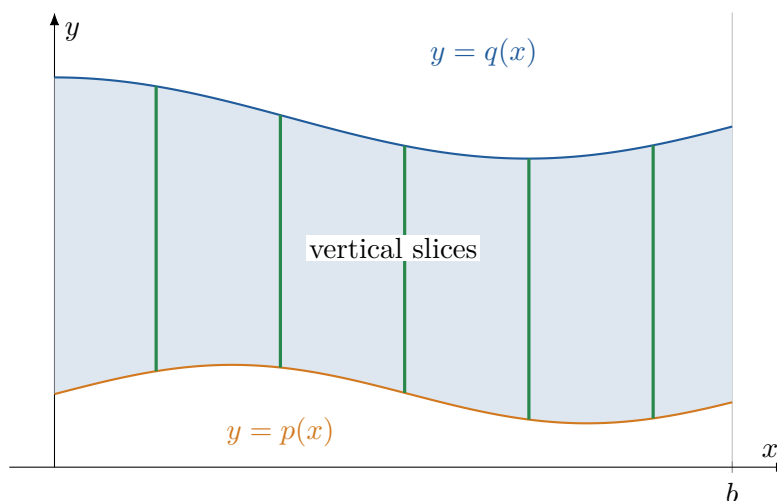


Figure 2: A Type I region is described by left-to-right limits and vertical slices.

Example 2.1: A region between a parabola and a line

Let D be the first-quadrant region bounded by $y = 4 - x^2$ and $y = 3x$. Compute

$$\iint_D 2xy \, dA.$$

Set the curves equal:

$$4 - x^2 = 3x \implies x^2 + 3x - 4 = 0 = (x + 4)(x - 1).$$

The first-quadrant intersection has $x = 1$ and $y = 3$. For $0 \leq x \leq 1$, the lower curve is $y = 3x$ and the upper curve is $y = 4 - x^2$. Thus

$$D = \{(x, y) : 0 \leq x \leq 1, 3x \leq y \leq 4 - x^2\}.$$

Therefore

$$\begin{aligned} \iint_D 2xy \, dA &= \int_0^1 \int_{3x}^{4-x^2} 2xy \, dy \, dx \\ &= \int_0^1 x[(4 - x^2)^2 - 9x^2] \, dx \\ &= \int_0^1 (16x - 17x^3 + x^5) \, dx = \frac{47}{12}. \end{aligned}$$

3 Type II regions: horizontal slicing

A plane region D is called a *Type II region* if it can be described as

$$D = \{(x, y) : c \leq y \leq d, g(y) \leq x \leq h(y)\},$$

where g and h are continuous functions and $g(y) \leq h(y)$ for $c \leq y \leq d$.

Geometrically, we move from bottom to top. For each fixed y , the variable x begins at the left curve $x = g(y)$ and ends at the right curve $x = h(y)$.

Double integral over a Type II region

If $D = \{(x, y) : c \leq y \leq d, g(y) \leq x \leq h(y)\}$ and f is continuous on D , then

$$\iint_D f(x, y) dA = \int_c^d \int_{g(y)}^{h(y)} f(x, y) dx dy.$$

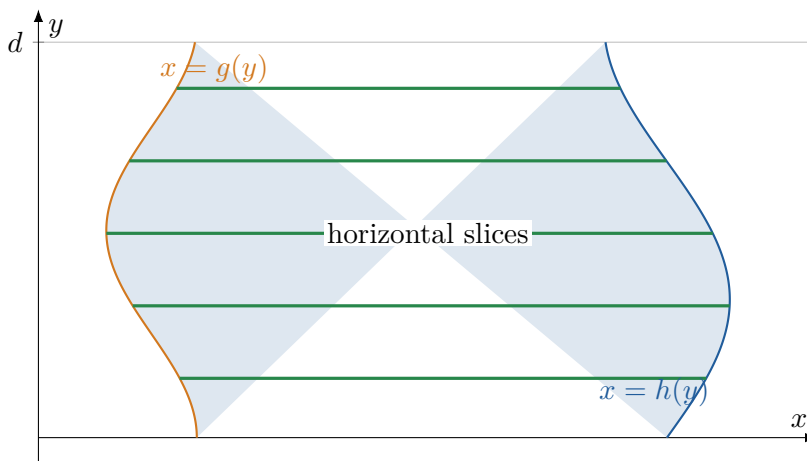


Figure 3: A Type II region is described by bottom-to-top limits and horizontal slices.

Example 3.1: A region between $x = y$ and $x = y^2$

Let D be the region bounded by $x = y$ and $x = y^2$. Compute

$$\iint_D (2xy + 6y^2) dA.$$

Intersections satisfy $y = y^2$, so $y = 0$ and $y = 1$. For $0 \leq y \leq 1$, we have $y^2 \leq x \leq y$. Hence

$$\begin{aligned} \iint_D (2xy + 6y^2) dA &= \int_0^1 \int_{y^2}^y (2xy + 6y^2) dx dy \\ &= \int_0^1 (7y^3 - 6y^4 - y^5) dy = \frac{23}{60}. \end{aligned}$$

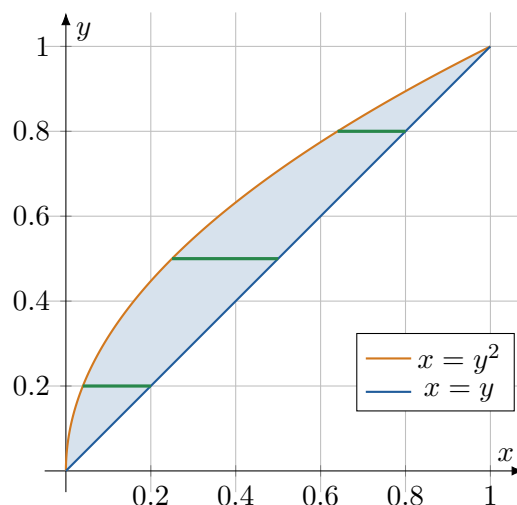


Figure 4: The region $y^2 \leq x \leq y$, $0 \leq y \leq 1$. Horizontal slices make the limits simple.

4 Choosing and reversing the order of integration

When setting up a double integral, first ask what a typical slice looks like. A vertical slice is useful if $x = \text{constant}$ enters and exits the region in one piece. A horizontal slice is useful if $y = \text{constant}$ enters and exits in one piece. Then ask which order makes the integrand easier.

Geometry first: algebra second

Many mistakes in double integrals are not integration mistakes. They are region mistakes. Before computing, check: the lower curve is really below the upper curve, the left curve is really left of the right curve, the intersection values are correct, and each slice stays inside the region.

Some iterated integrals are difficult in the given order but easy after reversing the order. Consider

$$\int_0^1 \int_y^1 e^{-x^2} dx dy.$$

The inner integral has no elementary antiderivative. The region is

$$D = \{(x, y) : 0 \leq y \leq 1, y \leq x \leq 1\}.$$

This is the triangle below $y = x$ inside the unit square. Rewriting in Type I form gives

$$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x\}.$$

Thus

$$\int_0^1 \int_y^1 e^{-x^2} dx dy = \int_0^1 \int_0^x e^{-x^2} dy dx = \int_0^1 x e^{-x^2} dx = \frac{1}{2}(1 - e^{-1}).$$

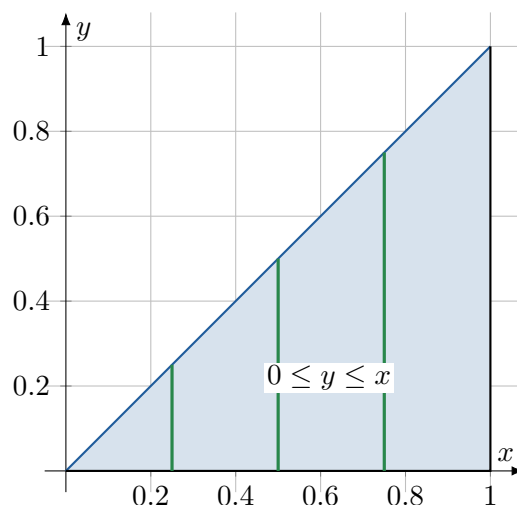


Figure 5: The triangle $0 \leq y \leq x \leq 1$. Reversing order turns a difficult integral into an elementary one.

Example 4.1: A reversed integral that requires splitting

Rewrite

$$\int_0^4 \int_{y/2}^{\sqrt{y+1}} f(x, y) \, dx \, dy$$

with the order $dy \, dx$. The given region is

$$0 \leq y \leq 4, \quad \frac{y}{2} \leq x \leq \sqrt{y+1}.$$

The boundaries become $y = 2x$ and $y = x^2 - 1$. A careful sketch shows that vertical slices change their lower or upper boundary, so the region splits:

$$\begin{aligned} \int_0^4 \int_{y/2}^{\sqrt{y+1}} f(x, y) \, dx \, dy &= \int_0^1 \int_0^{2x} f(x, y) \, dy \, dx \\ &\quad + \int_1^2 \int_{x^2-1}^{2x} f(x, y) \, dy \, dx \\ &\quad + \int_2^{\sqrt{5}} \int_{x^2-1}^4 f(x, y) \, dy \, dx. \end{aligned}$$

5 Area of a plane region

The area of a plane region D is computed by integrating the constant function 1:

Area by double integral

$$\text{Area}(D) = \iint_D 1 \, dA.$$

For a Type I region,

$$\text{Area}(D) = \int_a^b (q(x) - p(x)) dx.$$

For a Type II region,

$$\text{Area}(D) = \int_c^d (h(y) - g(y)) dy.$$

Example 5.1: Area between a line and a parabola

Find the area of the region bounded by $y = x - 1$ and $y^2 = 2x + 6$. Write both curves as x in terms of y :

$$y = x - 1 \iff x = y + 1, \quad x = \frac{y^2 - 6}{2}.$$

Intersections satisfy $y^2 = 2(y + 1) + 6$, so $(y - 4)(y + 2) = 0$. Thus $-2 \leq y \leq 4$. The parabola lies on the left and the line lies on the right:

$$\begin{aligned} \text{Area}(D) &= \int_{-2}^4 \int_{(y^2-6)/2}^{y+1} 1 dx dy \\ &= \int_{-2}^4 \left(y + 1 - \frac{y^2 - 6}{2} \right) dy = 18. \end{aligned}$$

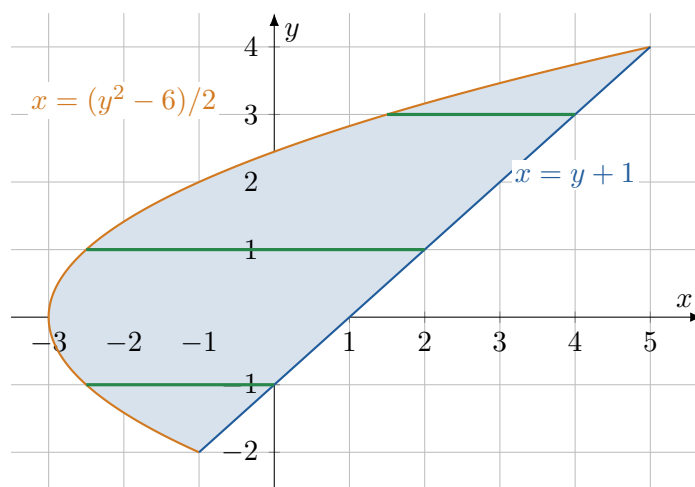


Figure 6: Area between the line $x = y + 1$ and the parabola $x = (y^2 - 6)/2$ by horizontal slices.

6 Splitting regions

Not every region is Type I or Type II in a single piece. Some regions must be split into simpler pieces. If $D = D_1 \cup D_2$ and the interiors do not overlap, then

$$\iint_D f(x, y) dA = \iint_{D_1} f(x, y) dA + \iint_{D_2} f(x, y) dA.$$

This is the two-dimensional version of splitting a one-variable integral at a point.

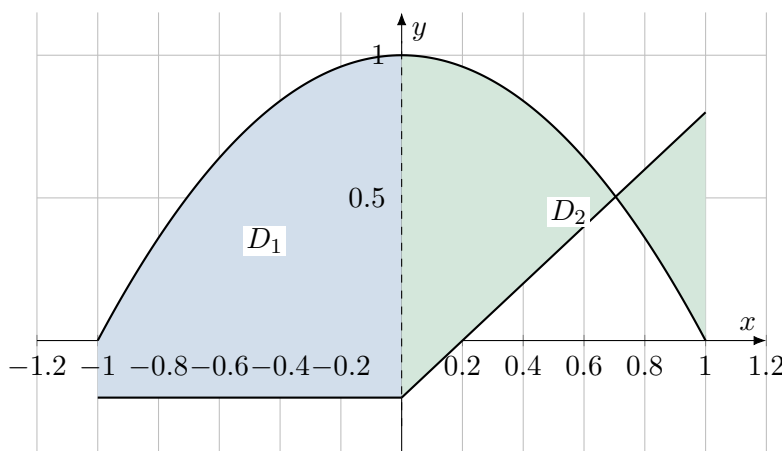


Figure 7: Some regions are best handled by splitting them into simpler Type I or Type II pieces.

When splitting is necessary

Splitting is needed when a single vertical or horizontal slice does not enter and exit the region through the same pair of boundary curves for the whole range. Splitting is not a failure; it is part of correct geometric modeling.

7 Volume under a surface over a curved base

If $f(x, y) \geq 0$ on a region D , then

$$V = \iint_D f(x, y) \, dA$$

is the volume of the solid above D and below the surface $z = f(x, y)$.

Example 7.1: Volume over a triangular region

Let D be the triangular region with vertices $(0, 0)$, $(1, 0)$, and $(1, 2)$. Find the volume under $z = 3 + x + y$ over D . The region is

$$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 2x\}.$$

Therefore

$$\begin{aligned} V &= \int_0^1 \int_0^{2x} (3 + x + y) \, dy \, dx \\ &= \int_0^1 (6x + 4x^2) \, dx = 3 + \frac{4}{3} = \frac{13}{3}. \end{aligned}$$

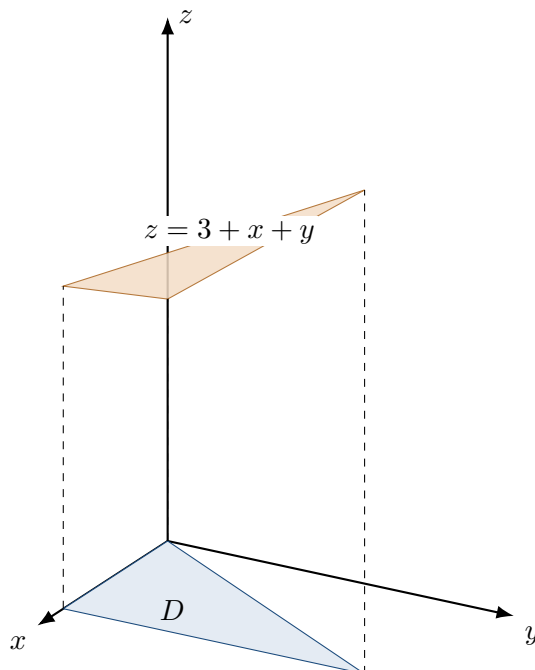


Figure 8: A surface over a triangular base region. The double integral gives volume.

8 Density, mass, and average value over general regions

If a thin plate occupies a region D and has density $\rho(x, y)$, then the mass is

$$m = \iint_D \rho(x, y) \, dA.$$

The average value of f over D is

$$f_{\text{avg}} = \frac{1}{\text{Area}(D)} \iint_D f(x, y) \, dA.$$

These formulas are the same as for rectangles, but now the geometry of D must be handled carefully.

Example 8.1: Average value on a triangle

Find the average value of $f(x, y) = x + 2y$ over

$$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}.$$

The area is $\text{Area}(D) = 1/2$. Compute

$$\begin{aligned} \iint_D (x + 2y) \, dA &= \int_0^1 \int_0^{1-x} (x + 2y) \, dy \, dx \\ &= \int_0^1 (1 - x) \, dx = \frac{1}{2}. \end{aligned}$$

Thus $f_{\text{avg}} = (1/2)/(1/2) = 1$.

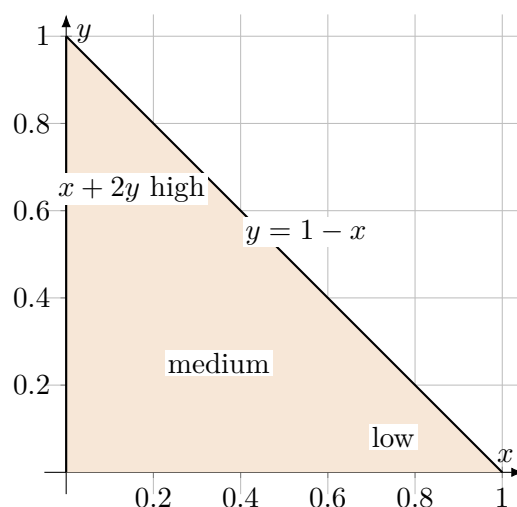


Figure 9: Average value over the triangular region $0 \leq x \leq 1$, $0 \leq y \leq 1 - x$.

9 Probability over a region

In statistics, a joint density $p(x, y)$ assigns probability density to points in the plane. If (X, Y) has joint density p , then

$$P((X, Y) \in D) = \iint_D p(x, y) dA.$$

This is one of the most important applications of double integrals over general regions.

Example 9.1: Probability over a curved event

Suppose (X, Y) is uniformly distributed over the unit square $[0, 1] \times [0, 1]$. Find $P(Y \leq X^2)$. The event is

$$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x^2\}.$$

Since the density is 1 on the unit square,

$$P(Y \leq X^2) = \int_0^1 \int_0^{x^2} 1 dy dx = \int_0^1 x^2 dx = \frac{1}{3}.$$

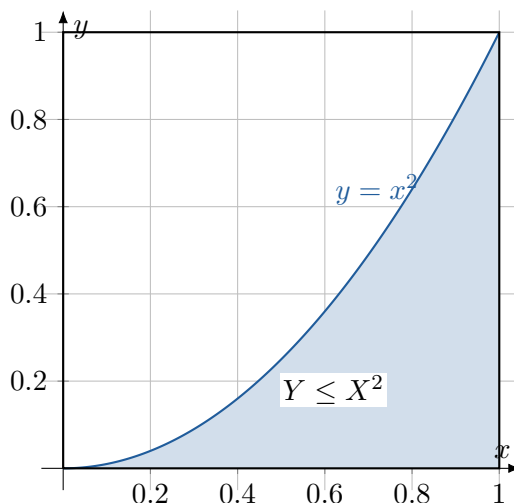


Figure 10: The probability $P(Y \leq X^2)$ for a uniform point in the unit square equals the area under $y = x^2$.

A nonuniform example: let $p(x, y) = 2$ on the triangle $T = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}$ and 0 outside T . Since $\text{Area}(T) = 1/2$, this is a density. For the event $Y \leq X$,

$$P(Y \leq X) = \int_0^{1/2} \int_0^x 2 \, dy \, dx + \int_{1/2}^1 \int_0^{1-x} 2 \, dy \, dx = \frac{1}{2}.$$

10 Numerical integration over irregular regions

For complicated regions or complicated functions, exact integration may be difficult. Two common numerical strategies are grid-based approximation and Monte Carlo integration.

Suppose D lies inside a rectangle R of area $A(R)$. Then

$$\iint_D f(x, y) \, dA = \iint_R f(x, y) \mathbf{1}_D(x, y) \, dA,$$

where $\mathbf{1}_D(x, y)$ is 1 inside D and 0 outside D .

A Monte Carlo approximation samples random points in R :

$$\iint_D f(x, y) \, dA \approx A(R) \cdot \frac{1}{N} \sum_{i=1}^N f(X_i, Y_i) \mathbf{1}_D(X_i, Y_i).$$

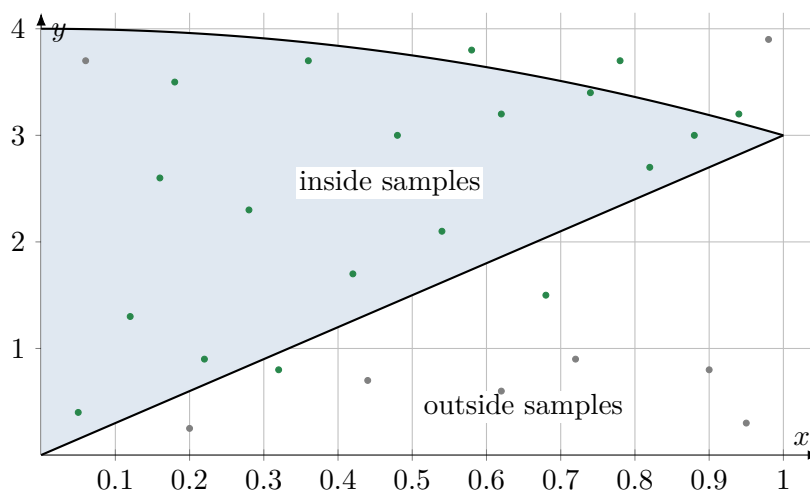


Figure 11: Monte Carlo integration over a curved region by sampling points in an enclosing rectangle.

Monte Carlo integration is not usually the most accurate method for low-dimensional smooth problems, but it scales naturally to high-dimensional problems. This is why it is central in statistics, Bayesian computation, and machine learning.

A common implementation mistake

When using numerical integration libraries, check the order of arguments. Some libraries call the integrand as $f(y, x)$ for an integral written in the order $dy dx$. The mathematical order and software argument order do not always look the same.

11 Checklist for setting up a double integral

1. Sketch the region and mark all boundary curves.
2. Find intersection points; these determine outer limits.
3. Choose vertical or horizontal slices. Decide whether the region is Type I, Type II, or must be split.
4. Write the inequalities. Do not compute until the region is correct.
5. Check lower/upper or left/right boundaries with a sample point.
6. Set up the integral, matching the inner variable to the inner differential.
7. Simplify before integrating; sometimes reversing order is better.
8. Interpret the result as area, volume, mass, average value, probability, or signed accumulation.

12 Modern applications

Statistics and probability

A joint density over a nonrectangular region appears when variables are constrained. Examples include $0 \leq y \leq x \leq 1$ for ordered random variables, $x + y \leq 1$ for compositional data, $y \leq x^2$ for nonlinear events, and feasible regions in Bayesian priors.

Machine learning

If $L(\theta_1, \theta_2)$ is a loss function and D is a feasible set, then

$$\frac{1}{\text{Area}(D)} \iint_D L(\theta_1, \theta_2) dA$$

is the average loss over D .

Applied mathematics

Regions may be determined by physical constraints: a membrane shape, a flow region, an obstacle boundary, a material interface, or a domain for a partial differential equation. Double integrals over such regions compute mass, energy, heat, charge, and probability.

13 Chapter summary

A double integral over a general region adds a function over a curved or constrained two-dimensional domain. The central task is translating geometry into limits.

- A Type I region has the form

$$D = \{(x, y) : a \leq x \leq b, p(x) \leq y \leq q(x)\},$$

and

$$\iint_D f(x, y) dA = \int_a^b \int_{p(x)}^{q(x)} f(x, y) dy dx.$$

- A Type II region has the form

$$D = \{(x, y) : c \leq y \leq d, g(y) \leq x \leq h(y)\},$$

and

$$\iint_D f(x, y) dA = \int_c^d \int_{g(y)}^{h(y)} f(x, y) dx dy.$$

- Area is computed by $\text{Area}(D) = \iint_D 1 dA$.
- Regions can be split: $\iint_D f dA = \iint_{D_1} f dA + \iint_{D_2} f dA$.
- Reversing order can turn a difficult integral into an elementary one.

The strongest students do not memorize limits. They learn to see slices.

14 Exercises

Conceptual exercises

1. In your own words, explain the difference between a Type I region and a Type II region.
2. Why do intersection points usually determine the outer limits of integration?
3. Explain why reversing the order of integration is a geometric operation before it is an algebraic operation.
4. Give an example of a region that is both Type I and Type II.
5. Give an example of a region that must be split to be described by vertical slices.
6. What does $\iint_D 1 \, dA$ compute?
7. If $f(x, y)$ is sometimes positive and sometimes negative on D , why is $\iint_D f \, dA$ not necessarily a volume?
8. Explain the role of the indicator function $\mathbf{1}_D(x, y)$ in numerical integration over an irregular region.

Skill exercises

9. Let $D = \{(x, y) : 0 \leq x \leq 2, x^2 \leq y \leq 2x\}$. Compute $\iint_D y \, dA$.
10. Let $D = \{(x, y) : 0 \leq y \leq 1, y \leq x \leq 2 - y\}$. Compute $\iint_D x \, dA$.
11. Find the area bounded by $y = x^2$ and $y = x + 2$.
12. Find the area bounded by $x = y^2$ and $x = 2 - y$.
13. Write the region bounded by $y = \sqrt{x}$ and $y = x^2$ as a Type I region and as a Type II region.
14. Evaluate $\int_0^1 \int_0^{1-x} (x + y) \, dy \, dx$.
15. Evaluate $\int_0^2 \int_{x/2}^1 xy \, dy \, dx$.
16. Set up, but do not evaluate, the volume under $z = x^2 + y^2$ over the region bounded by $y = x$ and $y = x^2$.

Reversing-order exercises

17. Reverse the order of integration: $\int_0^1 \int_0^x f(x, y) \, dy \, dx$.
18. Reverse the order of integration: $\int_0^4 \int_{\sqrt{y}}^2 f(x, y) \, dx \, dy$.
19. Evaluate by reversing order: $\int_0^1 \int_y^1 \cos(x^2) \, dx \, dy$.
20. Evaluate by reversing order: $\int_0^1 \int_x^1 e^{y^2} \, dy \, dx$.
21. Rewrite in the opposite order: $\int_0^1 \int_{x^2}^{\sqrt{x}} f(x, y) \, dy \, dx$.
22. Rewrite in the opposite order: $\int_0^1 \int_y^{2-y} f(x, y) \, dx \, dy$.

Applications and challenges

23. A plate occupies $D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}$ and has density $\rho(x, y) = 1 + x + y$. Find its mass.
24. Find the average value of $f(x, y) = xy$ over $D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x\}$.
25. A point is chosen uniformly from the triangle with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$. Find the probability that $Y \leq X$.
26. A point is chosen uniformly from the unit square. Find the probability that $X^2 \leq Y \leq X$.
27. Let D be the disk $x^2 + y^2 \leq 1$. Without using polar coordinates, explain why this region is not naturally Type I or Type II in one simple formula unless square roots are used.
28. Approximate $\iint_D e^{-(x^2+y^2)} dA$, where $D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x^2\}$.
29. Let D be the region bounded by $y = x^2$ and $y = 2x - x^2$. Compute $\iint_D x dA$.
30. Let $D = \{(x, y) : 0 \leq x \leq 1, x^2 \leq y \leq x\}$. Compute the average value of $f(x, y) = x + y$ over D .
31. Find a region D and an integrand f for which one order of integration is elementary and the other is not.
32. Let D be the region bounded by $x = 0$, $y = 1$, and $y = x^2$. Set up $\iint_D f(x, y) dA$ in both orders.

15 Python lab: slicing, switching, and probability

In the companion Python lab, you will plot Type I and Type II regions, approximate double integrals by grid sums, reverse integration order from a plotted region, compute area and average value over triangular and parabolic regions, estimate probabilities by Monte Carlo simulation, and compare symbolic, numerical, and simulation-based answers.

16 Looking ahead

This chapter still uses rectangular coordinates. In the next chapter, we introduce polar coordinates and general changes of variables. That will make circular and radial regions much easier, and it will introduce one of the most important ideas in multivariable integration: the area element changes when coordinates change.

Chapter 15. Polar Coordinates and Change of Variables in the Plane

Circular geometry, Jacobians, transformed regions, probability densities, and coordinate systems

Many two-dimensional integrals become simple only after we choose coordinates that match the geometry. Rectangular coordinates are natural for boxes and vertical slices, but circular and radial regions call for polar coordinates. More generally, a change of variables replaces a complicated region in the xy -plane by a simpler region in a new uv -plane, at the cost of multiplying by a Jacobian factor.

The central idea is

$$\text{new area element} = \text{local area scaling factor} \times \text{new coordinate area element}.$$

For polar coordinates this becomes

$$dA = r \, dr \, d\theta.$$

For a general transformation $(u, v) \mapsto (x, y)$, this becomes

$$dA = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du \, dv.$$

Learning objectives

After studying this chapter, you should be able to:

1. convert between rectangular coordinates (x, y) and polar coordinates (r, θ) ;
2. describe disks, annuli, sectors, and shifted circles using polar inequalities;
3. explain geometrically why $dA = r \, dr \, d\theta$;
4. set up and evaluate double integrals in polar coordinates;
5. compute and interpret the Jacobian determinant of a transformation;
6. apply the change-of-variables formula for double integrals;
7. connect coordinate transformations to probability densities and modern data modeling.

1 Why change coordinates?

A coordinate system is not the object itself. It is a language for describing the object. The disk

$$x^2 + y^2 \leq 1$$

is simple geometrically, but in rectangular coordinates it requires square roots:

$$-1 \leq x \leq 1, \quad -\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2}.$$

In polar coordinates, the same disk becomes

$$0 \leq r \leq 1, \quad 0 \leq \theta \leq 2\pi.$$

The geometry did not change; only the description changed. A good coordinate system reduces complexity.

This principle appears throughout applied mathematics: ellipses become circles after scaling, correlated Gaussian densities become independent after a linear transformation, and complicated constraints may become rectangles in new variables.

2 Polar coordinates

In polar coordinates, a point is described by distance r from the origin and angle θ from the positive x -axis. The conversion formulas are

$$x = r \cos \theta, \quad y = r \sin \theta,$$

and

$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}$$

when $x \neq 0$, with the quadrant determining the correct angle.

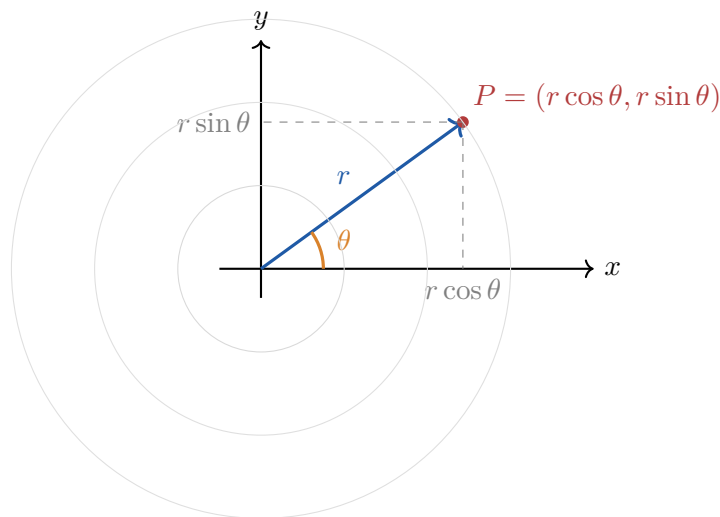


Figure 1: A point described by polar coordinates. The radius gives distance from the origin and the angle gives direction.

Nonuniqueness of polar coordinates

A rectangular coordinate pair (x, y) is unique. Polar coordinates are not unique. For example, $(1, 1)$ has polar coordinates

$$\left(\sqrt{2}, \frac{\pi}{4}\right), \quad \left(\sqrt{2}, \frac{\pi}{4} + 2\pi\right), \quad \left(\sqrt{2}, \frac{\pi}{4} + 4\pi\right),$$

and many more. The origin has $r = 0$ and any value of θ .

Practical convention

When setting up polar integrals, choose bounds so that almost every point in the region is counted exactly once. Boundaries may overlap, but overlapping boundaries have area zero and do not change a double integral.

3 Common polar regions

Polar coordinates are most useful when the region is described naturally by distance from the origin and angle.

Region	Rectangular description	Polar description
Disk	$x^2 + y^2 \leq a^2$	$0 \leq r \leq a, 0 \leq \theta \leq 2\pi$
Annulus	$a^2 \leq x^2 + y^2 \leq b^2$	$a \leq r \leq b, 0 \leq \theta \leq 2\pi$
Sector	circular wedge	$0 \leq r \leq a, \alpha \leq \theta \leq \beta$
Upper half-disk	$x^2 + y^2 \leq a^2, y \geq 0$	$0 \leq r \leq a, 0 \leq \theta \leq \pi$
Shifted circle	$(x - a)^2 + y^2 \leq a^2$	$0 \leq r \leq 2a \cos \theta, -\pi/2 \leq \theta \leq \pi/2$

For the shifted circle $(x - a)^2 + y^2 = a^2$, expanding gives $x^2 + y^2 = 2ax$. In polar coordinates this is $r^2 = 2ar \cos \theta$, so for $r \geq 0$ the boundary is $r = 2a \cos \theta$.

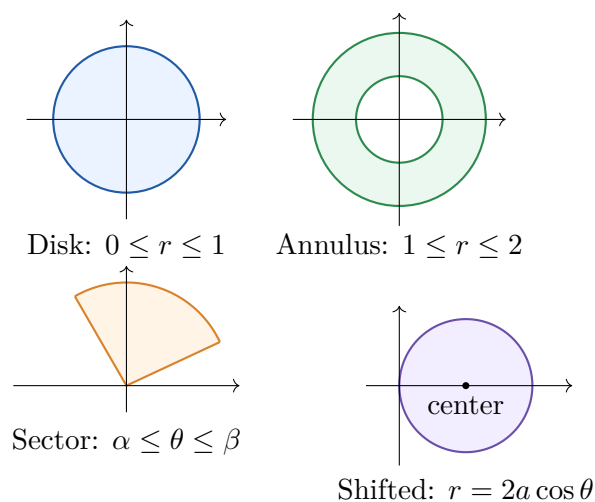


Figure 2: Disks, annuli, sectors, and shifted circles have compact descriptions in polar coordinates.

4 The polar area element

The most important formula in polar integration is

$$dA = r \, dr \, d\theta.$$

The factor r is not optional. It records the fact that a small angular change $d\theta$ sweeps out a longer arc when the point is farther from the origin.

Consider a small polar rectangle:

$$r \leq \rho \leq r + dr, \quad \theta \leq \phi \leq \theta + d\theta.$$

Its radial thickness is approximately dr . Its angular width is approximately arc length $r d\theta$. Therefore its area is approximately $(r d\theta)(dr) = r dr d\theta$.

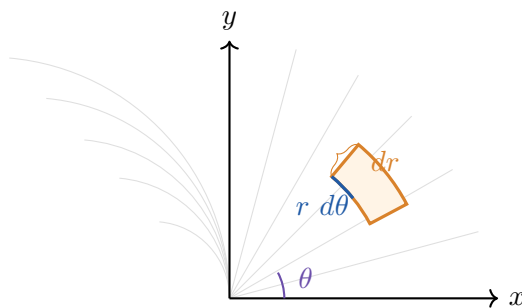


Figure 3: A small polar coordinate cell has radial thickness dr and angular width approximately $r d\theta$, so its area is approximately $r dr d\theta$.

Common mistake

Do not write $dA = dr d\theta$. The correct formula is $dA = r dr d\theta$. The factor r is the local stretching factor from angular movement.

5 Double integrals in polar coordinates

Suppose a region D is described in polar form as

$$\alpha \leq \theta \leq \beta, \quad h_1(\theta) \leq r \leq h_2(\theta).$$

Then

Polar double-integral formula

$$\iint_D f(x, y) dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) r dr d\theta.$$

The process has three steps: rewrite the region in polar bounds, substitute $x = r \cos \theta$ and $y = r \sin \theta$ into the integrand, and replace dA by $r dr d\theta$.

Worked example 5.1: Area of a disk

Find the area of the disk $D : x^2 + y^2 \leq a^2$.

Since D is $0 \leq r \leq a$, $0 \leq \theta \leq 2\pi$,

$$\text{Area}(D) = \int_0^{2\pi} \int_0^a r \, dr \, d\theta = \int_0^{2\pi} \frac{a^2}{2} \, d\theta = \pi a^2.$$

Worked example 5.2: Converting a rectangular-coordinate integral

Convert and evaluate

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} 4 \, dy \, dx.$$

The region is the upper half of the unit disk. In polar coordinates,

$$0 \leq r \leq 1, \quad 0 \leq \theta \leq \pi.$$

Therefore

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} 4 \, dy \, dx = \int_0^\pi \int_0^1 4r \, dr \, d\theta = 2\pi.$$

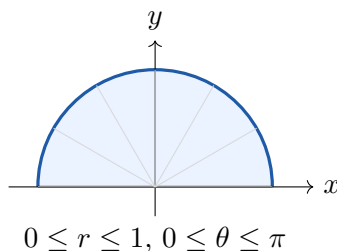


Figure 4: The region for the rectangular integral is the upper half of the unit disk.

Worked example 5.3: An annular integral

Evaluate

$$\iint_D \cos(x^2 + y^2) \, dA,$$

where D is the region between $r = 3$ and $r = 4$ in the upper half-plane.

The bounds are $3 \leq r \leq 4$ and $0 \leq \theta \leq \pi$. Since $x^2 + y^2 = r^2$,

$$\iint_D \cos(x^2 + y^2) \, dA = \int_0^\pi \int_3^4 \cos(r^2) r \, dr \, d\theta.$$

Let $u = r^2$, so $du = 2r \, dr$. Then

$$\int_3^4 \cos(r^2) r \, dr = \frac{1}{2} \int_9^{16} \cos u \, du = \frac{1}{2} (\sin 16 - \sin 9).$$

Thus

$$\iint_D \cos(x^2 + y^2) \, dA = \frac{\pi}{2} (\sin 16 - \sin 9).$$

6 Volume using polar coordinates

If a solid lies under $z = f(x, y)$ and above a region D , then its volume is

$$V = \iint_D f(x, y) \, dA.$$

When D is circular or the function depends on $x^2 + y^2$, polar coordinates often simplify the integral.

Worked example 6.1: Paraboloid over a quadrant

Find the volume under

$$z = 6 - 2x^2 - 2y^2$$

above the first-quadrant part of the region where $z \geq 0$.

The condition $z \geq 0$ gives $x^2 + y^2 \leq 3$. In the first quadrant,

$$0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq r \leq \sqrt{3}.$$

Therefore

$$V = \int_0^{\pi/2} \int_0^{\sqrt{3}} (6 - 2r^2)r \, dr \, d\theta.$$

The inner integral is

$$\int_0^{\sqrt{3}} (6r - 2r^3) \, dr = \left[3r^2 - \frac{1}{2}r^4 \right]_0^{\sqrt{3}} = \frac{9}{2}.$$

Hence $V = \frac{\pi}{2} \cdot \frac{9}{2} = \frac{9\pi}{4}$.

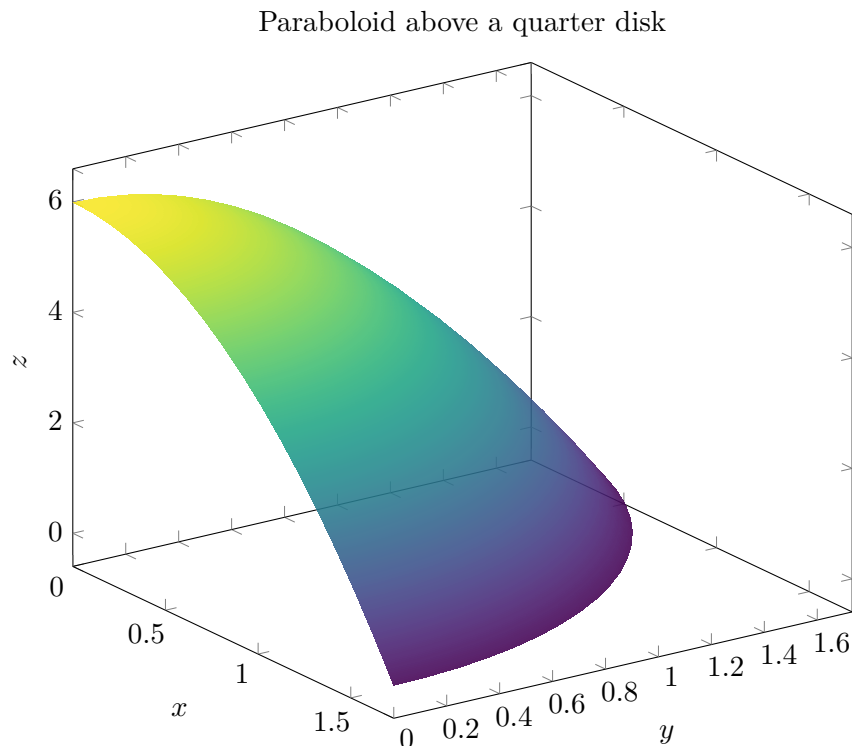


Figure 5: The surface $z = 6 - 2x^2 - 2y^2$ above a quarter disk. Polar coordinates match both the circular boundary and the radial integrand.

7 Shifted circular regions

Polar coordinates are not only for circles centered at the origin. They also work well for some shifted circles, especially when the circle passes through the origin.

Consider

$$D = \{(x, y) : (x - 2)^2 + y^2 \leq 4\}.$$

Expanding gives $x^2 + y^2 \leq 4x$. In polar coordinates this becomes $r^2 \leq 4r \cos \theta$, so for $r \geq 0$,

$$0 \leq r \leq 4 \cos \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}.$$

The area is

$$\int_{-\pi/2}^{\pi/2} \int_0^{4 \cos \theta} r \, dr \, d\theta = \int_{-\pi/2}^{\pi/2} 8 \cos^2 \theta \, d\theta = 4\pi,$$

which agrees with the area of a circle of radius 2.

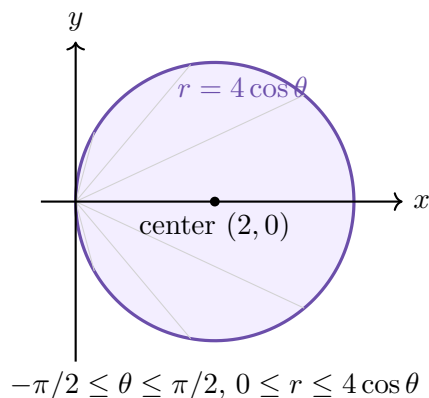


Figure 6: The shifted disk $(x - 2)^2 + y^2 \leq 4$ is described by $0 \leq r \leq 4 \cos \theta$, $-\pi/2 \leq \theta \leq \pi/2$.

8 Strategy for choosing polar coordinates

Polar coordinates are often a good choice when at least one of the following appears:

1. the region involves $x^2 + y^2$;
2. the boundary is a circle, disk, annulus, sector, or ray from the origin;
3. the integrand contains $x^2 + y^2$, $\sqrt{x^2 + y^2}$, or radial symmetry;
4. a rectangular-coordinate integral contains square roots such as $\sqrt{a^2 - x^2}$;
5. the geometry is best described by distance and angle.

Polar coordinates may not help if the region is rectangular and the integrand has no radial structure.

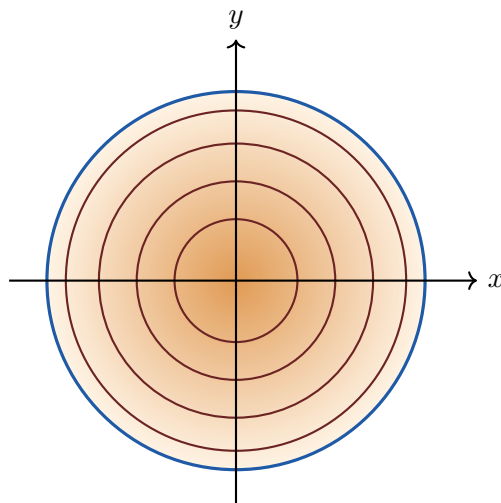
Worked example 8.1: A radial integrand over a disk

The integral

$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} e^{-(x^2+y^2)} dy dx$$

is hard in rectangular coordinates. In polar coordinates it becomes

$$\int_0^{2\pi} \int_0^1 e^{-r^2} r dr d\theta = 2\pi \left[-\frac{1}{2} e^{-r^2} \right]_0^1 = \pi(1 - e^{-1}).$$



level curves of $e^{-(x^2+y^2)}$ over a disk are circles

Figure 7: A radially symmetric integrand over a disk. Polar coordinates turn the integral into a radial integral times an angular factor.

9 General change of variables

Polar coordinates are one example of a broader idea. Let a transformation T map a point (u, v) in a new coordinate plane to a point (x, y) in the original plane:

$$T(u, v) = (x(u, v), y(u, v)).$$

The Jacobian determinant of the transformation is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}.$$

The absolute value of this determinant measures local area scaling.

Change-of-variables formula

If T maps a region S in the uv -plane to a region D in the xy -plane, and if the transformation behaves well on the region, then

$$\iint_D f(x, y) \, dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \, du \, dv.$$

For polar coordinates, $x = r \cos \theta$ and $y = r \sin \theta$. The Jacobian is

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r.$$

Thus the polar formula is just the general change-of-variables formula applied to the polar map.

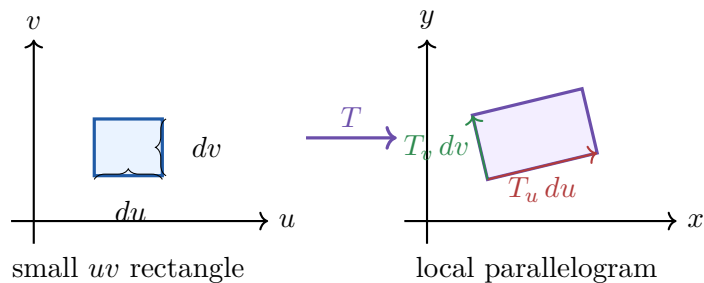


Figure 8: The Jacobian determinant measures how a small coordinate rectangle is transformed into a small parallelogram in the original plane.

10 Linear transformations and area scaling

A linear transformation

$$\begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} u \\ v \end{pmatrix}$$

scales every area by the constant factor $|\det A|$. For example,

$$x = 2u + v, \quad y = u + 3v,$$

so

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad \det A = 5.$$

Any region in the uv -plane has image area five times as large in the xy -plane.

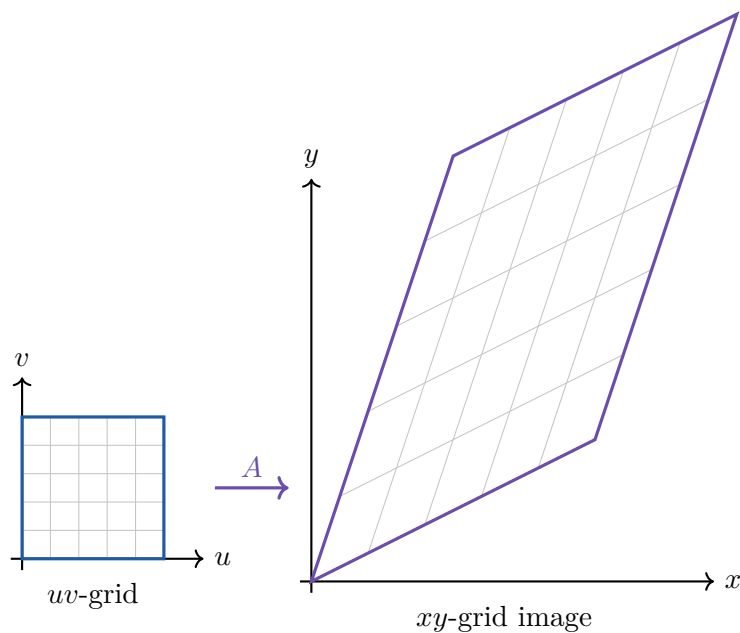


Figure 9: A square grid in the uv -plane transformed by a linear map. The determinant gives the constant area-scaling factor.

Worked example 10.1: Area of a parallelogram

Let $S = [0, 1] \times [0, 1]$ and define

$$x = 2u + v, \quad y = u + 3v.$$

The image $D = T(S)$ is a parallelogram. Since $|\det A| = 5$,

$$\text{Area}(D) = \iint_D 1 \, dA = \iint_S 1 \cdot 5 \, du \, dv = 5.$$

11 Nonlinear transformations

For nonlinear transformations, the Jacobian usually varies from point to point. The transformation may stretch one part of the region more than another.

For example,

$$x = u, \quad y = v + 0.4u^2.$$

The Jacobian is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 1 & 0 \\ 0.8u & 1 \end{vmatrix} = 1.$$

Although this transformation bends horizontal grid lines into parabolas, it preserves area locally because the Jacobian determinant is 1.

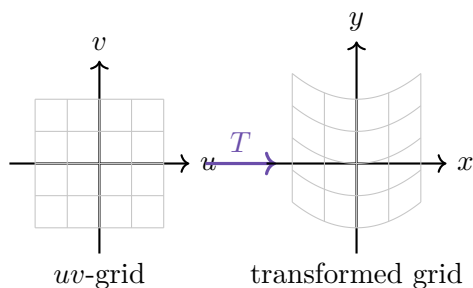


Figure 10: A nonlinear shear bends grid lines but has Jacobian determinant 1, so it preserves area locally.

Worked example 11.1: A triangular transformation

Let

$$x = u, \quad y = uv, \quad 1 \leq u \leq 2, \quad 0 \leq v \leq 1.$$

The Jacobian is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 1 & 0 \\ v & u \end{vmatrix} = u.$$

The image region satisfies $1 \leq x \leq 2$ and $0 \leq y \leq x$. Hence

$$\iint_D (x + y) \, dA = \int_1^2 \int_0^1 (u + uv)u \, dv \, du = \int_1^2 \frac{3}{2}u^2 \, du = \frac{7}{2}.$$

12 Change of variables in probability

Change of variables is essential in probability and statistics. If (X, Y) has joint density $f_{X,Y}(x, y)$ and $(X, Y) = T(U, V)$, then the density of (U, V) involves a Jacobian factor. Informally,

$$\text{probability mass} = \text{density} \times \text{area}.$$

When area changes, density must change in the opposite way so that probability is preserved.

Radial probability

Suppose a density depends only on radius:

$$f(x, y) = ce^{-(x^2+y^2)}.$$

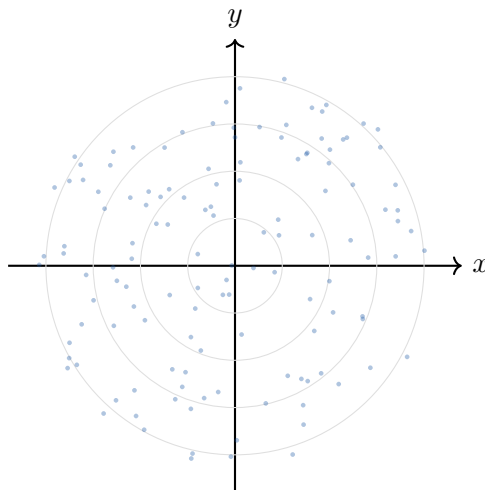
To find the normalizing constant c , require

$$1 = \iint_{\mathbb{R}^2} ce^{-(x^2+y^2)} dA.$$

Using polar coordinates,

$$1 = c \int_0^{2\pi} \int_0^\infty e^{-r^2} r dr d\theta = c(2\pi) \left(\frac{1}{2}\right) = c\pi.$$

Thus $c = 1/\pi$.



radial density: angle is uniform, radius controls concentration

Figure 11: Samples from a radial distribution. Polar coordinates separate radius and angle, which is useful for simulation and probability calculations.

13 Computational viewpoint

In computation, a change of variables is more than a symbolic trick. It is a method for transforming a difficult numerical problem into an easier one.

To approximate

$$\iint_{x^2+y^2 \leq 1} e^{-(x^2+y^2)} dA,$$

we can sample uniformly from the disk. A correct polar simulation must not sample r uniformly on $[0, 1]$. Since area in polar coordinates is $r dr d\theta$, the radius distribution must put more points near the outer part of the disk.

A uniform point in the unit disk can be generated by

$$\Theta \sim \text{Uniform}(0, 2\pi), \quad R = \sqrt{U},$$

where $U \sim \text{Uniform}(0, 1)$.

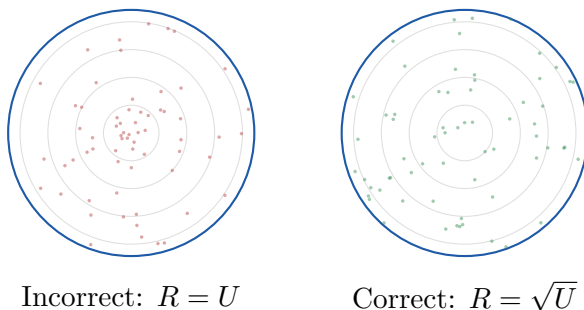


Figure 12: Uniform sampling in a disk requires $R = \sqrt{U}$, not $R = U$. The Jacobian factor r explains the difference.

The exact value of the disk integral is

$$\int_0^{2\pi} \int_0^1 e^{-r^2} r dr d\theta = \pi(1 - e^{-1}).$$

14 Applications in data, statistics, and AI

Radial basis functions

Many machine-learning models use radial basis functions such as

$$\phi(x, y) = e^{-\gamma(x^2+y^2)}.$$

These functions are constant on circles centered at the origin. Polar coordinates reveal their radial structure immediately.

Gaussian models

The bivariate normal density has elliptical level curves. After shifting, scaling, and rotating coordinates, those ellipses become circles. This is a change-of-variables idea. The determinant of the covariance matrix controls how volume changes.

Normalizing flows

In modern generative modeling, a normalizing flow transforms a simple latent variable z into a more complicated observed variable $x = T(z)$. The probability density changes by a Jacobian determinant. In two dimensions the formula is the same idea as this chapter; in high dimensions it becomes

$$\text{density correction} = |\det J_T|^{-1}.$$

Numerical integration

Monte Carlo methods, importance sampling, and coordinate transformations all rely on understanding how area or volume changes under a transformation. A poor coordinate system can make an integral numerically unstable; a good one can make the same problem nearly one-dimensional.

15 Chapter summary

The key polar formulas are

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r^2 = x^2 + y^2.$$

The polar area element is

$$dA = r \, dr \, d\theta.$$

Thus, for a polar region

$$D = \{(r, \theta) : \alpha \leq \theta \leq \beta, \, h_1(\theta) \leq r \leq h_2(\theta)\},$$

we have

$$\iint_D f(x, y) \, dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) r \, dr \, d\theta.$$

For a general transformation $T(u, v) = (x(u, v), y(u, v))$, the Jacobian determinant is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix},$$

and the change-of-variables formula is

$$\iint_D f(x, y) \, dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \, du \, dv.$$

The absolute value of the Jacobian determinant is the local area-scaling factor.

16 Exercises

Conceptual questions

1. Explain why polar coordinates are useful for disks and annuli.

2. Explain why the polar area element is $r \, dr \, d\theta$, not $dr \, d\theta$.
3. Why are polar coordinates not unique?
4. What does the Jacobian determinant measure geometrically?
5. Why do we use the absolute value of the Jacobian determinant in the change-of-variables formula?
6. Give an example of a region where rectangular coordinates are more convenient than polar coordinates.
7. Give an example of an integrand where polar coordinates simplify the algebra.
8. Explain why uniform sampling in a disk requires $R = \sqrt{U}$ instead of $R = U$.

Polar coordinate descriptions

9. Describe the disk $x^2 + y^2 \leq 9$ in polar coordinates.
10. Describe the annulus $4 \leq x^2 + y^2 \leq 25$ in polar coordinates.
11. Describe the upper half of the disk $x^2 + y^2 \leq 16$ in polar coordinates.
12. Describe the first-quadrant part of the annulus $1 \leq x^2 + y^2 \leq 4$ in polar coordinates.
13. Convert the circle $x^2 + y^2 = 6x$ to polar form.
14. Convert the circle $x^2 + y^2 = -4y$ to polar form.
15. Describe the region inside $r = 2 \cos \theta$.
16. Describe the region inside $r = 3 \sin \theta$.

Polar integrals

17. Evaluate $\iint_D 1 \, dA$ where D is the disk $x^2 + y^2 \leq 4$.
18. Evaluate $\iint_D (x^2 + y^2) \, dA$ where D is the disk $x^2 + y^2 \leq 1$.
19. Evaluate $\iint_D e^{-(x^2+y^2)} \, dA$ where D is the disk $x^2 + y^2 \leq 4$.
20. Evaluate $\iint_D \sqrt{x^2 + y^2} \, dA$ where D is the annulus $1 \leq x^2 + y^2 \leq 9$.
21. Evaluate $\iint_D y \, dA$ where D is the upper half of the disk $x^2 + y^2 \leq 1$.
22. Evaluate $\iint_D x^2 \, dA$ where D is the first-quadrant unit disk.
23. Convert $\int_{-2}^2 \int_0^{\sqrt{4-x^2}} f(x, y) \, dy \, dx$ to polar form.
24. Convert $\int_0^2 \int_0^{\sqrt{4-y^2}} f(x, y) \, dx \, dy$ to polar form.
25. Evaluate the volume under $z = 9 - x^2 - y^2$ above the disk $x^2 + y^2 \leq 9$.
26. Evaluate the volume under $z = 4 - x^2 - y^2$ above the first-quadrant part of $x^2 + y^2 \leq 4$.

Change of variables

27. Compute the Jacobian for $x = 3u$, $y = 2v$.
28. Compute the Jacobian for $x = u + v$, $y = u - v$.
29. Compute the Jacobian for $x = u^2 - v^2$, $y = 2uv$.
30. Compute the Jacobian for $x = e^u \cos v$, $y = e^u \sin v$.
31. Let $x = 2u + v$, $y = u + 4v$, and $S = [0, 1] \times [0, 2]$. Find the area of $T(S)$.
32. Let $x = u$, $y = uv$, where $1 \leq u \leq 3$ and $0 \leq v \leq 2$. Write the corresponding region in the xy -plane.
33. Use the change of variables $x = 2u$, $y = 3v$ to compute the area of the ellipse $x^2/4 + y^2/9 \leq 1$.
34. Use an appropriate change of variables to convert the ellipse $x^2/a^2 + y^2/b^2 \leq 1$ into a unit disk.

Applications and computation

35. Use polar coordinates to normalize the density $f(x, y) = ce^{-(x^2+y^2)}$ over the plane.
36. Find the probability that a point with density $f(x, y) = \pi^{-1}e^{-(x^2+y^2)}$ lies inside the unit disk.
37. Write code to approximate $\iint_{x^2+y^2 \leq 1} \sqrt{x^2+y^2} dA$ by Monte Carlo simulation.
38. Explain how the determinant appears in a two-dimensional normalizing flow.
39. Let $L(a, b) = \iint_D (a + br - y)^2 dA$ over a disk D . Explain why polar coordinates may simplify the geometry but not necessarily the integrand.
40. A sensor has circular range $x^2 + y^2 \leq R^2$. Write an integral for the average signal strength $S(x, y) = e^{-\lambda\sqrt{x^2+y^2}}$ over the range.

17 Python lab and looking ahead

In the companion Python lab, students plot polar grids and polar regions, convert rectangular integrals over disks into polar form, compute polar integrals numerically, visualize the polar area element, compute Jacobians of transformations, transform grids under linear and nonlinear maps, simulate uniform points in disks, and estimate radial probability integrals using Monte Carlo methods.

This chapter introduced coordinate transformations in the plane. The next chapters extend the same idea to three-dimensional integration. Cylindrical and spherical coordinates play the role that polar coordinates played here, and the Jacobian factors become volume-scaling factors.

Chapter 16. Triple Integrals and Geometry in $\mathbb{R}^3$

Volume, mass, probability, and continuous sums over solid regions

A double integral sums over a two-dimensional region. A triple integral sums over a three-dimensional solid. The change sounds small: replace tiny rectangles by tiny boxes. Conceptually, however, this is a major step. Triple integrals allow us to compute volume, mass, charge, probability, average value, and spatial accumulation over objects in space.

The central geometric skill in this chapter is **describing a solid region by inequalities**. Once the region is described correctly, the integral is often straightforward. Most errors in triple integrals are not integration errors; they are errors in reading the geometry.

This chapter develops triple integrals in rectangular coordinates. Cylindrical and spherical coordinates are treated in the next chapter. For now, the guiding idea is:

A triple integral is a continuous sum over a solid region.

Learning goals

After studying this chapter, you should be able to:

1. Interpret $\iiint_E f(x, y, z) dV$ as a continuous sum over a solid region E .
2. Compute volume as $\iiint_E 1 dV$.
3. Set up triple integrals using projections onto coordinate planes.
4. Describe solids with bounds such as $g_1(x, y) \leq z \leq g_2(x, y)$.
5. Change the order of integration in simple three-dimensional regions.
6. Compute mass, average value, and probability using triple integrals.
7. Use computation to visualize regions and approximate triple integrals numerically.

16.1 From double integrals to triple integrals

Recall the structure of a double integral over a region $R \subset \mathbb{R}^2$:

$$\iint_R f(x, y) dA.$$

It is the limit of sums of the form

$$\sum f(x_{ij}^*, y_{ij}^*) \Delta A.$$

For a triple integral over a solid region $E \subset \mathbb{R}^3$, we replace area pieces by volume pieces:

$$\iiint_E f(x, y, z) dV = \lim \sum f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \Delta V.$$

In rectangular coordinates,

$$\Delta V = \Delta x \Delta y \Delta z, \quad dV = dx dy dz.$$

For example,

$$\int_a^b \int_c^d \int_p^q f(x, y, z) \, dz \, dy \, dx$$

means that we integrate first with respect to z , then y , then x .

The guiding picture

A double integral sums tiny columns above an area region. A triple integral sums tiny boxes inside a solid region.

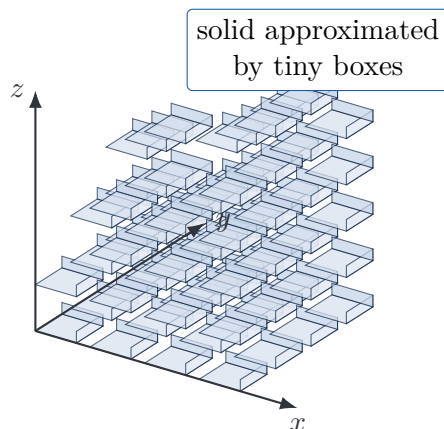


Figure 1: A solid can be approximated by many small rectangular boxes. A triple integral is the limit of such voxel sums.

16.2 Volume as a triple integral

The simplest triple integral occurs when the integrand is 1.

Volume of a solid

If $E \subset \mathbb{R}^3$ is a solid region, then

$$\text{Vol}(E) = \iiint_E 1 \, dV.$$

This formula says that volume is the sum of all infinitesimal volume elements dV inside the region. It is the three-dimensional analog of

$$\text{Area}(R) = \iint_R 1 \, dA.$$

Example 16.1: A rectangular box

Let

$$E = [0, 2] \times [1, 4] \times [-1, 3].$$

Then

$$\text{Vol}(E) = \int_0^2 \int_1^4 \int_{-1}^3 1 \, dz \, dy \, dx.$$

Evaluate from inside to outside:

$$\int_{-1}^3 1 \, dz = 4,$$

so

$$\text{Vol}(E) = \int_0^2 \int_1^4 4 \, dy \, dx = \int_0^2 12 \, dx = 24.$$

Of course, the box has side lengths 2, 3, and 4, so the volume is $2 \cdot 3 \cdot 4 = 24$.

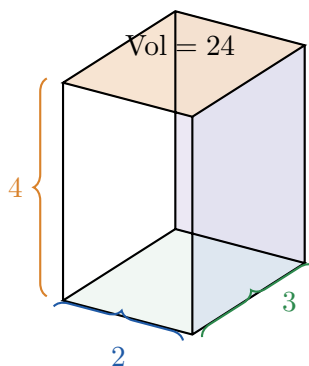


Figure 2: The triple integral of 1 over a rectangular box reproduces length times width times height.

16.3 Solids between two surfaces

A very common solid has the form

$$E = \{(x, y, z) : (x, y) \in D, \, g_1(x, y) \leq z \leq g_2(x, y)\}.$$

Here D is the projection of E onto the xy -plane. The solid lies above the lower surface $z = g_1(x, y)$ and below the upper surface $z = g_2(x, y)$. Then

$$\iiint_E f(x, y, z) \, dV = \iint_D \left[\int_{g_1(x, y)}^{g_2(x, y)} f(x, y, z) \, dz \right] dA.$$

If D is a Type I region,

$$D = \{(x, y) : a \leq x \leq b, \, h_1(x) \leq y \leq h_2(x)\},$$

then

$$\iiint_E f(x, y, z) \, dV = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{g_1(x, y)}^{g_2(x, y)} f(x, y, z) \, dz \, dy \, dx.$$

Projection-first method

To set up a triple integral, first choose a projection plane. Then write the outer two bounds for the projection and the inner bounds for the variable perpendicular to that projection.

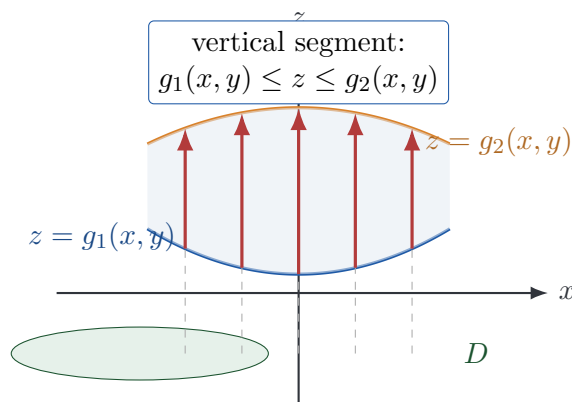


Figure 3: A solid between two surfaces is described by a projection region in the xy -plane and vertical z -bounds.

Example 16.2: Volume under a plane over a triangle

Let E be the tetrahedron in the first octant bounded by the coordinate planes and

$$x + y + z = 1.$$

The region is

$$E = \{(x, y, z) : x \geq 0, y \geq 0, z \geq 0, x + y + z \leq 1\}.$$

Project onto the xy -plane. Since z ranges from 0 to $1 - x - y$, the projection is

$$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}.$$

Thus

$$\text{Vol}(E) = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} 1 \, dz \, dy \, dx.$$

Evaluate:

$$\text{Vol}(E) = \int_0^1 \int_0^{1-x} (1 - x - y) \, dy \, dx = \int_0^1 \frac{(1-x)^2}{2} \, dx = \frac{1}{6}.$$

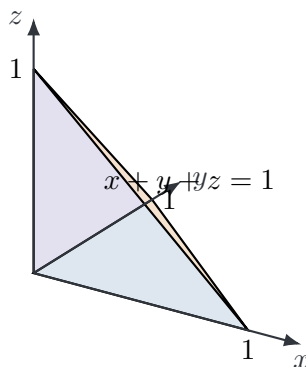


Figure 4: The tetrahedron $x + y + z \leq 1$ in the first octant. Its projection onto the xy -plane is a triangle.

16.4 Integrating functions over solids

Volume is only one example. If $f(x, y, z)$ measures a density, intensity, probability density, energy density, or cost density, then

$$\iiint_E f(x, y, z) dV$$

sums that quantity over the whole solid.

Example 16.3: A triple integral over the standard tetrahedron

Evaluate

$$\iiint_E y dV,$$

where

$$E = \{(x, y, z) : x \geq 0, y \geq 0, z \geq 0, x + y + z \leq 1\}.$$

Using the same bounds as before,

$$\iiint_E y dV = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} y dz dy dx.$$

Since y does not depend on z ,

$$\int_0^{1-x-y} y dz = y(1-x-y).$$

Thus

$$\iiint_E y dV = \int_0^1 \int_0^{1-x} y(1-x-y) dy dx = \int_0^1 \frac{(1-x)^3}{6} dx = \frac{1}{24}.$$

Because of symmetry,

$$\iiint_E x dV = \iiint_E y dV = \iiint_E z dV = \frac{1}{24}.$$

16.5 Mass and density in three dimensions

Suppose an object occupies a solid region E and has density $\rho = \rho(x, y, z)$. The mass of a small box is approximately

$$\Delta m \approx \rho(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \Delta V.$$

Summing and passing to a limit gives the mass formula.

Mass from density

If an object occupies $E \subset \mathbb{R}^3$ and has density $\rho(x, y, z)$, then

$$M = \iiint_E \rho(x, y, z) dV.$$

Example 16.4: Density increasing with height

Let E be the box

$$0 \leq x \leq 2, \quad 0 \leq y \leq 3, \quad 0 \leq z \leq 1,$$

and suppose the density is

$$\rho(x, y, z) = 1 + z.$$

Then

$$M = \int_0^2 \int_0^3 \int_0^1 (1 + z) \, dz \, dy \, dx.$$

First,

$$\int_0^1 (1 + z) \, dz = \frac{3}{2}.$$

Thus

$$M = \int_0^2 \int_0^3 \frac{3}{2} \, dy \, dx = \int_0^2 \frac{9}{2} \, dx = 9.$$

If the density were constant 1, the mass would be the volume 6. Since the density increases with height, the mass is larger.

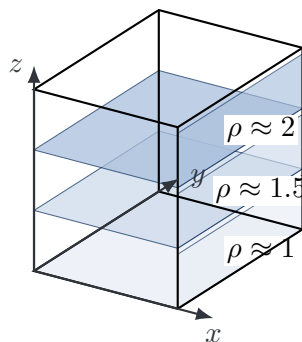


Figure 5: Density $\rho = 1 + z$ is larger in higher horizontal layers, so mass exceeds volume.

16.6 Average value over a solid

For a function $f(x, y, z)$ on a solid E , its average value over E is

$$f_{\text{avg}} = \frac{1}{\text{Vol}(E)} \iiint_E f(x, y, z) \, dV.$$

This is the three-dimensional version of the average value of a function over an interval or a plane region.

Example 16.5: Average height in a tetrahedron

For the standard tetrahedron

$$E = \{(x, y, z) : x, y, z \geq 0, \, x + y + z \leq 1\},$$

the average value of z is

$$z_{\text{avg}} = \frac{1}{\text{Vol}(E)} \iiint_E z \, dV.$$

By symmetry, $\iiint_E z \, dV = 1/24$ and $\text{Vol}(E) = 1/6$. Therefore

$$z_{\text{avg}} = \frac{1/24}{1/6} = \frac{1}{4}.$$

The centroid of the tetrahedron is

$$\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right).$$

This example previews the role of triple integrals in computing centers of mass.

16.7 Other projection directions

Projection onto the xy -plane is common, but not always best. A solid may also be described by projecting onto the yz -plane or the xz -plane.

Projection onto the yz -plane

If

$$E = \{(x, y, z) : (y, z) \in D, \, g_1(y, z) \leq x \leq g_2(y, z)\},$$

then

$$\iiint_E f(x, y, z) \, dV = \iint_D \left[\int_{g_1(y, z)}^{g_2(y, z)} f(x, y, z) \, dx \right] dA.$$

Projection onto the xz -plane

If

$$E = \{(x, y, z) : (x, z) \in D, \, g_1(x, z) \leq y \leq g_2(x, z)\},$$

then

$$\iiint_E f(x, y, z) \, dV = \iint_D \left[\int_{g_1(x, z)}^{g_2(x, z)} f(x, y, z) \, dy \right] dA.$$

Do not choose the projection too early

A difficult integral often becomes easier if you project in a different direction. Before writing bounds, sketch the solid and ask: which variable naturally runs between two simple surfaces?

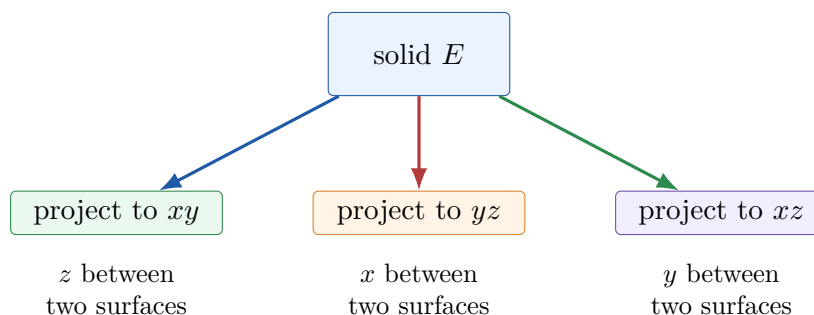


Figure 6: Projection choice determines which variable becomes the innermost integral.

16.8 Changing the order of integration

In double integrals, changing order means rewriting a region in the plane. In triple integrals, changing order means rewriting a solid in space. There are six possible orders:

$$dz \, dy \, dx, \quad dz \, dx \, dy, \quad dy \, dz \, dx, \quad dy \, dx \, dz, \quad dx \, dz \, dy, \quad dx \, dy \, dz.$$

The main principle is simple:

The innermost variable describes a line segment through the solid.

Example 16.6: The tetrahedron in another order

For the standard tetrahedron

$$E = \{(x, y, z) : x, y, z \geq 0, \, x + y + z \leq 1\},$$

we wrote

$$\iiint_E f(x, y, z) \, dV = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} f(x, y, z) \, dz \, dy \, dx.$$

Now write the order $dx \, dz \, dy$. For fixed y and z ,

$$0 \leq x \leq 1 - y - z.$$

The projection onto the yz -plane is

$$0 \leq y \leq 1, \quad 0 \leq z \leq 1 - y.$$

Therefore

$$\iiint_E f(x, y, z) \, dV = \int_0^1 \int_0^{1-y} \int_0^{1-y-z} f(x, y, z) \, dx \, dz \, dy.$$

Because the tetrahedron is symmetric, all six orders have similar bounds.

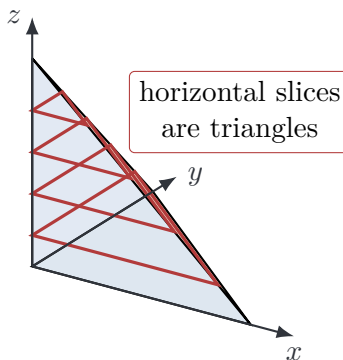


Figure 7: Horizontal slices of the first-octant tetrahedron are triangles. Slicing helps rewrite bounds in a new order.

16.9 Triple integrals and probability

A joint probability density for three continuous random variables X, Y, Z is a nonnegative function $p(x, y, z)$ such that

$$\iiint_{\mathbb{R}^3} p(x, y, z) dV = 1.$$

For a solid event $E \subset \mathbb{R}^3$,

$$P((X, Y, Z) \in E) = \iiint_E p(x, y, z) dV.$$

This interpretation is essential in statistics and machine learning. Many models use high-dimensional integrals, but the geometric meaning begins here: probability is accumulated density over a region.

Example 16.7: Uniform distribution in a box

Suppose (X, Y, Z) is uniformly distributed in the box

$$0 \leq x \leq 2, \quad 0 \leq y \leq 1, \quad 0 \leq z \leq 3.$$

The volume is 6, so the density is $p(x, y, z) = 1/6$ inside the box and 0 outside. The probability of the sub-box

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1, \quad 1 \leq z \leq 3$$

is

$$\int_0^1 \int_0^1 \int_1^3 \frac{1}{6} dz dy dx = \frac{2}{6} = \frac{1}{3}.$$

16.10 Numerical triple integration

In practice, many triple integrals cannot be evaluated exactly. Numerical integration is then necessary. For a rectangular box $[a, b] \times [c, d] \times [p, q]$, a midpoint rule approximation is

$$\iiint_E f(x, y, z) dV \approx \sum_{i=1}^m \sum_{j=1}^n \sum_{k=1}^{\ell} f(\bar{x}_i, \bar{y}_j, \bar{z}_k) \Delta x \Delta y \Delta z.$$

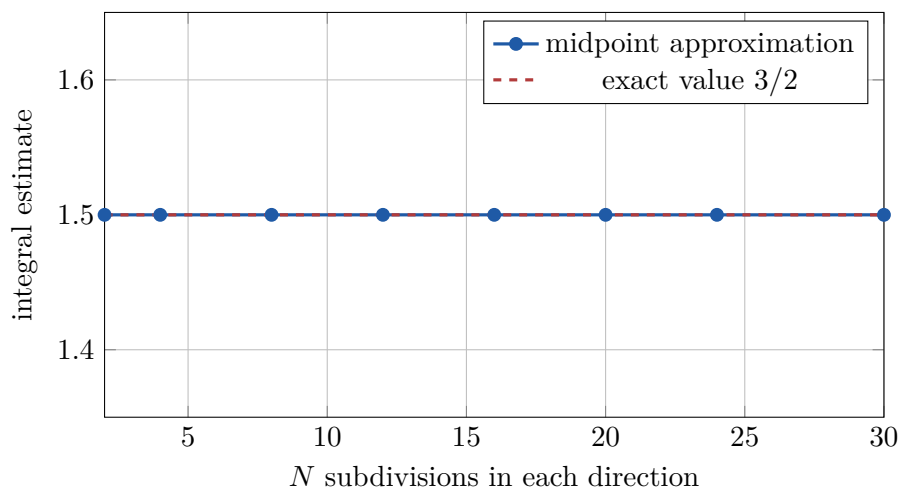


Figure 8: For $f(x, y, z) = x + y + z$ over the unit cube, the midpoint rule is exact for every grid size because the function is linear.

For nonlinear functions, the approximations generally improve as the grid is refined.

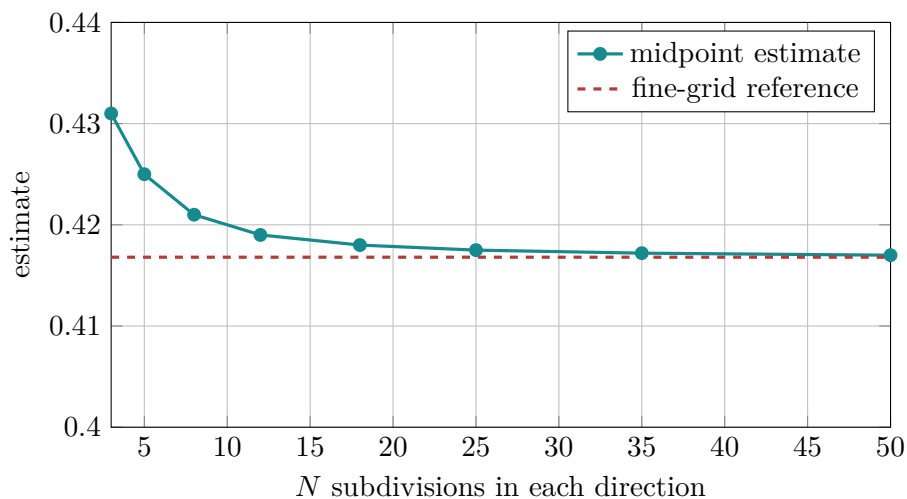


Figure 9: Midpoint approximations for a nonlinear function over the unit cube stabilize as the grid is refined.

16.11 Monte Carlo integration in three dimensions

Monte Carlo integration estimates an integral by random sampling. Suppose E is a box of volume V and we want

$$I = \iiint_E f(x, y, z) dV.$$

If $P_1, \dots, P_N$ are uniformly sampled from E , then

$$I \approx V \cdot \frac{1}{N} \sum_{i=1}^N f(P_i).$$

Monte Carlo methods become especially important in high-dimensional statistics, Bayesian inference, uncertainty quantification, and machine learning.

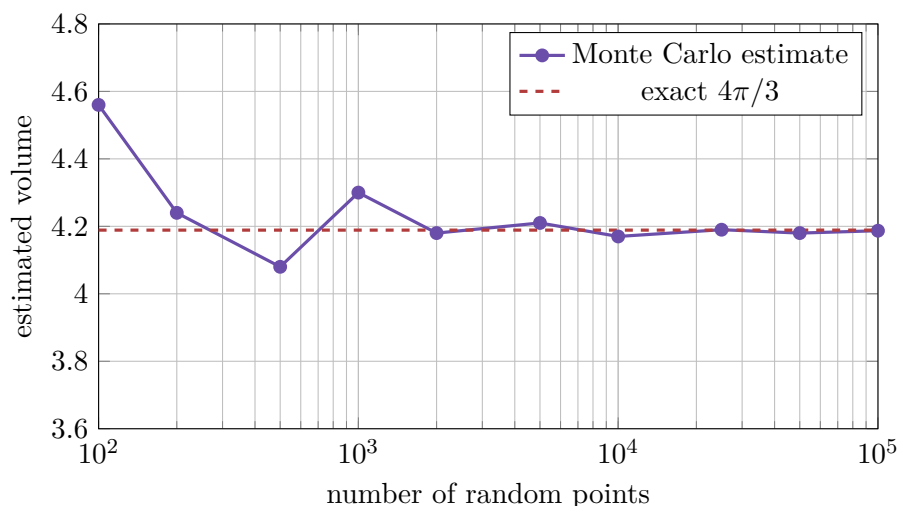


Figure 10: Monte Carlo estimation of the volume of the unit ball. Random estimates fluctuate but tend to stabilize with many samples.

16.12 Moments and centers of mass

Triple integrals also compute balance points. Suppose an object occupies E with density $\rho(x, y, z)$ and mass

$$M = \iiint_E \rho(x, y, z) dV.$$

The moments about the coordinate planes are

$$M_{yz} = \iiint_E x\rho(x, y, z) dV, \quad M_{xz} = \iiint_E y\rho(x, y, z) dV, \quad M_{xy} = \iiint_E z\rho(x, y, z) dV.$$

The center of mass is

$$(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{M_{yz}}{M}, \frac{M_{xz}}{M}, \frac{M_{xy}}{M} \right).$$

If ρ is constant, the center of mass is the centroid of the region.

Example 16.8: Centroid of the standard tetrahedron

For the standard tetrahedron, by symmetry and the computation above,

$$\bar{x} = \bar{y} = \bar{z} = \frac{1}{4}.$$

Thus the centroid is

$$\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right).$$

This point is closer to the origin than to the opposite face because the tetrahedron is widest near the coordinate planes.

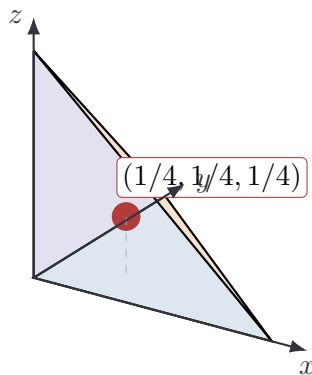


Figure 11: The centroid of the first-octant tetrahedron is $(1/4, 1/4, 1/4)$.

16.13 Triple integrals in the age of data and AI

Triple integrals appear naturally in modern applied mathematics, statistics, and machine learning.

Spatial data and voxel models

Medical imaging, climate models, fluid simulations, and three-dimensional computer vision often represent space using voxels. A voxel is a small three-dimensional box. A triple integral is the mathematical limit of summing quantities over voxels.

Examples include total mass in a scanned object, total heat in a region, average intensity in a three-dimensional image, and probability that a random spatial point lies in a target region.

Probability and Bayesian inference

For three parameters $\theta_1, \theta_2, \theta_3$, a posterior density $p(\theta_1, \theta_2, \theta_3 \mid \text{data})$ satisfies

$$\iiint p(\theta_1, \theta_2, \theta_3 \mid \text{data}) d\theta_1 d\theta_2 d\theta_3 = 1.$$

A probability statement about the parameters is a triple integral over a region in parameter space.

Continuous loss and risk

In statistical learning, expected loss may be written as an integral over feature space. In three variables, a risk functional may look like

$$R(w) = \iiint L(w; x, y, z) p(x, y, z) dV.$$

This is a triple integral of loss weighted by a probability density.

16.14 Common mistakes

Mistake 1: Forgetting the projection

Bounds such as $0 \leq z \leq g(x, y)$ are incomplete unless the allowed (x, y) values are described.

Mistake 2: Using the wrong upper surface

Always check which surface is above on the projection region.

Mistake 3: Mixing the order of bounds and differentials

In $dz dy dx$, the z -bounds come first, then y -bounds, then x -bounds.

Mistake 4: Treating dependent variables as constants incorrectly

When integrating with respect to z , expressions involving x and y are constant, but the z -bounds may depend on x and y .

Mistake 5: Assuming every solid is a box

Most real regions require variable bounds.

Mistake 6: Not checking units

If f has units of density, then $f dV$ has units of mass.

16.15 Summary

A triple integral is a continuous sum over a solid region:

$$\iiint_E f(x, y, z) dV.$$

The most important special cases are:

$$\text{Vol}(E) = \iiint_E 1 dV, \quad M = \iiint_E \rho(x, y, z) dV, \quad f_{\text{avg}} = \frac{1}{\text{Vol}(E)} \iiint_E f(x, y, z) dV.$$

The most common rectangular-coordinate setup is

$$E = \{(x, y, z) : (x, y) \in D, \ g_1(x, y) \leq z \leq g_2(x, y)\},$$

which gives

$$\iiint_E f(x, y, z) dV = \iint_D \left[\int_{g_1(x, y)}^{g_2(x, y)} f(x, y, z) dz \right] dA.$$

The key skill is not merely integration. The key skill is **turning geometry into correct bounds**.

Exercises

Conceptual exercises

1. Explain why $\iiint_E 1 \, dV$ gives the volume of E .
2. What is the projection of a solid onto the xy -plane?
3. In the integral

$$\int_0^2 \int_0^{4-x} \int_{x+y}^5 f(x, y, z) \, dz \, dy \, dx,$$

describe the solid region in words.

4. Why might it be easier to project a region onto the yz -plane instead of the xy -plane?
5. Explain the difference between mass and volume in terms of triple integrals.
6. What does it mean for $p(x, y, z)$ to be a joint probability density?

Computational exercises

7. Evaluate $\int_0^1 \int_0^2 \int_0^3 (x + y + z) \, dz \, dy \, dx$.
8. Find the volume of the box $-1 \leq x \leq 2$, $0 \leq y \leq 4$, $1 \leq z \leq 5$.
9. Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} 1 \, dz \, dy \, dx$.
10. Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} x \, dz \, dy \, dx$.
11. Let E be the solid under $z = 4 - x^2 - y^2$ and above the rectangle $0 \leq x \leq 1$, $0 \leq y \leq 1$. Set up a triple integral for the volume of E .
12. Let E be the solid between $z = x^2 + y^2$ and $z = 4$. Set up the volume integral in rectangular coordinates using projection onto the xy -plane. Do not evaluate.
13. Let E be the tetrahedron $x, y, z \geq 0$, $x + y + z \leq 2$. Find its volume.
14. Let $E = [0, 1]^3$ and $\rho(x, y, z) = 1 + x + y + z$. Find the mass of E .
15. Find the average value of $f(x, y, z) = z$ over the unit cube $[0, 1]^3$.

Geometry and setup exercises

16. Describe the solid represented by

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1 - x, \quad 0 \leq z \leq 2 - x - y.$$

17. Rewrite the tetrahedron integral

$$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} f(x, y, z) \, dz \, dy \, dx$$

in the order $dy \, dz \, dx$.

18. Set up, but do not evaluate, an integral for the volume of the solid above $z = 0$, below $z = 9 - x^2 - y^2$, and above the first quadrant of the xy -plane.
19. Set up a triple integral for the mass of the solid in Exercise 18 if $\rho(x, y, z) = z + 1$.
20. Let E be the region bounded by $z = 0$, $z = 1 - y$, $x = 0$, $x = 2$, and $y = 0$. Write $\iiint_E f(x, y, z) dV$ as an iterated integral.

Computational lab exercises

21. Use a midpoint rule on a grid to approximate

$$\int_0^1 \int_0^1 \int_0^1 e^{-(x^2+y^2+z^2)} dV.$$

22. Use random sampling to estimate the volume of the region $x^2 + y^2 + z^2 \leq 1$.
23. Use random sampling to estimate the volume of the tetrahedron $x, y, z \geq 0$, $x + y + z \leq 1$.
24. Plot horizontal slices of the solid under $z = 4 - x^2 - y^2$ and above the xy -plane.
25. Write a function that estimates $\iiint_E f dV$ over a box using the midpoint rule.

Python lab

A companion Python lab for this chapter should include:

- plotting basic solid regions,
- voxel approximations of volume,
- midpoint approximation for triple integrals,
- Monte Carlo estimation of volumes,
- probability integrals over 3D regions,
- center-of-mass computations for simple solids.

Chapter 17

Cylindrical and Spherical Coordinates

Triple integrals with symmetry, radial geometry, and modern applications

Chapter overview

Rectangular coordinates are natural for boxes, planes, and regions bounded by simple inequalities in x , y , and z . Many important three-dimensional regions are not rectangular. Cylinders, cones, spheres, shells, balls, hemispheres, and rotationally symmetric solids are often much easier to describe using coordinates adapted to their geometry.

This chapter develops two coordinate systems for triple integrals:

- **cylindrical coordinates** (r, θ, z) , which use polar coordinates in the xy -plane and keep the vertical coordinate z ;
- **spherical coordinates** (ρ, θ, ϕ) , which describe a point by distance from the origin and two angles.

The goal is not simply to memorize formulas. The goal is to learn how to choose coordinates that make geometry, integration, probability, and computation simpler.

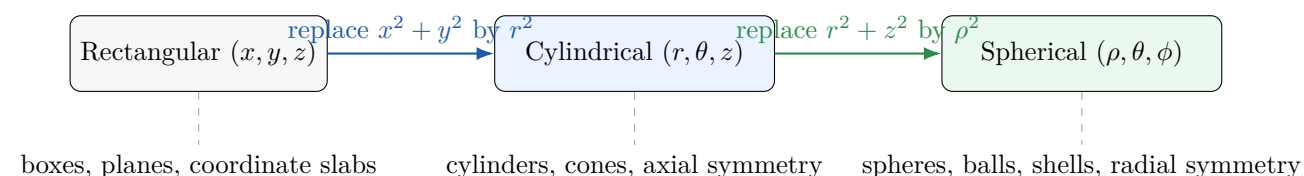


Figure 17.1: Coordinate systems as languages for geometry. The best language is usually the one whose coordinate surfaces match the boundary surfaces.

17.1 Learning goals

After studying this chapter, you should be able to:

1. convert points and equations among rectangular, cylindrical, and spherical coordinates;
2. recognize common coordinate surfaces: cylinders, cones, spheres, and half-planes;
3. use the volume elements $dV = r \, dz \, dr \, d\theta$ and $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$;

4. set up and evaluate triple integrals in cylindrical coordinates;
5. set up and evaluate triple integrals in spherical coordinates;
6. choose an efficient coordinate system by reading the symmetry of the region and integrand;
7. use coordinate transformations in probability, statistics, and simulation.

17.2 Why new coordinates are needed

A coordinate system is a language for describing points. Rectangular coordinates describe a point by three signed distances (x, y, z) . This is perfect for a rectangular box such as

$$0 \leq x \leq 2, \quad 1 \leq y \leq 4, \quad -1 \leq z \leq 3.$$

But the cylinder $x^2 + y^2 \leq 4$ and the sphere $x^2 + y^2 + z^2 \leq 9$ already contain distance expressions. That is the signal that radial coordinates may simplify the problem.

Coordinate choice principle

Use a coordinate system whose coordinate surfaces resemble the boundary surfaces of the region.

- Cylinders suggest cylindrical coordinates.
- Spheres suggest spherical coordinates.
- Cones often work well in either cylindrical or spherical coordinates.
- Planes and boxes often work well in rectangular coordinates.

17.3 Cylindrical coordinates

Cylindrical coordinates

Cylindrical coordinates combine polar coordinates in the xy -plane with the usual vertical coordinate z :

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

Here $r \geq 0$ and usually $0 \leq \theta \leq 2\pi$. The inverse relations are

$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}, \quad z = z.$$

The number r is the distance from (x, y, z) to the z -axis, not the distance to the origin in space.

Cylindrical equation	Geometric meaning	Rectangular form
$r = r_0$	cylinder around the z -axis	$x^2 + y^2 = r_0^2$
$\theta = \theta_0$	vertical half-plane	$y = (\tan \theta_0)x$ with orientation
$z = z_0$	horizontal plane	$z = z_0$
$z = cr$	cone	$z = c\sqrt{x^2 + y^2}$

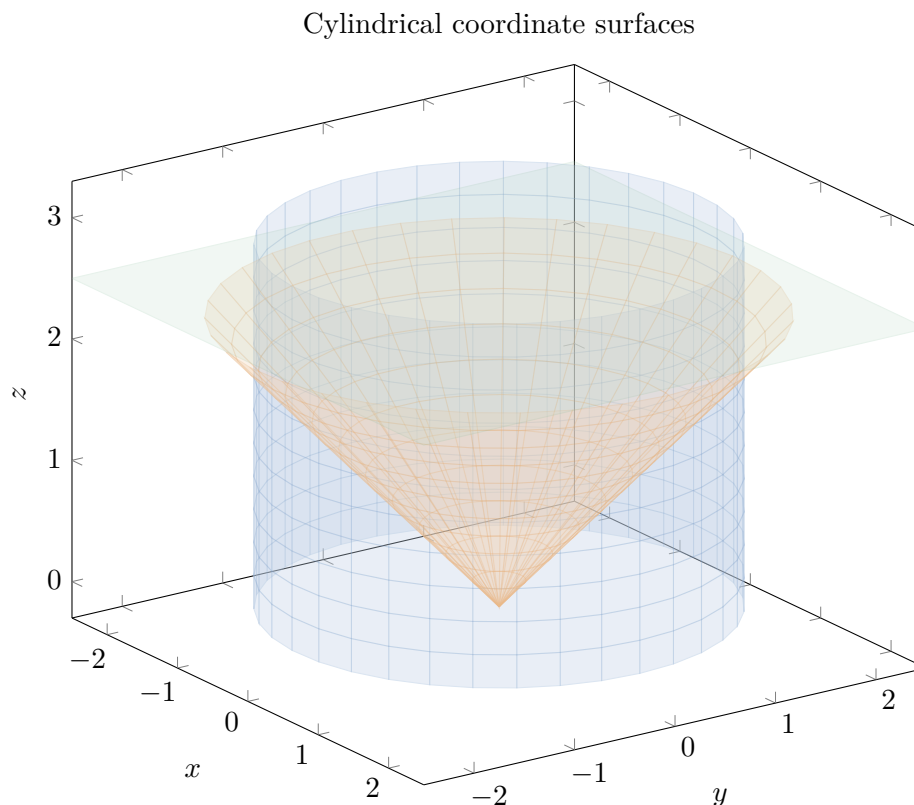


Figure 17.2: Basic cylindrical coordinate surfaces: a cylinder $r = 2$, a cone $z = r$, and a horizontal plane $z = 2.5$.

17.4 The cylindrical volume element

In rectangular coordinates, a small box has volume $dV = dx dy dz$. In cylindrical coordinates, a small coordinate box has dimensions approximately

$$dr, \quad r d\theta, \quad dz.$$

Therefore

$$dV = r dz dr d\theta.$$

The factor r accounts for the fact that an angular change $d\theta$ sweeps out a longer arc when r is larger.

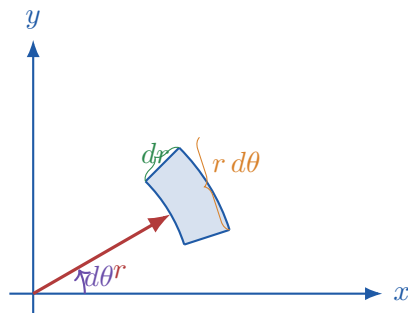


Figure 17.3: The polar base of a cylindrical volume element. Multiplying the base dimensions dr and $r d\theta$ by height dz gives $dV = r dz dr d\theta$.

Cylindrical integration formula

If $x = r \cos \theta$, $y = r \sin \theta$, and $z = z$, then

$$\iiint_E f(x, y, z) dV = \int_{\theta=\alpha}^{\beta} \int_{r=h_1(\theta)}^{h_2(\theta)} \int_{z=g_1(r, \theta)}^{g_2(r, \theta)} f(r \cos \theta, r \sin \theta, z) r dz dr d\theta.$$

The extra factor r is not optional. It is part of the volume element.

17.5 Worked examples in cylindrical coordinates

Example 1: Convert a point

Find rectangular coordinates for $(r, \theta, z) = \left(2, \frac{2\pi}{3}, 3\right)$. Using $x = r \cos \theta$ and $y = r \sin \theta$,

$$x = 2 \cos \left(\frac{2\pi}{3}\right) = -1, \quad y = 2 \sin \left(\frac{2\pi}{3}\right) = \sqrt{3},$$

and $z = 3$. Thus the rectangular coordinates are $(-1, \sqrt{3}, 3)$.

Example 2: A cone below a plane

Find the volume of the solid above the cone $z = r$ and below the plane $z = 1$. The cone and plane intersect when $r = 1$. The region is

$$0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq 1, \quad r \leq z \leq 1.$$

Thus

$$V = \int_0^{2\pi} \int_0^1 \int_r^1 r dz dr d\theta = 2\pi \int_0^1 r(1-r) dr = \frac{\pi}{3}.$$

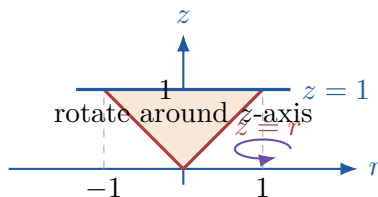


Figure 17.4: A cross-section of the solid above $z = r$ and below $z = 1$. Cylindrical coordinates turn the solid into $0 \leq r \leq 1$, $r \leq z \leq 1$.

Example 3: A cylinder above a paraboloid

Let E be the solid inside $x^2 + y^2 \leq 1$, above the paraboloid $z = 1 - x^2 - y^2$, and below the plane $z = 3$. Since $x^2 + y^2 = r^2$, the bounds are

$$0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq 1, \quad 1 - r^2 \leq z \leq 3.$$

Hence

$$V = \int_0^{2\pi} \int_0^1 \int_{1-r^2}^3 r \, dz \, dr \, d\theta = 2\pi \int_0^1 r(2 + r^2) \, dr = \frac{5\pi}{2}.$$

Cylinder, paraboloid, and plane

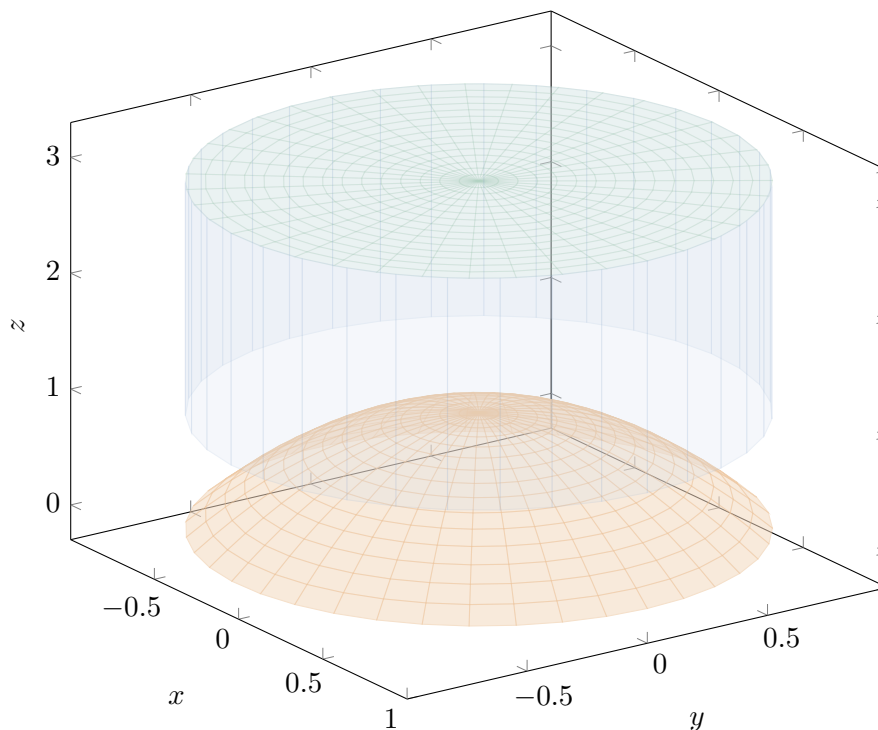


Figure 17.5: A solid inside the cylinder $r = 1$, above $z = 1 - r^2$, and below $z = 3$.

Example 4: Mass with radial density

For the same solid E , suppose $\delta(x, y, z) = \sqrt{x^2 + y^2}$. In cylindrical coordinates, $\delta = r$. Mass is density integrated over volume:

$$M = \int_0^{2\pi} \int_0^1 \int_{1-r^2}^3 r \cdot r \, dz \, dr \, d\theta.$$

The first r is the density and the second r is the Jacobian factor. Therefore

$$M = 2\pi \int_0^1 r^2(2 + r^2) \, dr = 2\pi \left(\frac{2}{3} + \frac{1}{5} \right) = \frac{26\pi}{15}.$$

Example 5: Converting a rectangular integral

Evaluate

$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^1 (x^2 + y^2) \, dz \, dy \, dx.$$

The projection onto the xy -plane is the unit disk. The lower surface is the cone $z = \sqrt{x^2 + y^2} = r$, the upper surface is $z = 1$, and the integrand becomes r^2 . Therefore

$$\int_0^{2\pi} \int_0^1 \int_r^1 r^2 \, dz \, dr \, d\theta = 2\pi \int_0^1 r^3(1 - r) \, dr = \frac{\pi}{10}.$$

17.6 Spherical coordinates

Spherical coordinates

Spherical coordinates describe a point by (ρ, θ, ϕ) , where ρ is the distance from the origin, θ is the angle in the xy -plane measured from the positive x -axis, and ϕ is the angle from the positive z -axis. The conversion formulas are

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi.$$

Usually $\rho \geq 0$, $0 \leq \theta \leq 2\pi$, and $0 \leq \phi \leq \pi$. The relations with cylindrical coordinates are

$$r = \rho \sin \phi, \quad z = \rho \cos \phi, \quad \theta = \theta.$$

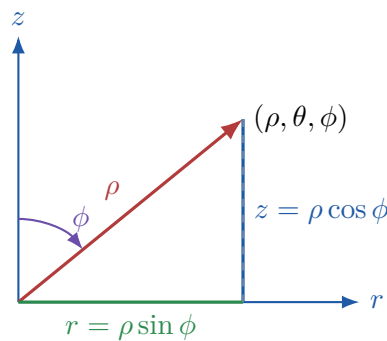


Figure 17.6: In a vertical cross-section through the z -axis, spherical coordinates reduce to right-triangle geometry: $r = \rho \sin \phi$ and $z = \rho \cos \phi$.

Spherical equation	Geometric meaning	Rectangular or cylindrical form
$\rho = \rho_0$	sphere centered at origin	$x^2 + y^2 + z^2 = \rho_0^2$
$\theta = \theta_0$	vertical half-plane	same as cylindrical angle
$\phi = \phi_0$	cone with vertex at origin	$z = r \cot \phi_0$

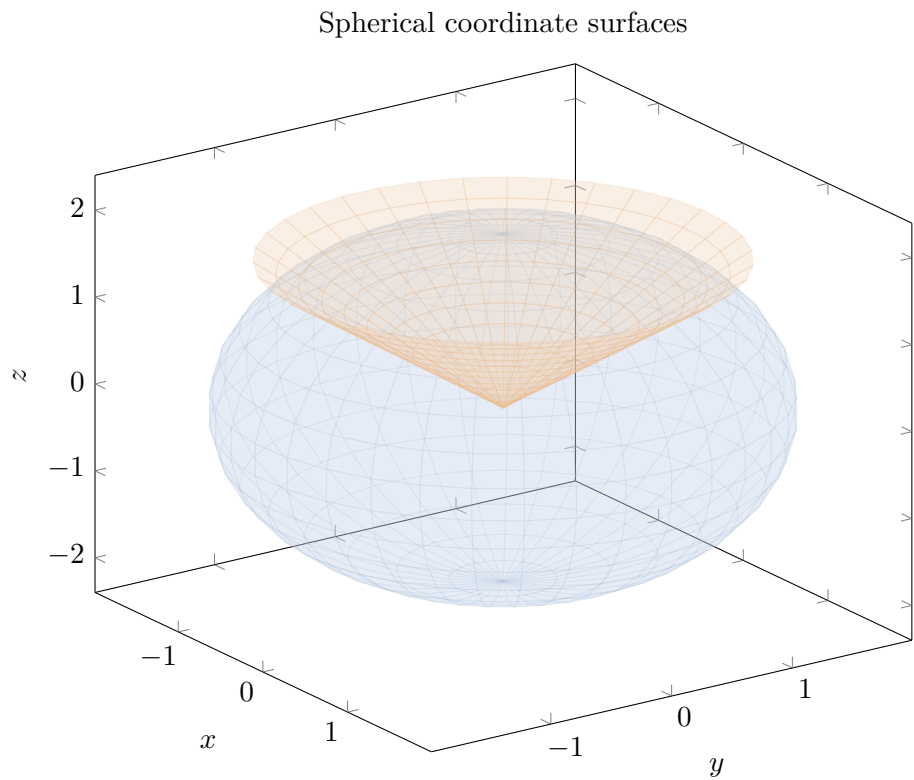


Figure 17.7: Basic spherical coordinate surfaces: the sphere $\rho = 2$ and the cone $\phi = \pi/4$.

17.7 The spherical volume element

A small spherical coordinate box has approximate side lengths

$$d\rho, \quad \rho d\phi, \quad \rho \sin \phi d\theta.$$

Therefore

$$dV = \rho^2 \sin \phi d\rho d\phi d\theta.$$

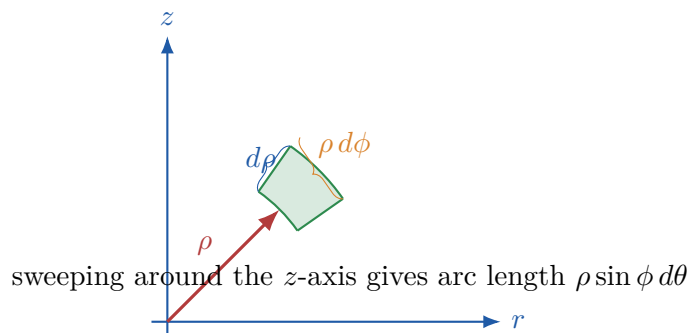


Figure 17.8: The spherical volume factor is a product of three local lengths: $d\rho$, $\rho d\phi$, and $\rho \sin \phi d\theta$.

Spherical integration formula

If

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi,$$

then

$$\iiint_E f(x, y, z) dV = \iiint f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi d\rho d\phi d\theta.$$

The factor $\rho^2 \sin \phi$ is the three-dimensional Jacobian factor for spherical coordinates.

17.8 Worked examples in spherical coordinates

Example 6: Volume of the unit ball

For the unit ball $x^2 + y^2 + z^2 \leq 1$, spherical bounds are

$$0 \leq \rho \leq 1, \quad 0 \leq \phi \leq \pi, \quad 0 \leq \theta \leq 2\pi.$$

Thus

$$V = \int_0^{2\pi} \int_0^\pi \int_0^1 \rho^2 \sin \phi d\rho d\phi d\theta = (2\pi)(2) \left(\frac{1}{3} \right) = \frac{4\pi}{3}.$$

Example 7: Inside a sphere and above a cone

Find the volume inside $x^2 + y^2 + z^2 = 8$ and above $z = \sqrt{x^2 + y^2}$. The sphere is $\rho = \sqrt{8}$. The cone is $z = r$, so

$$\rho \cos \phi = \rho \sin \phi,$$

which gives $\phi = \pi/4$ away from the origin. Hence

$$0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \frac{\pi}{4}, \quad 0 \leq \rho \leq \sqrt{8}.$$

Therefore

$$V = \int_0^{2\pi} \int_0^{\pi/4} \int_0^{\sqrt{8}} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \frac{32\pi}{3}(\sqrt{2} - 1).$$

Sphere-cone solid as a spherical box

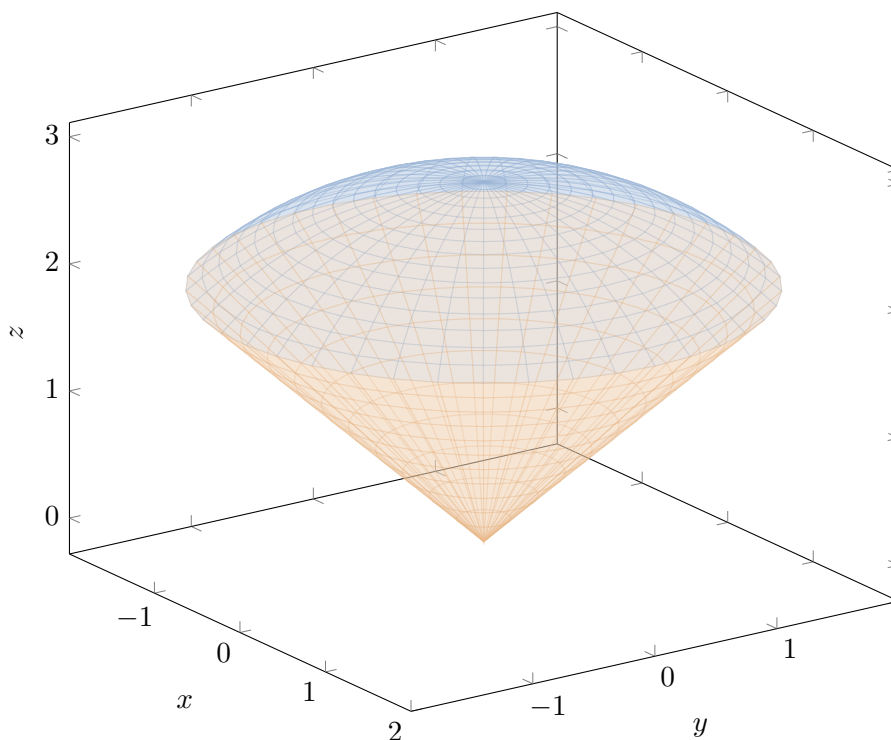


Figure 17.9: The region inside a sphere and above a cone becomes a simple rectangular box in (ρ, ϕ, θ) -space.

Example 8: A spherical shell in the first octant

Let E be the part of the spherical shell $1 \leq x^2 + y^2 + z^2 \leq 4$ in the first octant. The first octant means $x, y, z \geq 0$, so

$$0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq \phi \leq \frac{\pi}{2}.$$

The shell condition gives $1 \leq \rho \leq 2$. Hence

$$V = \int_0^{\pi/2} \int_0^{\pi/2} \int_1^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \left(\frac{\pi}{2}\right) (1) \left(\frac{2^3 - 1^3}{3}\right) = \frac{7\pi}{6}.$$

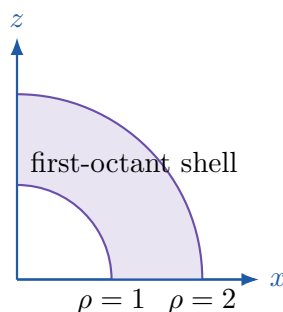


Figure 17.10: A cross-section of the first-octant spherical shell. The full three-dimensional region has $0 \leq \theta \leq \pi/2$ and $0 \leq \phi \leq \pi/2$.

17.9 Choosing among rectangular, cylindrical, and spherical coordinates

Coordinate choice is part of problem solving. A good choice makes the region and integrand simple. A poor choice may make a straightforward problem unnecessarily complicated.

Geometry or expression	Recommended coordinates	Reason
Box with constant x, y, z bounds	rectangular	boundaries already align with coordinate planes
Cylinder $x^2 + y^2 = R^2$	cylindrical	becomes $r = R$
Cone $z = c\sqrt{x^2 + y^2}$	cylindrical or spherical	becomes $z = cr$ or $\phi = \text{constant}$
Sphere $x^2 + y^2 + z^2 = R^2$	spherical	becomes $\rho = R$
Radial integrand $f(x^2 + y^2)$	cylindrical	becomes $f(r^2)$
Radial integrand $f(x^2 + y^2 + z^2)$	spherical	becomes $f(\rho^2)$
First octant of a ball	spherical	constant angle bounds
Solid of revolution around the z -axis	cylindrical	natural r and z description

A practical workflow

1. Sketch or identify the boundary surfaces.
2. Translate the boundary equations into candidate coordinate systems.
3. Choose the system where most bounds become constants or simple functions.

4. Rewrite the integrand.
5. Include the correct volume element.

17.10 Numerical integration in cylindrical and spherical coordinates

The coordinate formulas are useful not only for exact integration but also for computation. A numerical method works best when the region has simple bounds. For the sphere-cone solid from Example 7,

$$0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \frac{\pi}{4}, \quad 0 \leq \rho \leq \sqrt{8}.$$

A grid-based spherical-coordinate sum approximates the volume by

$$\sum \rho_{ijk}^2 \sin \phi_j \Delta \rho \Delta \phi \Delta \theta.$$

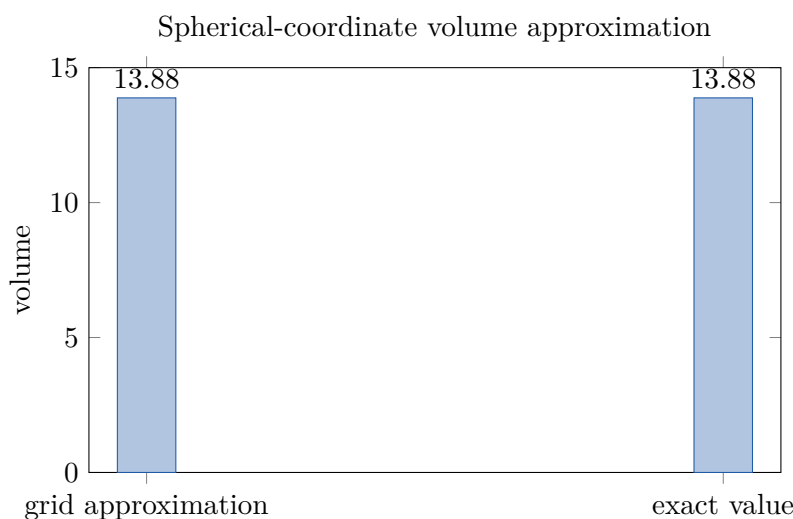


Figure 17.11: For regions with spherical bounds, a grid in (ρ, ϕ, θ) directly matches the geometry.

17.11 Coordinate transformations and probability

Triple integrals are also probability integrals. If $f(x, y, z)$ is a joint density, then

$$P((X, Y, Z) \in E) = \iiint_E f(x, y, z) dV.$$

When the density is radially symmetric, spherical coordinates are natural. For example, the three-dimensional standard normal density is

$$f(x, y, z) = \frac{1}{(2\pi)^{3/2}} e^{-(x^2+y^2+z^2)/2}.$$

Since $x^2 + y^2 + z^2 = \rho^2$, spherical coordinates give

$$f = \frac{1}{(2\pi)^{3/2}} e^{-\rho^2/2}.$$

The probability mass in a ball of radius R is

$$P(\rho \leq R) = \int_0^{2\pi} \int_0^\pi \int_0^R \frac{1}{(2\pi)^{3/2}} e^{-\rho^2/2} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

After integrating out the angles,

$$P(\rho \leq R) = \sqrt{\frac{2}{\pi}} \int_0^R \rho^2 e^{-\rho^2/2} \, d\rho.$$

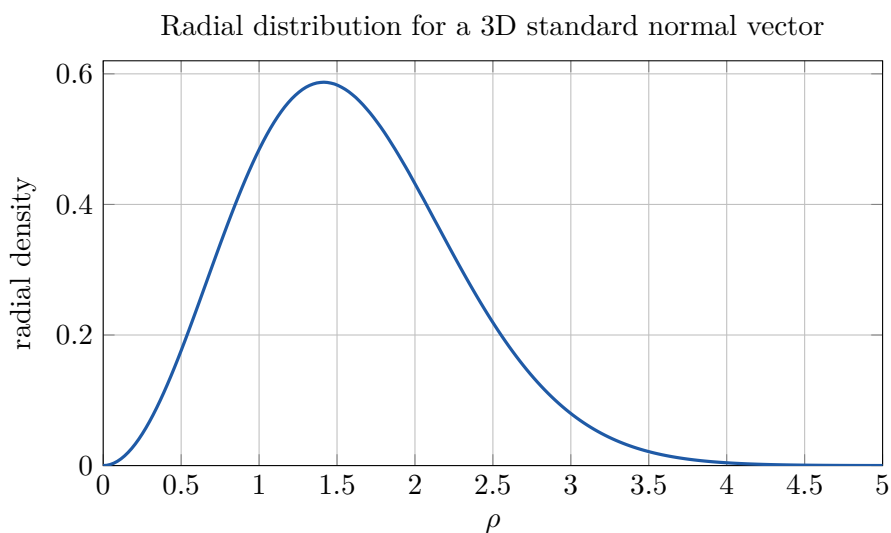


Figure 17.12: A rotationally symmetric three-dimensional density becomes a one-dimensional radial density after integrating out angular variables.

17.12 Sampling uniformly from a ball

Coordinate transformations matter in simulation. To sample uniformly from the unit ball in $\mathbb{R}^3$, it is not correct to choose ρ uniformly from $[0, 1]$. The volume element contains ρ^2 , so more volume lies near the boundary.

A correct method is

$$\theta = 2\pi U_1, \quad \cos \phi = 2U_2 - 1, \quad \rho = U_3^{1/3},$$

where U_1, U_2, U_3 are independent uniform random variables on $[0, 1]$. The power $1/3$ appears because the volume of a ball of radius r is proportional to r^3 .

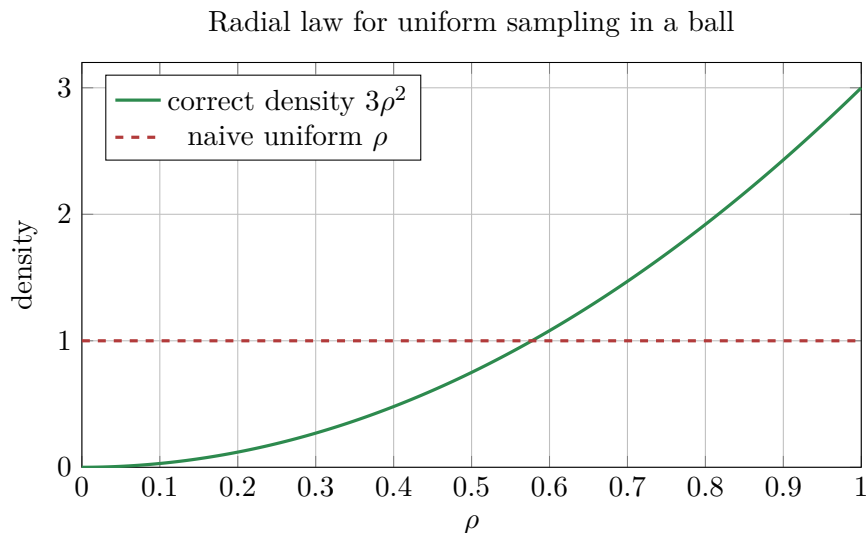


Figure 17.13: Uniform volume sampling has radial density $3\rho^2$, not a flat radial density.

17.13 Common mistakes

Mistake 1: Forgetting the Jacobian factor

In cylindrical coordinates, the volume element is $r \, dz \, dr \, d\theta$, not just $dz \, dr \, d\theta$. In spherical coordinates, the volume element is $\rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$, not just $d\rho \, d\phi \, d\theta$.

Mistake 2: Confusing r and ρ

In cylindrical coordinates, $r = \sqrt{x^2 + y^2}$. In spherical coordinates, $\rho = \sqrt{x^2 + y^2 + z^2}$. Thus r is distance to the z -axis, while ρ is distance to the origin.

Mistake 3: Confusing θ and ϕ

The angle θ is measured in the xy -plane. The angle ϕ is measured down from the positive z -axis.

Mistake 4: Writing bounds before understanding the geometry

Do not begin by guessing bounds. First translate the boundary surfaces. Then decide which coordinate system makes them simple.

17.14 Chapter summary

Cylindrical coordinates are defined by

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z,$$

with volume element

$$dV = r \, dz \, dr \, d\theta.$$

Spherical coordinates are defined by

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi,$$

with volume element

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

The main problem-solving skill is choosing coordinates that match the region and integrand. Cylindrical coordinates simplify cylinders, cones, and axial symmetry. Spherical coordinates simplify spheres, balls, shells, cones, and radial functions.

17.15 Practice problems

1. Convert the cylindrical point $(4, \pi/6, 2)$ to rectangular coordinates.
2. Convert the rectangular point $(1, 1, 3)$ to cylindrical coordinates.
3. Describe the surface $r = 5$ in rectangular coordinates.
4. Describe the surface $z = 2r$ in rectangular coordinates.
5. Find the volume inside $x^2 + y^2 = 9$ between $z = 0$ and $z = 4$.
6. Find the mass of the solid in Problem 5 if the density is $\delta(x, y, z) = x^2 + y^2$.
7. Set up a cylindrical-coordinate integral for the volume under $z = 4 - x^2 - y^2$ and above the xy -plane.
8. Evaluate the integral from Problem 7.
9. Convert the spherical point $(\rho, \theta, \phi) = (2, \pi/3, \pi/4)$ to rectangular coordinates.
10. Describe the surface $\rho = 3$ in rectangular coordinates.
11. Describe the surface $\phi = \pi/3$ in rectangular coordinates.
12. Find the volume of the ball $x^2 + y^2 + z^2 \leq 27$.
13. Find the volume of the upper hemisphere $x^2 + y^2 + z^2 \leq 9, z \geq 0$.
14. Set up a spherical-coordinate integral for the volume inside $x^2 + y^2 + z^2 = 16$ and above the cone $z = \sqrt{3(x^2 + y^2)}$.
15. Evaluate the integral in Problem 14.
16. Let E be the first-octant part of the ball $x^2 + y^2 + z^2 \leq 8$. Set up and evaluate a spherical-coordinate integral for its volume.
17. Explain why choosing ρ uniformly from $[0, 1]$ does not produce uniform points in the unit ball.
18. For the density $f(x, y, z) = Ce^{-(x^2+y^2+z^2)}$, write the normalization condition for C using spherical coordinates.

17.16 Python lab and interactive page

The lab for this chapter asks you to plot cylindrical and spherical coordinate surfaces, compare descriptions of the same solid, approximate triple integrals with coordinate grids, simulate points uniformly in a ball, and compare radial distributions.

- Python lab: `../labs/chapter-17-cylindrical-and-spherical-coordinates.ipynb`
- Interactive page: `../htmls/chapter-17-cylindrical-and-spherical-coordinates.html`

Chapter 18. Multivariable Taylor Approximation and Quadratic Forms

Second-order geometry for optimization, statistics, and machine learning

Linear approximation gives the best flat approximation to a differentiable function near a point. It answers a first-order question: if the input changes a little, what is the leading-order change in the output? Second-order approximation asks a deeper question: how does the function curve near the point?

The central object is the **Hessian matrix**. The gradient gives a vector of first derivatives. The Hessian gives a matrix of second derivatives. Near a point, the Hessian controls the quadratic part of a function and therefore controls the local geometry of bowls, hills, saddles, flat directions, least-squares losses, likelihoods, and Newton steps.

Learning goals

By the end of this chapter, you should be able to:

1. write first- and second-order Taylor approximations for functions of several variables;
2. compute and interpret the Hessian matrix;
3. connect quadratic forms $h^T H h$ with local curvature;
4. classify quadratic forms as positive definite, negative definite, indefinite, or semidefinite;
5. use eigenvalues of the Hessian to understand bowls, hills, saddles, and flat directions;
6. explain why second-order approximation is important in optimization, statistics, and machine learning.

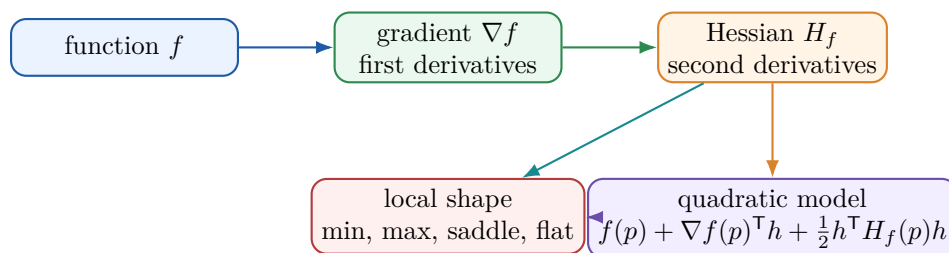


Figure 1: The flow of second-order thinking: first derivatives build the gradient; second derivatives build the Hessian; the Hessian controls the local quadratic model and local shape.

1 From linear approximation to quadratic approximation

For a one-variable function, the Taylor approximation at $x = a$ begins as

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{1}{2} f''(a)(x - a)^2.$$

The first two terms form the tangent-line approximation. The third term adds curvature.

For a two-variable function $f(x, y)$ near (a, b) , the linear approximation is

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

The second-order approximation adds all possible second-order terms:

$$\begin{aligned} T_2(x, y) = & f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b) \\ & + \frac{1}{2}f_{xx}(a, b)(x - a)^2 + f_{xy}(a, b)(x - a)(y - b) + \frac{1}{2}f_{yy}(a, b)(y - b)^2. \end{aligned}$$

The mixed term appears with coefficient $f_{xy}(a, b)$ because the symmetric matrix form contains two equal mixed terms and the factor $1/2$ balances them.

Compact vector notation

Let

$$p = \begin{bmatrix} a \\ b \end{bmatrix}, \quad x = \begin{bmatrix} x \\ y \end{bmatrix}, \quad h = x - p = \begin{bmatrix} x - a \\ y - b \end{bmatrix}.$$

Then the quadratic Taylor approximation is

$$f(p + h) \approx f(p) + \nabla f(p)^\top h + \frac{1}{2}h^\top H_f(p)h.$$

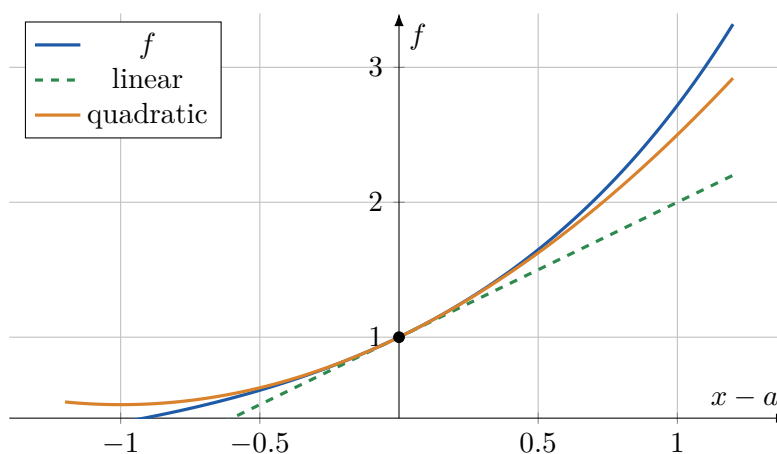


Figure 2: The one-variable picture: the tangent line gives first-order information; the quadratic approximation adds curvature.

2 The Hessian matrix

For a function $f(x, y)$ with continuous second partial derivatives, the **Hessian matrix** is

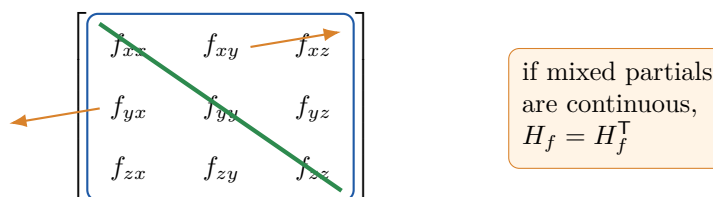
$$H_f(x, y) = \begin{bmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{yx}(x, y) & f_{yy}(x, y) \end{bmatrix}.$$

When the mixed partial derivatives are continuous, Clairaut's theorem gives $f_{xy} = f_{yx}$, so the Hessian is symmetric.

For $f(x_1, \dots, x_n)$, the Hessian is the $n \times n$ matrix

$$H_f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{bmatrix}.$$

The Hessian is the derivative of the gradient. The gradient tells which direction increases f fastest. The Hessian tells how that gradient changes from point to point.



Hessian: all second partial derivatives organized as a curvature matrix

Figure 3: The Hessian packages second derivatives into a matrix. Symmetry comes from equality of mixed partial derivatives under standard smoothness assumptions.

3 Worked example: a second-order Taylor polynomial

Example 3.1: Taylor approximation for $e^{x-y} + xy$

Consider

$$f(x, y) = e^{x-y} + xy.$$

Find the second-order Taylor approximation at $(0, 0)$.

First,

$$f(0, 0) = 1, \quad f_x = e^{x-y} + y, \quad f_y = -e^{x-y} + x.$$

Thus $f_x(0, 0) = 1$ and $f_y(0, 0) = -1$. The second derivatives are

$$f_{xx} = e^{x-y}, \quad f_{xy} = -e^{x-y} + 1, \quad f_{yy} = e^{x-y}.$$

At $(0, 0)$,

$$f_{xx}(0, 0) = 1, \quad f_{xy}(0, 0) = 0, \quad f_{yy}(0, 0) = 1.$$

Therefore

$$T_2(x, y) = 1 + x - y + \frac{1}{2}x^2 + \frac{1}{2}y^2.$$

The first-order terms create a tilted plane. The quadratic terms bend the approximation upward in both coordinate directions.

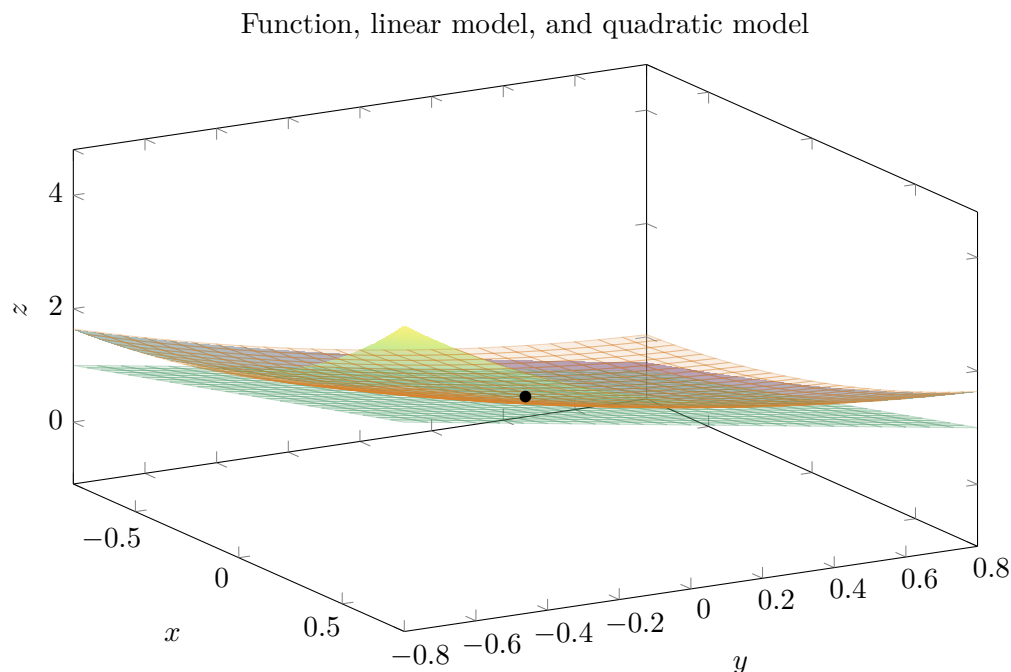


Figure 4: Near the expansion point, the quadratic Taylor model follows the true surface more closely than the tangent plane.

4 Quadratic forms

A **quadratic form** in two variables has the form

$$Q(h_1, h_2) = ah_1^2 + 2bh_1h_2 + ch_2^2.$$

In matrix form,

$$Q(h) = h^T Ah, \quad h = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}, \quad A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}.$$

The quadratic term in Taylor's formula is exactly a quadratic form:

$$\frac{1}{2}h^T H_f(p)h.$$

Thus local second-order geometry is linear algebra. The shape of a function near a point is controlled by $H_f(p)$.

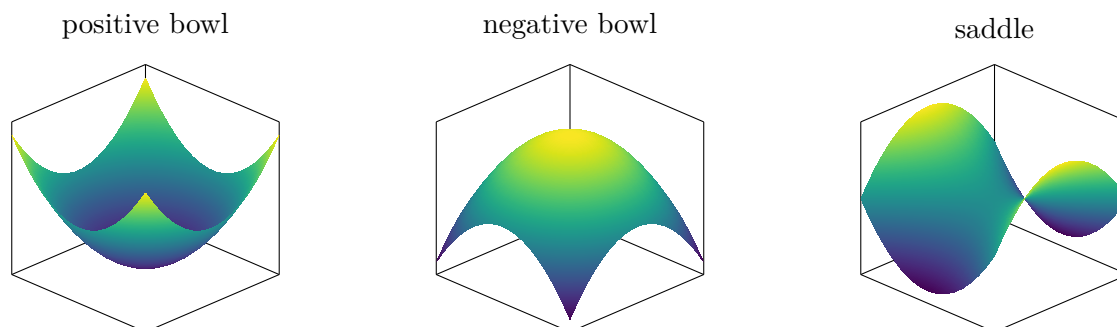


Figure 5: Three basic quadratic forms: positive definite, negative definite, and indefinite.

5 Definiteness and local shape

A symmetric matrix A defines a quadratic form $Q(h) = h^T A h$.

- A is **positive definite** if $h^T A h > 0$ for every nonzero vector h .
- A is **negative definite** if $h^T A h < 0$ for every nonzero vector h .
- A is **indefinite** if $h^T A h$ is positive in some directions and negative in others.
- A is **positive semidefinite** if $h^T A h \geq 0$ for every vector h , but may be zero for some nonzero h .
- A is **negative semidefinite** if $h^T A h \leq 0$ for every vector h , but may be zero for some nonzero h .

This classification is geometric: positive definite means bowl-shaped, negative definite means upside-down bowl-shaped, indefinite means saddle-shaped, and semidefinite means curved in some directions but flat in others.

6 Eigenvalues and principal directions

For a symmetric matrix A , the eigenvectors give the principal directions of the quadratic form, and the eigenvalues give the curvature in those directions. If A has orthonormal eigenvectors $u_1, \dots, u_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$, and if

$$h = c_1 u_1 + \dots + c_n u_n,$$

then

$$h^T A h = \lambda_1 c_1^2 + \dots + \lambda_n c_n^2.$$

This formula is one of the most important bridges between calculus and linear algebra.

Hessian eigenvalues and local shape

At a critical point p , the first-order term disappears because $\nabla f(p) = 0$. Then

$$f(p+h) - f(p) \approx \frac{1}{2} h^T H_f(p) h.$$

Therefore the eigenvalues of $H_f(p)$ determine the local shape:

- all positive eigenvalues: local minimum;
- all negative eigenvalues: local maximum;
- both positive and negative eigenvalues: saddle point;
- at least one zero eigenvalue: higher-order analysis may be needed.

Contours and eigen-directions

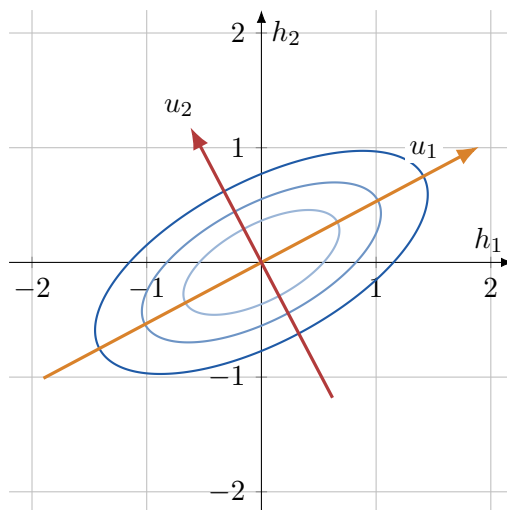


Figure 6: For a symmetric Hessian, eigenvectors give principal axes of the contour ellipses and eigenvalues give curvature along those axes.

7 The two-variable determinant test revisited

For a two-variable function at a critical point (a, b) ,

$$H_f(a, b) = \begin{bmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{xy}(a, b) & f_{yy}(a, b) \end{bmatrix}.$$

Let

$$D = f_{xx}(a, b)f_{yy}(a, b) - (f_{xy}(a, b))^2.$$

The determinant test is a convenient two-variable version of the eigenvalue test:

- If $D > 0$ and $f_{xx}(a, b) > 0$, then $H_f(a, b)$ is positive definite and f has a local minimum.
- If $D > 0$ and $f_{xx}(a, b) < 0$, then $H_f(a, b)$ is negative definite and f has a local maximum.
- If $D < 0$, then $H_f(a, b)$ is indefinite and f has a saddle point.
- If $D = 0$, the test is inconclusive.

The determinant measures whether the two eigenvalues have the same sign or opposite signs. The entry f_{xx} then tells whether the same sign is positive or negative.

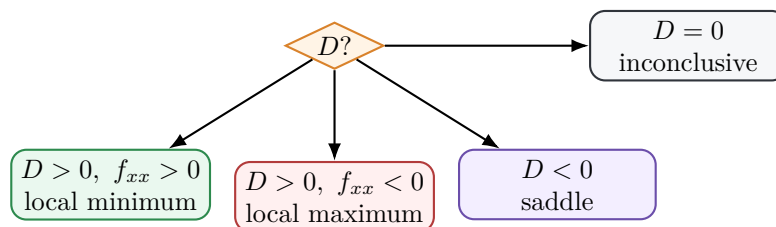


Figure 7: The determinant test is a quick way to read the two eigenvalue signs of a 2×2 Hessian.

8 Worked example: classify critical points

Example 8.1: Classifying critical points by quadratic approximation

Let

$$f(x, y) = x^3 + y^3 - 3xy + 4.$$

The gradient is

$$f_x = 3x^2 - 3y, \quad f_y = 3y^2 - 3x.$$

Critical points satisfy $y = x^2$ and $x = y^2$. Hence $x = x^4$, so $x = 0$ or $x = 1$. The critical points are $(0, 0)$ and $(1, 1)$.

The Hessian is

$$H_f(x, y) = \begin{bmatrix} 6x & -3 \\ -3 & 6y \end{bmatrix}.$$

At $(0, 0)$,

$$H_f(0, 0) = \begin{bmatrix} 0 & -3 \\ -3 & 0 \end{bmatrix}, \quad D = -9 < 0,$$

so $(0, 0)$ is a saddle point. At $(1, 1)$,

$$H_f(1, 1) = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}, \quad D = 27 > 0, \quad f_{xx}(1, 1) = 6 > 0,$$

so $(1, 1)$ is a local minimum, with value $f(1, 1) = 3$.

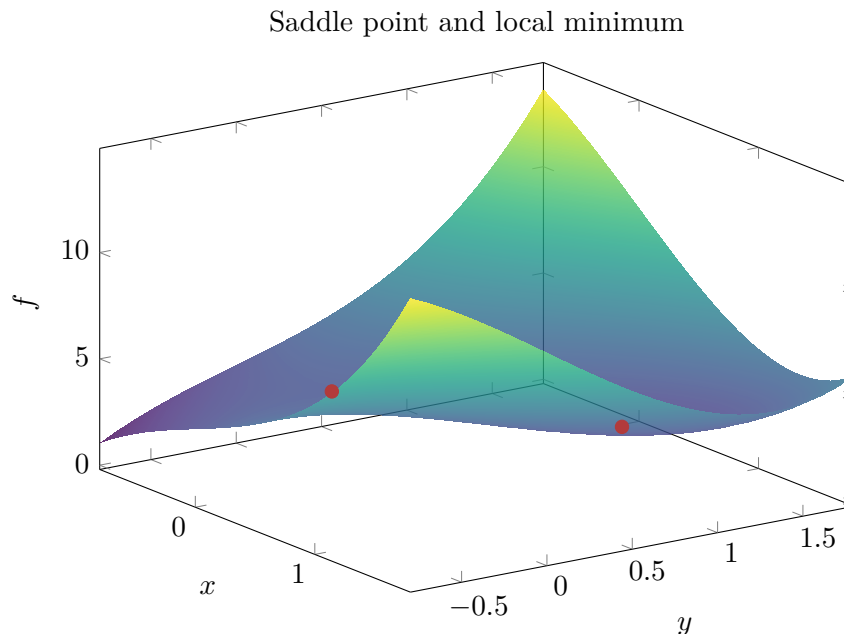


Figure 8: Near a critical point, the quadratic Taylor approximation predicts whether the surface behaves like a saddle or a bowl.

9 Taylor approximation in $\mathbb{R}^n$

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the second-order Taylor approximation near p is

$$f(p+h) \approx f(p) + \nabla f(p)^\top h + \frac{1}{2} h^\top H_f(p) h.$$

This formula is independent of dimension. The only difference is that h has n components and $H_f(p)$ is an $n \times n$ matrix.

A useful way to read the formula is

$$\text{new value} \approx \text{base value} + \text{linear change} + \text{curvature correction}.$$

The first-order term dominates for very small h unless $\nabla f(p) = 0$. At a critical point, the quadratic term becomes the leading term.

Example 9.1: Second-order approximation in three variables

Let

$$f(x, y, z) = x^2 + 2y^2 + 3z^2 + xy - yz + 2x.$$

At $p = (0, 0, 0)$,

$$f(p) = 0, \quad \nabla f(p) = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}.$$

The Hessian is constant:

$$H_f = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 4 & -1 \\ 0 & -1 & 6 \end{bmatrix}.$$

For $h = (h_1, h_2, h_3)^\top$,

$$f(h) = 2h_1 + \frac{1}{2}h^\top H_f h.$$

The Taylor approximation is exact because f is already a quadratic polynomial plus a linear term.

Slice $z = 0.4$ of a three-variable quadratic

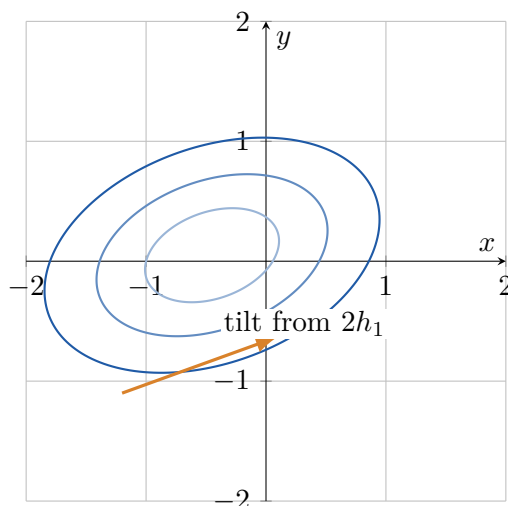


Figure 9: A two-dimensional slice of a three-variable quadratic model: contours reveal curvature and the linear term tilts the landscape.

10 Least squares as a quadratic form

Second-order geometry is especially clear in least squares. Suppose X is a data matrix, y is a response vector, and β is a parameter vector. The least-squares loss is

$$L(\beta) = \|y - X\beta\|^2.$$

Expanding,

$$L(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta.$$

This is a quadratic function of β . Its gradient is

$$\nabla L(\beta) = 2X^\top(X\beta - y),$$

and its Hessian is

$$H_L(\beta) = 2X^\top X.$$

The Hessian does not depend on β . The eigenvalues of $2X^T X$ determine the shape of the loss landscape: large eigenvalues mean steep curvature, small eigenvalues mean flat directions, and zero eigenvalues mean non-unique least-squares solutions.

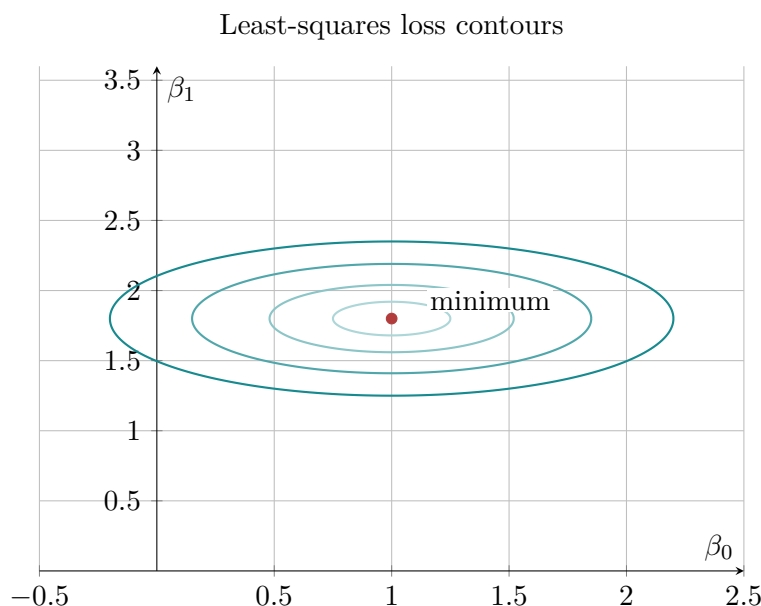


Figure 10: A least-squares loss is a quadratic surface in the model parameters; its contours are ellipses when the Hessian is positive definite.

11 Newton's method from Taylor approximation

Second-order Taylor approximation also explains Newton's method for optimization. Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is smooth and we are at x_k . Let h be a small step. The quadratic model is

$$m_k(h) = f(x_k) + \nabla f(x_k)^T h + \frac{1}{2} h^T H_f(x_k) h.$$

To minimize this quadratic model, set its gradient with respect to h equal to zero:

$$\nabla f(x_k) + H_f(x_k) h = 0.$$

Thus the Newton step satisfies

$$H_f(x_k) h = -\nabla f(x_k), \quad x_{k+1} = x_k + h.$$

When the Hessian is positive definite and the starting point is close enough to a local minimum, Newton's method can converge very quickly. When the Hessian is indefinite or nearly singular, the Newton step may point toward a saddle or become unstable.

Optimization paths on a quadratic model

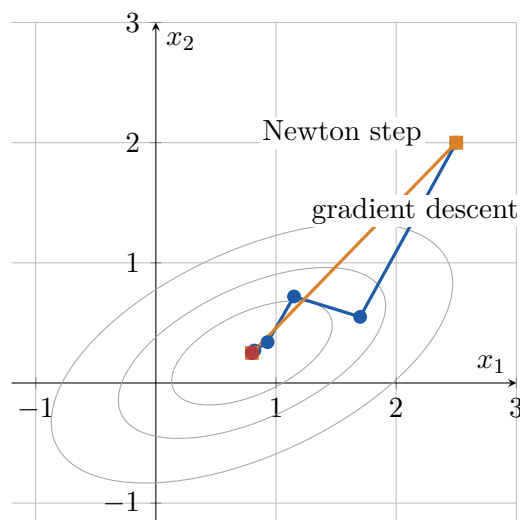


Figure 11: On a quadratic model, Newton's step jumps to the minimizer in one exact step, while gradient descent follows the negative-gradient direction repeatedly.

12 Conditioning and narrow valleys

A positive definite Hessian may still create numerical difficulty. If the eigenvalues of H have very different sizes, then the quadratic bowl is narrow and stretched. The ratio

$$\kappa(H) = \frac{\lambda_{\max}}{\lambda_{\min}}$$

is the **condition number** of the Hessian when all eigenvalues are positive.

A large condition number means the function is much steeper in one direction than another. Gradient descent may zigzag in narrow valleys. Newton's method corrects for curvature by multiplying by H^{-1} , but this can be expensive or unstable in high dimensions.

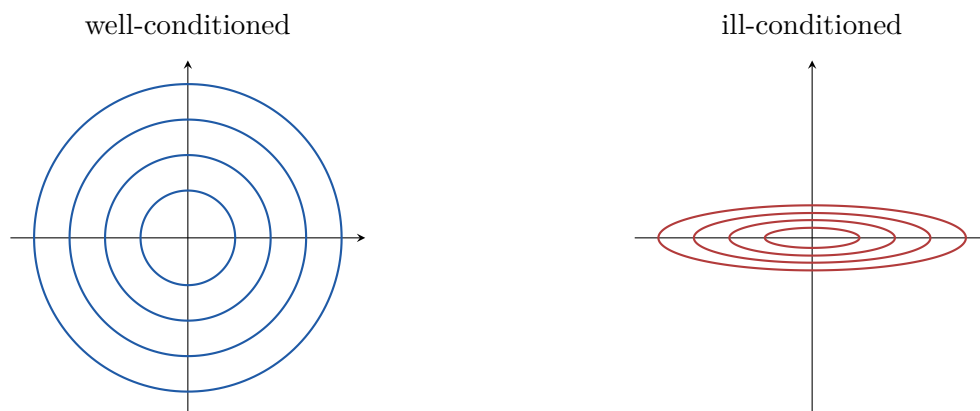


Figure 12: A large Hessian condition number produces narrow, stretched contours; this is a geometric source of slow gradient descent.

13 Statistical interpretation: likelihood and uncertainty

In statistics, second-order approximation appears in the local behavior of a log-likelihood. Suppose $\ell(\theta)$ is a log-likelihood and $\hat{\theta}$ is a maximum likelihood estimate. Since $\hat{\theta}$ is a critical point,

$$\nabla \ell(\hat{\theta}) = 0.$$

The second-order Taylor approximation is

$$\ell(\theta) \approx \ell(\hat{\theta}) + \frac{1}{2}(\theta - \hat{\theta})^\top H_\ell(\hat{\theta})(\theta - \hat{\theta}).$$

At a local maximum, the Hessian is negative definite. The negative Hessian

$$-H_\ell(\hat{\theta})$$

measures local curvature of the log-likelihood. Sharper curvature means the likelihood drops quickly away from the estimate, indicating smaller uncertainty. Flatter curvature means larger uncertainty.

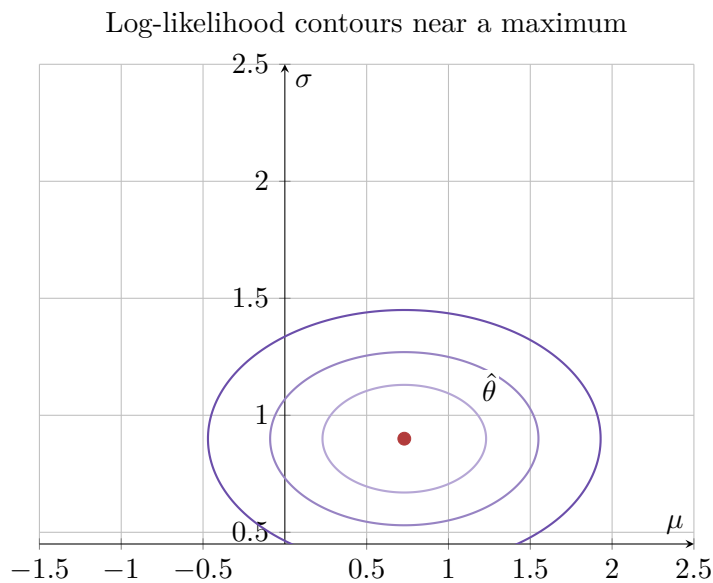


Figure 13: Near a maximum, a smooth log-likelihood is often well approximated by a concave quadratic. The negative Hessian describes local precision.

14 Beyond second order and degenerate examples

The second-order Taylor approximation is powerful, but it has limitations:

1. If the Hessian is singular, second-order information may not classify a critical point.
2. If the point is far from the center of approximation, higher-order terms may dominate.
3. If the function is not twice continuously differentiable, the Hessian may not describe local geometry reliably.

4. In high-dimensional models, the Hessian may be too large to store or invert directly.

The function $f(x, y) = x^4 + y^4$ has a local minimum at $(0, 0)$, but its Hessian at the origin is the zero matrix. The second derivative test gives no information, even though the function is clearly nonnegative. The function $g(x, y) = x^4 - y^4$ also has zero Hessian at the origin, but the origin is a saddle point. When the Hessian is degenerate, higher-order terms matter.

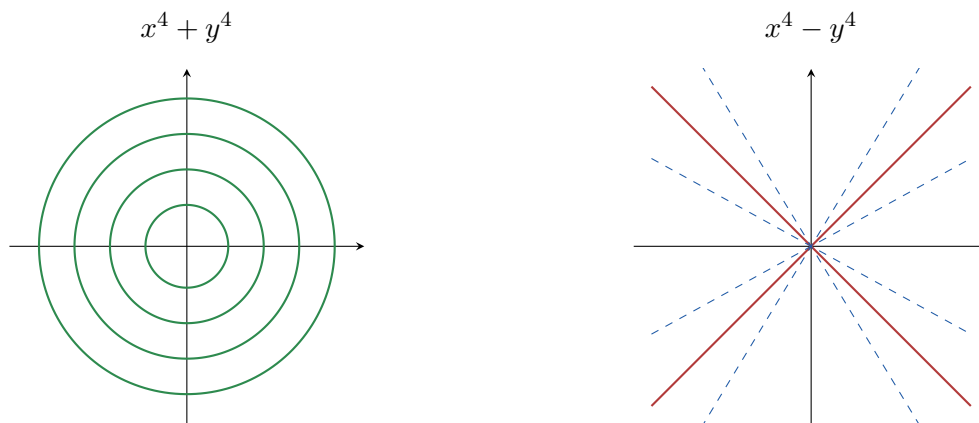


Figure 14: Degenerate critical points may require higher-order terms beyond the Hessian. Both examples have zero Hessian at the origin, but their local behavior is different.

15 Algorithmic summary

To use second-order approximation in practice:

1. Choose a center point p .
2. Compute $f(p)$, $\nabla f(p)$, and $H_f(p)$.
3. Form the local model

$$T_2(p + h) = f(p) + \nabla f(p)^\top h + \frac{1}{2} h^\top H_f(p) h.$$

4. At a critical point, classify the Hessian using eigenvalues or definiteness tests.
5. Interpret the eigenvectors as principal directions and eigenvalues as curvatures.
6. Use the approximation only locally unless the function is exactly quadratic.

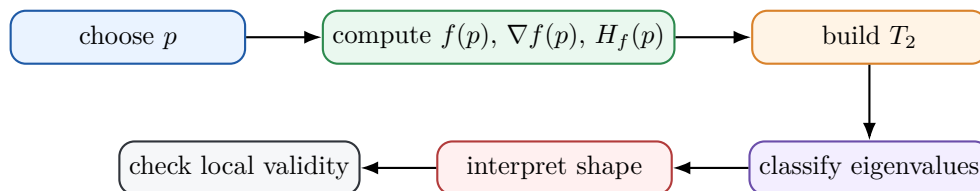


Figure 15: A practical workflow for second-order approximation and Hessian-based classification.

16 Python lab preview

The companion notebook for this chapter asks students to build a small second-order approximation toolkit:

- input a symbolic or numerical function;
- compute or approximate the gradient;
- compute or approximate the Hessian;
- build a quadratic Taylor model;
- plot the true function and approximation;
- classify a critical point by Hessian eigenvalues.

A finite-difference Hessian can approximate local curvature when symbolic derivatives are unavailable. For a numerical function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the central-difference formula

$$H_{ij}(p) \approx \frac{f(p + \varepsilon e_i + \varepsilon e_j) - f(p + \varepsilon e_i - \varepsilon e_j) - f(p - \varepsilon e_i + \varepsilon e_j) + f(p - \varepsilon e_i - \varepsilon e_j)}{4\varepsilon^2}$$

provides an approximation to the (i, j) entry of the Hessian.

17 Exercises

Conceptual questions

1. Explain the difference between a linear approximation and a quadratic approximation.
2. What does the Hessian matrix measure that the gradient does not measure?
3. Why does the quadratic term dominate the local shape at a critical point?
4. Explain why a positive definite Hessian produces a local minimum.
5. Explain why an indefinite Hessian produces a saddle point.
6. What does it mean geometrically for a Hessian to have a zero eigenvalue?
7. Why are eigenvectors of the Hessian called principal directions?
8. In least squares, why is the loss function quadratic in the parameters?
9. Why can Newton's method be faster than gradient descent near a well-conditioned minimum?
10. Why can a degenerate critical point require higher-order Taylor terms?

Computational practice

11. Find the second-order Taylor approximation of $f(x, y) = \sin(x + y)$ at $(0, 0)$.
12. Find the second-order Taylor approximation of $f(x, y) = e^x \cos y$ at $(0, 0)$.

13. Find the Hessian of $f(x, y) = x^2 + 4xy + 3y^2$ and classify its quadratic form.
14. Find the Hessian of $f(x, y) = x^2 - y^2 + 2xy$ and classify its quadratic form.
15. For $f(x, y) = x^3 + y^3 - 3xy + 4$, verify all critical points and classify them.
16. For $f(x, y) = x^4 + y^4 - 4xy$, find all critical points and use the Hessian test where possible.
17. Let $f(x, y, z) = x^2 + 2y^2 + 3z^2 + xy - yz$. Compute H_f and its eigenvalues.
18. Let $A = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$. Determine whether $h^\top Ah$ is positive definite.
19. Let $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$. Determine whether $h^\top Ah$ is positive definite, negative definite, or indefinite.
20. Show that if $A = X^\top X$, then A is positive semidefinite.

Applications and modeling

21. For the least-squares loss $L(\beta) = \|y - X\beta\|^2$, derive the gradient and Hessian.
22. Explain how the eigenvalues of $X^\top X$ affect gradient descent for least squares.
23. Suppose a log-likelihood has Hessian $\begin{bmatrix} -10 & 0 \\ 0 & -1 \end{bmatrix}$ at its maximum. Which parameter direction is more uncertain?
24. For a two-parameter model, design a quadratic loss with a narrow valley and plot its contours.
25. Implement a finite-difference Hessian for a function of two variables and test it on $f(x, y) = e^{x-y} + xy$.

Challenge problems

26. Show that if all eigenvalues of a symmetric Hessian at a critical point are positive, then the critical point is a strict local minimum.
27. Give an example where the Hessian test is inconclusive but the critical point is a local minimum.
28. Give an example where the Hessian test is inconclusive but the critical point is a saddle point.
29. Let $f(x, y) = x^2 + 2cxy + y^2$. For which values of c is the origin a local minimum?
30. Let $f(x, y) = x^2 + 2cxy + 4y^2$. For which values of c is the Hessian positive definite?

18 Chapter summary

Second-order Taylor approximation extends tangent-plane approximation by adding curvature. In several variables, curvature is encoded by the Hessian matrix. The quadratic term

$$\frac{1}{2}h^\top H_f(p)h$$

is a quadratic form, so the local geometry of a function is controlled by linear algebra. At a critical point, the gradient term disappears and the Hessian determines whether the function locally behaves like a bowl, an upside-down bowl, a saddle, or a degenerate flat shape.

This chapter is a bridge from calculus to modern applied mathematics. Quadratic approximations explain second derivative tests, Newton's method, least-squares geometry, local likelihood approximations, uncertainty quantification, and the curvature of loss landscapes.

19 Python lab and interactive page

Companion resources

The companion resources are the Chapter 18 notebook lab and the Chapter 18 interactive HTML page in the book repository.

Chapter 19. Vector Fields and Gradient Fields

From local arrows to global motion, force, and optimization

Scalar functions assign numbers to points. A temperature map assigns a temperature to each location, a density function assigns a density to each point in space, and a loss function assigns an error value to each parameter vector. Vector fields are different: they assign an arrow to each point.

This chapter begins the vector-calculus part of the book. We have already used vectors to describe geometry, curves, tangent directions, gradients, and optimization. Now vectors themselves become the output of a function. A vector field is the mathematical language of *local direction*. It answers the question: if I stand at this point, which way does the field point, and how strongly?

Learning goals

After studying this chapter, you should be able to:

1. define and visualize vector fields in $\mathbb{R}^2$, $\mathbb{R}^3$, and $\mathbb{R}^n$;
2. distinguish scalar fields, vector fields, and vector-valued parameterized curves;
3. sketch basic vector fields such as constant, radial, rotational, and gradient fields;
4. compute gradient vector fields from scalar potential functions;
5. recognize conservative vector fields and potential functions in simple cases;
6. interpret vector fields as velocity fields, force fields, gradient flows, and optimization dynamics;
7. use computation to visualize vector fields, gradient fields, and trajectories.

Why this chapter matters

Multiple integrals accumulate scalar quantities over regions. Vector calculus studies how vector quantities move through curves and surfaces. Before line integrals, flux, Green's theorem, Stokes' theorem, or the divergence theorem, we need a precise understanding of vector fields.

1 Scalar fields versus vector fields

A **scalar field** assigns one number to each point. For example,

$$f(x, y) = x^2 + y^2$$

assigns a scalar height, temperature, loss, or density value to each point (x, y) .

A **vector field** assigns a vector to each point. For example,

$$F(x, y) = \langle -y, x \rangle$$

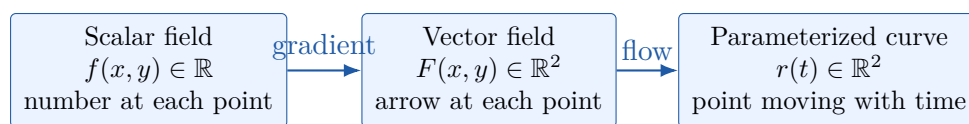
assigns the vector $\langle -y, x \rangle$ to the point (x, y) . At $(1, 0)$, it gives $\langle 0, 1 \rangle$. At $(0, 1)$, it gives $\langle -1, 0 \rangle$.

Three different objects

Do not confuse these objects:

- a scalar field has the form $f : \mathbb{R}^n \rightarrow \mathbb{R}$;
- a vector field has the form $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$;
- a parameterized curve has the form $r : I \rightarrow \mathbb{R}^n$.

A curve uses one parameter to trace points. A vector field places an arrow at many points.



vector calculus studies how scalar information, local arrows, and motion interact

Figure 1: Three related but distinct objects. A gradient transforms a scalar field into a vector field, and a vector field can generate motion.

2 Vector fields in the plane

A vector field on a region $D \subseteq \mathbb{R}^2$ is a function

$$F(x, y) = \langle P(x, y), Q(x, y) \rangle$$

that assigns to each point $(x, y) \in D$ a two-dimensional vector. Here P is the horizontal component and Q is the vertical component. In unit-vector notation,

$$F(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}.$$

Example 2.1: Constant, downward, and rotational fields

The field $F(x, y) = \langle 1, 0 \rangle$ places the same right-pointing vector at every point. The field $G(x, y) = \langle 0, -1 \rangle$ places the same downward vector at every point. The field

$$R(x, y) = \langle -y, x \rangle$$

is tangent to circles centered at the origin because

$$\langle -y, x \rangle \cdot \langle x, y \rangle = -xy + xy = 0.$$

This field rotates counterclockwise around the origin.

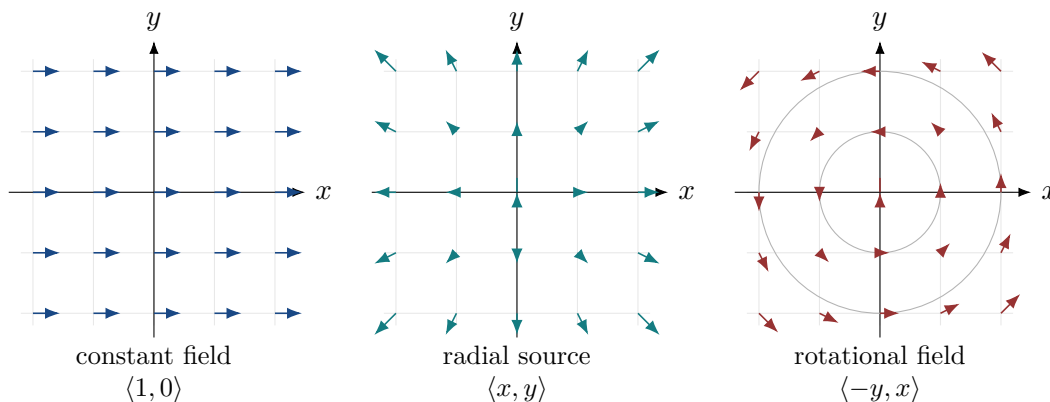


Figure 2: Constant, radial, and rotational vector fields in the plane. The same grid of points can carry very different local directions.

3 Vector fields in space and high dimensions

A vector field in three-dimensional space has the form

$$F(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle.$$

For example, $F(x, y, z) = \langle x, y, z \rangle$ points radially outward from the origin, while $G(x, y, z) = \langle -y, x, 0 \rangle$ rotates around the z -axis.

More generally, in $\mathbb{R}^n$ a vector field has the form

$$F(x_1, \dots, x_n) = \langle F_1(x), F_2(x), \dots, F_n(x) \rangle.$$

High-dimensional vector fields appear in optimization and machine learning, where a point represents a parameter vector and the arrow represents a parameter update direction.

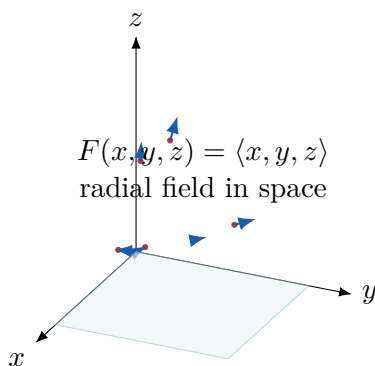


Figure 3: A vector field in $\mathbb{R}^3$ assigns a three-dimensional arrow to each point in space.

4 Gradient fields

If f is a scalar field, then its gradient is a vector field. In the plane,

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle.$$

In space,

$$\nabla f(x, y, z) = \langle f_x(x, y, z), f_y(x, y, z), f_z(x, y, z) \rangle.$$

A vector field F is called a **gradient field** if there exists a scalar function f such that

$$F = \nabla f.$$

The function f is called a **potential function** for F .

Geometry of a gradient field

A gradient vector field is perpendicular to level curves or level surfaces of its potential. The arrows point uphill; the negative gradient points downhill.

Example 4.1: A quadratic potential

Let

$$f(x, y) = \frac{1}{2}x^2 + y^2.$$

Then

$$\nabla f(x, y) = \langle x, 2y \rangle.$$

The level curves of f are ellipses. The gradient vectors are perpendicular to those ellipses and point outward toward larger values of f .

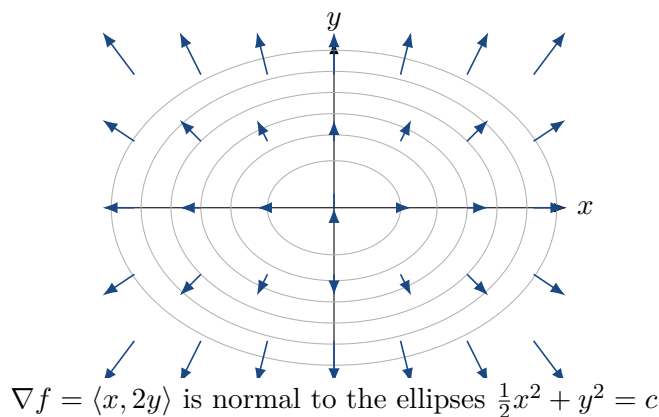


Figure 4: Gradient field of a quadratic potential. The arrows are perpendicular to the level curves.

Example 4.2: Computing gradient fields

For

$$f(x, y) = \sin(xy) + e^y,$$

we have

$$f_x(x, y) = y \cos(xy), \quad f_y(x, y) = x \cos(xy) + e^y.$$

Therefore

$$\nabla f(x, y) = \langle y \cos(xy), x \cos(xy) + e^y \rangle.$$

For a three-dimensional example, if $f(x, y, z) = xyz$, then

$$\nabla f(x, y, z) = \langle yz, xz, xy \rangle, \quad \nabla f(1, 2, 3) = \langle 6, 3, 2 \rangle.$$

5 Gradient fields and optimization

In optimization, the scalar function is often a loss function $L(\theta)$, where θ is a vector of parameters. The gradient field

$$\nabla L(\theta)$$

points in the direction of fastest increase of the loss. Gradient descent follows the negative gradient:

$$\theta_{k+1} = \theta_k - \alpha \nabla L(\theta_k).$$

This is also a discrete approximation to a continuous vector field flow:

$$\frac{d\theta}{dt} = -\nabla L(\theta).$$

Thus optimization can be viewed as motion in a vector field.

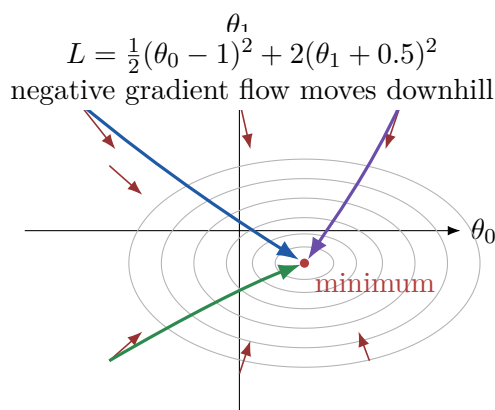


Figure 5: Gradient descent follows the negative gradient field toward smaller values of the loss.

6 Conservative vector fields

A vector field F is **conservative** if it is the gradient of some potential function:

$$F = \nabla f.$$

For example,

$$F(x, y) = \langle 2x, 2y \rangle$$

is conservative because $F = \nabla(x^2 + y^2)$. The potential function is not unique: adding a constant does not change the gradient. Thus $x^2 + y^2 + C$ has the same gradient field for every constant C .

How to find a potential function

Suppose

$$F(x, y) = \langle P(x, y), Q(x, y) \rangle.$$

To find a potential function f , we want

$$f_x = P, \quad f_y = Q.$$

A practical method is:

1. integrate P with respect to x ;
2. include an unknown function of y ;
3. differentiate with respect to y and compare with Q .

Example 6.1: Finding a potential

Let

$$F(x, y) = \langle y^2 - 3x^2, 2xy \rangle.$$

We seek f such that $f_x = y^2 - 3x^2$. Integrating with respect to x gives

$$f(x, y) = xy^2 - x^3 + g(y),$$

where $g(y)$ is unknown. Differentiate with respect to y :

$$f_y = 2xy + g'(y).$$

Since we need $f_y = 2xy$, we get $g'(y) = 0$. Therefore g is constant, and one potential function is

$$f(x, y) = xy^2 - x^3.$$

7 A quick test in the plane

For many fields on simple domains, a useful test is based on mixed partial derivatives. If

$$F(x, y) = \langle P(x, y), Q(x, y) \rangle$$

is a gradient field, then $P = f_x$ and $Q = f_y$. Therefore

$$P_y = f_{xy}, \quad Q_x = f_{yx}.$$

When the relevant second partial derivatives are continuous, equality of mixed partials gives

$$P_y = Q_x.$$

This condition is necessary for being a gradient field, and on many simple regions it is also sufficient. A complete treatment comes later when we study line integrals and Green's theorem.

Example 7.1: Radial fields versus rotational fields

For $F(x, y) = \langle -y, x \rangle$, we have $P_y = -1$ and $Q_x = 1$. Since $P_y \neq Q_x$, this field is not a gradient field on the plane. By contrast, for $F(x, y) = \langle x, y \rangle$, we have $P_y = 0$ and $Q_x = 0$, and indeed

$$F = \nabla \left(\frac{1}{2}x^2 + \frac{1}{2}y^2 \right).$$

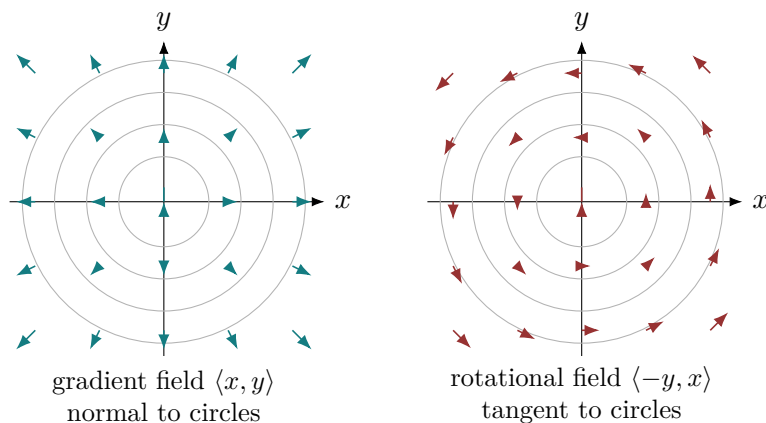


Figure 6: A gradient field has arrows normal to level curves, while a rotational field circulates around them.

8 Vector fields as differential equations

A vector field can define motion. If

$$F(x, y) = \langle P(x, y), Q(x, y) \rangle,$$

then a trajectory following the field satisfies

$$\frac{dx}{dt} = P(x, y), \quad \frac{dy}{dt} = Q(x, y).$$

In vector form,

$$\frac{dr}{dt} = F(r(t)).$$

The vector field tells the particle's velocity at each point. The trajectory is a curve whose tangent vector agrees with the field.

Example 8.1: Rotation around the origin

For $F(x, y) = \langle -y, x \rangle$, the motion equations are

$$\frac{dx}{dt} = -y, \quad \frac{dy}{dt} = x.$$

The solutions move around circles centered at the origin. The vector field is tangent to those circles.

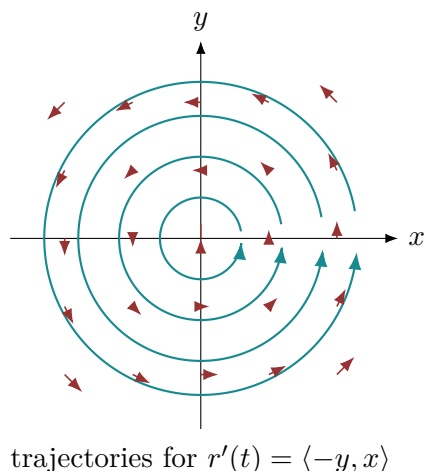


Figure 7: Trajectories following the rotational field move around circles.

9 Sources, sinks, and radial flow

The radial field $F(x, y) = \langle x, y \rangle$ points away from the origin. It behaves like a **source**: trajectories move outward. The negative radial field $F(x, y) = \langle -x, -y \rangle$ points toward the origin. It behaves like a **sink**: trajectories move inward.

In optimization, a sink often represents a local minimum of a loss function. If $L(x, y) = \frac{1}{2}(x^2 + y^2)$, then

$$-\nabla L(x, y) = \langle -x, -y \rangle.$$

So the negative gradient flow moves toward the minimizer at the origin.

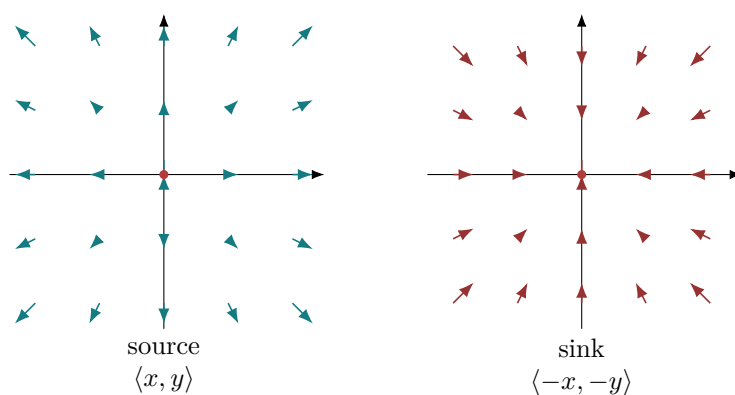


Figure 8: Source and sink vector fields. The source points outward; the sink points inward.

10 Vector fields in probability and statistics

Vector fields also appear in probability and statistics. For a positive density $p(x)$ on $\mathbb{R}^n$, the vector field

$$s(x) = \nabla \log p(x)$$

is called the **score field**. It points toward directions where the log-density increases.

For the standard Gaussian density in two dimensions,

$$p(x, y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2},$$

we have

$$\log p(x, y) = C - \frac{1}{2}(x^2 + y^2),$$

so

$$\nabla \log p(x, y) = \langle -x, -y \rangle.$$

Thus the score field points toward the high-density region at the origin.

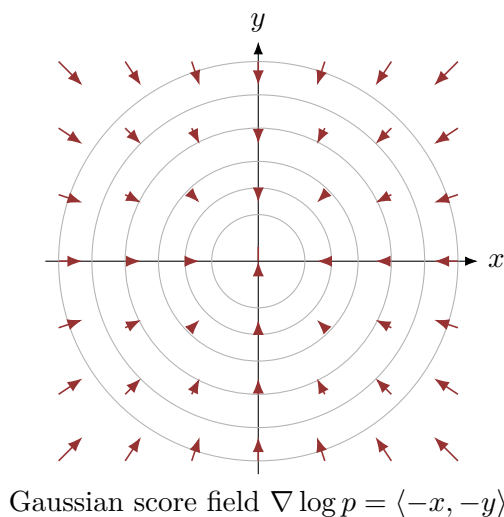


Figure 9: The score field of a standard bivariate Gaussian points toward the center of highest density.

This is one reason vector fields are increasingly important in modern statistics and machine learning: they describe how probability mass, particles, parameters, and samples move.

11 A field guide to common vector fields

Field	Formula	Main behavior
Constant field	$\langle a, b \rangle$	uniform direction and magnitude
Radial source	$\langle x, y \rangle$	points away from origin
Radial sink	$\langle -x, -y \rangle$	points toward origin
Rotational field	$\langle -y, x \rangle$	circulates counterclockwise
Gradient field	∇f	points toward fastest increase of f
Negative gradient field	$-\nabla f$	points toward fastest decrease of f
Singular radial field	$\langle x, y \rangle / (x^2 + y^2)$	radial with singularity at origin

12 Worked examples

Example 12.1: Classify and interpret a field

Consider

$$F(x, y) = \langle 2x, 8y \rangle.$$

We look for f such that $f_x = 2x$ and $f_y = 8y$. Integrating $f_x = 2x$ with respect to x gives

$$f(x, y) = x^2 + g(y).$$

Then $f_y = g'(y)$, so we need $g'(y) = 8y$. Hence $g(y) = 4y^2 + C$, and one potential is

$$f(x, y) = x^2 + 4y^2.$$

The level curves $x^2 + 4y^2 = c$ are ellipses. The vector field points outward and is perpendicular to these ellipses. The negative field $-F(x, y) = \langle -2x, -8y \rangle$ would drive points toward the minimum at $(0, 0)$.

Example 12.2: Testing conservative behavior

Let $F(x, y) = \langle y, x + y \rangle$. Here $P_y = 1$ and $Q_x = 1$, so the mixed-partial test passes. Find a potential by solving $f_x = y$. Integrating gives

$$f(x, y) = xy + g(y).$$

Then $f_y = x + g'(y)$. Since $f_y = x + y$, we need $g'(y) = y$, so $g(y) = \frac{1}{2}y^2$. Hence

$$f(x, y) = xy + \frac{1}{2}y^2.$$

For $H(x, y) = \langle -y, x \rangle$, however, $P_y = -1$ and $Q_x = 1$, so the field cannot be a gradient field on the plane.

What comes next

In the next chapter, we integrate functions along curves. For a scalar field, a line integral measures accumulated density along a curve. For a vector field, a line integral measures accumulated tangential work. Conservative fields will become powerful because their work integrals depend only on endpoints.

13 Python lab: building and visualizing vector fields

The Python lab for this chapter focuses on:

- comparing scalar fields, vector fields, and trajectories;
- plotting constant, radial, rotational, and singular vector fields;
- computing gradient vector fields from potential functions;

- visualizing negative gradient flow and gradient descent paths;
- exploring score fields from probability models.

A reusable helper has the form

$$(P, Q) \mapsto \text{a quiver plot of } F(x, y) = \langle P(x, y), Q(x, y) \rangle.$$

For example, plotting $F(x, y) = \langle x - y, x + y \rangle$ reveals a combination of stretching and rotation.

14 Chapter summary

A vector field assigns an arrow to each point. In the plane it has the form

$$F(x, y) = \langle P(x, y), Q(x, y) \rangle,$$

and in space it has the form

$$F(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle.$$

The most important special case is a gradient field:

$$F = \nabla f.$$

Gradient fields arise from scalar potentials. They are perpendicular to level curves and level surfaces, and they point in the direction of fastest increase. Negative gradient fields describe descent, optimization, stable motion toward minima, and score-based movement toward high-density regions.

Vector fields prepare us for line integrals, work, circulation, flux, curl, divergence, Green's theorem, Stokes' theorem, and the divergence theorem.

15 Exercises

Conceptual exercises

1. Explain the difference between a scalar field and a vector field.
2. Give one physical example and one data-science example of a vector field.
3. Why is $F(x, y) = \langle -y, x \rangle$ tangent to circles centered at the origin?
4. Explain why a gradient field is perpendicular to level curves of its potential.
5. What does it mean for a vector field to have a singularity?
6. In optimization, why do we usually follow $-\nabla L$ instead of ∇L ?

Computational exercises

1. Sketch $F(x, y) = \langle x, 0 \rangle$ by hand and describe its behavior.

2. Sketch $F(x, y) = \langle 0, y \rangle$ by hand and describe its behavior.
3. Sketch $F(x, y) = \langle y, x \rangle$. Is it radial, rotational, or neither?
4. Compute the gradient field of $f(x, y) = x^2 + xy + y^2$.
5. Compute the gradient field of $f(x, y, z) = x^2y + yz^2$.
6. Find a potential function for $F(x, y) = \langle 3x^2 + y, x + 2y \rangle$.
7. Decide whether $F(x, y) = \langle y^2, 2xy \rangle$ is a gradient field. If so, find a potential.
8. Decide whether $F(x, y) = \langle x - y, x + y \rangle$ can be a gradient field on the plane.
9. Let $F(x, y) = \langle -x, -y \rangle$. Solve $r'(t) = F(r(t))$ with $r(0) = (a, b)$.
10. Let $F(x, y) = \langle -y, x \rangle$. Show that $x(t)^2 + y(t)^2$ is constant along trajectories.

Applied exercises

1. Suppose $L(\theta_0, \theta_1) = (\theta_0 - 2)^2 + 3(\theta_1 + 1)^2$. Compute the negative gradient field and identify the minimizer.
2. For the density $p(x, y) = Ce^{-(x^2+4y^2)/2}$, compute the score field $\nabla \log p$.
3. A velocity field is $F(x, y) = \langle 1 + y, 1 - x \rangle$. Compute the velocity at $(0, 0)$, $(1, 0)$, and $(0, 1)$.
4. A force field is $F(x, y, z) = \langle x, y, z \rangle$. What is the magnitude of the force at $(1, 2, 2)$?
5. Plot $F(x, y) = \langle \sin y, \cos x \rangle$ and describe its visible behavior.
6. Plot the negative gradient field of $f(x, y) = x^4 + y^4 - 4xy$. Locate possible stable points visually.

Challenge exercises

1. Let $F = \langle P, Q \rangle$ be a gradient field with potential f . Explain why the vector field is zero at every critical point of f .
2. Find all points where $F(x, y) = \langle x^2 - y^2, 2xy \rangle$ is zero.
3. Let $F(x, y) = \langle ax + by, cx + dy \rangle$. Find a condition on a, b, c, d that is necessary for F to be a gradient field.
4. Suppose $F = \nabla f$ and f has a strict local minimum at p . What does the negative gradient field usually do near p ?
5. Consider $F(x, y) = \langle -y/(x^2 + y^2), x/(x^2 + y^2) \rangle$ on $\mathbb{R}^2 \setminus \{0\}$. Check the mixed-partial condition and describe what the field looks like.

16 Chapter project: vector fields behind gradient descent

Choose a differentiable loss function $L(\theta_0, \theta_1)$ with at least one local minimum. Analyze the vector field

$$F(\theta_0, \theta_1) = -\nabla L(\theta_0, \theta_1).$$

Your report should include:

1. the formula for L ;
2. the formula for ∇L ;
3. a contour plot of L ;
4. a quiver plot of $-\nabla L$;
5. several gradient descent paths from different starting points;
6. a short explanation of what the field reveals about the geometry of the loss surface.

This project connects multivariable calculus directly to optimization in statistics, machine learning, and artificial intelligence.

Chapter 20

Line Integrals and Work

Accumulating scalar quantities and vector forces along curves

Chapter overview

A definite integral accumulates a quantity along an interval. A double integral accumulates a quantity over a region. A triple integral accumulates a quantity through a solid. A **line integral** accumulates a quantity along a curve.

Line integrals are the first place where the geometry of a path becomes part of the integral. A curve can bend, twist, move through changing density, or pass through a force field whose direction changes from point to point. The value of the integral may depend not only on the endpoints, but also on the route taken between them.

This chapter develops two closely related ideas:

1. **Scalar line integrals**, written $\int_C f \, ds$, which accumulate a scalar field along a curve.
2. **Vector line integrals**, written $\int_C F \cdot dr$, which measure the work done by a vector field along an oriented curve.

The second kind is one of the central objects of vector calculus. It connects geometry, physics, conservative fields, Green's theorem, Stokes' theorem, and the dynamics of optimization algorithms.

The main idea

A line integral is a limit of small contributions along a curve. A curve is chopped into short pieces. On each piece, the field is approximately constant and the curve is approximately straight. The line integral is the limiting sum of all these small contributions.

Learning goals

After studying this chapter, you should be able to:

1. define and compute scalar line integrals with respect to arc length;
2. compute line integrals with respect to x , y , and z ;
3. compute vector line integrals using a parameterization of the curve;
4. interpret $\int_C F \cdot dr$ as work done by a force field;
5. understand orientation, path dependence, and additivity over piecewise curves;
6. approximate line integrals numerically from sampled points;

7. connect line integrals to motion, optimization paths, probability flows, and machine-learning loss landscapes.

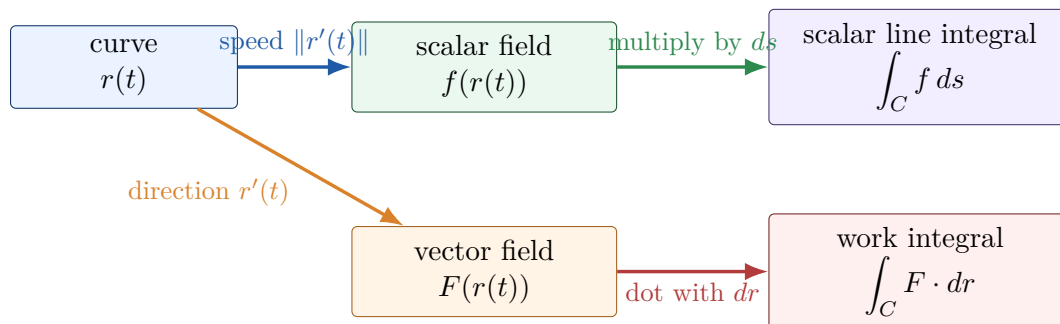


Figure 20.1: Two line integrals use the same path but accumulate different local quantities. Scalar line integrals use length; vector line integrals use oriented displacement.

20.1 Oriented curves and parameterizations

A curve C in the plane or space is usually described by a vector-valued function

$$r(t) = \langle x(t), y(t) \rangle, \quad a \leq t \leq b,$$

or

$$r(t) = \langle x(t), y(t), z(t) \rangle, \quad a \leq t \leq b.$$

The direction in which t increases gives the **orientation** of the curve. The same geometric curve can have the opposite orientation if we traverse it in reverse.

The velocity vector is $r'(t) = \langle x'(t), y'(t) \rangle$ in the plane, or $r'(t) = \langle x'(t), y'(t), z'(t) \rangle$ in space. Its magnitude is the speed along the curve. The arc length differential is

$$ds = \|r'(t)\| dt.$$

For a space curve,

$$\|r'(t)\| = \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}.$$

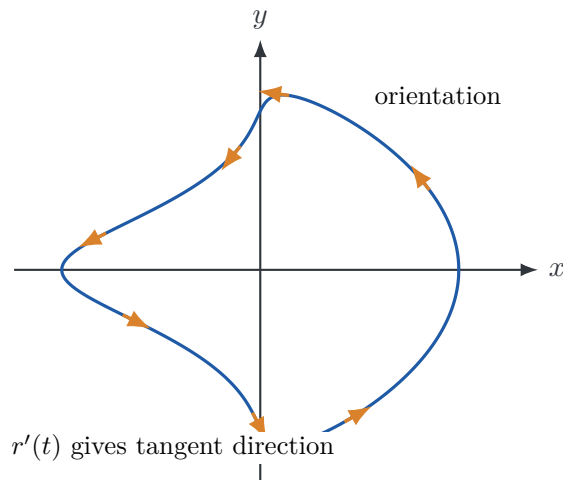


Figure 20.2: An oriented curve with tangent vectors. The arrows show the direction in which the parameter increases.

20.2 Scalar line integrals with respect to arc length

Suppose $f(x, y)$ is a scalar field defined on a curve C . The scalar line integral of f along C with respect to arc length is

$$\int_C f(x, y) ds.$$

If C is parameterized by $r(t) = \langle x(t), y(t) \rangle$ for $a \leq t \leq b$, then

$$\int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \sqrt{[x'(t)]^2 + [y'(t)]^2} dt.$$

For a space curve $r(t) = \langle x(t), y(t), z(t) \rangle$,

$$\int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2} dt.$$

The factor $ds = \|r'(t)\| dt$ converts the parameter interval into actual distance traveled along the curve.

Scalar line integrals do not depend on orientation

Because ds measures length, reversing the orientation of the same curve does not change $\int_C f ds$. The curve is traversed in the opposite direction, but the same small lengths are added.

Example 1: scalar line integral over a quarter circle

Evaluate

$$\int_C (3 - xy^2) ds,$$

where C is the first-quadrant part of the unit circle $x^2 + y^2 = 1$. Use

$$r(t) = \langle \cos t, \sin t \rangle, \quad 0 \leq t \leq \frac{\pi}{2}.$$

Then

$$r'(t) = \langle -\sin t, \cos t \rangle, \quad \|r'(t)\| = 1.$$

Therefore

$$\begin{aligned} \int_C (3 - xy^2) ds &= \int_0^{\pi/2} (3 - \cos t \sin^2 t) dt \\ &= \frac{3\pi}{2} - \frac{1}{3}. \end{aligned}$$

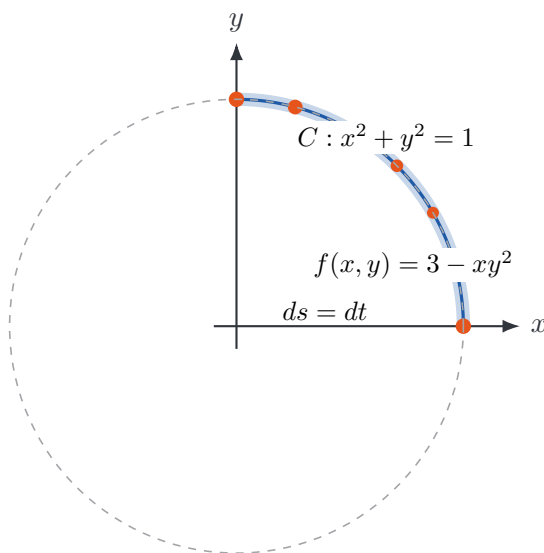


Figure 20.3: A scalar field sampled along a quarter circle. A scalar line integral accumulates field values weighted by arc length. Larger dots indicate larger sampled field values.

20.3 Mass of a wire

A major interpretation of a scalar line integral is the mass of a thin wire. Suppose a wire has shape C and density $\rho(x, y, z)$ measured in mass per unit length. Then the mass is

$$M = \int_C \rho(x, y, z) ds.$$

If the density is constant, this becomes density times length. If the density changes from point to point, the line integral accumulates the varying density along the wire.

Example 2: mass of a helical wire

Let C be the helix

$$r(t) = \langle \sin t, \cos t, t \rangle, \quad 0 \leq t \leq \pi.$$

Suppose the density is $\rho(x, y, z) = 2x \sin z$. Then $\rho(r(t)) = 2 \sin^2 t$. Also

$$r'(t) = \langle \cos t, -\sin t, 1 \rangle, \quad \|r'(t)\| = \sqrt{2}.$$

Thus

$$\int_C 2x \sin z ds = \int_0^\pi 2 \sin^2 t \sqrt{2} dt = \pi \sqrt{2}.$$

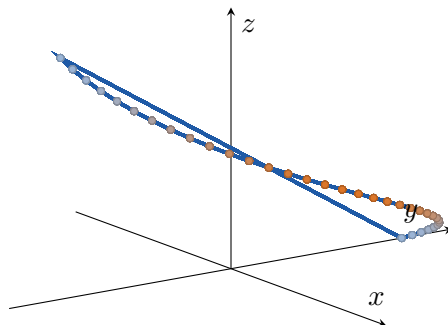


Figure 20.4: A helix with scalar density sampled along the curve. The scalar line integral gives the mass of a wire with this density.

20.4 Line integrals with respect to coordinates

Line integrals can also be taken with respect to coordinates. In the plane, we may see

$$\int_C f(x, y) dx, \quad \int_C g(x, y) dy.$$

If C is parameterized by $x = x(t)$ and $y = y(t)$, then

$$\int_C f(x, y) dx = \int_a^b f(x(t), y(t)) x'(t) dt,$$

and

$$\int_C g(x, y) dy = \int_a^b g(x(t), y(t)) y'(t) dt.$$

In space, similarly,

$$\int_C f(x, y, z) dz = \int_a^b f(x(t), y(t), z(t)) z'(t) dt.$$

Unlike ds , the coordinate differentials dx , dy , and dz can be positive or negative. Therefore coordinate line integrals depend on orientation.

ds versus dx, dy, and dz

The symbol ds is a length element and is always nonnegative. The symbols dx , dy , and dz are signed coordinate changes. Reversing the orientation of a curve leaves $\int_C f ds$ unchanged, but changes the sign of $\int_C f dx$, $\int_C f dy$, and $\int_C f dz$.

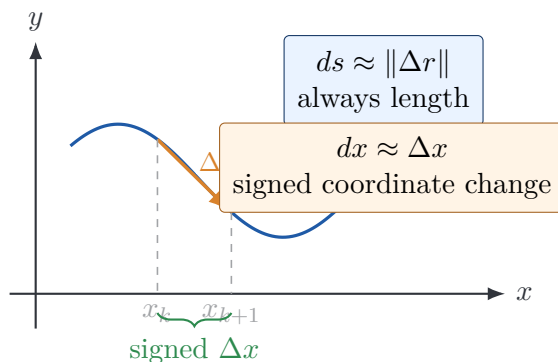


Figure 20.5: The arc length element ds measures actual distance along the curve, while dx measures signed horizontal change.

20.5 Vector line integrals and work

Suppose an object is displaced by a vector d while a constant force F acts on it. The work done is

$$W = F \cdot d.$$

If the force varies from point to point and the object moves along a curve C , we chop the curve into small displacement vectors Δr . On each small piece, the work is approximately $F \cdot \Delta r$. Adding these contributions and taking a limit gives the vector line integral

$$\int_C F \cdot dr.$$

If $F(x, y) = \langle P(x, y), Q(x, y) \rangle$ and $r(t) = \langle x(t), y(t) \rangle$, then

$$\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) dt.$$

Equivalently,

$$\int_C F \cdot dr = \int_C P dx + Q dy.$$

In space, if $F(x, y, z) = \langle P, Q, R \rangle$, then

$$\int_C F \cdot dr = \int_C P dx + Q dy + R dz.$$

The computation recipe

To compute $\int_C F \cdot dr$:

1. parameterize the curve as $r(t)$;
2. compute $r'(t)$;
3. substitute the parameterization into the field to get $F(r(t))$;
4. take the dot product $F(r(t)) \cdot r'(t)$;

5. integrate with respect to t over the parameter interval.

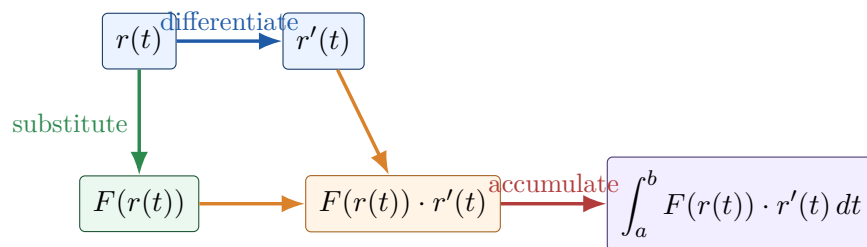


Figure 20.6: The standard workflow for a vector line integral.

Example 3: work along a circular arc

Let

$$F(x, y) = \langle y^2, -xy \rangle,$$

and let C be the curve

$$r(t) = \langle \sin t, \cos t \rangle, \quad 0 \leq t \leq \frac{\pi}{2}.$$

Then $r'(t) = \langle \cos t, -\sin t \rangle$. Substituting into the vector field gives

$$F(r(t)) = \langle \cos^2 t, -\sin t \cos t \rangle.$$

The dot product is

$$F(r(t)) \cdot r'(t) = \cos^3 t + \sin^2 t \cos t = \cos t.$$

Therefore

$$\int_C F \cdot dr = \int_0^{\pi/2} \cos t \, dt = 1.$$

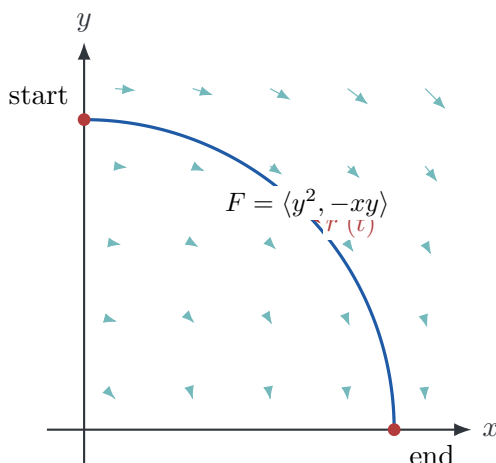


Figure 20.7: Work along a circular arc in a vector field. The integrand is the tangential component of the field times speed.

20.6 Orientation matters for vector line integrals

Let $-C$ denote the same curve as C but with the opposite orientation. Then

$$\int_{-C} F \cdot dr = - \int_C F \cdot dr.$$

This agrees with the work interpretation. Pushing an object forward along a path may do positive work; moving the same object backward through the same field reverses the displacement direction and reverses the sign of the work.

Example 4: reversing a path

For the field and curve in Example 3,

$$\int_C F \cdot dr = 1.$$

If the same arc is traversed from $(1, 0)$ back to $(0, 1)$, then

$$\int_{-C} F \cdot dr = -1.$$

This sign change is one of the most common sources of mistakes in vector line integrals.

20.7 Piecewise smooth curves

Many useful curves are not given by one formula. They are built from several smooth pieces. If

$$C = C_1 \cup C_2 \cup \cdots \cup C_m$$

with compatible orientations, then

$$\int_C F \cdot dr = \int_{C_1} F \cdot dr + \int_{C_2} F \cdot dr + \cdots + \int_{C_m} F \cdot dr.$$

The same additivity holds for scalar line integrals.

Example 5: a piecewise line integral in space

Evaluate

$$\int_C y \, dx + z \, dy + x \, dz,$$

where $C = C_1 \cup C_2$, C_1 is the line segment from $(3, 4, 0)$ to $(3, 4, 5)$, and C_2 is the line segment from $(3, 4, 5)$ to $(2, 0, 0)$.

For C_1 , use $r_1(t) = \langle 3, 4, t \rangle$, $0 \leq t \leq 5$. Then $dx = 0$, $dy = 0$, and $dz = dt$, so

$$\int_{C_1} y \, dx + z \, dy + x \, dz = \int_0^5 3 \, dt = 15.$$

For C_2 , use $r_2(s) = \langle 3-s, 4-4s, 5-5s \rangle$, $0 \leq s \leq 1$. Then $dx = -ds$, $dy = -4ds$, and $dz = -5ds$. Therefore

$$\begin{aligned} \int_{C_2} y \, dx + z \, dy + x \, dz &= \int_0^1 [(4-4s)(-1) + (5-5s)(-4) + (3-s)(-5)] \, ds \\ &= \int_0^1 (-39 + 29s) \, ds = -\frac{49}{2}. \end{aligned}$$

Thus

$$\int_C y \, dx + z \, dy + x \, dz = 15 - \frac{49}{2} = -\frac{19}{2}.$$

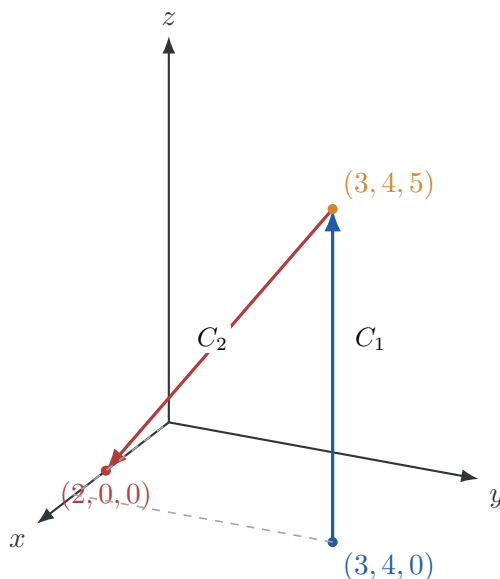


Figure 20.8: A piecewise curve in space. Line integrals over piecewise curves are computed one segment at a time.

20.8 Path dependence

A major difference between line integrals and ordinary definite integrals is that path matters. Two different curves with the same starting and ending points can produce different line integrals.

Example 6: same endpoints, different paths

Let $F(x, y) = \langle y, 0 \rangle$. We compute the work from $(0, 0)$ to $(1, 1)$ along two paths.

Path A: straight line. Let $r_A(t) = \langle t, t \rangle$, $0 \leq t \leq 1$. Then $F(r_A(t)) = \langle t, 0 \rangle$ and $r'_A(t) = \langle 1, 1 \rangle$. Therefore

$$\int_{C_A} F \cdot dr = \int_0^1 t \, dt = \frac{1}{2}.$$

Path B: horizontal then vertical. First go from $(0, 0)$ to $(1, 0)$, then from $(1, 0)$ to $(1, 1)$. On the horizontal segment, $y = 0$, so $F = \langle 0, 0 \rangle$. On the vertical segment, $dx = 0$, and

$F \cdot dr = y dx = 0$. Hence

$$\int_{C_B} F \cdot dr = 0.$$

The endpoints are the same, but the work is different. This field is path-dependent.

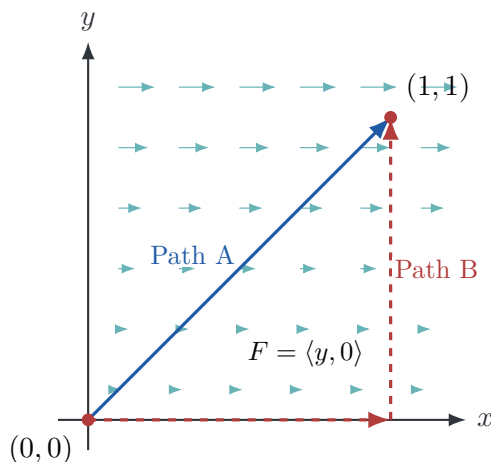


Figure 20.9: Two paths with the same endpoints can produce different vector line integrals.

20.9 A three-dimensional work integral

Let

$$F(x, y, z) = \langle xy, yz, zx \rangle$$

and let C be parameterized by

$$r(t) = \langle t, t^2, t^3 \rangle, \quad 0 \leq t \leq 1.$$

Then $r'(t) = \langle 1, 2t, 3t^2 \rangle$. Substituting into F gives

$$F(r(t)) = \langle t^3, t^5, t^4 \rangle.$$

Thus

$$F(r(t)) \cdot r'(t) = t^3 + 2t^6 + 3t^6 = t^3 + 5t^6.$$

Therefore

$$\int_C F \cdot dr = \int_0^1 (t^3 + 5t^6) dt = \frac{1}{4} + \frac{5}{7} = \frac{27}{28}.$$

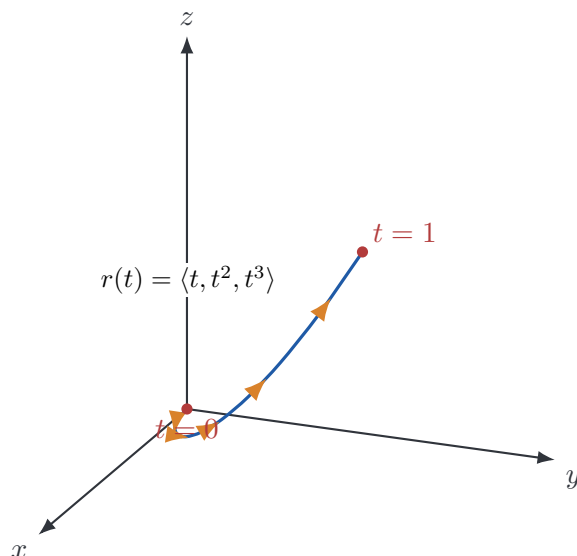


Figure 20.10: A space curve used for a three-dimensional work integral, with tangent directions shown along the path.

20.10 Numerical approximation of vector line integrals

If a curve is sampled at points

$$r_0, r_1, \dots, r_N,$$

then a numerical approximation to the work integral is

$$\int_C F \cdot dr \approx \sum_{k=0}^{N-1} F\left(\frac{r_k + r_{k+1}}{2}\right) \cdot (r_{k+1} - r_k).$$

This is the midpoint rule along a curve. It is useful when the curve is known only from data, simulation output, robot trajectories, or numerical solutions of differential equations.

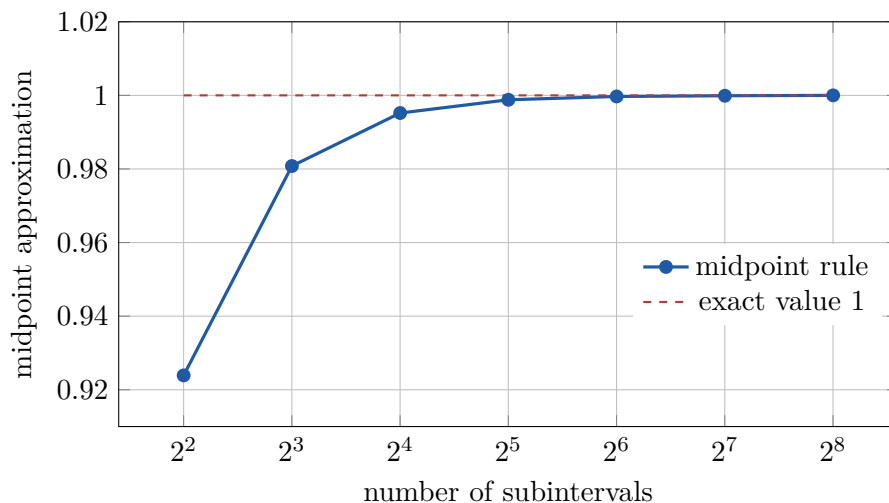


Figure 20.11: Numerical midpoint approximations converge to the vector line integral as the curve sampling becomes finer.

20.11 Line integrals along optimization paths

In machine learning and statistics, we often study a loss function $L(\theta)$, where θ is a vector of parameters. The gradient $\nabla L(\theta)$ is a vector field on parameter space. An optimization algorithm produces a path

$$\theta_0, \theta_1, \theta_2, \dots$$

through that space.

The work-like quantity

$$\int_C \nabla L \cdot d\theta$$

measures accumulated change of the loss along the path. For a smooth path, the chain rule gives

$$\frac{d}{dt} L(\theta(t)) = \nabla L(\theta(t)) \cdot \theta'(t),$$

so line integrals of gradients are closely tied to change in objective values. This idea becomes the Fundamental Theorem for Line Integrals in the next chapter.

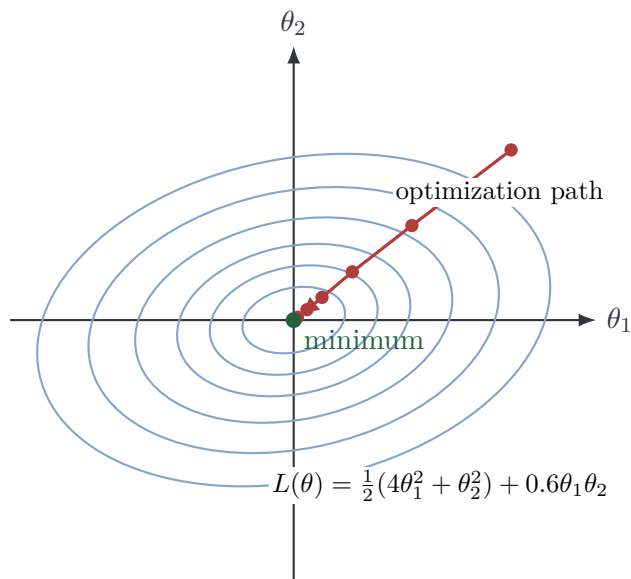


Figure 20.12: A path through a quadratic loss landscape. Line integrals can measure accumulated change along paths in parameter space.

20.12 Work, circulation, and flow

The vector line integral $\int_C F \cdot dr$ measures the component of the field tangent to the curve. If the field points mostly along the direction of motion, the integral is positive. If it points mostly opposite the direction of motion, the integral is negative. If the field is mostly perpendicular to the motion, the contribution is small.

For a closed curve, the integral

$$\oint_C F \cdot dr$$

is called the **circulation** of F around C . Circulation measures how much the vector field tends to push around the loop. It is a central idea behind Green's theorem and Stokes' theorem.

Example 7: circulation of a rotational field

Let $F(x, y) = \langle -y, x \rangle$. On the unit circle with counterclockwise orientation,

$$r(t) = \langle \cos t, \sin t \rangle, \quad 0 \leq t \leq 2\pi.$$

Then

$$F(r(t)) = \langle -\sin t, \cos t \rangle, \quad r'(t) = \langle -\sin t, \cos t \rangle.$$

Thus $F(r(t)) \cdot r'(t) = 1$, so

$$\oint_C F \cdot dr = \int_0^{2\pi} 1 \, dt = 2\pi.$$

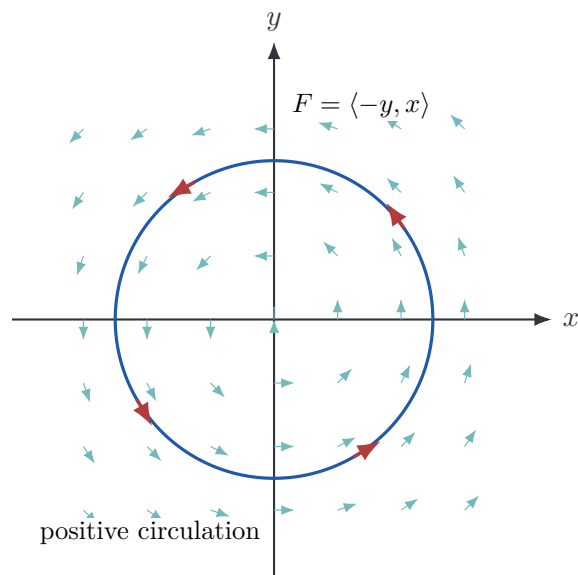


Figure 20.13: Positive circulation around the unit circle for the rotational field $F = \langle -y, x \rangle$.

20.13 Common mistakes

Mistake 1: forgetting $\|r'(t)\|$ in scalar line integrals

For scalar line integrals, the formula is

$$\int_C f \, ds = \int_a^b f(r(t)) \|r'(t)\| \, dt.$$

The factor $\|r'(t)\|$ is essential unless the parameterization has unit speed.

Mistake 2: using $\|r'(t)\|$ in vector line integrals

For vector line integrals, the formula is

$$\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) \, dt.$$

Do not replace $r'(t)$ by $\|r'(t)\|$. The direction of motion matters.

Mistake 3: ignoring orientation

Vector line integrals depend on orientation. Reversing the path changes the sign.

20.14 Chapter summary

A line integral accumulates quantities along curves.

For scalar fields,

$$\int_C f \, ds = \int_a^b f(r(t)) \|r'(t)\| \, dt.$$

This measures total mass, charge, cost, or exposure along a curve when f is a density per unit length.

For vector fields,

$$\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) \, dt.$$

This measures work done by a force field or accumulated tangential flow along an oriented curve. In coordinates,

$$\int_C F \cdot dr = \int_C P \, dx + Q \, dy$$

in the plane, and

$$\int_C F \cdot dr = \int_C P \, dx + Q \, dy + R \, dz$$

in space.

The essential themes are parameterization, orientation, tangential components, and path dependence. The next chapter studies the special vector fields for which line integrals depend only on endpoints.

20.15 Exercises

Conceptual exercises

1. Explain in your own words the difference between $\int_C f \, ds$ and $\int_C F \cdot dr$.
2. Why does reversing the orientation of a curve not change $\int_C f \, ds$?
3. Why does reversing the orientation of a curve change the sign of $\int_C F \cdot dr$?
4. What does it mean for a vector field to do positive work along a curve?
5. Give a physical interpretation of $\int_C \rho \, ds$ when ρ is a density along a wire.
6. Give a data-science interpretation of integrating a cost or risk function along a trajectory.

Computational exercises

7. Evaluate $\int_C (x + y) \, ds$, where C is the line segment from $(0, 0)$ to $(2, 1)$.
8. Evaluate $\int_C x^2 \, ds$, where C is the unit circle parameterized counterclockwise.
9. Evaluate $\int_C y \, dx + x \, dy$, where C is the line segment from $(0, 0)$ to $(1, 2)$.
10. Evaluate $\int_C F \cdot dr$ for $F(x, y) = \langle x, y \rangle$ and $r(t) = \langle \cos t, \sin t \rangle$, $0 \leq t \leq 2\pi$.
11. Evaluate $\int_C F \cdot dr$ for $F(x, y) = \langle -y, x \rangle$ and $r(t) = \langle 2 \cos t, 2 \sin t \rangle$, $0 \leq t \leq 2\pi$.
12. Evaluate $\int_C F \cdot dr$ for $F(x, y, z) = \langle z, x, y \rangle$ and $r(t) = \langle t, t^2, t^3 \rangle$, $0 \leq t \leq 1$.
13. Evaluate $\int_C z \, ds$ for the helix $r(t) = \langle \cos t, \sin t, t \rangle$, $0 \leq t \leq 2\pi$.

14. Let $F(x, y) = \langle y, 0 \rangle$. Compute the work from $(0, 0)$ to $(1, 1)$ along the curve $y = x^2$.
15. Let $F(x, y) = \langle 0, x \rangle$. Compute the work from $(0, 0)$ to $(1, 1)$ along the straight line and along the path that goes vertically first and then horizontally.

Computational lab exercises

16. Write a function that approximates $\int_C F \cdot dr$ from sampled curve points.
17. Use numerical approximation to estimate the work done by $F(x, y) = \langle \sin y, \cos x \rangle$ along $r(t) = \langle t, t^2 \rangle$, $0 \leq t \leq 1$.
18. Compare the numerical approximation from Exercise 17 for $N = 10, 20, 40, 80, 160$ subintervals.
19. Plot the vector field $F(x, y) = \langle -y, x \rangle$ and three closed curves. Estimate the circulation around each.
20. Generate a random polygonal path from $(0, 0)$ to $(1, 1)$ and estimate $\int_C \langle y, 0 \rangle \cdot dr$.

Challenge exercises

21. Find a vector field F and two paths from A to B such that the work along one path is positive and the work along the other path is negative.
22. Let $F(x, y) = \langle ax + by, cx + dy \rangle$. Determine conditions on a, b, c, d so that $\int_C F \cdot dr$ has the same value for every path from $(0, 0)$ to $(1, 1)$.
23. For $F(x, y) = \langle -y, x \rangle$, compute the circulation around a circle of radius a .
24. Let C be any circle centered at the origin, oriented counterclockwise. Explain why $F(x, y) = \langle x, y \rangle$ has zero circulation around C .
25. Suppose a curve is reparameterized by a smooth increasing change of variables. Prove that $\int_C F \cdot dr$ is unchanged.
26. Suppose a curve is reparameterized by a smooth decreasing change of variables. Prove that $\int_C F \cdot dr$ changes sign.

Chapter 21

Conservative Fields, Potentials, and Green's Theorem

Path independence, circulation, flux, and the bridge from local derivatives to global integrals

The previous chapter introduced line integrals. We learned how to integrate a scalar field along a curve and how to integrate a vector field along an oriented path. The main structural question is now:

When does a line integral depend only on endpoints, and when does it depend on the whole path?

This question is not merely a computational shortcut. In physics, it separates conservative forces from forces with circulation. In geometry, it connects local derivatives to global behavior around closed curves. In optimization, it explains why gradients of loss functions behave differently from rotational vector fields. In data science and machine learning, it gives a language for energy functions, potential landscapes, and dynamics that may or may not be driven by a scalar objective.

This chapter develops two major results:

1. the **Fundamental Theorem for Line Integrals**, which says that line integrals of gradient fields depend only on endpoints;
2. **Green's Theorem**, which converts a line integral around a closed curve into a double integral over the enclosed region.

Together, these results form the two-dimensional foundation for Stokes' theorem and the divergence theorem.

21.1 Learning goals

After studying this chapter, you should be able to:

1. recognize gradient fields and conservative vector fields;
2. find potential functions for conservative fields;
3. use the Fundamental Theorem for Line Integrals;
4. understand path independence and closed-loop integrals;
5. test whether a plane vector field is conservative on a suitable domain;
6. state and apply Green's theorem in circulation form;

7. use Green's theorem to compute areas and circulations;
8. understand the role of orientation, holes, and singularities;
9. connect conservative fields and circulation to optimization, probability flow, and learning dynamics.

The main theme

A derivative is local: it describes behavior near one point. An integral is global: it accumulates behavior over a curve or a region. This chapter explains when local derivative information completely controls global line integrals.

21.2 Gradient fields and conservative vector fields

A scalar function $f(x, y)$ has gradient field

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle.$$

A scalar function $f(x, y, z)$ has gradient field

$$\nabla f(x, y, z) = \langle f_x(x, y, z), f_y(x, y, z), f_z(x, y, z) \rangle.$$

A vector field F is called **conservative** if it is the gradient of some scalar function f :

$$F = \nabla f.$$

The function f is called a **potential function** for F .

This terminology is natural in physical examples. A gravitational or electrostatic field often points in the direction in which potential energy changes most rapidly. In optimization, the vector field $-\nabla L$ points in the direction of steepest decrease of the loss function L .

Potential function

If $F = \langle P, Q \rangle$ is conservative in the plane, then there is a scalar function f such that

$$P = f_x, \quad Q = f_y.$$

The scalar function f is the potential. The vector field F is the gradient field generated by f .

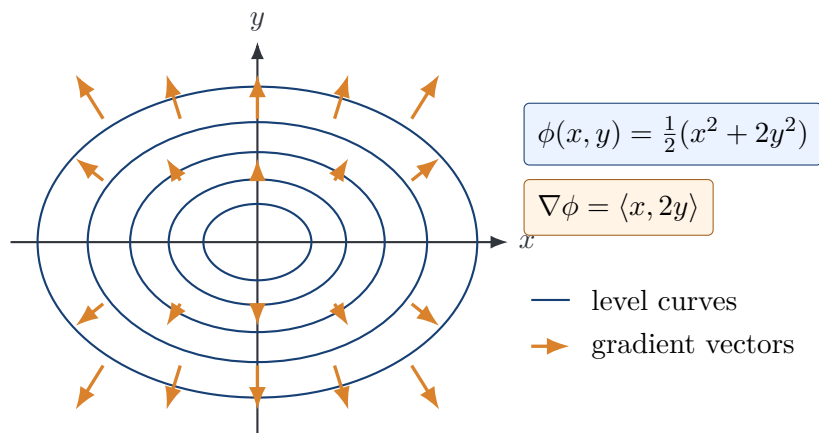


Figure 21.1: A gradient field is organized by the level curves of a potential function. The arrows point perpendicular to the level curves.

21.3 The Fundamental Theorem for Line Integrals

The one-variable Fundamental Theorem of Calculus says that if $G'(x) = g(x)$, then

$$\int_a^b g(x) dx = G(b) - G(a).$$

The multivariable version has the same spirit. If a vector field is a gradient field, then its line integral is the change in the potential.

Fundamental Theorem for Line Integrals

Let C be a smooth curve from point A to point B , and let f be differentiable on a region containing C . Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A).$$

Therefore, the integral depends only on the endpoints, not on the path.

21.3.1 Why the theorem is true

Suppose C is parameterized by

$$\mathbf{r}(t) = \langle x(t), y(t) \rangle, \quad a \leq t \leq b.$$

Then

$$\int_C \nabla f \cdot d\mathbf{r} = \int_a^b \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt.$$

By the chain rule,

$$\frac{d}{dt} f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t).$$

Therefore,

$$\int_C \nabla f \cdot d\mathbf{r} = \int_a^b \frac{d}{dt} f(\mathbf{r}(t)) dt = f(\mathbf{r}(b)) - f(\mathbf{r}(a)).$$

The theorem is the chain rule plus the one-variable Fundamental Theorem of Calculus.

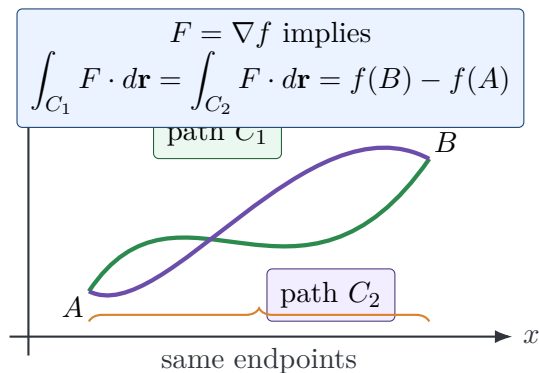


Figure 21.2: For a gradient field, different paths with the same start and end give the same line integral.

Use a potential instead of parameterizing the path

Let

$$F(x, y) = \langle 2xy, x^2 + 3y^2 \rangle.$$

Compute $\int_C F \cdot d\mathbf{r}$ for any smooth curve C from $A = (1, 0)$ to $B = (2, 1)$.

We look for a potential f such that

$$f_x = 2xy, \quad f_y = x^2 + 3y^2.$$

Integrating $f_x = 2xy$ with respect to x gives

$$f(x, y) = x^2y + g(y).$$

Then $f_y = x^2 + g'(y)$. Matching $f_y = x^2 + 3y^2$ gives $g'(y) = 3y^2$ and $g(y) = y^3 + C$. One potential is

$$f(x, y) = x^2y + y^3.$$

By the Fundamental Theorem for Line Integrals,

$$\int_C F \cdot d\mathbf{r} = f(2, 1) - f(1, 0) = (4 + 1) - 0 = 5.$$

No parameterization of C is needed.

21.4 Path independence

A vector line integral is **path independent** if any two curves with the same starting point and ending point give the same value. For a conservative field,

$$\int_C F \cdot d\mathbf{r} = f(B) - f(A),$$

so the integral is path independent. The converse is also true under standard connectedness assumptions: if line integrals of F are path independent on a region, then F is conservative on that region.

Equivalent ideas

For a well-behaved vector field on a suitable region, the following ideas are essentially equivalent:

1. F is conservative.
2. $F = \nabla f$ for some potential function f .
3. $\int_C F \cdot d\mathbf{r}$ depends only on endpoints.
4. $\int_C F \cdot d\mathbf{r} = 0$ for every closed curve C .

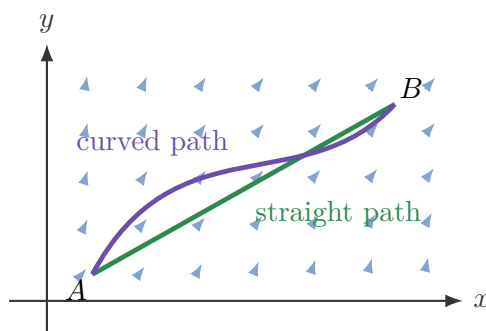


Figure 21.3: Two different paths from the same start point to the same endpoint. In a conservative field, both paths give the same line integral.

21.5 How to find a potential function

Suppose

$$F(x, y) = \langle P(x, y), Q(x, y) \rangle.$$

To find a potential function f :

1. integrate $P = f_x$ with respect to x ;
2. include an unknown function of y ;
3. differentiate the result with respect to y ;
4. compare with $Q = f_y$;

5. solve for the unknown function.

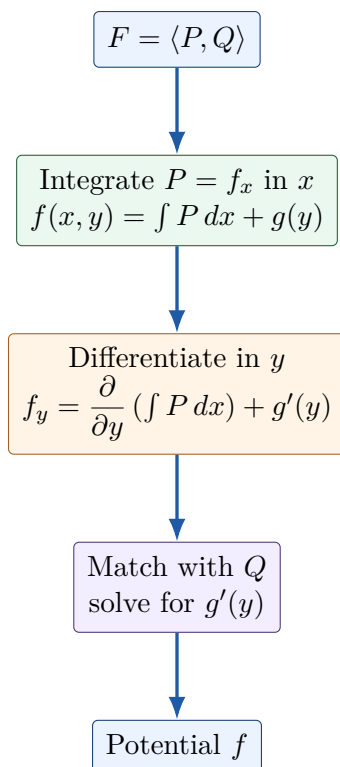


Figure 21.4: A practical workflow for finding a potential function in the plane.

A conservative field

Let

$$F(x, y) = \langle y \cos(xy) + 2x, x \cos(xy) + 4y^3 \rangle.$$

We want $f_x = y \cos(xy) + 2x$. Integrating with respect to x gives

$$f(x, y) = \sin(xy) + x^2 + g(y).$$

Then $f_y = x \cos(xy) + g'(y)$. Matching the second component gives

$$g'(y) = 4y^3, \quad g(y) = y^4 + C.$$

So one potential is

$$f(x, y) = \sin(xy) + x^2 + y^4.$$

21.6 The component test in the plane

For a plane vector field $F = \langle P, Q \rangle$, if $F = \nabla f$, then

$$P = f_x, \quad Q = f_y.$$

If the second partial derivatives are continuous, then

$$P_y = f_{xy} = f_{yx} = Q_x.$$

Therefore, every conservative field satisfies

$$P_y = Q_x.$$

On a simply connected region, this condition is also sufficient.

A necessary condition is not always sufficient

The equation $P_y = Q_x$ is a local test. It guarantees conservativeness only when the domain has no holes and the field is defined throughout the region. Singularities and holes can create nonzero circulation even when $P_y = Q_x$ away from the singularity.

Not conservative

Let $F(x, y) = \langle -y, x \rangle$. Here

$$P = -y, \quad Q = x.$$

Then

$$P_y = -1, \quad Q_x = 1.$$

Since $P_y \neq Q_x$, the field is not conservative. This field rotates counterclockwise around the origin. Closed-loop integrals are generally nonzero.

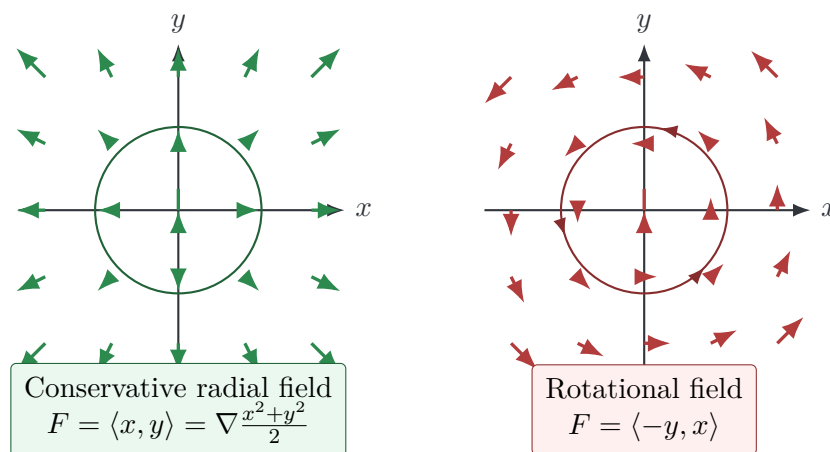


Figure 21.5: A conservative radial field and a rotational field. The rotational field has circulation around closed curves.

21.7 Closed curves and circulation

A curve C is **closed** if its endpoint is the same as its starting point. For a plane field $F = \langle P, Q \rangle$, the line integral around a closed curve is often written

$$\oint_C F \cdot d\mathbf{r} = \oint_C P dx + Q dy.$$

This is called the **circulation** of the vector field around C .

If F is conservative, then every closed-loop integral is zero:

$$\oint_C \nabla f \cdot d\mathbf{r} = 0.$$

If a closed-loop integral is not zero, then the field cannot be conservative.

Circulation around a circle

Let $F(x, y) = \langle -y, x \rangle$, and let C be the circle

$$\mathbf{r}(t) = \langle R \cos t, R \sin t \rangle, \quad 0 \leq t \leq 2\pi.$$

Then

$$d\mathbf{r} = \langle -R \sin t, R \cos t \rangle dt, \quad F(\mathbf{r}(t)) = \langle -R \sin t, R \cos t \rangle.$$

Thus

$$F(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = R^2 \sin^2 t + R^2 \cos^2 t = R^2.$$

Therefore,

$$\oint_C F \cdot d\mathbf{r} = \int_0^{2\pi} R^2 dt = 2\pi R^2.$$

The circulation is twice the area of the disk.

21.8 Green's theorem: circulation form

Green's theorem converts circulation around a boundary curve into a double integral over the enclosed region. Let D be a plane region with positively oriented boundary curve C . Positive orientation means that the region stays on the left as the curve is traversed; for a simple outer boundary, this is counterclockwise orientation.

Green's Theorem: circulation form

Let $F = \langle P, Q \rangle$, where P and Q have continuous first partial derivatives on a region containing D . If C is the positively oriented boundary of D , then

$$\oint_C P dx + Q dy = \iint_D (Q_x - P_y) dA.$$

The quantity $Q_x - P_y$ is the two-dimensional scalar curl of the field. Green's theorem says:

Circulation around the boundary equals total curl inside the region.

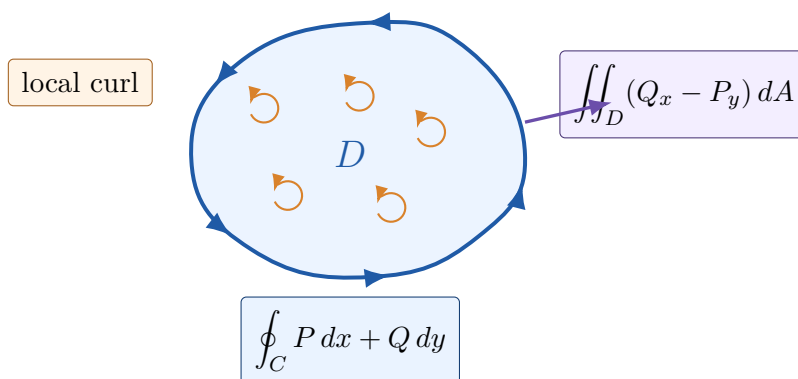


Figure 21.6: Green's theorem relates circulation around a positively oriented boundary to total curl over the enclosed region.

Use Green's theorem on a rectangle

Let $F(x, y) = \langle -y, x \rangle$. Compute $\oint_C F \cdot d\mathbf{r}$ where C is the positively oriented boundary of the rectangle $0 \leq x \leq a$, $0 \leq y \leq b$.

Here $P = -y$ and $Q = x$. Then

$$Q_x - P_y = 1 - (-1) = 2.$$

Therefore,

$$\oint_C -y dx + x dy = \iint_D 2 dA = 2ab.$$

The answer is twice the area of the rectangle.

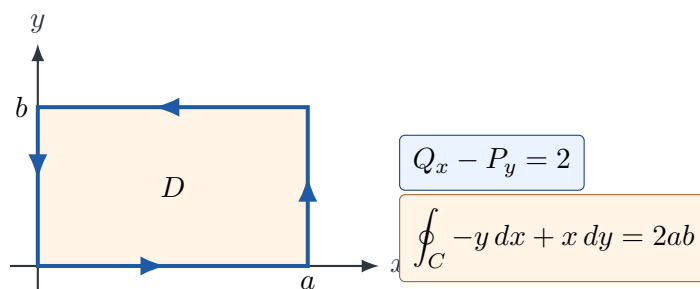


Figure 21.7: For $F = \langle -y, x \rangle$, circulation around a rectangle is twice the area of the rectangle.

21.9 Area from a line integral

Green's theorem can be used to compute area. If we choose P and Q so that $Q_x - P_y = 1$, then

$$\text{Area}(D) = \iint_D 1 dA = \oint_C P dx + Q dy.$$

Three common choices are

$$P = 0, \quad Q = x, \quad \text{Area}(D) = \oint_C x dy,$$

$$P = -y, \quad Q = 0, \quad \text{Area}(D) = \oint_C -y \, dx,$$

and the symmetric formula

$$\text{Area}(D) = \frac{1}{2} \oint_C x \, dy - y \, dx.$$

Area of an ellipse

Let C be the ellipse

$$\mathbf{r}(t) = \langle a \cos t, b \sin t \rangle, \quad 0 \leq t \leq 2\pi.$$

Using $\text{Area}(D) = \frac{1}{2} \oint_C x \, dy - y \, dx$, we have

$$x = a \cos t, \quad y = b \sin t, \quad dx = -a \sin t \, dt, \quad dy = b \cos t \, dt.$$

Thus

$$x \, dy - y \, dx = ab \cos^2 t \, dt + ab \sin^2 t \, dt = ab \, dt.$$

Therefore,

$$\text{Area}(D) = \frac{1}{2} \int_0^{2\pi} ab \, dt = \pi ab.$$

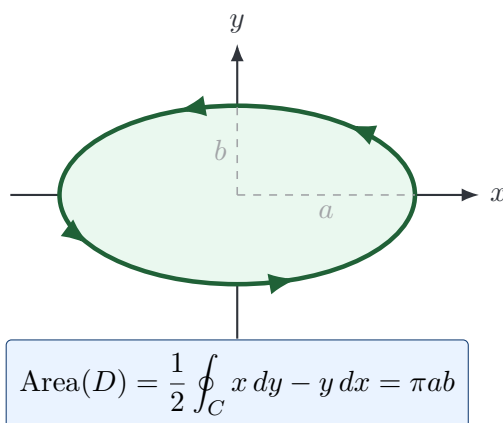


Figure 21.8: Area can be computed from boundary data alone using Green's theorem.

21.10 Green's theorem: flux form

Green's theorem also has a flux form. Let $F = \langle P, Q \rangle$. The outward flux across C is

$$\oint_C F \cdot \mathbf{n} \, ds,$$

where $\mathbf{n}$ is the outward unit normal. For a positively oriented plane curve, the outward normal satisfies

$$\mathbf{n} \, ds = \langle dy, -dx \rangle.$$

Therefore,

$$F \cdot \mathbf{n} \, ds = P \, dy - Q \, dx.$$

Green's theorem in flux form is

$$\oint_C F \cdot \mathbf{n} \, ds = \iint_D (P_x + Q_y) \, dA.$$

The quantity $P_x + Q_y$ is the two-dimensional divergence of F .

Flux interpretation

Outward flux through the boundary equals total divergence inside the region. This result foreshadows the divergence theorem in three dimensions.

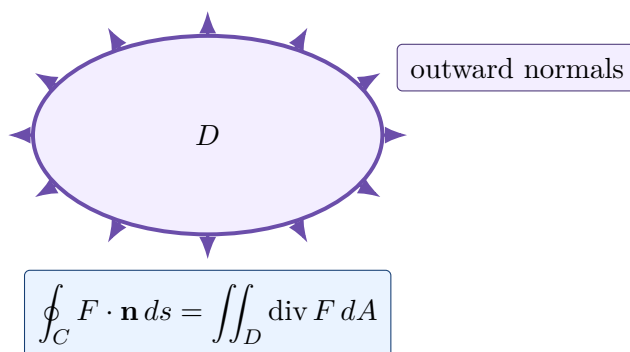


Figure 21.9: Green's theorem in flux form relates outward flux to total divergence in the region.

Outward flux of a radial field

Let $F(x, y) = \langle x, y \rangle$. Then

$$\operatorname{div} F = P_x + Q_y = 1 + 1 = 2.$$

If C bounds a region D , then

$$\oint_C F \cdot \mathbf{n} \, ds = \iint_D 2 \, dA = 2 \operatorname{Area}(D).$$

For a disk of radius R , this gives

$$\oint_C F \cdot \mathbf{n} \, ds = 2\pi R^2.$$

21.11 Holes, singularities, and why hypotheses matter

The field

$$F(x, y) = \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle$$

is defined everywhere except the origin. Away from the origin, one can check that $P_y = Q_x$. However, the field is not conservative on any region that loops around the origin. On the unit circle $\mathbf{r}(t) = \langle \cos t, \sin t \rangle$, we have

$$F(\mathbf{r}(t)) = \langle -\sin t, \cos t \rangle = \mathbf{r}'(t),$$

so

$$\oint_C F \cdot d\mathbf{r} = \int_0^{2\pi} 1 \, dt = 2\pi.$$

The origin is a singularity. The domain has a hole. This example warns that local derivative tests do not always determine global behavior.

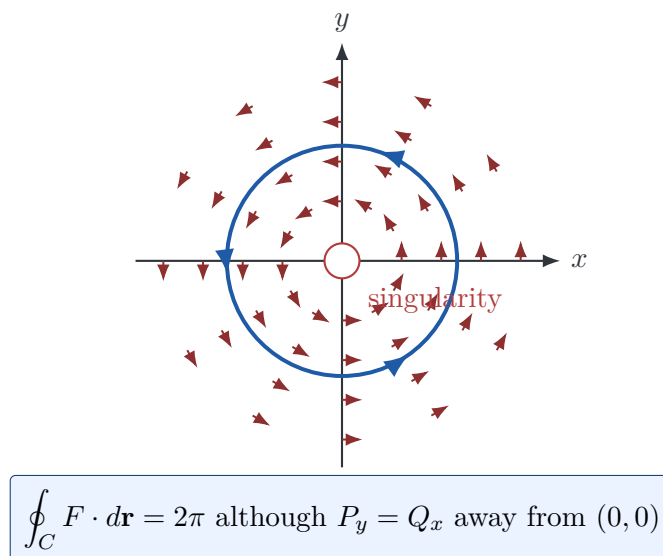


Figure 21.10: A field with a singularity at the origin. The local curl is zero away from the origin, but the circulation around the origin is nonzero.

21.12 Numerical curl and discrete Green's theorem intuition

Green's theorem says that circulation around a boundary measures accumulated curl inside. In computational settings, one often estimates derivatives on a grid. The scalar curl of a plane vector field $F = \langle P, Q \rangle$ is

$$\text{curl}_{2D} F = Q_x - P_y.$$

A gradient field has zero scalar curl when second partial derivatives commute. A rotational field has visible nonzero scalar curl.

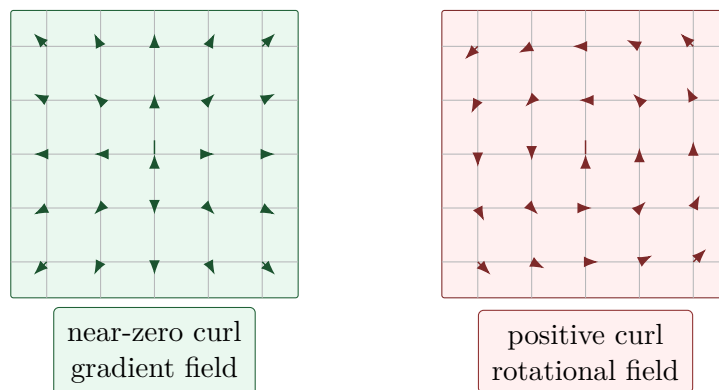


Figure 21.11: Numerical scalar curl can be estimated on a grid. Gradient fields have near-zero curl; rotational fields do not.

21.13 Modern interpretation: potentials, losses, and dynamics

In applied mathematics, statistics, data science, and machine learning, a scalar objective function is often the central object:

$$L(\theta_1, \dots, \theta_n).$$

The gradient field ∇L points in the direction of steepest increase, while $-\nabla L$ points in the direction of steepest decrease. Gradient descent follows a trajectory determined by a conservative field:

$$\frac{d\theta}{dt} = -\nabla L(\theta).$$

Along such a path,

$$\frac{d}{dt}L(\theta(t)) = \nabla L(\theta(t)) \cdot \theta'(t) = -\|\nabla L(\theta(t))\|^2 \leq 0.$$

So a pure gradient flow cannot cycle around a closed loop while strictly decreasing the loss. In contrast, a rotational component can create circulation and oscillation. This distinction matters in dynamical systems, adversarial optimization, non-equilibrium probability flows, and training dynamics for models with coupled parameters.

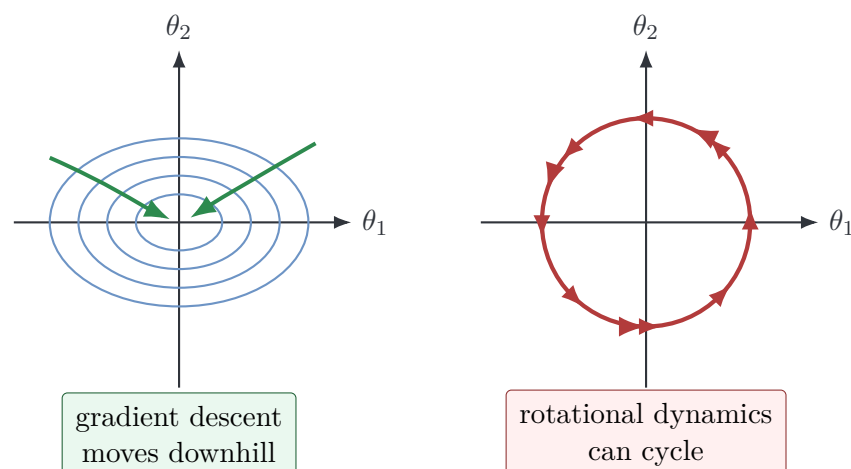


Figure 21.12: A pure gradient flow descends a potential landscape, while rotational dynamics can create persistent circulation.

Conservative versus rotational learning dynamics

A pure gradient field comes from a scalar objective. A rotational field does not. Many modern algorithms can be viewed as combining gradient-like motion with rotational or momentum-like effects. Green's theorem gives a mathematical language for detecting and measuring circulation in two-dimensional slices of a high-dimensional system.

21.14 Problem-solving guide

When faced with a line integral of a plane vector field:

1. **Check whether the field is conservative.** Try to find a potential function.
2. **If the curve is open and the field is conservative, use endpoints.**
3. **If the curve is closed, check Green's theorem.** The boundary integral may become an easier double integral.
4. **Check orientation.** Counterclockwise outer boundaries are positive. Reversing orientation changes the sign.
5. **Check the domain.** Holes or singularities can invalidate a simple conservative-field conclusion.
6. **Choose the easier representation.** Some problems are easier as line integrals; others are easier as double integrals.

21.15 Common mistakes

Watch for these errors

- Treating $P_y = Q_x$ as sufficient without checking the domain.
- Forgetting that Green's theorem requires a closed boundary curve.
- Using the wrong orientation and missing a minus sign.
- Confusing circulation form with flux form.
- Forgetting that a potential function is determined only up to an additive constant.
- Applying Green's theorem across a singularity where the field is not defined.

21.16 Exercises

Conceptual exercises

1. Explain in your own words what it means for a vector field to be conservative.
2. Why does a gradient field have zero line integral around every closed curve?
3. What is the difference between path independence and parameterization independence?
4. What does Green's theorem say geometrically?
5. Why does a hole in the domain matter for the conservative-field test?
6. Explain why the field $F = \langle -y, x \rangle$ should have nonzero circulation around the unit circle.

Computational exercises

7. Find a potential function for

$$F(x, y) = \langle 3x^2y + 2, x^3 + 4y \rangle.$$

8. Evaluate $\int_C F \cdot d\mathbf{r}$ from $(0, 0)$ to $(1, 2)$ for the field in Exercise 7.
9. Determine whether

$$F(x, y) = \langle y^2 + e^x, 2xy + \cos y \rangle$$

is conservative. If it is, find a potential function.

10. Compute

$$\oint_C -y \, dx + x \, dy$$

where C is the positively oriented boundary of the triangle with vertices $(0, 0)$, $(2, 0)$, and $(0, 3)$.

11. Use Green's theorem to compute

$$\oint_C (x^2 - y) dx + (x + y^2) dy$$

where C is the positively oriented boundary of the rectangle $0 \leq x \leq 2$, $0 \leq y \leq 1$.

12. Use a boundary integral to find the area enclosed by the parametric curve

$$\mathbf{r}(t) = \langle 3 \cos t, 2 \sin t \rangle, \quad 0 \leq t \leq 2\pi.$$

13. Let

$$F(x, y) = \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle.$$

Compute the line integral around the circle of radius 3 centered at the origin. Explain why the answer does not contradict the local equality $P_y = Q_x$ away from the origin.

Applied and computational exercises

14. Let $L(x, y) = x^2 + 4y^2$. Compute the work done by ∇L along any curve from $(1, 1)$ to $(2, 0)$.
15. For the vector field $F = \langle -y, x \rangle$, numerically approximate the circulation around the ellipse $x = 2 \cos t$, $y = \sin t$. Compare your answer with Green's theorem.
16. Let $F = \langle P, Q \rangle$ be sampled on a grid. Describe how finite differences could be used to estimate whether F is approximately conservative.
17. In a learning problem, suppose parameter dynamics show persistent closed-loop motion in a two-dimensional slice of parameter space. Explain why this suggests that the dynamics are not simply pure gradient descent on a scalar loss.
18. Use Green's theorem to derive the symmetric area formula

$$\text{Area}(D) = \frac{1}{2} \oint_C x dy - y dx.$$

21.17 Chapter summary

This chapter developed the bridge from line integrals to global vector-calculus theorems.

- A vector field is conservative if it is the gradient of a potential function.
- Line integrals of conservative fields depend only on endpoints.
- The Fundamental Theorem for Line Integrals is the chain rule plus the one-variable Fundamental Theorem of Calculus.
- Closed-loop integrals of conservative fields are zero.
- In the plane, the condition $P_y = Q_x$ is the local conservative-field test, but the domain matters.
- Green's theorem converts circulation around a closed curve into a double integral of scalar curl over the enclosed region.

- Green's theorem also has a flux form involving divergence.
- Area can be computed from a boundary integral.
- Singularities and holes explain why local derivative information may not determine global circulation.

In the next chapter, these ideas move from plane regions to surfaces in three-dimensional space. Curl, divergence, surface integrals, and flux are the ingredients needed for Stokes' theorem and the divergence theorem.

Chapter 22. Curl, Divergence, and Surface Integrals

Local rotation, local expansion, oriented surfaces, and flux

The previous chapters developed line integrals and Green's theorem. A line integral measures how a scalar or vector field accumulates along a curve. Green's theorem showed that circulation or flux around a closed plane curve can be computed by integrating a local derivative over the enclosed region.

This chapter moves that idea into three-dimensional space. Instead of curves enclosing planar regions, we now study surfaces in space. Instead of asking only how much a vector field pushes along a path, we ask how much it rotates near a point, how much it expands out of a point, and how much it flows through a surface.

The central objects are curl, divergence, scalar surface integrals, and flux integrals. They prepare the way for Stokes' theorem and the divergence theorem.

Learning goals

After studying this chapter, you should be able to:

1. compute the curl of a vector field in space;
2. interpret curl as local rotational tendency;
3. compute and interpret divergence;
4. use the identities $\nabla \cdot (\nabla \times F) = 0$ and $\nabla \times (\nabla f) = 0$ when appropriate;
5. parameterize surfaces and compute tangent vectors;
6. compute surface area using $|r_u \times r_v|$;
7. compute scalar surface integrals;
8. orient a surface and compute flux integrals;
9. use graph formulas for surface area and flux;
10. approximate curl, divergence, surface area, and flux numerically.

The main theme

Curl and divergence are local differential measurements. Surface integrals and flux integrals are global accumulations. The great theorems of vector calculus explain when local rotation or local expansion determines a global boundary integral.

1 The differential operator ∇

For a scalar function $f(x, y, z)$, the gradient is

$$\nabla f = \langle f_x, f_y, f_z \rangle.$$

For a vector field

$$F(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle,$$

we use the formal differential operator

$$\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle.$$

This notation organizes three operations:

$$\nabla f \quad \text{gradient}, \quad \nabla \cdot F \quad \text{divergence}, \quad \nabla \times F \quad \text{curl}.$$

The notation is formal: ∇ is not an ordinary vector of numbers. Its components are derivative operators.

2 Curl

Let $F = \langle P, Q, R \rangle$ be a vector field in space. The *curl* of F is

$$\nabla \times F = \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle.$$

Equivalently,

$$\nabla \times F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}.$$

Curl measures the infinitesimal tendency of the field to rotate. Its direction is the axis of rotation determined by the right-hand rule, and its magnitude measures the local strength of rotation.

Curl in the plane

A two-dimensional vector field $F(x, y) = \langle P(x, y), Q(x, y) \rangle$ can be treated as the three-dimensional field $\langle P, Q, 0 \rangle$. Then

$$\nabla \times F = \langle 0, 0, Q_x - P_y \rangle.$$

The scalar quantity $Q_x - P_y$ is the scalar curl of the plane field. It is exactly the integrand in Green's theorem in circulation form.

Example 2.1: A rotating field

Let

$$F(x, y, z) = \langle -y, x, 0 \rangle.$$

Then $P = -y$, $Q = x$, and $R = 0$. Hence

$$\nabla \times F = \langle 0 - 0, 0 - 0, 1 - (-1) \rangle = \langle 0, 0, 2 \rangle.$$

The field rotates counterclockwise in the xy -plane. The curl points in the positive z -direction.

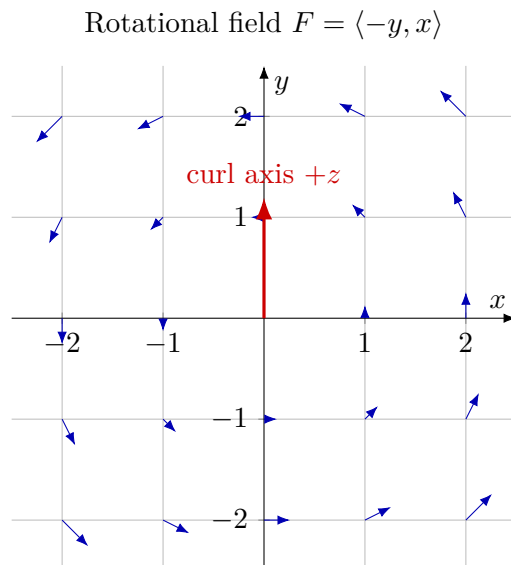


Figure 1: The field $F = \langle -y, x \rangle$ rotates counterclockwise. Its curl points in the positive z -direction.

Example 2.2: Curl of a three-dimensional field

Let

$$F(x, y, z) = \langle yz, xz, xy + e^z \rangle.$$

Then $P = yz$, $Q = xz$, and $R = xy + e^z$. We have

$$R_y = x, \quad Q_z = x, \quad P_z = y, \quad R_x = y, \quad Q_x = z, \quad P_y = z.$$

Hence

$$\nabla \times F = \langle x - x, y - y, z - z \rangle = \langle 0, 0, 0 \rangle.$$

Indeed, $F = \nabla f$ for $f(x, y, z) = xyz + e^z$.

3 Divergence

Let $F = \langle P, Q, R \rangle$. The *divergence* of F is the scalar field

$$\nabla \cdot F = P_x + Q_y + R_z.$$

Divergence measures the local tendency of a vector field to flow outward from a point. Positive divergence indicates source-like behavior, negative divergence indicates sink-like behavior, and zero divergence indicates local incompressibility.

Divergence is a scalar field

Curl returns a vector field in three dimensions. Divergence returns a scalar field:

$$\nabla \times F \in \mathbb{R}^3, \quad \nabla \cdot F \in \mathbb{R}.$$

Example 3.1: A source field and an incompressible rotational field

For $F(x, y, z) = \langle x, y, z \rangle$,

$$\nabla \cdot F = 1 + 1 + 1 = 3.$$

The field points away from the origin and has positive divergence.

For $F(x, y, z) = \langle -y, x, 0 \rangle$,

$$\nabla \cdot F = 0 + 0 + 0 = 0.$$

The field rotates, but it does not locally expand or compress volume.

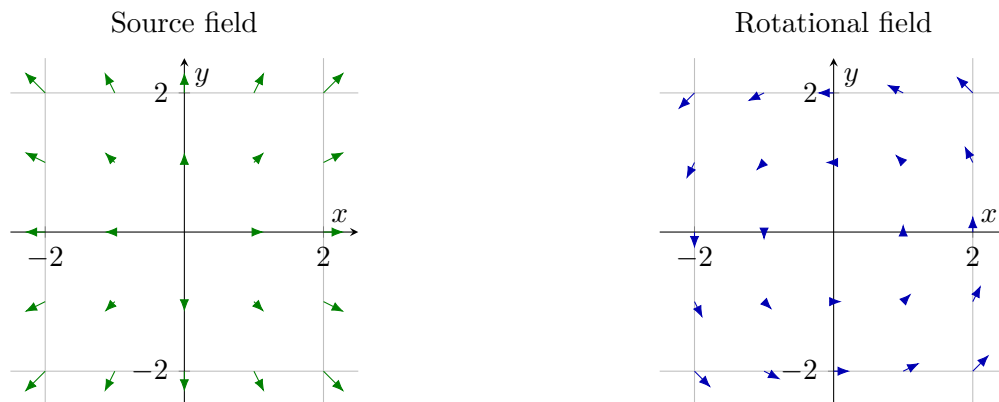


Figure 2: A radial source field has positive divergence. A rotational field can have zero divergence but nonzero curl.

4 Two fundamental identities

For sufficiently smooth functions and fields, two identities occur throughout vector calculus:

$$\nabla \times (\nabla f) = 0, \quad \nabla \cdot (\nabla \times F) = 0.$$

The first says that gradient fields have no local rotation. It is the three-dimensional version of the conservative-field test. The second says that a curl field has no local source or sink.

Smoothness and domain matter

These identities require enough differentiability. Also, equations such as $\nabla \times F = 0$ guarantee a global potential only under suitable assumptions on the domain. Holes and singularities can create global circulation even when local curl is zero away from the singularity.

5 Numerical curl and divergence on a grid

In applications, a vector field may be sampled rather than given by a formula. For a plane field $F = \langle P, Q \rangle$, finite differences approximate

$$\nabla \cdot F = P_x + Q_y, \quad \text{curl}_z F = Q_x - P_y.$$

This is used for fluid simulation, weather models, image motion, and learned dynamical systems.

A sampled field with divergence and scalar curl

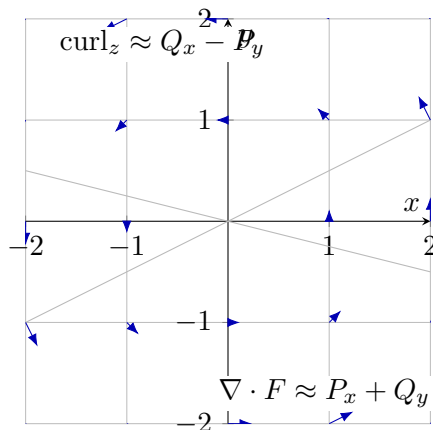


Figure 3: Finite differences approximate divergence and scalar curl from neighboring values on a grid.

6 Surfaces and parameterizations

A curve is described by one parameter:

$$r(t) = \langle x(t), y(t), z(t) \rangle.$$

A surface is described by two parameters:

$$r(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle, \quad (u, v) \in D.$$

The parameter domain D lies in the uv -plane. The map r bends and stretches this flat parameter region into a surface S in space.

For a parameterized surface, the tangent vectors are

$$r_u = \frac{\partial r}{\partial u}, \quad r_v = \frac{\partial r}{\partial v}.$$

The cross product $r_u \times r_v$ is normal to the surface. Its magnitude gives the local area scaling factor:

$$dS = |r_u \times r_v| du dv.$$

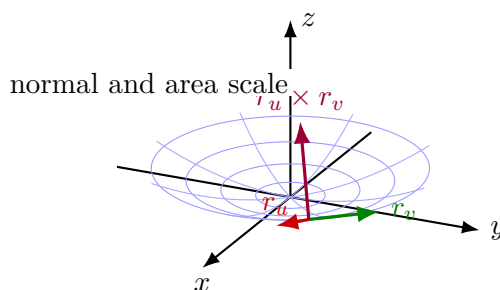


Figure 4: For a parameterized surface, $r_u \times r_v$ gives a normal direction and the local area scale.

7 Surface area

The area of a parameterized surface $r(u, v)$, $(u, v) \in D$, is

$$A(S) = \iint_D |r_u \times r_v| dA.$$

This is the surface analog of arc length. For curves, $|r'(t)|$ measures how much the parameter stretches length. For surfaces, $|r_u \times r_v|$ measures how much the two-parameter map stretches area.

Graph surfaces

If a surface is a graph $z = g(x, y)$ over D , parameterize it as

$$r(x, y) = \langle x, y, g(x, y) \rangle.$$

Then

$$r_x = \langle 1, 0, g_x \rangle, \quad r_y = \langle 0, 1, g_y \rangle,$$

so

$$r_x \times r_y = \langle -g_x, -g_y, 1 \rangle.$$

Therefore

$$dS = \sqrt{1 + g_x^2 + g_y^2} dA, \quad A(S) = \iint_D \sqrt{1 + g_x^2 + g_y^2} dA.$$

Example 7.1: Area of a plane patch

Find the area of the part of the plane

$$z = 8 + 2x + 5y$$

above the rectangle $0 \leq x \leq 3$, $1 \leq y \leq 6$. Here $g_x = 2$ and $g_y = 5$, so

$$\sqrt{1 + g_x^2 + g_y^2} = \sqrt{1 + 4 + 25} = \sqrt{30}.$$

The rectangle has area 15, so

$$A(S) = 15\sqrt{30}.$$

8 Scalar surface integrals

If $f(x, y, z)$ is a scalar function defined on a surface S , then the scalar surface integral is

$$\iint_S f(x, y, z) dS.$$

For a parameterized surface $r(u, v)$,

$$\iint_S f dS = \iint_D f(r(u, v)) |r_u \times r_v| du dv.$$

If f is a surface density, this integral gives total mass. If $f = 1$, it gives surface area.

Example 8.1: Surface integral over a paraboloid

Let S be the graph $z = x^2 + y^2$ over the disk $x^2 + y^2 \leq 1$. Compute

$$\iint_S z \, dS.$$

Since $g(x, y) = x^2 + y^2$, we have $g_x = 2x$ and $g_y = 2y$, so

$$dS = \sqrt{1 + 4x^2 + 4y^2} \, dA.$$

The integral becomes

$$\iint_D (x^2 + y^2) \sqrt{1 + 4x^2 + 4y^2} \, dA.$$

In polar coordinates,

$$\iint_S z \, dS = \int_0^{2\pi} \int_0^1 r^2 \sqrt{1 + 4r^2} \, r \, dr \, d\theta = 2\pi \int_0^1 r^3 \sqrt{1 + 4r^2} \, dr.$$

Paraboloid over the unit disk

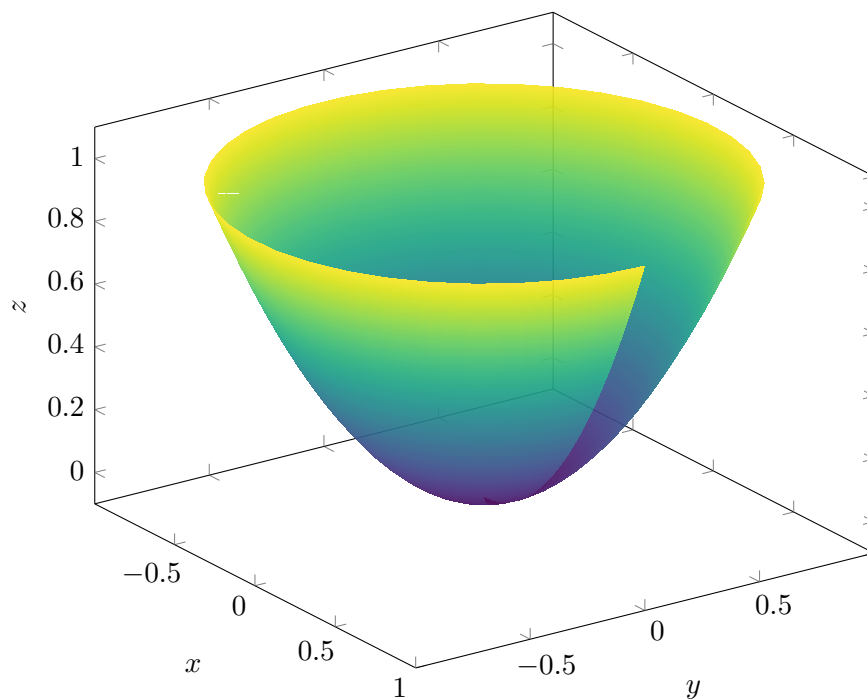


Figure 5: A scalar surface integral accumulates density over a curved surface. Here the density is $f(x, y, z) = z$ on $z = x^2 + y^2$.

9 Orientation of surfaces

A surface is *orientable* if it is possible to choose a continuous unit normal vector n at every point of the surface. Choosing one of the two possible normal directions gives an *orientation*.

For a parameterized surface $r(u, v)$, the vector $r_u \times r_v$ defines one orientation. Reversing the parameter order changes the normal direction:

$$r_v \times r_u = -(r_u \times r_v).$$

For a closed surface, the standard orientation is outward. For an open surface, the orientation must be specified, for example upward, downward, outward, or inward.

Orientation affects flux

Scalar surface integrals do not depend on orientation because dS is positive. Flux integrals do depend on orientation because changing n to $-n$ changes the sign of $F \cdot n$.

10 Flux integrals

Let F be a vector field and let S be an oriented surface with unit normal vector n . The *flux* of F through S is

$$\iint_S F \cdot n \, dS.$$

This measures the net amount of the field passing through the surface in the chosen normal direction.

Using a parameterization $r(u, v)$, we have

$$n \, dS = (r_u \times r_v) \, du \, dv$$

for the orientation determined by $r_u \times r_v$. Therefore

$$\iint_S F \cdot n \, dS = \iint_D F(r(u, v)) \cdot (r_u \times r_v) \, du \, dv.$$

Graph formula for flux

Suppose $S : z = g(x, y)$ over D , with upward orientation. Then

$$r_x \times r_y = \langle -g_x, -g_y, 1 \rangle.$$

If $F = \langle P, Q, R \rangle$, then

$$\iint_S F \cdot n \, dS = \iint_D (-Pg_x - Qg_y + R) \, dA,$$

where P, Q, R are evaluated at $(x, y, g(x, y))$. For downward orientation, the sign changes.

Example 10.1: Flux through a graph

Let $F(x, y, z) = \langle x, y, z \rangle$, and let S be the upper paraboloid

$$z = 1 - x^2 - y^2, \quad x^2 + y^2 \leq 1,$$

with upward orientation. Here $g_x = -2x$ and $g_y = -2y$. The upward flux integrand is

$$-Pg_x - Qg_y + R = -x(-2x) - y(-2y) + (1 - x^2 - y^2) = 1 + x^2 + y^2.$$

Therefore

$$\iint_S F \cdot n \, dS = \iint_D (1 + x^2 + y^2) \, dA.$$

In polar coordinates,

$$\int_0^{2\pi} \int_0^1 (1 + r^2) r \, dr \, d\theta = 2\pi \left[\frac{r^2}{2} + \frac{r^4}{4} \right]_0^1 = \frac{3\pi}{2}.$$

Projected upward normals on $z = 1 - x^2 - y^2$

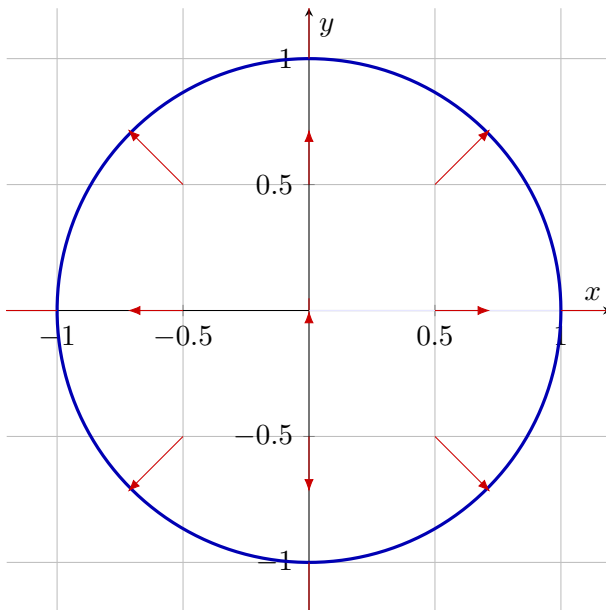


Figure 6: For the upward-oriented graph $z = 1 - x^2 - y^2$, the projected normal arrows in the xy -plane point outward and the normal has positive z -component.

11 Flux through standard surfaces

Sphere

The sphere of radius a centered at the origin can be parameterized by

$$r(\phi, \theta) = \langle a \sin \phi \cos \theta, a \sin \phi \sin \theta, a \cos \phi \rangle, \quad 0 \leq \phi \leq \pi, \quad 0 \leq \theta \leq 2\pi.$$

The outward unit normal is

$$n = \frac{1}{a} \langle x, y, z \rangle.$$

If $F = \langle x, y, z \rangle$, then on the sphere,

$$F \cdot n = \langle x, y, z \rangle \cdot \frac{1}{a} \langle x, y, z \rangle = \frac{x^2 + y^2 + z^2}{a} = a.$$

Since the surface area is $4\pi a^2$, the flux is

$$\iint_S F \cdot n \, dS = a(4\pi a^2) = 4\pi a^3.$$

For the sphere $x^2 + y^2 + z^2 = 4$, $a = 2$, so the flux is 32π .

Radial outward normals: cross-section

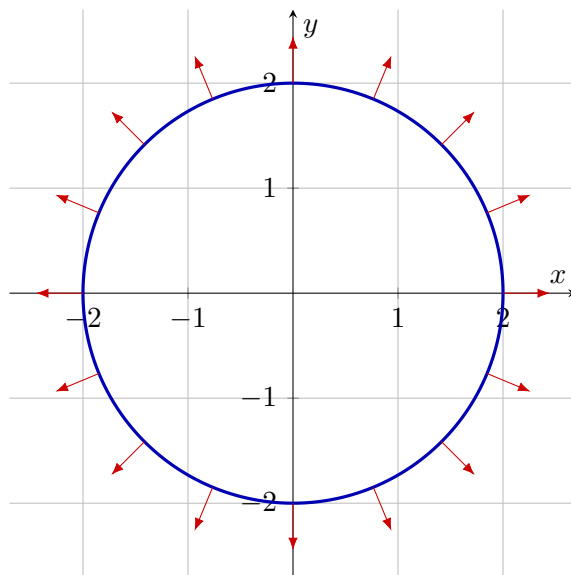


Figure 7: For a sphere centered at the origin, outward normals are radial. The cross-section shows the same orientation idea.

Cylinder

A circular cylinder of radius a can be parameterized by

$$r(\theta, z) = \langle a \cos \theta, a \sin \theta, z \rangle.$$

Then

$$r_\theta = \langle -a \sin \theta, a \cos \theta, 0 \rangle, \quad r_z = \langle 0, 0, 1 \rangle.$$

The cross product

$$r_z \times r_\theta = \langle -a \cos \theta, -a \sin \theta, 0 \rangle$$

points inward, while

$$r_\theta \times r_z = \langle a \cos \theta, a \sin \theta, 0 \rangle$$

points outward. Always check orientation.

12 Numerical surface integration

For computational work, a parameterized surface integral is usually approximated by sampling a grid in parameter space. If $r(u, v)$, $(u, v) \in D$, then

$$\iint_S f \, dS \approx \sum_{i,j} f(r(u_i, v_j)) |r_u(u_i, v_j) \times r_v(u_i, v_j)| \Delta u \Delta v.$$

Similarly,

$$\iint_S F \cdot n \, dS \approx \sum_{i,j} F(r(u_i, v_j)) \cdot (r_u(u_i, v_j) \times r_v(u_i, v_j)) \Delta u \Delta v.$$

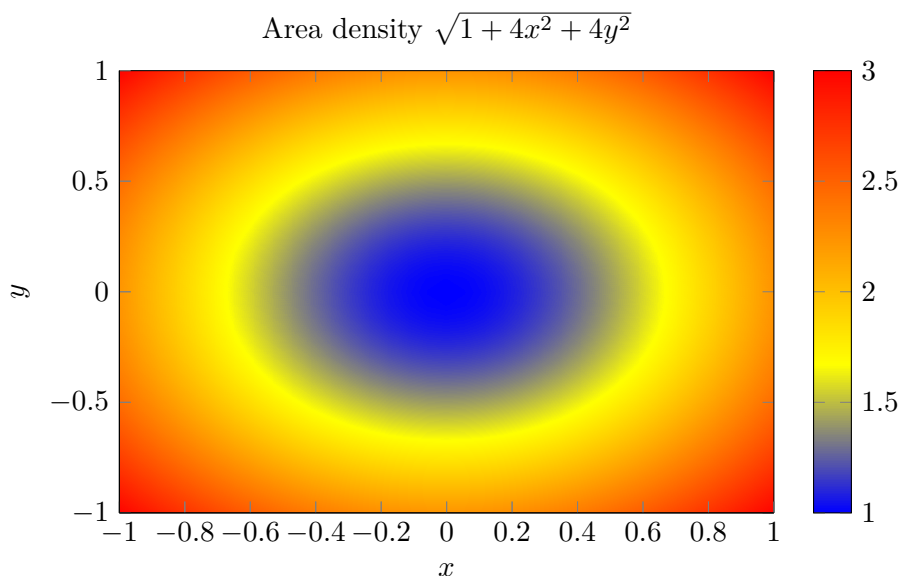


Figure 8: Numerical surface integration samples the parameter domain and multiplies by the local area scale.

The plot shows the area-density factor for $z = x^2 + y^2$ over a square. Summing this density over a fine grid is a Riemann-sum approximation to surface area.

13 Curl, divergence, and flux in applied modeling

Fluid velocity fields

If F is a fluid velocity field, then $\nabla \times F$ measures local swirling, $\nabla \cdot F$ measures local expansion or compression, and $\iint_S F \cdot n \, dS$ measures the rate of flow through a surface. These ideas appear in incompressible flow, fluid simulation, weather models, and diffusion processes.

Probability flow

In probability and statistics, a time-dependent density can move under a velocity field. Divergence helps describe whether probability mass is locally spreading or concentrating. In continuous normalizing flows and score-based generative modeling, vector fields and Jacobians are central tools.

Optimization and machine learning

Gradient descent is driven by a vector field $-\nabla L$. Such a field has zero curl when L is smooth. More general training dynamics may include momentum, stochastic noise, constraints, or game-

theoretic interactions. These dynamics can show rotational behavior, naturally detected by curl-like measurements.

Geometry and data

Surface integrals appear when data lie on curved manifolds rather than flat Euclidean spaces. Area elements and Jacobian factors explain why densities change under parameterization and why sampling uniformly in parameter coordinates does not always sample uniformly on a surface.

14 Common mistakes

1. Forgetting that curl is a vector in space. In three dimensions, $\nabla \times F$ has three components.
2. Confusing scalar curl with full curl. In the plane, $Q_x - P_y$ is the z -component of the three-dimensional curl of $\langle P, Q, 0 \rangle$.
3. Forgetting the area factor. The factor $|r_u \times r_v|$ is essential in a surface integral.
4. Using the wrong orientation. Flux changes sign when the normal direction is reversed.
5. Using graph formulas outside their setting. The formula for $z = g(x, y)$ applies only when the surface is represented as a graph over the xy -plane.
6. Treating dS like dA . For curved surfaces, dS includes an area-stretching factor.

15 Chapter exercises

Conceptual exercises

1. Explain the difference between curl and divergence.
2. Give an example of a vector field with nonzero curl and zero divergence.
3. Give an example of a vector field with zero curl and nonzero divergence.
4. Explain why scalar surface integrals do not depend on orientation but flux integrals do.
5. What does the vector $r_u \times r_v$ represent geometrically?
6. Why does reversing the order of the cross product reverse the orientation?
7. Explain why $\nabla \times (\nabla f) = 0$ should be expected geometrically.
8. Explain why $\nabla \cdot (\nabla \times F) = 0$ suggests that curl fields have no sources or sinks.

Computational exercises

9. Compute the curl of $F(x, y, z) = \langle xz, yz, xy \rangle$.
10. Compute the divergence of $F(x, y, z) = \langle x^2y, y^2z, z^2x \rangle$.
11. Determine whether $F(x, y, z) = \langle yz, xz, xy \rangle$ has zero curl.

12. Let $F(x, y, z) = \langle -y, x, z \rangle$. Compute $\nabla \times F$ and $\nabla \cdot F$.
13. Let S be the graph $z = x^2 + y^2$ over $0 \leq x \leq 1$, $0 \leq y \leq 1$. Set up the surface area integral.
14. Compute the area of the plane $z = 1 + 2x - 2y$ above the rectangle $0 \leq x \leq 3$, $0 \leq y \leq 2$.
15. Let S be the sphere of radius 3 centered at the origin. Compute the outward flux of $F = \langle x, y, z \rangle$ through S .
16. Let S be the graph $z = 4 - x - y$ over the triangle with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$. Compute the upward flux of $F = \langle 0, 0, z \rangle$ through S .
17. Parameterize the cylinder $x^2 + y^2 = 9$, $0 \leq z \leq 4$, and compute $|r_\theta \times r_z|$.
18. Let $r(u, v) = \langle u, v, u^2 - v^2 \rangle$, $0 \leq u \leq 1$, $0 \leq v \leq 1$. Compute $r_u \times r_v$ and set up the surface area integral.

Applied and computational exercises

19. Approximate the surface area of $z = \sin(xy)$ over $[-1, 1] \times [-1, 1]$.
20. Approximate the flux of $F = \langle x, y, z \rangle$ through the upward-oriented graph $z = 1 - x^2 - y^2$ over the unit disk.
21. Suppose a measured two-dimensional velocity field has approximately zero divergence but nonzero scalar curl. Give an applied interpretation.
22. Suppose a learned dynamical system in parameter space has visible rotational motion around a point. Which derivative measurement from this chapter would detect that behavior?
23. A surface is parameterized by a neural network $r(u, v)$. Explain why $|r_u \times r_v|$ is the correct local area distortion factor.
24. In a probability flow model, why is divergence related to local density change?

16 Chapter summary

This chapter introduced the local derivative operations and surface integrals needed for the final theorems of vector calculus.

- Curl measures local rotation.
- Divergence measures local source or sink behavior.
- In the plane, scalar curl is $Q_x - P_y$.
- In space, curl is $\nabla \times F$ and divergence is $\nabla \cdot F$.
- The identity $\nabla \times (\nabla f) = 0$ says gradient fields have no local rotation.
- The identity $\nabla \cdot (\nabla \times F) = 0$ says curl fields have no local source or sink.
- A parameterized surface $r(u, v)$ has tangent vectors r_u and r_v .
- The vector $r_u \times r_v$ gives a normal direction and an area scale.

- Scalar surface integrals accumulate scalar quantities over curved surfaces.
- Flux integrals measure how much a vector field passes through an oriented surface.
- Orientation matters for flux but not for scalar surface integrals.

The next chapter uses these tools to state and apply Stokes' theorem and the divergence theorem, the three-dimensional successors of Green's theorem.

Chapter 23. Stokes Theorem and the Divergence Theorem

Boundary measurements, local derivatives, circulation, and flux

The previous chapters introduced vector fields, line integrals, curl, divergence, surface integrals, and flux. This chapter puts those ideas together into two central global theorems of vector calculus:

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$$

and

$$\iint_{\partial E} \mathbf{F} \cdot \mathbf{n} dS = \iiint_E \nabla \cdot \mathbf{F} dV.$$

The first theorem says that **circulation around a boundary** can be measured by **curl through a surface**. The second says that **outward flux through a closed boundary** can be measured by **divergence inside the solid**.

global boundary measurement = accumulated local derivative inside.

Bridge to applied mathematics, statistics, and AI

Stokes theorem and the Divergence theorem teach a powerful modeling habit: do not only compute pointwise quantities. Ask how local behavior accumulates globally. This perspective reappears in conservation laws, PDEs, diffusion models, probability flux, score-based generative models, graph flows, and numerical methods on grids and meshes.

Learning goals

After reading this chapter, you should be able to:

1. state Stokes theorem and identify the surface, boundary curve, orientation, curl, and circulation;
2. state the Divergence theorem and identify the solid, closed boundary surface, outward normal, divergence, and flux;
3. choose between a direct line integral, a direct surface integral, Stokes theorem, and the Divergence theorem;
4. use orientation rules correctly;
5. recognize the common structure behind the Fundamental Theorem of Calculus, Green's theorem, Stokes theorem, and the Divergence theorem;
6. verify these theorems numerically in simple examples;
7. explain how these ideas support conservation laws, flow models, and computational applied mathematics.

1 The family of fundamental theorems

The one-variable Fundamental Theorem of Calculus says

$$\int_a^b f'(x) dx = f(b) - f(a).$$

The left side accumulates a derivative over an interval. The right side measures boundary values at the endpoints:

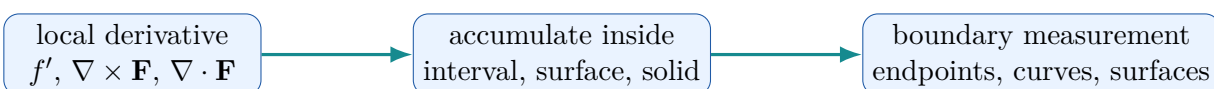
$$\text{total change inside} = \text{boundary difference}.$$

For vector calculus, intervals become curves, surfaces, and solids.

Theorem	Interior quantity	Boundary quantity
Fundamental Theorem of Calculus	derivative on an interval	endpoint values
Fundamental Theorem for Line Integrals	gradient along a curve	potential values at endpoints
Green's theorem	curl or divergence over a plane region	circulation or flux around boundary
Stokes theorem	curl through a surface	circulation around boundary curve
Divergence theorem	divergence inside a solid	flux through closed boundary surface

One guiding sentence

Stokes theorem converts **boundary circulation** into **surface curl**, while the Divergence theorem converts **closed-surface flux** into **solid divergence**.



Global theorems turn local rates into boundary information.

Figure 1: The common structure behind the global theorems of calculus.

2 Orientation and boundary direction

Orientation is the most common source of mistakes in this chapter. For a surface S , choosing a unit normal vector $\mathbf{n}$ gives an orientation. If S has a boundary curve ∂S , then the orientation of ∂S is determined by the right-hand rule:

- point the thumb of your right hand in the direction of $\mathbf{n}$;
- your curled fingers give the positive direction around ∂S .

For a closed surface ∂E , such as a sphere or the boundary of a box, the standard orientation is **outward**.

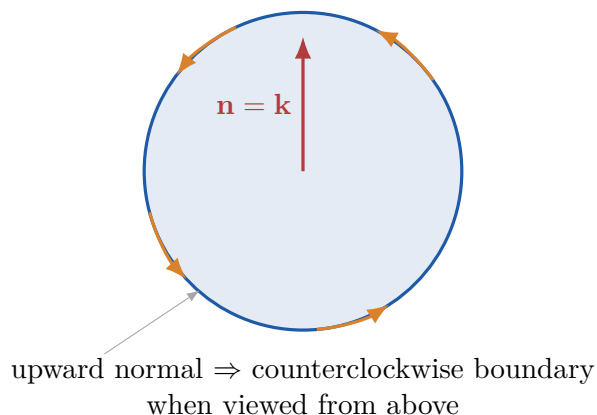


Figure 2: Boundary orientation for an upward oriented disk. Reversing the normal reverses the boundary direction.

For the disk in the xy -plane with upward normal $\mathbf{n} = \mathbf{k}$, the positive boundary orientation is counterclockwise when viewed from above. If the normal is changed to $-\mathbf{k}$, the positive boundary orientation becomes clockwise.

3 Stokes theorem

Stokes theorem

Let S be an oriented smooth surface with boundary curve $C = \partial S$. Let $\mathbf{F}$ be a vector field with continuous first partial derivatives near S . Then

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS.$$

Here $C = \partial S$ is the boundary curve, $d\mathbf{r}$ is tangent displacement along the curve, $\mathbf{F} \cdot d\mathbf{r}$ measures circulation, $\nabla \times \mathbf{F}$ is the curl, and $(\nabla \times \mathbf{F}) \cdot \mathbf{n}$ measures the component of curl passing through the surface.

Surface independence

If two oriented surfaces have the same boundary curve and compatible orientations, then Stokes theorem gives the same circulation integral over both surfaces. The boundary controls the answer.

4 Example: circulation around a circle

Circulation around a circle

Let

$$\mathbf{F}(x, y, z) = \langle -y, x, 0 \rangle.$$

Let C be the unit circle $x^2 + y^2 = 1$, $z = 0$, oriented counterclockwise as viewed from above. Parameterize

$$\mathbf{r}(t) = \langle \cos t, \sin t, 0 \rangle, \quad 0 \leq t \leq 2\pi.$$

Then

$$\mathbf{r}'(t) = \langle -\sin t, \cos t, 0 \rangle, \quad \mathbf{F}(\mathbf{r}(t)) = \langle -\sin t, \cos t, 0 \rangle.$$

Thus

$$\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = \sin^2 t + \cos^2 t = 1,$$

and

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} 1 \, dt = 2\pi.$$

The curl is $\nabla \times \mathbf{F} = \langle 0, 0, 2 \rangle$. Using the unit disk with upward normal,

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_S 2 \, dA = 2\pi.$$

The direct line integral and Stokes theorem agree.

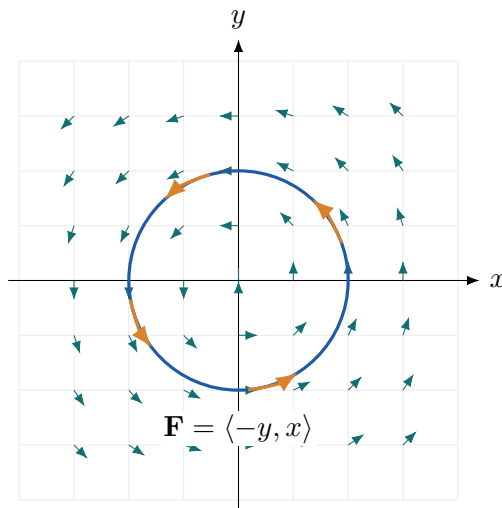


Figure 3: Rotational field and circulation around the unit circle. The field has constant upward curl $\langle 0, 0, 2 \rangle$.

5 Stokes theorem as a computational shortcut

Stokes theorem is useful in two directions. Sometimes the line integral is difficult but the curl integral is simple. Sometimes the surface integral is difficult but the boundary curve is simple. The

theorem lets us choose the easier calculation.

Replacing a tilted surface by a flat disk

Let

$$\mathbf{F}(x, y, z) = \langle -y, x, z \rangle.$$

Let S be any smooth oriented surface whose boundary is

$$C : x^2 + y^2 = a^2, \quad z = 0,$$

with upward-compatible orientation. The curl is still

$$\nabla \times \mathbf{F} = \langle 0, 0, 2 \rangle.$$

By Stokes theorem, instead of integrating over the possibly complicated surface S , we may use the flat disk D bounded by C :

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D 2 \, dA = 2\pi a^2.$$

The value depends on the boundary curve and orientation, not on the particular spanning surface.

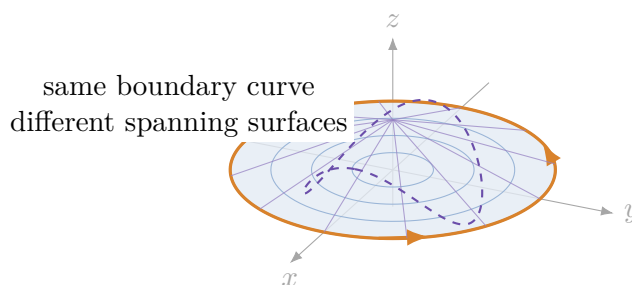


Figure 4: Different spanning surfaces may share the same boundary curve. Stokes theorem lets us use the convenient one.

6 Orientation signs in Stokes theorem

Changing the orientation changes the sign. If S has normal $\mathbf{n}$, then

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS.$$

If we reverse the normal to $-\mathbf{n}$, then the boundary orientation also reverses, and both sides become negative:

$$\oint_{-\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot (-\mathbf{n}) \, dS.$$

For $\mathbf{F} = \langle -y, x, 0 \rangle$ on the unit circle, counterclockwise orientation gives 2π , while clockwise orientation gives -2π .

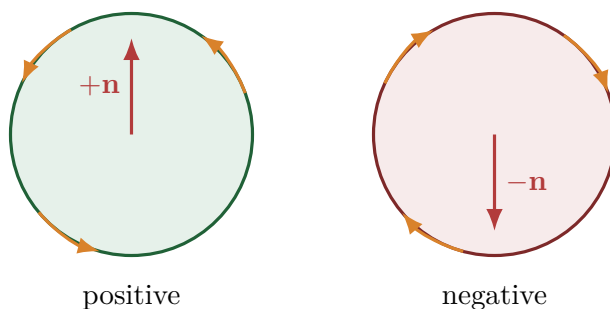


Figure 5: Reversing boundary orientation reverses the sign of circulation.

7 The Divergence theorem

The Divergence theorem

Let E be a solid region in $\mathbb{R}^3$ with closed boundary surface ∂E , oriented outward. If $\mathbf{F}$ is a vector field with continuous first partial derivatives near E , then

$$\iint_{\partial E} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_E \nabla \cdot \mathbf{F} \, dV.$$

Here ∂E is the closed boundary surface, $\mathbf{n}$ is the outward unit normal, $\mathbf{F} \cdot \mathbf{n}$ measures outward flux density, $\nabla \cdot \mathbf{F}$ measures local source strength, and the triple integral gives total source strength inside the solid.

outward flux through boundary = total divergence inside.

Closed surface required

The Divergence theorem applies to a closed surface, meaning the full boundary of a solid. A single disk, a single hemisphere, or the side of a cylinder alone is not closed. If the surface is not closed, add the missing pieces or use a direct surface integral.

8 Example: flux through a sphere

Outward radial field

Let

$$\mathbf{F}(x, y, z) = \langle x, y, z \rangle.$$

Let S be the sphere $x^2 + y^2 + z^2 = R^2$, oriented outward. On the sphere, the outward unit normal is

$$\mathbf{n} = \frac{1}{R} \langle x, y, z \rangle.$$

Thus

$$\mathbf{F} \cdot \mathbf{n} = \langle x, y, z \rangle \cdot \frac{1}{R} \langle x, y, z \rangle = \frac{x^2 + y^2 + z^2}{R} = R.$$

Therefore

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_S R \, dS = R(4\pi R^2) = 4\pi R^3.$$

The divergence is

$$\nabla \cdot \mathbf{F} = 1 + 1 + 1 = 3.$$

Hence

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_E 3 \, dV = 3 \cdot \frac{4\pi R^3}{3} = 4\pi R^3.$$

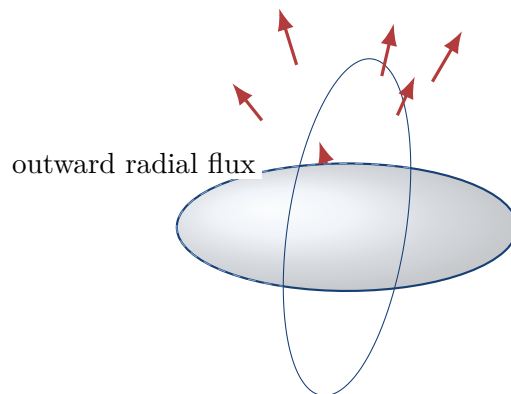


Figure 6: Outward radial field on a sphere. For $\mathbf{F} = \langle x, y, z \rangle$, every vector points normal to the sphere.

9 Example: outward flux through a box

Flux through a rectangular box

Let

$$\mathbf{F}(x, y, z) = \langle x^2, y^2, z^2 \rangle, \quad E = [0, a] \times [0, b] \times [0, c].$$

The divergence is

$$\nabla \cdot \mathbf{F} = 2x + 2y + 2z.$$

By the Divergence theorem,

$$\iint_{\partial E} \mathbf{F} \cdot \mathbf{n} \, dS = \int_0^a \int_0^b \int_0^c (2x + 2y + 2z) \, dz \, dy \, dx.$$

Computing term by term gives

$$\iint_{\partial E} \mathbf{F} \cdot \mathbf{n} \, dS = a^2bc + ab^2c + abc^2 = abc(a + b + c).$$

For a unit cube, the answer is 3.

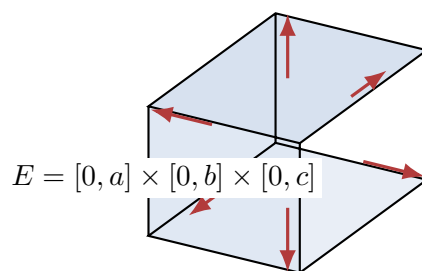


Figure 7: The Divergence theorem sums local source strength inside the box and equals outward flux through all six faces.

10 Choosing the right theorem

When a problem asks for a line integral, surface integral, circulation, or flux, choose the method that simplifies the geometry or lowers the dimension.

Situation	Strategy
Line integral over a curve	If the field is conservative, use a potential. If the curve is closed and bounds a surface, consider Stokes theorem. If it is planar, Green's theorem may be simpler.
Flux through a surface	If the surface is closed, consider the Divergence theorem. If it is open, add missing pieces or compute directly.
Curl through a surface	If the boundary curve is simple, use Stokes theorem. If the surface is simple, integrate directly.
Orientation uncertainty	Draw the normal and boundary direction before computing. Reversing orientation changes the sign.

Computational strategy

Use the theorem that lowers the dimension or simplifies the geometry. A line integral may become a surface integral, a surface integral may become a triple integral, and a complicated surface may be replaced by a simpler surface with the same boundary.

11 Example: using Stokes theorem on a tilted triangle

A triangular Stokes calculation

Let

$$\mathbf{F}(x, y, z) = \langle y, z, x \rangle.$$

Let C be the boundary of the triangle with vertices

$$A = (1, 0, 0), \quad B = (0, 1, 0), \quad C_0 = (0, 0, 1),$$

oriented so that the normal direction is compatible with

$$\mathbf{n} = \frac{1}{\sqrt{3}}\langle 1, 1, 1 \rangle.$$

The curl is

$$\nabla \times \mathbf{F} = \left\langle \frac{\partial x}{\partial y} - \frac{\partial z}{\partial z}, \frac{\partial y}{\partial z} - \frac{\partial x}{\partial x}, \frac{\partial z}{\partial x} - \frac{\partial y}{\partial y} \right\rangle = \langle -1, -1, -1 \rangle.$$

The triangle lies in the plane $x + y + z = 1$. Its oriented vector area is

$$\frac{1}{2}(B - A) \times (C_0 - A) = \frac{1}{2}\langle 1, 1, 1 \rangle.$$

Therefore

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS = \langle -1, -1, -1 \rangle \cdot \frac{1}{2}\langle 1, 1, 1 \rangle = -\frac{3}{2}.$$

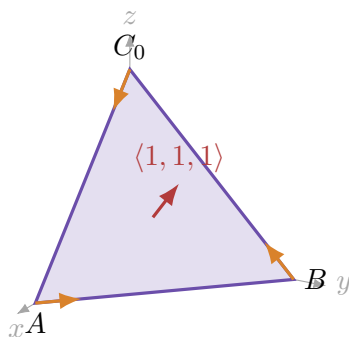


Figure 8: Oriented triangular surface for a Stokes theorem calculation. The boundary orientation is chosen by the normal direction.

12 Example: open surface plus missing cap

Hemisphere with a disk cap

Let

$$\mathbf{F}(x, y, z) = \langle x, y, z \rangle.$$

Let S be the upper hemisphere

$$x^2 + y^2 + z^2 = R^2, \quad z \geq 0,$$

oriented outward. The hemisphere is not a closed surface. Add the disk

$$D : x^2 + y^2 \leq R^2, \quad z = 0.$$

The closed surface $S \cup D$ bounds the upper half-ball. By the Divergence theorem,

$$\iint_{S \cup D} \mathbf{F} \cdot \mathbf{n} dS = \iiint_E 3 dV = 3 \cdot \frac{1}{2} \cdot \frac{4\pi R^3}{3} = 2\pi R^3.$$

On the disk D , the outward normal for the half-ball is $\mathbf{n} = -\mathbf{k}$, and $z = 0$, so

$$\mathbf{F} \cdot \mathbf{n} = \langle x, y, 0 \rangle \cdot \langle 0, 0, -1 \rangle = 0.$$

Thus the flux through the upper hemisphere is

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = 2\pi R^3.$$

For $R = 1$, this is 2π .

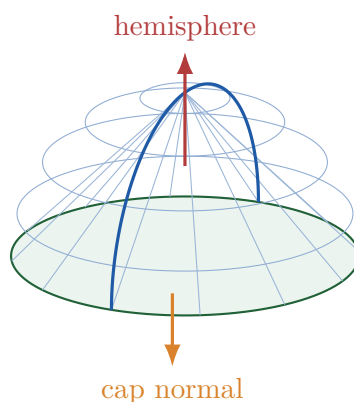


Figure 9: A hemisphere becomes a closed surface after adding its disk cap. The cap flux may be easier to compute.

13 Numerical verification of Stokes theorem

For $\mathbf{F} = \langle -y, x, 0 \rangle$ and a circle of radius R , Stokes theorem predicts

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi R^2.$$

A polygonal line-integral approximation converges to this value as the curve is refined.

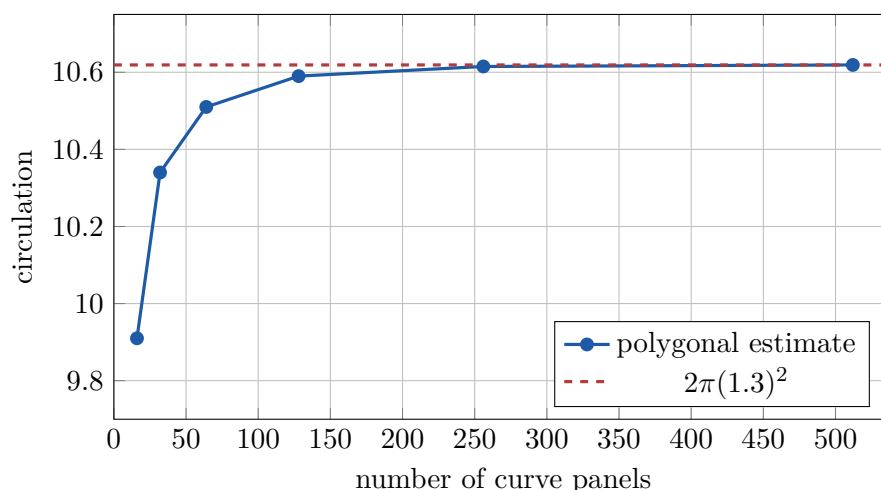


Figure 10: Numerical line-integral approximation agrees with Stokes theorem as the boundary curve approximation is refined.

14 Numerical verification of the Divergence theorem

For $\mathbf{F} = \langle x, y, z \rangle$ and the unit ball, the Divergence theorem predicts

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_E 3 dV = 4\pi.$$

One can approximate the right side by Monte Carlo sampling inside the cube $[-1, 1]^3$ and accepting points inside the unit ball.

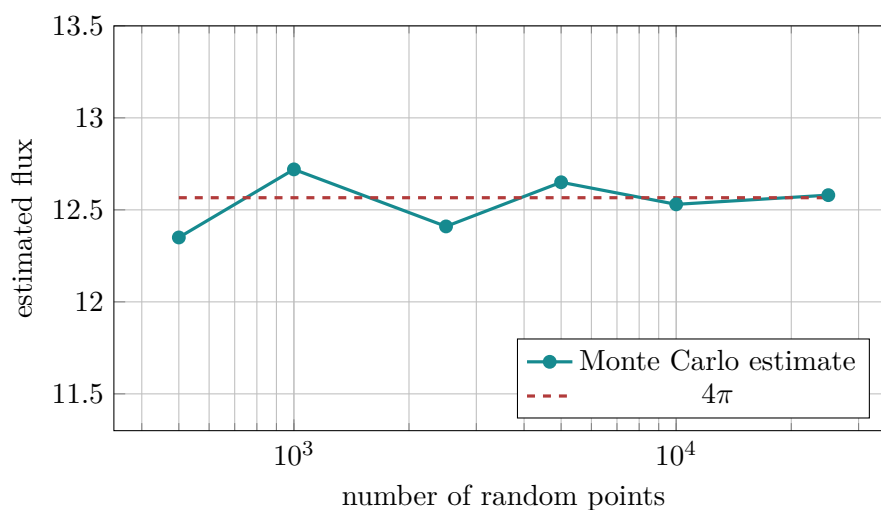


Figure 11: Monte Carlo estimate of total divergence inside the unit ball. Random estimates fluctuate but center near 4π .

15 Conservation laws and applied interpretation

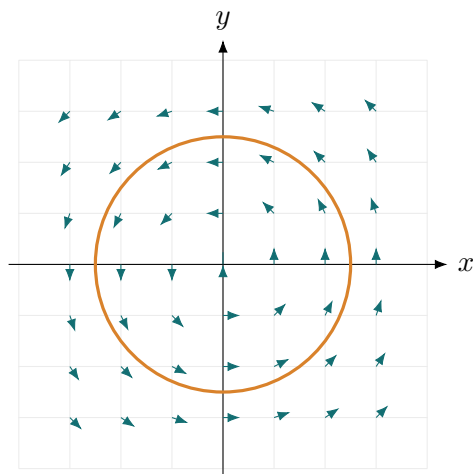
The Divergence theorem is the mathematical language of conservation laws. Suppose $\mathbf{F}$ represents a flow field. Then:

- $\mathbf{F} \cdot \mathbf{n}$ is the amount of flow crossing a boundary per unit area;
- $\iint_{\partial E} \mathbf{F} \cdot \mathbf{n} dS$ is the net flow leaving the region;
- $\nabla \cdot \mathbf{F}$ is the local rate of expansion or source strength;
- $\iiint_E \nabla \cdot \mathbf{F} dV$ is the total source strength inside.

If $\nabla \cdot \mathbf{F} = 0$, the field is incompressible or source-free. Then every closed surface has zero net outward flux:

$$\iint_{\partial E} \mathbf{F} \cdot \mathbf{n} dS = 0.$$

This does not mean the field is zero. It means whatever flows into a closed region must also flow out.



$\mathbf{F} = \langle -y, x \rangle$: zero divergence but nonzero circulation

Figure 12: A rotational field can have zero divergence but nonzero curl. Divergence and curl measure different kinds of behavior.

16 Summary of the four global theorems

Theorem	Formula	Meaning
Fundamental Theorem for Line Integrals	$\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)$	gradient work depends only on endpoints
Green's theorem	$\oint_C P dx + Q dy = \iint_R (Q_x - P_y) dA$	plane circulation equals total scalar curl
Stokes theorem	$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$	boundary circulation equals total curl through surface
Divergence theorem	$\iint_{\partial E} \mathbf{F} \cdot \mathbf{n} dS = \iiint_E \nabla \cdot \mathbf{F} dV$	boundary flux equals total divergence inside

A good way to remember the pattern is

boundary integral = interior integral of a derivative.

17 Common mistakes

Common mistakes

1. **Using the Divergence theorem on an open surface.** The surface must be closed. If it is open, add missing pieces or compute directly.
2. **Forgetting orientation.** Reversing orientation changes the sign of circulation or flux.
3. **Mixing up curl and divergence.** Curl is a vector in three dimensions; divergence is a scalar.
4. **Using Stokes theorem without a boundary curve.** A closed surface has no boundary, so Stokes theorem over a closed surface gives zero boundary circulation.
5. **Forgetting surface replacement.** Any convenient surface with the same boundary and compatible orientation may be used.
6. **Ignoring singularities.** The vector field must be sufficiently smooth on a region containing the surface or solid.

18 Exercises

Conceptual exercises

1. Explain the difference between circulation and flux.
2. State Stokes theorem in words without using formulas.

3. State the Divergence theorem in words without using formulas.
4. Why does reversing the normal vector in Stokes theorem reverse the boundary orientation?
5. Why does the Divergence theorem require a closed surface?
6. Give an example of a vector field with zero divergence but nonzero curl.
7. Give an example of a vector field with nonzero divergence but zero curl.
8. Explain how Green's theorem is related to Stokes theorem.

Computational exercises

1. Let $\mathbf{F} = \langle -y, x, 0 \rangle$ and let C be the circle $x^2 + y^2 = 9$, $z = 0$, oriented counterclockwise as viewed from above. Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$.
2. Let $\mathbf{F} = \langle y, z, x \rangle$ and let S be the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$, oriented by $\langle 1, 1, 1 \rangle$. Use Stokes theorem to compute $\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}$.
3. Let $\mathbf{F} = \langle x, y, z \rangle$ and let S be the sphere of radius 2 centered at the origin. Compute the outward flux through S .
4. Let $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$ and let $E = [0, 2] \times [0, 1] \times [0, 3]$. Use the Divergence theorem to compute the outward flux through ∂E .
5. Let $\mathbf{F} = \langle yz, xz, xy \rangle$. Compute $\nabla \cdot \mathbf{F}$. What does the Divergence theorem say about the outward flux through any closed surface?
6. Let $\mathbf{F} = \langle z, x, y \rangle$. Compute $\nabla \times \mathbf{F}$ and use Stokes theorem over a convenient surface for a given circular boundary.
7. Compute the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ through the upper hemisphere $x^2 + y^2 + z^2 = 4$, $z \geq 0$.
8. Use the Divergence theorem to compute the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ through the unit ball.

Computational lab exercises

1. Modify the numerical Stokes theorem code for a circle of radius $R = 2$ and compare with the exact value.
2. Replace $\mathbf{F} = \langle -y, x, 0 \rangle$ by $\mathbf{F} = \langle -2y, 3x, 0 \rangle$ and determine the new circulation around the unit circle.
3. Use Monte Carlo sampling to estimate the flux of $\mathbf{F} = \langle x, y, z \rangle$ through a ball of radius 2.
4. Write a function that estimates $\iiint_E \nabla \cdot \mathbf{F} dV$ over a rectangular box using the midpoint rule.
5. Visualize a vector field with zero divergence and nonzero curl.
6. Visualize a vector field with nonzero divergence and zero curl.

19 Python lab and interactive page

- Notebook lab: `../labs/Chapter23_Stokes_Theorem_and_the_Divergence_Theorem.ipynb`
- Interactive HTML page: `../htmls/chapter-23.html`

20 Chapter summary

Stokes theorem and the Divergence theorem are global versions of differentiation. Stokes theorem says that circulation around a boundary equals total curl through a spanning surface. The Divergence theorem says that outward flux through a closed boundary equals total divergence inside the solid. Both theorems convert difficult boundary measurements into interior integrals, or difficult interior measurements into boundary integrals. Their common structure is the backbone of vector calculus and prepares students for conservation laws, partial differential equations, computational physics, geometric data analysis, and modern applied mathematics.

Chapter 24

Multivariable Calculus for Probability and Statistics

Densities, expectations, transformations, likelihood, and local approximation

Probability and statistics give one of the most important modern uses of multivariable calculus. A probability model is often a function on a space of possible outcomes. A statistical model is often a function on a space of unknown parameters. In both cases, we need the language of multivariable calculus: domains, level curves, multiple integrals, gradients, Hessians, Jacobians, and change of variables.

This chapter is a bridge from the core calculus of several variables to later courses in probability, mathematical statistics, data science, and machine learning. The central idea is simple.

Central idea

A density is a function whose integral gives probability. A likelihood is a function whose maximum gives an estimate.

The first sentence uses multiple integrals. The second uses gradients and optimization.

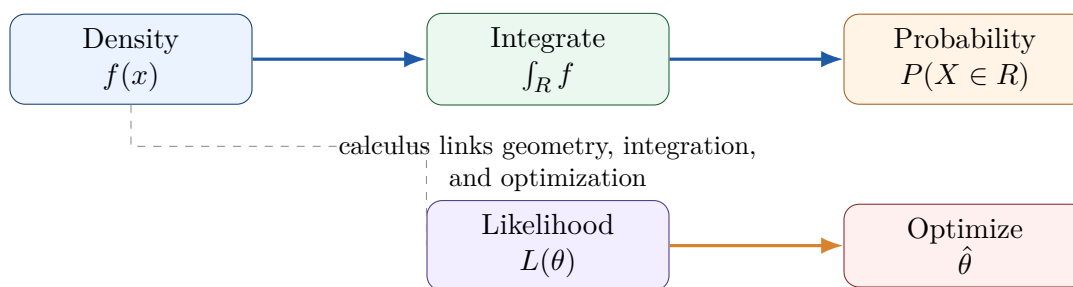


Figure 24.1: Two central roles of multivariable calculus in statistics: integration over outcome space and optimization over parameter space.

Learning goals

After studying this chapter, you should be able to:

1. interpret a joint density $f_{X,Y}(x,y)$ geometrically and probabilistically;
2. compute probabilities and expectations using double and triple integrals;
3. find marginal and conditional densities from a joint density;
4. use change of variables and Jacobians to transform random variables;
5. interpret covariance matrices through ellipses and quadratic forms;
6. use gradients and Hessians to understand likelihood functions;
7. connect Taylor approximation to standard errors and uncertainty quantification;
8. use numerical integration and simulation to check analytic calculations.

Throughout the chapter, keep two pictures in mind. A joint density is like a surface whose total volume is 1. A log-likelihood is like a landscape whose highest point gives the best-fitting parameter.

24.1 Random variables as functions and data as points

A random variable is a function from a sample space to numbers. A random vector is a function from a sample space to a vector space.

For example, if we observe height and weight from a randomly selected person, then the outcome can be written as a vector

$$X = (X_1, X_2) = (\text{height}, \text{weight}) \in \mathbb{R}^2.$$

If we observe n measurements, the dataset can be written as a point in $\mathbb{R}^n$:

$$x = (x_1, x_2, \dots, x_n).$$

If we fit a model with parameters $\theta_1, \dots, \theta_p$, the parameter vector is a point in parameter space:

$$\theta = (\theta_1, \dots, \theta_p) \in \mathbb{R}^p.$$

This is why multivariable calculus appears everywhere in statistics. We integrate over outcome spaces, differentiate over parameter spaces, and use linear and quadratic approximations to measure uncertainty.

Three spaces in statistics

A statistical problem often involves three different spaces:

- **Sample space:** where random outcomes live.
- **Data space:** where observed datasets live.
- **Parameter space:** where unknown model parameters live.

Multivariable calculus helps us move among all three.

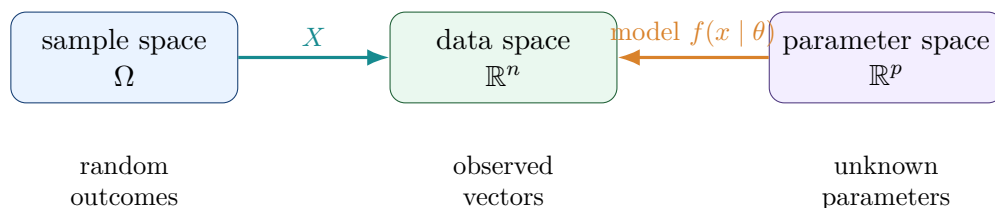


Figure 24.2: Statistics moves among sample space, data space, and parameter space. Multivariable calculus supplies the geometry and computation.

24.2 Joint densities and probability as volume

A continuous random vector (X, Y) has a joint density $f_{X,Y}(x, y)$ if probabilities are computed by double integrals:

$$\mathbb{P}((X, Y) \in R) = \iint_R f_{X,Y}(x, y) dA.$$

For $f_{X,Y}$ to be a valid density, it must satisfy

$$f_{X,Y}(x, y) \geq 0, \quad \iint_{\mathbb{R}^2} f_{X,Y}(x, y) dA = 1.$$

The probability of a region is the volume under the density surface above that region.

Example 24.1: A simple density on a square

Let

$$f(x, y) = x + y$$

on the unit square $0 \leq x \leq 1$, $0 \leq y \leq 1$, and let $f(x, y) = 0$ outside the square. This is a probability density because

$$\int_0^1 \int_0^1 (x + y) dy dx = \int_0^1 \left(x + \frac{1}{2} \right) dx = 1.$$

Now compute $\mathbb{P}(X + Y \leq 1)$. The region is the triangle $0 \leq x \leq 1$, $0 \leq y \leq 1 - x$. Thus

$$\mathbb{P}(X + Y \leq 1) = \int_0^1 \int_0^{1-x} (x + y) dy dx.$$

The inner integral is

$$\int_0^{1-x} (x + y) dy = x(1 - x) + \frac{(1 - x)^2}{2}.$$

Therefore,

$$\mathbb{P}(X + Y \leq 1) = \int_0^1 \left[x(1 - x) + \frac{(1 - x)^2}{2} \right] dx = \frac{1}{2}.$$

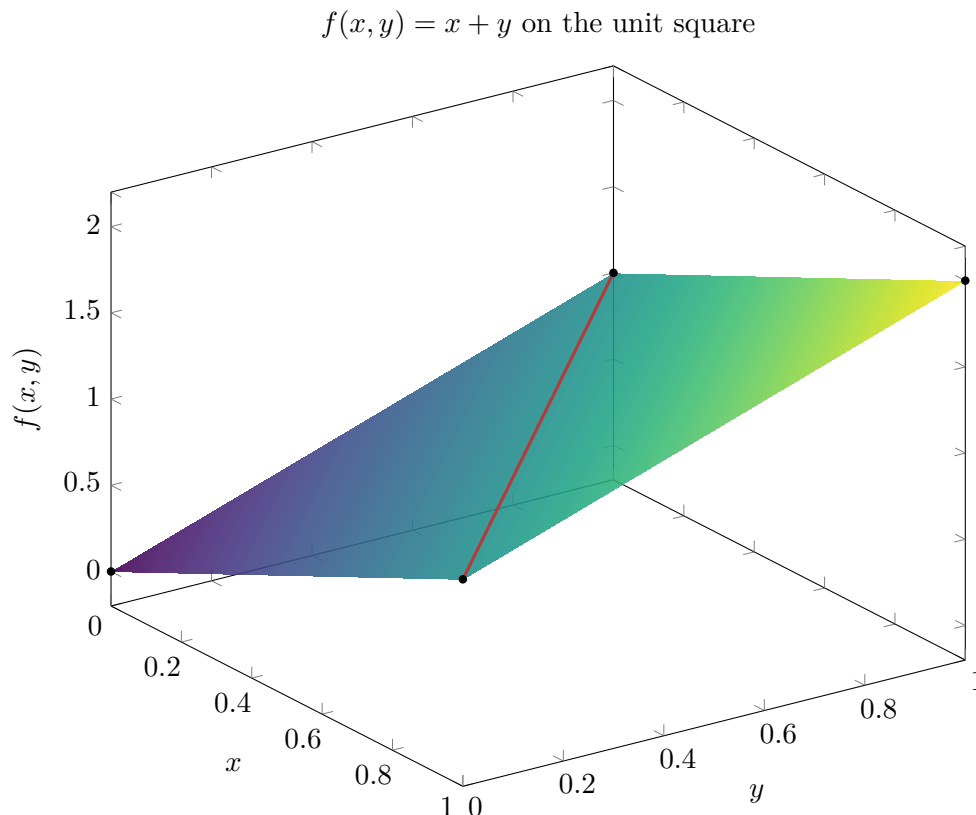


Figure 24.3: The density $f(x, y) = x + y$ on the unit square. Probability over a region is volume under the surface over that region.

24.3 Marginal densities

A joint density describes the simultaneous behavior of two random variables. A marginal density describes one variable alone. If (X, Y) has joint density $f_{X,Y}$, then the marginal density of X is

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy.$$

Similarly,

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx.$$

The operation is geometric: we collapse the density surface in one direction by integrating out the other variable.

Example 24.2: Marginals for $f(x, y) = x + y$

On the unit square,

$$f_X(x) = \int_0^1 (x + y) dy = x + \frac{1}{2}.$$

Similarly,

$$f_Y(y) = \int_0^1 (x + y) dx = y + \frac{1}{2}.$$

Check normalization:

$$\int_0^1 f_X(x) dx = \int_0^1 \left(x + \frac{1}{2}\right) dx = 1.$$

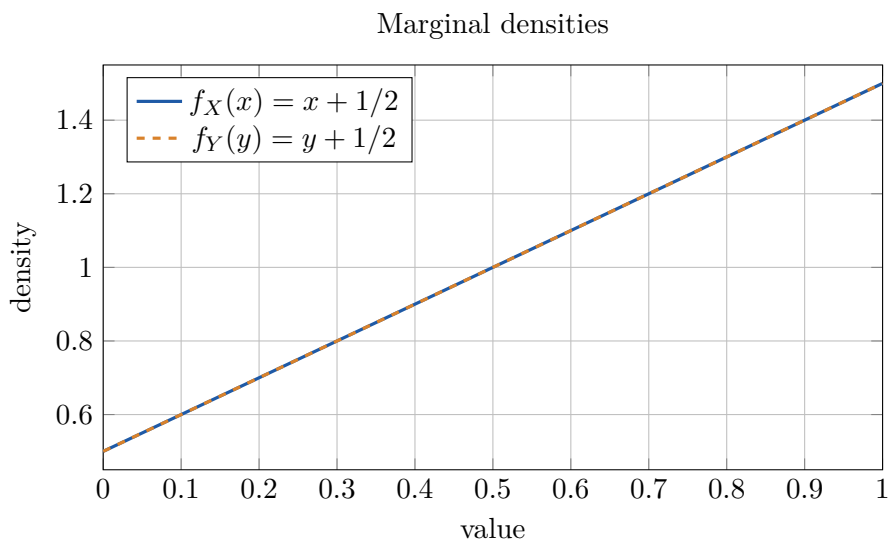


Figure 24.4: Marginal densities are obtained by integrating the joint density in one variable.

Integrating out variables

In statistics and machine learning, the phrase **integrate out** means to remove a variable by integration. Marginalization is integration.

24.4 Conditional densities and independence

The conditional density of Y given $X = x$ is

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)},$$

provided $f_X(x) > 0$. This is the continuous analogue of conditional probability:

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Example 24.3: Conditional density from a joint density

For $f(x, y) = x + y$ on the unit square,

$$f_{Y|X}(y | x) = \frac{x + y}{x + 1/2}, \quad 0 \leq y \leq 1.$$

For fixed x , this is a density in y :

$$\int_0^1 \frac{x+y}{x+1/2} dy = \frac{x+1/2}{x+1/2} = 1.$$

The conditional mean of Y given $X = x$ is

$$\mathbb{E}[Y | X = x] = \int_0^1 y \frac{x+y}{x+1/2} dy = \frac{x/2 + 1/3}{x+1/2}.$$

This conditional mean depends on x , so knowing X changes the distribution of Y .

Random variables X and Y are independent if

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

for all relevant (x,y) . For $f(x,y) = x+y$, we have

$$f_X(x)f_Y(y) = \left(x + \frac{1}{2}\right) \left(y + \frac{1}{2}\right),$$

which is not equal to $x+y$. Therefore, X and Y are not independent.

Conditional densities change with x

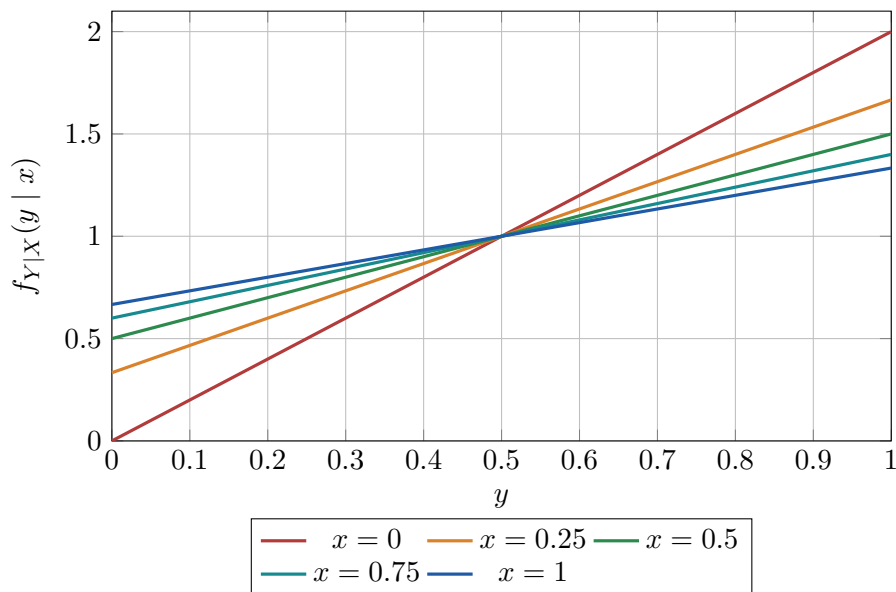


Figure 24.5: Conditional densities $f_{Y|X}(y | x)$ for several fixed values of x .

24.5 Expectation as integration

For a function $g(X,Y)$, the expected value is

$$\mathbb{E}[g(X,Y)] = \iint_{\mathbb{R}^2} g(x,y) f_{X,Y}(x,y) dA.$$

Important special cases include

$$\mathbb{E}[X] = \iint x f_{X,Y}(x, y) dA, \quad \mathbb{E}[Y] = \iint y f_{X,Y}(x, y) dA,$$

$$\mathbb{E}[XY] = \iint xy f_{X,Y}(x, y) dA.$$

The covariance is

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

The covariance matrix of a random vector $X = (X_1, \dots, X_n)$ is

$$\Sigma = \mathbb{E}[(X - \mu)(X - \mu)^T].$$

It is a matrix version of variance.

Example 24.4: Means and covariance for a simple joint density

For $f(x, y) = x + y$ on the unit square,

$$\mathbb{E}[X] = \int_0^1 \int_0^1 x(x + y) dy dx = \frac{7}{12},$$

and by symmetry,

$$\mathbb{E}[Y] = \frac{7}{12}.$$

Also,

$$\mathbb{E}[XY] = \int_0^1 \int_0^1 xy(x + y) dy dx = \frac{1}{3}.$$

Thus

$$\text{Cov}(X, Y) = \frac{1}{3} - \frac{7}{12} \cdot \frac{7}{12} = -\frac{1}{144}.$$

The covariance is slightly negative. This is a small but real dependence.

24.6 Numerical integration on grids

Many integrals in probability and statistics do not have closed-form antiderivatives. A practical alternative is numerical integration. Suppose $f(x, y)$ is evaluated on a rectangular grid. Then

$$\iint_R f(x, y) dA \approx \sum_{i,j} f(x_i, y_j) \Delta x \Delta y.$$

This is the same Riemann-sum idea from multiple integration, now used for probabilities and expectations.

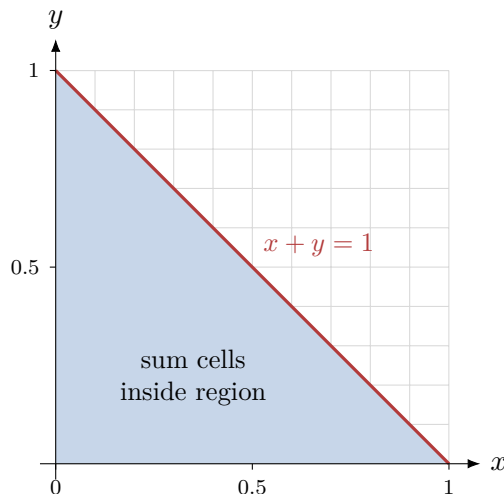


Figure 24.6: Numerical integration estimates the probability of a region by summing density values over grid cells. The exact answer in Example 24.1 is $1/2$.

24.7 Change of variables for random variables

Change of variables is one of the most important applications of the Jacobian. Suppose (U, V) has density $f_{U,V}(u, v)$, and suppose

$$(x, y) = T(u, v).$$

If T is one-to-one on the relevant region and has nonzero Jacobian determinant, then the density of $(X, Y) = T(U, V)$ is

$$f_{X,Y}(x, y) = f_{U,V}(u(x, y), v(x, y)) \left| \det \frac{\partial(u, v)}{\partial(x, y)} \right|.$$

Equivalently, if we integrate in (u, v) -coordinates, then

$$dx dy = \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

The Jacobian measures local area distortion.

Example 24.5: Uniform distribution on a disk

A common mistake is to sample a radius uniformly when trying to sample uniformly from a disk. If R is uniform on $[0, 1]$ and Θ is uniform on $[0, 2\pi]$, then points concentrate near the center. For uniform sampling in the unit disk, use

$$R = \sqrt{U}, \quad \Theta = 2\pi V,$$

where U, V are independent uniform random variables on $[0, 1]$. Then

$$X = R \cos \Theta, \quad Y = R \sin \Theta.$$

Why the square root? Because the area of a disk of radius r is proportional to r^2 , so

$$\mathbb{P}(R \leq r) = r^2.$$

Thus R^2 should be uniform on $[0, 1]$.

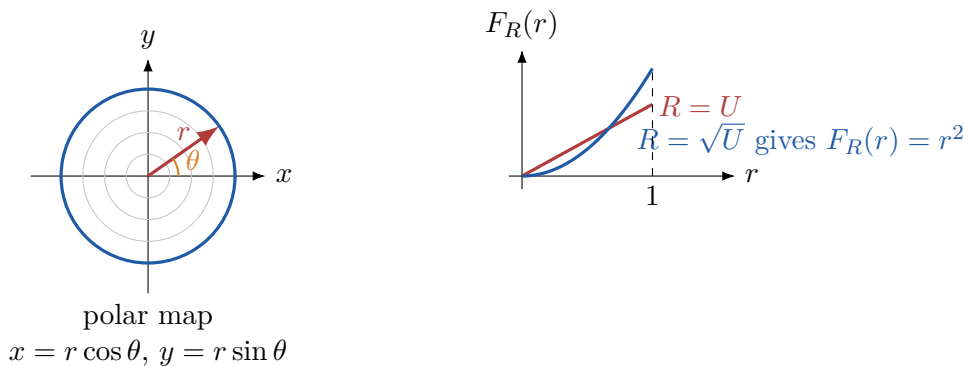


Figure 24.7: Uniform area in a disk requires the radial distribution $\mathbb{P}(R \leq r) = r^2$, hence $R = \sqrt{U}$.

24.8 The bivariate normal distribution

The multivariate normal distribution is the most important continuous distribution in multivariable statistics. A random vector $X \in \mathbb{R}^n$ has multivariate normal density with mean vector μ and covariance matrix Σ if

$$f(x) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right).$$

For $n = 2$, the level curves of the density are ellipses:

$$(x - \mu)^T \Sigma^{-1} (x - \mu) = c.$$

This is a quadratic form. The eigenvectors of Σ determine the principal directions of the ellipses, and the eigenvalues determine their squared spread in those directions.

Geometry of covariance

The covariance matrix is not just a table of numbers. It is a geometric object. It determines the directions and scales of uncertainty.

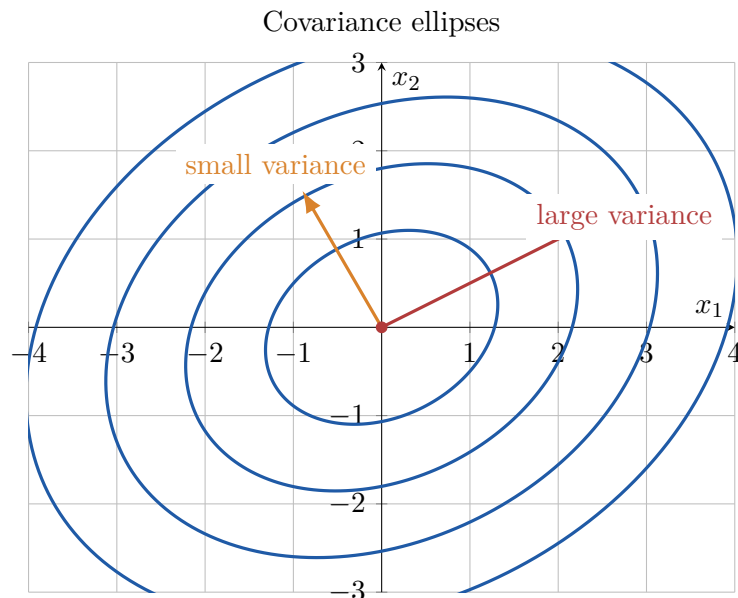


Figure 24.8: Level curves of a bivariate normal density are ellipses. The covariance matrix controls the directions and scales.

24.9 Likelihood functions as multivariable functions

In statistics, the unknown parameter is often a vector

$$\theta = (\theta_1, \dots, \theta_p).$$

Given data $x_1, \dots, x_n$, the likelihood function is

$$L(\theta) = f(x_1, \dots, x_n \mid \theta).$$

The log-likelihood is

$$\ell(\theta) = \log L(\theta).$$

Maximum likelihood estimation asks us to find

$$\hat{\theta} = \operatorname{argmax}_{\theta} \ell(\theta).$$

This is a multivariable optimization problem.

Example 24.6: Normal model with unknown mean and variance

Suppose $x_1, \dots, x_n$ are modeled as independent normal observations with mean μ and standard deviation $\sigma > 0$. The log-likelihood is, up to an additive constant,

$$\ell(\mu, \sigma) = -n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

The maximum occurs at

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2.$$

This is a two-variable optimization problem on the half-plane $\sigma > 0$.

A normal log-likelihood landscape

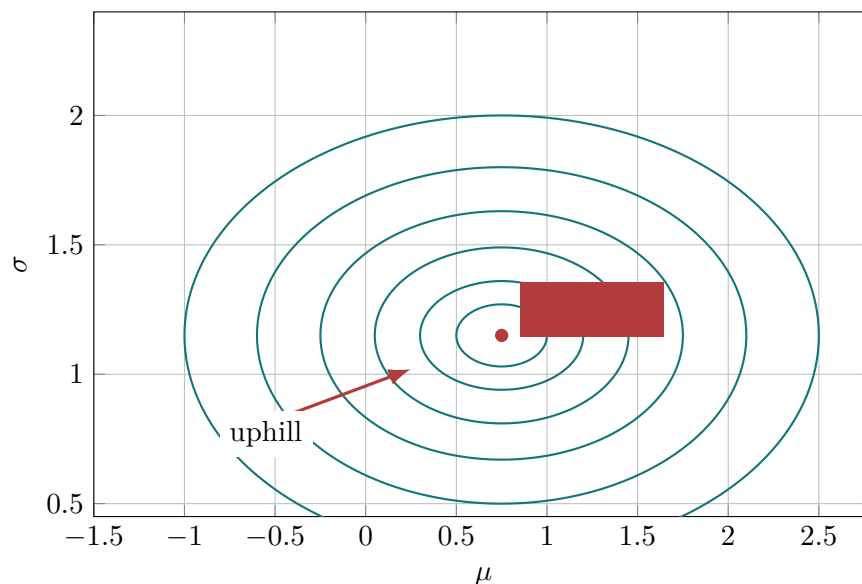


Figure 24.9: A log-likelihood is a function on parameter space. Its maximum gives the maximum likelihood estimate.

24.10 Score, Hessian, and information

The gradient of the log-likelihood is called the **score**:

$$S(\theta) = \nabla \ell(\theta).$$

At an interior maximum likelihood estimate, the score is zero:

$$\nabla \ell(\hat{\theta}) = 0.$$

The Hessian matrix is

$$H(\theta) = \nabla^2 \ell(\theta).$$

At a local maximum, the Hessian is usually negative definite. The negative Hessian

$$-\nabla^2 \ell(\hat{\theta})$$

measures local curvature of the log-likelihood near the maximum. In many statistical models, this curvature is connected to information and standard errors.

A second-order Taylor approximation gives

$$\ell(\theta) \approx \ell(\hat{\theta}) + \frac{1}{2}(\theta - \hat{\theta})^T \nabla^2 \ell(\hat{\theta})(\theta - \hat{\theta}).$$

Since the Hessian is negative definite near a maximum, the log-likelihood looks locally like an upside-down quadratic bowl.

Curvature and uncertainty

A sharply curved log-likelihood means the parameter is estimated precisely. A flat log-likelihood means the data do not strongly identify the parameter.

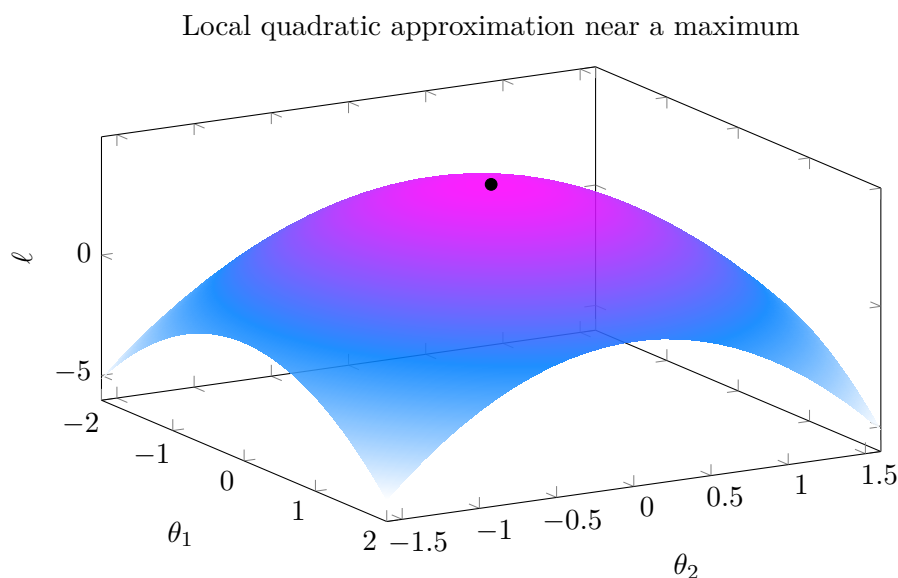


Figure 24.10: Near an interior maximum, the second-order Taylor approximation makes the log-likelihood look like an upside-down quadratic bowl.

24.11 The delta method as linear approximation

Suppose an estimator $\hat{\theta}$ is approximately normal:

$$\hat{\theta} \approx N(\theta, \Sigma).$$

If we want to understand $g(\hat{\theta})$ for a differentiable function g , we use linear approximation:

$$g(\hat{\theta}) \approx g(\theta) + \nabla g(\theta)^T (\hat{\theta} - \theta).$$

Therefore,

$$\text{Var}(g(\hat{\theta})) \approx \nabla g(\theta)^T \Sigma \nabla g(\theta).$$

This is the multivariable delta method. It is just the linearization from Chapter 9, applied to statistical uncertainty.

Example 24.7: Ratio of two estimates

Suppose $\hat{\theta} = (\hat{a}, \hat{b})$ has covariance matrix Σ , and we want the uncertainty of

$$g(a, b) = \frac{a}{b}.$$

The gradient is

$$\nabla g(a, b) = \left(\frac{1}{b}, -\frac{a}{b^2} \right).$$

So

$$\text{Var} \left(\frac{\hat{a}}{\hat{b}} \right) \approx \begin{bmatrix} 1/b & -a/b^2 \end{bmatrix} \Sigma \begin{bmatrix} 1/b \\ -a/b^2 \end{bmatrix}.$$

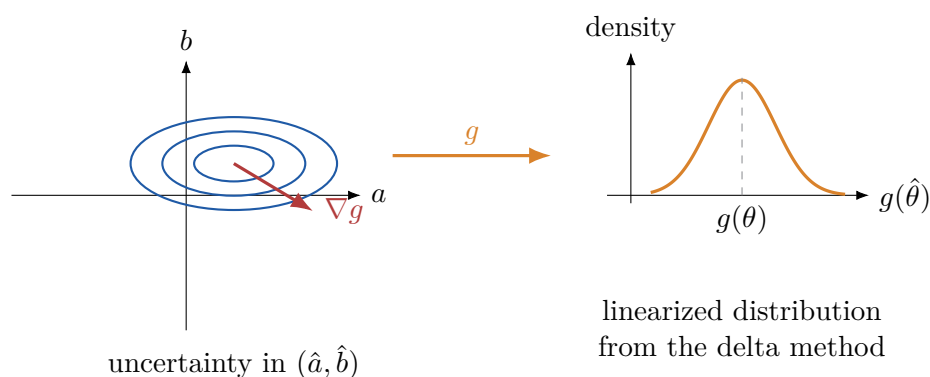


Figure 24.11: The delta method approximates the distribution of a nonlinear transformation by linearizing the transformation near the mean.

24.12 Monte Carlo integration

Monte Carlo integration uses random samples to estimate integrals. If $U_1, \dots, U_N$ are uniformly sampled from a region R with area $A(R)$, then

$$\iint_R f(x, y) dA \approx A(R) \frac{1}{N} \sum_{i=1}^N f(U_i).$$

If $X_1, \dots, X_N$ are sampled from a density $p(x)$, then

$$\mathbb{E}[g(X)] \approx \frac{1}{N} \sum_{i=1}^N g(X_i).$$

This is one of the key computational ideas behind simulation-based statistics, Bayesian computation, and machine learning.

Example 24.8: Estimating an integral by simulation

Estimate

$$I = \int_0^1 \int_0^1 e^{-(x^2+y^2)} dy dx.$$

If (U, V) is uniform on the unit square, then

$$I = \mathbb{E}[e^{-(U^2+V^2)}].$$

So we can approximate I by a sample average.

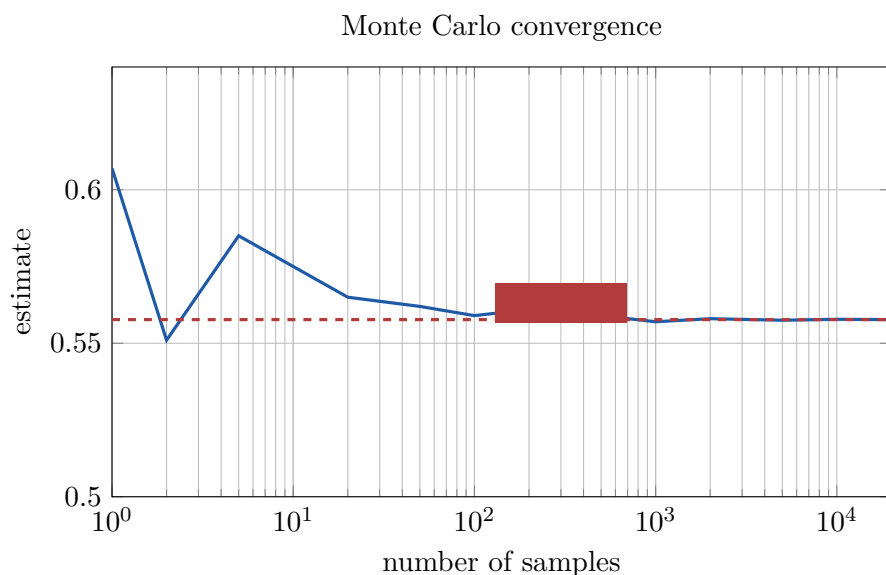


Figure 24.12: Monte Carlo integration estimates an integral by averaging random function values. The estimate stabilizes as the sample size grows.

24.13 Bayesian posterior as a normalized density

Bayesian statistics combines a likelihood and a prior distribution. If θ is a parameter and x is observed data, Bayes' rule gives

$$p(\theta | x) = \frac{p(x | \theta)p(\theta)}{p(x)}.$$

The denominator

$$p(x) = \int p(x | \theta)p(\theta) d\theta$$

is a normalizing constant. In multivariable problems,

$$p(x) = \int_{\Theta} p(x | \theta)p(\theta) d\theta$$

may be a high-dimensional integral. This is why numerical integration, Monte Carlo methods, and optimization are so important in modern Bayesian statistics.

Example 24.9: A two-parameter posterior landscape

Suppose a posterior density for two parameters is proportional to

$$\exp \left[-\frac{1}{2}(\theta - \mu)^T A(\theta - \mu) \right].$$

The contours are ellipses. The matrix A is a precision matrix, and A^{-1} is the covariance matrix.

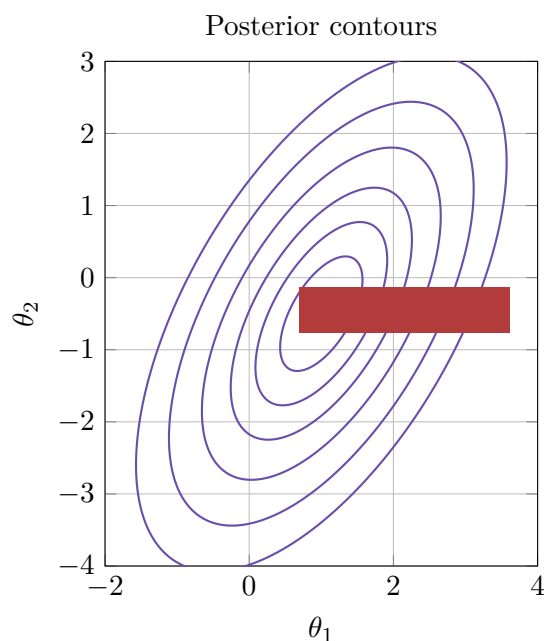


Figure 24.13: A two-parameter posterior density. The highest point is the posterior mode, and the contour shape describes uncertainty.

24.14 What to remember

Multivariable calculus gives statistics its geometry and computation.

- A joint density is a nonnegative function with total integral 1.
- A probability is an integral over a region.
- A marginal density is obtained by integrating out variables.
- A conditional density is a normalized slice of the joint density.
- An expectation is an integral of a function against a density.
- A covariance matrix is a geometric object describing spread and direction.
- A Jacobian accounts for local volume distortion under a change of variables.
- A likelihood is a function on parameter space.
- A score is the gradient of a log-likelihood.

- A Hessian describes local curvature and uncertainty.
- Monte Carlo methods approximate integrals by random averages.

24.15 Python lab

In the accompanying lab, you will:

1. plot joint densities as surfaces and contour maps;
2. compute marginal and conditional densities numerically;
3. simulate points from a disk using the correct radial transformation;
4. visualize covariance ellipses of a bivariate normal distribution;
5. plot a normal log-likelihood surface for unknown mean and variance;
6. approximate a nonlinear estimator using the delta method;
7. estimate probabilities and expectations using Monte Carlo simulation.

Chapter resources

- Lab notebook: [../labs/chapter-24-multivariable-calculus-for-probability-and-statistics.ipynb](#)
- Interactive page: [../htmls/chapter-24-multivariable-calculus-for-probability-and-statistics.html](#)

24.16 Exercises

Conceptual questions

1. Explain why a joint density must integrate to 1.
2. In your own words, explain the difference between a marginal density and a conditional density.
3. Why is a covariance matrix a geometric object?
4. Why does a change of variables formula need a Jacobian factor?
5. What is the difference between likelihood and probability?
6. Explain why the gradient of the log-likelihood is zero at an interior maximum likelihood estimate.
7. Explain how the Hessian of the log-likelihood is related to uncertainty.
8. What does Monte Carlo integration estimate, and why does it work?

Computation and derivation

1. Let $f(x, y) = 2$ on the triangle $0 \leq y \leq x \leq 1$ and 0 elsewhere. Verify that f is a joint density.

2. For the density in Problem 1, compute $f_X(x)$ and $f_Y(y)$.
3. For the density in Problem 1, compute $\mathbb{P}(Y \leq 1/2)$.
4. Let $f(x, y) = 4xy$ on $0 \leq x \leq 1$, $0 \leq y \leq 1$. Find the marginal densities. Are X and Y independent?
5. Let $f(x, y) = c(x^2 + y^2)$ on the unit disk. Find c .
6. For independent uniform random variables U, V on $[0, 1]$, define $X = U + V$. Find the density of X by geometric reasoning.
7. Let X, Y have joint density $f(x, y) = x + y$ on the unit square. Compute $\mathbb{E}[X^2]$.
8. Let X, Y have joint density $f(x, y) = x + y$ on the unit square. Compute $\text{Var}(X)$.
9. Let X, Y be independent standard normal random variables. Use polar coordinates to evaluate

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} dx dy.$$

10. Suppose X has density $f_X(x) = 2x$ on $0 \leq x \leq 1$, and $Y = X^2$. Find the density of Y .

Likelihood and approximation

1. For Bernoulli data with k successes in n trials, the log-likelihood is

$$\ell(p) = k \log p + (n - k) \log(1 - p).$$

Find the maximum likelihood estimate of p .

2. Compute the second derivative of the Bernoulli log-likelihood and evaluate it at $\hat{p} = k/n$.
3. For normal data with known variance σ^2 , derive the maximum likelihood estimate of μ .
4. For normal data with unknown mean and variance, explain why the log-likelihood is a function of two variables.
5. Let $g(a, b) = a/b$. Compute the linear approximation of g at $(2, 1)$.
6. Let $\Sigma = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$. For $g(a, b) = a/b$, approximate the variance of $g(\hat{a}, \hat{b})$ at $(a, b) = (2, 1)$ using the delta method.
7. Let $\ell(\theta_1, \theta_2) = -(\theta_1 - 1)^2 - 4(\theta_2 + 2)^2$. Find the maximum and the Hessian.
8. Explain how the contours of the function in Problem 7 describe uncertainty.

Python exercises

1. Numerically verify that $f(x, y) = x + y$ integrates to 1 on the unit square.
2. Use Monte Carlo simulation to estimate $\mathbb{P}(X + Y \leq 1)$ for the density $f(x, y) = x + y$ on the unit square. Compare with the exact answer.
3. Write code to plot the bivariate normal density for several covariance matrices.

4. Simulate uniform points in the unit disk using both $R = U$ and $R = \sqrt{U}$. Compare the radial histograms.
5. Generate normal data and plot the log-likelihood contours for (μ, σ) .
6. Simulate the delta method example for different covariance matrices.
7. Estimate $\int_0^1 \int_0^1 \sin(xy) \, dy \, dx$ by Monte Carlo simulation.
8. Use finite differences to approximate the gradient and Hessian of a two-parameter log-likelihood.

24.17 Chapter summary

This chapter showed how the main ideas of multivariable calculus become tools for probability and statistics. Densities are functions integrated over regions. Expectations are weighted integrals. Marginalization is integration over unused variables. Change of variables requires Jacobians. Likelihood functions are multivariable functions on parameter space, and their maxima are studied using gradients and Hessians. Taylor approximation explains why many estimators are approximately normal and why curvature measures uncertainty. These ideas prepare us for statistical theory, Bayesian inference, data science, and machine learning.

Chapter 25

Multivariable Calculus for Data Science and Machine Learning

Chapter overview

Machine learning training is an optimization problem on a high-dimensional parameter space. A model turns inputs into predictions, a loss function turns prediction errors into a single number, and calculus tells us how that number changes as the parameters move. The main objects are familiar: gradients, Jacobians, Hessians, chain rules, tangent approximations, and level sets. What changes in modern data science is scale: instead of two or three variables, the parameter vector may contain thousands, millions, or billions of components.

Learning goals

By the end of this chapter, you should be able to:

1. translate a learning problem into a multivariable function of parameters;
2. identify loss functions, empirical risk, gradients, Jacobians, and Hessians;
3. derive gradients for linear regression, logistic regression, and simple neural-network models;
4. interpret gradient descent as motion on a high-dimensional surface;
5. explain the role of regularization in optimization and statistical learning;
6. use computational visualization to study loss landscapes and gradient-based training;
7. connect calculus ideas to automatic differentiation, backpropagation, and modern AI workflows.

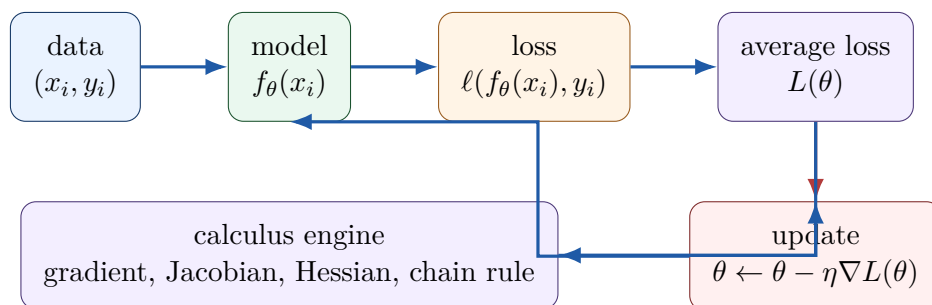


Figure 25.1: A supervised learning workflow viewed as multivariable calculus. The parameter vector θ is adjusted by information from derivatives of the loss.

25.1 Why machine learning is multivariable calculus

A supervised learning problem begins with data

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n),$$

where each input x_i is often a vector in $\mathbb{R}^d$ and each output y_i may be a number, a class label, or another vector. A model is a family of functions

$$f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^m,$$

where $\theta \in \mathbb{R}^p$ is a vector of parameters. Training the model means choosing θ so that $f_\theta(x_i)$ is close to y_i for the observed data and hopefully remains accurate on new data.

The central object is a *loss function*

$$L(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(f_\theta(x_i), y_i),$$

a function from parameter space to the real line:

$$L : \mathbb{R}^p \rightarrow \mathbb{R}.$$

So training is not mysterious from the viewpoint of calculus: it is optimization for a function of many variables.

Main principle

A model turns inputs into predictions. A loss function turns predictions into a number measuring error. Optimization changes the parameters to reduce that number.

The phrase *high-dimensional calculus* is not just a metaphor. In a small linear regression model, the parameter vector may have a few components. In a modern neural network, it may have many more. The mathematical objects remain the same.

Calculus object	Machine learning role
Point $\theta \in \mathbb{R}^p$	Current parameter values
Function $L(\theta)$	Loss or objective
Gradient $\nabla L(\theta)$	Direction of steepest increase
Negative gradient $-\nabla L(\theta)$	Direction of steepest local decrease
Hessian $H_L(\theta)$	Local curvature of the loss
Jacobian $J_f(\theta)$	Sensitivity of a vector-valued model
Chain rule	Backpropagation and automatic differentiation

25.2 A first loss function: least squares

Suppose the data are approximately linear:

$$y_i \approx \beta_0 + \beta_1 x_i.$$

The model has two parameters $\theta = (\beta_0, \beta_1)$. For a single data point, the prediction error is

$$e_i(\beta_0, \beta_1) = y_i - (\beta_0 + \beta_1 x_i).$$

The least-squares loss is

$$L(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2.$$

This is a surface over the (β_0, β_1) -plane. Training means finding the lowest point on this surface.

Worked example: gradient of least-squares loss

For

$$L(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2,$$

let

$$r_i = y_i - \beta_0 - \beta_1 x_i.$$

Then

$$L = \frac{1}{n} \sum_{i=1}^n r_i^2.$$

Differentiate with respect to β_0 :

$$\frac{\partial L}{\partial \beta_0} = \frac{1}{n} \sum_{i=1}^n 2r_i \frac{\partial r_i}{\partial \beta_0} = -\frac{2}{n} \sum_{i=1}^n r_i.$$

Differentiate with respect to β_1 :

$$\frac{\partial L}{\partial \beta_1} = \frac{1}{n} \sum_{i=1}^n 2r_i \frac{\partial r_i}{\partial \beta_1} = -\frac{2}{n} \sum_{i=1}^n x_i r_i.$$

Therefore

$$\nabla L(\beta_0, \beta_1) = \left\langle -\frac{2}{n} \sum_{i=1}^n r_i, -\frac{2}{n} \sum_{i=1}^n x_i r_i \right\rangle.$$

At a critical point the residuals satisfy

$$\sum_{i=1}^n r_i = 0, \quad \sum_{i=1}^n x_i r_i = 0.$$

Geometrically, the residual vector is orthogonal to the columns of the design matrix.

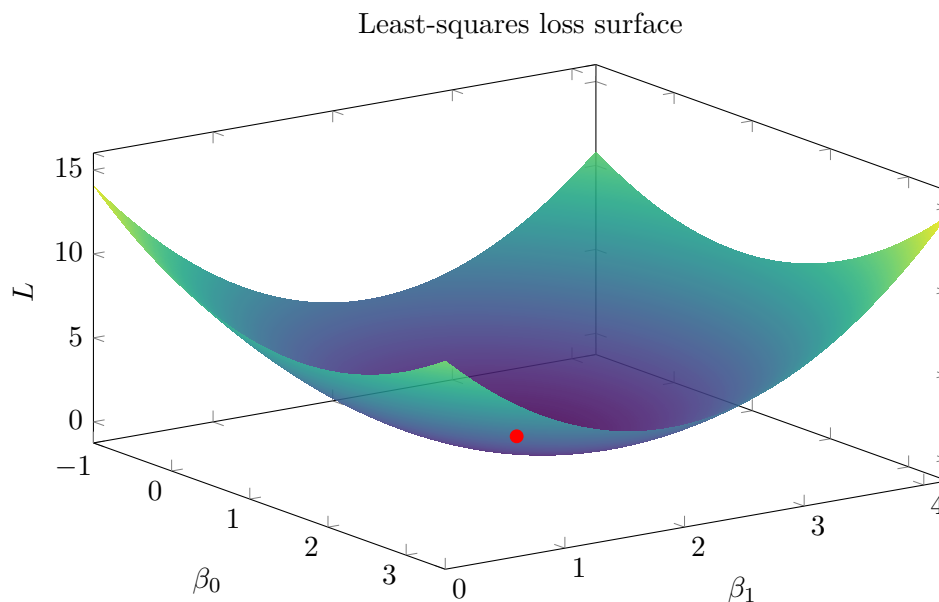


Figure 25.2: A least-squares loss surface. Training searches for the lowest point in parameter space.

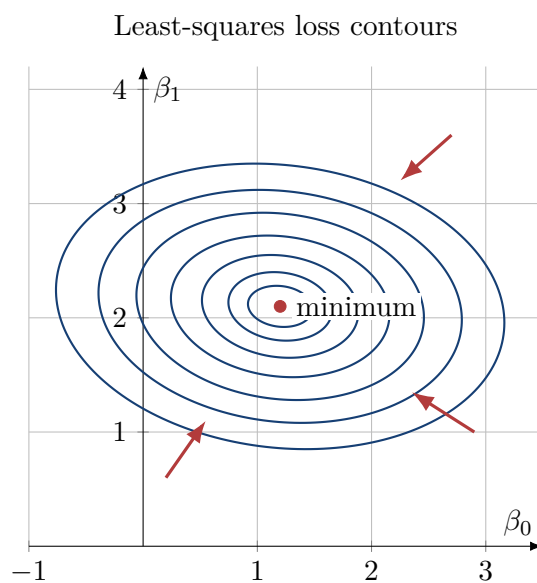


Figure 25.3: Contour lines of the least-squares loss. The gradient is perpendicular to the contours; the negative gradient points downhill.

25.3 Matrix form and the geometry of linear models

For multiple features, place the data in a design matrix

$$X = \begin{bmatrix} - & x_1^\top & - \\ - & x_2^\top & - \\ & \vdots & \\ - & x_n^\top & - \end{bmatrix}, \quad w \in \mathbb{R}^d, \quad \hat{y} = Xw.$$

The mean squared error loss is

$$L(w) = \frac{1}{n} \|Xw - y\|^2.$$

The gradient is

$$\nabla L(w) = \frac{2}{n} X^\top (Xw - y).$$

This formula is a bridge between multivariable calculus and linear algebra. The residual vector $Xw - y$ lives in data space $\mathbb{R}^n$, while the gradient lives in parameter space $\mathbb{R}^d$.

At a critical point,

$$X^\top (Xw - y) = 0.$$

Thus the residual vector is orthogonal to every column of X . This is the projection geometry behind least squares.

Geometry of least squares

The least-squares solution chooses Xw to be the orthogonal projection of y onto the column space of X . The calculus condition $\nabla L(w) = 0$ is the same as the linear algebra condition that the residual is orthogonal to the model space.

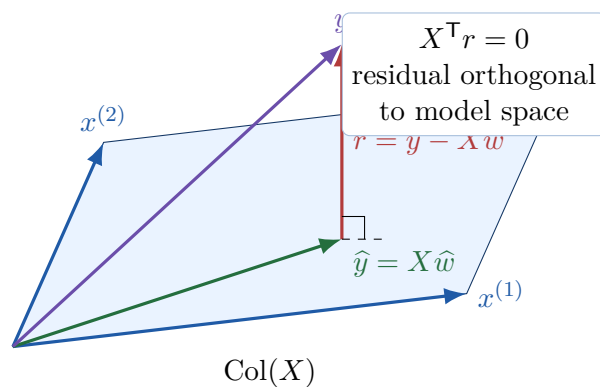


Figure 25.4: Projection geometry of least squares. The best prediction $X\hat{w}$ is the projection of y onto the column space of X .

Gradient descent for linear regression

Gradient descent updates parameters by

$$w_{k+1} = w_k - \eta \nabla L(w_k),$$

where $\eta > 0$ is the learning rate. For mean squared error,

$$w_{k+1} = w_k - \eta \frac{2}{n} X^\top (Xw_k - y).$$

If the learning rate is too small, training is slow. If it is too large, the method may oscillate or diverge.

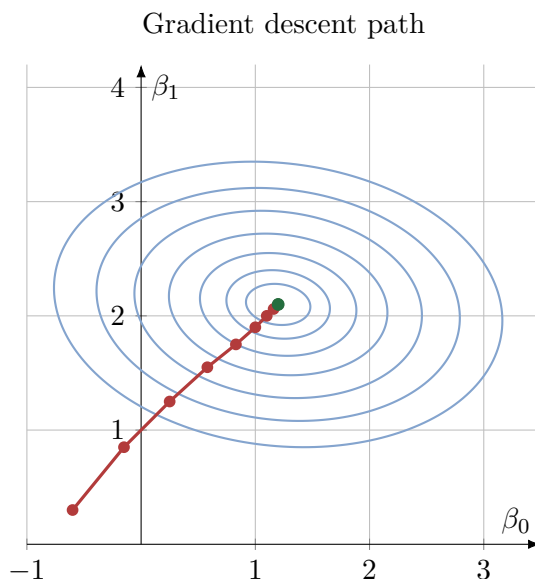


Figure 25.5: Gradient descent follows the negative gradient direction across loss contours.

25.4 Empirical risk and the structure of training problems

Most supervised learning problems can be written as empirical risk minimization:

$$\min_{\theta} L(\theta) = \min_{\theta} \frac{1}{n} \sum_{i=1}^n \ell(f_{\theta}(x_i), y_i).$$

There are three layers:

1. **Model layer:** $f_{\theta}(x)$ converts an input into a prediction.
2. **Loss layer:** $\ell(f_{\theta}(x_i), y_i)$ measures the error on one data point.
3. **Optimization layer:** the average loss is minimized over θ .

For a scalar-output model, the chain rule gives the gradient of one training example:

$$\nabla_{\theta} \ell(f_{\theta}(x_i), y_i) = \frac{\partial \ell}{\partial \hat{y}_i} \nabla_{\theta} f_{\theta}(x_i), \quad \hat{y}_i = f_{\theta}(x_i).$$

For a vector-output model $f_{\theta}(x_i) \in \mathbb{R}^m$, the Jacobian appears:

$$\nabla_{\theta} \ell(f_{\theta}(x_i), y_i) = J_{f_{\theta}}(x_i)^\top \nabla_{\hat{y}_i} \ell(\hat{y}_i, y_i).$$

This is the central calculus identity behind backpropagation.

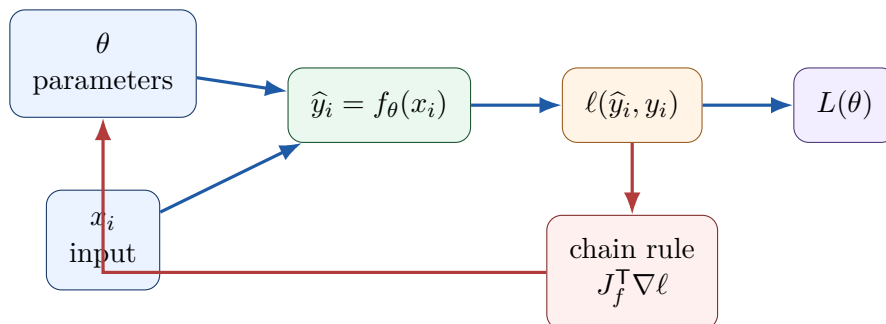


Figure 25.6: Empirical risk has a model layer, a loss layer, and an optimization layer. The chain rule transports derivatives backward to the parameters.

25.5 Logistic regression and classification

For binary classification, labels are often $y_i \in \{0, 1\}$. Logistic regression uses

$$p_i = \sigma(w^\top x_i + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

The output p_i is interpreted as a probability that $y_i = 1$. The binary cross-entropy loss is

$$L(w, b) = -\frac{1}{n} \sum_{i=1}^n [y_i \log p_i + (1 - y_i) \log(1 - p_i)].$$

Let $z_i = w^\top x_i + b$ and $p_i = \sigma(z_i)$. A useful derivative is

$$\frac{d}{dz} \sigma(z) = \sigma(z)(1 - \sigma(z)).$$

For the cross-entropy loss, the derivative simplifies:

$$\frac{\partial L}{\partial z_i} = \frac{1}{n} (p_i - y_i).$$

Therefore

$$\nabla_w L(w, b) = \frac{1}{n} \sum_{i=1}^n (p_i - y_i) x_i, \quad \frac{\partial L}{\partial b} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i).$$

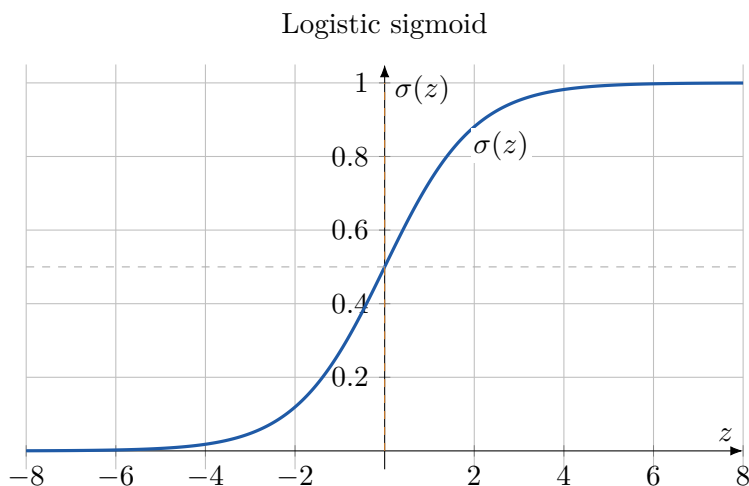


Figure 25.7: The logistic sigmoid transforms real-valued scores into probabilities between 0 and 1.

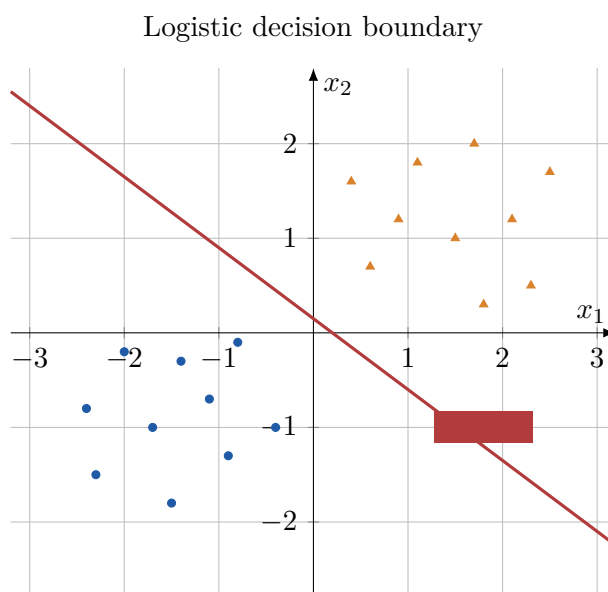


Figure 25.8: A logistic regression model learns a linear decision boundary in feature space.

25.6 Regularization and the bias-variance tradeoff

A loss function measures fit to data. A regularization term measures model complexity. A common objective is

$$J(\theta) = L(\theta) + \lambda R(\theta),$$

where $\lambda \geq 0$ controls the strength of regularization.

For ridge regression,

$$R(w) = \|w\|^2.$$

The regularized objective is

$$J(w) = \frac{1}{n} \|Xw - y\|^2 + \lambda \|w\|^2.$$

Its gradient is

$$\nabla J(w) = \frac{2}{n} X^\top (Xw - y) + 2\lambda w.$$

The term $2\lambda w$ pulls the parameter vector toward the origin. This often improves generalization by preventing unstable parameter estimates.

Calculus interpretation of regularization

Regularization changes the geometry of the loss surface. It adds curvature that penalizes large parameter values, often making the optimization problem better conditioned.

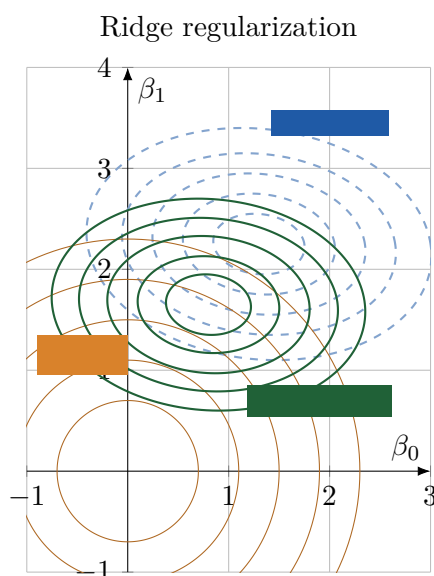


Figure 25.9: Ridge regularization adds circular penalty contours to the data-fitting loss, shifting the optimum toward smaller coefficients.

25.7 Stochastic gradient descent

For large data sets, computing the full gradient

$$\nabla L(\theta) = \frac{1}{n} \sum_{i=1}^n \nabla_{\theta} \ell_i(\theta)$$

may be expensive. Stochastic gradient descent replaces the full average by a random estimate:

$$\theta_{k+1} = \theta_k - \eta_k g_k,$$

where g_k is computed from one data point or a mini-batch. A mini-batch gradient has the form

$$g_k = \frac{1}{|B_k|} \sum_{i \in B_k} \nabla_{\theta} \ell_i(\theta_k).$$

This introduces noise into the path. The noise can slow convergence near a minimum, but it can also help training move through complicated landscapes.

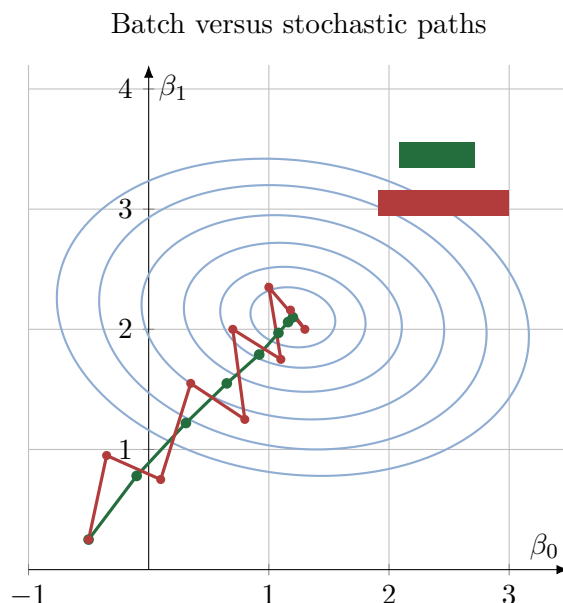


Figure 25.10: Batch gradient descent follows a smooth path; stochastic gradient descent follows a noisier path.

25.8 Neural networks and the chain rule

A neural network is a composition of functions. A simple one-hidden-layer network has the form

$$f_{\theta}(x) = W_2 \phi(W_1 x + b_1) + b_2,$$

where ϕ is an activation function applied componentwise. The parameters are

$$\theta = (W_1, b_1, W_2, b_2).$$

The main calculus idea is not new: the derivative of a composition is computed by the chain rule. What changes is scale. A network may have many layers:

$$x \mapsto h_1 \mapsto h_2 \mapsto \cdots \mapsto h_K \mapsto \hat{y} \mapsto L.$$

Backpropagation computes derivatives from the output backward through the computational graph.

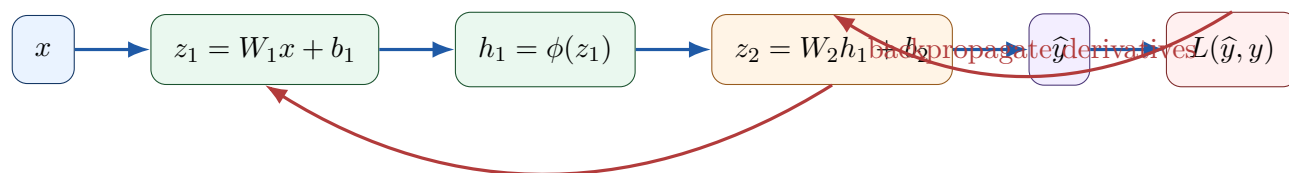


Figure 25.11: A neural network is a composition. Backpropagation is the chain rule applied backward through the computational graph.

A scalar example

Let

$$\hat{y} = a \phi(wx + b), \quad L = \frac{1}{2}(\hat{y} - y)^2.$$

Let $z = wx + b$ and $h = \phi(z)$. Then $\hat{y} = ah$. By the chain rule,

$$\frac{\partial L}{\partial a} = (\hat{y} - y)h,$$

$$\frac{\partial L}{\partial w} = (\hat{y} - y)a\phi'(z)x,$$

and

$$\frac{\partial L}{\partial b} = (\hat{y} - y)a\phi'(z).$$

Every large backpropagation computation is built from this same pattern.

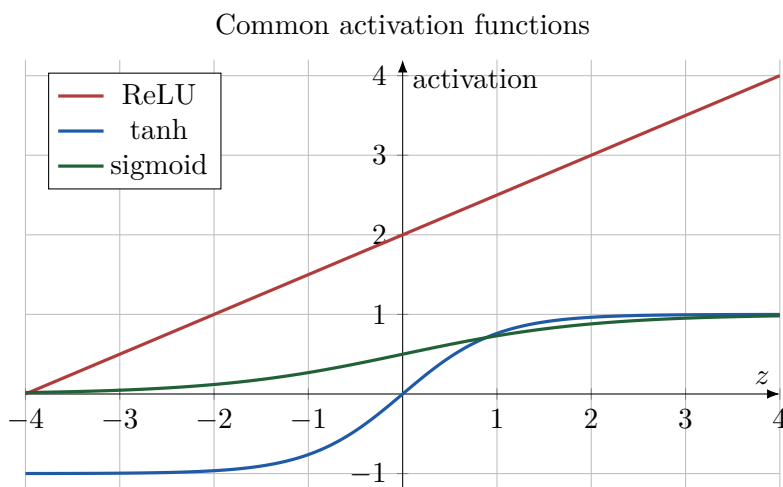


Figure 25.12: Common activation functions shape how neural networks bend space.

25.9 Numerical gradient checking

Even when gradients are computed automatically, it is useful to understand numerical checks. For a differentiable function $L : \mathbb{R}^p \rightarrow \mathbb{R}$, the central difference approximation to the j th partial derivative is

$$\frac{\partial L}{\partial \theta_j}(\theta) \approx \frac{L(\theta + he_j) - L(\theta - he_j)}{2h}.$$

This checks whether an analytic or automatic gradient is plausible.

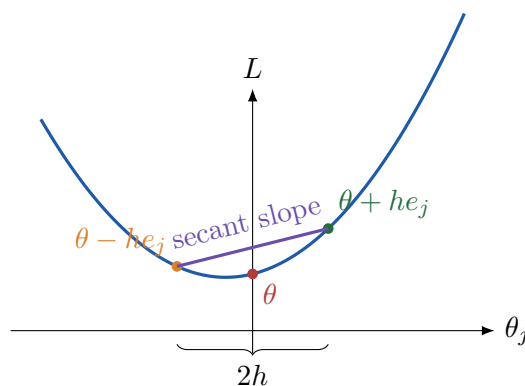


Figure 25.13: A central difference uses two nearby function values to approximate one component of the gradient.

A small difference between the analytic and numerical gradients gives confidence that the derivative was implemented correctly. A large difference suggests an algebra error, a coding error, or a nondifferentiability issue.

25.10 Hessians, curvature, and learning rates

Near a point θ_0 , a twice differentiable loss has the second-order approximation

$$L(\theta_0 + u) \approx L(\theta_0) + \nabla L(\theta_0)^\top u + \frac{1}{2} u^\top H_L(\theta_0) u.$$

The Hessian tells us how curved the loss is in different directions. If the Hessian has eigenvalues with very different sizes, the level sets are long and narrow. Gradient descent may zigzag.

For least squares,

$$L(w) = \frac{1}{n} \|Xw - y\|^2,$$

the Hessian is constant:

$$H_L(w) = \frac{2}{n} X^\top X.$$

The eigenvalues of $X^\top X$ control the conditioning of the optimization problem.



Figure 25.14: Poor conditioning creates narrow valleys, making gradient descent sensitive to learning rate and prone to zigzagging.

25.11 Feature maps and nonlinear models

A model can be nonlinear in the input but linear in parameters. For example,

$$f_w(x) = w_0 + w_1x + w_2x^2.$$

This is linear regression after replacing x by the feature vector

$$\phi(x) = (1, x, x^2).$$

More generally, a feature map

$$\phi : \mathbb{R}^d \rightarrow \mathbb{R}^m$$

transforms the original input into a space where a simpler model may work better.

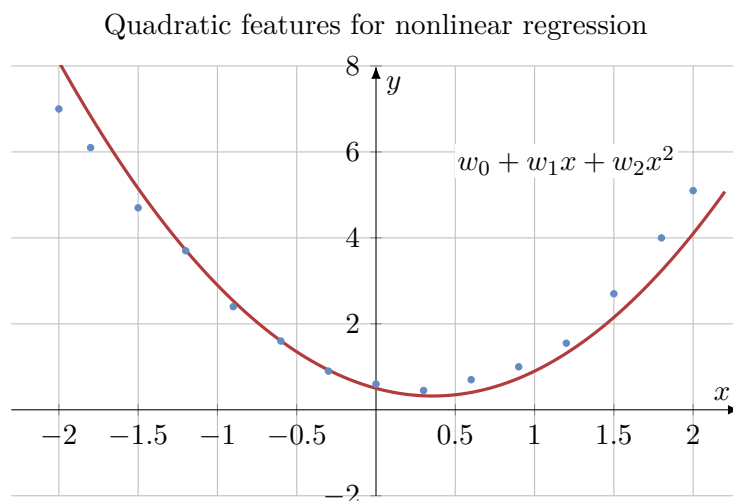


Figure 25.15: A nonlinear feature map can turn a curved pattern into a linear problem in transformed coordinates.

This example separates two ideas:

- *nonlinear in the input* means the graph may bend;
- *linear in the parameters* means the loss may still be a convex quadratic.

Neural networks are more complicated because they are usually nonlinear in both inputs and parameters.

25.12 What calculus tells us about machine learning

Multivariable calculus gives a precise language for several major questions in machine learning.

Sensitivity

How much does the output change when the input or parameters change? This is measured by derivatives and Jacobians.

Training

How should parameters be changed to reduce the loss? This is guided by gradients and optimization algorithms.

Curvature

Why is one learning problem easy and another hard? Hessians, eigenvalues, and conditioning provide part of the answer.

Generalization

Why might a model fit training data but fail on new data? Regularization, model complexity, and stability are connected to the geometry of the parameter space.

Computation

How can derivatives be computed for huge composed functions? The answer is the chain rule organized as automatic differentiation.

The big picture

Machine learning does not replace calculus. It scales calculus. The same gradient, Jacobian, Hessian, and chain-rule ideas from this book become computational tools for learning from data.

25.13 Common mistakes

Common mistakes

1. **Confusing feature space and parameter space.** Inputs x live in feature space; parameters θ live in parameter space. Gradients of the loss are usually taken with respect to parameters.
2. **Using the gradient direction instead of the negative gradient direction.** The gradient points toward steepest increase. Gradient descent moves in the negative gradient direction.
3. **Choosing a learning rate without checking behavior.** A learning rate can be too small, useful, unstable, or divergent.
4. **Assuming every loss is convex.** Least squares is convex in linear parameters, but neural-network losses are usually nonconvex.
5. **Ignoring scaling.** Feature scaling changes the geometry of the loss surface and can strongly affect optimization.
6. **Trusting code without gradient checks.** Numerical derivatives are useful for debugging small models.

25.14 Practice problems

Conceptual problems

1. Explain the difference between feature space and parameter space for a linear model $f_w(x) = w^\top x$.
2. What does the loss surface represent in linear regression?
3. Why is the negative gradient direction used in gradient descent?
4. Explain why the chain rule is the mathematical foundation of backpropagation.
5. In logistic regression, why is the output interpreted as a probability?
6. What is the purpose of regularization?
7. Why can poor conditioning slow down gradient descent?
8. What is the difference between batch gradient descent and stochastic gradient descent?

Computational problems

9. For $L(w_1, w_2) = 3w_1^2 + 2w_1w_2 + w_2^2$, compute ∇L and perform three steps of gradient descent starting at $(1, 2)$ with learning rate $\eta = 0.1$.
10. Let

$$L(\beta_0, \beta_1) = \frac{1}{3} \left[(1 - \beta_0)^2 + (2 - \beta_0 - \beta_1)^2 + (4 - \beta_0 - 2\beta_1)^2 \right].$$

Compute ∇L and solve $\nabla L = 0$.

11. For logistic regression with one feature, $p_i = \sigma(wx_i + b)$, show that

$$\frac{\partial L}{\partial w} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)x_i.$$

12. For ridge regression,

$$J(w) = \frac{1}{n} \|Xw - y\|^2 + \lambda \|w\|^2,$$

compute the Hessian.

13. Let $f_w(x) = w_0 + w_1x + w_2x^2$. Write the design matrix for data points $x_1, \dots, x_n$.

14. Suppose $L(\theta) = \frac{1}{2}(f(\theta) - y)^2$. Show that

$$\nabla L(\theta) = (f(\theta) - y)\nabla f(\theta).$$

Python problems

15. Modify the gradient descent code in this chapter to use a smaller learning rate and compare the path.
16. Modify the logistic regression simulation so that the two classes overlap more. What happens to the decision boundary?
17. Implement ridge regression by gradient descent and compare the result for $\lambda = 0$, $\lambda = 0.1$, and $\lambda = 1$.
18. Write a numerical gradient checker for the logistic regression loss.
19. Simulate stochastic gradient descent with different mini-batch sizes.
20. Create a polynomial regression example with features $1, x, x^2, x^3$ and compare the fit to a linear model.

Project problems

21. **Loss landscape project.** Choose a two-parameter model and create a contour plot of its loss surface. Overlay several gradient descent paths with different learning rates.
22. **Classification project.** Generate a two-class data set, train logistic regression from scratch, and visualize the learned decision boundary.
23. **Regularization project.** Fit polynomial models of several degrees and compare training error, testing error, and coefficient size.
24. **Backpropagation project.** Implement the scalar neural-network example from Section 25.8 and compare analytic gradients with numerical gradients.

25.15 Python lab and interactive page

- Notebook lab: `Chapter25_Multivariable_Calculus_for_Data_Science_and_Machine_Learning.ipynb`
- Interactive page: `chapter-25.html`

25.16 Chapter summary

This chapter connected multivariable calculus to data science and machine learning. A model is a function with parameters, and training is the problem of minimizing a loss function on parameter space. Gradients guide local improvement, Hessians describe curvature, Jacobians measure sensitivity, and the chain rule powers backpropagation. Linear regression, logistic regression, regularization, stochastic gradient descent, and neural networks all use the same core calculus ideas developed throughout this book.

Chapter 26. Multivariable Calculus for Applied Mathematics

Fields, flows, conservation laws, and numerical models

Applied mathematics is the art of turning a real system into a mathematical model, studying the model, computing with it, and interpreting the result. Multivariable calculus is one of the main languages of this process. It tells us how quantities change in space and time, how material flows through regions, how heat diffuses, how forces produce motion, and how energy principles lead to equations.

The previous chapters developed gradients, Jacobians, Hessians, multiple integrals, vector fields, line integrals, surface integrals, and the major integral theorems. This chapter brings those ideas together from an applied mathematics point of view. The goal is not to replace a future course in differential equations, numerical analysis, or partial differential equations. The goal is to show how multivariable calculus becomes the foundation for those subjects.

In this chapter we study four recurring applied themes:

1. **Local change:** derivatives measure how quantities change at a point.
2. **Accumulation:** integrals measure total amount over curves, surfaces, and regions.
3. **Balance laws:** divergence and flux express conservation of mass, heat, probability, or charge.
4. **Energy principles:** many equations arise by minimizing an energy.

Learning goals

After completing this chapter, you should be able to:

- interpret scalar fields, vector fields, gradient fields, divergence, curl, and the Laplacian in applied models;
- derive simple conservation laws from flux balance;
- connect the heat equation to diffusion, smoothing, and gradient-based dynamics;
- use finite differences to approximate gradients, divergence, and the Laplacian on a grid;
- explain how continuous models lead to numerical algorithms;
- recognize variational and energy-based formulations;
- connect applied calculus ideas to data science, image processing, probability flow, and optimization.

1 The applied mathematics pipeline

A useful applied mathematics workflow is

system $\longrightarrow$ variables $\longrightarrow$ model $\longrightarrow$ analysis $\longrightarrow$ computation $\longrightarrow$ interpretation.

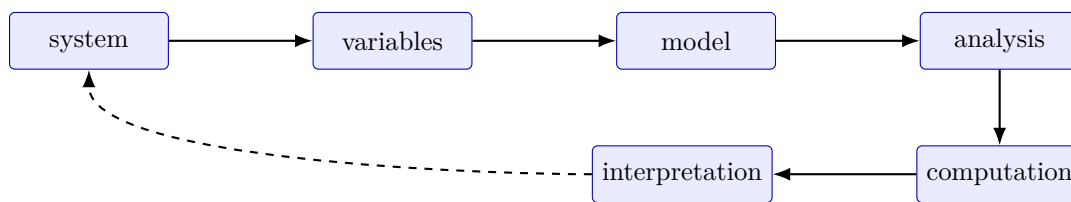


Figure 1: The applied mathematics pipeline. Interpretation often feeds back into the choice of variables and model.

For example, if $u(x, t)$ is the temperature of a thin rod at position x and time t , a physical law says heat flows from hot regions to cold regions. A calculus model converts this statement into a differential equation. A numerical method approximates the equation on a finite grid, and a simulation helps us understand the temperature evolution.

Applied setting	Typical field	Calculus object
heat flow	$u(x, y, t)$	scalar field
fluid velocity	$F(x, y, z, t)$	vector field
probability density	$p(x, t)$	scalar field with integral 1
image intensity	$I(x, y)$	scalar field on a pixel grid
model loss	$L(\theta)$	scalar field on parameter space
gradient descent	$\theta'(t) = -\nabla L(\theta)$	vector field on parameter space

A key message of this chapter is that the same calculus operators appear across many disciplines.

2 Scalar fields, vector fields, and operators

A scalar field assigns a number to each point. Examples include temperature, pressure, density, altitude, potential energy, image brightness, or a loss function:

$$u = u(x, y, z).$$

A vector field assigns a vector to each point. Examples include velocity, force, heat flux, electric field, gradient direction, or parameter update direction:

$$F(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle.$$

The most important differential operators in applied mathematics are

$$\nabla u = \langle u_x, u_y, u_z \rangle, \quad \nabla \cdot F = P_x + Q_y + R_z,$$

$$\nabla \times F = \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle, \quad \Delta u = \nabla \cdot (\nabla u) = u_{xx} + u_{yy} + u_{zz}.$$

The operator Δ is called the *Laplacian*. It is one of the central operators in applied mathematics.

Interpretation of the operators

- ∇u points in the direction of fastest increase of u .
- $\nabla \cdot F$ measures local expansion or compression of a vector field.
- $\nabla \times F$ measures local rotation.
- Δu compares a value with nearby values and measures local curvature or imbalance.

Example 2.1: Gradient, divergence, curl, and Laplacian

Let $u(x, y, z) = x^2 + y^2 + z^2$. Then

$$\nabla u = \langle 2x, 2y, 2z \rangle, \quad \Delta u = u_{xx} + u_{yy} + u_{zz} = 2 + 2 + 2 = 6.$$

If $F(x, y, z) = \langle x, y, z \rangle$, then

$$\nabla \cdot F = 1 + 1 + 1 = 3, \quad \nabla \times F = \langle 0, 0, 0 \rangle.$$

This vector field expands outward from the origin but does not rotate.

Contours and gradient field of $u = x^2 + 0.5y^2$

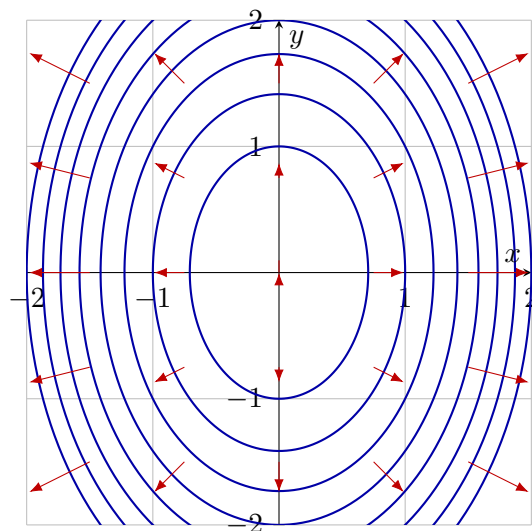


Figure 2: A scalar field, its contours, and its gradient vectors. The gradient points uphill and is perpendicular to level curves.

3 Balance laws and conservation

Many applied models begin with a balance principle:

$$\text{rate of change inside a region} = \text{inflow} - \text{outflow} + \text{sources}.$$

Suppose $\rho(x, y, z, t)$ is the density of a quantity and $J(x, y, z, t)$ is its flux field. For a fixed region E , the total amount inside E is

$$\int_E \rho dV.$$

The outward flux through the boundary ∂E is

$$\iint_{\partial E} J \cdot n dS.$$

If there are no internal sources, conservation says

$$\frac{d}{dt} \int_E \rho dV = - \iint_{\partial E} J \cdot n dS.$$

Using the Divergence Theorem,

$$\iint_{\partial E} J \cdot n dS = \int_E \nabla \cdot J dV.$$

Therefore

$$\int_E \rho_t dV = - \int_E \nabla \cdot J dV.$$

Since this should hold for every region E , the local form is

$$\rho_t + \nabla \cdot J = 0.$$

This is called a *continuity equation*.

Conservation law template

A large class of applied models has the form

$$\rho_t + \nabla \cdot J = s,$$

where ρ is a density, J is a flux, and s is a source term.

Example 3.1: Source, sink, and incompressible fields

In the plane, consider

$$F_1(x, y) = \langle x, y \rangle, \quad F_2(x, y) = \langle -x, -y \rangle, \quad F_3(x, y) = \langle -y, x \rangle.$$

Their divergences are

$$\nabla \cdot F_1 = 2, \quad \nabla \cdot F_2 = -2, \quad \nabla \cdot F_3 = 0.$$

The first field is a source, the second is a sink, and the third is rotation without local expansion.

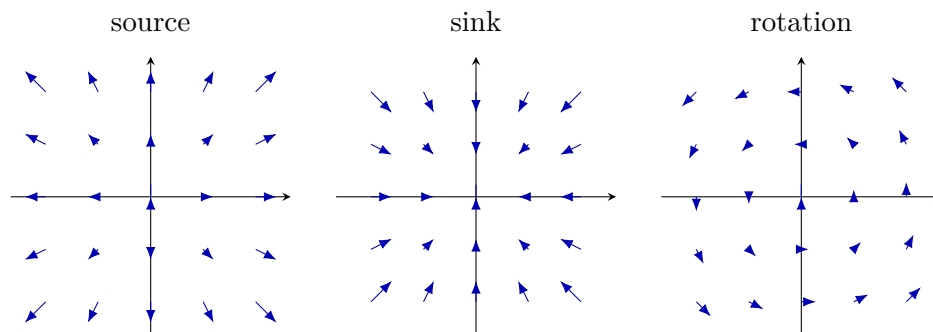


Figure 3: Three basic vector fields: source, sink, and rotational field. Divergence distinguishes expansion from compression.

4 Diffusion and the heat equation

One of the most important applied models is the heat equation. Suppose $u(x, t)$ is temperature along a rod. Heat flows from high temperature to low temperature. A common model for heat flux is Fourier's law:

$$J = -k\nabla u,$$

where $k > 0$ is the conductivity. The minus sign means heat flows downhill, opposite the direction of increasing temperature.

If heat is conserved and $J = -k\nabla u$, then

$$u_t + \nabla \cdot J = 0.$$

Substituting the flux law gives

$$u_t - k\nabla \cdot (\nabla u) = 0,$$

so

$$u_t = k\Delta u.$$

This is the *heat equation*. In one spatial dimension, $u_t = ku_{xx}$. In two spatial dimensions, $u_t = k(u_{xx} + u_{yy})$.

Why the Laplacian appears

The Laplacian measures whether a point is higher or lower than its neighbors. If a point is hotter than its neighbors, heat diffuses away. If it is colder than its neighbors, heat diffuses in. This local averaging behavior is exactly what the heat equation models.

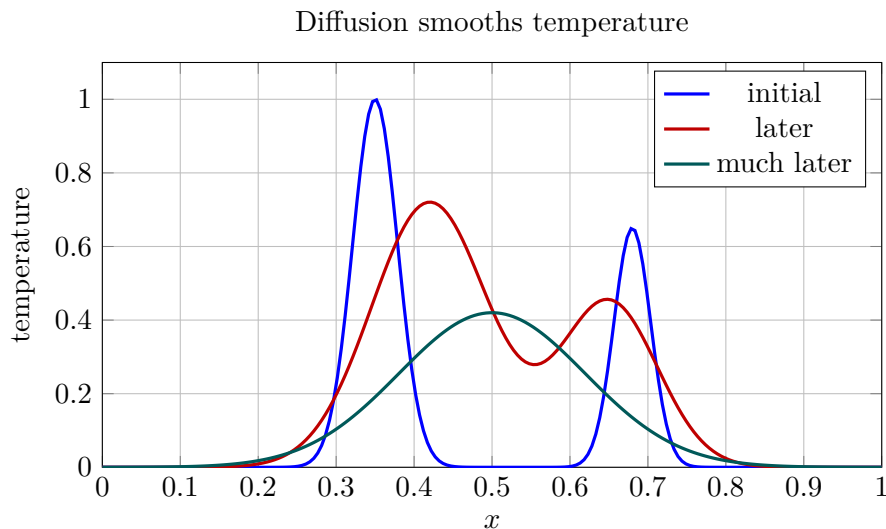


Figure 4: A localized temperature profile spreads and smooths over time, as predicted by the one-dimensional heat equation.

5 The Laplacian as local averaging

On a grid, the two-dimensional Laplacian is often approximated by

$$\Delta u(x_i, y_j) \approx \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}}{h^2}.$$

This formula compares a grid value to its four nearest neighbors. It appears in numerical partial differential equations, image processing, graph learning, and spatial statistics.

	$u_{i,j+1}$	
$u_{i-1,j}$	$u_{i,j}$	$u_{i+1,j}$
	$u_{i,j-1}$	

$$\Delta u \approx \frac{\text{neighbor sum} - 4 \cdot \text{center}}{h^2}$$

Figure 5: The five-point stencil for the two-dimensional discrete Laplacian.

6 Advection and transport

Diffusion describes spreading. Advection describes transport by a velocity field. Suppose $\rho(x, y, t)$ is the density of a substance moving with velocity field

$$v(x, y) = \langle P(x, y), Q(x, y) \rangle.$$

The flux is $J = \rho v$, and the conservation law becomes

$$\rho_t + \nabla \cdot (\rho v) = 0.$$

If the velocity field is incompressible, meaning $\nabla \cdot v = 0$, then

$$\rho_t + v \cdot \nabla \rho = 0.$$

The term $v \cdot \nabla \rho$ is a directional derivative: it measures how density changes in the direction of the flow.

Example 6.1: Rotation transports density

For $v(x, y) = \langle -y, x \rangle$, particles rotate around the origin. Since

$$\nabla \cdot v = \frac{\partial(-y)}{\partial x} + \frac{\partial x}{\partial y} = 0,$$

this field preserves area locally.

Advection by $v = \langle -y, x \rangle$

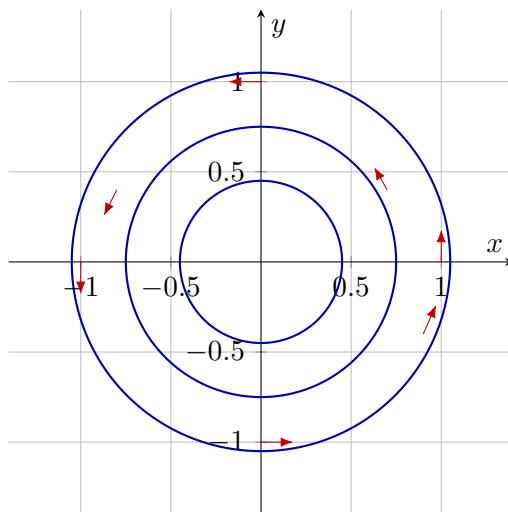


Figure 6: Particle trajectories in a rotational velocity field. Advection transports quantities along flow lines.

7 Steady states and Laplace's equation

A steady state is a solution that does not change with time. For the heat equation $u_t = k\Delta u$, a steady state satisfies

$$\Delta u = 0.$$

This is *Laplace's equation*. Functions satisfying $\Delta u = 0$ are called *harmonic functions*. Laplace's equation appears in heat conduction, electrostatics, gravitational potential, fluid flow, and image interpolation.

Example 7.1: A harmonic function

Let $u(x, y) = x^2 - y^2$. Then $u_{xx} = 2$ and $u_{yy} = -2$, so

$$\Delta u = u_{xx} + u_{yy} = 0.$$

Thus $u(x, y) = x^2 - y^2$ is harmonic.

Level curves and gradient field for $u = x^2 - y^2$

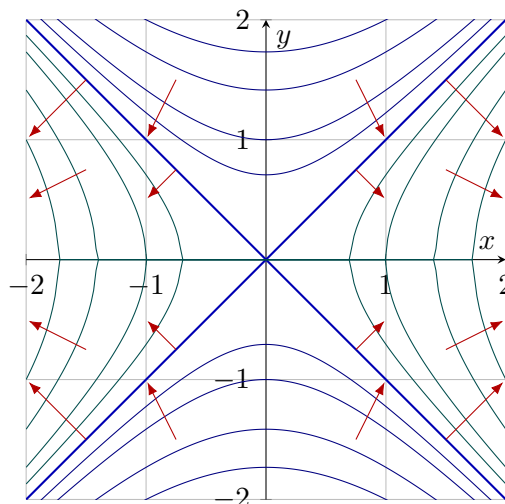


Figure 7: The harmonic function $u = x^2 - y^2$ has zero Laplacian. It bends upward in one direction and downward in the other.

8 Numerical derivatives on grids

In applied mathematics, functions are often known only through sampled values. For grid spacing h , common finite-difference approximations are

$$u_x(x_i, y_j) \approx \frac{u_{i+1,j} - u_{i-1,j}}{2h}, \quad u_y(x_i, y_j) \approx \frac{u_{i,j+1} - u_{i,j-1}}{2h},$$

$$\Delta u(x_i, y_j) \approx \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}}{h^2}.$$

These approximations convert calculus into linear algebra. A differential operator becomes a matrix or a stencil.

Continuous-discrete dictionary

Continuous calculus	Discrete computation
function $u(x, y)$	array of values
derivative u_x	finite difference
gradient ∇u	pair of arrays
Laplacian Δu	stencil or sparse matrix
integral	sum times area element
partial differential equation	update rule or linear system

9 Variational principles and energy minimization

Many equations in applied mathematics come from minimizing an energy. A simple example is the Dirichlet energy

$$E[u] = \frac{1}{2} \int_D |\nabla u|^2 dA.$$

A function that minimizes this energy subject to fixed boundary values satisfies Laplace's equation

$$\Delta u = 0$$

inside the domain.

The intuition is that minimizing $\int |\nabla u|^2$ favors smooth functions. The minimizer is as smooth as possible while still matching boundary constraints. This idea appears in elastic membranes, electrostatic potentials, image denoising and interpolation, smoothing splines, graph-based semi-supervised learning, and regularization in inverse problems.

Example 9.1: Smoothing by energy minimization

Suppose we observe noisy data y_i and want a smooth curve u_i . A discrete energy is

$$E(u) = \sum_i (u_i - y_i)^2 + \lambda \sum_i (u_{i+1} - u_i)^2.$$

The first term keeps u close to the data. The second term penalizes roughness. The parameter λ controls the tradeoff.

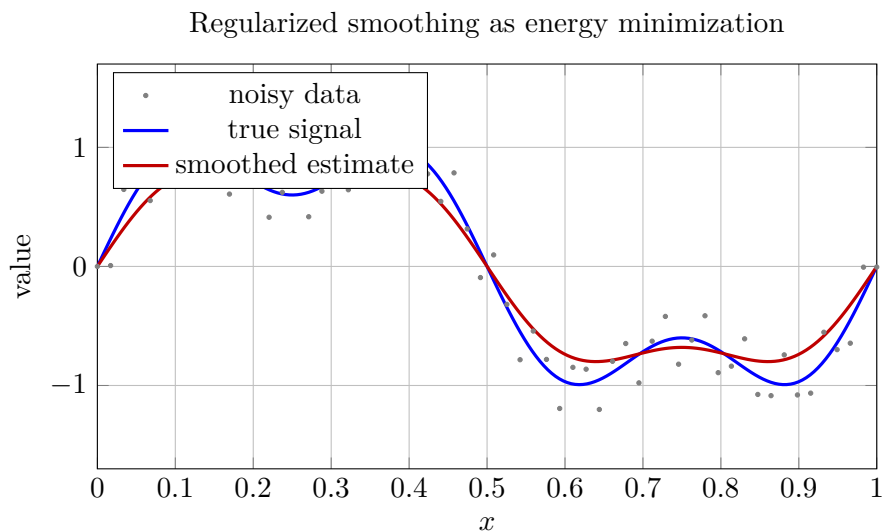


Figure 8: Energy-based smoothing balances fidelity to noisy data with a penalty on roughness.

10 Gradient flows

Optimization can also be viewed as an applied dynamical system. Given an energy or loss function $E(x)$, gradient flow is the differential equation

$$\frac{dx}{dt} = -\nabla E(x).$$

This equation moves downhill as quickly as possible in continuous time. Gradient descent is a discrete approximation:

$$x_{k+1} = x_k - \eta \nabla E(x_k).$$

Thus optimization algorithms are closely related to differential equations.

Example 10.1: Gradient flow toward a minimum

For an energy landscape $E(x, y)$, gradient flow follows the vector field $-\nabla E$. The motion crosses contours in the downhill direction and tends to move toward local minima.

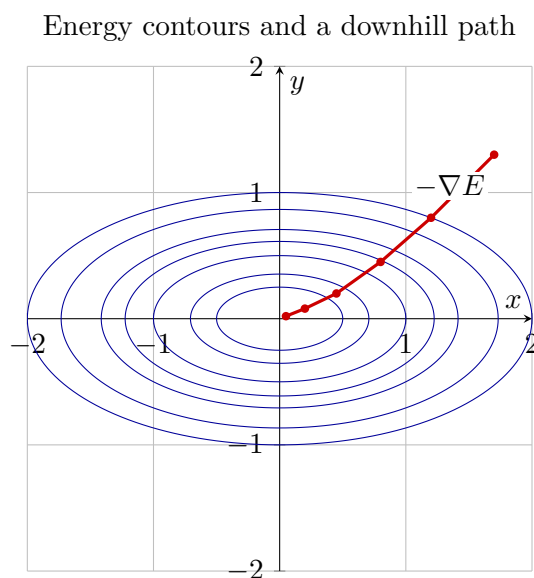


Figure 9: Gradient flow follows the vector field $-\nabla E$ and moves downhill on an energy landscape.

11 From continuous models to numerical algorithms

Applied mathematics often converts a continuous model into a finite-dimensional computation. The common steps are:

1. Choose variables. Decide what the unknown function represents.
2. State a law. Use balance, geometry, probability, or energy.
3. Write equations. Express the law using derivatives and integrals.
4. Choose boundary and initial conditions. Models need enough information to determine a solution.

5. Discretize. Replace derivatives by finite differences, finite elements, or spectral approximations.
6. Compute. Solve a system of algebraic equations or run an update rule.
7. Validate. Compare against data, units, limiting cases, or known solutions.

For the one-dimensional heat equation $u_t = ku_{xx}$, a simple explicit method is

$$u_i^{n+1} = u_i^n + r(u_{i+1}^n - 2u_i^n + u_{i-1}^n), \quad r = \frac{k\Delta t}{(\Delta x)^2}.$$

This formula says: update each value using the difference between the value and its neighbors.

Stability matters

A numerical formula can be mathematically reasonable but computationally unstable. For the explicit heat equation scheme, the time step must be sufficiently small relative to the spatial grid size. This is a first glimpse of numerical analysis.

12 Applied mathematics examples

Image denoising

An image can be viewed as a scalar field $I(x, y)$. Diffusion smooths the image by reducing sharp local variations. A simple denoising model is $I_t = \Delta I$. This smooths noise but may also blur edges. More advanced models modify the diffusion coefficient so smoothing happens more strongly in flat regions and less strongly near edges.

Probability diffusion

A probability density $p(x, t)$ can evolve by diffusion:

$$p_t = Dp_{xx}.$$

Under appropriate boundary conditions this equation preserves total probability:

$$\int p(x, t) dx = 1.$$

This connects calculus to stochastic processes, Brownian motion, and statistical physics.

Inverse problems

In an inverse problem, we infer hidden causes from observed effects. A common regularized model minimizes

$$\|Au - y\|^2 + \lambda\|Lu\|^2,$$

where A models the measurement process, y is observed data, and L measures roughness.

Conservation in networks

Even when a system is not spatially continuous, conservation ideas remain. In a network, mass or probability may flow along edges. The graph divergence at a node measures net outflow. The graph Laplacian plays the role of the continuous Laplacian. This is why Laplacian ideas appear in spectral clustering, graph signal processing, semi-supervised learning, and diffusion maps.

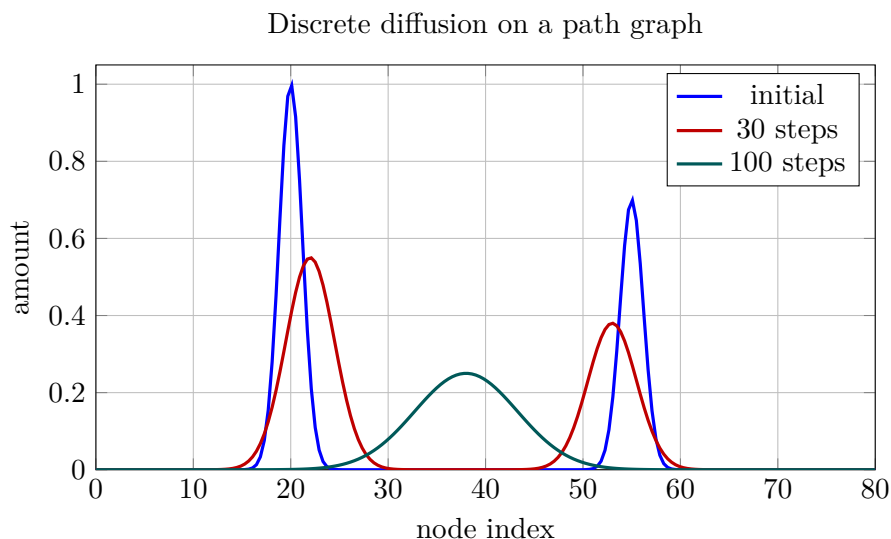


Figure 10: Discrete diffusion on a one-dimensional network. Repeated averaging behaves like a graph version of the heat equation.

13 Common modeling mistakes

Applied mathematics is powerful, but models require judgment. Common mistakes include:

- confusing density with total amount;
- forgetting units when defining flux or source terms;
- using a local equation without appropriate boundary conditions;
- assuming a numerical solution is accurate without checking stability or convergence;
- treating a model as reality rather than as a simplified representation;
- ignoring whether a vector field is conservative, divergence-free, or rotational;
- applying a theorem without checking orientation, smoothness, or region assumptions.

A good model should be mathematically consistent, computationally feasible, and interpretable in context.

14 Computational lab: finite differences and diffusion

The following functions form a small numerical calculus toolkit for scalar fields on a grid:

```

def central_gradient(U, h):
    Ux[:,1:-1] = (U[:,2:] - U[:, :-2])/(2*h)
    Uy[1:-1,:] = (U[2:,:] - U[ :-2,:])/(2*h)
    return Ux, Uy

def five_point_laplacian(U, h):
    Lap[1:-1,1:-1] = (
        U[2:,1:-1] + U[ :-2,1:-1]
        + U[1:-1,2:] + U[1:-1, :-2]
        - 4*U[1:-1,1:-1]
    )/(h**2)
    return Lap

```

For $u(x, y) = x^2 + y^2$, the exact value is $\Delta u = 4$. The numerical value should be close to 4 on interior grid points.

15 Chapter summary

This chapter connected multivariable calculus to applied mathematics. The central ideas are:

- scalar fields describe quantities such as temperature, density, potential, image intensity, and loss;
- vector fields describe forces, velocities, fluxes, and gradient flows;
- conservation laws combine time derivatives with flux integrals;
- the Divergence Theorem converts global flux balance into local differential equations;
- diffusion and the heat equation arise from conservation plus a flux law;
- the Laplacian measures local imbalance and acts like a continuous averaging operator;
- finite differences turn derivatives into computational formulas;
- energy minimization leads to differential equations and regularized algorithms;
- gradient flow connects optimization with dynamical systems.

The main conceptual bridge is

$$\text{calculus operators} \longrightarrow \text{models} \longrightarrow \text{algorithms}.$$

16 Exercises

Conceptual exercises

1. Explain the difference between a scalar field and a vector field. Give two examples of each from applied mathematics.
2. What does ∇u measure? Why is it perpendicular to level curves?

3. What does $\nabla \cdot F$ measure physically?
4. What does Δu measure intuitively?
5. Explain why heat should flow in the direction $-\nabla u$ rather than ∇u .
6. What is the difference between diffusion and advection?
7. Why does a conservation law naturally involve flux through the boundary of a region?
8. Explain why a steady-state heat equation leads to $\Delta u = 0$.
9. Why do numerical models require boundary conditions?
10. Explain the connection between gradient descent and gradient flow.

Computational exercises

11. For $u(x, y) = e^{-x^2-y^2}$, compute ∇u and Δu .
12. For $F(x, y) = \langle x^2, xy \rangle$, compute $\nabla \cdot F$ and interpret the result.
13. For $F(x, y) = \langle -y, x \rangle$, compute divergence and curl.
14. Use finite differences to approximate u_x and u_y for $u(x, y) = \sin x \cos y$ on a grid.
15. Use the five-point stencil to approximate Δu for $u(x, y) = x^2 + y^2$.
16. Simulate one-dimensional diffusion with a different initial condition.
17. Simulate two-dimensional diffusion from a square-shaped hot region.
18. Compare numerical diffusion using two different time step sizes.
19. Implement gradient descent for $E(x, y) = x^2 + 4y^2$ and plot the path.
20. Solve a smoothing problem using $E(u) = \sum_i (u_i - y_i)^2 + \lambda \sum_i (u_{i+1} - u_i)^2$ for several values of λ .

Applied exercises

21. Derive the heat equation from $\rho_t + \nabla \cdot J = 0$ and Fourier's law $J = -k \nabla u$.
22. Suppose $p(x, t)$ is a probability density satisfying $p_t = D p_{xx}$ on a closed interval with no-flux boundary conditions. Explain why total probability is conserved.
23. A pollutant concentration $c(x, y, t)$ moves with velocity $v(x, y)$ and diffuses with coefficient D . Explain the model $c_t + v \cdot \nabla c = D \Delta c$ term by term.
24. In image processing, why might ordinary diffusion blur edges? How might one modify the model to reduce edge blurring?
25. In a loss landscape $L(\theta)$, explain the difference between gradient flow and Newton's method.

Challenge exercises

26. Let $u(x, y) = \log \sqrt{x^2 + y^2}$ for $(x, y) \neq (0, 0)$. Show that $\Delta u = 0$ away from the origin.
27. Let $F = \nabla u$. Show that $\nabla \cdot F = \Delta u$.
28. Let $v(x, y) = \langle -y, x \rangle$. Show that the flow preserves $x^2 + y^2$.
29. Derive the finite-difference update $u_i^{n+1} = u_i^n + r(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$ from $u_t = ku_{xx}$.
30. For $E(u) = \|u - y\|^2 + \lambda \|Du\|^2$, show that the minimizer satisfies $(I + \lambda D^T D)u = y$.

Chapter 27. Review, Synthesis, and Bridge to Graduate Study

A final map of geometry, derivatives, integrals, vector calculus, computation, and graduate preparation

Multivariable calculus is not a list of formulas. It is a language for describing how quantities change, accumulate, interact, flow, and optimize in many dimensions. This final chapter organizes the whole book into a small number of durable ideas that carry into applied mathematics, statistics, data science, machine learning, physics, engineering, and quantitative modeling.

The chapters have developed four connected viewpoints:

1. **Geometry:** points, vectors, curves, surfaces, regions, and fields.
2. **Local change:** derivatives, gradients, Jacobians, Hessians, and approximations.
3. **Accumulation:** double integrals, triple integrals, surface integrals, line integrals, and change of variables.
4. **Global structure:** conservative fields, Green's theorem, Stokes theorem, the Divergence theorem, and conservation laws.

Modern computational work uses all four viewpoints at once. A statistical model has parameters in a high-dimensional space. A learning algorithm follows gradients of a loss function. A probability distribution is a density whose mass is computed by integration. A physical system is described by fields, fluxes, and conservation laws.

Learning goals

After studying this chapter, you should be able to:

- organize multivariable calculus by object type: scalar functions, vector functions, regions, curves, surfaces, and vector fields;
- decide whether a problem is asking for local change, accumulation, optimization, or global flow;
- use gradients, Jacobians, and Hessians as the main derivative objects in modern computation;
- connect double and triple integrals to mass, probability, expectation, and volume;
- connect line and surface integrals to work, circulation, flux, and conservation;
- explain how the major integral theorems generalize the Fundamental Theorem of Calculus;
- implement a small numerical toolkit for gradients, Jacobians, Hessians, and integrals;
- prepare for graduate courses in applied mathematics, statistics, data science, AI, and machine learning.

The central habit

Before computing, identify the object. Is it a scalar field, vector field, curve, surface, region, density, loss function, or constraint? The correct formula usually becomes clear once the mathematical object is clear.

27.1 The big map of the subject

A compact way to remember the book is by object type.

Object	Typical notation	Question	Main tools
Point or vector	$x \in \mathbb{R}^n$	Where are we? What direction?	norm, dot product, projection
Scalar field	$f : \mathbb{R}^n \rightarrow \mathbb{R}$	How does a quantity change?	gradient, Hessian, Taylor expansion
Vector function	$r(t)$	How does a curve move?	velocity, speed, acceleration, arc length
Region	$R \subset \mathbb{R}^2, E \subset \mathbb{R}^3$	How much accumulates?	double and triple integrals
Coordinate map	$T(u, v)$ or $T(u, v, w)$	How does area or volume transform?	Jacobian determinant
Vector field	$F : \mathbb{R}^n \rightarrow \mathbb{R}^n$	How does a field push or flow?	line integrals, flux, curl, divergence
Constraint	$g(x) = c$	What is optimal under restriction?	Lagrange multipliers
Model or loss	$L(\theta)$	How do we fit or optimize?	gradients, Hessians, descent methods

The most important distinction is between **local** and **global** information. A derivative tells us what happens near a point. An integral tells us what accumulates over an interval, region, curve, or surface. Theorems such as Green's theorem and the Divergence theorem connect local information to global information.

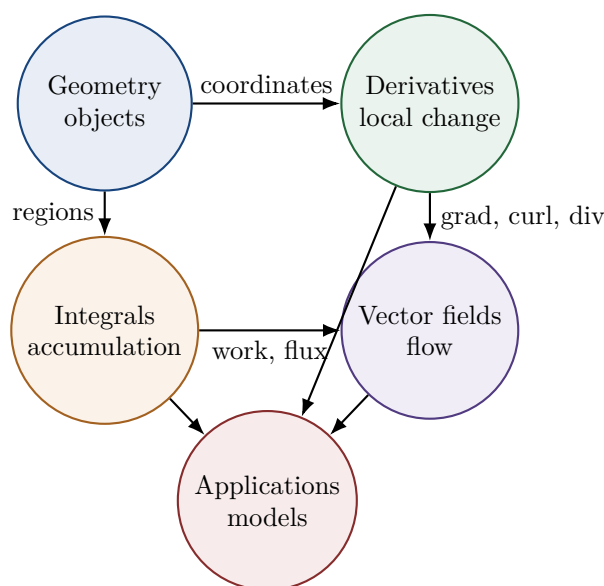


Figure 1: A schematic map of the main objects in multivariable calculus.

27.2 Geometry: the coordinate language

The first part of the book developed the coordinate language of space. The essential objects are points, vectors, distances, angles, projections, lines, planes, and surfaces. For $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$,

$$\|x - y\| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}.$$

The dot product is

$$x \cdot y = x_1 y_1 + \cdots + x_n y_n,$$

and it measures alignment:

$$x \cdot y = \|x\| \|y\| \cos \theta.$$

Projection is the geometry behind least squares and many data methods. The projection of x onto the direction of a nonzero vector v is

$$\text{proj}_v x = \frac{x \cdot v}{v \cdot v} v.$$

A hyperplane in $\mathbb{R}^n$ has equation

$$w \cdot x = b.$$

This equation appears in tangent planes, constraints, regression, support vector machines, and linear classifiers.

Example 27.1: Projection as a least-squares approximation

Let $x = (3, 4)$ and $v = (1, 2)$. The best approximation to x among all multiples of v is

$$\text{proj}_v x = \frac{x \cdot v}{v \cdot v} v = \frac{3 + 8}{1 + 4} (1, 2) = \frac{11}{5} (1, 2).$$

The residual $r = x - \text{proj}_v x$ is orthogonal to v . This is the same geometry used when fitting a linear model by least squares: the residual is orthogonal to the space of fitted values.

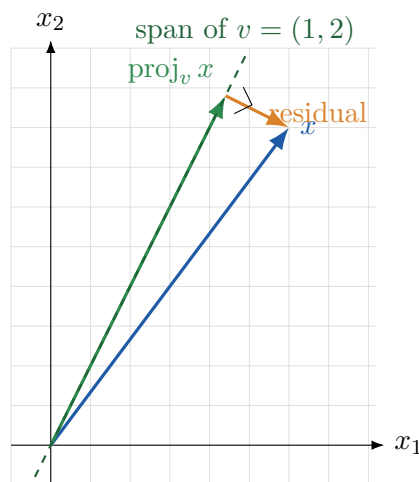


Figure 2: Projection decomposes a vector into a component along a direction and an orthogonal residual.

27.3 Derivatives: local change and local models

For a scalar function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the first derivative is the gradient

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right).$$

The gradient gives the best linear approximation, gives directional derivatives by dot product, and points in the direction of fastest increase. Near a ,

$$f(a + h) \approx f(a) + \nabla f(a) \cdot h.$$

The directional derivative in a unit direction u is

$$D_u f(a) = \nabla f(a) \cdot u.$$

The second derivative is organized by the Hessian matrix

$$H_f(a) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

The second-order Taylor approximation is

$$f(a + h) \approx f(a) + \nabla f(a) \cdot h + \frac{1}{2} h^T H_f(a) h.$$

This is the local geometry behind optimization, uncertainty approximation, and many algorithms in statistics and machine learning.

Example 27.2: Local quadratic geometry

For $f(x, y) = x^2 + 4y^2$, we have $\nabla f(0, 0) = 0$ and

$$H_f(0, 0) = \begin{bmatrix} 2 & 0 \\ 0 & 8 \end{bmatrix}.$$

The eigenvalues are positive, so the origin is a local minimum. The curvature is larger in the y -direction than in the x -direction.

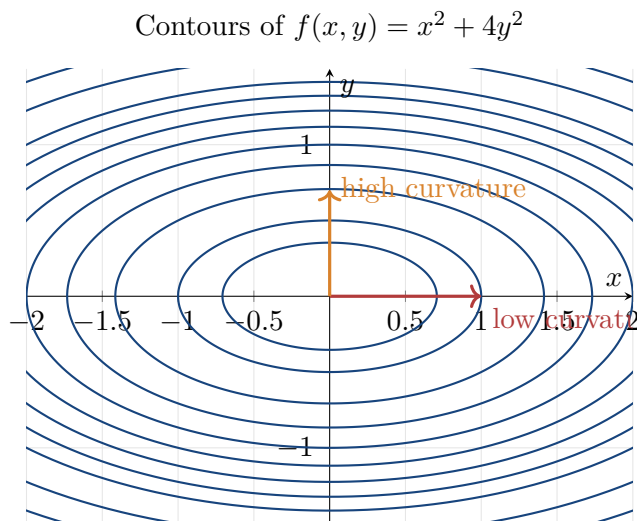


Figure 3: A quadratic function is controlled by its Hessian matrix. Narrower contours correspond to larger curvature.

27.4 Jacobians: derivatives of vector-valued maps

For a map $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the derivative is the Jacobian matrix

$$J_F(x) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \cdots & \frac{\partial F_m}{\partial x_n} \end{bmatrix}.$$

Near a ,

$$F(a + h) \approx F(a) + J_F(a)h.$$

The Jacobian is the linear map used in the multivariable chain rule. If $F = G \circ H$, then

$$J_F(x) = J_G(H(x))J_H(x).$$

In computation, this is the matrix form of backpropagation. In integration, the determinant of the Jacobian measures how area or volume changes under a coordinate transformation.

Example 27.3: The polar-coordinate Jacobian

The polar-coordinate map is

$$T(r, \theta) = (r \cos \theta, r \sin \theta).$$

Its Jacobian matrix is

$$J_T(r, \theta) = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}, \quad |\det J_T(r, \theta)| = r.$$

That is why $dA = r \, dr \, d\theta$.

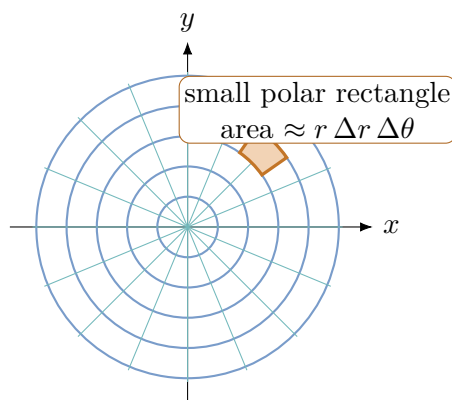


Figure 4: The polar-coordinate map stretches small rectangles by the Jacobian factor r .

27.5 Integrals: accumulation over geometric objects

A definite integral accumulates over an interval. Multivariable calculus extends this idea to regions, solids, curves, and surfaces.

Integral	Object	Common meaning
$\int_a^b f(x) \, dx$	interval	signed area, total change
$\iint_R f(x, y) \, dA$	plane region	volume, mass, probability
$\iiint_E f(x, y, z) \, dV$	solid region	mass, volume, expectation
$\int_C f \, ds$	curve	mass of a wire
$\int_C \mathbf{F} \cdot d\mathbf{r}$	oriented curve	work, circulation
$\iint_S f \, dS$	surface	mass of a shell
$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$	oriented surface	flux

In statistics, integrals compute probability, expectation, variance, and normalization constants. In applied mathematics, integrals compute mass, energy, flow, and conservation quantities. In machine learning, integrals appear in expected risk, Bayesian inference, generative models, and continuous approximations.

Example 27.4: Probability as a double integral

Suppose a joint density is

$$f(x, y) = 4xy, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1.$$

Then $\iint_{[0,1]^2} 4xy \, dA = 1$, so it is a valid density. The probability of the event $x + y \leq 1$ is

$$\int_0^1 \int_0^{1-x} 4xy \, dy \, dx = \int_0^1 2x(1-x)^2 \, dx = \frac{1}{6}.$$

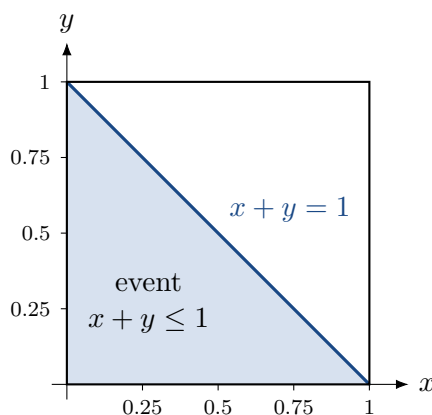


Figure 5: A probability over a two-dimensional event is an integral over a region.

27.6 Optimization: from critical points to algorithms

Optimization is one of the main bridges from calculus to data science and machine learning. The central objects are the gradient ∇f , the Hessian H_f , and constraint gradients ∇g_i .

For unconstrained optimization,

$$\nabla f(x) = 0.$$

For one constraint,

$$\nabla f(x) = \lambda \nabla g(x), \quad g(x) = c.$$

For two constraints,

$$\nabla f(x) = \lambda \nabla g(x) + \mu \nabla h(x), \quad g(x) = c, \quad h(x) = d.$$

Gradient descent uses

$$x_{k+1} = x_k - \alpha \nabla f(x_k),$$

while Newton's method uses curvature:

$$x_{k+1} = x_k - H_f(x_k)^{-1} \nabla f(x_k).$$

The graduate-level habit is to ask not only where the optimum is, but also whether the problem is well-conditioned, whether the Hessian is positive definite, and whether constraints or regularization change the geometry.

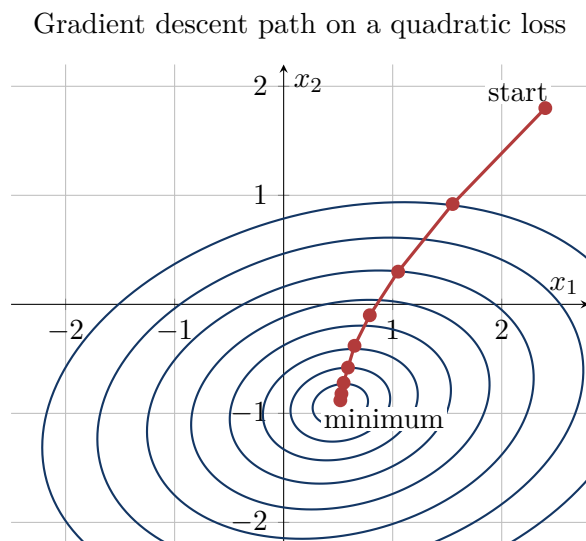


Figure 6: Gradient descent follows local negative-gradient directions on a loss landscape.

27.7 Vector fields and the global theorems

Vector calculus studies fields that assign a vector to each point. The main differential operators are

$$\operatorname{div} F = \nabla \cdot F, \quad \operatorname{curl} F = \nabla \times F.$$

Divergence measures local source or sink behavior. Curl measures local rotation. The global theorems convert integrals over boundaries into integrals over interiors.

The four Fundamental-Theorem patterns

Theorem	Boundary integral	Interior integral	Meaning
FTC	$f(b) - f(a)$	$\int_a^b f'(x) dx$	endpoint change equals accumulated derivative
Line integral FTC	$f(B) - f(A)$	$\int_C \nabla f \cdot dr$	potential difference equals work
Green's theorem	$\oint_C P dx + Q dy$	$\iint_R (Q_x - P_y) dA$	boundary circulation equals interior curl
Stokes theorem	$\oint_{\partial S} F \cdot dr$	$\iint_S (\nabla \times F) \cdot n dS$	boundary circulation equals surface curl flux
Divergence theorem	$\iint_{\partial E} F \cdot n dS$	$\iiint_E \nabla \cdot F dV$	boundary flux equals interior source

The pattern is powerful: a global boundary measurement is controlled by a local derivative inside.

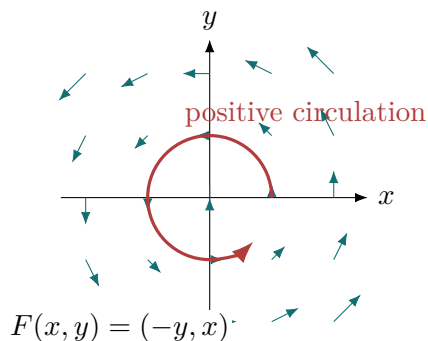


Figure 7: A rotational vector field illustrates circulation around closed curves.

27.8 A problem-solving decision tree

A good problem-solving strategy is often more useful than a memorized formula.

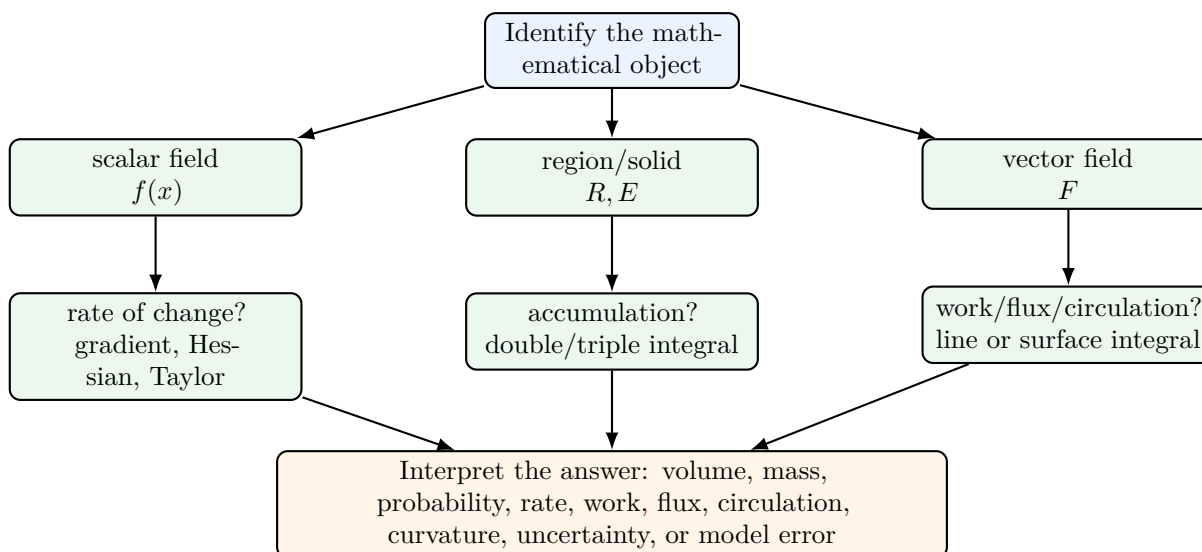


Figure 8: Decision tree: identify the object, identify the action, then interpret the result.

27.9 A compact formula sheet with meaning

Derivatives

$$\begin{aligned}
 D_u f(a) &= \nabla f(a) \cdot u, \quad \|u\| = 1, \quad f(a+h) \approx f(a) + \nabla f(a) \cdot h, \\
 f(a+h) &\approx f(a) + \nabla f(a) \cdot h + \frac{1}{2} h^T H_f(a) h, \quad J_{G \circ F}(x) = J_G(F(x)) J_F(x).
 \end{aligned}$$

Optimization

$$\nabla f = 0, \quad \nabla f = \lambda \nabla g, \quad x_{k+1} = x_k - \alpha \nabla f(x_k).$$

Multiple integrals and coordinate factors

$$\iint_R f(x, y) dA, \quad \iiint_E f(x, y, z) dV,$$

$$dA = r dr d\theta, \quad dV = r dr d\theta dz, \quad dV = \rho^2 \sin \phi d\rho d\phi d\theta.$$

Vector calculus

$$\int_C F \cdot dr, \quad \iint_S F \cdot n dS, \quad \nabla \cdot F, \quad \nabla \times F.$$

$$\oint_{\partial R} P dx + Q dy = \iint_R (Q_x - P_y) dA,$$

$$\oint_{\partial S} F \cdot dr = \iint_S (\nabla \times F) \cdot n dS, \quad \iint_{\partial E} F \cdot n dS = \iiint_E \nabla \cdot F dV.$$

27.10 Computational capstone: a mini calculus toolkit

This section builds a small numerical toolkit. The goal is not to replace exact mathematics, but to make the formulas executable. A graduate student should be able to move between exact formulas, numerical approximations, and visual checks.

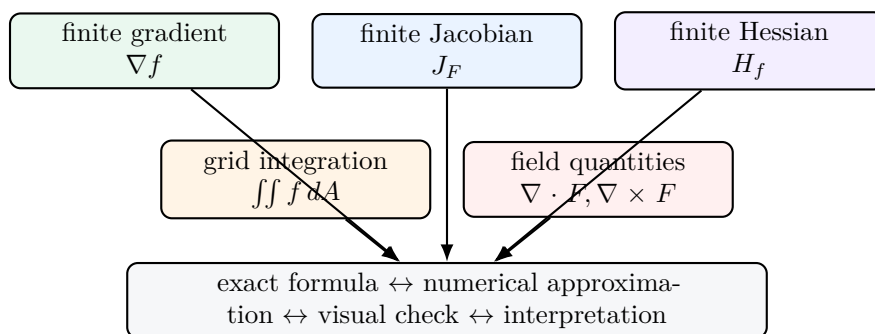


Figure 9: A mini calculus toolkit turns derivative and integral formulas into executable approximations.

Listing 1: Finite differences for gradients and Hessians.

```

def finite_gradient(f, x, h=1e-6):
    g = zeros_like(x)
    for j in range(len(x)):
        e = coordinate_vector(j)
        g[j] = (f(x + h*e) - f(x - h*e))/(2*h)
    return g
  
```

```
def finite_hessian(f, x, h=1e-4):
    H = zeros((len(x), len(x)))
    for i in range(len(x)):
        for j in range(len(x)):
            H[i,j] = centered_second_difference(f, x, i, j, h)
    return H
```

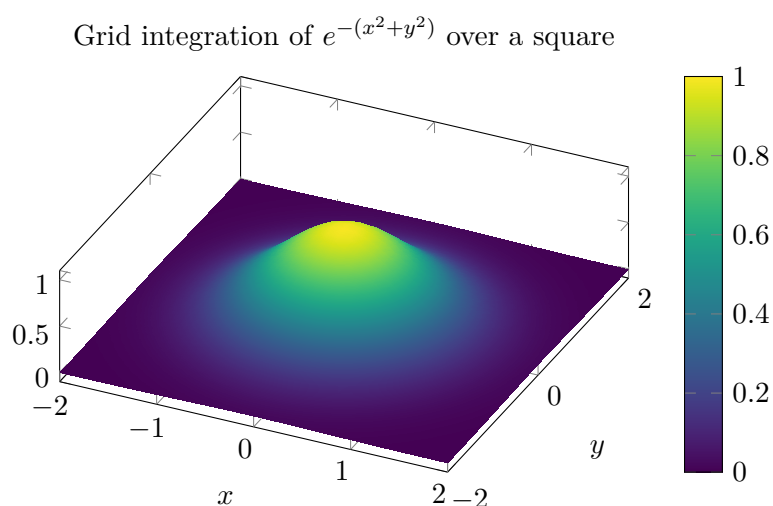


Figure 10: Grid integration approximates accumulation over a rectangle.

Numerical computation is not proof

A numerical approximation can suggest a pattern, check an answer, or make a concept visible. It does not replace a proof. In graduate work, the best habit is to combine computation with exact reasoning: use computation to explore, and use mathematics to explain.

27.11 Graduate-study bridge

The following table shows how the main ideas of this book reappear in later courses.

Later area	Multivariable calculus foundation
Real analysis	limits, continuity, differentiability, rigorous approximation
Linear algebra	vector spaces, projections, hyperplanes, quadratic forms
Optimization	gradients, Hessians, convexity, constraints, descent methods
Probability	joint densities, transformations, expectation, covariance
Mathematical statistics	likelihood, score, Hessian, Fisher information, delta method
Machine learning	loss functions, backpropagation, gradient descent, regularization
Numerical analysis	finite differences, numerical integration, error analysis
Differential equations	vector fields, gradient flow, divergence, curl, conservation laws
Partial differential equations	gradient, divergence, Laplacian, flux, boundary conditions
Geometry and topology	curves, surfaces, orientation, Stokes-type theorems

The most important preparation is not memorizing every formula. It is learning how to translate a new problem into the correct mathematical object.

27.12 Capstone project: build a calculus engine

A strong final project for this book is to build a small package or notebook called `calculus_engine`. It should include:

1. `finite_gradient(f, x)`
2. `finite_jacobian(F, x)`
3. `finite_hessian(f, x)`
4. `integrate_rectangle(f, xlim, ylim)`
5. `monte_carlo_integral(f, sampler)`
6. `line_integral(F, r, a, b)`
7. `surface_flux(F, r, u_range, v_range)`
8. `gradient_descent(f, grad, x0, alpha)`
9. `newton_step(f, grad, hessian, x)`
10. `plot_contours(f)`

The project should include examples from probability, optimization, and vector calculus, together with written interpretations of each output.

Example project theme: one dataset, many calculus ideas

Choose a simple dataset with input-output pairs (x_i, y_i) . Fit a linear regression model

$$\hat{y}_i = \beta_0 + \beta_1 x_i.$$

Then study the least-squares loss

$$L(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

This one example uses vectors for data, a scalar field for the loss landscape, gradients for optimization, Hessians for curvature, contour plots for visualization, linear approximation for sensitivity, and probability if errors are modeled as random noise.

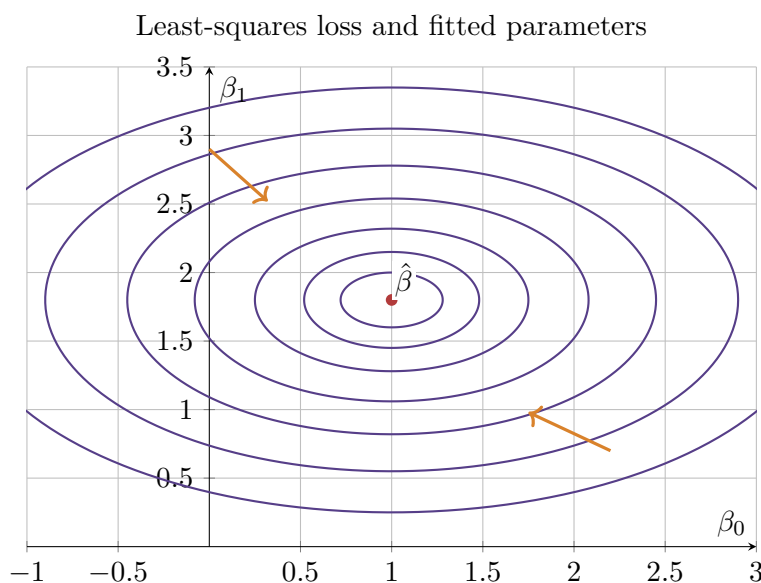


Figure 11: A single least-squares loss landscape connects vectors, gradients, Hessians, optimization, and statistics.

27.13 Common mistakes to avoid

1. **Confusing points and vectors.** A point gives location; a vector gives displacement. Coordinates can represent either, but the meaning differs.
2. **Forgetting that multivariable limits require all paths.** Checking one path is not enough to prove a limit exists.
3. **Using the gradient without normalizing the direction.** Directional derivatives use unit vectors.
4. **Forgetting boundary cases in optimization.** Critical points inside a region are not enough for absolute extrema.
5. **Missing the Jacobian factor.** Coordinate changes require area or volume scaling.
6. **Using the wrong orientation.** Line and surface integrals over oriented objects depend on direction.
7. **Applying Green, Stokes, or Divergence theorem without checking hypotheses.** Holes, singularities, and missing boundaries matter.

8. **Treating numerical output as exact.** Computation supports reasoning but does not replace it.

27.14 Cumulative worked example

A single field using many tools

Consider

$$f(x, y) = x^2 + xy + 2y^2.$$

The gradient is

$$\nabla f(x, y) = (2x + y, x + 4y), \quad \nabla f(1, -1) = (1, -3).$$

In the unit direction $u = \frac{1}{\sqrt{5}}(1, 2)$,

$$D_u f(1, -1) = (1, -3) \cdot \frac{1}{\sqrt{5}}(1, 2) = -\sqrt{5}.$$

The Hessian is

$$H_f = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}.$$

It is positive definite because its leading principal minors are positive. Therefore f is strictly convex and has a unique global minimum where $\nabla f = 0$, namely $(0, 0)$.

The average value over $[-1, 1] \times [-1, 1]$ is

$$\frac{1}{4} \int_{-1}^1 \int_{-1}^1 (x^2 + xy + 2y^2) dy dx.$$

By symmetry, the xy term integrates to zero, so the average is

$$\frac{1}{4} \left(\frac{4}{3} + \frac{8}{3} \right) = 1.$$

This one example uses gradients, directional derivatives, Hessians, optimization, symmetry, and double integrals.

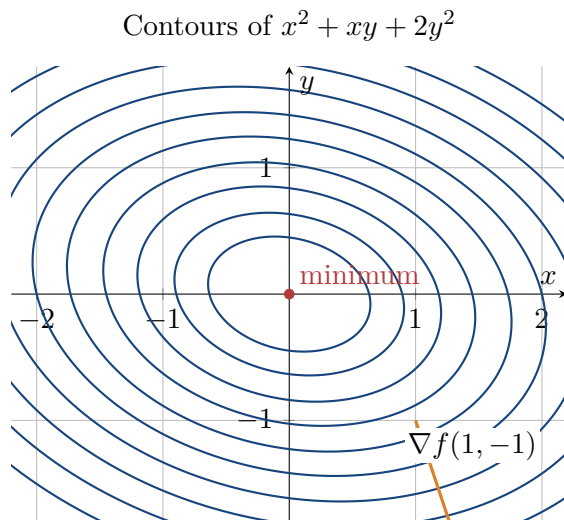


Figure 12: The cumulative example combines contour geometry, gradients, Hessians, and optimization.

27.15 Practice problems

Conceptual review

1. Explain the difference between a scalar field and a vector field. Give one example of each.
2. Explain why the gradient is perpendicular to level curves.
3. Explain the difference between a double integral over a region and a line integral over a curve.
4. Explain why the Jacobian determinant appears in change of variables.
5. Compare Green's theorem, Stokes theorem, and the Divergence theorem in one paragraph.

Core skills

6. Let $f(x, y) = x^2 + 3xy + y^2$. Compute ∇f , H_f , and the directional derivative at $(1, 2)$ in the direction $(3, 4)$.
7. Find the tangent plane to $z = x^2 + y^2$ at $(1, 2, 5)$.
8. Compute $\iint_R xy \, dA$, where $R = [0, 2] \times [1, 3]$.
9. Change $\int_0^1 \int_y^1 e^{x^2} \, dx \, dy$ to the opposite order.
10. Use polar coordinates to evaluate $\iint_R (x^2 + y^2) \, dA$, where R is the disk $x^2 + y^2 \leq 4$.

Vector calculus

11. For $F(x, y) = (-y, x)$, compute $\oint_C F \cdot dr$ where C is the unit circle oriented counterclockwise.

12. For $F(x, y, z) = (x, y, z)$, compute the flux through the sphere $x^2 + y^2 + z^2 = a^2$ using the Divergence theorem.
13. Find a potential function for $F(x, y) = (2xy, x^2 + 3y^2)$ if one exists.
14. Compute $\nabla \cdot F$ and $\nabla \times F$ for $F(x, y, z) = (yz, xz, xy)$.

Statistics, data science, and applied mathematics

15. Let $L(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$. Derive the gradient of L .
16. Let $f(x, y) = ce^{-(x^2+y^2)}$. Find c so that f is a probability density on $\mathbb{R}^2$.
17. For a differentiable loss function $L(\theta)$, explain how the Hessian near a minimum is related to curvature and uncertainty.
18. Write pseudocode for gradient descent and explain the role of the step size.
19. Explain why a conservation law naturally involves divergence.
20. Design one computational experiment that illustrates a major theorem from vector calculus.

27.16 Python lab and interactive page

- Notebook lab: `Chapter27_Review_Synthesis_and_Bridge_to_Graduate_Study.ipynb`
- Interactive HTML page: `../htmls/chapter-27.html`

Chapter summary

Final synthesis

This chapter organized multivariable calculus as a coherent language for modern quantitative work. Geometry describes the spaces and objects. Derivatives describe local change. Integrals describe accumulation. Optimization turns derivatives into algorithms. Vector calculus connects local field behavior with global boundary behavior. Probability and statistics reinterpret integrals as probabilities and expectations, and machine learning reinterprets gradients and Hessians as tools for fitting models.

The deepest lesson is that the same mathematical structures appear across many subjects. A tangent plane, a linear approximation, a least-squares projection, a gradient descent step, a probability density, a flux integral, and a conservation law are not isolated facts. They are connected parts of one multivariable calculus language.